Evidence edition · Historical analysis, not a forecast

Japan power: prices, fleets, regions and system stress

Regional physical systems, commercial obligations and the historical mechanisms behind price separation.

- Japan combines a nine-area wholesale market with ten regional physical systems. The eastern 50 Hz and western 60 Hz networks meet through frequency converters, so western surplus is not an unlimited substitute for eastern generation.
- Area prices separate by route and season: in summer 2026 Tohoku→Tokyo carried a mean difference of +4.05 JPY/kWh while Tokyo→Chubu was −0.90. Tokyo's northern and western exposures cannot be compressed into a single import premium.
- Load alone does not set the price. Tokyo's winter 2026 residual demand was higher than summer's, yet summer was dearer: 20.38 versus 11.48 JPY/kWh.
- January 2021 remains the extreme: Tokyo's monthly mean reached 66.53 JPY/kWh and an interval hit 252.00 JPY/kWh on 15 January, a month-long fuel-endurance episode rather than one afternoon's capacity shortfall.
- Eight event studies keep dated chronology, physical response, all-nine prices and matched references separate; an observed price split alone does not establish that a route bound the auction or that a single asset moved the price.

Coverage

Tokyo · Chubu · Kansai
10 regional systems
15 company profiles
8 event studies

Observation window

1 April 2024 – 31 October 2026
JST interval-start

Public-source historical analysis. Source-image credits and underlying rights remain with their publishers; public republication permission is not asserted.

Contents

Evidence and scope

Standard modern observation window: 2024-04-01 to 2026-10-31 inclusive, JST interval-start. Each exhibit retains its own observation clock and qualified history. Later documentary milestones are dated separately.

Fundamentals

Japan market structure, geography and price formation 4
Demand, weather and residual-load shapes 15
Supply, nuclear, renewables, storage and outages 39
LNG infrastructure, fuel procurement and generation economics 63
Companies, ownership and generation footprints 73
Regional markets, prices and transmission 93
Capacity, short-term products and commercial market structure 131
Environmental markets and obligations 144

Events and back matter

Eight cases: evidence before explanation p. 149

Historical cases

February 2026 northern cold shock 150
August 2026 heat and support phases 156
January 2021 energy and fuel stress 162
March 2022 compound eastern shock 168
February 2025 winter price episode 174
May 2026 Kyushu solar control 180
September 2024 holiday basis 186
September 2026 Mihama return and interruption 192

Registers, methods and sources

Cross-case observations 198
Specialist glossary 199
Equations and methods 200
Factual registers 204
Sources and reading guide 214

Region and company finder

Physical page links include each attributed exhibit across chapters.

Regional exhibits

Hokkaido 15 · 16 · 17 · 22 · 27 · 32 · 37 · 38 · 39 · 40 · 45 · 46 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 56 · 61 · 62 · 63 · 69 · 73 · 74 · 82 · 93 · 112 · 113 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 150 · 151 · 152 · 153 · 154 · 155 · 159 · 165 · 171 · 174 · 175 · 176 · 177 · 178 · 179 · 183 · 189 · 195

Tohoku 15 · 16 · 17 · 22 · 27 · 32 · 37 · 38 · 39 · 40 · 45 · 46 · 48 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 56 · 61 · 62 · 63 · 69 · 73 · 74 · 83 · 93 · 114 · 115 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 150 · 151 · 152 · 153 · 154 · 155 · 159 · 165 · 168 · 169 · 170 · 171 · 172 · 173 · 177 · 183 · 189 · 195

Tokyo 15 · 16 · 18 · 23 · 28 · 32 · 33 · 34 · 37 · 38 · 39 · 41 · 45 · 46 · 47 · 48 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 57 · 61 · 62 · 63 · 65 · 66 · 73 · 74 · 78 · 91 · 93 · 94 · 95 · 96 · 97 · 98 · 99 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 145 · 153 · 159 · 162 · 163 · 164 · 165 · 166 · 167 · 168 · 169 · 170 · 171 · 172 · 173 · 174 · 175 · 176 · 177 · 178 · 179 · 183 · 186 · 187 · 188 · 189 · 190 · 191 · 195

Chubu 15 · 16 · 18 · 23 · 28 · 32 · 33 · 35 · 37 · 38 · 39 · 41 · 45 · 46 · 47 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 57 · 61 · 62 · 63 · 67 · 73 · 74 · 76 · 79 · 93 · 100 · 101 · 102 · 103 · 104 · 105 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 153 · 156 · 157 · 158 · 159 · 160 · 161 · 165 · 168 · 169 · 170 · 171 · 172 · 173 · 177 · 183 · 186 · 187 · 188 · 189 · 190 · 191 · 192 · 193 · 194 · 195 · 196 · 197

Hokuriku 15 · 16 · 19 · 24 · 29 · 32 · 37 · 38 · 39 · 42 · 45 · 46 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 58 · 61 · 62 · 63 · 69 · 73 · 74 · 84 · 93 · 116 · 117 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 153 · 159 · 165 · 171 · 177 · 183 · 189 · 195

Kansai 15 · 16 · 19 · 24 · 29 · 32 · 33 · 36 · 37 · 38 · 39 · 42 · 45 · 46 · 47 · 48 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 58 · 61 · 62 · 63 · 68 · 73 · 74 · 80 · 81 · 93 · 106 · 107 · 108 · 109 · 110 · 111 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 153 · 156 · 157 · 158 · 159 · 160 · 161 · 162 · 163 · 164 · 165 · 166 · 167 · 171 · 177 · 183 · 189 · 192 · 193 · 194 · 195 · 196 · 197

Chugoku 15 · 16 · 20 · 25 · 30 · 32 · 37 · 38 · 39 · 43 · 45 · 46 · 48 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 59 · 61 · 62 · 63 · 69 · 73 · 74 · 85 · 93 · 118 · 119 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 153 · 156 · 157 · 158 · 159 · 160 · 161 · 165 · 171 · 177 · 180 · 181 · 182 · 183 · 184 · 185 · 189 · 192 · 193 · 194 · 195 · 196 · 197

Shikoku 15 · 16 · 20 · 25 · 30 · 32 · 37 · 38 · 39 · 43 · 45 · 46 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 59 · 61 · 62 · 63 · 69 · 73 · 74 · 86 · 93 · 120 · 121 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 153 · 159 · 165 · 171 · 177 · 183 · 189 · 195

Kyushu 15 · 16 · 21 · 26 · 31 · 32 · 37 · 38 · 39 · 44 · 45 · 46 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 60 · 61 · 62 · 63 · 69 · 73 · 74 · 87 · 93 · 122 · 123 · 126 · 127 · 128 · 129 · 130 · 135 · 136 · 153 · 159 · 162 · 163 · 164 · 165 · 166 · 167 · 171 · 174 · 175 · 176 · 177 · 178 · 179 · 180 · 181 · 182 · 183 · 184 · 185 · 189 · 195

Okinawa 15 · 16 · 21 · 26 · 31 · 32 · 37 · 38 · 39 · 44 · 46 · 49 · 50 · 51 · 52 · 53 · 54 · 55 · 60 · 61 · 62 · 63 · 69 · 73 · 74 · 88 · 93 · 124 · 125 · 126 · 127 · 128 · 129 · 130

Company exhibits

TEPCO 73 · 74 · 75

Chubu 73 · 74 · 76

JERA 73 · 74 · 77 · 78 · 79

Kansai 73 · 74 · 80 · 81

Hokkaido 73 · 74 · 82

Tohoku 73 · 74 · 83

Hokuriku 73 · 74 · 84

Chugoku 73 · 74 · 85

Shikoku 73 · 74 · 86

Kyushu 73 · 74 · 87

Okinawa 73 · 74 · 88

J-POWER 73 · 74 · 89

JAPC 73 · 74 · 90

Tokyo Gas 73 · 74 · 91

Daigas 73 · 74 · 92

Japan market structure, geography and price formation

Key features of Japan's power market

Japan combines a nine-area wholesale market with ten regional physical systems. Okinawa is outside the mainland interconnection and has no JEPX area-price series. Within the interconnected system, the eastern 50 Hz and western 60 Hz networks meet through frequency converters: western surplus is therefore not an unlimited substitute for eastern generation. [14] [15]

- Geography makes neighbouring supply valuable—but not interchangeable
- The system price is not a national average of area prices
- Support depends on the route and the duration of the shortfall

Market feature	Why it matters in Japan
Regional rather than nodal spot prices	Inter-area constraints can separate JEPX prices; local congestion is managed through network and redispatch arrangements.
Different nuclear and renewable footprints	A national fuel price does not produce the same residual demand or thermal requirement in every area.
Coastal thermal supply and imported fuel	Cargo, terminal and gas-network conditions matter alongside fuel benchmarks and the yen.
Large demand centres connected to smaller systems	Tokyo's northern imports and the central frequency-converter route expose it to different neighbouring balances.
Energy, capacity and balancing obligations coexist	A capacity award is not a spot-energy purchase, and a balancing offer creates its own availability duties.

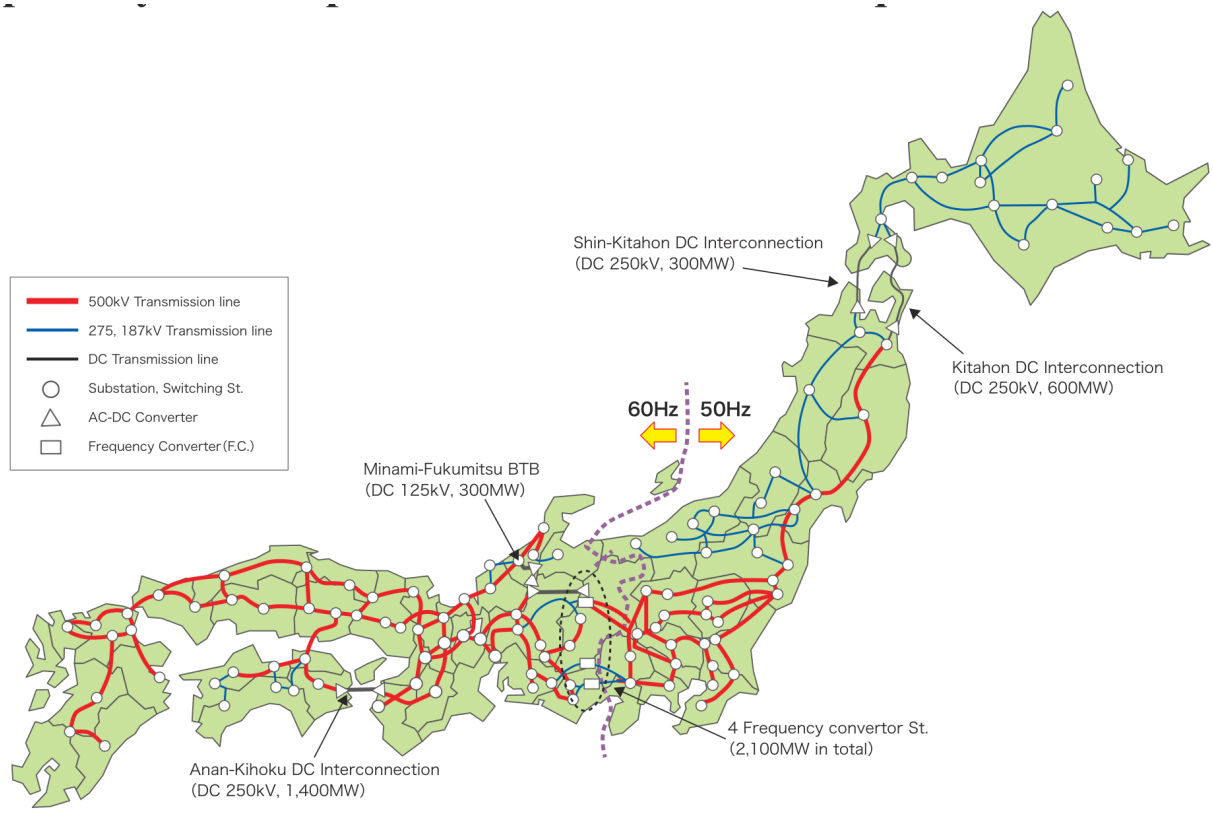
The price geography is economically substantial. Across September 2025–August 2026, Tokyo averaged **JPY 15.51/kWh**, Chubu **14.44**, Kansai **12.72** and Shikoku **9.68**. Those differences reflect the combination of regional offers and constrained exchange, not a different national fuel benchmark for each region. The regional chapters examine how nuclear, solar, demand and network conditions accompanied the differences. [1]

The chapter follows three questions: **where power can travel**, **who operates and clears the markets**, and **which obligations remain after procurement**. These features recur throughout the demand, fleet and event analysis; they are the starting observations, not a set of trading forecasts. [16] [217]

Geography makes neighbouring supply valuable—but not interchangeable

Japan's main grid follows the archipelago from Hokkaido to Kyushu. Demand is concentrated in the large metropolitan and industrial corridors, while major nuclear, hydro and renewable resources need not sit beside those centres. Long-distance transfers connect these different balances, but coastal generation, mountain terrain and the available routes give each region a distinct exposure. [17]

Exhibit 1: Japan grid geography in the supplied 2025 reference



Note: This extract from page 34 of JEPIC's July 2025 publication retains the dates shown in the original. The ratings reflect the publication's source edition and should not be read as verified operating limits at this edition's cutoff. Fine annotations can be enlarged in the accompanying original PDF. Credit: General Electricity Transmission and Distribution Utility Website. © 2025 JEPIC. All rights reserved.

Source: [17]

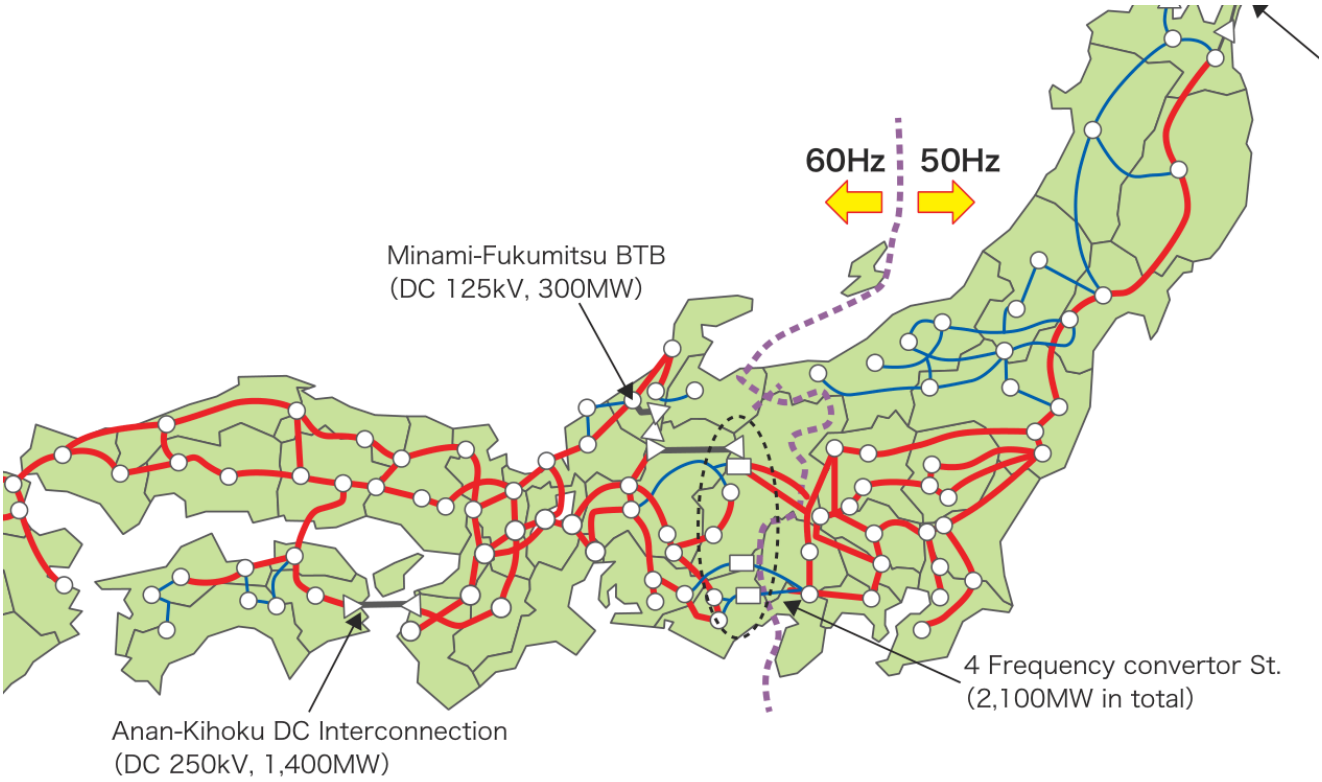
Tokyo can draw on Tohoku to the north and the frequency-converter route to the west; those routes reach different generation fleets. Kyushu reaches the mainland through Chugoku, while Shikoku has separate connections to Chugoku and Kansai. A surplus in one of these systems can depress its local price without eliminating a premium several boundaries away. [14] [329]

Okinawa's separation is more fundamental: neither a mainland import nor a mainland JEPX area price is available to resolve its island-system balance. Its cooling demand, local generation and fuel logistics therefore receive their own treatment rather than a fabricated tenth wholesale-price panel. [15]

The frequency boundary shapes Tokyo–Chubu–Kansai exchange

The central network joins Japan's two frequency systems through conversion facilities in Nagano and Shizuoka. East–west exchange therefore depends on controlled conversion capacity as well as the AC networks on either side. Tokyo, Chubu and Kansai are commercially adjacent price areas, but they are not one unconstrained synchronous pool. [14]

Exhibit 2: Enlarged central-Japan grid detail



Note: This extract from page 34 of JEPIC's July 2025 publication retains the dates shown in the original. The crop shows central Japan rather than an electrical-area boundary, with red lines for 500 kV, green for 275 or 187 kV, and dark lines for DC connections. The July 2025 ratings do not indicate how much supply is available. Credit: General Electricity Transmission and Distribution Utility Website. © 2025 JEPIC. All rights reserved.

Source: [17]

Chubu occupies the central position: its eastern basis is against Tokyo, while its western connections reach Kansai and Hokuriku. The value of supply on one side can change without the same movement on the other. This is why Chubu's price history belongs beside both its industrial demand pattern and the changing regional supply around it. [17] [1]

Electrical geography also cuts across administrative labels. A plant's connection determines the regional balance it supplies; its prefecture or parent company alone does not. Operating fences can constrain several paths together and must not be counted as additional physical cables. The regional transmission analysis separates individual connections, their directions and the central publication changes. [330]

Japan separates policymaking, market operation and physical dispatch

OCCTO coordinates the interconnected system, but it is not a single vertically integrated national dispatcher. Regional transmission and distribution operators retain network-operation responsibilities. JEPX clears wholesale energy and other specified products; EPRX operates the balancing-capability market. A market award and an operational instruction consequently come from different institutional functions. [18] [19] [16]

Exhibit 3: Who makes rules, clears markets and operates the system

Who makes rules, clears markets and operates the system

METI / ANRE	Energy policy and market design Surveillance commission: competition and trading conduct
NRA	Nuclear safety regulation Safety permission is distinct from a generator's operating decision
OCCTO	Cross-regional coordination and capacity market Network planning, adequacy assessment and support coordination
Regional TSOs	Network operation and balancing delivery Connection, congestion management and physical dispatch instructions
JEPX / EPRX	Energy exchange / balancing-capability exchange Separate market operators; neither replaces the TSOs
Market participants	Generators, storage, DR aggregators and retailers Offers, supply plans, performance and payments under each contract

Note: Japan-specific responsibilities and contract sequence; applicable delivery-year rules govern exceptions.
Source: [18–20]

For a trader, the division matters when interpreting an announcement. An NRA safety decision concerns the regulatory position of a reactor; it is not the same event as first generation. An OCCTO support instruction concerns system coordination; a JEPX price is the result of the exchange auction. The same physical stress can appear in all three channels on different dates. [20] [21]

Generators and aggregators offer resources under product-specific conditions. Retailers procure for customer demand and bear the corresponding planning and settlement obligations. These commercial roles cross the inherited regional-company boundaries, even though the networks and regional fleets remain important to price formation. [18]

Liberalisation changed the participants, not Japan's geography

The reforms separated functions without removing the regional network inheritance. OCCTO began in April 2015, full retail entry liberalisation followed in April 2016, and legal unbundling of transmission and distribution followed in April 2020. The practical consequence is that a familiar utility name no longer identifies a single business with all the historical functions. [18]

Change	Lasting commercial consequence
Cross-regional coordination, 2015	Network planning and coordination have a national institution distinct from individual operators.
Full retail entry, 2016	Customers can contract with new retailers; the local network operator still delivers electricity.
Legal network unbundling, 2020	Network operation must be distinguished from the affiliated generation and retail businesses.
Exchange-based procurement alongside bilateral supply	Exchange turnover describes transactions, not the whole of customer consumption or every retailer's procurement mix.

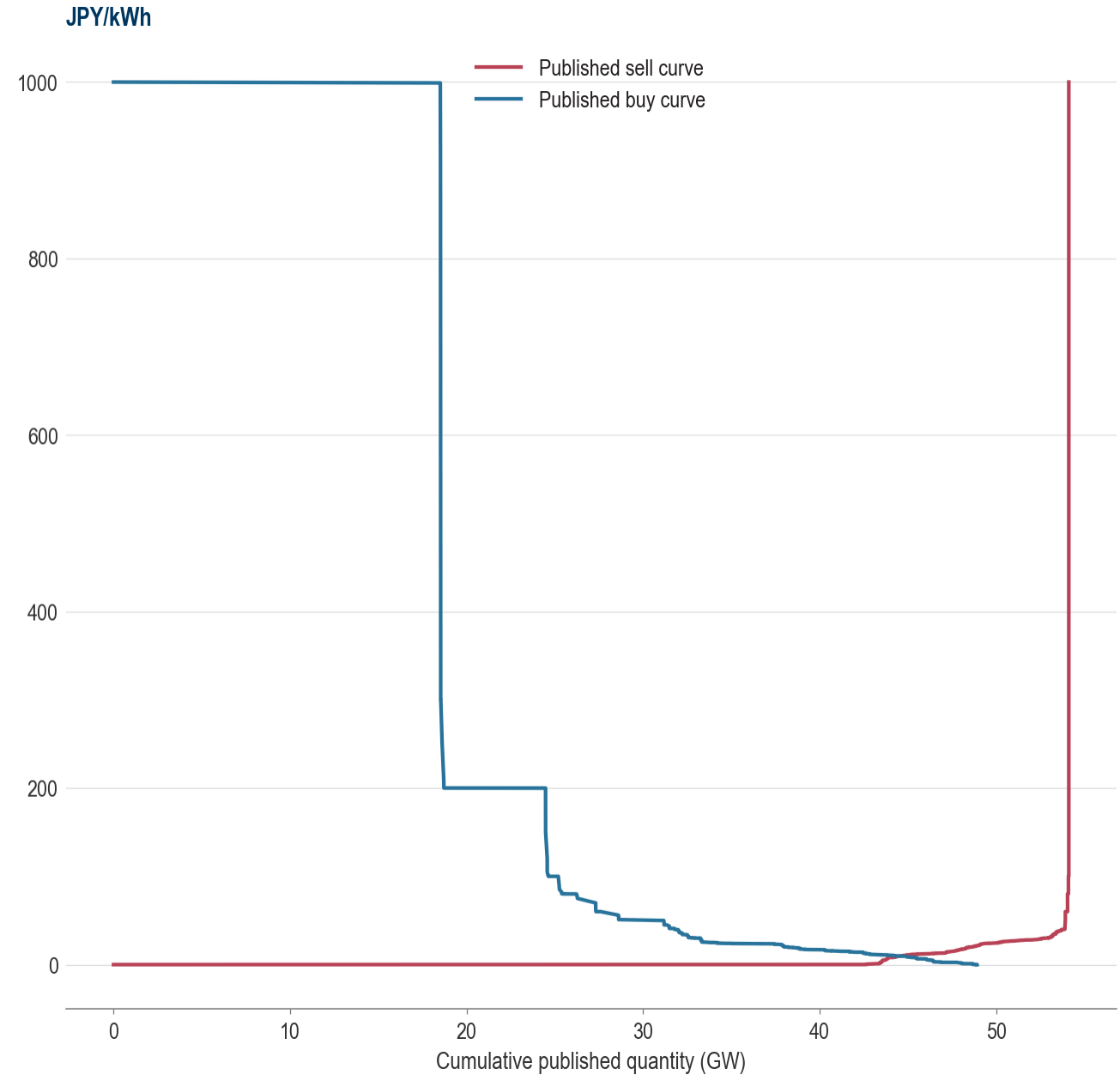
TEPCO's holding, network and retail entities should not be treated as interchangeable with JERA's generation business. Similarly, Chubu's direct generation footprint is not JERA's entire joint-venture fleet. The company chapter identifies operating entities and ownership relationships without counting the same physical MW again under both parents. [20]

Gross bidding and changes in trading arrangements can increase exchange volume without an equivalent increase in electricity consumed. Conversely, a large affiliated portfolio does not reveal how much a retailer purchases through JEPX on a particular day. The useful distinction is between the observed exchange, the physical regional balance and the actual entities holding the obligations—not an assumed confidential procurement share. [20] [16]

JEPX clears 48 half-hour products through a single-price auction

JEPX day-ahead is a **zonal market, not bus-level LMP**. Ordinary bids close at **10:00 on the preceding day**. Each order specifies its delivery area and planning code, with 50 kWh quantity increments and JPY 0.01/kWh price increments. Accepted trades settle at the applicable area price. [22]

Exhibit 4: Reading the published auction curves



Note: The two lines reproduce the frozen system-price curve for 27 September 2026 at noon. Their quantity axis describes published exchange orders, not national available generation; block treatment and area splitting prevent a plant-level reading.

Source: [218] · Chart data

JEPX aggregates orders into supply and demand curves. If the preliminary allocation breaches interconnection capacity notified by OCCTO or another applicable constraint, it clears again with those constraints and area prices can separate. Network access is part of market clearing, not a private physical path secured by each exchange buyer. [22]

Block orders connect consecutive products. Acceptance depends on quantity-weighted prices across the block; an apparently profitable block may remain unaccepted. The published curve shows offered quantities and prices for its stated interval, but neither every participant's commitment decision nor a specific marginal generator. The following pages examine the resulting area-price geography. [22]

The system price is not a national average of area prices

JEPX's **system price is the auction result calculated without market splitting**. It is not the average of the nine area prices. The actual area settlement prices incorporate the constrained clearing result, so a national system-price hedge and a local procurement position can remain materially different. [16]

Region	Mean area price	Area − system	Unequal intervals
Hokkaido	13.06	-0.23	93.5%
Tohoku	13.15	-0.14	91.6%
Tokyo	15.51	+2.22	92.5%
Chubu	14.44	+1.14	91.4%
Hokuriku	12.98	-0.32	91.6%
Kansai	12.72	-0.58	91.6%
Chugoku	11.90	-1.40	91.6%
Shikoku	9.68	-3.61	93.8%
Kyushu	11.10	-2.19	91.7%

Prices and differences are JPY/kWh; September 2025–August 2026, all 17,520 half-hours per area. The system-price mean was **13.29**. [1]

Tokyo's mean premium to system was **2.22**, while Kansai's mean difference was **−0.58** and Shikoku's **−3.61**. Those persistent differences show why national price exposure is not a complete description of a retailer's or generator's regional position. The direction is consistent with differing fleet and demand balances constrained by the network; the relative contributions are examined with the physical histories rather than inferred from the price difference alone.

Most areas differed from system in more than 90% of these intervals, but that statistic is not a count of independently congested lines. The unsplit calculation can differ from a whole group of equal area prices. Nor does equal pricing establish that a physical route was unconstrained. Internal congestion is addressed through the network framework without automatically creating a plant-specific JEPX price. [16] [23]

Typical neighbouring spreads change with the season

Tokyo's premium over Tohoku was much larger in spring and summer 2026 than in winter. Across the central boundary, Chubu was cheaper than Tokyo on average in every displayed season, while Kansai was cheaper again than Chubu. The pattern is not a fixed tariff between regions: the sign, persistence and tail all matter. [1]

Origin → destination	Winter	Spring	Summer	Autumn
Hokkaido → Tohoku	-1.10/+0.00	+0.15/+0.00	+1.94/+0.00	-0.67/+0.00
Tohoku → Tokyo	+0.32/+0.00	+3.81/+0.00	+4.05/+0.00	+1.20/+0.00
Tokyo → Chubu	-0.48/+0.00	-1.92/+0.00	-0.90/+0.00	-0.99/+0.00
Chubu → Hokuriku	-0.51/+0.00	-2.48/+0.00	-2.29/-0.51	-0.53/+0.00
Chubu → Kansai	-0.74/+0.00	-2.51/+0.00	-2.67/-1.49	-0.91/+0.00
Hokuriku → Kansai	-0.23/+0.00	-0.04/+0.00	-0.38/+0.00	-0.38/+0.00
Kansai → Chugoku	-0.03/+0.00	-1.11/+0.00	-1.84/+0.00	-0.29/+0.00
Chugoku → Kyushu	-0.25/+0.00	-1.77/+0.00	-0.86/+0.00	-0.27/+0.00
Chugoku → Shikoku	-1.27/+0.00	-4.20/-1.85	-2.14/+0.00	-1.21/+0.00
Kansai → Shikoku	-1.30/+0.00	-5.31/-3.09	-3.97/+0.00	-1.50/+0.00

Each cell is **mean / median**, destination minus origin, JPY/kWh. Winter = December 2025–February 2026; spring/summer = MAM/JJA 2026; autumn = SON 2025. All four are complete seasons. [1]

A zero median often coexists with substantial tail exposure. Tohoku → Tokyo's summer p10–p90 range was **0.00 to 13.61**; Chubu → Kansai's was **−7.13 to 0.00**. Kansai → Shikoku's spring range reached **−13.77 to 0.00**, consistent with a much cheaper Shikoku balance in many daytime intervals. Fleet composition, demand and renewable timing explain why these differences should not be treated as one nationwide fuel move.

The regional and boundary chapters connect these observations to the physical conditions and publish the full distributions. A price separation alone does not establish which line bound or in which direction power actually flowed. [16]

Japanese procurement combines portfolio supply with exchange adjustment

A retailer's supply can come from affiliated generation, bilateral contracts, PPAs, baseload-market procurement and JEPX purchases. These channels coexist rather than forming a single mandatory path through the day-ahead exchange. The procurement mix determines how much short-term price and shape exposure remains, while the physical delivery area determines which regional balance serves the customer. [20] [219]

Exhibit 5: Japan's procurement chain and remaining obligations

Japan's procurement chain and remaining obligations

1 · Contract supply	Affiliated generation, bilateral contracts, PPAs Baseload market access and exchange procurement complement these
2 · Day-ahead	JEPX: 48 half-hours; ordinary cutoff D−1 at 10:00 Orders identify delivery area, quantity, price and planning code
3 · Register plans	Accepted sales / procurement enter OCCTO plans The planning code connects the trade to delivery responsibilities
4 · Adjust & deliver	Intraday adjustment; TSO balancing in real time Uncovered deviations enter the applicable imbalance settlement
5 · Pay & comply	JEPX settlement: two banking business days after notice Network charges, capacity contributions and attributes remain separate

Note: Japan-specific responsibilities and contract sequence; applicable delivery-year rules govern exceptions.
Source: [16] [217] [220]

JEPX accepted quantities enter sales and procurement plans under the registered planning codes. Later changes in demand, renewable output or plant availability can require intraday adjustment; residual deviations face the applicable imbalance rules. Buying day-ahead energy therefore does not extinguish the obligation to keep the delivery plan aligned with actual consumption. [16]

Cross-area procurement adds a separate regional exposure. Indirect transmission rights settle eligible directional price differences under their terms; they are not a reserved physical cable. An affiliated or long-term supply contract may reduce spot procurement, but neither it nor a financial hedge alone guarantees delivery through a constrained network. The detailed market-products chapter examines the actual Japanese instruments and obligations. [24]

Cash timing and non-energy obligations remain material for retailers

Under the captured JEPX rules, ordinary spot settlement falls **two banking business days after the trade notification**. Deposits are required, and failure to provide the prescribed deposit can suspend trading. A retailer can consequently face a funding need before customer receipts arrive even when its longer-dated commercial position is hedged. [16]

Obligation	Japan-specific responsibility
Exchange energy and fees	JEPX settlement follows the exchange notice and banking calendar.
Delivery plans and deviations	Accepted trades enter plans; deviations are handled under the applicable imbalance arrangements.
Network service	The relevant network contract governs delivery and associated charges.
Capacity contribution	OCCTO's applicable delivery-year contribution rules allocate the funding obligation.
Environmental claims	Non-fossil instruments and PPA/FIT/FIP arrangements determine which attributes can be claimed.

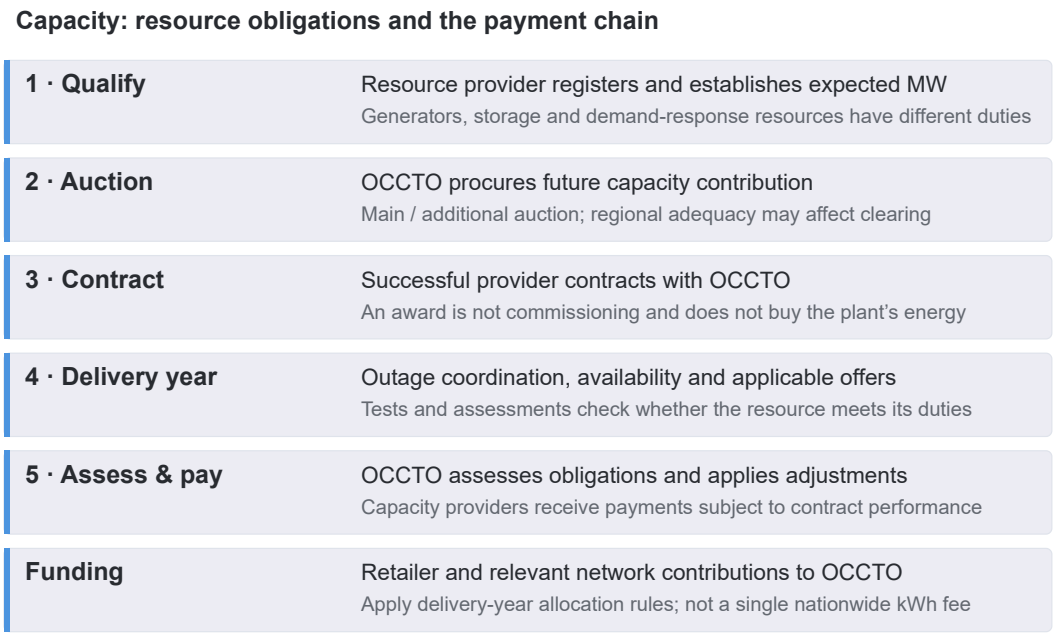
These cash flows do not share a single invoice date or price basis. A capacity contribution funds qualifying future capability; it does not purchase the retailer's immediate energy. A non-fossil certificate supports a permitted environmental claim; it does not cure an energy shortfall. The separate obligations explain why retail economics can deteriorate even when the system continues to supply customers. [217] [220]

Fuel-adjustment arrangements add another timing difference because the applicable regulated mechanism refers to earlier import-price averages rather than the current day's spot price. Contract-specific pass-through matters: a wholesale shock need not reach customer tariffs immediately or uniformly. The fuel chapter connects that lag to Japan's LNG and imported-fuel exposure. [30]

Capacity procurement connects resource providers to future obligations

Japan's capacity market procures a future contribution to supply adequacy through OCCTO, alongside—not instead of—the energy market. The connected nine-area market excludes Okinawa. Regional adequacy and transmission deliverability can affect outcomes, so the system should not be read as either ten independent capacity auctions or an unconditional single price for every resource. [217] [221]

Exhibit 6: Capacity: resource obligations and the payment chain



Note: Japan-specific responsibilities and contract sequence; applicable delivery-year rules govern exceptions.

Source: [217] [221–223]

The providers include generation, storage and demand-response resources, with different qualification and performance conditions. Retailers fund capacity through the applicable contribution arrangements; they are not buying a specified parcel of next-day energy from the awarded provider. Regional network responsibilities remain part of the delivery-year assessment and coordination framework. [222] [223]

For regional price analysis, the distinction between awarded expected capacity and operating plant is decisive. A future award can support investment or retention, but it is not extra commissioned MW today. The supply chapters trace operating assets; the capacity-results pages separately compare auction outcomes, resource classes and delivery years.

An awarded resource still has to perform in the delivery year

The ordinary capacity sequence begins with resource registration and expected-capacity determination, followed by bidding, clearing and a contract with OCCTO. Before and during the delivery year, outage coordination and the resource's availability, market-offer or activation obligations apply. Assessments and adjustments then determine whether the contracted contribution was actually provided. [221] [222]

Participant	Typical process and commercial exposure
Generator	Qualifies the resource, contracts expected MW and meets applicable availability and market-offer obligations; maintenance timing can affect compliance.
Storage provider	Must meet the applicable resource and duration requirements; connected power rating alone is not unlimited deliverable energy.
Demand-response aggregator	Contracts qualifying demand reduction and must demonstrate delivery under the relevant tests and activation conditions.
Retailer	Pays contributions under delivery-year regional allocation and adjustment rules.
OCCTO / regional networks	Coordinate adequacy and relevant resource information; assessment and payments follow the applicable manuals.

Main and additional auctions serve different procurement stages. Long-term decarbonisation auctions add a distinct investment-support framework and conditions; their awarded projects must not be mixed into present-day operating capacity. Auction year, delivery year and commissioning date are separate events. [217] [221]

This structure links the capacity market to the maintenance and availability cycle without making capacity payments a mechanical spot-price setter. A reactor inspection, a thermal outage and a demand-response test can matter differently for future capacity obligations and for today's energy offers. The later product chapter gives the dated auction results and payment details rather than assuming the same rules applied in every historical year. [222]

Non-firm access expands entry while retaining curtailment exposure

Japan's Connect & Manage reforms allow fuller use of existing networks, but a connection does not promise unrestricted generation in every interval. Non-firm access began for the specified congested main-grid population on 13 January 2021, expanded to the specified main-grid voltage applications from April 2022 and reached local networks from April 2023. Redispatch broadened from adjustment resources in December 2022 to the prescribed wider order in December 2023. [23]

Constraint	Physical response and commercial consequence
Regional supply surplus	Thermal reductions, pumping and feasible exports help absorb output; the priority rules govern further control.
Internal network congestion	Generation may be reduced in a constrained location even when demand remains elsewhere in the area.
Inter-area constraint	Limited exchange can accompany different JEPX area prices; it does not create a price at every internal bus.
Conditional non-firm access	The project can connect subject to control, leaving output-volume and capture exposure.

Balance curtailment and network curtailment can coincide, but they are not the same mechanism. ANRE's priority sequence reflects the technical characteristics of the resources, rather than assuming all thermal, nuclear and hydro output can be reduced instantaneously. Online proxy control also separates the plant making the physical reduction from the allocation of its economic consequences. [25]

The market implication is locational: rapid renewable entry can make the midday balance cheaper while connection-specific restrictions still limit a project's saleable output. Kyushu's solar-control case and the regional renewable histories examine those conditions alongside pumping and export opportunities, rather than treating every low price as proof of curtailment. [23]

Support depends on the route and the duration of the shortfall

OCCTO's cross-regional coordination works through the resources and networks that are available at the time. Regional operators must also respect internal congestion, which can restrict delivery even when a neighbouring area has generation to spare. The August 2024 Kansai network discussion illustrates why an area-wide reserve figure alone cannot establish that every load centre can receive that surplus. [21]

System condition	Response that can matter
Short-lived residual-load ramp	Flexible thermal output, hydro/storage and feasible inter-area adjustment.
Persistent high energy requirement	Fuel replenishment and sustained generation, not only a high turbine rating.
Local network restriction	Redispatch and route-specific operating measures.
Broad weather or outage shock	Cross-regional coordination, demand-side actions and recovery of affected resources.

The historical cases distinguish the response channels. January 2021 involved fuel and energy endurance; March 2022 combined damaged generation with cold and weak solar; the February and August 2026 episodes include documented support actions with their own regional and timing constraints. Support can prevent a worse outcome, so its use need not coincide with the highest daily average price. [26] [21]

For a trader, the recurring question is which part of the system can adjust next, over what period and through which route. The next chapter begins with the regional demand and residual-load shapes that create those requirements: heating, cooling, industrial calendars and the withdrawal of solar are not uniform across Japan. [31]

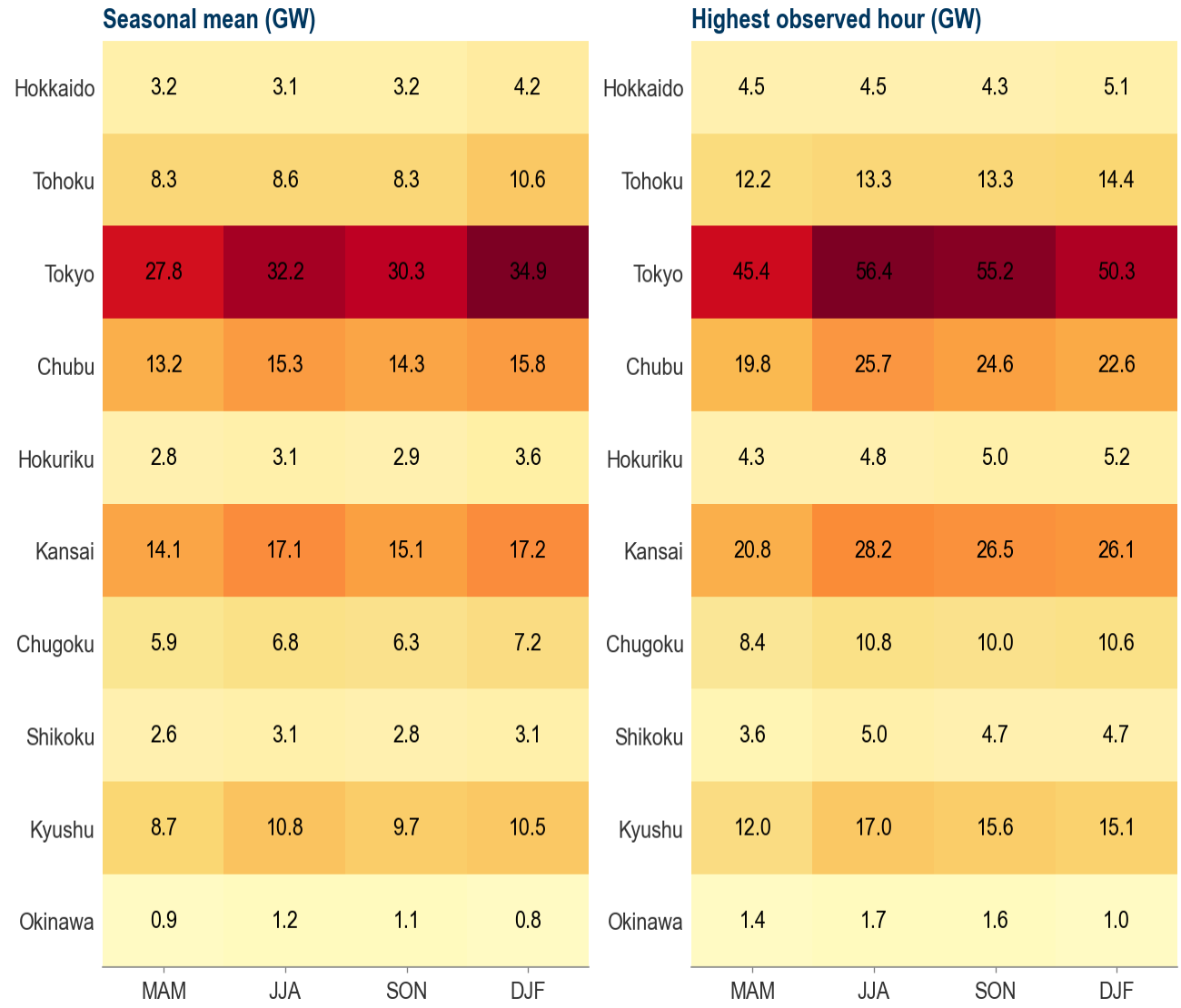
Demand, weather and residual-load shapes

Demand, weather and the shape of residual load

Japan's largest demand area is not a miniature version of its smallest. Tokyo averaged **32.16 GW** in summer 2026 against **1.20 GW** in Okinawa; Hokkaido's winter mean of **4.15 GW** exceeded its summer mean of **3.07 GW**. Cooling, heating and the industrial calendar create different seasonal exposures before generation availability enters the balance. [2] [3]

- Ten regional histories: levels and peaks
- Kyushu and Okinawa: similar warmth, unequal seasonality
- Load factors and peak clocks: average utilisation is not evening security

Exhibit 7: Japan's ten demand areas across four seasons



Note: Latest eligible complete season in each area; year-specific results and peak timestamps in extract. A mean-profile maximum is not the daily or seasonal peak.

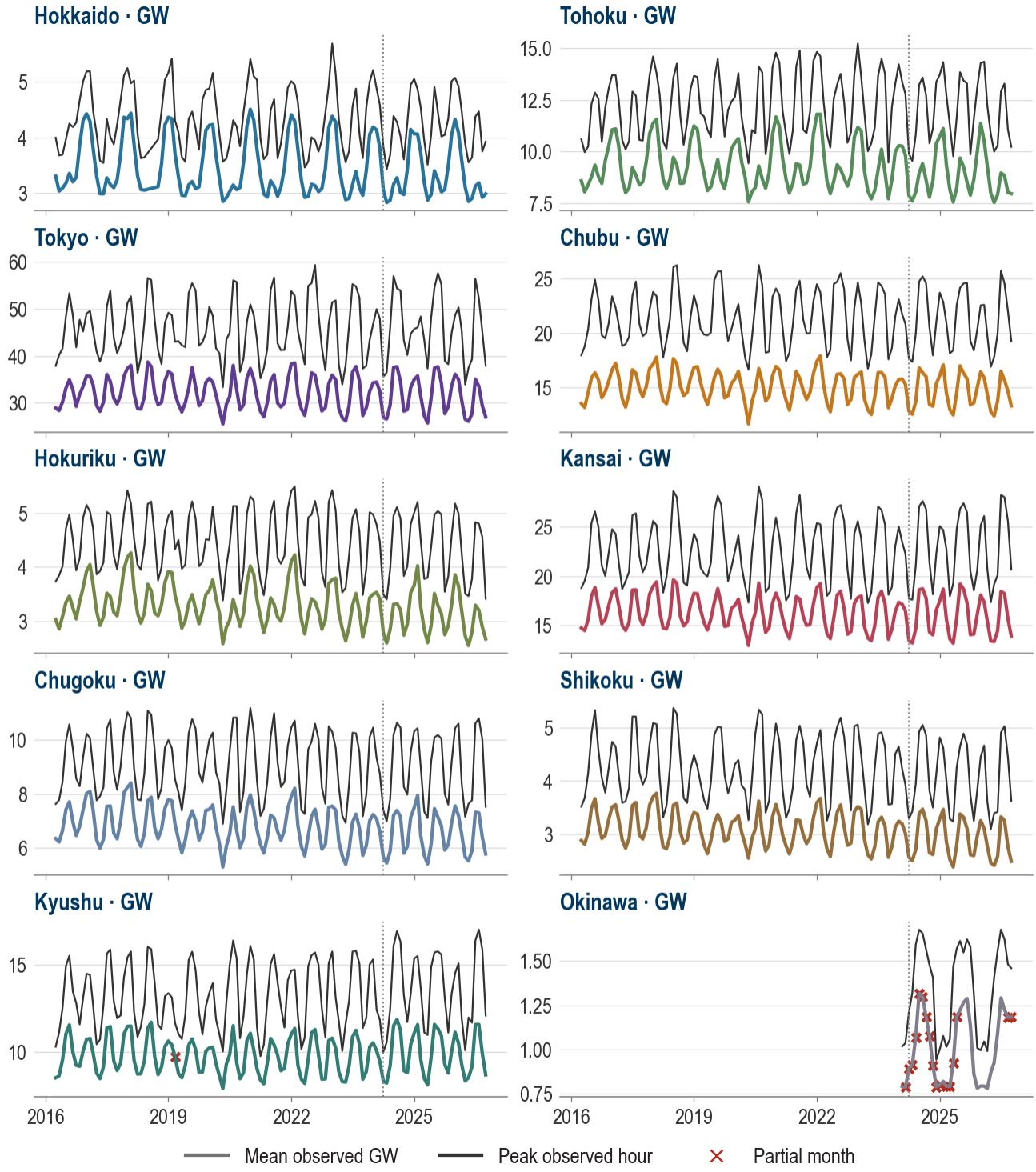
Source: [2–4] [32] · Chart data

Section route: regional levels and daily shape; weather and the Japanese holiday calendar; residual load and evening flexibility. Seasonal means and highest observed hours above are individual-area quantities, not coincident national peaks. Winter 2026 means December 2025–February 2026; autumn uses the latest completed SON, 2025.

Ten regional histories: levels and peaks

Summer 2026 hourly peaks versus seasonal means were **56.38/32.16 GW** in Tokyo, **28.24/17.06 GW** in Kansai and **25.71/15.35 GW** in Chubu. These different peak-to-average relationships affect capacity requirements before solar is deducted. [2] [3]

Exhibit 8: Monthly regional demand and peaks, longest qualified history



Note: Dotted line marks April 2024 reporting transition. Partial months retain observed GW, never extrapolated TWh; no Okinawa legacy trace is invented.

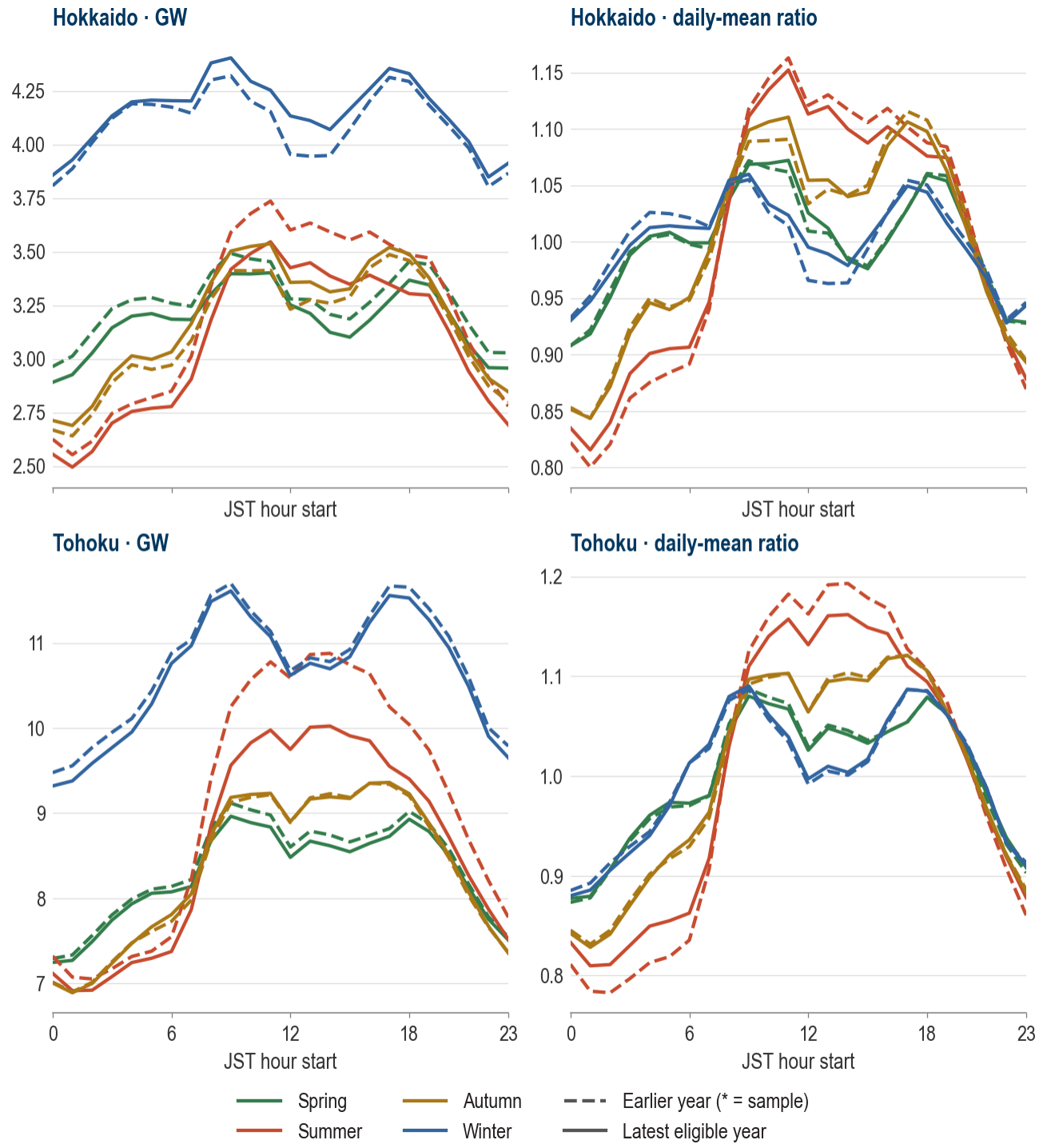
Source: [2–4] [32] · Chart data

Official Chugoku hourly history spans April 2016–March 2024. Okinawa has no admitted pre-2024 hourly history; consumer uses accounted for **83.1%** of its FY2023 demand, against **60.5%** elsewhere. That mix makes mainland industrial shutdowns a poor island benchmark. [32] [33]

Northern demand: winter level, daytime shape

Hokkaido's winter 2026 mean demand was 4.15 GW, 35% above summer's 3.07 GW; Tohoku's corresponding means were 10.63 and 8.59 GW. The northern winter exposure is therefore present in the energy level, not merely in isolated daily maxima. [2] [3]

Exhibit 9: Hokkaido / Tohoku: level and daily shape



Note: Dashed = earlier, solid = latest; normalized ratios use each day's own mean. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

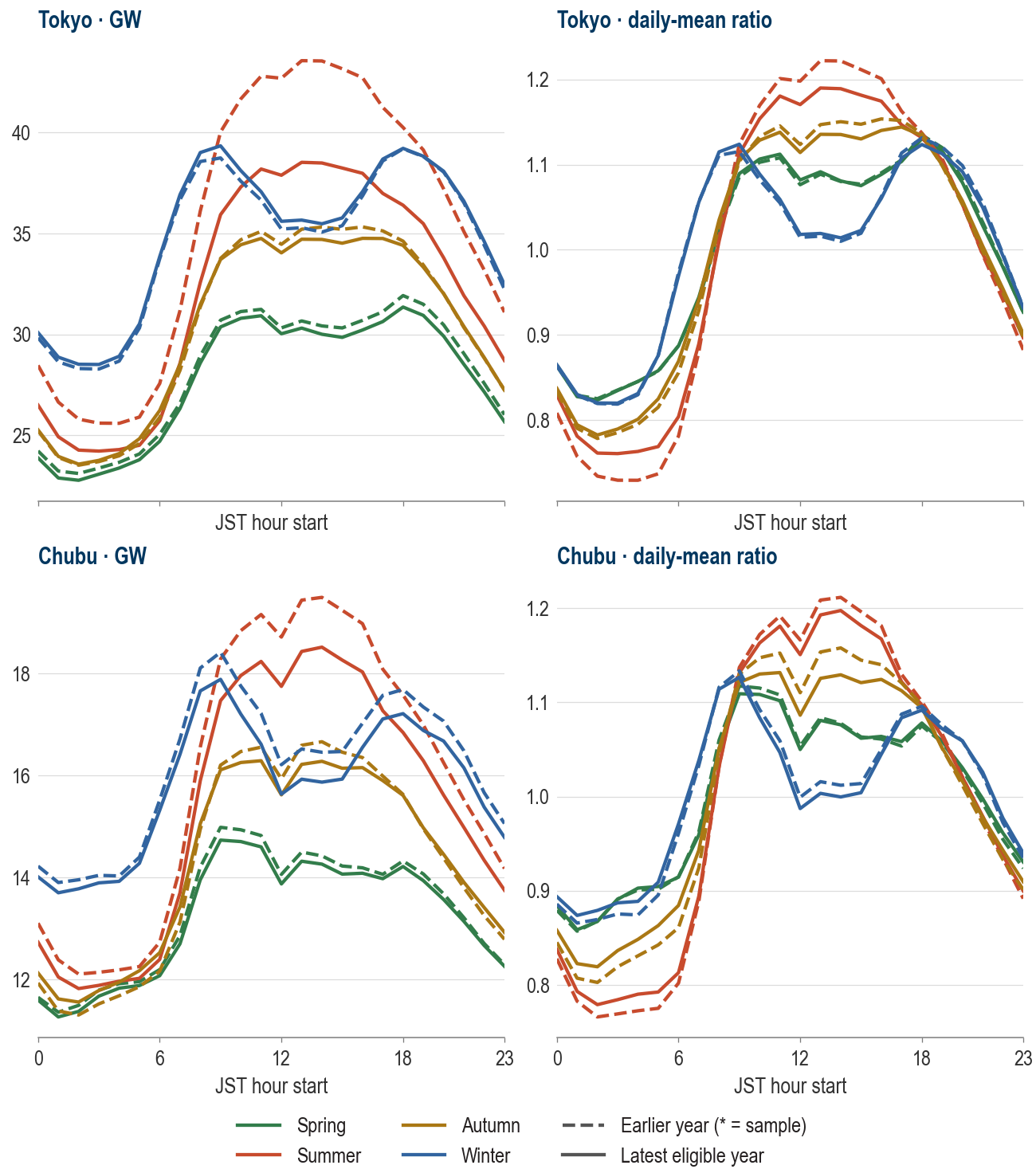
Source: [2–4] [32] · Chart data

Summer noon demand averaged 1.113 times the daily mean in Hokkaido and 1.132 in Tohoku. Winter noon ratios were only 0.995 and 0.997: a flatter winter profile can still impose a much larger absolute requirement. Heating is a plausible contributor, but the regional totals do not identify heating technology or individual customers.

Tokyo and Chubu: scale and industrial-day shape

Tokyo averaged 27.75 GW in spring 2026 and 32.16 GW in summer; Chubu moved from 13.20 to 15.35 GW. Their summer maximum hourly demands, 56.38 and 25.71 GW, greatly exceeded either seasonal mean. [2] [3]

Exhibit 10: Tokyo / Chubu: level and daily shape



Note: Dashed = earlier, solid = latest; normalized ratios use each day's own mean. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

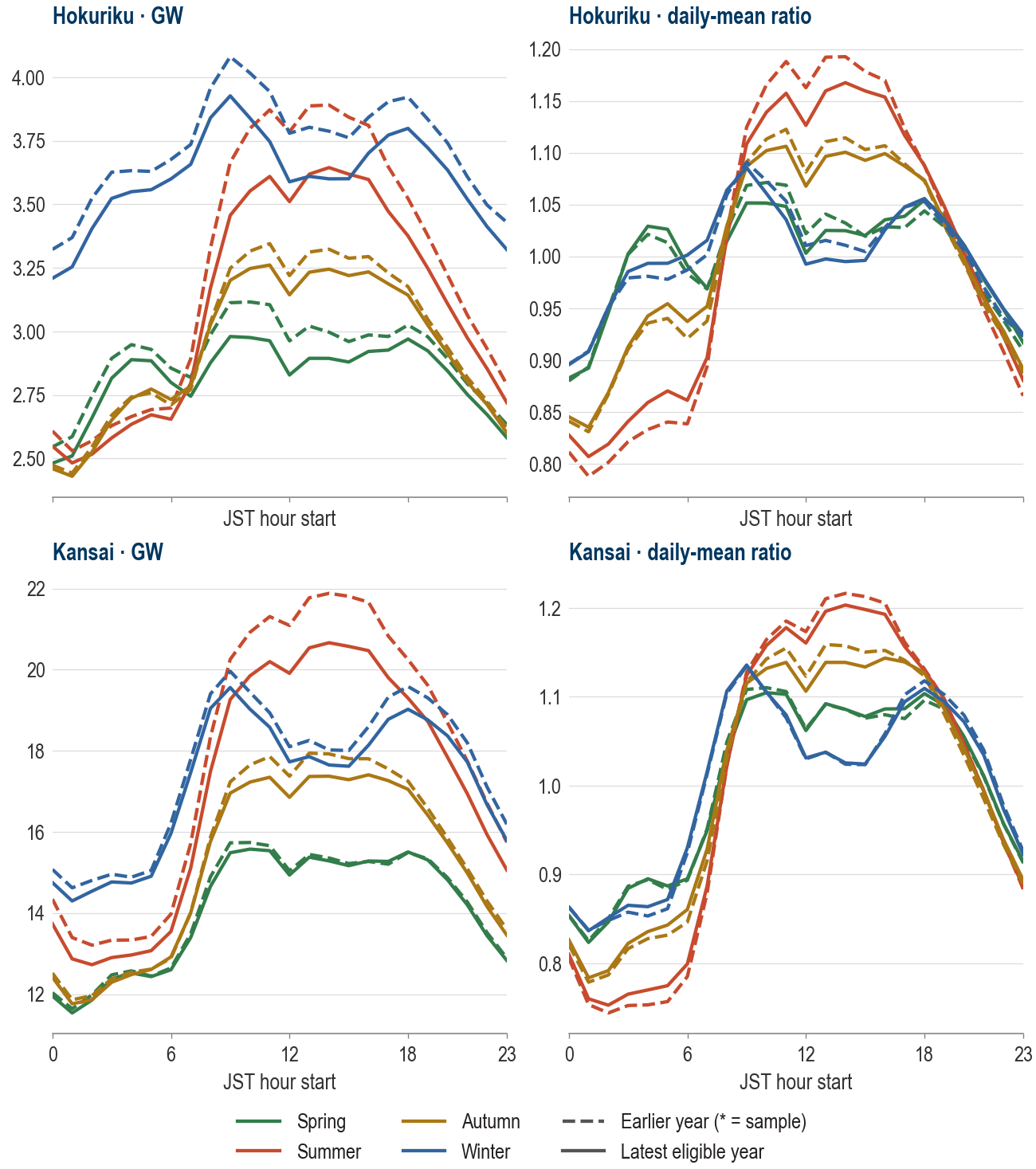
Source: [2–4] [32] · Chart data

Tokyo's summer noon ratio was 1.170 versus Chubu's 1.151. In winter both areas' mean-demand profiles peaked at 09:00, whereas summer peaks shifted to 13:00 in Tokyo and 14:00 in Chubu. Cooling and working-day activity can reinforce each other; the stronger Chubu holiday contraction on page 36 separately exposes calendar sensitivity.

Hokuriku and Kansai: similar clocks, different scale

Hokuriku averaged 3.60 GW in winter 2026 against 3.10 GW in summer. Kansai's winter and summer means were much closer, 17.16 and 17.06 GW, although summer's highest hour reached 28.24 GW versus 26.10 GW in winter. [2] [3]

Exhibit 11: Hokuriku / Kansai: level and daily shape



Note: Dashed = earlier, solid = latest; normalized ratios use each day's own mean. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

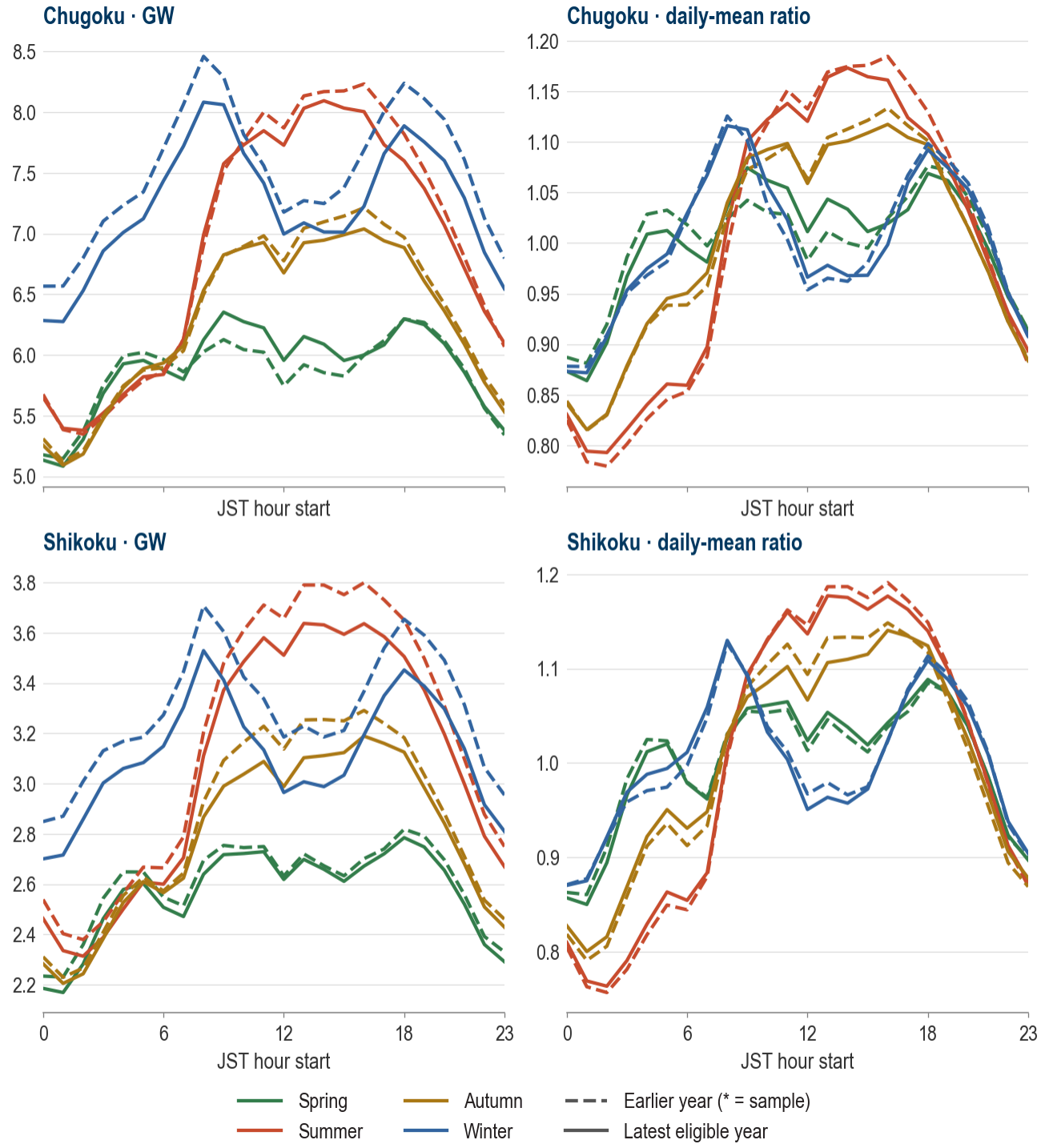
Source: [2–4] [32] · Chart data

Summer noon ratios were 1.126 in Hokuriku and 1.161 in Kansai. The sharper Kansai daytime concentration is consistent with commercial activity and cooling, while Hokuriku's larger winter level leaves less seasonal relief. Hydro and nuclear affect the resources meeting these shapes; they do not turn observed customer demand into generation.

Chugoku and Shikoku: evening persistence

Chugoku averaged 6.85 GW in summer 2026; Shikoku averaged 3.07 GW. Their summer 18:00 daily-normalized ratios were 1.107 and 1.140 respectively, so Shikoku retained a larger relative evening requirement despite its smaller absolute load. [2] [3]

Exhibit 12: Chugoku / Shikoku: level and daily shape



Note: Dashed = earlier, solid = latest; normalized ratios use each day's own mean. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

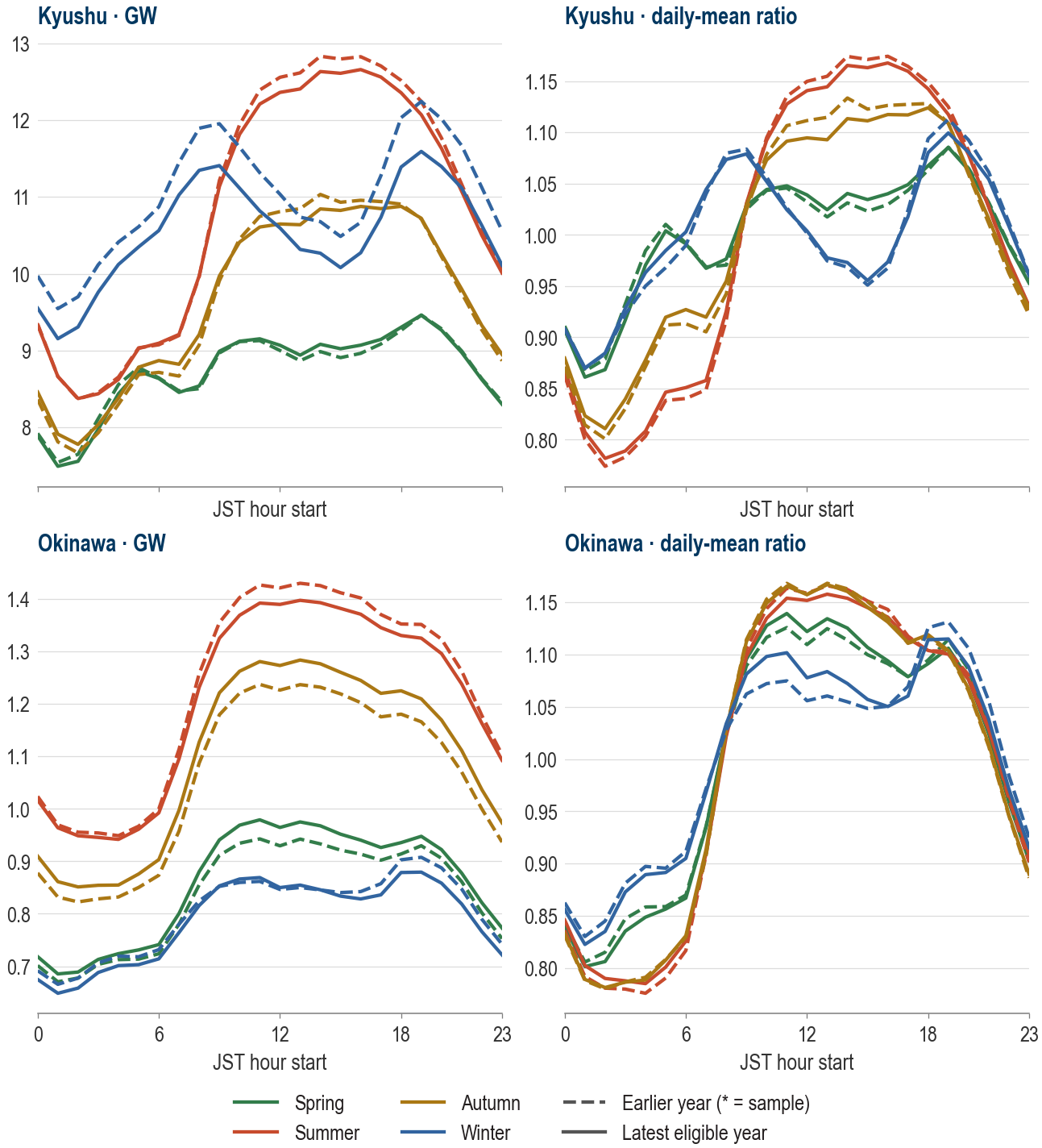
Source: [2–4] [32] · Chart data

Winter morning demand peaks occurred at 08:00 in both mean profiles. In summer Chugoku's mean profile peaked at 14:00, Shikoku's at 16:00. Regional aggregation cannot identify which industrial customers closed or reopened, but these clocks distinguish their flexibility requirement from a uniform national afternoon peak. [32]

Kyushu and Okinawa: different seasons

Summer 2026 mean demand was **10.78 GW** in Kyushu and **1.20 GW** in Okinawa; winter means were **10.54 GW** and **0.79 GW**. Cooling-led island seasonality is much stronger. [2] [3]

Exhibit 13: Kyushu / Okinawa: level and daily shape



Note: Dashed = earlier, solid = latest; normalized ratios use each day's own mean. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25. Exceptions: Okinawa: MAM: 2025* (2025-03-01–2025-05-29) vs 2026; JJA: 2025* (2025-06-01–2025-08-31) vs 2026; SON: 2024* (2024-09-01–2024-11-29) vs 2025; DJF: 2025* (2024-12-01–2025-02-28) vs 2026. * = incomplete-season complete-day sample; dated ranges are inclusive.

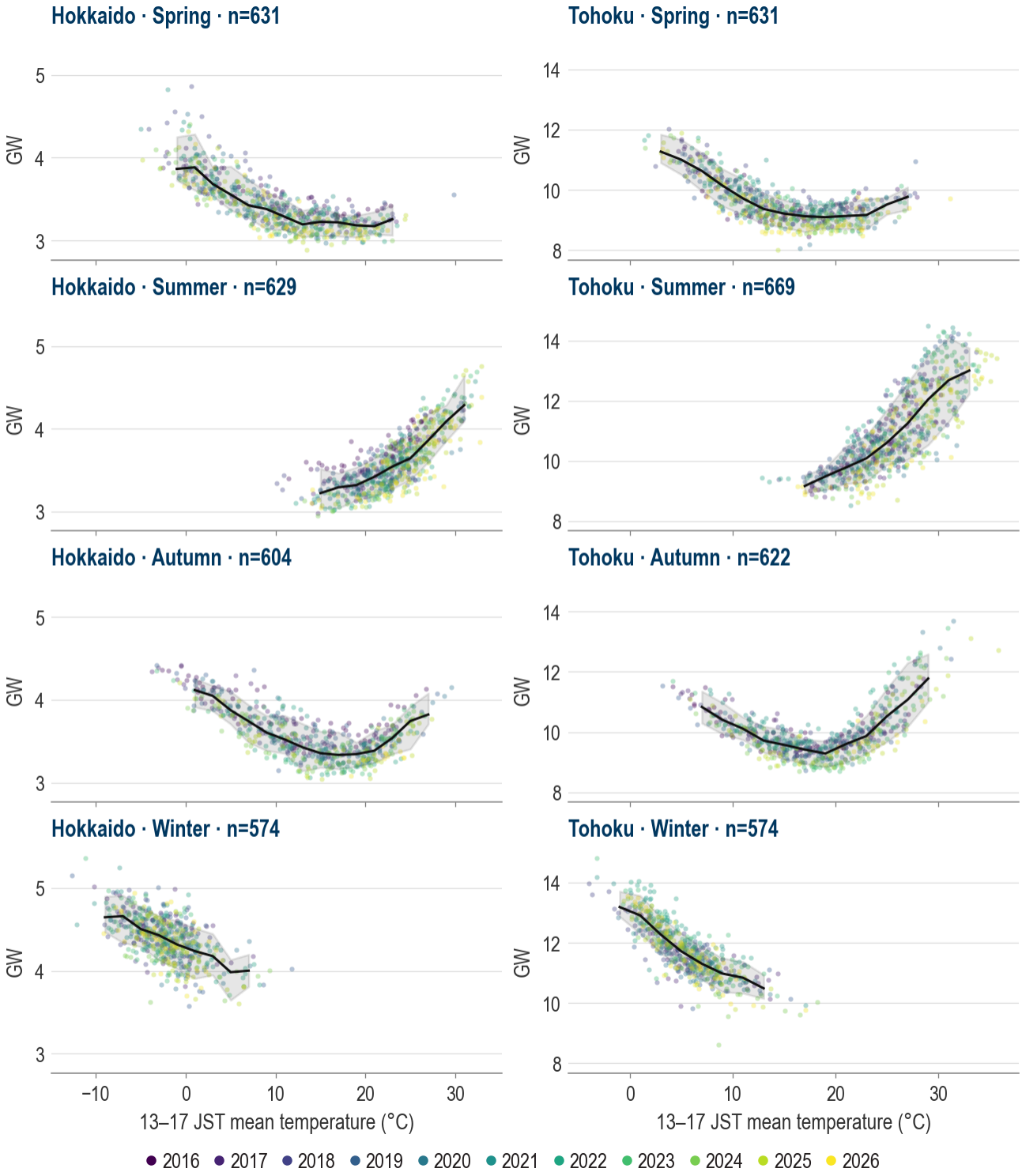
Source: [2–4] [32] · Chart data

Okinawa has one complete season-year per season; earlier samples remain separately dated, not complete-season comparators. Its operator describes eleven isolated, consumer-heavy systems: no mainland transfer can cover their evening peaks. [33]

Northern temperature exposure is asymmetric

On ordinary winter 2026 weekdays, Hokkaido's afternoon demand averaged **4.38 GW** over a representative-point temperature range of **−9.1 to 6.5°C**. Tohoku averaged **11.74 GW** over **−0.3 to 17.1°C**. Those populations differ in weather as well as size; they are not comparable fitted response coefficients. [2] [3] [5]

Exhibit 14: Hokkaido / Tohoku: four-season temperature–demand scatter



Note: Complete ordinary weekdays, 13–17 JST; representative-point reanalysis. Black median/p10–p90: 2°C bins with ≥10 days. Year colours retain all observed points.

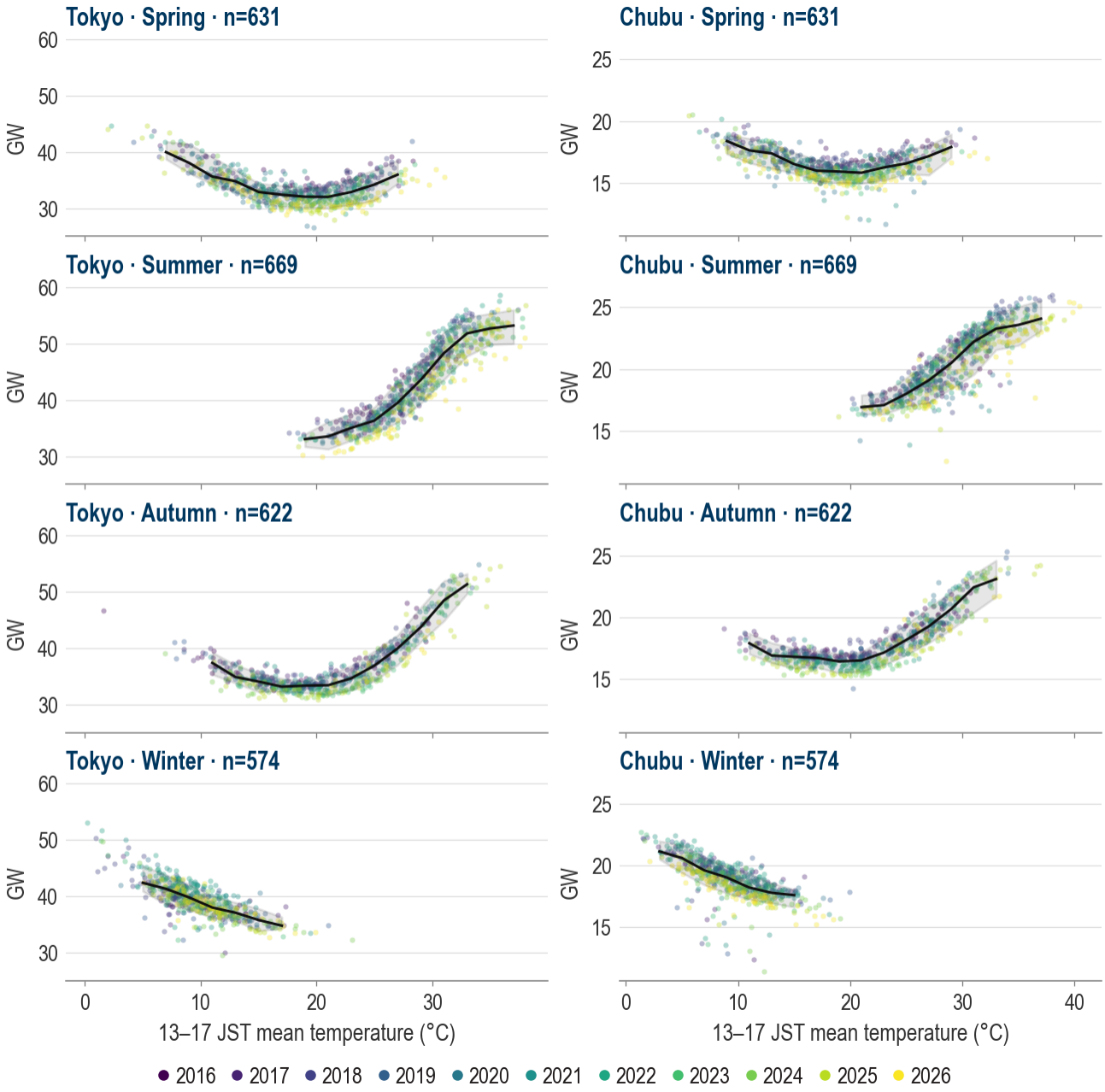
Source: [2–5] [32] [34] · Chart data

Summer afternoon averages were 3.56 GW and 10.63 GW. The four seasonal point clouds retain heating/cooling asymmetry and year shifts; an unusually warm spring day is not relabelled as a summer observation.

Tokyo and Chubu: weather and clock

Summer cooling and working-day activity coincide; night demand remains materially lower on the same ordinary weekdays. [2] [3] [5]

Exhibit 15: Tokyo / Chubu: four-season temperature–demand scatter



Note: Complete ordinary weekdays, 13–17 JST; representative-point reanalysis. Black median/p10–p90: 2°C bins with ≥10 days. Year colours retain all observed points.

Source: [2–5] [32] [34] · Chart data

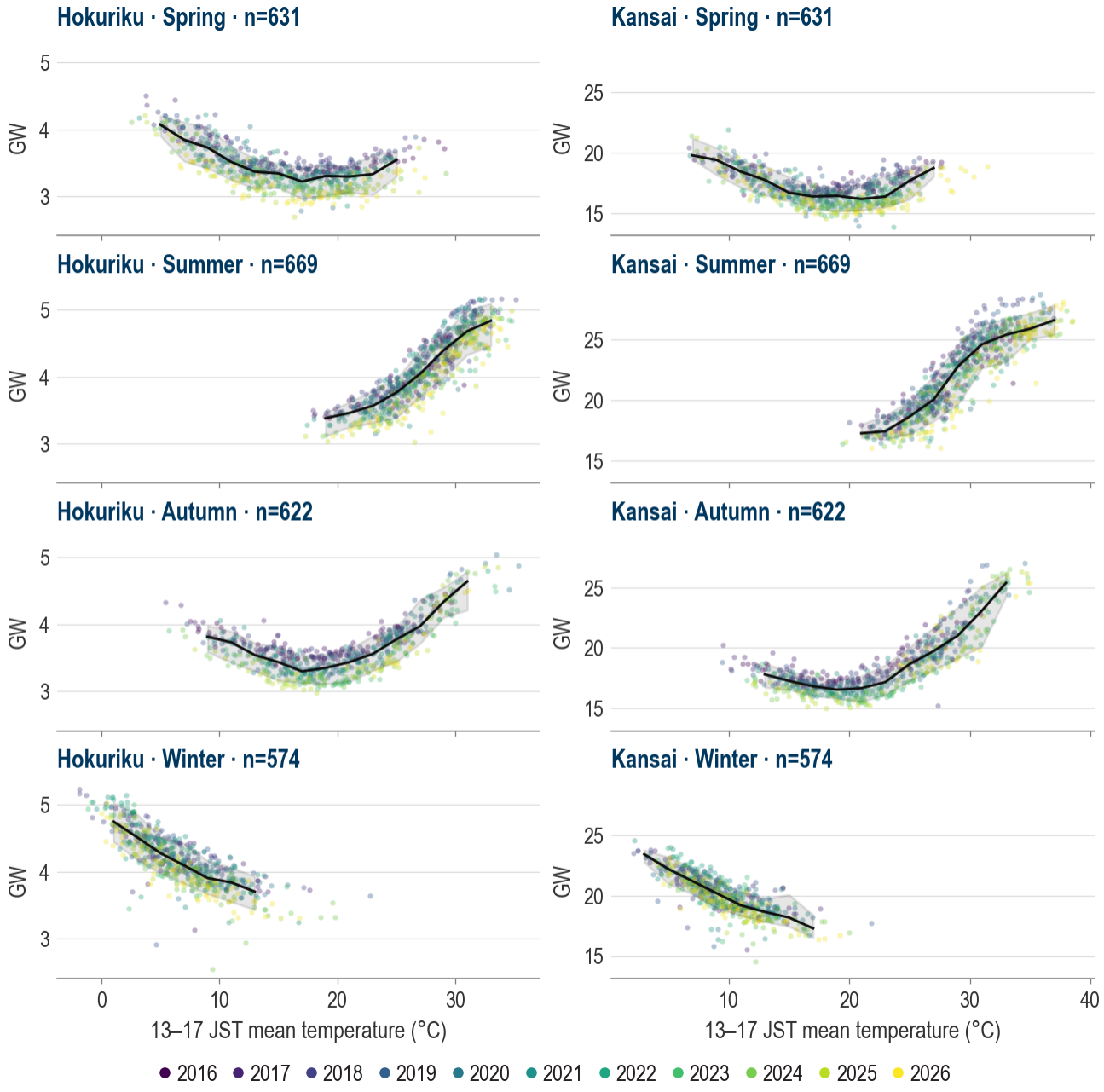
Season/year (days)	Night/afternoon GW	Afternoon °C range	Bin Δ GW
Spring 2026 (58)	23.52/32.54; 12.00/15.97	7.4–31.0; 9.4–32.2	7.00; 3.12
Summer 2026 (62)	24.61/40.48; 12.28/20.07	18.8–38.0; 21.3–40.5	13.01; 6.06–9.66
Autumn 2025 (60)	24.28/37.17; 12.13/17.93	11.1–35.8; 11.3–36.9	unsupported; unsupported
Winter 2026 (57)	29.07/38.57; 14.32/17.93	5.1–19.9; 2.1–18.4	9.11–9.49; 3.36–3.71

Each cell: **Tokyo; Chubu**, matched days. Night 01:00–04:59; afternoon 13:00–16:59 JST. Bin Δ: afternoon-minus-night mean GW across supported 2°C afternoon bins (≥10 days/bin); not a causal coefficient.

Hokuriku and Kansai: seasonal exposure

Kansai's ordinary-weekday afternoon means were **21.86 GW** in summer 2026 and **19.34 GW** in winter; Hokuriku's were **3.92 GW** and **3.99 GW**. Summer working-day exposure is stronger in Kansai. [2] [3] [5]

Exhibit 16: Hokuriku / Kansai: four-season temperature–demand scatter



Note: Complete ordinary weekdays, 13–17 JST; representative-point reanalysis. Black median/p10–p90: 2°C bins with ≥10 days. Year colours retain all observed points.

Source: [2–5] [32] [34] · Chart data

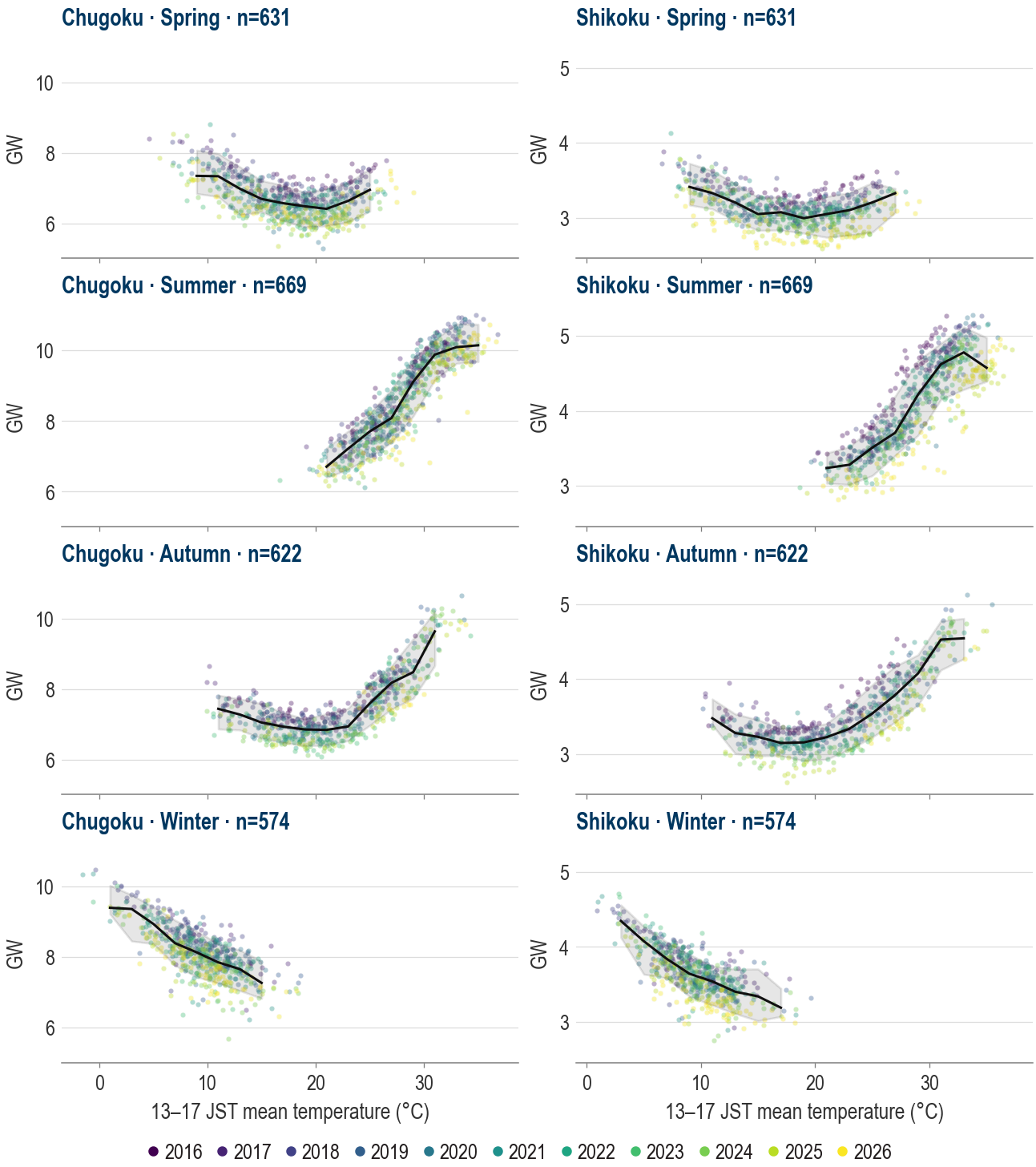
Kansai season/year (days)	Night/afternoon GW	Afternoon °C range	Bin Δ GW
Spring 2026 (58)	12.37/16.69	9.0–31.4	3.03
Summer 2026 (62)	13.00/21.86	21.9–37.7	7.61–11.19
Autumn 2025 (60)	12.30/18.77	12.0–34.8	unsupported
Winter 2026 (57)	14.86/19.34	4.6–19.2	4.15–4.55

Matched days; night 01:00–04:59, afternoon 13:00–16:59 JST. Bin Δ: afternoon-minus-night mean GW across supported 2°C afternoon bins (≥10 days/bin); not a causal coefficient.

Western Honshu and Shikoku: temperature does not erase the calendar

Chugoku's summer 2026 ordinary-weekday afternoons averaged **8.56 GW**, against **5.58 GW** at night. Shikoku's corresponding means were **3.86 GW** and **2.41 GW**. Both retained a substantial working-day increment even though their absolute sizes differ. [2] [3] [5]

Exhibit 17: Chugoku / Shikoku: four-season temperature–demand scatter



Note: Complete ordinary weekdays, 13–17 JST; representative-point reanalysis. Black median/p10–p90: 2°C bins with ≥10 days. Year colours retain all observed points.

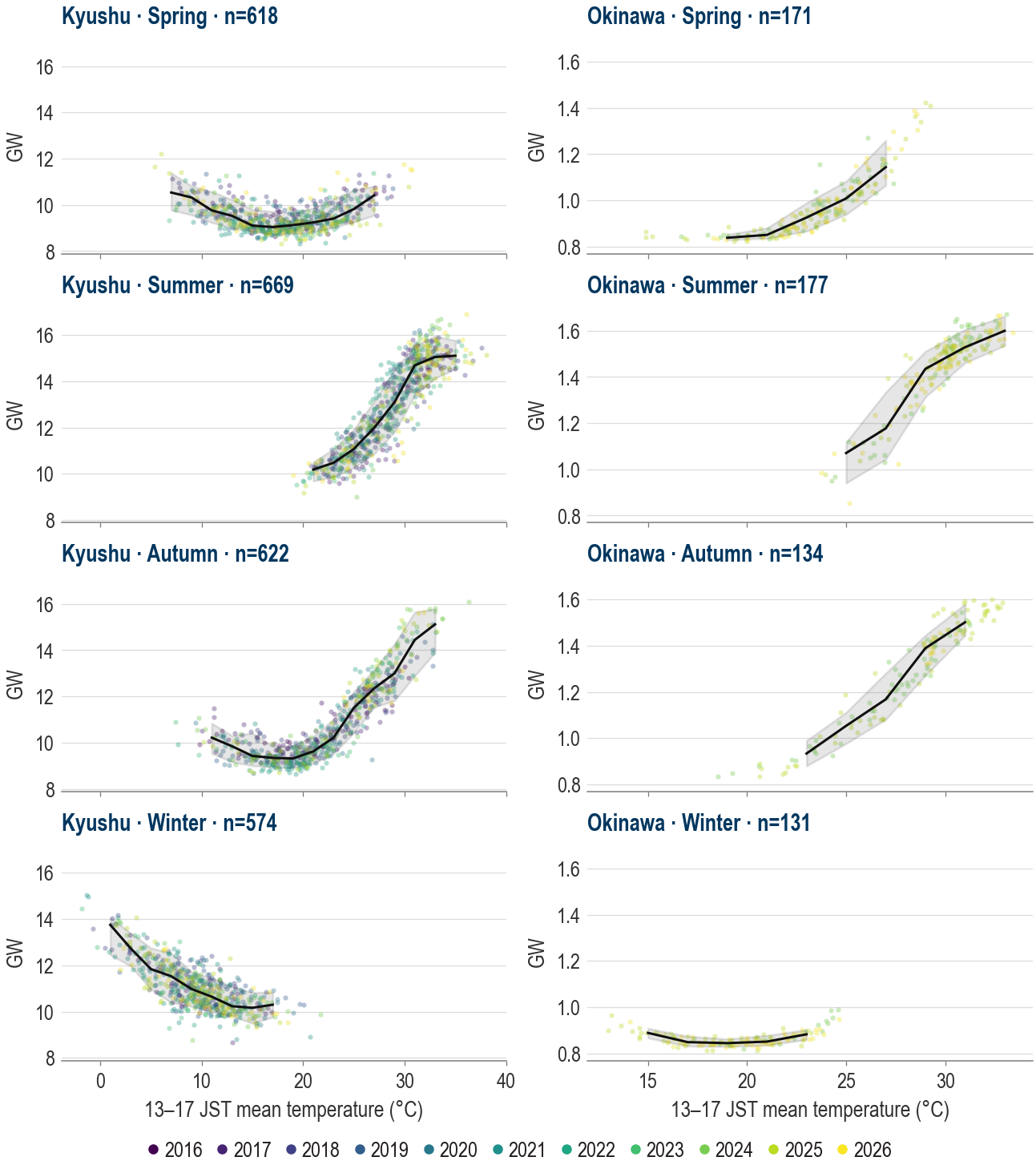
Source: [2–5] [32] [34] · Chart data

Their winter night-to-afternoon increments were smaller: 0.83 GW in Chugoku and 0.35 GW in Shikoku. Cooling plausibly steepens summer daytime exposure, but humidity, behind-the-meter solar and customer activity remain competing contributors. [32]

Kyushu and Okinawa: similar warmth, unequal seasonality

Okinawa's summer 2026 ordinary-weekday afternoon mean was **1.42 GW**, compared with **0.86 GW** in winter. Kyushu's corresponding means were **13.13 GW** and **10.79 GW**. Okinawa's matched winter afternoons still ranged from **14.1 to 24.6°C**. [2] [3] [5]

Exhibit 18: Kyushu / Okinawa: four-season temperature–demand scatter



Note: Complete ordinary weekdays, 13–17 JST; representative-point reanalysis. Black median/p10–p90: 2°C bins with ≥10 days. Year colours retain all observed points.

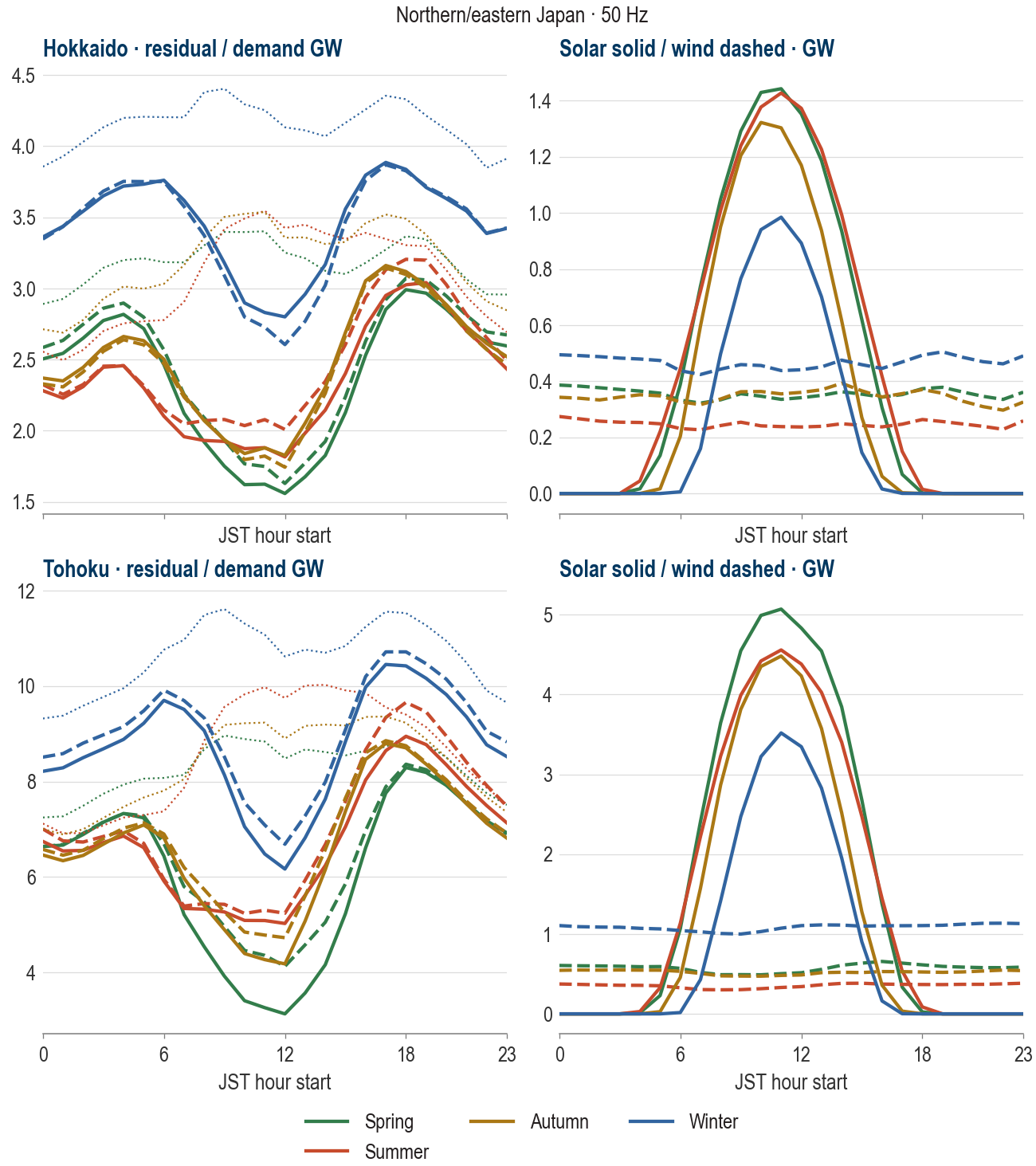
Source: [2–5] [32] [34] · Chart data

The island's consumer-use share and tourism exposure make temperature and service activity commercially relevant together. Modelled weather at one point does not capture every island; no unsupported historical Okinawa demand points are manufactured. [33]

Northern residual load shifts the pressure into evening

In spring 2026 Hokkaido's mean noon residual was **1.56 GW**, rising to **2.99 GW** at 18:00. Tohoku moved from **3.14 GW** to **8.29 GW**. Noon solar output averaged 1.35 GW and 4.83 GW respectively. [2] [3] [4]

Exhibit 19: Hokkaido / Tohoku: residual and renewable profiles



Note: Left: residual latest solid/earlier dashed; demand dotted. Right: latest solar solid/wind dashed. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

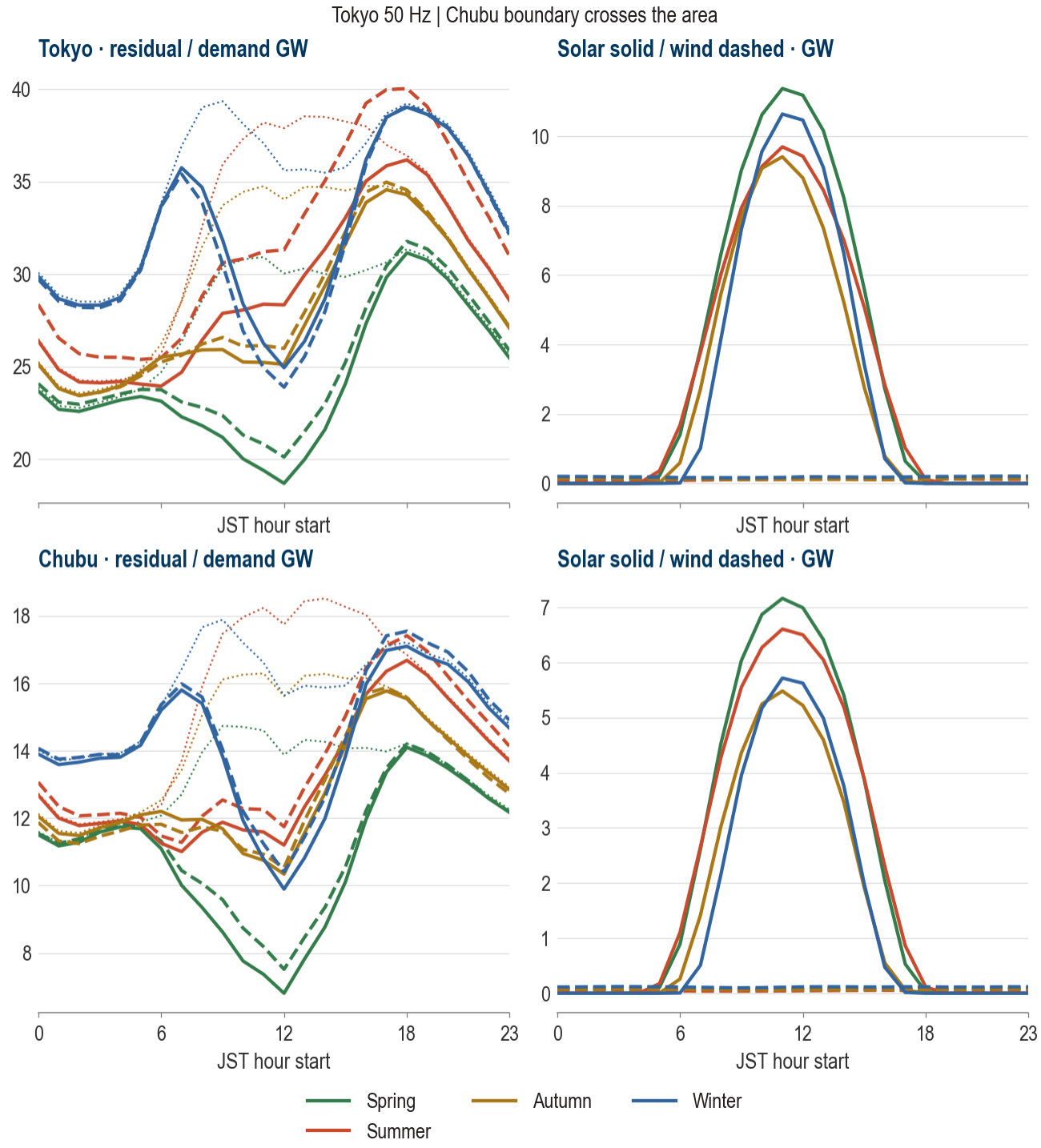
Source: [2–4] [32] · Chart data

The mean demand profile peaked at 11:00 in Hokkaido and 09:00 in Tohoku, but both residual profiles peaked at 18:00. Wind contributes materially in the north—winter means were 0.47 GW and 1.09 GW—but its mean output does not guarantee availability on the steepest evening.

Tokyo and Chubu: solar changes the required clock

Tokyo's spring 2026 noon residual averaged **18.70 GW**, against **31.14 GW** at 18:00. Chubu's corresponding values were **6.81 GW** and **14.10 GW**. The spring noon solar contribution was **11.18 GW** in Tokyo and **6.99 GW** in Chubu. [2] [3] [4]

Exhibit 20: Tokyo / Chubu: residual and renewable profiles



Note: Left: residual latest solid/earlier dashed; demand dotted. Right: latest solar solid/wind dashed. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

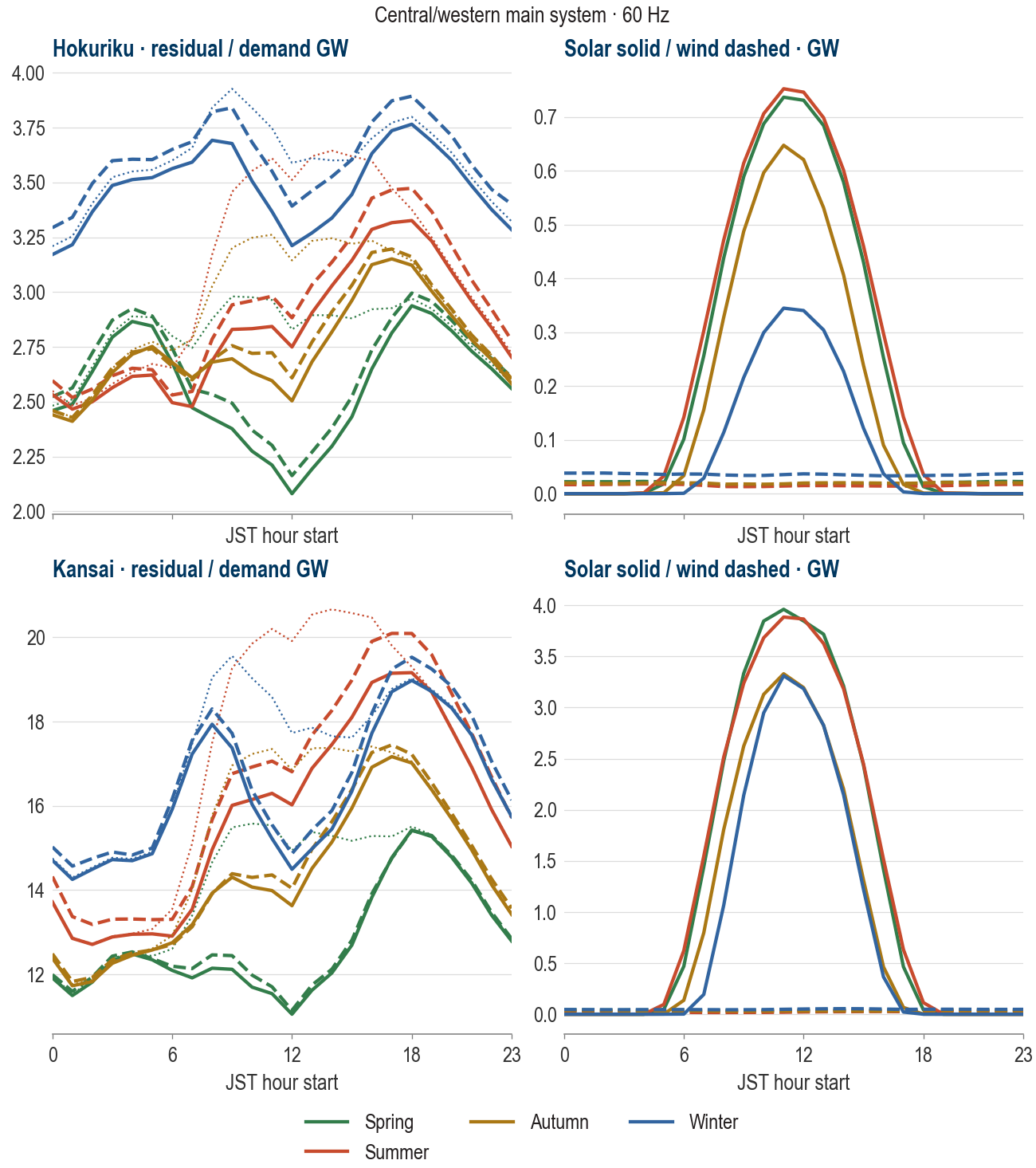
Source: [2–4] [32] · Chart data

Chubu's mean demand peaked at 09:00 while its residual peaked at 18:00. The larger evening residual is a call on thermal, hydro, storage and transfers jointly, not a claim that one fuel follows every GW. The frequency-converter boundary is geographically inside the wider central-Japan comparison, not proof that all Chubu locations are 60 Hz.

Hokuriku and Kansai: unequal solar subtraction

Hokuriku's spring 2026 noon solar averaged **0.73 GW**, compared with **3.85 GW** in Kansai. Noon-to-18:00 residual increased from **2.08 to 2.94 GW** in Hokuriku and from **11.05 to 15.42 GW** in Kansai. [2] [3] [4]

Exhibit 21: Hokuriku / Kansai: residual and renewable profiles



Note: Left: residual latest solid/earlier dashed; demand dotted. Right: latest solar solid/wind dashed. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25.

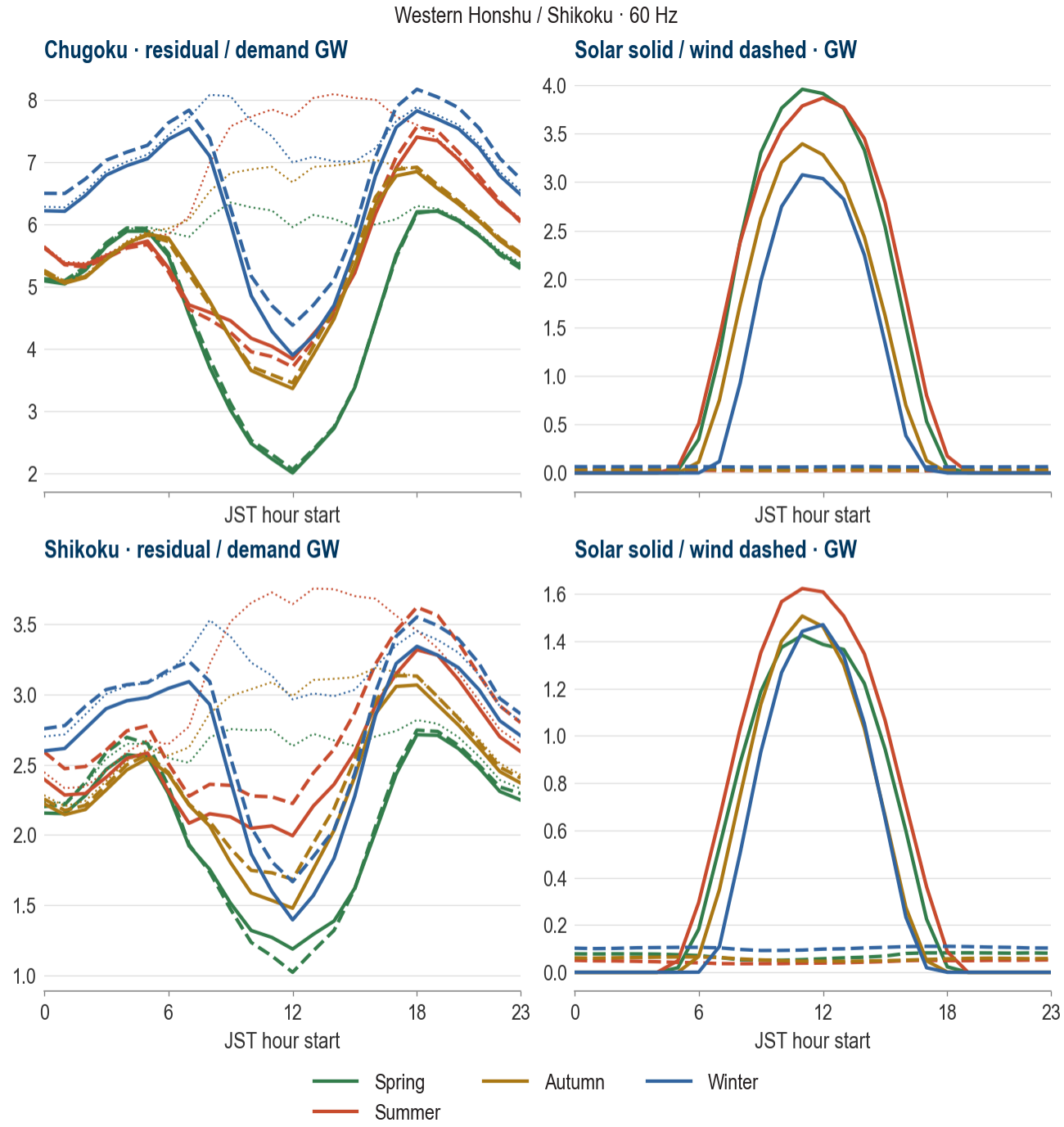
Source: [2–4] [32] · Chart data

Both residual mean profiles peaked at 18:00, while their demand peaks were 09:00 and 10:00. The timing is shared, but generation needed to cover it depends on hydro conditions and the nuclear/thermal fleet discussed in the supply chapter. Nuclear is deliberately not subtracted from this demand-minus-solar-minus-wind measure.

Chugoku and Shikoku: deep midday relief is not spare evening supply

Chugoku's spring 2026 mean noon solar was **3.91 GW** against noon residual of **2.00 GW**. By 18:00 residual was **6.20 GW**. Its demand mean profile peaked at 09:00; the residual profile peaked at 19:00. [2] [3] [4]

Exhibit 22: Chugoku / Shikoku: residual and renewable profiles



Note: Left: residual latest solid/earlier dashed; demand dotted. Right: latest solar solid/wind dashed. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25. Exceptions: Shikoku: MAM: 2023 vs 2025; JJA: 2022 vs 2023.

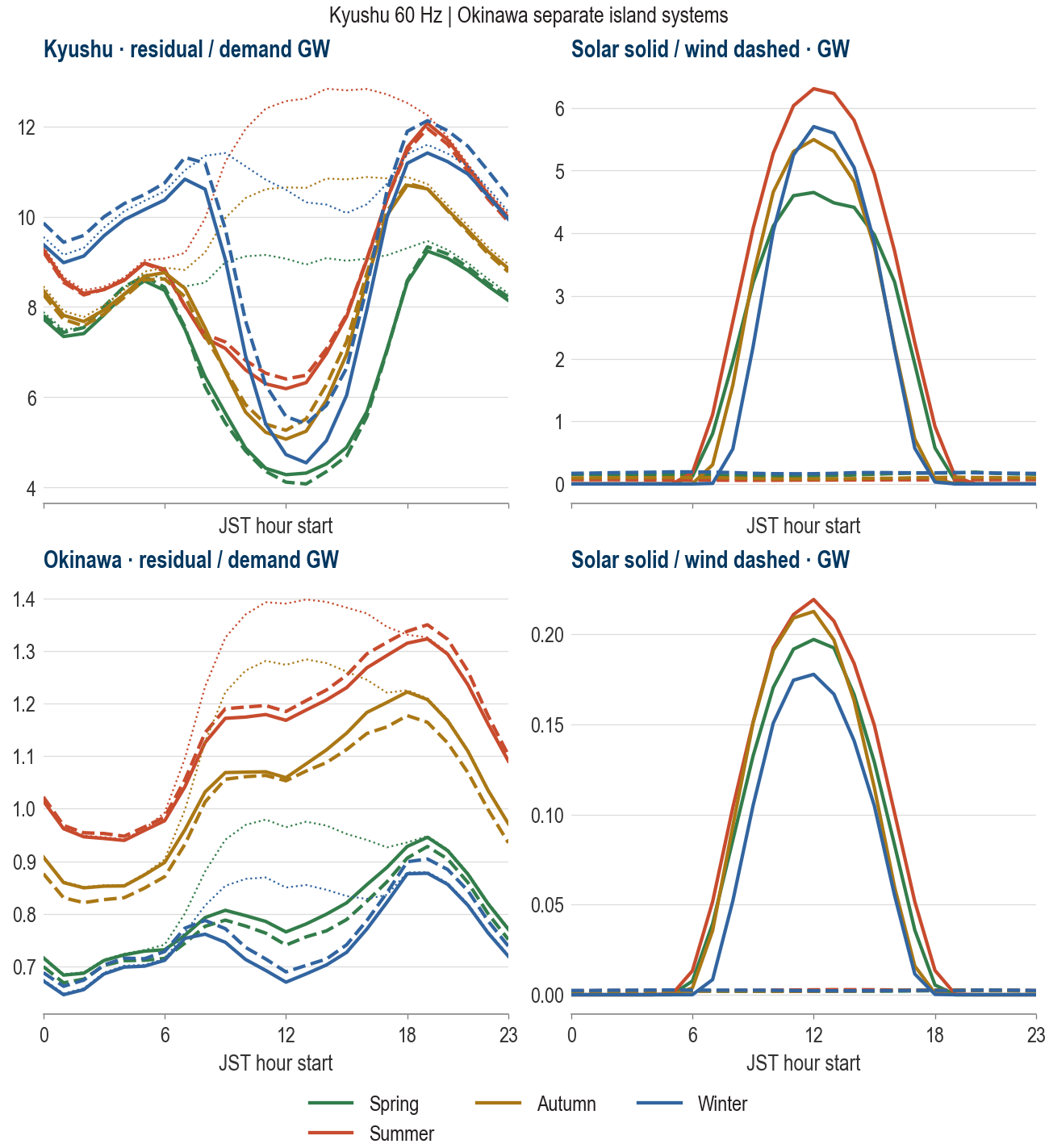
Source: [2–4] [32] · Chart data

Shikoku's spring 2025 mean residual rose from **1.19 GW** at noon to **2.71 GW** at 18:00. It has complete demand seasons that fail residual eligibility because individual solar/wind observations are missing or inadmissible. Its latest complete spring residual comparison ends in 2025 and latest summer in 2023; no missing renewable output is silently set to zero. The separate clocks preserve the western evening requirement without pretending both histories have equal coverage. [32] [35] [36] [37]

Kyushu and Okinawa: evening needs cannot be inferred from noon

Kyushu's spring 2026 mean residual increased from **4.29 GW** at noon to **8.55 GW** at 18:00; noon accepted solar averaged **4.65 GW**. Okinawa's corresponding residual values were **0.77 and 0.93 GW**, with **0.20 GW** of noon solar. [2] [3] [4]

Exhibit 23: Kyushu / Okinawa: residual and renewable profiles



Note: Left: residual latest solid/earlier dashed; demand dotted. Right: latest solar solid/wind dashed. Season-years: MAM/JJA/DJF 2025–26; SON 2024–25. Exceptions: Kyushu: JJA: 2024 vs 2025. Okinawa: MAM: 2025* (2025-03-01–2025-05-29) vs 2026; JJA: 2025* (2025-06-01–2025-08-31) vs 2026; SON: 2024* (2024-09-01–2024-11-29) vs 2025; DJF: 2025* (2024-12-01–2025-02-28) vs 2026. * = incomplete-season complete-day sample; dated ranges are inclusive.

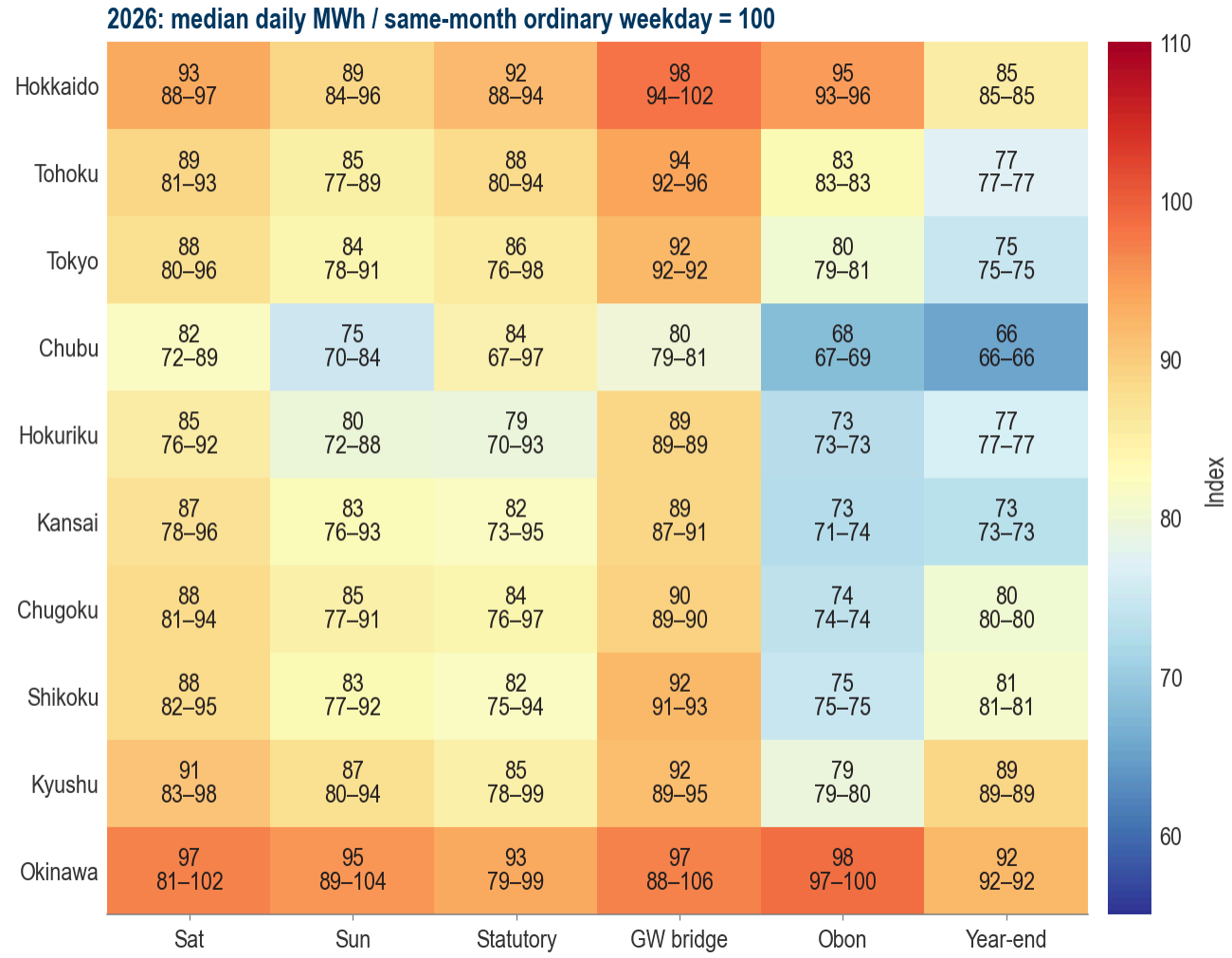
Source: [2–4] [32] · Chart data

Both mean residual profiles peaked at 19:00. Kyushu's exports, pumping and curtailment can accompany its solar-rich midday; Okinawa has no mainland interconnector. Okinawa's earlier starred curves remain observed-day samples. [33]

Japan's holiday calendar

Chubu's 2026 Obon-weekday energy index was **68.3**, against **98.4** in Okinawa (ordinary weekday = 100; two days each): the same calendar produces different demand shocks. [34] [2] [3]

Exhibit 24: Japan's calendar and regional demand behaviour



Note: Median index / daily p10–p90; same-month ordinary weekday = 100. Observed 2026 days only. Statutory holidays precede weekend/customary labels.

Source: [2–4] [32] [34] · Chart data

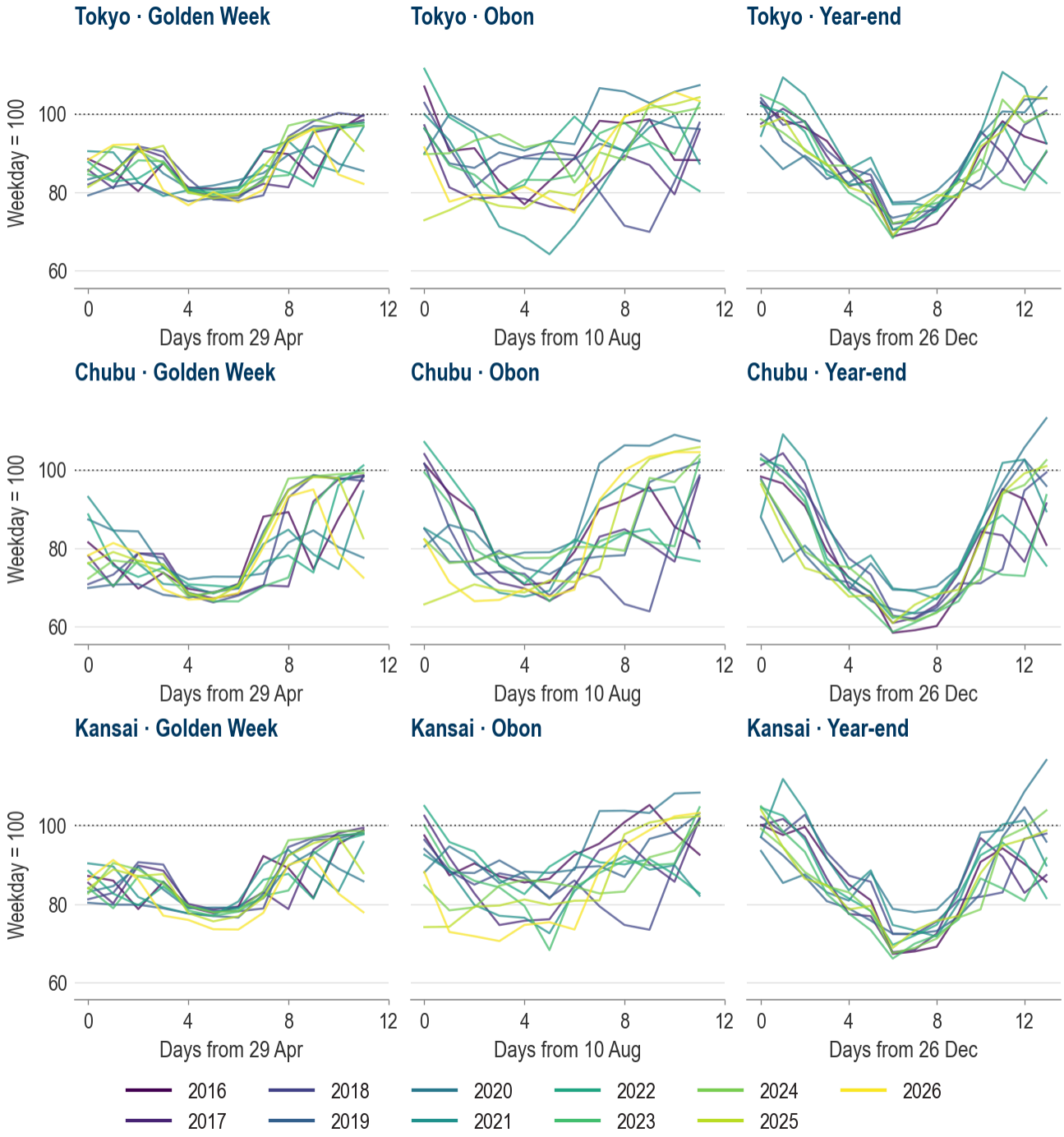
2026 date	English holiday	2026 date	English holiday
01 Jan	New Year's Day · S	06 May	Substitute holiday · R
12 Jan	Coming of Age Day · S	20 Jul	Marine Day · S
11 Feb	National Foundation Day · S	11 Aug	Mountain Day · S
23 Feb	Emperor's Birthday · S	21 Sep	Respect for the Aged Day · S
20 Mar	Vernal Equinox Day · S	22 Sep	Citizens' holiday · C
29 Apr	Showa Day · S	23 Sep	Autumnal Equinox Day · S
03 May	Constitution Memorial Day · S	12 Oct	Sports Day · S
04 May	Greenery Day · S	03 Nov	Culture Day · S
05 May	Children's Day · S	23 Nov	Labour Thanksgiving Day · S

S: statutory; **R:** substitute; **C:** citizens' holiday. November dates are calendar facts, outside observed demand. **Customary:** Golden Week bridge 30 Apr–2 May; Obon 13–16 Aug; year-end 29 Dec–3 Jan. Classification precedence: statutory → weekend → customary; customary windows exclude statutory dates.

Shutdown and reopening: Chubu is not Tokyo, and neither is an island

In the two Obon weekdays of 2026, Chubu's average daily demand was **283,756 MWh** and its index **68.3**; Tokyo averaged **686,885 MWh** at **80.2**, and Kansai **341,141 MWh** at **72.7**. The contraction is consistent with industrial and commercial closure exposure, not proof of any named factory's operating schedule. [2] [3] [34]

Exhibit 25: Shutdown and reopening paths across years



Note: Daily energy index, including weekends and statutory days, in fixed comparison windows; these are not universal factory closure dates. Year-end windows continue into January. All available years remain visible.

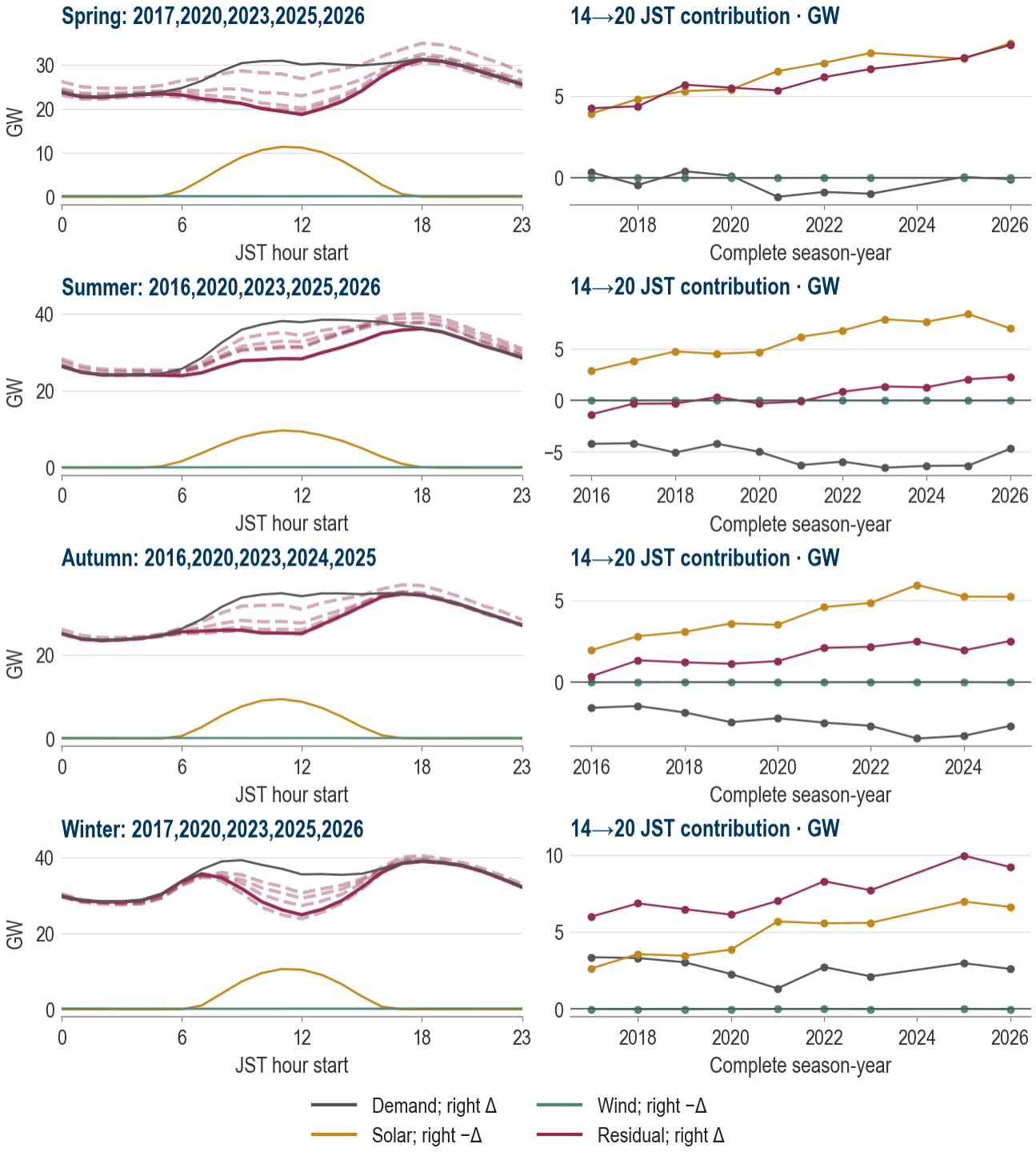
Source: [2–4] [32] [34] · Chart data

The January 2026 customary year-end weekday had indices **65.6** in Chubu, **74.8** in Tokyo and **73.3** in Kansai. These are one-day observations, not a ten-day sample or a universal shutdown rule. Weather and the weekday on which work resumes also matter: the reopening paths retain each actual calendar year, including the January continuation of December windows.

Tokyo: the evening component balance

Tokyo's spring 2026 mean residual rose **8.11 GW** from 14:00 to 20:00 even though demand changed by **-0.09 GW**. Solar fell **8.22 GW**; wind changed **+0.022 GW**. This is an observed component balance, not a hypothetical ramp example. [2] [3] [4]

Exhibit 26: Tokyo: seasonal evening transition across years



Note: Left: latest residual solid; faint earlier-year traces are unkeyed context (years in headings). Other components: latest year. Right: all complete years; 14–20 JST ΔResidual = ΔDemand − ΔSolar − ΔWind.

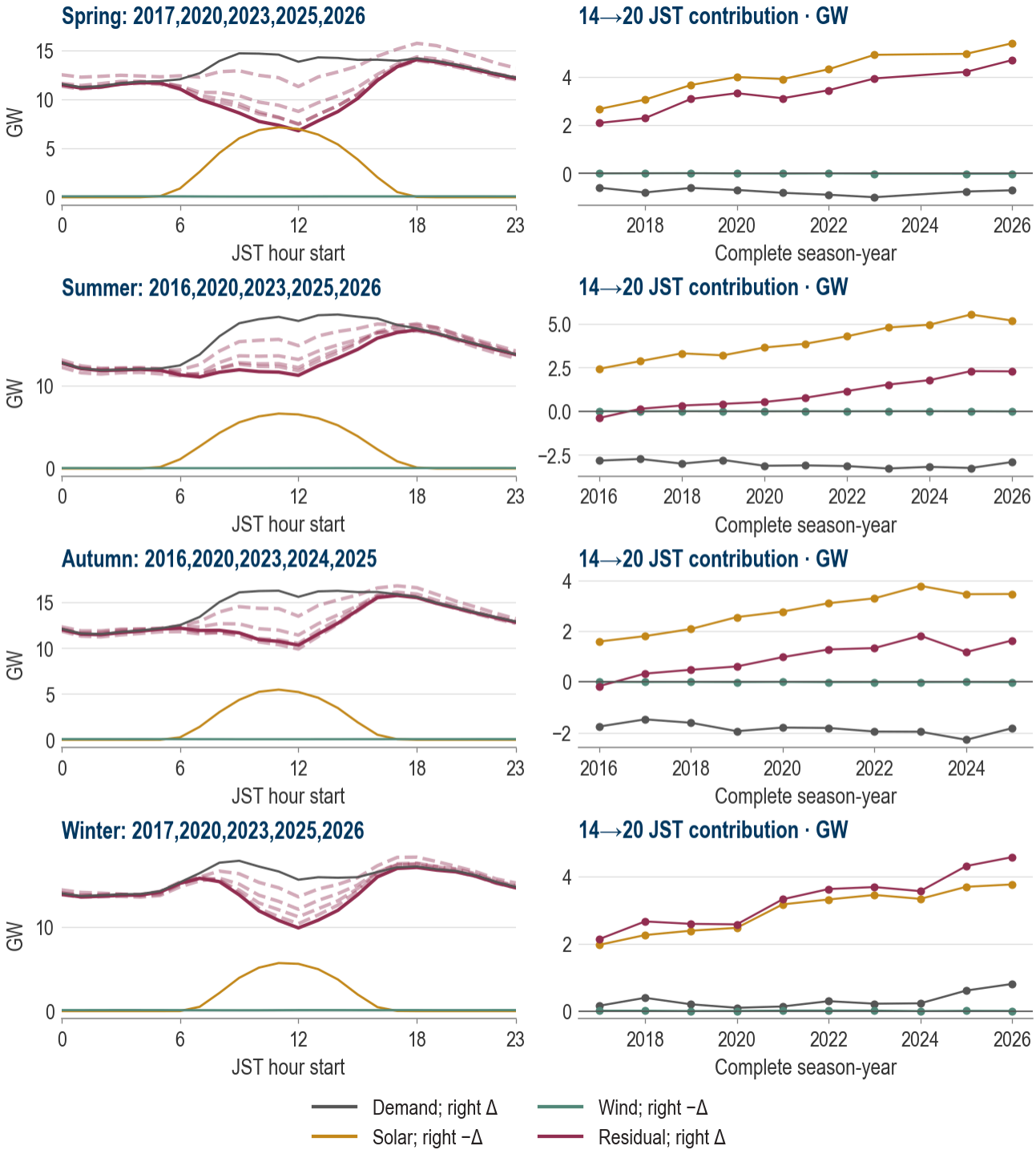
Source: [2–4] [32] · Chart data

Summer's corresponding residual increase was **2.31 GW**, with demand falling **4.69 GW** as solar withdrew. Winter's increase was **9.21 GW**: demand rose **2.60 GW** while solar fell **6.62 GW**. The six-hour change is distinct from the largest adjacent-hour ramp; dispatchable resources face both the transition's speed and its total energy requirement.

Chubu: the evening component balance

Chubu's spring 2026 mean residual rose **4.71 GW** from 14:00 to 20:00 even though demand changed by **-0.69 GW**. Solar fell **5.40 GW**; wind changed **+0.005 GW**. This is an observed component balance, not a hypothetical ramp example. [2] [3] [4]

Exhibit 27: Chubu: seasonal evening transition across years



Note: Left: latest residual solid; faint earlier-year traces are unkeyed context (years in headings). Other components: latest year. Right: all complete years; 14–20 JST ΔResidual = ΔDemand − ΔSolar − ΔWind.

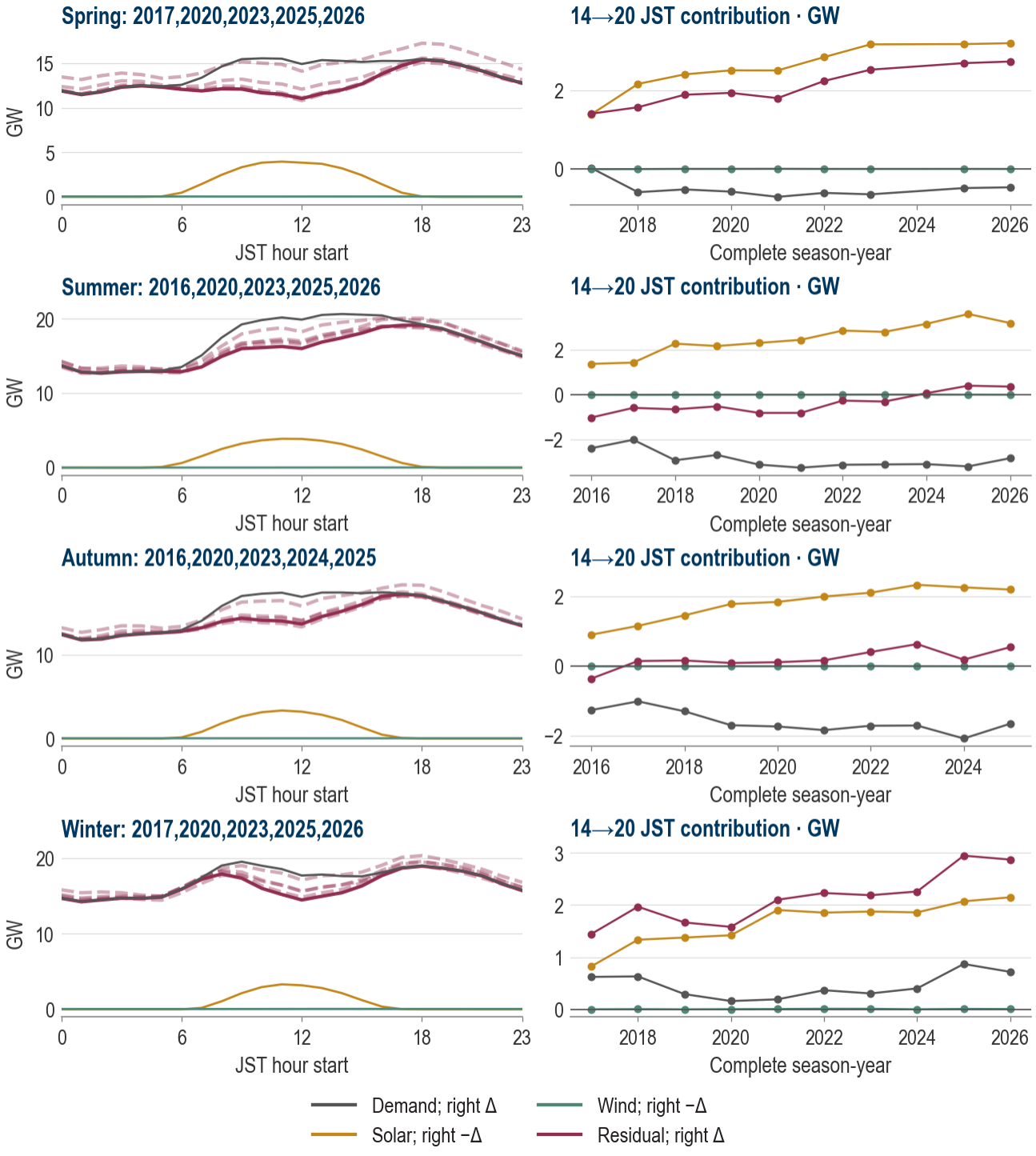
Source: [2–4] [32] · Chart data

Summer's corresponding residual increase was **2.29 GW**, with demand falling **2.90 GW** as solar withdrew. Winter's increase was **4.57 GW**: demand rose **0.81 GW** while solar fell **3.76 GW**. The six-hour change is distinct from the largest adjacent-hour ramp; dispatchable resources face both the transition's speed and its total energy requirement.

Kansai: the evening component balance

Kansai's spring 2026 mean residual rose **2.74 GW** from 14:00 to 20:00 even though demand changed by **-0.47 GW**. Solar fell **3.21 GW**; wind changed **-0.000 GW**. This is an observed component balance, not a hypothetical ramp example. [2] [3] [4]

Exhibit 28: Kansai: seasonal evening transition across years



Note: Left: latest residual solid; faint earlier-year traces are unkeyed context (years in headings). Other components: latest year. Right: all complete years; 14–20 JST ΔResidual = ΔDemand − ΔSolar − ΔWind.

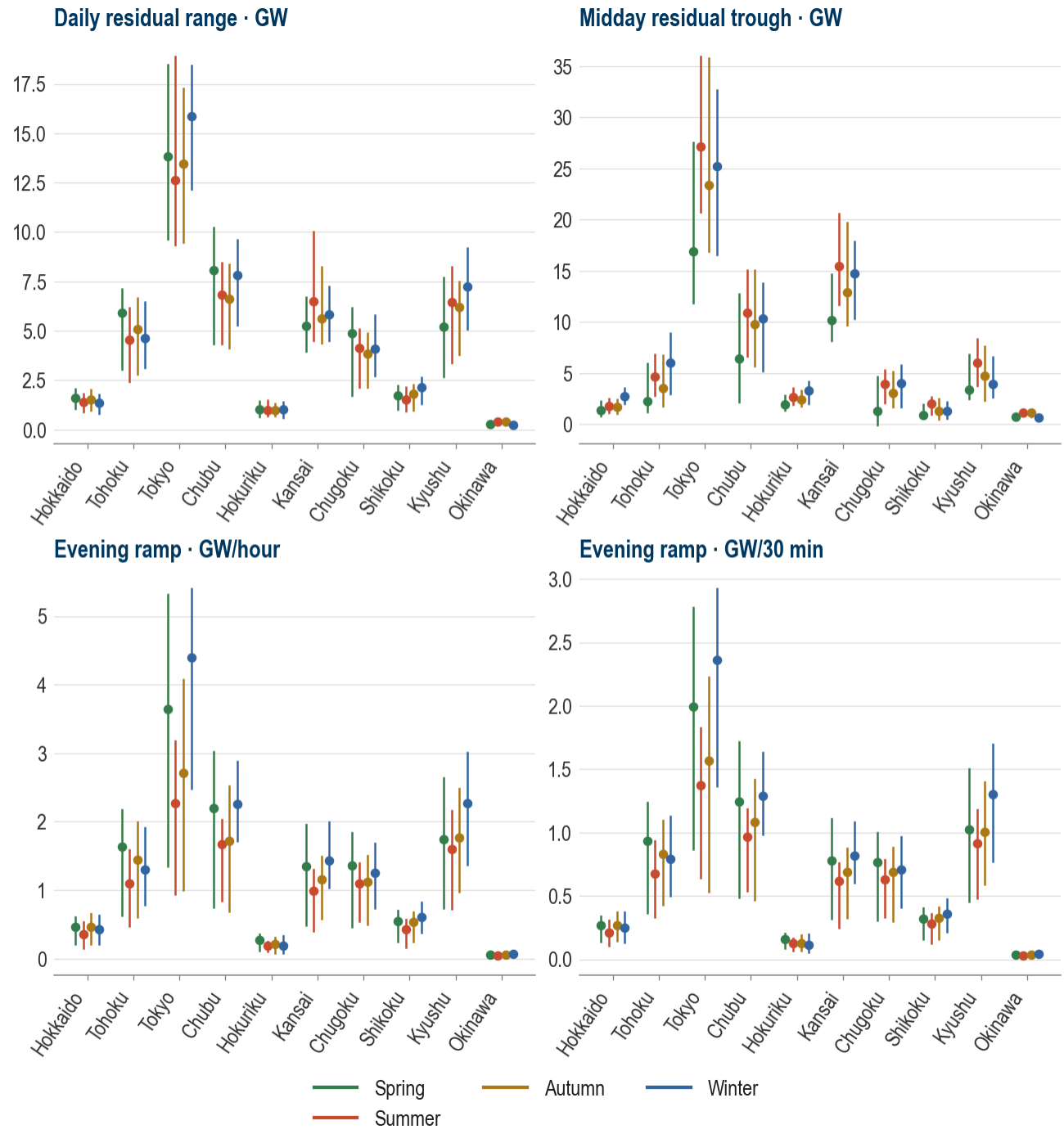
Source: [2–4] [32] · Chart data

Summer's corresponding residual increase was **0.36 GW**, with demand falling **2.82 GW** as solar withdrew. Winter's increase was **2.87 GW**: demand rose **0.72 GW** while solar fell **2.15 GW**. The six-hour change is distinct from the largest adjacent-hour ramp; dispatchable resources face both the transition's speed and its total energy requirement.

Ramps and daily ranges: flexibility is regional and seasonal

Tokyo's winter 2026 median daily residual range was **15.87 GW**; the p90 maximum positive evening ramp was **5.41 GW/hour**. Chubu's corresponding figures were **7.83 GW** and **2.88 GW/hour**. Kansai's were **5.82 GW** and **1.99 GW/hour**. These differences persist even though all three face the same sunset cycle. [2] [3] [4]

Exhibit 29: All-region residual ranges and evening ramps



Note: Latest complete season separately by resolution; full dates/year in extract. Shikoku native summer is an observed-day sample when no complete modern summer exists. Whiskers are daily p10–p90 with median dots; clock panels use frequency-weighted dots and earliest tied peak. Nonpositive residual maxima have no load factor.

Source: [2–4] [32] · Chart data

Spring p90 evening ramps were **2.17 GW/hour** in Tohoku and **1.84 GW/hour** in Chugoku, against **0.60 GW/hour** in Hokkaido. Solar withdrawal, wind variation and customer demand jointly determine the ramp. The modern half-hour panel retains GW/30 min, not a doubled hourly rate. Shikoku's incomplete modern summers remain a dated observed-day sample; complete historical seasons stay separate.

Load factors and peak clocks: average utilisation is not evening security

Using hourly observations, spring 2026 Tokyo's median daily demand load factor was **0.867**, but residual load factor was **0.764**. Chugoku moved from **0.898** to **0.701**; Hokuriku from **0.904** to **0.851**. Subtracting variable renewables reshapes utilisation more in some regions than others. [2] [3] [4]

Exhibit 30: Regional peak timing and load factors

Note: Latest complete season separately by resolution; full dates/year in extract. Shikoku native summer is an observed-day sample when no complete modern summer exists. Whiskers are daily p10–p90 with median dots; clock panels use frequency-weighted dots and earliest tied peak. Nonpositive residual maxima have no load factor.

Source: [2–4] [32] · Chart data

Hourly clock distributions use the earliest tied daily peak, not an arithmetic average of times around midnight. Tokyo's winter daily demand peak was most often 09:00, while residual most often peaked at 18:00; Okinawa's winter modes were both 19:00. A low midday residual does not remove evening capacity needs. The next chapter connects these regional requirements to actual generation, storage and outage histories. [33]

Supply, nuclear, renewables, storage and outages

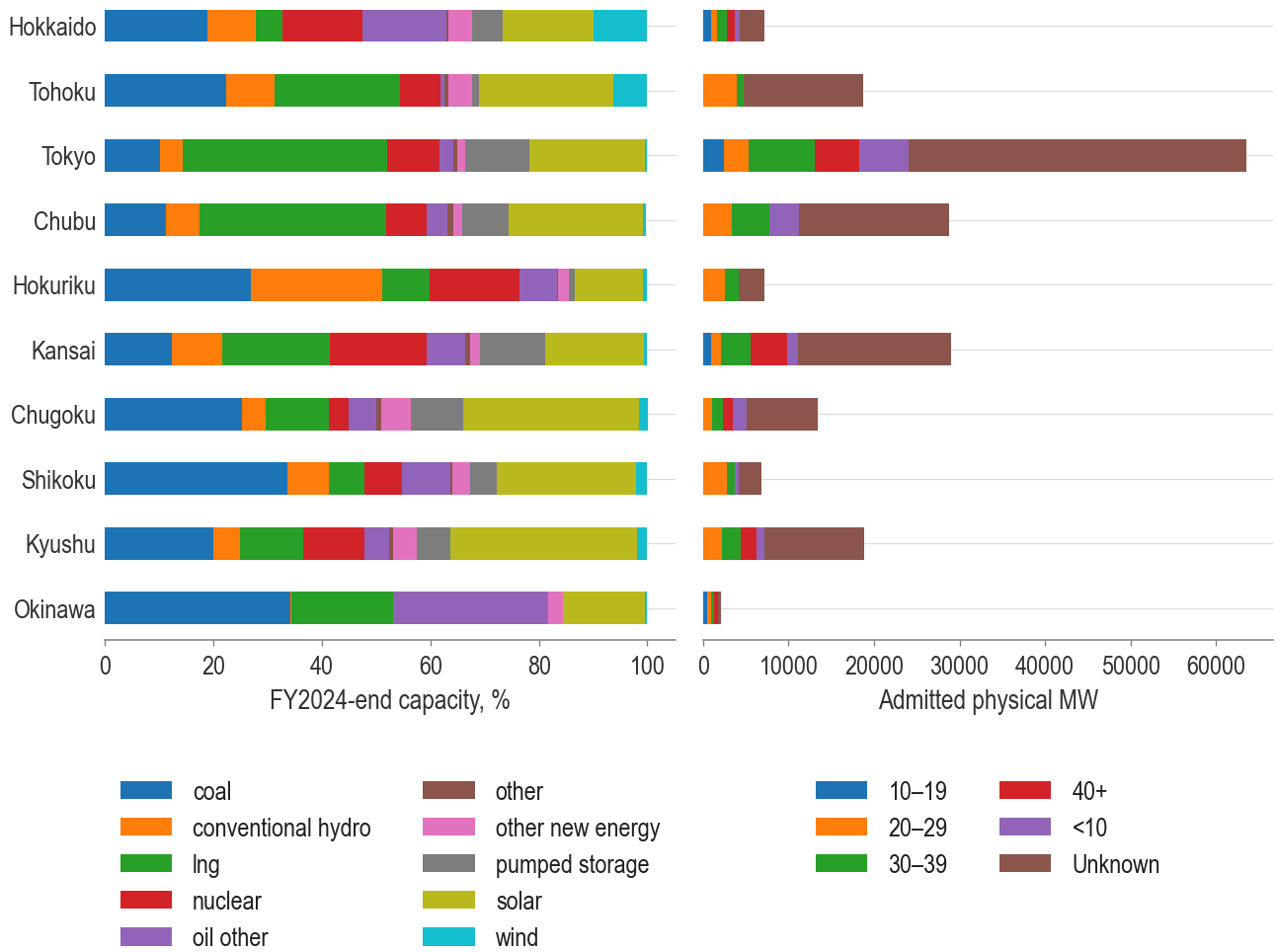
Installed supply is not available supply

The official FY2024-end composition includes idle thermal capacity and 33 non-retired reactors; the NRA recorded only 13 commercial reactors operating on 29 October 2026. An installed nuclear fleet can therefore contribute no contemporaneous generation. [27] [38]

- Hokkaido / Tohoku: winter thermal needs
- Renewables: output and seasonal control
- Evening flexibility is a joint thermal, water and storage response

The compilation reported 324.16 GW nationally: 147.96 GW thermal, 33.08 GW nuclear, 22.03 GW conventional hydro and 27.34 GW pumped storage. Solar nameplate was 76.70 GW, but daylight energy cannot substitute one-for-one for evening capability. These capacity categories do not establish availability. [27]

Exhibit 31: Installed assets, operating state and age are different quantities



Note: Left: official FY2024-end composition, rounding retained. Right: admitted-cohort age at 7 November 2026; unknown dates and decade-boundary ambiguity stay separate. Neither measures availability.

Source: [27] [28] [331] · Chart data

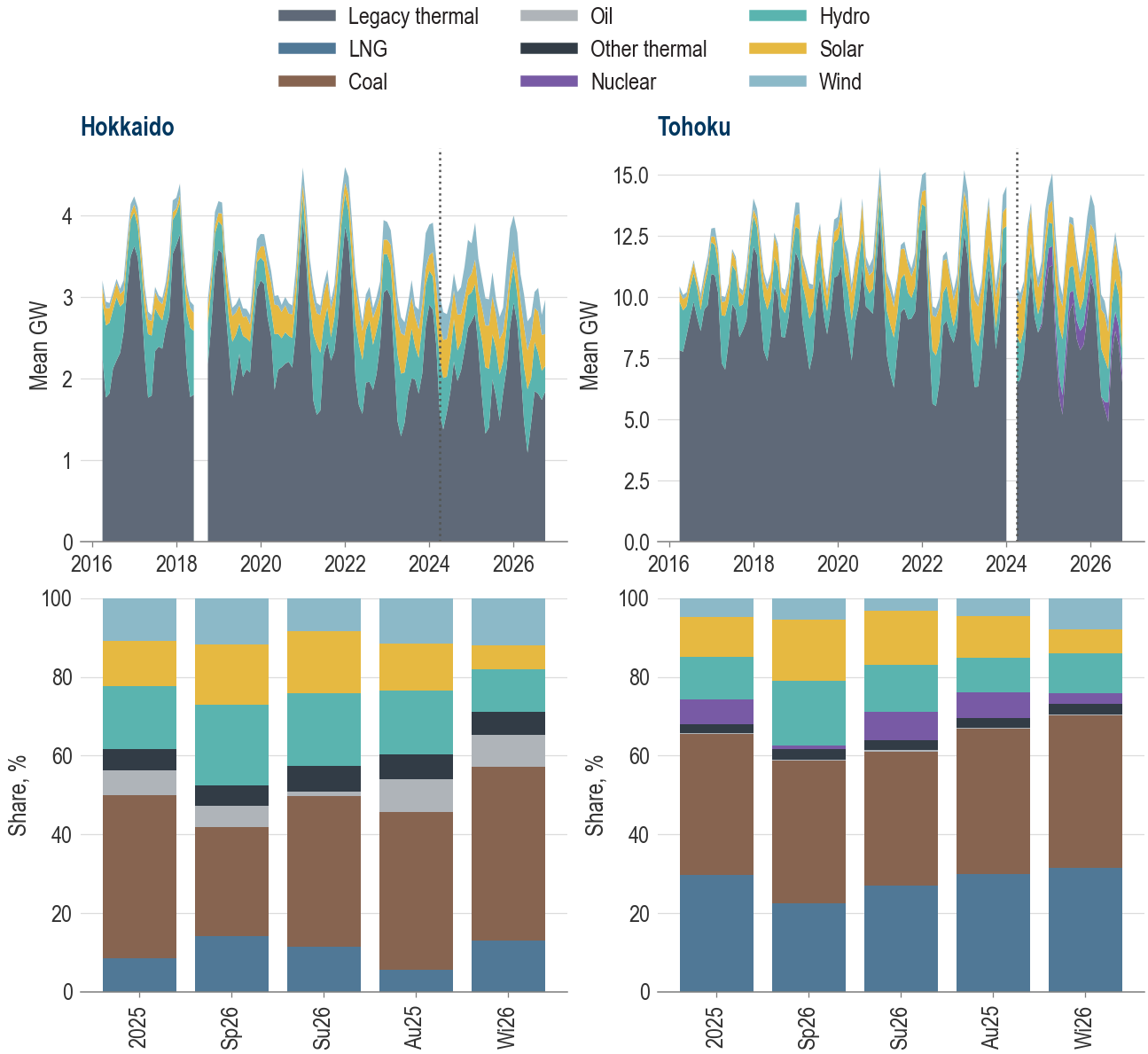
Age is not an outage probability. Nuclear operation-start dates are documented; month/year commissioning evidence establishes age bounds. A decade-boundary crossing stays ambiguous, while unknown thermal commissioning dates remain unknown—not a spurious young fleet. [39] [331]

Hokkaido / Tohoku: winter thermal needs

Hokkaido's thermal output averaged 2.76 GW in winter 2026, against 1.58 GW in spring. Winter wind was stronger—0.47 versus 0.36 GW—but solar fell from 0.46 to 0.23 GW. Wind therefore moderates the winter requirement without eliminating the cold-weather thermal exposure. Tomari remained suspended in the October reactor snapshot. [4] [38]

Tohoku's winter thermal mean was 10.04 GW, versus 6.56 GW in spring. Its 1.09 GW winter wind contribution was almost twice spring's 0.57 GW; nuclear output nevertheless varied sharply between seasonal windows. Onagawa's commercial return did not make regional nuclear output constant. [4]

Exhibit 32: Hokkaido and Tohoku: generation through time



Note: Top: monthly observed GW; gray thermal is aggregate; dotted line: April 2024. Bottom: modern shares, complete years and latest Sp/Su/Au/Wi (year suffix); * marks partial seasons. Full annual/seasonal history is in the extract.

Source: [3] [4] · Chart data

Spring 2026 observed thermal means, GW; all 4,416 native intervals:

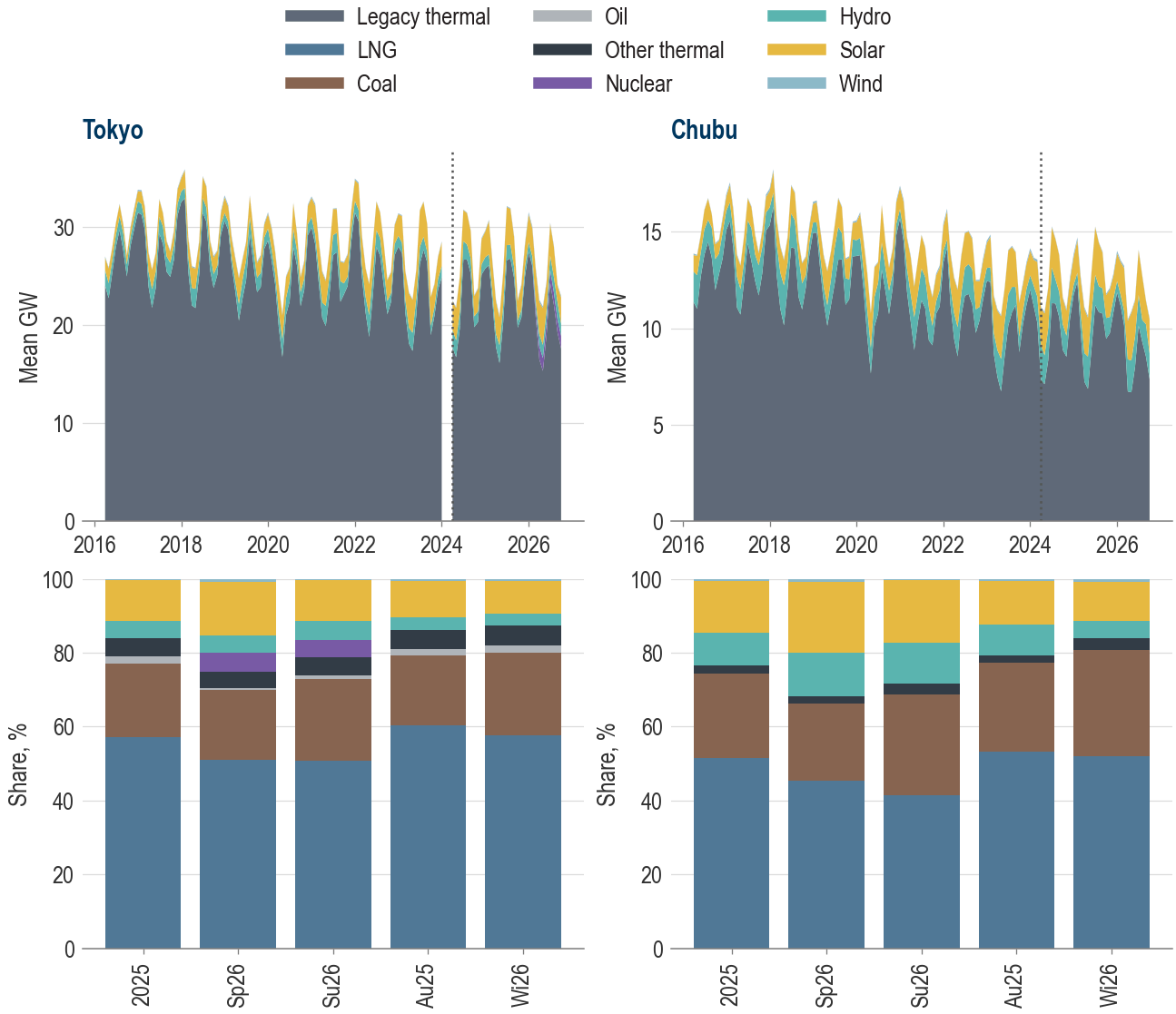
Area	LNG	Coal	Oil	Other thermal
Hokkaido	0.42	0.83	0.16	0.16
Tohoku	2.39	3.85	0.03	0.29

Tokyo / Chubu: LNG and nuclear states

Tokyo's winter 2026 thermal mean was 26.10 GW, compared with 17.55 GW in spring; Chubu's corresponding values were 11.28 and 7.87 GW. Solar helps both regions during daylight, but its spring contribution—3.40 GW in Tokyo and 2.23 GW in Chubu—is not firm evening supply. [4]

Tokyo's nuclear mean rose to 1.17 GW in spring and 1.30 GW in summer 2026. Kashiwazaki-Kariwa 6's commercial return was confirmed on 16 April. Chubu still reported zero nuclear output: Hamaoka 3–5 remained suspended. The same imported-fuel shock therefore reaches materially different residual thermal requirements. [40] [38]

Exhibit 33: Tokyo and Chubu: generation through time



Note: Top: monthly observed GW; gray thermal is aggregate; dotted line: April 2024. Bottom: modern shares, complete years and latest Sp/Su/Au/Wi (year suffix); * marks partial seasons. Full annual/seasonal history is in the extract.

Source: [3] [4] · Chart data

Spring 2026 observed thermal means, GW; all 4,416 native intervals:

Area	LNG	Coal	Oil	Other thermal
Tokyo	11.93	4.44	0.14	1.04
Chubu	5.23	2.40	0.00	0.23

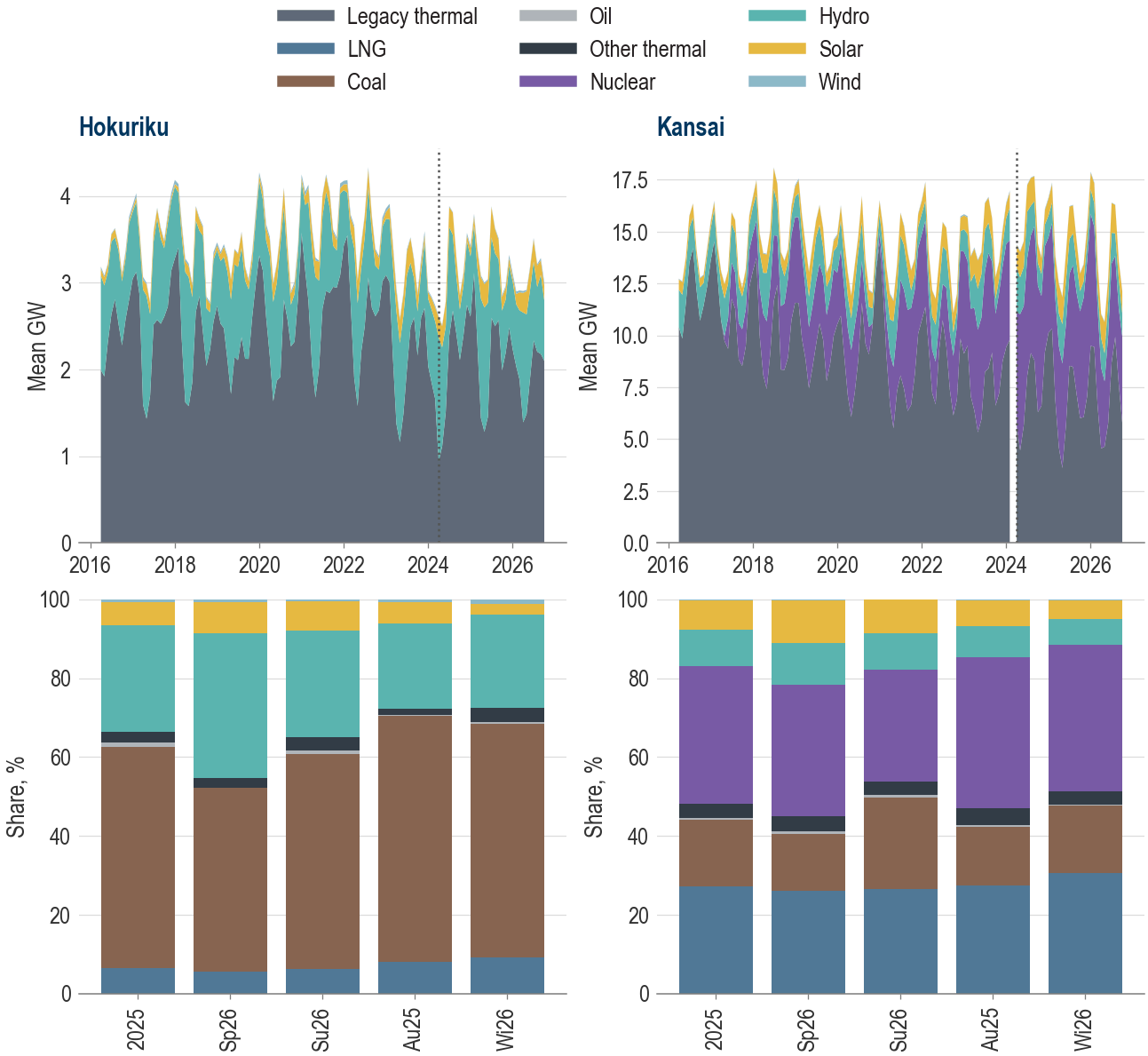
These are electrical reporting areas: a reactor's physical prefecture is not its electricity destination. [4]

Hokuriku / Kansai: water and nuclear cycles

Hokuriku's spring 2026 conventional hydro mean was 1.07 GW, against 1.59 GW thermal. Hydro supplied 36.7% of the five consistently reported generation categories. Shika's suspended reactors contributed no nuclear generation, so nameplate nuclear capacity does not explain the region's observed spring supply. [4] [38]

Kansai's nuclear mean moved from 3.93 GW in spring 2026 to 4.30 GW in summer; winter 2026 had averaged 6.24 GW. Thermal output also rose from spring to summer, from 5.29 to 8.16 GW. Higher nuclear output need not coincide with lower thermal output when cooling demand and the rest of the balance change. [4]

Exhibit 34: Hokuriku and Kansai: generation through time



Note: Top: monthly observed GW; gray thermal is aggregate; dotted line: April 2024. Bottom: modern shares, complete years and latest Sp/Su/Au/Wi (year suffix); * marks partial seasons. Full annual/seasonal history is in the extract.

Source: [3] [4] · Chart data

Spring 2026 observed thermal means, GW; all 4,416 native intervals:

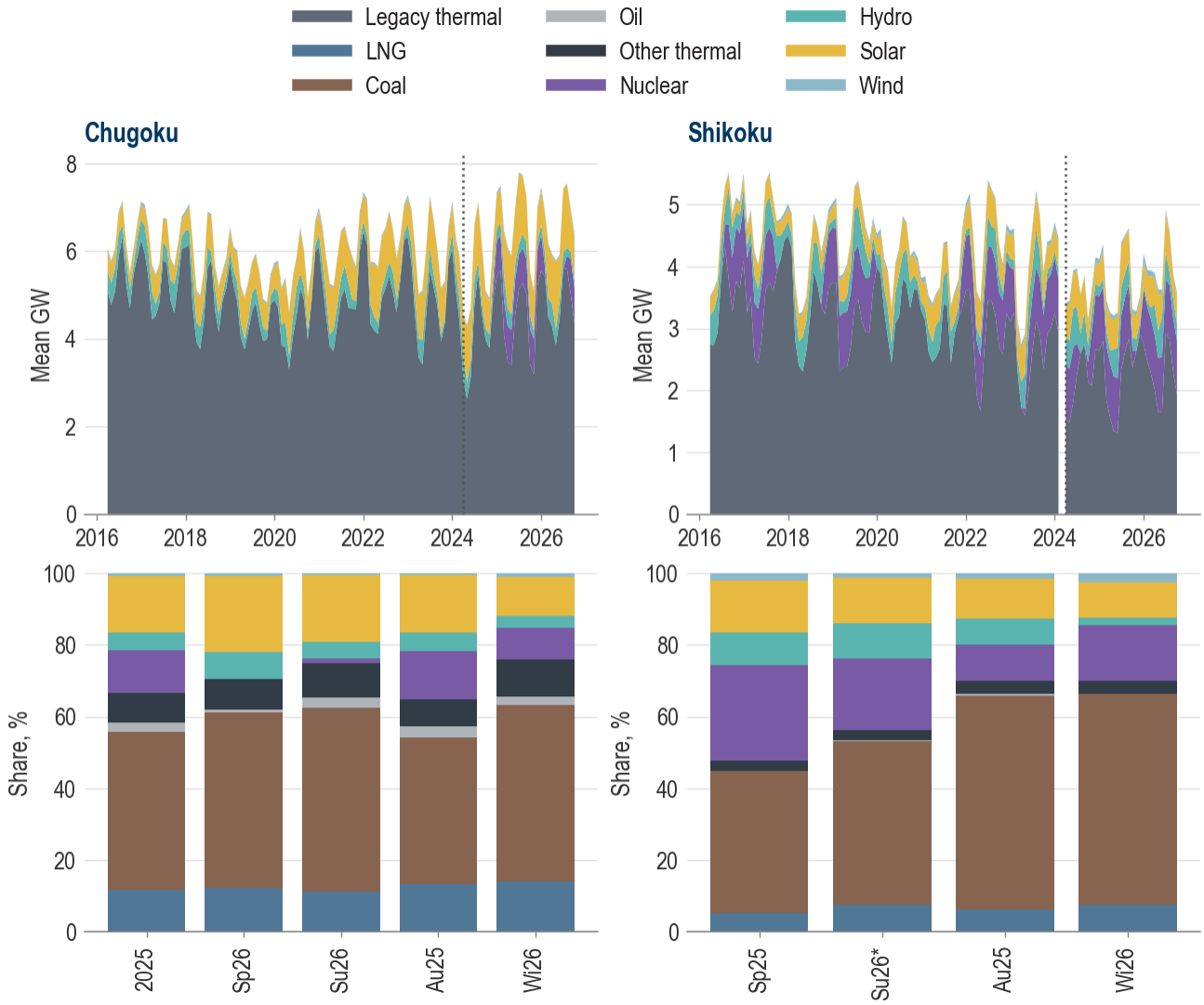
Area	LNG	Coal	Oil	Other thermal
Hokuriku	0.16	1.36	0.00	0.07
Kansai	3.08	1.70	0.06	0.45

Chugoku / Shikoku: single-reactor exposure

Shimane 2 resumed commercial operation on 10 January 2025, yet Chugoku's spring 2026 regional nuclear series was zero. Its autumn 2025 mean had been 0.80 GW. A restart milestone creates a new operating possibility, not a permanent flat injection. Chugoku's spring thermal mean was 4.21 GW and solar 1.28 GW. [41] [4]

Shikoku's Ikata 3 is a large block relative to its regional demand. Nuclear output averaged 0.38 GW in autumn 2025 and 0.61 GW in winter 2026. Spring and summer 2026 have incomplete common wind/generation days; their samples are not complete-season comparators. [4]

Exhibit 35: Chugoku and Shikoku: generation through time



Note: Top: monthly observed GW; gray thermal is aggregate; dotted line: April 2024. Bottom: modern shares, complete years and latest Sp/Su/Au/Wi (year suffix); * marks partial seasons. Full annual/seasonal history is in the extract.

Source: [3] [4] · Chart data

Spring 2026 thermal-only observed means, GW; all 4,416 native thermal intervals:

Area	LNG	Coal	Oil	Other thermal
Chugoku	0.73	2.91	0.05	0.51
Shikoku	0.26	1.62	0.00	0.10

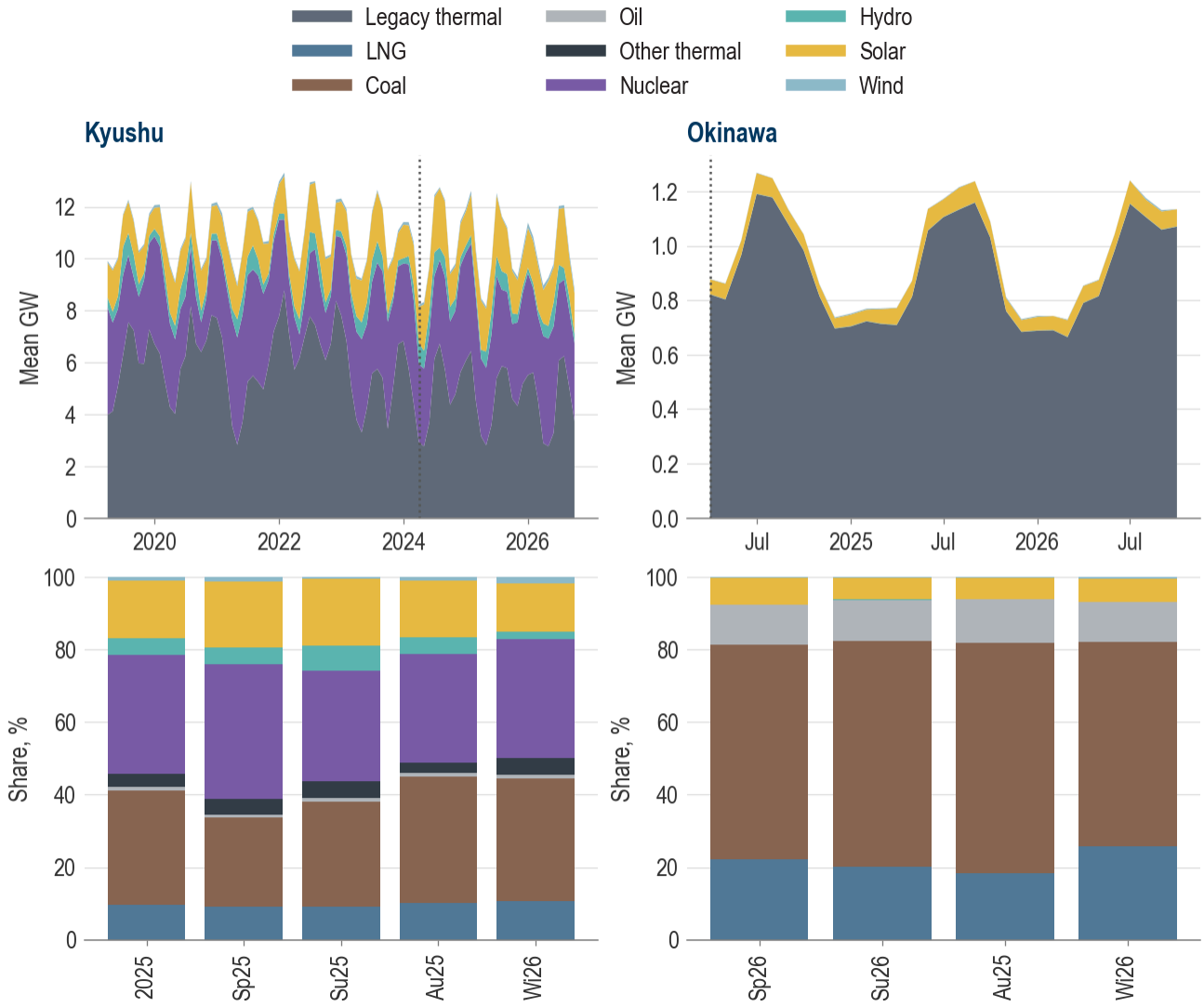
Coal dominates these thermal slices; the two Shikoku connections and Chugoku's western junction remain distinct routes, not unlimited pooling. [42]

Kyushu / Okinawa: different supply constraints

Kyushu's spring 2026 means were 3.84 GW nuclear, 3.41 GW thermal and 1.59 GW accepted solar. Nuclear supplied 40.9% of the five long-history categories. This coexistence explains why a low-demand sunny interval can need pumping, exports or control even while thermal generation remains important outside daylight. [4]

Okinawa's thermal mean rose from 0.69 GW in winter 2026 to 1.08 GW in summer. Nuclear was zero, and the islands have no mainland interconnector. Its cooling-season fuel requirement cannot be interpreted through a fictitious JEPX area price or a mainland import response. [4] [42]

Exhibit 36: Kyushu and Okinawa: generation through time



Note: Top: monthly observed GW; gray thermal is aggregate; dotted line: April 2024. Bottom: modern shares, complete years and latest Sp/Su/Au/Wi (year suffix); * marks partial seasons. Full annual/seasonal history is in the extract.

Source: [3] [4] · Chart data

Spring 2026 observed thermal means, GW; all 4,416 native intervals:

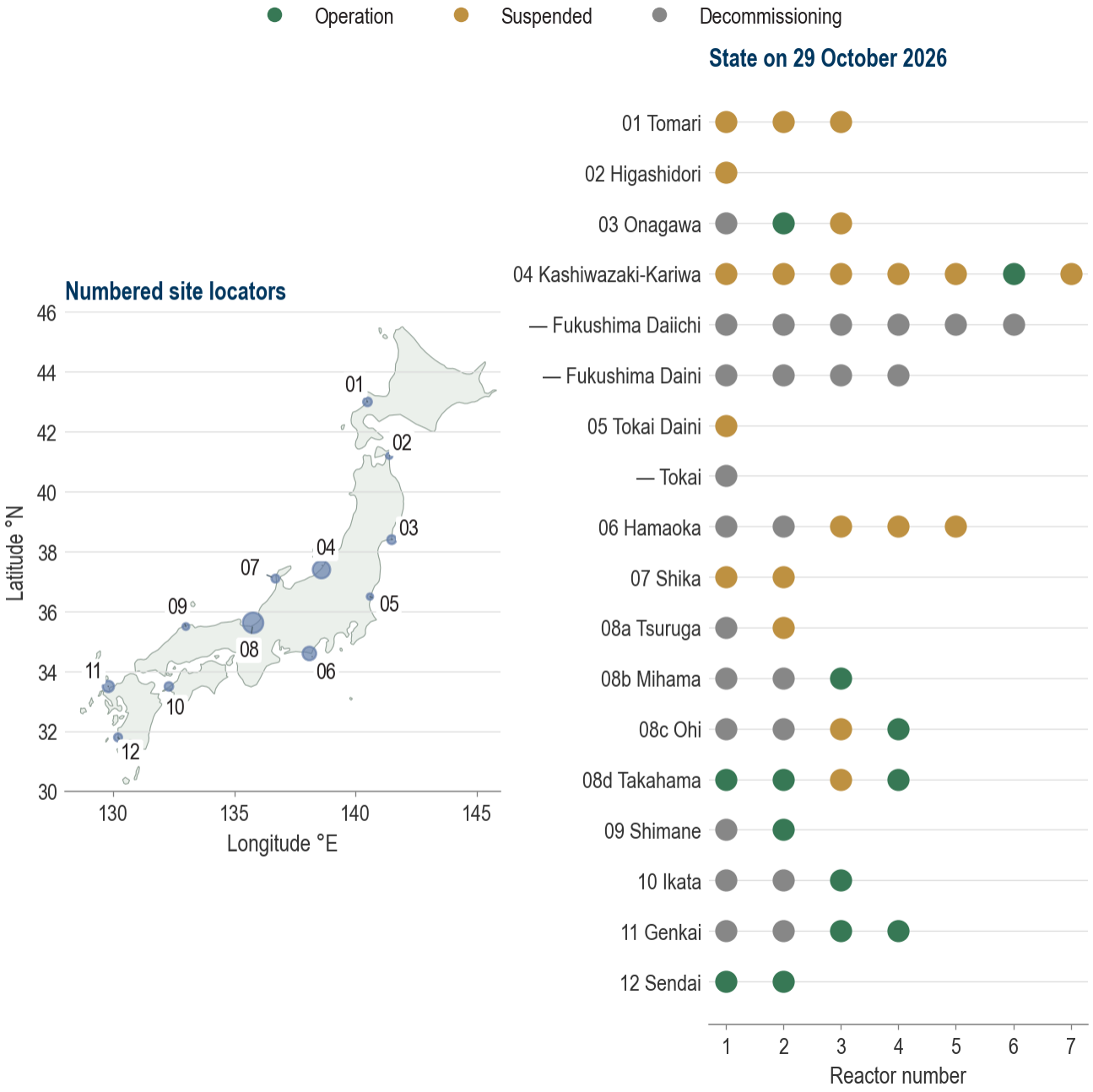
Area	LNG	Coal	Oil	Other thermal
Kyushu	0.58	2.27	0.08	0.48
Okinawa	0.18	0.49	0.09	0.00

Kyushu's legacy history starts in 2019; Okinawa's qualified interval history is shorter. Neither is backfilled from national generation. [3]

Nuclear atlas: 57 commercial reactors

On 29 October 2026 the NRA listed 13 operating and 20 suspended commercial reactors. Another 24 commercial reactors were in decommissioning, including Fukushima Daiichi's six units. Research reactors are outside these counts. Kansai and Kyushu dominate the operating western fleet; Tokyo's Kashiwazaki-Kariwa 6, Tohoku's Onagawa 2 and Chugoku's Shimane 2 widen the geographical contribution. [38]

Exhibit 37: Current commercial-reactor atlas



Note: 13 operating, 20 suspended, 24 decommissioning reactors. Numbers match matrix labels; a–d share the Fukui locator; — means no corroborated coordinate. Map point area is nameplate, not output. Coast: Natural Earth.

Source: [38] [39] [98] [332] · Chart data

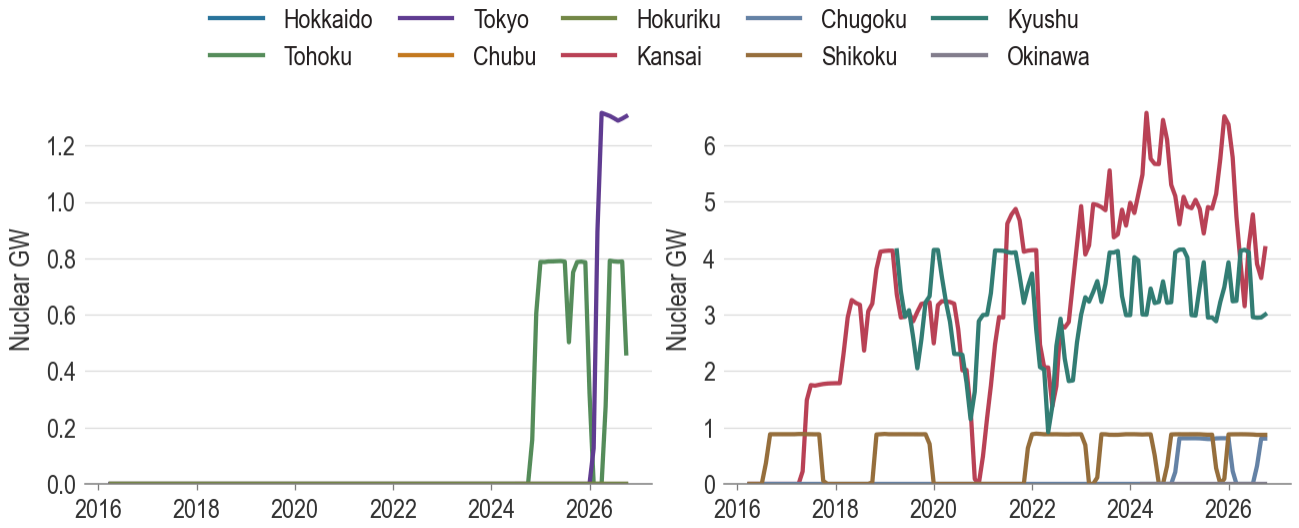
Chubu's Hamaoka 3–5 and Hokuriku's Shika 1–2 were suspended, as were Tomari's three reactors. Suspended does not specify whether a reactor is in a routine inspection, awaiting regulatory approval or facing another restriction. A permitted or restarted reactor can subsequently stop. [38]

Rated MW and original operation-start dates do not establish dispatch. Regulatory approval, first generation and commercial operation are distinct clocks; none allocates an area total to an individual reactor. [39]

Nuclear output: regions and reactors

The regional history includes genuine zero-output periods in Chubu, Hokuriku and Hokkaido, rather than treating their installed reactors as generating assets. Kansai's multi-reactor cycle contrasts with the pronounced single-block effects in Shikoku and Chugoku. [4]

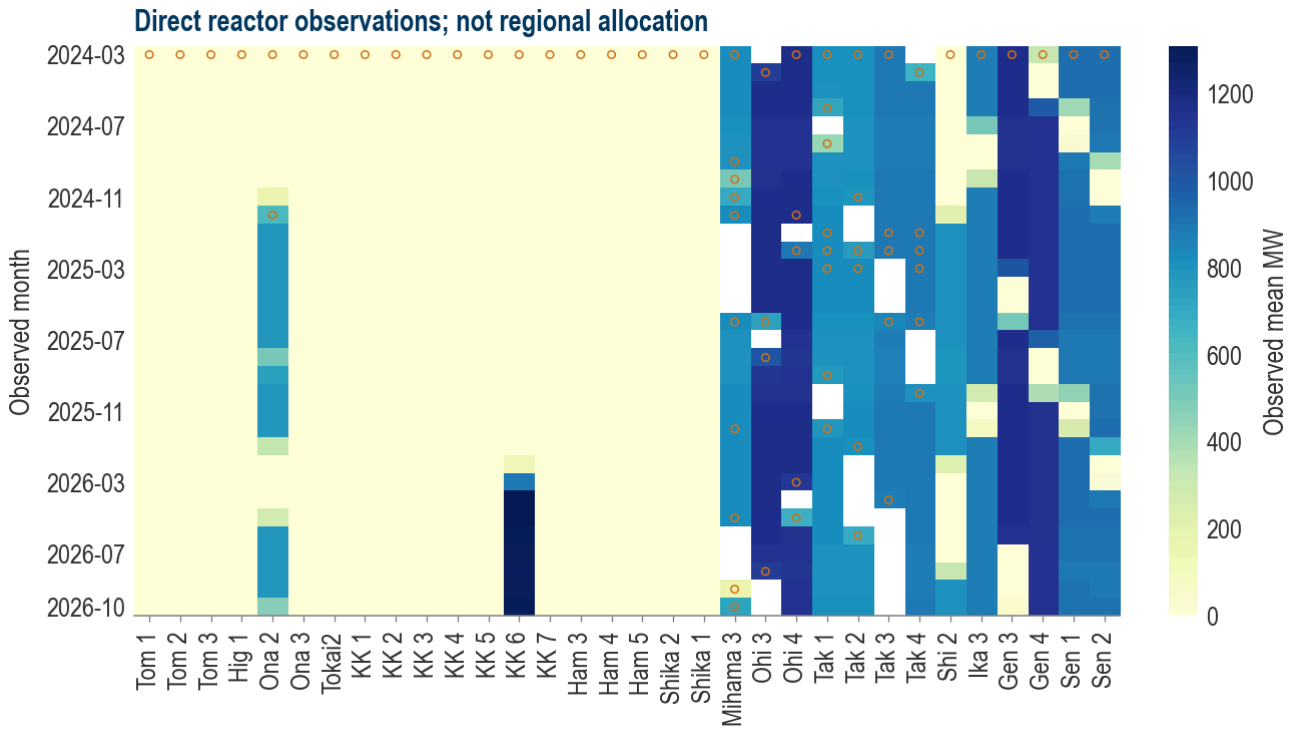
Exhibit 38: Ten-area nuclear generation history



Note: Regional nuclear GW, independent scales. Inspection cycles and permanent non-generation are distinct from missing observations; complete monthly energy is in the extract.

Source: [3] [4] · Chart data

Exhibit 39: Direct reactor-by-month output evidence



Note: Every observed month and intervening gap remains on the grid; orange rings mark partial months, blank cells unavailable output. Abbreviated plant labels resolve in the extract; a reported zero is not an inferred inspection reason.

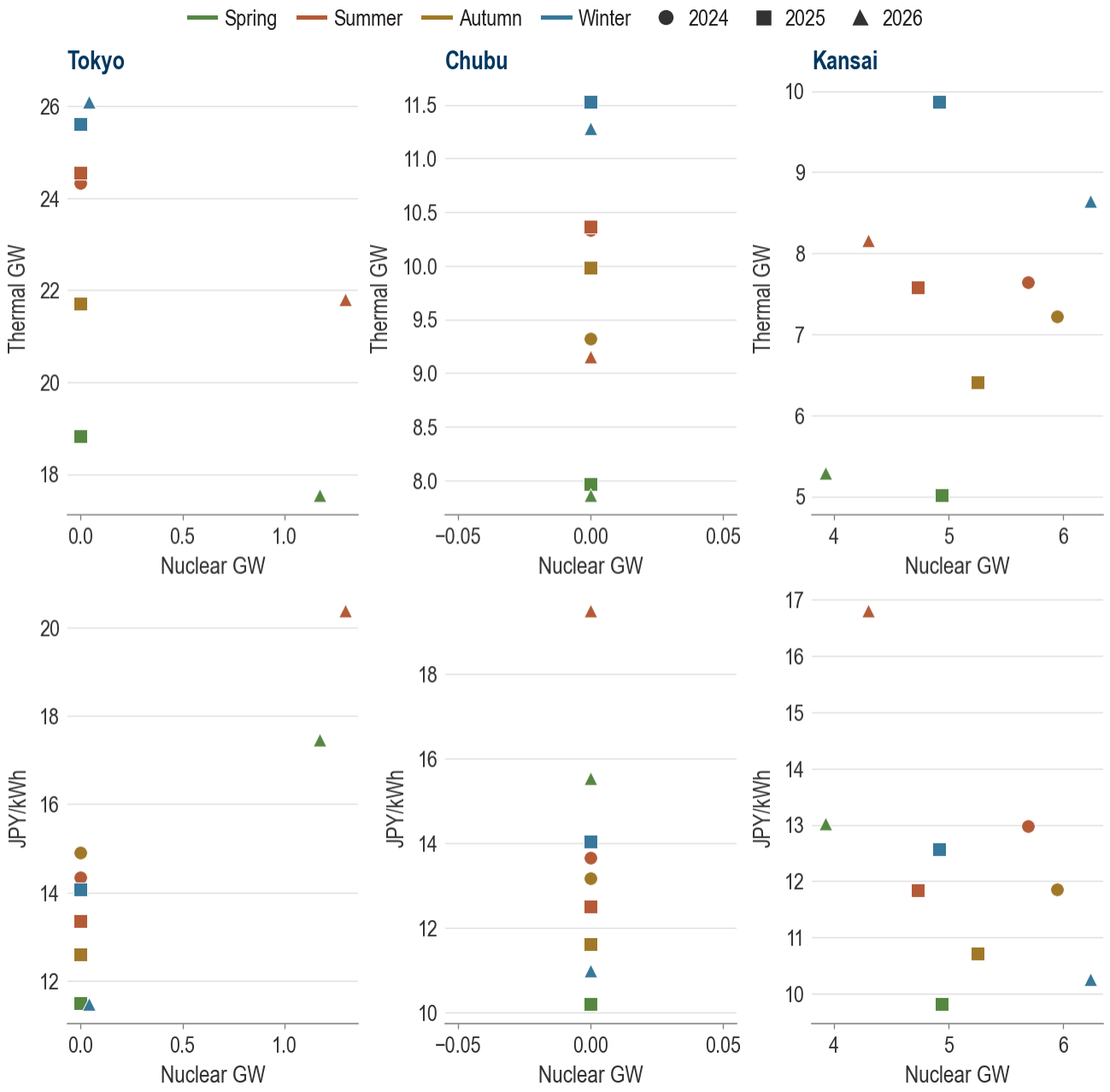
Source: [6] [43] · Chart data

Direct reactor observations are independent evidence, not a decomposition of regional totals. A partial month supports an observed-period MW mean, not a full month's energy or an assumed capacity factor. Fukushima's retired fleet stays in the status atlas, not in current generating capability. [6] [43]

Tokyo, Chubu and Kansai: nuclear does not explain price alone

Tokyo's spring 2026 nuclear mean of 1.17 GW coincided with 17.55 GW thermal generation; summer brought 1.30 GW nuclear and 21.79 GW thermal. A reactor return can reduce the thermal requirement relative to an unobserved no-return case while actual thermal output still rises with demand. The observed association is not that counterfactual. [4]

Exhibit 40: Nuclear and thermal substitution under changing residual demand



Note: Colour identifies season; marker shape identifies year. Every matched observation remains at its measured coordinates, including Chubu nuclear zero; no jitter. Residual demand and prices accompany the extract, not a reactor-level counterfactual.

Source: [1] [2] [4] · Chart data

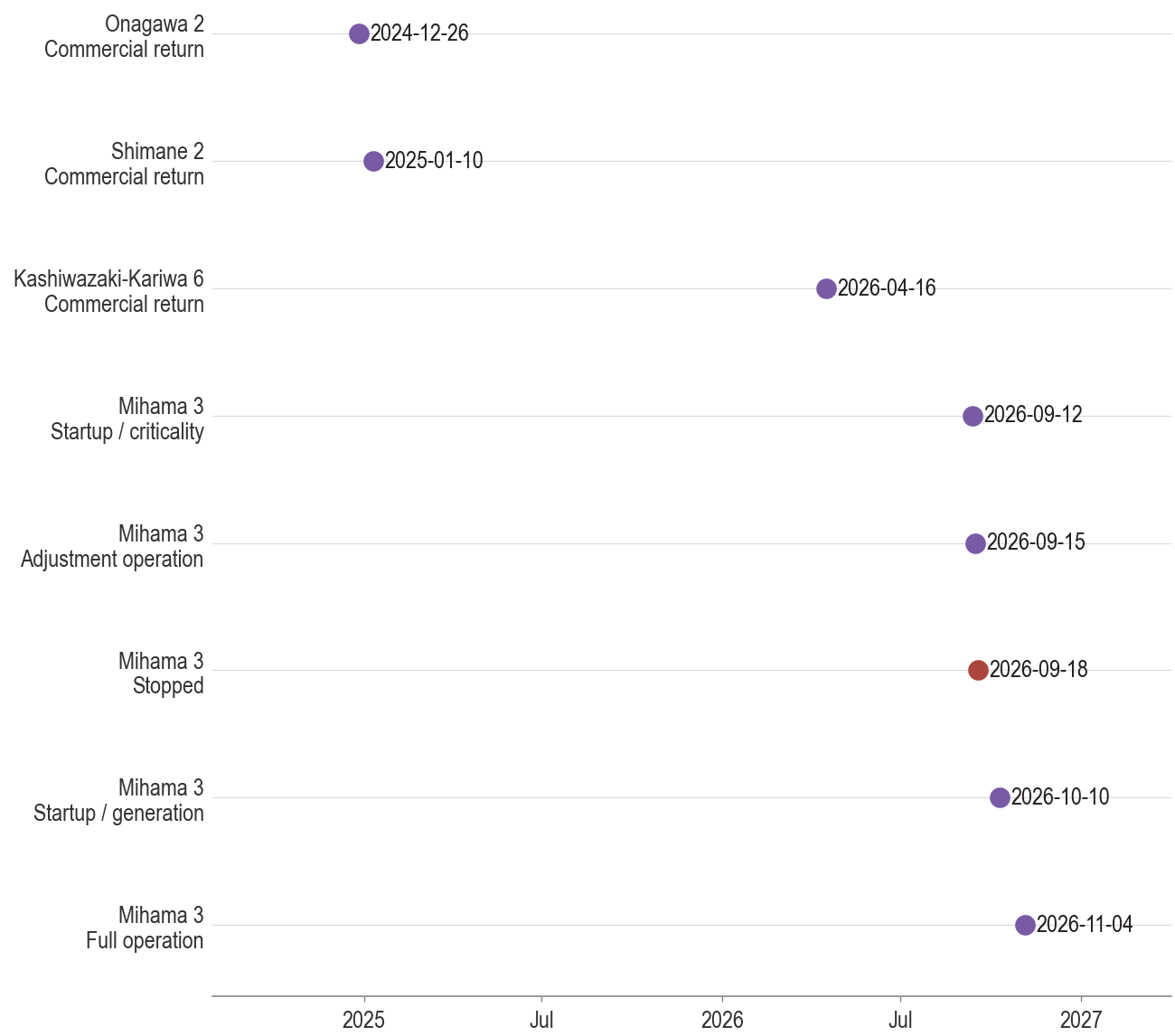
Kansai's nuclear mean varied from 3.93 GW in spring 2026 to 6.24 GW in winter 2026. Its thermal output varied in the same direction across those windows. Weather, residual demand, water availability, neighbouring supply and fuel prices compete with the reactor explanation. [4]

Chubu's nuclear observations remain at zero across the comparison. Its exposure is therefore partly to neighbouring nuclear availability through the network, not to a locally observed Hamaoka return. Regional spread analysis belongs alongside these physical comparisons, not in a claim that one reactor independently set the price. [38]

Recent nuclear returns and interruptions

Commercial returns: Onagawa 2, 26 December 2024 at 16:00 JST (825 MW); Shimane 2, 10 January 2025 at 15:10 (820 MW); Kashiwazaki-Kariwa 6, 16 April 2026 at 16:00 (1,356 MW). Ratings are not measured restart steps. [44] [41] [40] [39]

Exhibit 41: Confirmed nuclear return and interruption milestones



Note: Onagawa 2, Shimane 2 and Kashiwazaki-Kariwa 6 commercial returns precede the latest Mihama 3 interrupted-return sequence. Dates are confirmations, not price forecasts.

Source: [40] [41] [44] [333] · Chart data

Mihama 3 (826 MW) reached startup/criticality on 12 September 2026 and adjustment operation on 15 September, then stopped on 18 September. It restarted with criticality and adjustment operation on 10 October; full operation followed on 4 November at 15:00 JST. November's confirmation lies beyond the October analytical cutoff—not a retroactive September assumption. [333]

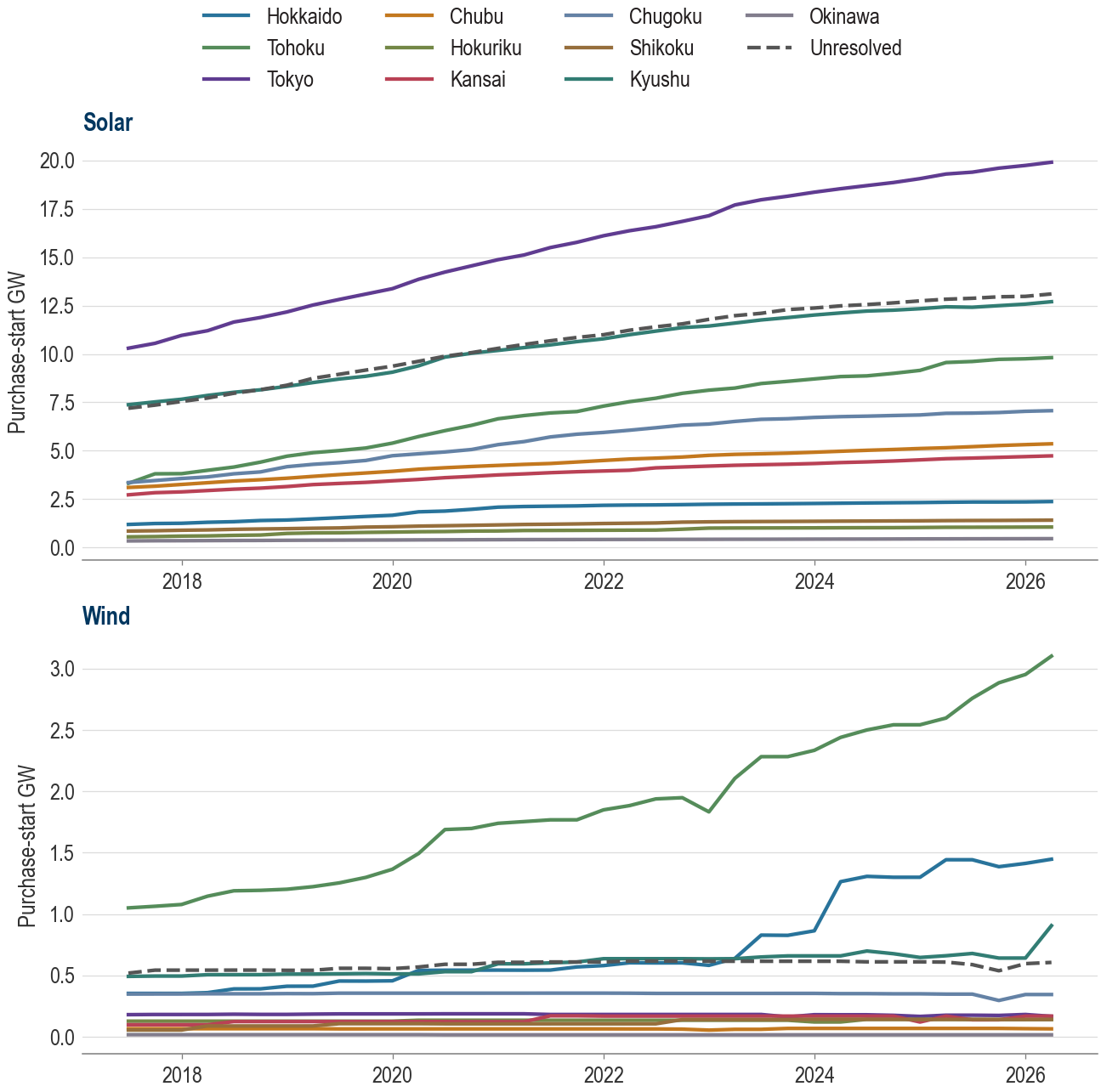
Tsuruga 2 faces a regulatory constraint. The NRA's September 2025 draft report records refusal of its permit-change application in November 2024: continuity between the relevant fault and the fracture zone beneath the reactor building could not be ruled out. Its 1,160 MW rating establishes no return date. [39]

The Mihama case separates coincident demand, neighbouring generation and price changes.

Renewable buildout: scheme capacity is not total installed capacity

FIT/FIP purchase-start solar capacity rose from 40.34 GW in June 2017 to 78.00 GW in March 2026; wind rose from 3.31 to 7.11 GW. These are scheme stocks after purchases began, not the larger pipeline of certified projects. New and transition certification are mutually exclusive; nested size or offshore subsets do not create additional capacity. [45] [46]

Exhibit 42: Regional renewable buildout without assigning split prefectures



Note: Quarterly stocks, June 2017–March 2026. Dashed unresolved geography includes seven split prefectures; detached systems remain in their published utility territory, not a mainland output denominator.

Source: [42] [45] [47–82] · Chart data

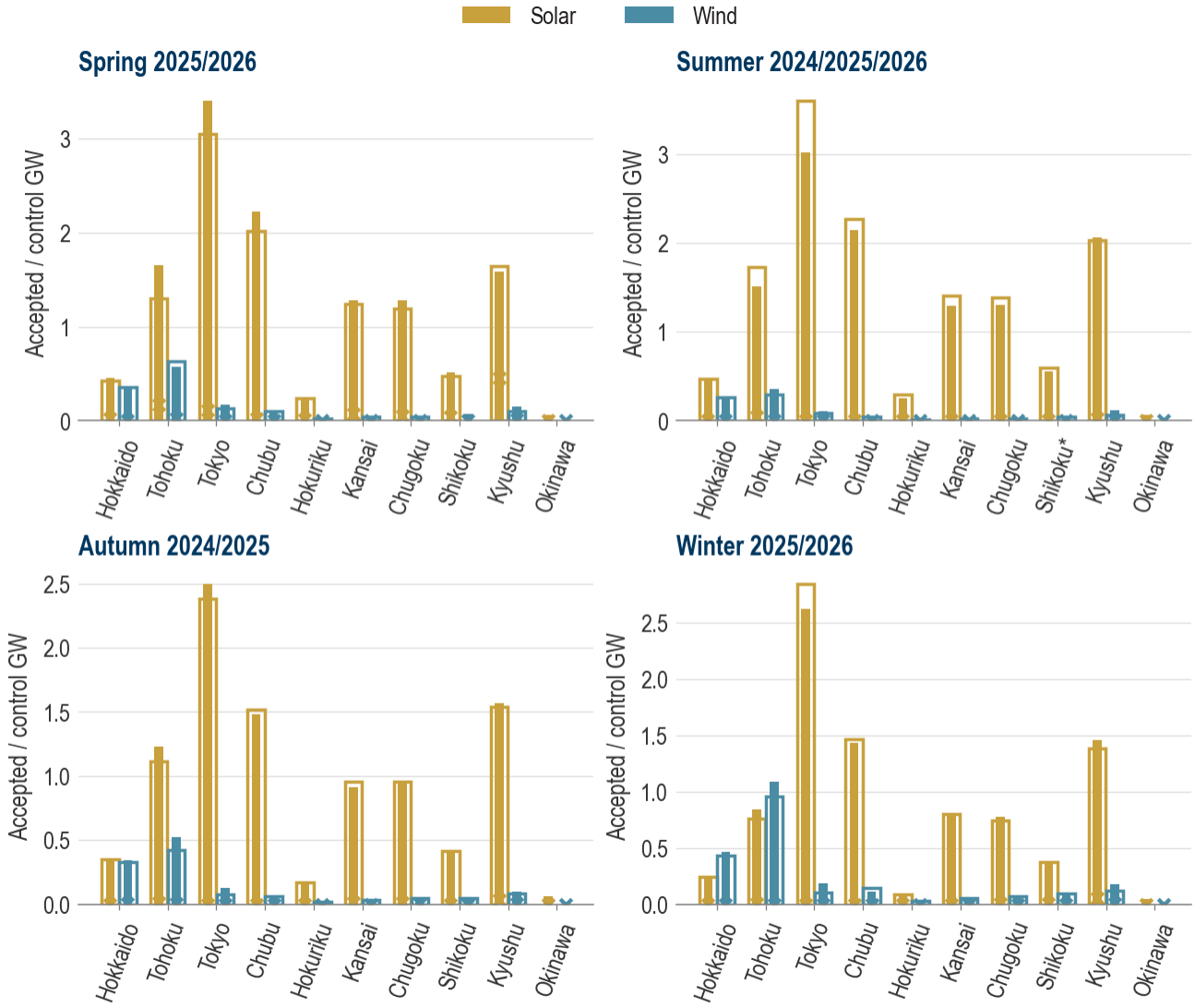
The verified whole-prefecture assignments locate 3.10 GW of wind in Tohoku and 1.45 GW in Hokkaido in March 2026. Solar is more geographically dispersed. Seven split prefectures leave 13.10 GW solar and 0.61 GW wind unresolved; those quantities are not silently awarded to the largest adjoining area. [45] [42]

FIT provides a regulated purchase arrangement; FIP combines market sale with a premium and exposes the producer to the practical value of timing and balancing. Neither label gives unrestricted network access. Environmental attributes and the electricity sale remain separate contractual questions. [224] [220]

Renewables: output and seasonal control

Winter/summer 2026 wind means were 0.47/0.25 GW in Hokkaido and 1.09/0.35 GW in Tohoku. Hokkaido solar moved oppositely, from 0.23 to 0.47 GW: renewable growth changes the timing of thermal needs. [4]

Exhibit 43: Accepted renewables and reported control across all regions and seasons



Note: Solid: latest eligible year; hollow: preceding year; ×: latest paired control. Panel dates identify years; * marks partial seasons. All years and paired fractions remain in the extract.

Source: [4] · Chart data

Mean solar-control profile peaks, JST. Sp/Su: spring/summer 2026; Au: autumn 2025; Wi: winter 2026. None = reported zero; — = no complete paired season.

Area	Sp	Su	Au	Wi	Area	Sp	Su	Au	Wi
Hokkaido	10:00	12:30	08:00	None	Kansai	12:30	09:30	12:00	12:00
Tohoku	12:00	10:30	10:30	13:00	Chugoku	12:00	12:00	12:00	12:30
Tokyo	12:00	None	None	None	Shikoku	12:00	12:00	None	12:00
Chubu	12:00	10:00	None	None	Kyushu	13:00	—	13:00	13:00
Hokuriku	12:00	13:00	None	None	Okinawa	12:30	None	None	11:30

Control concentrates in daylight hours, not a constant nameplate deduction. Fractions require matched accepted/controlled populations; control or low prices alone cannot prove an external interconnector bound. [4] [83]

Capture prices: solar is concentrated in the cheaper hours

In spring 2026, accepted solar's generation-weighted price was 11.68 JPY/kWh in Tokyo, 9.40 in Chubu, 9.32 in Kansai and 3.95 in Kyushu. Wind's corresponding values were 18.13, 15.12, 13.63 and 11.30. These matched-period associations reveal different delivery shapes; they are not guaranteed premiums available to a particular project. [1] [4]

Exhibit 44: Renewable capture differs from the area-average price

Note: Latest eligible year for each region/season. All year observations, equal-time price means and midday/evening contrasts accompany the plotted extract.

Source: [1] [2] [4] · Chart data

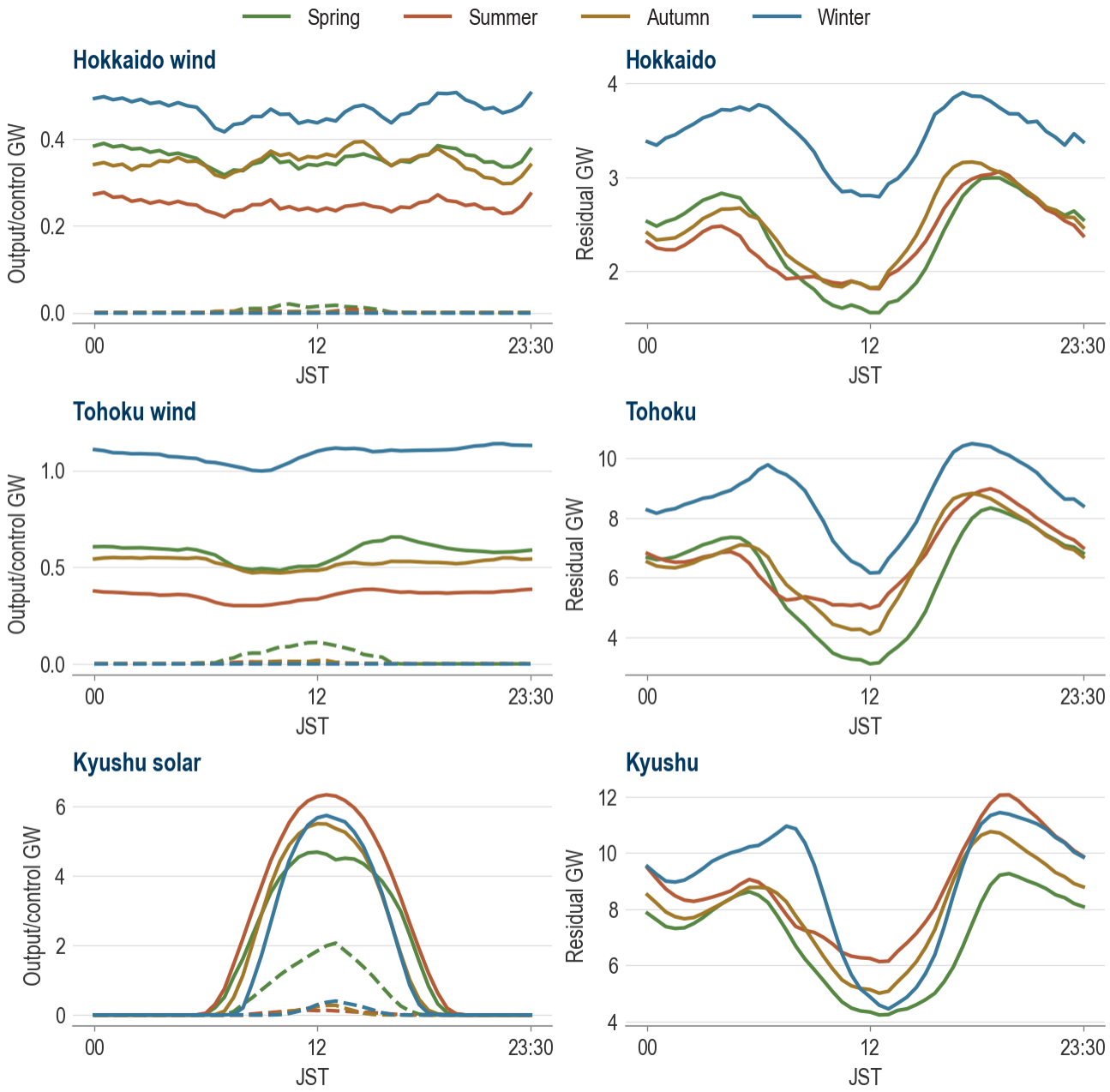
Kyushu's combination of nuclear output, concentrated solar and limited opportunities to move surplus energy can depress daytime value without forcing every evening interval lower. Hokkaido and Tohoku's winter wind contribution occurs under a different demand and transfer regime. Tokyo, Chubu and Kansai have larger demand centres but are not immune to a widening midday/evening contrast. [83] [2]

FIT receipts, a FIP premium, balancing costs and certificate revenues are not included in these capture prices. There is no Okinawa JEPX capture price. [224] [220]

Renewable surplus and its constraints

Kyushu's FY2026 mainland control windows are 07:00–17:00 in April–June and 08:00–16:00 in July–March. Online facilities are controlled only for necessary periods within those windows—not every sunny hour or low-price interval. [84]

Exhibit 45: Northern wind and southern solar create different residual shapes



Note: Four-season latest eligible profiles. Left: solid accepted output and dashed reported control; right: residual demand. Matched pumping and area net imports remain in the extract, not inferred bilateral transfers.

Source: [2] [4] · Chart data

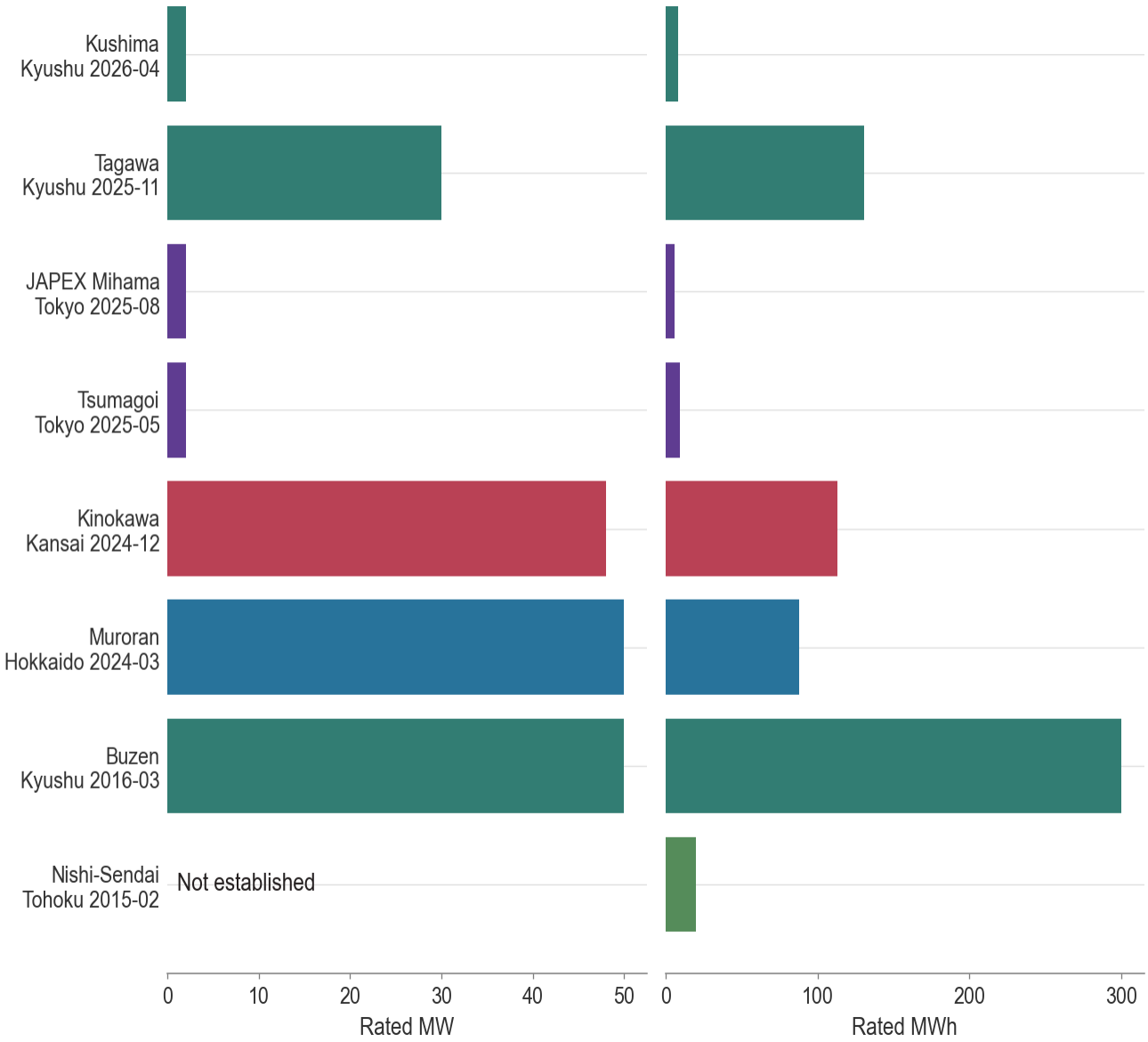
Priority dispatch first uses the applicable flexibility measures, including thermal output reduction and storage, before renewable control where surplus remains. Export headroom is valuable but finite; area net interchange does not reveal which bilateral route bound. Nor does non-firm connection mean that all control is caused by the external interconnector. [83]

Northern wind supports winter energy but can still create low-demand surplus; Kyushu solar withdrawal follows daylight. Charging shifts energy but cannot eliminate sunset. Demand, nuclear/thermal operation and deliverable transfers jointly determine the consequence; FIT/FIP status alone establishes no universal effective control order. [4]

Commissioned batteries: power, duration and business model differ

Muroran's commissioned grid battery was reported at 50 MW and 88 MWh, while Kinokawa entered operation at 48 MW and 113 MWh. Similar power ratings therefore do not imply the same stored-energy quantity. Kinokawa's operating vehicle and E-Flow's control role are also distinct from its sponsors' wider fleets. [85] [86] [87]

Exhibit 46: Commissioned batteries: geography, power and energy



Note: Project commissioning observations, not applications. MW and MWh are separately sourced; Nishi-Sendai MW is unestablished by the captured release. No assumed duration; documentary precision retained in the extract.

Source: [85] [86] [88–93] · Chart data

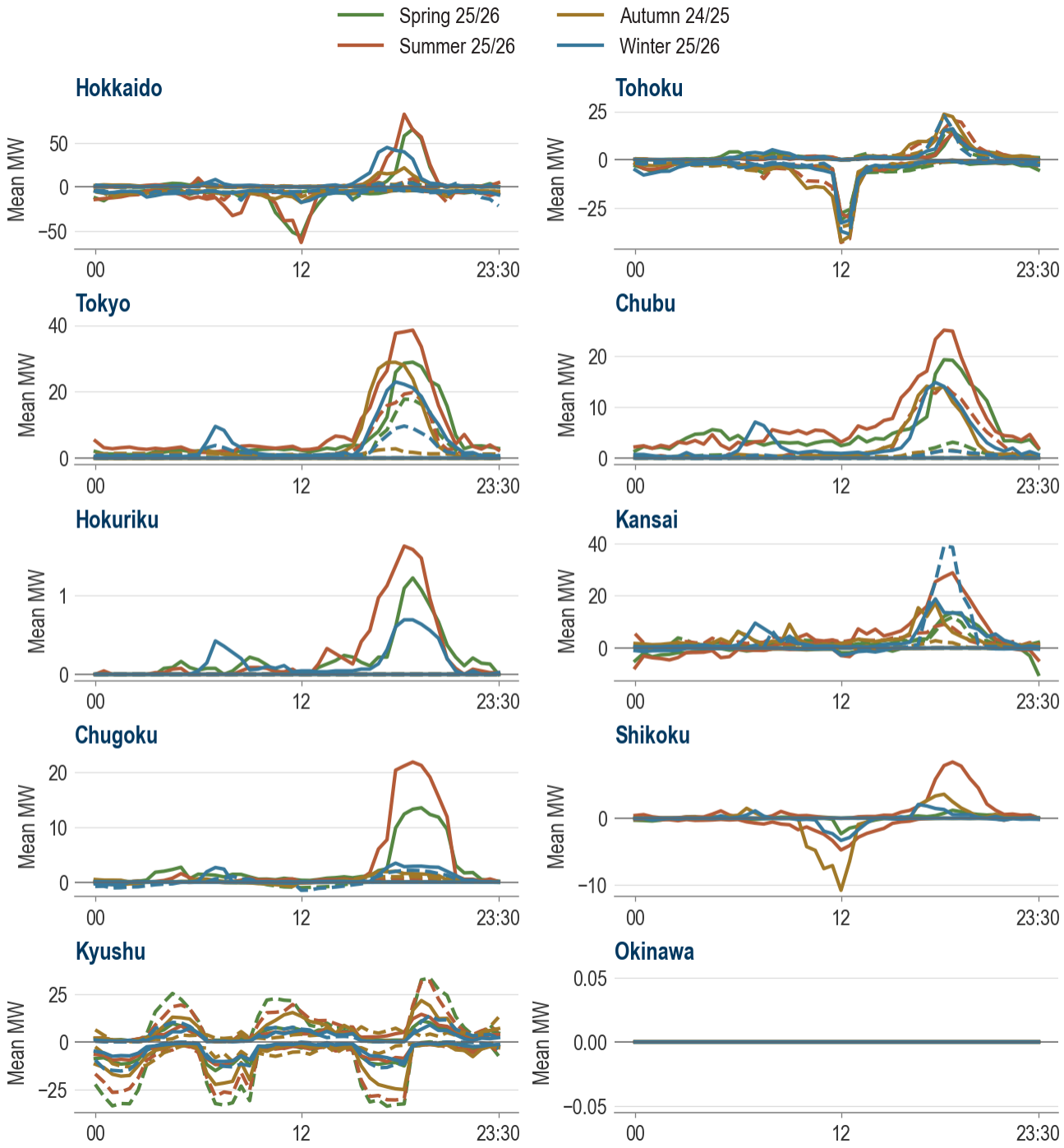
Tagawa's November 2025 operating start was confirmed in March 2026: 29.97 MW and 130.39 MWh. It enters the commissioned cohort because operation was confirmed, not because an auction award existed. JAPEX's Chiba-city Mihama battery is a different asset from Kansai's Mihama reactor. [88] [89]

Nishi-Sendai and Buzen served utility frequency/balance applications, not merchant projects. Nishi-Sendai's 20 MWh and February 2015 operation are documented; its MW rating is not established by the captured release. Tsumagoi's commercial start and later market entry are distinct milestones. These eight projects are not a national MW/MWh census. [90] [91] [92]

Battery nets conceal gross movements

Okinawa's retained battery observations are genuinely zero. Hokkaido, Tohoku, Kansai and Kyushu report both negative and positive activity; Tokyo and Chubu's regional nets are not evidence of all the charging occurring inside individual sites. Regional netting can conceal simultaneous charging and discharging. [4]

Exhibit 47: Ten-region four-season net battery activity



Note: Solid: latest eligible year; dashed: preceding year; *: partial season. Sign branches share the full observed denominator. Regional nets cannot recover simultaneous gross flows; year/coverage detail is in the extract.

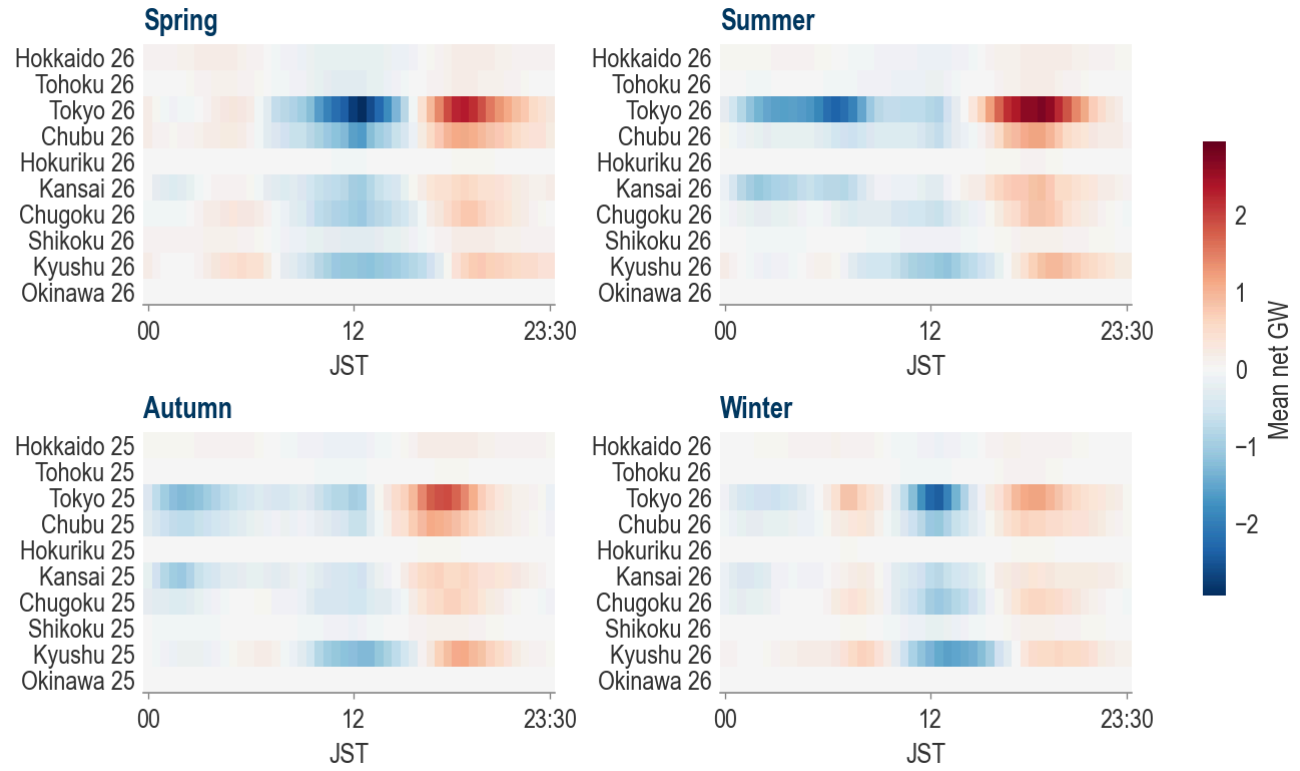
Source: [4] · Chart data

A quiet observed interval contributes zero to the inactive direction; an absent interval contributes nothing to either direction. Consequently, these are typical net-positive/net-negative clock profiles, not the much larger conditional mean of active dispatch. Full four-season and multiyear patterns remain separate from commercial commissioning announcements. [4]

Pumped storage and batteries

Pumped storage remains a separate large-scale balancing resource. In spring 2026 Tokyo, Chubu and Kansai's mean reported pumped power was negative: approximately −128, −60 and −111 MW. Net pumping over a season does not disclose the upper reservoir's usable energy or the next evening's availability. [4]

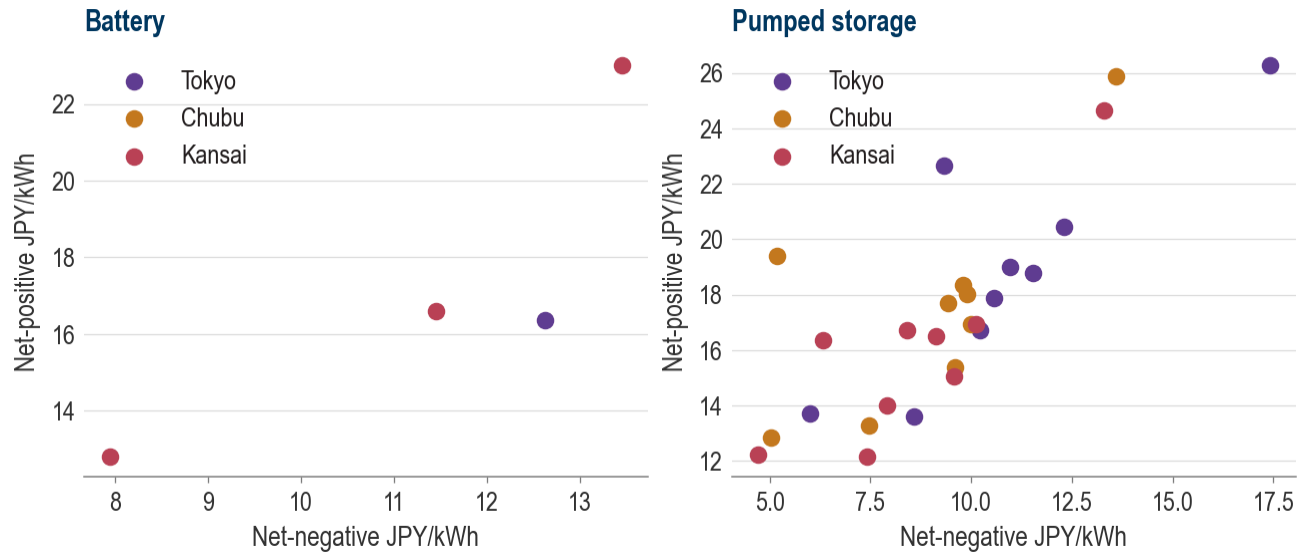
Exhibit 48: Pumped storage across ten regions and four seasons



Note: Blue: net pumping; red: net generation. Labels retain region/year; * denotes a partial-season sample. Missing observations are masked, not zero.

Source: [4] · Chart data

Exhibit 49: Storage-weighted area-price context



Note: Each point is a complete season-year. Opposite directions use the same eligible interval population; zero-energy directions have no weighted price.

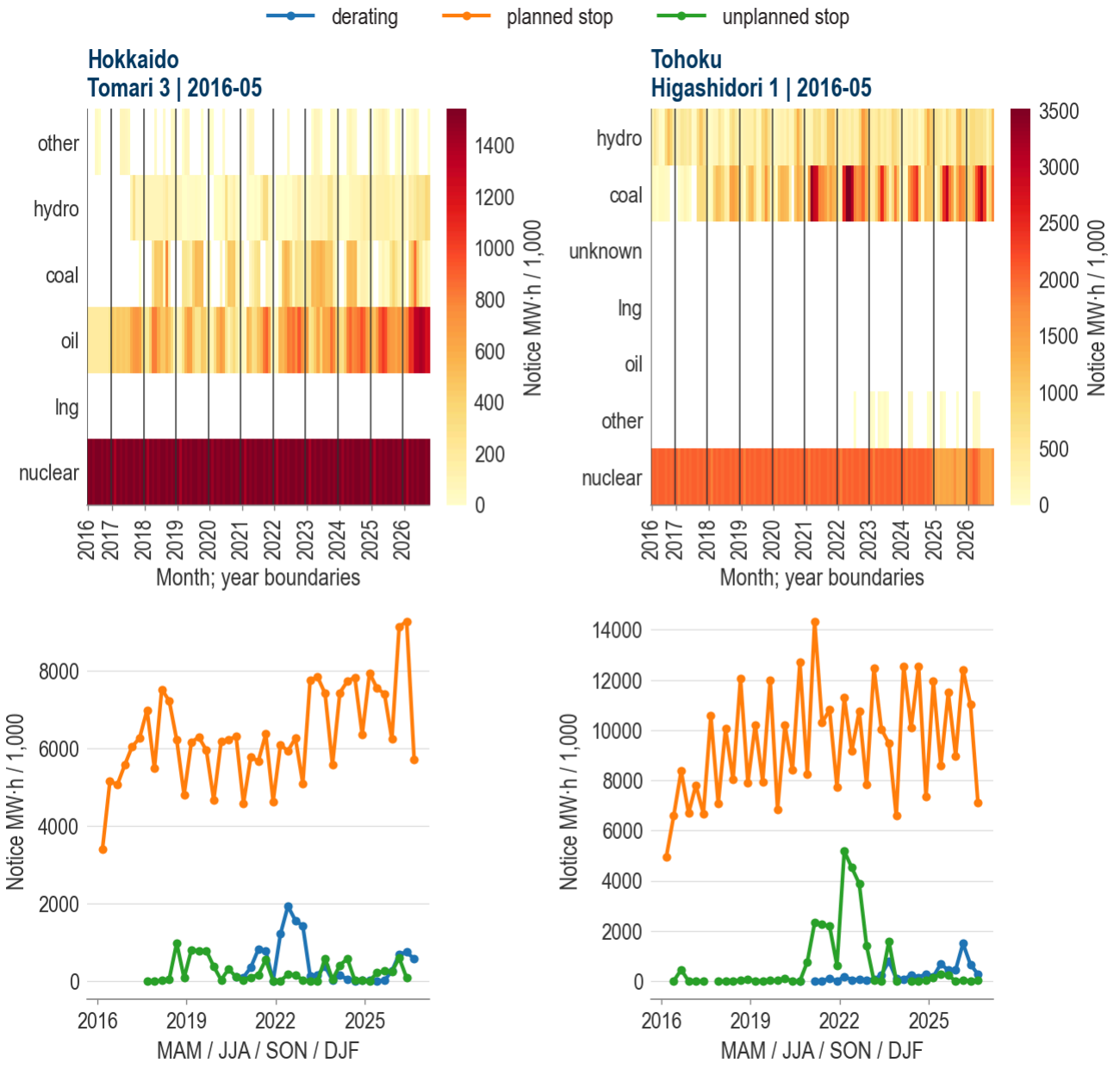
Source: [1] [4] · Chart data

Storage-weighted prices describe when net activity occurred—not profit or efficiency. They exclude fees, losses, other markets and unit-level dispatch. Actual energy limits mattered during the 2021 and 2022 stress episodes. [1] [4]

Hokkaido and Tohoku: cold exposure is not an outage-rate denominator

Hokkaido entered winter 2026 with no nuclear generation and higher thermal output than in spring. Tohoku's nuclear contribution changed within a broader coal/LNG/wind system. A reported thermal stop therefore interacts with cold demand and the northern transfer chain; it cannot be ranked by lost MW without considering the size and timing of the exposed fleet. [4]

Exhibit 50: Hokkaido / Tohoku: fuel-specific reported exposure



Note: Top: fuel × month; bottom: four-season/year notice categories. Subtitle: largest allocated expected unit-month, not confirmed loss. Blank is unavailable; restoration targets never certify operation.

Source: [7] [8] [94] · Chart data

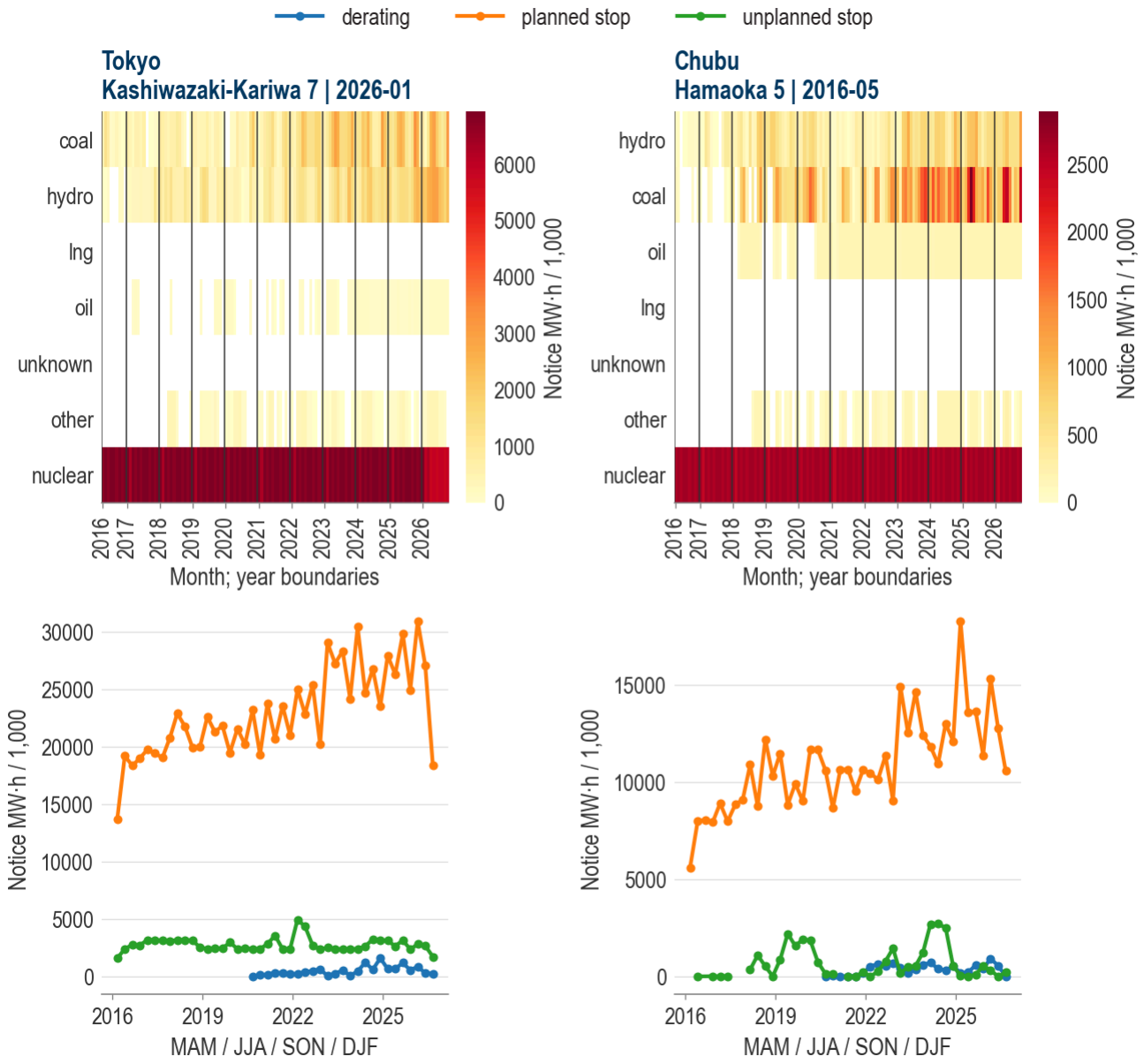
HJKS requires registration for units with authorised output of at least 100 MW. The current linked manual also calls for timely disclosure when a reduction of at least 100 MW is clearly expected to persist for at least 24 hours. It specifies event-driven disclosure, not a fixed hourly or daily publication SLA. [94] [95]

Northern dependent gas-turbine/steam-turbine components cannot be added to a whole combined-cycle block as if they were independent losses. Monthly notice exposure is retrospective expected capability-time; a restoration target is not confirmation that a unit produced electricity. [7]

Tokyo and Chubu: LNG outage exposure depends on physical blocks

Tokyo's spring 2026 LNG output averaged 11.93 GW and Chubu's 5.23 GW. Their maintenance exposure is therefore materially different from a coal-heavy regional fleet, but the largest absolute LNG exposure is not automatically the worst reliability performance. A gas-turbine component, its steam train and a complete combined-cycle block are not additive independent plants. [4]

Exhibit 51: Tokyo / Chubu: fuel-specific reported exposure



Note: Top: fuel × month; bottom: four-season/year notice categories. Subtitle: largest allocated expected unit-month, not confirmed loss. Blank is unavailable; restoration targets never certify operation.

Source: [7] [8] [94] · Chart data

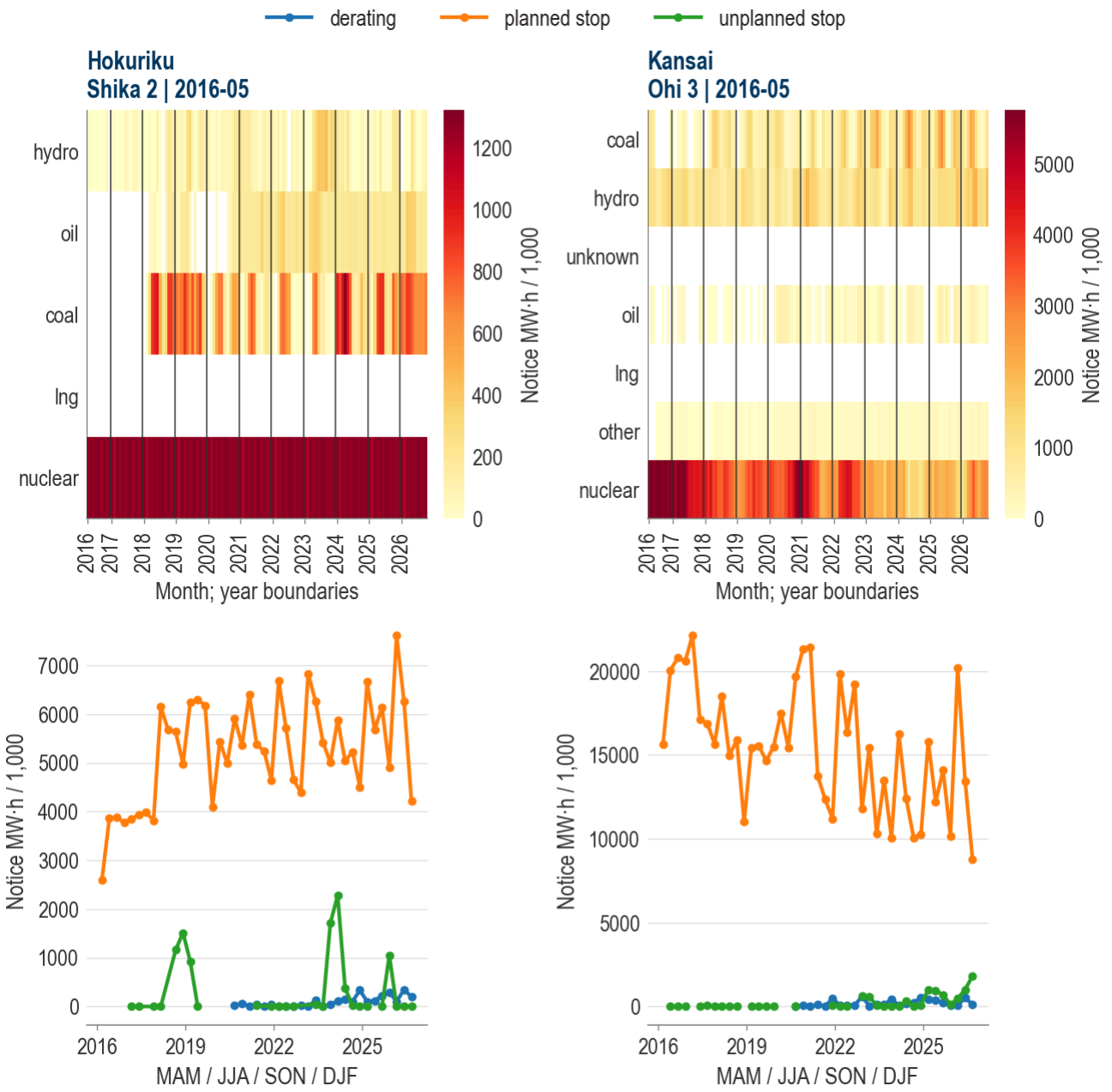
Joetsu's electrical destination must remain separate from its Niigata location. Older and replacement blocks also require different identities: a newer machine's commissioning date cannot reset the age of its predecessor or establish that the predecessor was available until that date. [331]

Planned stops, unplanned stops and reductions stay separate; long-term suspension is an additional dimension. Because the historical capture population is not a complete unit-year denominator, capacity-normalised reliability rankings are not established. Confirmed output and operator returns are stronger evidence of operation than subtracting expected notices from installed MW. [94] [96]

Hokuriku and Kansai: inspection cycles and suspended nuclear differ

Shika 1–2 were suspended in the latest NRA snapshot; Kansai had five operating reactors, with Ohi 3 and Takahama 3 suspended at that observation date. This is not one homogeneous nuclear outage category. Long suspension, periodic inspection and an interrupted return have different commercial persistence. [38]

Exhibit 52: Hokuriku / Kansai: fuel-specific reported exposure



Note: Top: fuel × month; bottom: four-season/year notice categories. Subtitle: largest allocated expected unit-month, not confirmed loss. Blank is unavailable; restoration targets never certify operation.

Source: [7] [8] [94] · Chart data

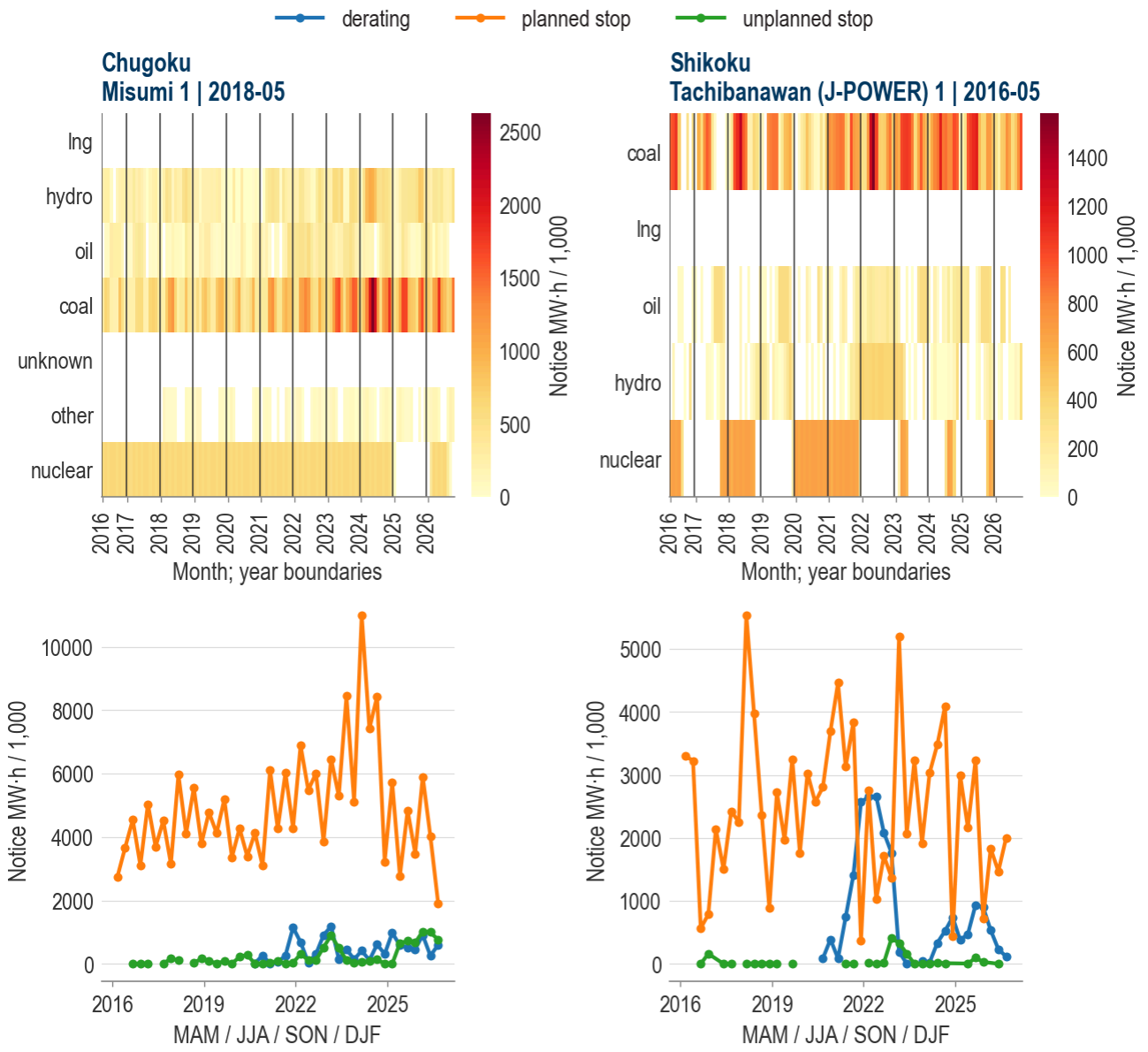
Kansai's older reactors make the original operation-start date relevant: Mihama 3 dates to December 1976, Takahama 1 to November 1974 and Takahama 2 to November 1975. Those dates establish age, not an assumed failure rate. Regulatory life-extension permission and actual power production remain separate. [39]

Mihama 3's September 2026 stop after adjustment operation demonstrates why expected restoration cannot be treated as an uninterrupted nameplate return. The confirmed full-operation milestone was 4 November. Hokuriku's hydro can change contemporaneous thermal needs without proving any thermal outage ended. [333] [4]

Chugoku and Shikoku: large-unit events meet a small regional balance

Chugoku's Shimane 2 and Shikoku's Ikata 3 are rated at 820 and 890 MW. One inspection or interruption can therefore change the regional nuclear contribution markedly. Yet the same event need not cause the same price outcome under different demand, thermal availability, solar or neighbour conditions. [39]

Exhibit 53: Chugoku / Shikoku: fuel-specific reported exposure



Note: Top: fuel × month; bottom: four-season/year notice categories. Subtitle: largest allocated expected unit-month, not confirmed loss. Blank is unavailable; restoration targets never certify operation.

Source: [7] [8] [94] · Chart data

The spring 2026 thermal means were coal-heavy: 2.91 GW coal in Chugoku and 1.62 GW in Shikoku. Nuclear status and coal notices should therefore be read against the entire local fuel balance, not a national average maintenance curve. [4]

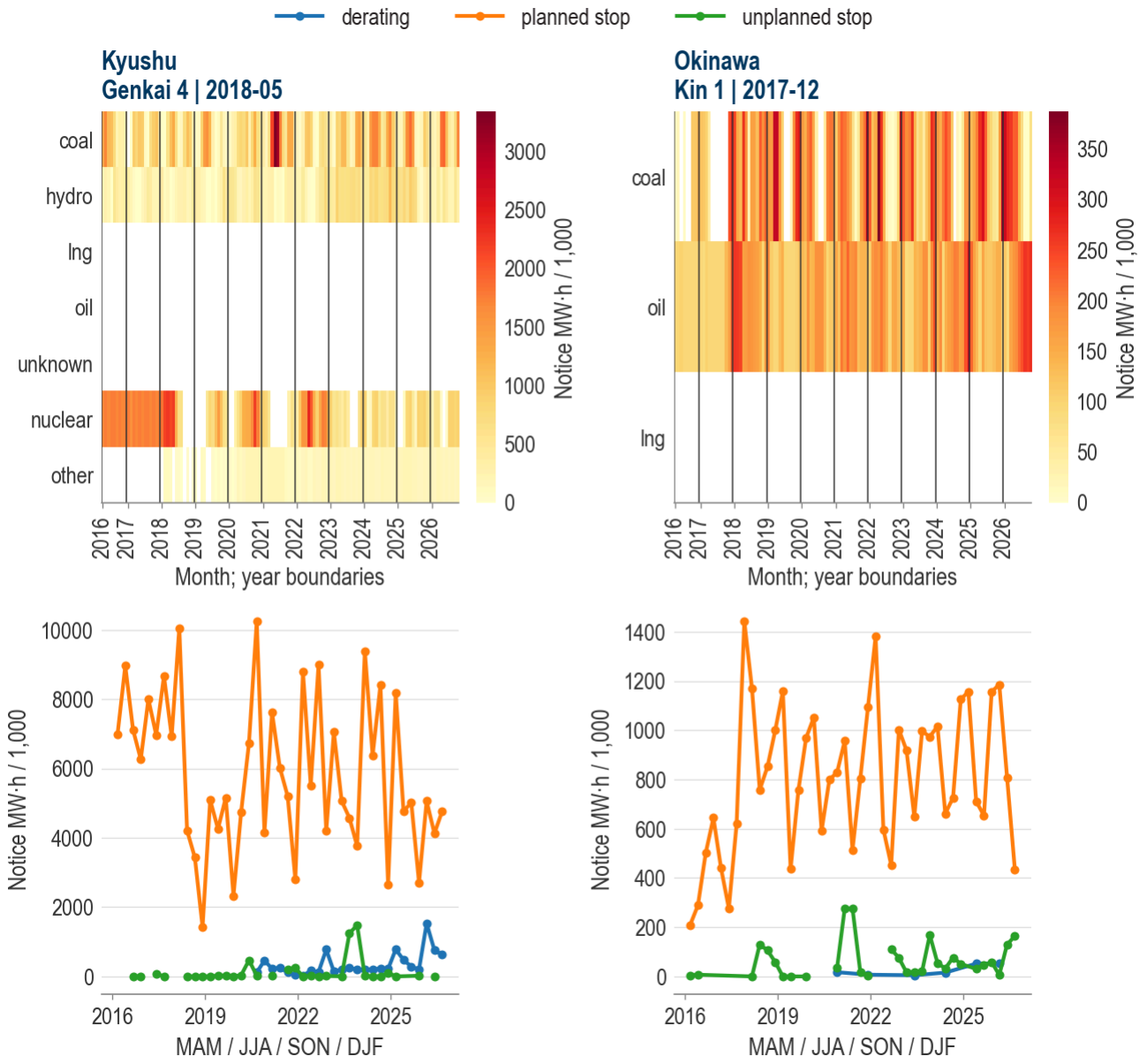
Replacement and alias decisions at Saijo must preserve physical identity and valid rating. Overlapping notices are not extra missing MW. Day-only targets bound total expected exposure, not measured duration; extending an endpoint can reassign exposure between overlapping categories, so each category is not independently bounded. [331] [94]

Shikoku's AC Honshi route and HVDC Anan–Kihoku route have different counterparties and constraints; neither is an unlimited outage hedge. [42]

Kyushu and Okinawa: outage timing changes the usable flexibility

Kyushu combines four operating nuclear reactors in the October snapshot with solar, thermal generation and pumping. A thermal outage can be less visible in a sunny low-demand midday balance yet matter during the evening solar decline. A nuclear inspection can reduce surplus-control pressure while increasing later thermal energy needs; neither effect is a price forecast. [38] [4]

Exhibit 54: Kyushu / Okinawa: fuel-specific reported exposure



Note: Top: fuel × month; bottom: four-season/year notice categories. Subtitle: largest allocated expected unit-month, not confirmed loss. Blank is unavailable; restoration targets never certify operation.

Source: [7] [8] [94] · Chart data

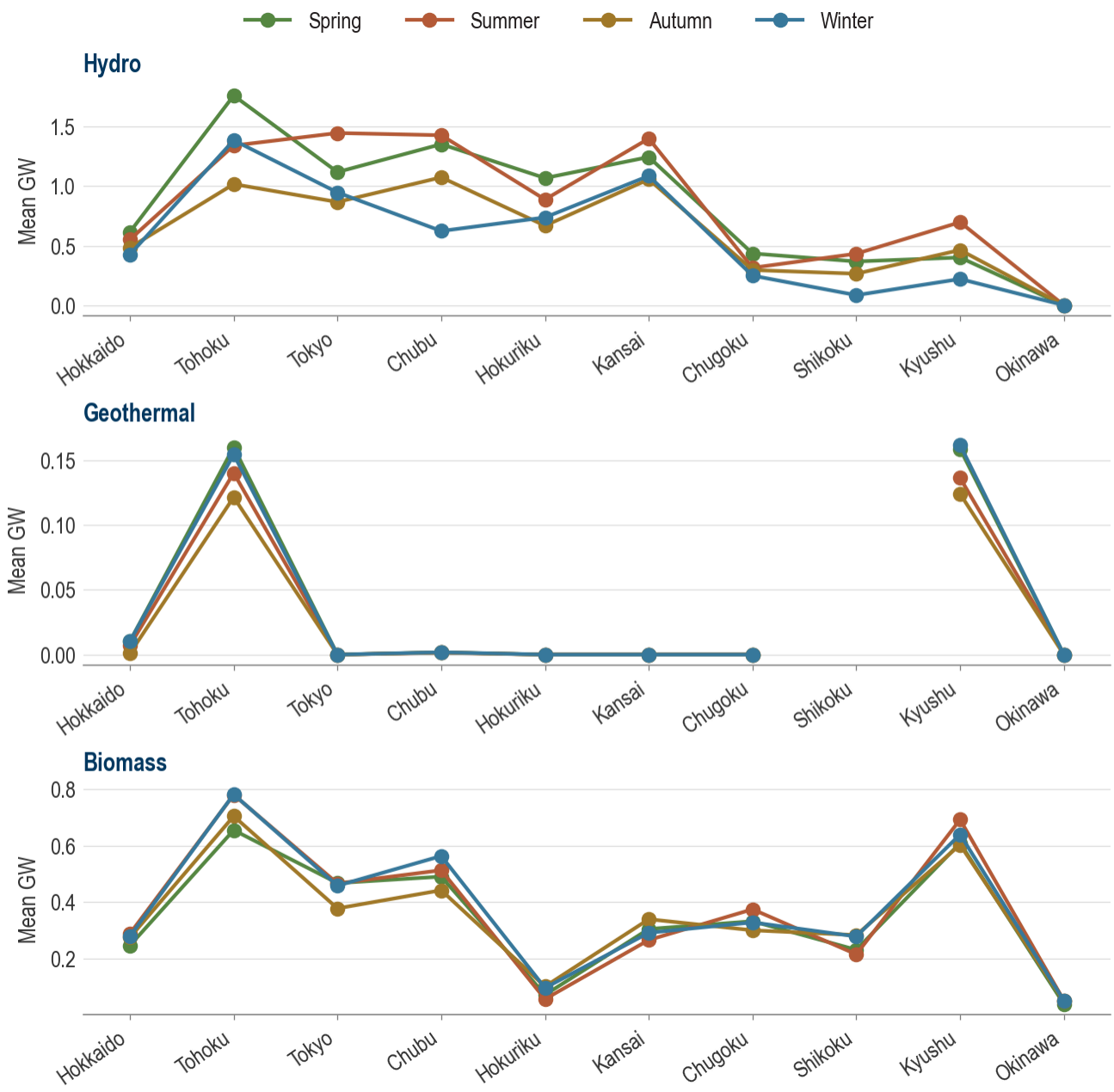
Okinawa's summer thermal mean of 1.08 GW exceeded winter's 0.69 GW. With no mainland interconnector, local unit state and fuel logistics cannot be replaced by assumed emergency imports. Small equipment below the reporting threshold and incomplete capture history prevent treating the notice inventory as a complete island reliability census. [4] [42] [94]

An open notice remains censored at the analytical cutoff; it is not proof that the unit stayed offline until then. Where historical unit-year and time-valid capacity denominators are unknown, seasonal exposure is shown without an invented availability percentage. [96]

Hydro, geothermal and biomass have different seasonal constraints

Conventional hydro was a substantial spring 2026 contributor: 1.76 GW in Tohoku, 1.35 GW in Chubu, 1.25 GW in Kansai and 1.07 GW in Hokuriku. Hokuriku's hydro was large relative to its 1.59 GW thermal mean. Water conditions can therefore reshape thermal demand even when installed hydro capacity has not changed. [4]

Exhibit 55: Hydro, geothermal and biomass are not interchangeable flexibility



Note: Latest eligible four-season observations; regional reported zeros are preserved. Incomplete fallback samples are explicitly identified in the extract.

Source: [4] · Chart data

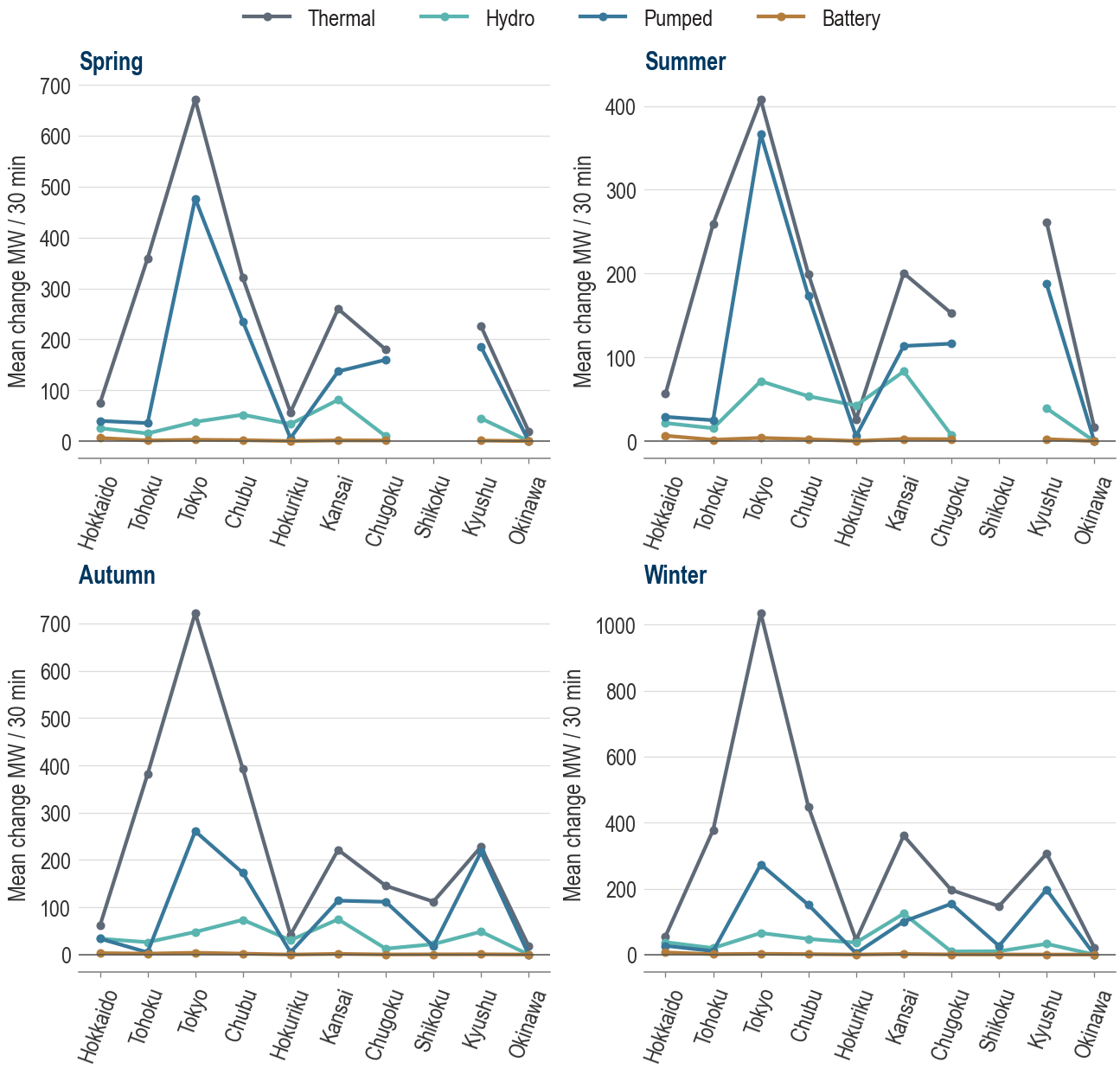
Geothermal and biomass are not interchangeable with reservoir hydro. Their observed production reflects different physical and fuel constraints, and a stable output trace does not demonstrate unused ramping headroom. Biomass is also not automatically equivalent to an unqualified environmental attribute. [4] [220]

Pumped storage is excluded from conventional hydro here because it transfers energy across time. Solar absorption, turbine output and reservoir replenishment can change independently. Okinawa's small local non-thermal output is retained rather than forced into a mainland hydro analogy. Unknown technology observations remain unknown, not evidence of absent equipment. [4]

Evening flexibility is a joint thermal, water and storage response

The evening withdrawal of solar moves the balancing requirement toward other resources even when total demand is flat or declining. Japan's response is region-specific: LNG-heavy Tokyo and Chubu differ from nuclear-rich Kansai and Kyushu, hydro-sensitive Hokuriku and the isolated Okinawa system. [4] [42]

Exhibit 56: The observed supply response to evening residual ramps



Note: Thermal, ordinary hydro, pumped storage and battery responses remain separate. Nuclear and high-residual output context are retained in the extract.

Source: [2] [4] · Chart data

A positive thermal change alongside a rising residual load is observed response, not proof that the unit set the area price. Pumped storage and batteries can respond in the same interval, but their future energy depends on previous charging and constraints not disclosed by a regional net series. High-residual output and evening-ramp activity are different stress populations. [4]

Four-season comparisons also prevent spring's low-demand solar pattern from becoming a year-round dispatch assumption. Winter heating, summer cooling, autumn maintenance and spring daytime surplus put different demands on the same fleet. The next chapter follows the imported-fuel infrastructure behind that thermal response: terminal connections and procurement clocks matter as well as turbine MW. [2]

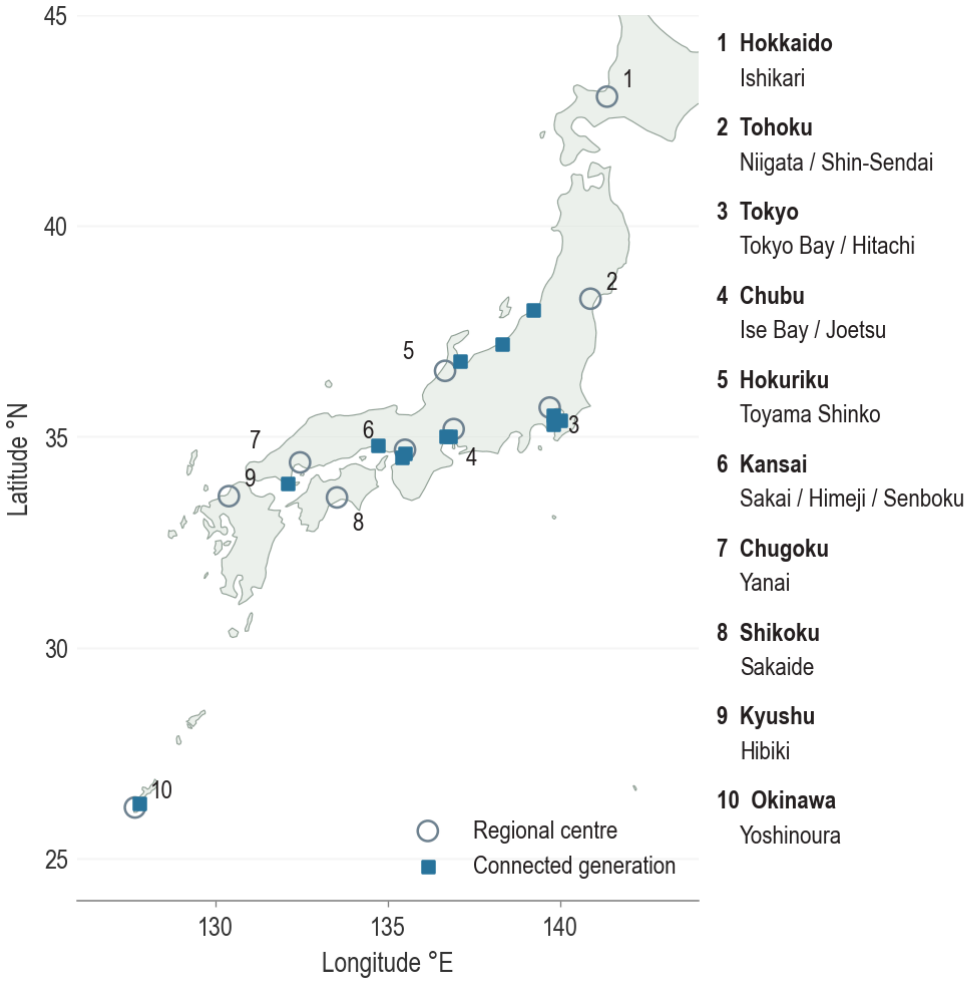
LNG infrastructure, fuel procurement and generation economics

LNG infrastructure: coastal chains, regional consequences

Japan's LNG exposure is not one national tank. Cargoes reach coastal receiving systems whose storage, vaporizers and gas networks serve particular power fleets and city-gas customers. JERA identifies interconnected Tokyo Bay and Ise Bay groups, not one pipeline joining those electrical regions. A common import benchmark can therefore coexist with different local fuel constraints. [99]

- LNG origins: national diversification is not a cargo book
- Ise Bay and Joetsu: Chubu's two fuel geographies
- Fuel security, tariff adjustment and taxation

Exhibit 57: Japan LNG receiving-system geography



Note: Hollow markers locate regional centres, not terminals; squares locate only qualified connected generation sites. Unlocated systems remain in the key. INPEX Naoetsu is a separate gas network, with no assigned power destination.

Source: [98] [99] [101–114] [332] [334] [335] · Chart data

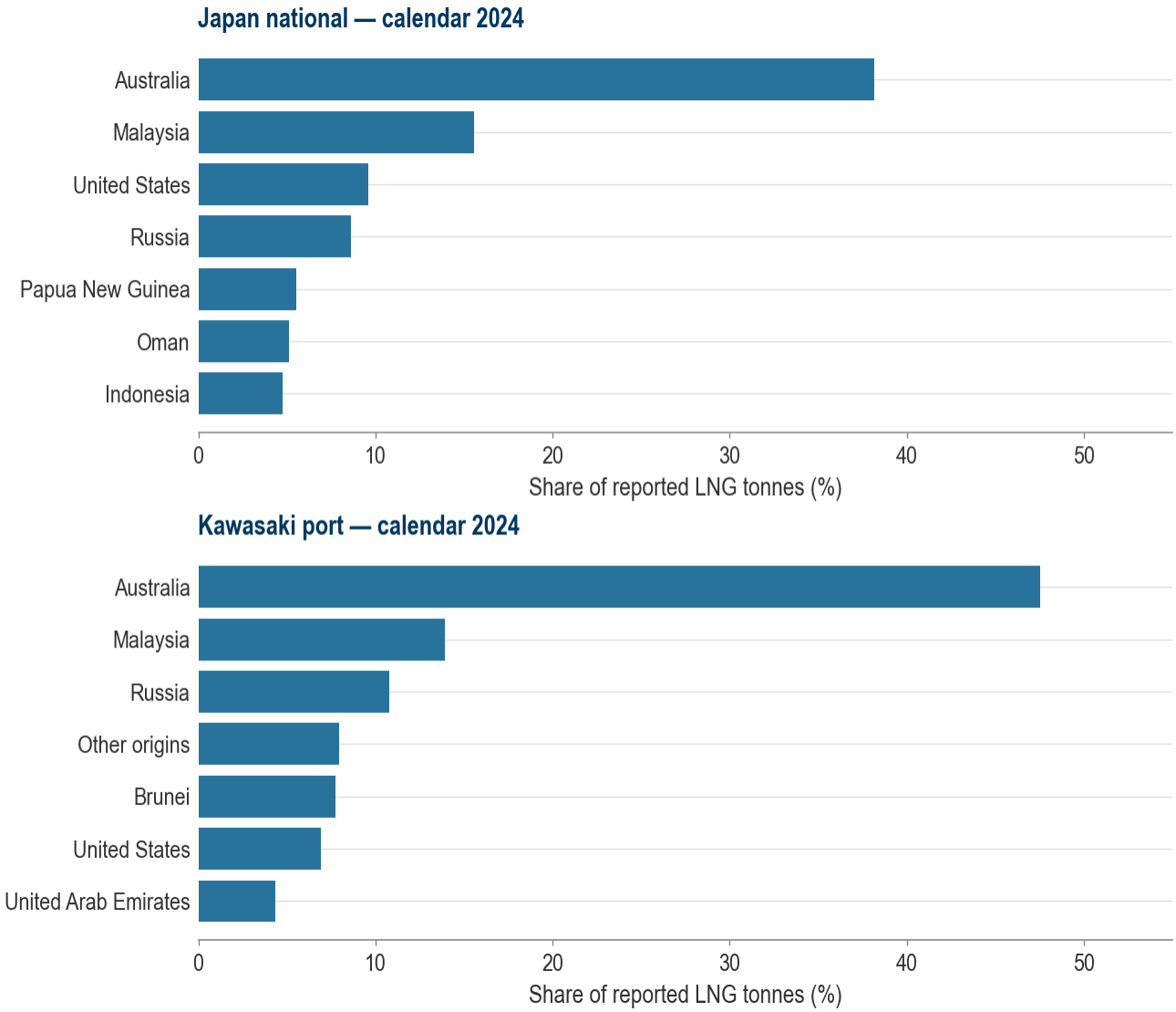
JERA Joetsu stands in Niigata but connects to Shin-Hokushin in Nagano and supplies Chubu. Nearby INPEX Naoetsu instead supplies a domestic gas network; proximity establishes no shared fuel connection. [100] [101]

Storage ratings describe dated equipment, not usable inventory. JERA reports tonnes; other operators report liquid kL or m^3. Joint-site scope, standby equipment and third-party access remain distinct. Natural Earth's generalized coastline provides geographic context: hollow markers are regional centres; squares are verified connected generation sites—not terminal tanks or surveyed pipelines. [98]

LNG origins: national diversification is not a cargo book

Japan imported **65.891 million tonnes in 2024**: Australia supplied 38.15%, Malaysia 15.55%, the United States 9.62% and Russia 8.63%. MOF Customs' national mix does not establish an individual generator's supplier concentration. [115]

Exhibit 58: LNG origins: national mix versus Kawasaki port



Note: CY2024 tonnes; seven largest origin categories. National: MOF Customs via METI. Kawasaki: March 2025 preliminary release. Port imports are not company cargo books.
Source: [115] [116] · Chart data

Kawasaki recorded **8.313 million tonnes**, with Australia's quantity share at 47.51%. Its five largest listed suppliers were Australia, Malaysia, Russia, Brunei and the United States. Quantity and value shares differ; neither identifies a nearby company's cargo receipts. [116]

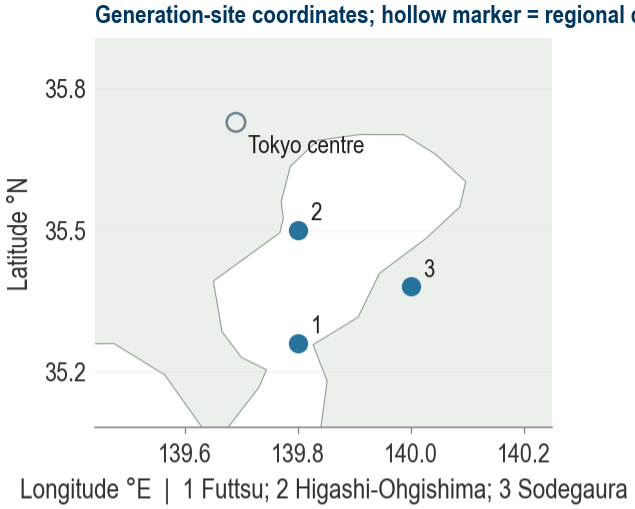
The physical chain is contract/nomination → liquefaction/shipping → berth acceptance → storage → vaporization/network delivery → generation. Sakaide names Bintulu, Malaysia, as its principal loading source, roughly one week's transport away. Okinawa's May 2020 disclosure names Osaka Gas for a 27-year contract beginning FY2012, about 400,000 tonnes/year, principally Gorgon supply and ex-ship delivery. These are dated arrangements, not current realized volumes. [110] [114]

Long-term sourcing reduces replenishment uncertainty without removing weather, berth, quality or tank constraints. National diversification does not make every cargo immediately substitutable at every terminal.

Tokyo Bay: two connected systems, not one operator

JERA confirms pipeline interconnection among Futtsu, Higashi-Ohgishima and Sodegaura; the arrows retain its named station-supply relationships. Its December 2021 storage disclosure gives **612,000, 243,000, 536,000 and 61,000 tonnes** respectively for Futtsu, Higashi-Ohgishima, Sodegaura and the separate Negishi joint-site scope. [99]

Exhibit 59: Tokyo Bay: separate JERA and Tokyo Gas networks



Futtsu Futtsu, Chiba	→	Futtsu; Chiba
Higashi-Ohgishima Kawasaki, Kanagawa	→	Higashi-Ohgishima; Yokohama; Kawasaki
Sodegaura Sodegaura, Chiba	→	Sodegaura; Anegasaki
Negishi Yokohama, Kanagawa	→	Minami-Yokohama
Tokyo Gas Hitachi Hitachi, Ibaraki	→	city gas network; individual plant not specified
Tokyo Gas Negishi Yokohama, Kanagawa	→	city gas network; individual plant not specified
Tokyo Gas Sodegaura Sodegaura, Chiba	→	city gas network; individual plant not specified
Tokyo Gas Ohgishima Yokohama, Kanagawa	→	city gas network; individual plant not specified

Note: Dated operator supply links. Arrows are schematic, not surveyed pipelines; points are rounded generation-site locations, not terminal pins. Network-only rows imply no individual plant edge; no distance scale.

Source: [98] [99] [109] [112] [332] · Chart data

Tokyo Gas's separate network links Negishi, Sodegaura, Ohgishima and Hitachi; the 2021 Ibaraki trunk line completed mutual backup. Its September 2026 account reports two 230,000-kL Hitachi tanks. **Tokyo Gas Ohgishima is not JERA Higashi-Ohgishima.** Shared sites do not make all tanks exclusively JERA assets. [112]

Tokyo Gas's 250,000-m^3 Ohgishima figure covers **tank 4 only**. The 26 March 2026 date is world-record recognition, not commissioning. [109]

Group interconnection establishes neither every pairwise pipe nor available throughput. Backup can redistribute accessible gas within operating constraints; it cannot replenish LNG during a cargo delay. Joint-site liquid volumes must not be added to JERA's differently scoped tonnes.

Tokyo LNG demand changes faster than cargo replenishment

Tokyo's gas fleet follows residual electricity demand while fuel arrives in discrete cargoes. Winter heating, summer cooling, nuclear availability, solar output and imports change burn requirements. Chubu and Kansai face the same international fuel market but different fleets and load profiles. [4]

Exhibit 60: Tokyo, Chubu and Kansai LNG generation through the seasons

Note: Observed monthly LNG-generation mean GW, all modern months and seasons. Incomplete months are not extrapolated; output is not terminal fuel consumption.

Source: [4] · Chart data

Electrical output is not terminal tonnes consumed: efficiency, city-gas deliveries and cargo schedules prevent a unique conversion. Monthly averages also conceal short delivery constraints.

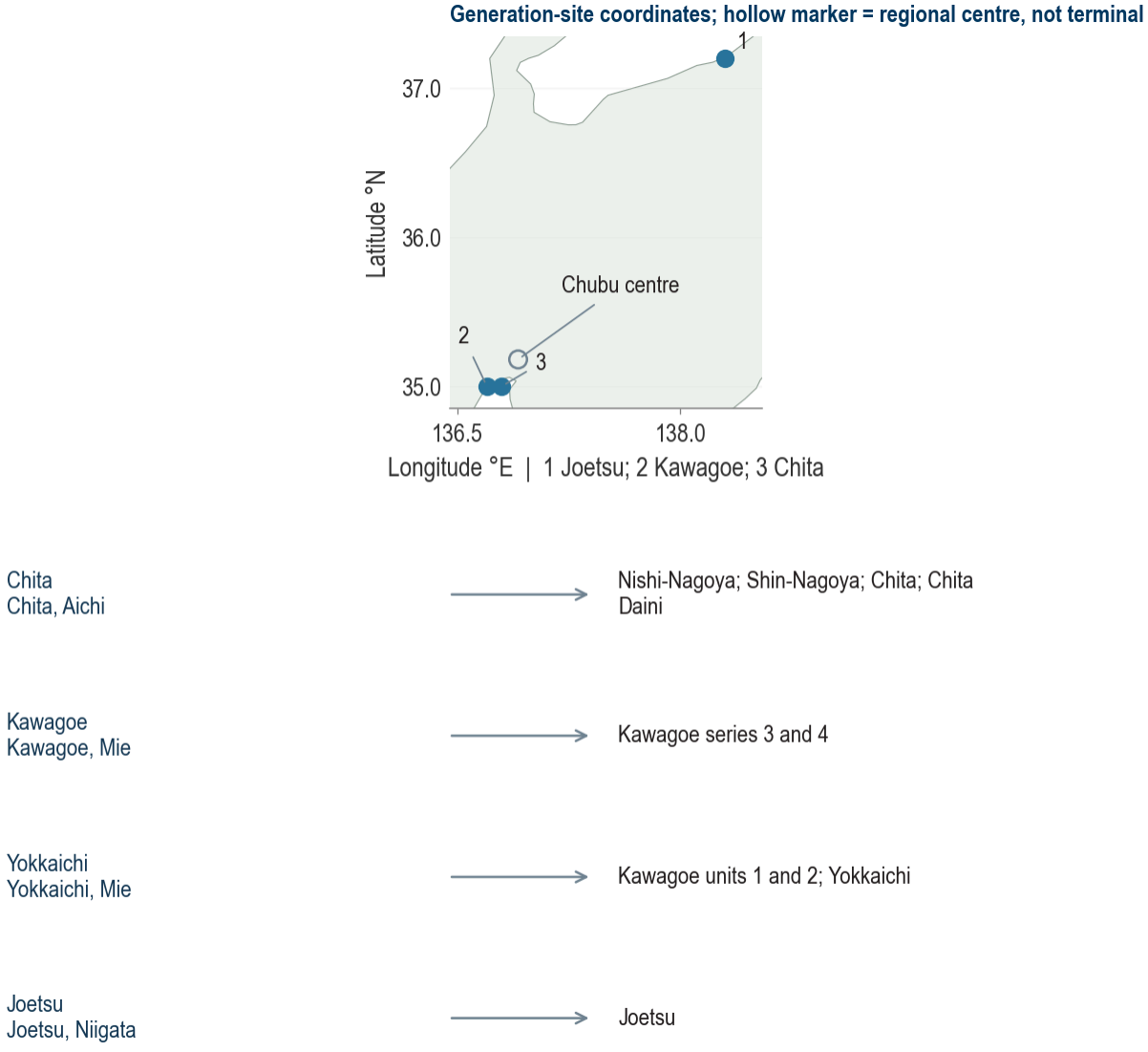
JERA's operating minimum starts where tank send-out limitations could force associated generators to stop, then adds arrival-delay risk: ordinarily two days' consumption, but seven winter days at Joetsu. Sodegaura and Negishi follow joint-terminal gas-company management. Actual lower limits depend on individual tank conditions, not an invariant site total. [99]

A cold spell, outages elsewhere and constrained imports can increase Tokyo LNG requirements together. Conversely, lower residual demand conserves fuel even when spot LNG rises. These mechanisms explain exposure, not a claim that one benchmark determines every power price. National stock observations on page 75 cannot resolve individual-terminal headroom.

Ise Bay and Joetsu: Chubu's two fuel geographies

JERA's interconnected Chita–Kawagoe–Yokkaichi group serves the explicitly named stations below. December 2021 reported storage was **702,000 tonnes at Chita, 378,000 at Kawagoe and 144,000 at Yokkaichi**, versus **243,000 at geographically separate Joetsu**. Group membership is not a complete pairwise pipeline topology. [99]

Exhibit 61: Ise Bay and Joetsu: fuel networks and electrical destination



Note: Dated operator supply links. Arrows are schematic, not surveyed pipelines; points are rounded generation-site locations, not terminal pins. Network-only rows imply no individual plant edge; no distance scale.

Source: [98] [99] [332] · Chart data

Chubu's 8 June 2011 release establishes the 62.7-km, two-circuit, 275-kV Joetsu line to Shin-Hokushin. Joetsu's Niigata address does not assign its electricity to Tohoku; no gas connection to Ise Bay is inferred. [100]

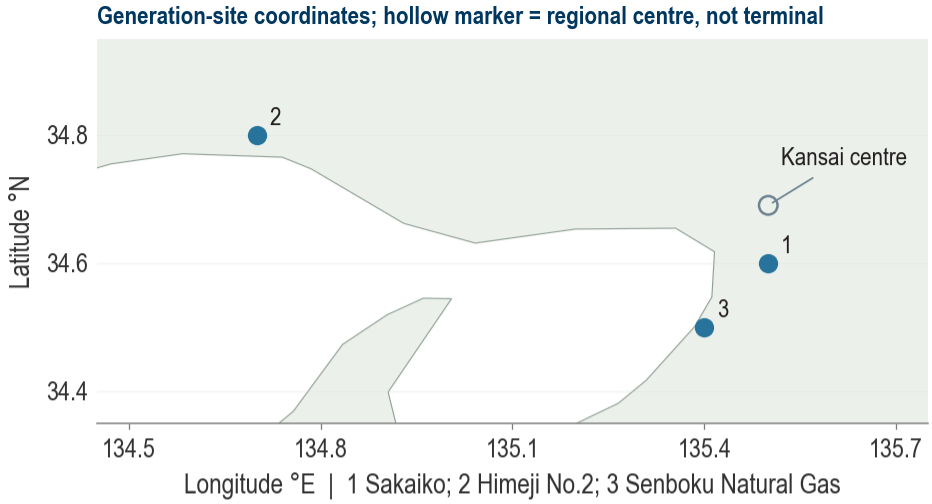
INPEX Naoetsu is separate: two 180,000-kL tanks and maximum production of 8.25 million m^3/day, revised April 2024. It combines regasified imported LNG with Minami-Nagaoka domestic gas for pipeline customers, not a verified JERA Joetsu supply edge. [101]

Chubu thus combines industrial-region electricity demand with distinct coastal delivery risks. JERA's seven-day winter arrival-delay allowance at Joetsu, versus two ordinarily, is a risk-management basis—not an observed outage or guaranteed usable fuel at every output level.

Kansai: KEPCO and Daigas must remain separate

KEPCO Himeji supplies Himeji No.1 and No.2: deliveries began in June 1994 and June 1979 respectively. Its use of Osaka Gas pipelines for gas sales does not merge the terminal with Daigas Himeji. [106]

Exhibit 62: Kansai: KEPCO and Daigas supply chains



Sakai LNG center Sakai, Osaka	→	Sakaiko
KEPCO Himeji Himeji, Hyogo	→	Himeji No.1 and Himeji No.2
Daigas Senboku Sakai and Takaishi, Osaka	→	Senboku Natural Gas Power Generation Co.
Daigas Himeji Himeji, Hyogo	→	Himeji Natural Gas Power Generation Co.

Note: Dated operator supply links. Arrows are schematic, not surveyed pipelines; points are rounded generation-site locations, not terminal pins. Network-only rows imply no individual plant edge; no distance scale.

Source: [98] [102] [106–108] [332] [334] · Chart data

KEPCO's May 2026 disclosure gives **520,000 kL at Himeji and 560,000 kL at Sakai**. Himeji's 19,000 m^3/hour is **spare gas-production capacity**, not total send-out; Sakai had no available third-party gas-production headroom. [108]

Sakai began operation in January 2006, became wholly owned by KEPCO in April 2022 and merged into KEPCO in July 2023. It supplies Sakaiko and other customers, with trucks serving off-network destinations. [107]

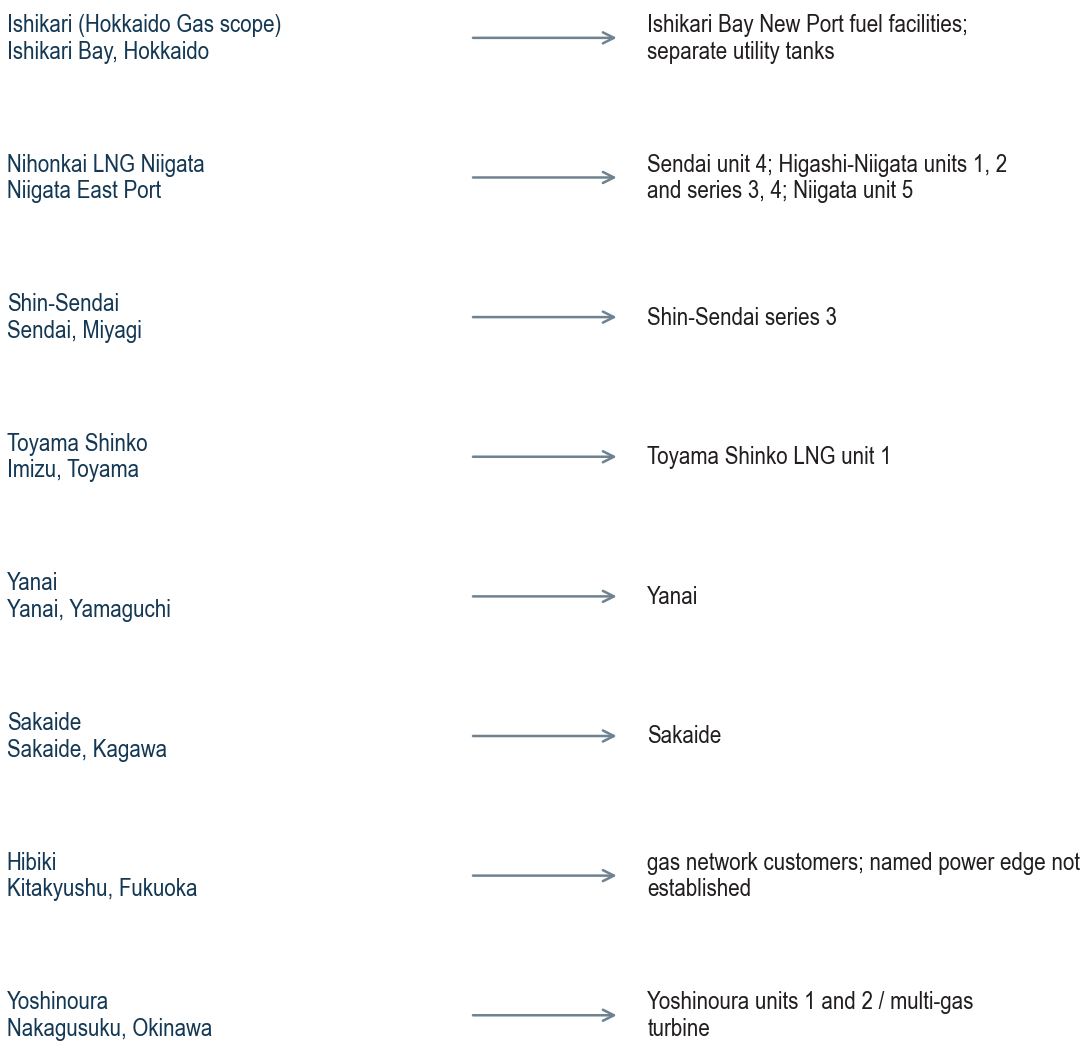
Daigas separately supplies Senboku's 1,109-MW plant and Himeji Natural Gas Power Generation's 1,245.2-MW plant; the latter's units started in January and May 2026. [102]

Nuclear operation and inspections change Kansai's thermal substitution and gas burn—not terminal storage or third-party access.

Receiving systems beyond the three priority regions

The seven other regions combine receiving hubs, dedicated station chains and an isolated island system. Confirmed station links remain distinct from network-only destinations.

Exhibit 63: Seven other regions: receiving systems and destinations



Note: Dated operator supply links. Arrows are schematic, not surveyed pipelines; points are rounded generation-site locations, not terminal pins. Network-only rows imply no individual plant edge; no distance scale.

Source: [103–105] [110] [111] [113] [114] · Chart data

Hokkaido Gas's **380,000 kL** covers Ishikari tanks 1–2, excluding the utility's separate tanks. Tohoku's December 2024 disclosure gives **720,000 kL at Nihonkai Niigata and 320,000 at Shin-Sendai**, with named station links. Niigata allows five winter arrival-delay days versus two otherwise. [105] [111]

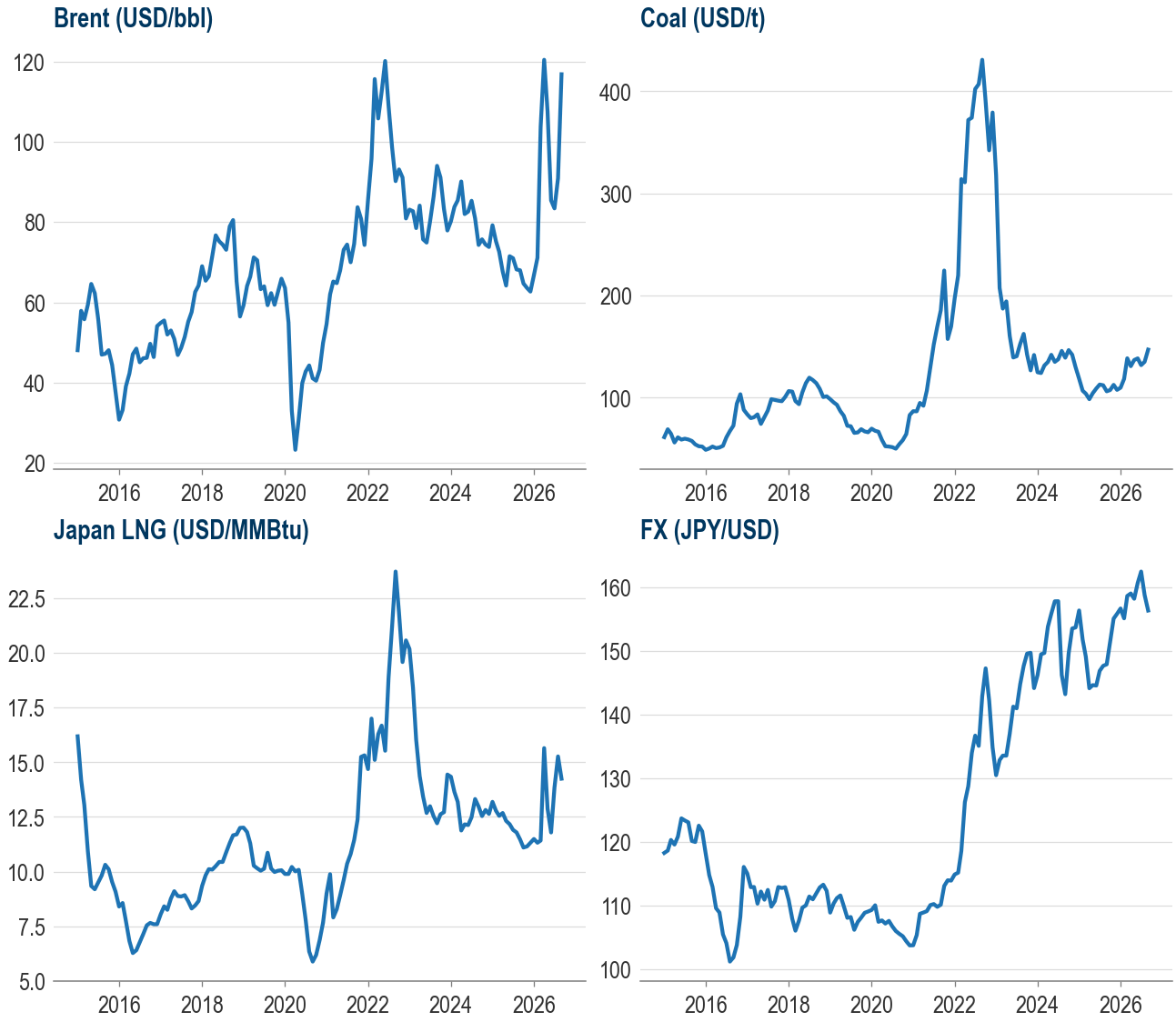
Toyama Shinko LNG started on 21 November 2018 with one **180,000-m^3-class tank**. Yanai's July 2018 disclosure gives **six 80,000-kL tanks and 275 t/hour generation vaporization**. Sakaide pipes fuel to its named station; announced expansion is not operating capacity. [104] [113] [110]

Hibiki has **two operating 180,000-kL tanks**. One high-pressure 55-t/hour vaporizer is shared standby, so equipment ratings cannot all be added; the extra 230,000-kL tank is scheduled for FY2029. Yoshinoura's May 2020 disclosure gives **two 140,000-kL tanks**. Mainland electricity support cannot replace Okinawa fuel endurance. [103] [114]

Fuel references, oil indexation and the yen

The observed Japan LNG import reference, coal, Brent and yen measure different procurement exposures. Import averages mix delivered cargoes and contract prices; Brent is not a generator's delivered fuel-oil invoice. Dollar fuel and FX movements can reinforce or offset before entering retail tariffs. [9]

Exhibit 64: Fuel references and the yen



Note: Monthly World Bank fuel references and ECB FX, 2015-01–2026-09, retain their own units and vertical scales. Japan LNG denotes the import reference, not a spot cargo quotation; missing observations remain gaps.

Source: [9] · Chart data

JCC is Japan's crude-import basket, not Brent. Oil-linked procurement follows a different clock from contemporary spot cargoes; private slopes, constants, lags, destination clauses and price reviews cannot be recovered from a national average. No universal contract formula is assigned to the fleet.

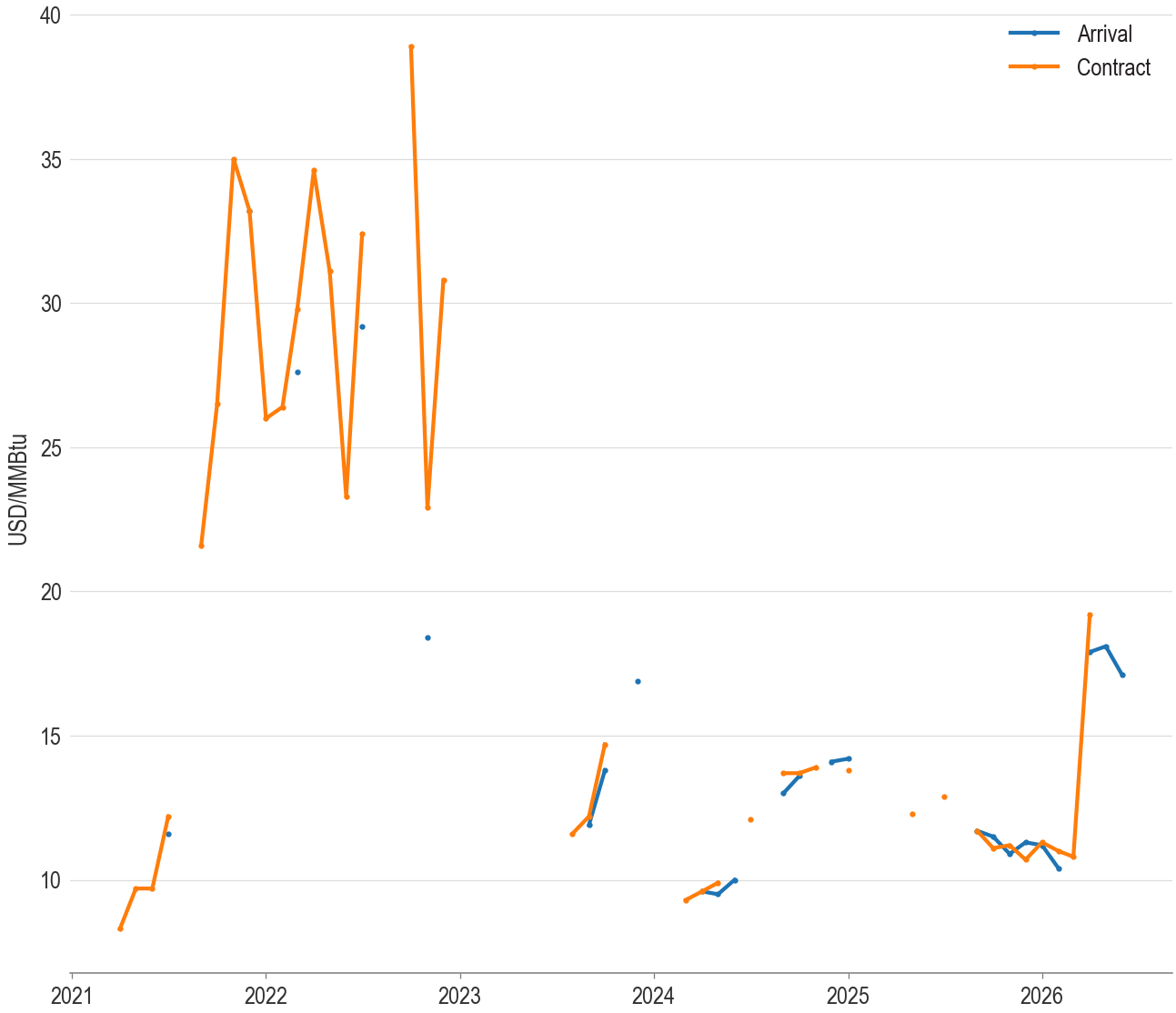
JOGMEC's contract-month and arrival-month surveys cover disclosed spot transactions, not the complete import basket or stored-inventory marks. Neither is JKM. The next page retains those clocks rather than filling missing quotations.

Tokyo and Chubu's gas-generation exposure contrasts with Kansai's nuclear/thermal substitution cycle; page 70 supplies the observed output context. Monthly cost bands cannot identify a marginal unit or confidential procurement share. Benchmark quality, delivery basis, efficiency and tax inclusion must be resolved before estimating a plant cost; tax already in the invoice must not be added twice.

Spot LNG and national stocks retain different clocks

A contract date and cargo arrival date represent different transaction populations. JOGMEC's survey can therefore differ between bases in the same month. Withheld values remain missing, not zero-price cargoes or unchanged prices; neither series replaces JKM. [117]

Exhibit 65: Disclosed Japanese spot LNG survey, not JKM



Note: Contract and arrival observations retain their own month. Dashes are withheld or unavailable observations, not zero; no interpolation or substitute JKM series.

Source: [117] [119–182] · Chart data

ANRE's 4 November 2026 release reports **1.92 million tonnes at 1 November**, versus **2.15 million on 25 October and 2.64 million on 11 October**. It revises 13 September to 2.50 million from the local 2.49. These national major-utility observations exclude dead stock; they are not terminal gauges. [118]

Tokyo's dense networks, Chubu's separate Joetsu/Ise Bay chains and Kansai's KEPCO/Daigas systems leave different replenishment risks. Lower aggregate stock can reflect ordinary seasonal burn and cargo timing; weather, outages and replacement-supply restrictions require separate evidence.

National stock cannot identify the short terminal or operating minimum. Equipment capacity does not reveal current inventory or third-party send-out; regional electrical output does not uniquely determine fuel burn. The register's unjoined stock chart preserves observation dates without interpolation or invented company days of cover.

Fuel security, tariff adjustment and taxation

Strategic Buffer LNG preserves procurement and redirection capability rather than creating a universal oil-style LNG tank reserve. The December 2023 policy describes certified supply-security arrangements whose surplus cargoes are ordinarily sold when no emergency occurs. JOGMEC's emergency procurement function is a separate government-request mechanism. Neither policy reveals present terminal inventory or guarantees that an individual generator has a replacement cargo ready. [183] [184]

Framework	Affected actors and commercial consequence
Gas Business Act	Defined gas businesses: safety and supply obligations, not a universal LNG stock quota
Electricity Business Act	Electricity-business supply obligations; no inferred terminal inventory
Economic Security Promotion Act and designation order	Certified critical-material supply-security plans, including LNG arrangements
JOGMEC law	Separate emergency fuel procurement under government request
Oil Stockpiling Act	Specified oil businesses and public stockpiles; not a blanket LNG requirement

These distinctions follow the consolidated laws captured at the edition cutoff. [185] [186] [187] [188] [189]

ANRE's regulated fuel-adjustment example averages January–March trade observations for June billing. A wholesale replacement cost can move before that retail adjustment or the customer's cash collection. Competitive contracts have their own formula and caps; the regulated example is not a universal retailer hedge. [190]

Ordinary combined petroleum/coal tax rates are JPY2,800/kL for crude and petroleum products, JPY1,860/t for gaseous hydrocarbons and JPY1,370/t for coal, before relevant exemptions or refunds. These already include the climate addition phased in by April 2016; adding a further JPY289/tCO_2 would double count it. Actual tax incidence and special measures must be resolved before attributing the rate to a generator's invoice. [191] [192] [193]

The next chapter connects these fuel systems to actual operating companies rather than assuming that group ownership reveals a private procurement book.

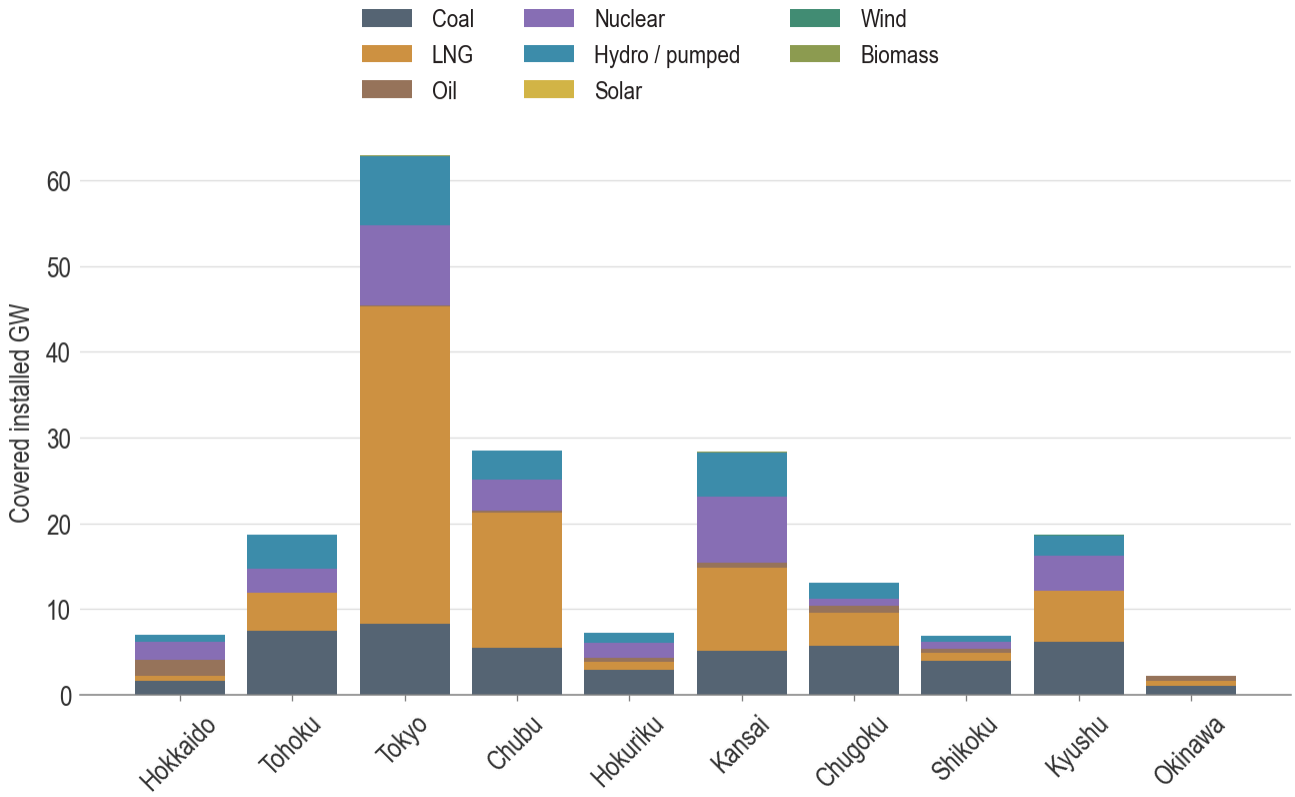
Companies, ownership and generation footprints

Companies: physical assets before brands

Generator, legal equipment holder, O&M contractor, JV shareholder and retailer may be different entities. Count the physical asset once, then record commercial relationships separately. The fifteen profiles are not the full competitive population: independent, industrial and project-company assets remain in regional totals. [331]

- Company by electrical area
- Hokkaido: concentrated thermal and nuclear
- Daigas: Himeji and a diversified portfolio

Exhibit 66: Regional capacity includes independents and project companies



Note: Installed composition is not a trading-market share. Independent, industrial and project-company assets remain in each area's denominator.

Source: [331] · Chart data

Denominator: installed admitted MW, not offers, firm capacity, equity MW or market share. Suspended reactors do not supply electricity; pumped storage requires charging. Shareholder stakes cannot create additional physical capacity.

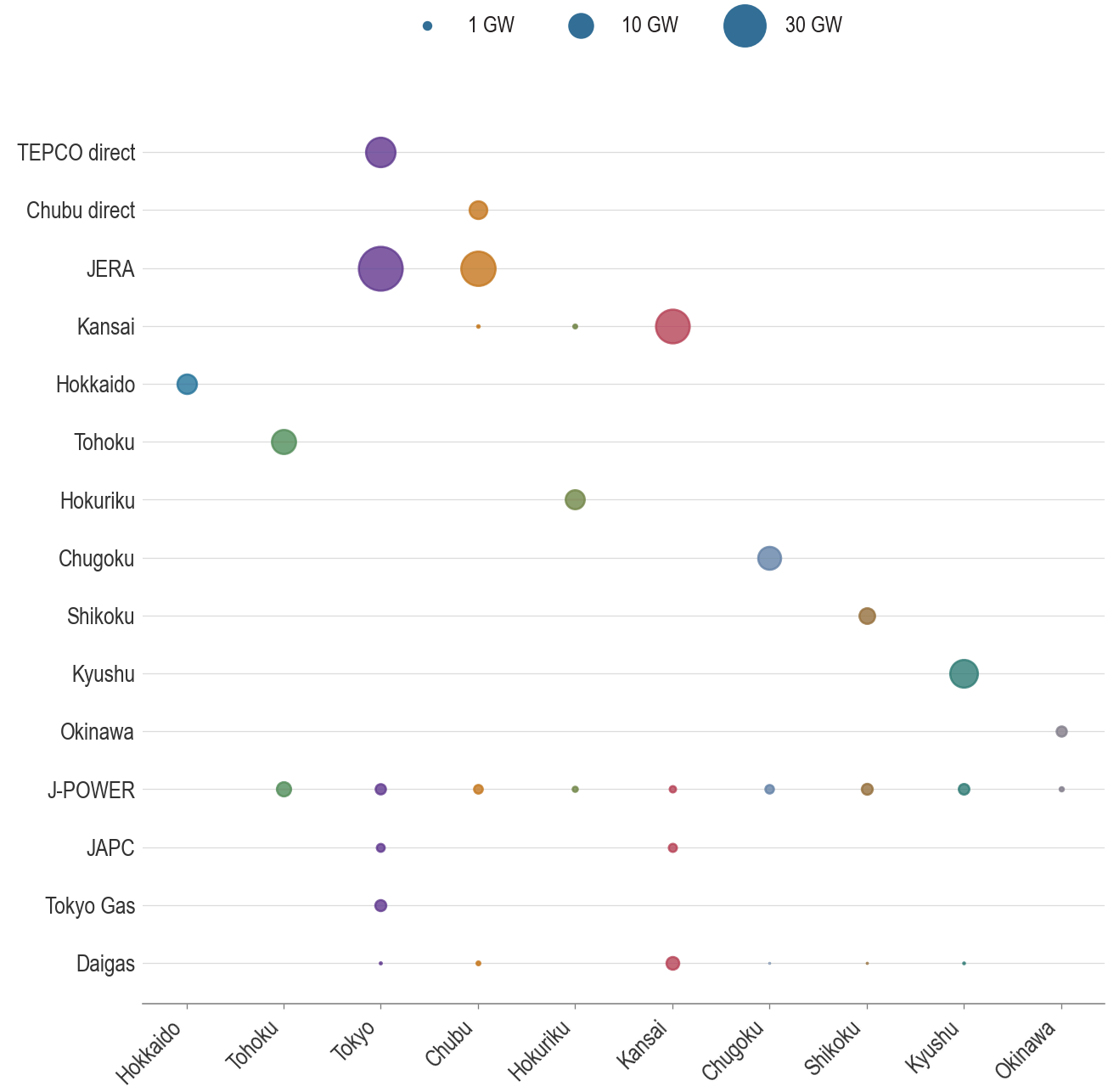
Each profile retains four components: corroborated geographic footprint, electrical-area/fuel totals, largest physical units and qualified longer operation. Markers are approximate stations/municipality locators—not headquarters, LNG terminals or surveyed connections. Unverified coordinates are omitted from maps, not capacity ledgers. [332]

Admitted large-unit totals differ from dated complete company inventories, which can include small hydro, islands, suspended plant or construction. Unknown dates stay unknown. Monthly output is the current admitted cohort measured historically, not reconstructed historical ownership. [336]

Company by electrical area

Headquarters and station prefecture do not determine electrical-area assignment. The matrix sums unique admitted records; Okinawa has physical assets but no invented JEPX price. Full regional/fuel extracts retain independent competitors beyond these fifteen rows. [336]

Exhibit 67: Fifteen companies across ten electrical areas



Note: Blank means no admitted record, not proof of no assets. The full competitor extract includes independent and unresolved operators.

Source: [331] · Chart data

JERA spans Tokyo/Chubu; J-POWER spans Japan. Kansai hydro extends beyond its home demand geography, while TEPCO's Niigata reactors retain their documented Tokyo electrical assignment.

No parent duplication: TEPCO direct and Chubu direct exclude JERA. JAPC reactors are not multiplied across shareholders. Gas-project gross MW count once; dated equity stakes and O&M roles remain separate. [337] [338]

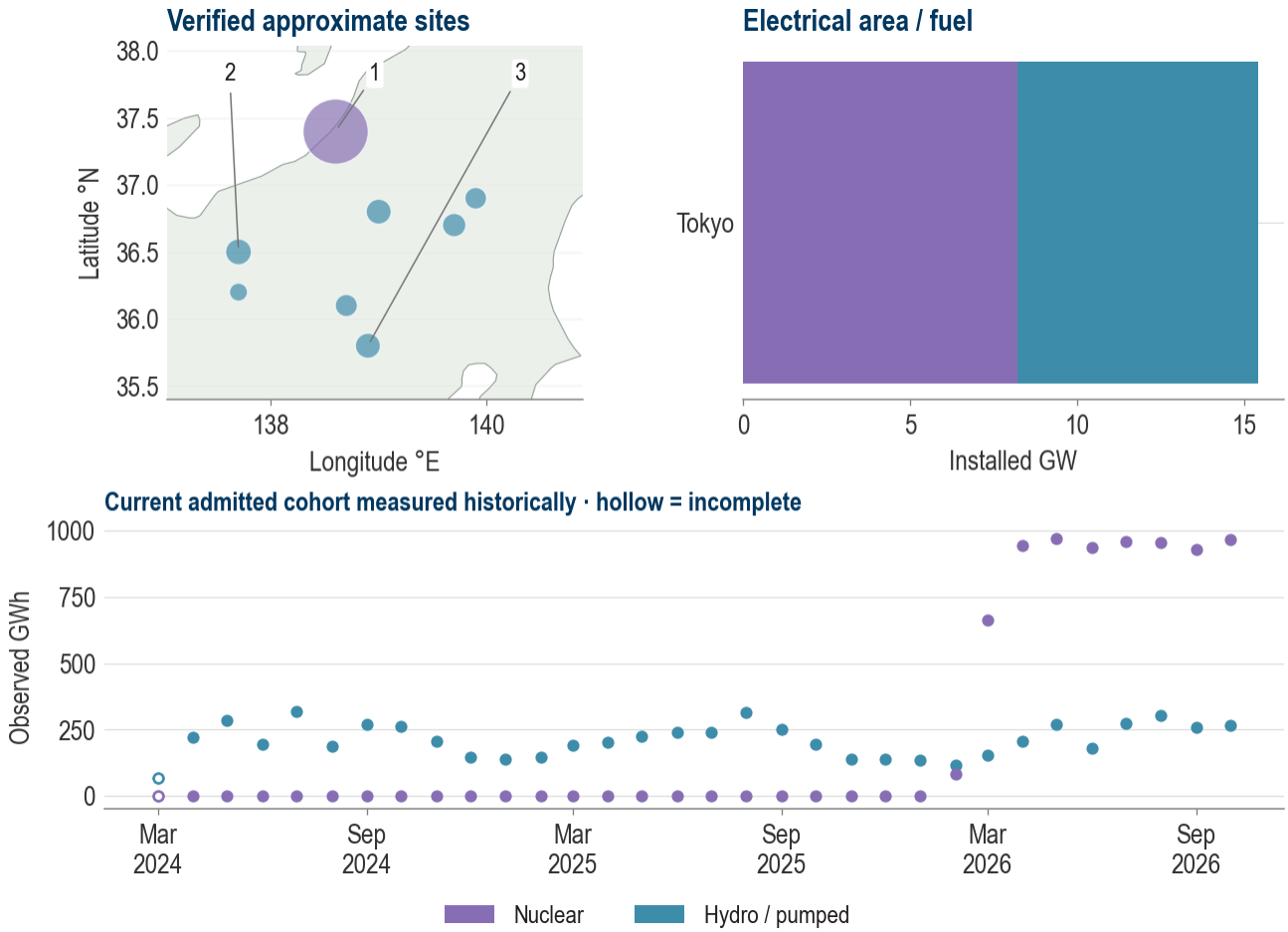
This is installed composition, not hourly dispatch. Use current outages, observed seasons, fuel access and neighboring systems before translating a footprint into a trading view. Small-asset and disclosure-date gaps remain explicit.

TEPCO: direct nuclear and storage

Kashiwazaki-Kariwa is in Niigata but belongs to the Tokyo electrical footprint. TEPCO Holdings owns the reactors; Renewable Power holds hydro/storage. Energy Partner is the retailer and Power Grid the network operator.

Admitted footprint: 15,405.0 MW in 32 records. Regional/fuel totals are shown below. Complete dated disclosure: 18,060.5 MW (HD nuclear plus RP hydro/new energy; excludes PG; 31 March 2025 unless stated), a different population. [339] [336]

Exhibit 68: TEPCO direct: sites, electrical footprint and measured operation



Note: 15,405.0 MW in the admitted footprint; 0.0 MW unmapped. Site locators: 1 Kashiwazaki-Kariwa; 2 Shin-Takasegawa; 3 Kazunogawa. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [339] [341] · Chart data

Physical unit	MW	First commercial operation
Kashiwazaki-Kariwa 6	1,356.0	1996-11-07
Kashiwazaki-Kariwa 7	1,356.0	1997-07-02
Kashiwazaki-Kariwa 1	1,100.0	1985-09-18
Kashiwazaki-Kariwa 2	1,100.0	1990-09-28
Kashiwazaki-Kariwa 3	1,100.0	1993-08-11

JERA remains outside this physical sum: shareholder interests create no extra MW. The seven reactors total 8,212 MW, but installed capacity does not certify operation. Pumped storage shifts energy and requires charging; it is not an additional primary-energy source.

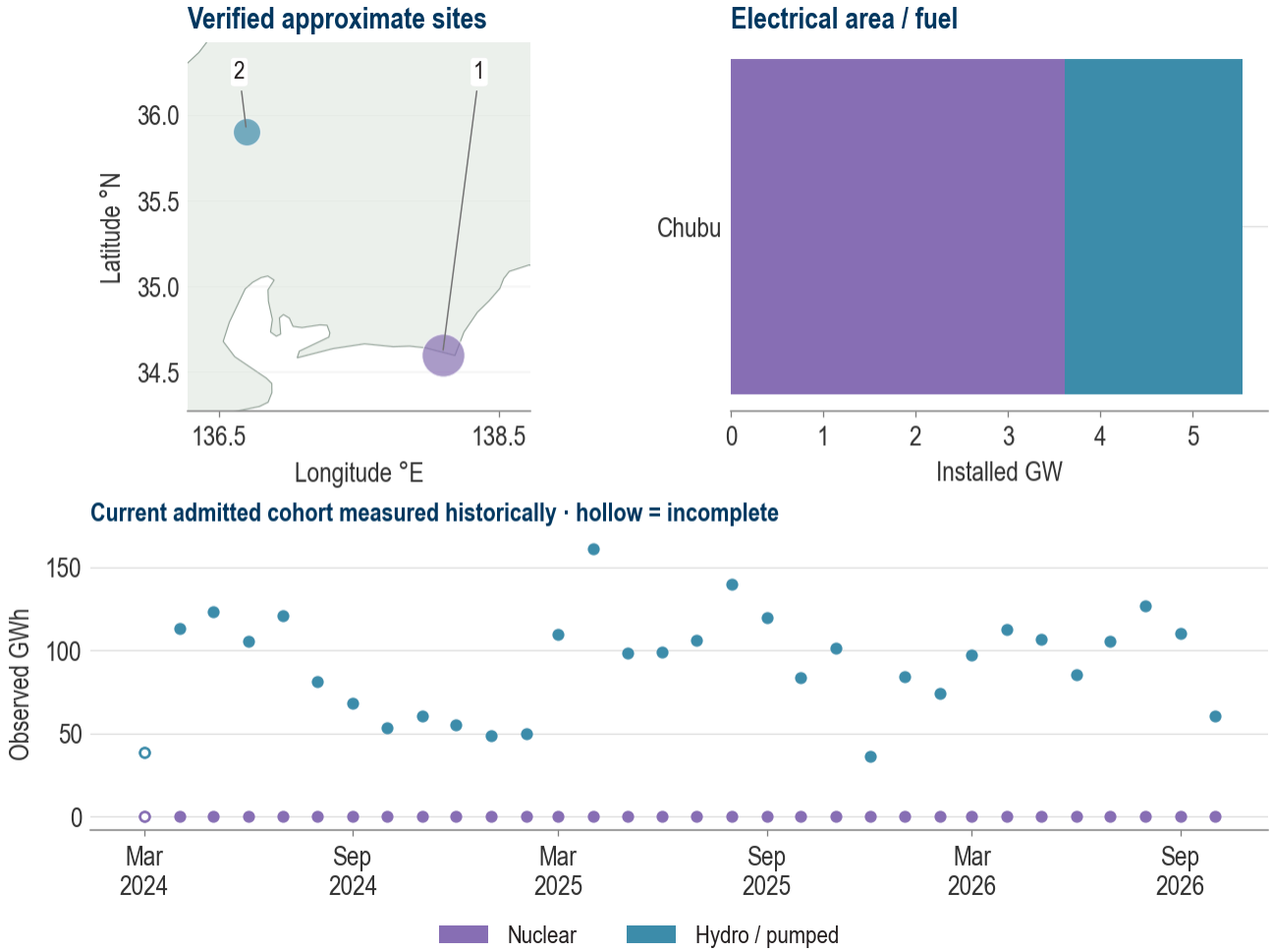
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Chubu: Hamaoka and hydro

The direct portfolio retains Hamaoka and hydro, not JERA's regional thermal fleet. Miraiz is the retail/procurement business; Power Grid manages networks and frequency conversion.

Admitted footprint: 5,544.0 MW in 12 records. Regional/fuel totals are shown below. Complete dated disclosure: 9,193.1 MW (own company; 31 March 2025 unless stated), a different population. [342] [336]

Exhibit 69: Chubu direct: sites, electrical footprint and measured operation



Note: 5,544.0 MW in the admitted footprint; 427.0 MW unmapped. Site locators: 1 Hamaoka; 2 Okumino. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [342] · Chart data

Physical unit	MW	First commercial operation
Hamaoka 5	1,380.0	2005-01-18
Hamaoka 4	1,137.0	1993-09-03
Hamaoka 3	1,100.0	1987-08-28
Okumino 1	250.0	Unit date unresolved
Okumino 2	250.0	Unit date unresolved

The own-equipment report lists Hamaoka 3,617 MW and hydro 5,477.290 MW. Hamaoka 3–5 are suspended in the dated reactor snapshot. Smaller hydro omitted from the admitted register explains part of the gap to the complete inventory; it is not inferred retirement.

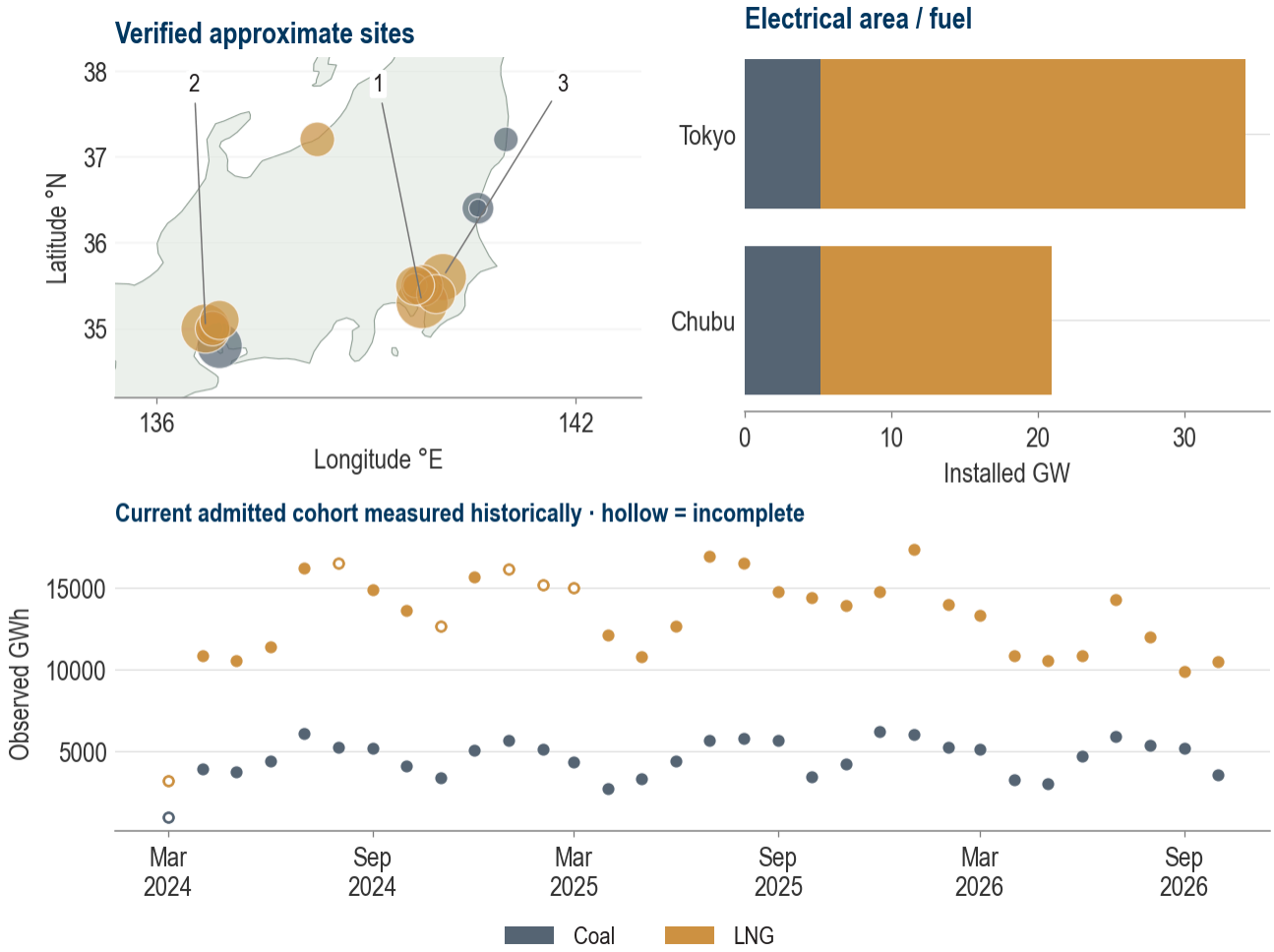
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

JERA: cross-regional thermal generation

JERA spans Tokyo and Chubu; Joetsu retains its verified electrical-area assignment, not a prefecture-based guess. Legal equipment holders include JERA and the separate Taketoyo, Anegasaki, Yokosuka, Goi United and Hitachinaka project entities.

Admitted footprint: 55,072.1 MW in 119 records. Regional/fuel totals are shown below. [336]

Exhibit 70: JERA: sites, electrical footprint and measured operation



Note: 55,072.1 MW in the admitted footprint; 9,635.7 MW unmapped. Site locators: 1 Futtsu; 2 Kawagoe; 3 Chiba. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [337] [341] [343] · Chart data

Physical unit	MW	First commercial operation
Taketoyo 5	1,070.0	2022-08
Hekinan 4	1,000.0	2001-11
Hekinan 5	1,000.0	2002-11
Higashi-Ohgishima (JERA) 1	1,000.0	1987-09
Higashi-Ohgishima (JERA) 2	1,000.0	1991-03

Nishi-Nagoya's 1,188.2 MW series are counted once but excluded from physical-unit rankings. Neither parent duplicates JERA capacity. Official generation fell 1.57%, from 230.870 TWh FY 2023 to 227.235 TWh FY 2024. The June 2025 disclosure 58.61 GW includes construction and excludes joint thermal, unlike the admitted cohort.

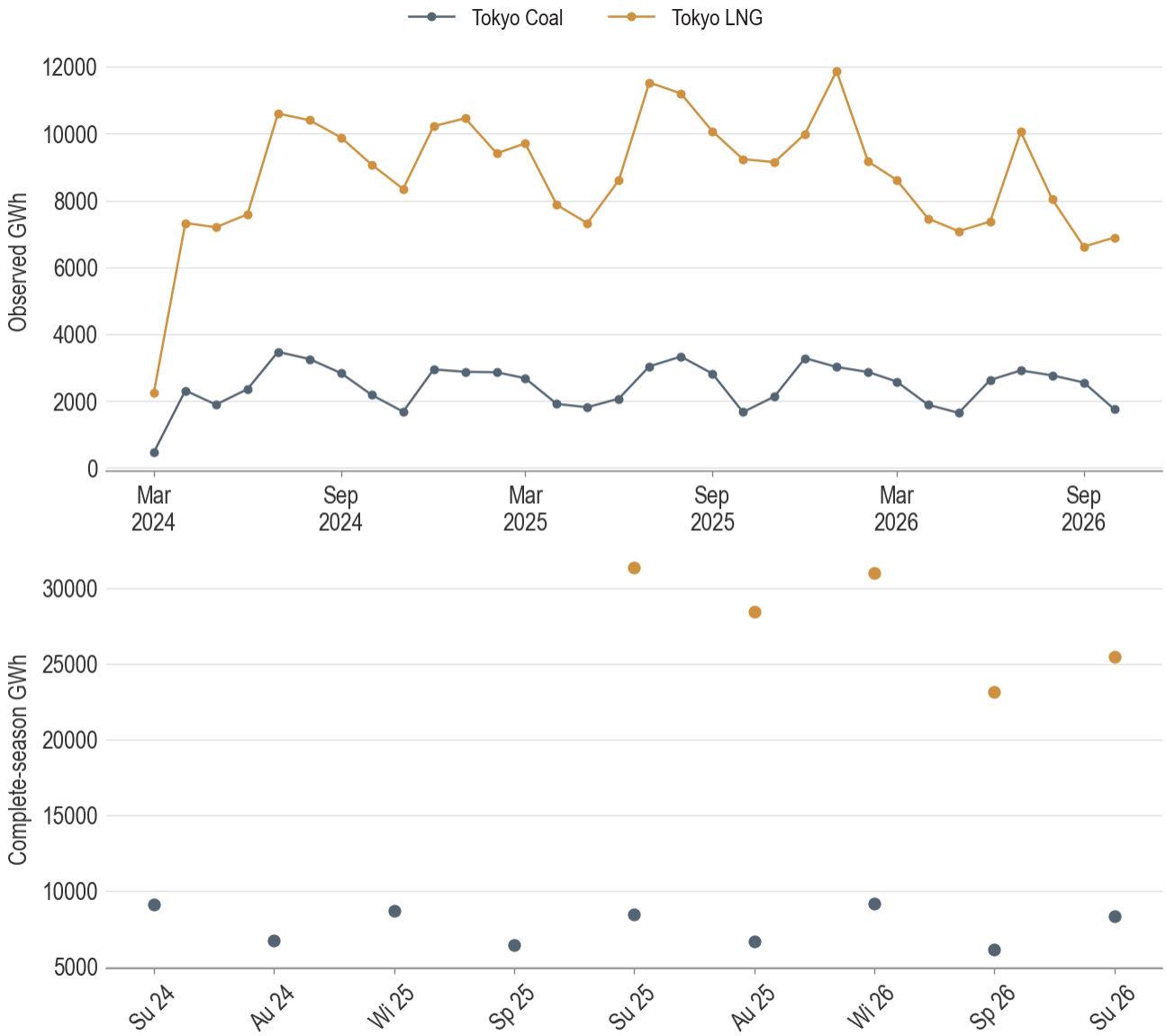
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [337] [340]

JERA in Tokyo: seasonal operation

Tokyo includes LNG trains at Futtsu/Chiba/Kawasaki/Yokohama, steam units at Higashi-Ohgishima/Sodegaura and coal at Hitachinaka/Yokosuka. Whole stations are not single outage units.

Admitted area fleet: 34,138.7 MW. Coal 5,150.0 MW; LNG 28,988.7 MW. This is not the company's global or equity-accounted total. [336]

Exhibit 71: JERA: monthly and four-season measured energy / Tokyo



Note: Only complete three-month seasons with unchanged membership enter the lower panel. Winter is December of the preceding year through February.

Source: [6] [331] · Chart data

New Goi units began August 2024, November 2024 and March 2025. The current cohort therefore cannot be extrapolated backward as an unchanged fleet. [337]

The upper panel retains monthly observations; the lower sums winter/spring/summer/autumn only with three complete months and unchanged measured membership. Winter runs December–February. Incomplete months are not extrapolated; missing seasons are not zero output.

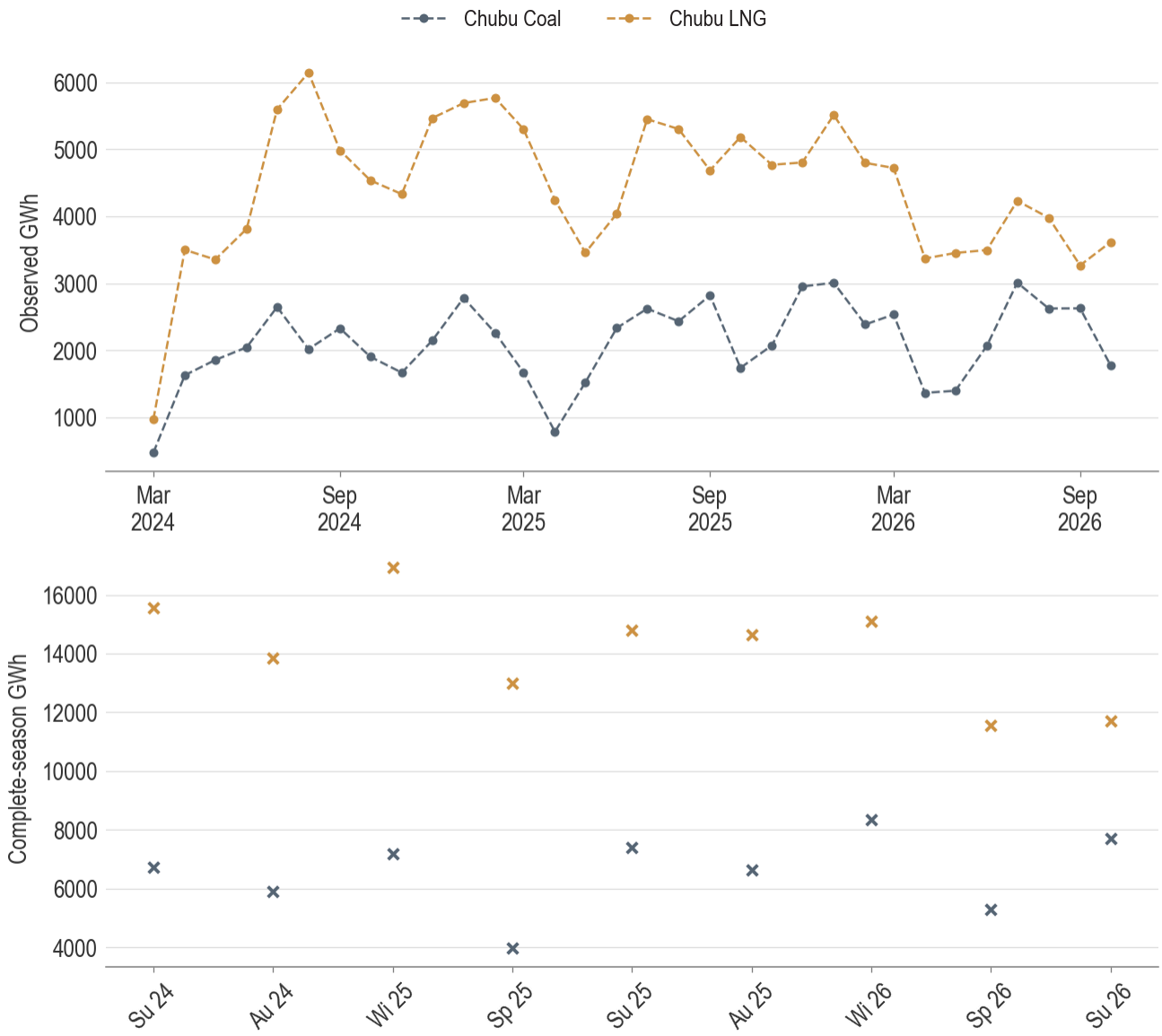
These are current-cohort GWh, not historical ownership or fuel capacity factors. Population and interval counts distinguish missing observations/new units from dispatch changes. Date-valid capacity, outage scope and consistent net/gross output are required for utilization comparisons. [331]

JERA in Chubu: seasonal operation

Hekinan/Taketoyo coal complements Kawagoe/Shin-Nagoya/Nishi-Nagoya/Chita LNG and Joetsu's verified electrical assignment. Nearby fuel infrastructure does not prove unrestricted gas interchangeability.

Admitted area fleet: 20,933.4 MW. Coal 5,170.0 MW; LNG 15,763.4 MW. This is not the company's global or equity-accounted total. [336]

Exhibit 72: JERA: monthly and four-season measured energy / Chubu



Note: Only complete three-month seasons with unchanged membership enter the lower panel. Winter is December of the preceding year through February.

Source: [6] [331] · Chart data

Nishi-Nagoya 7-1/7-2 began September 2017/March 2018; each 1,188.2 MW series is not a verified single generator. Taketoyo 5 began August 2022; Hekinan 4/5 November 2001/2002. [337]

The upper panel retains monthly observations; the lower sums winter/spring/summer/autumn only with three complete months and unchanged measured membership. Winter runs December–February. Incomplete months are not extrapolated; missing seasons are not zero output.

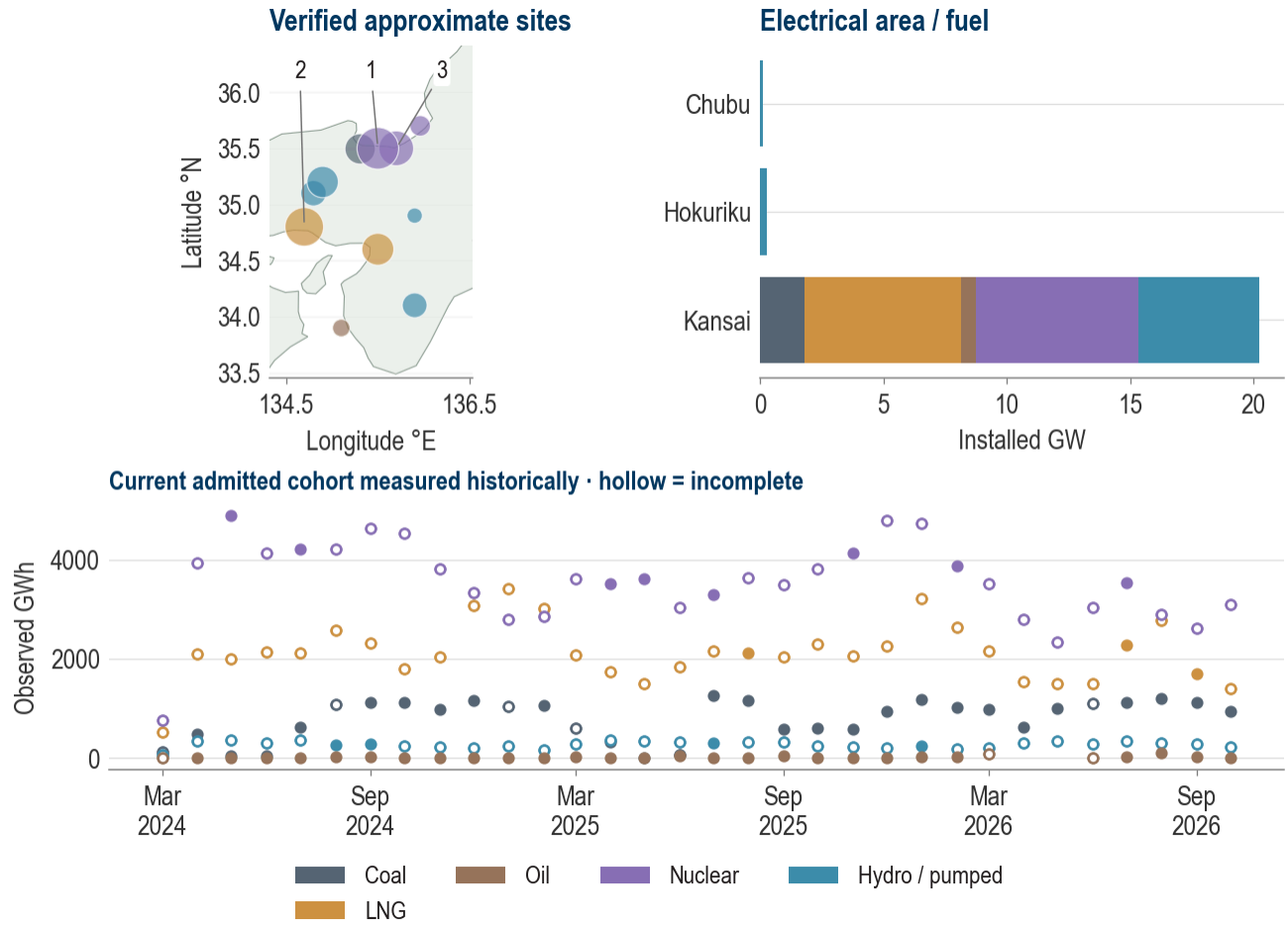
These are current-cohort GWh, not historical ownership or fuel capacity factors. Population and interval counts distinguish missing observations/new units from dispatch changes. Date-valid capacity, outage scope and consistent net/gross output are required for utilization comparisons. [331]

Kansai: nuclear, LNG and storage

The footprint combines Fukui reactors, coastal thermal and inland storage; it is wider than Osaka demand. Own-equipment disclosure separates Kansai's assets from network subsidiaries.

Admitted footprint: 20,605.0 MW in 44 records. Regional/fuel totals are shown below. Complete dated disclosure: 26,049.2 MW (own company before later announced retirements; 31 March 2025 unless stated), a different population. [344] [336]

Exhibit 73: Kansai: sites, electrical footprint and measured operation



Note: 20,605.0 MW in the admitted footprint; 1,824.0 MW unmapped. Site locators: 1 Takahama; 2 Himeji No.2; 3 Ohi. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [39] [98] [331–333] [341] [344] · Chart data

Physical unit	MW	First commercial operation
Ohi 3	1,180.0	1991-12-18
Ohi 4	1,180.0	1993-02-02
Maizuru 1	900.0	2004-08
Maizuru 2	900.0	2010-08
Takahama 3	870.0	1985-01-17

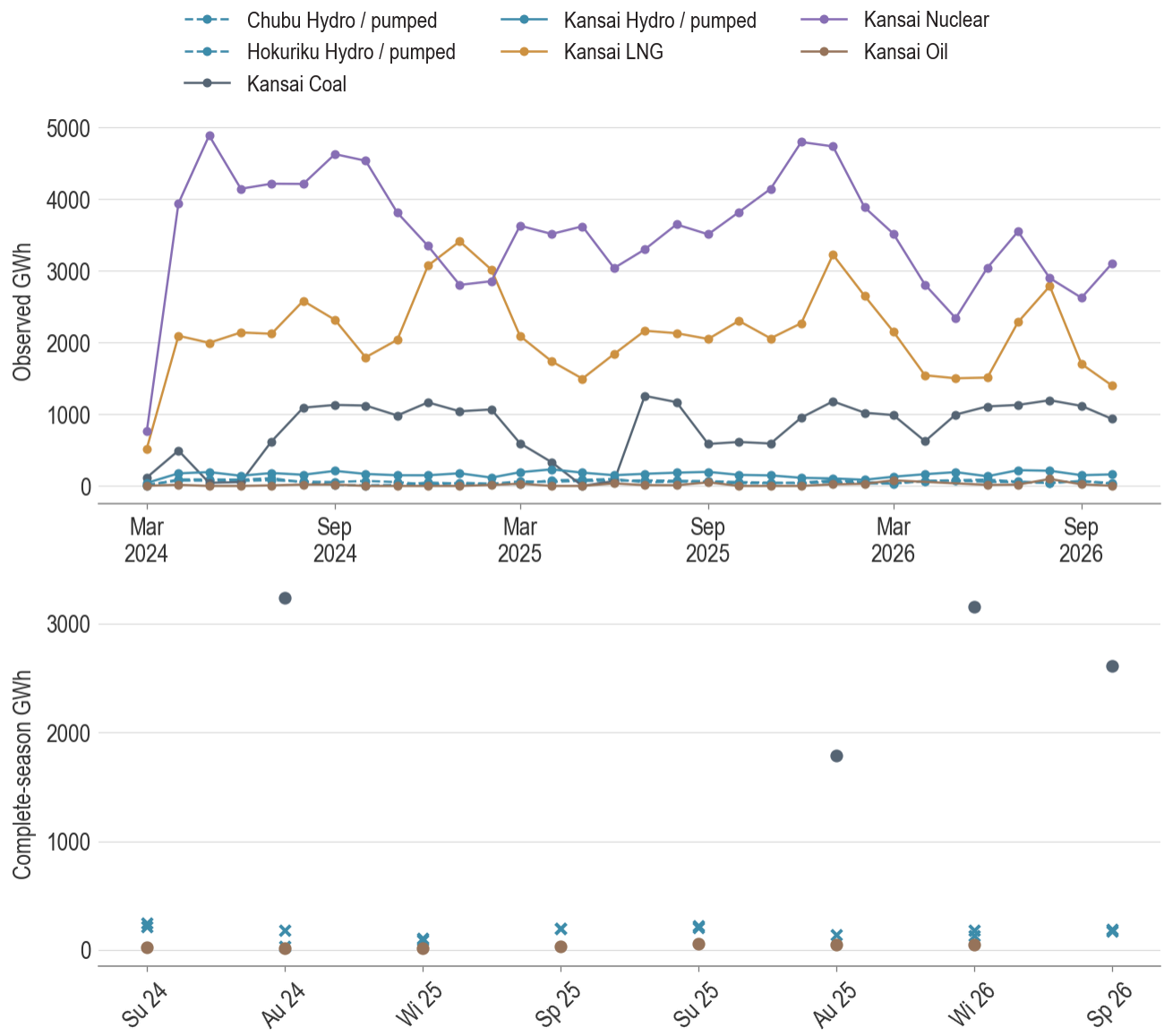
Mihama 3 is the resolved 826 MW reactor, not its ambiguous raw zero-capacity duplicate. Startup/criticality September 12 and first adjustment operation September 15 preceded the September 18 stop. Restart October 10 and full commercial return November 4 are separate actual milestones, not a completed September return.

Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Kansai: nuclear energy and thermal replacement

Reactors supply sustained energy when operating; thermal meets residual requirements; pumped storage shifts energy with a charging cost. The geographic profile and measured seasonal history answer different questions. [344]

Exhibit 74: Kansai: monthly and four-season measured energy



Note: Only complete three-month seasons with unchanged membership enter the lower panel. Winter is December of the preceding year through February.

Source: [6] [331] · Chart data

Installed nuclear: March 2025 equipment totals 6,578 MW—Mihama 826, Ohi 2,360 and Takahama 3,392—not guaranteed simultaneous output. Mihama 3's ambiguous zero-capacity master record is resolved to one physical reactor, not a second unit. [331]

In August 2026, early/late Kansai nuclear averages were 4.443/3.607 GW while demand was 19.247/19.328 GW; the cooler middle phase concealed this supply difference in monthly means. [285]

Mihama startup/criticality September 12, adjustment September 15 and stop September 18 were followed by actual resumption October 10 and full commercial return November 4. September was not a completed return. [333]

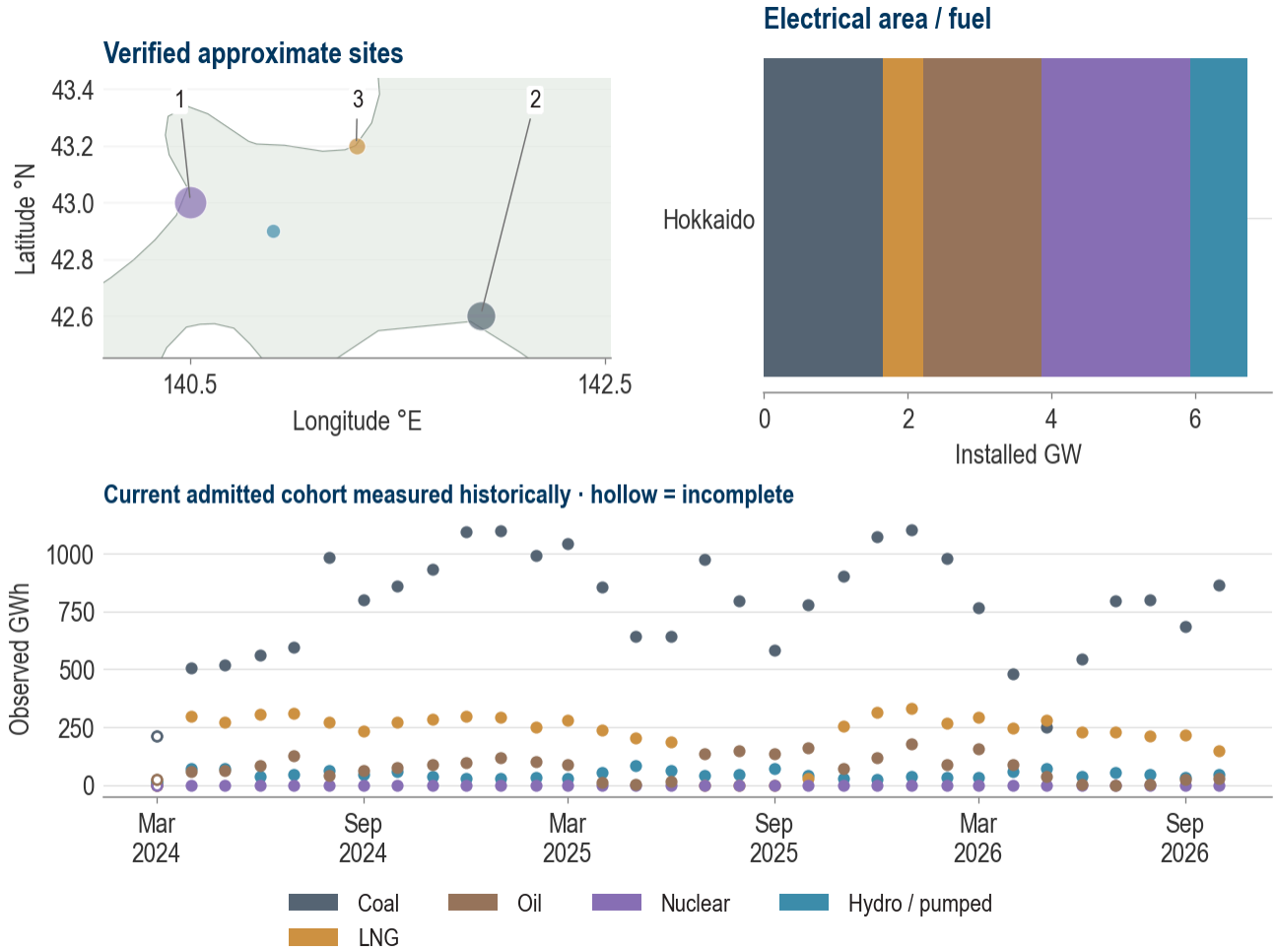
Seasonal sums require three complete months with unchanged measured membership. They are current-cohort observations, not retrospective ownership or capacity factors. Pumping and source-native net/gross boundaries remain distinct.

Hokkaido: concentrated thermal and nuclear

Tomari nuclear, Tomatoh-Atsuma coal and coastal thermal dominate. Ishikariwan Shinko is correctly located at Otaru/Zenibako: its 569.4 MW first unit started February 2019. Proposed later units are not operating capacity.

Admitted footprint: 6,739.4 MW in 18 records. Regional/fuel totals are shown below. Complete dated disclosure: 8,178.9 MW (own company including suspended thermal; 31 March 2025 unless stated), a different population. [345] [336]

Exhibit 75: Hokkaido: sites, electrical footprint and measured operation



Note: 6,739.4 MW in the admitted footprint; 2,050.0 MW unmapped. Site locators: 1 Tomari; 2 Tomatoh-Atsuma; 3 Ishikariwan Shinko. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [345] [346] · Chart data

Physical unit	MW	First commercial operation
Tomari 3	912.0	2009-12-22
Tomatoh-Atsuma 4	700.0	2002-06
Tomatoh-Atsuma 2	600.0	1985-10
Tomari 1	579.0	1989-06-22
Tomari 2	579.0	1991-04-12

Own-equipment disclosure includes suspended assets: Date 1 from December 2023 and Date 2 from April 2024. Tomari's 2,070 MW is suspended in the dated reactor snapshot. Installed ownership does not certify available supply; the February 2026 case separately documents actual support and physical response.

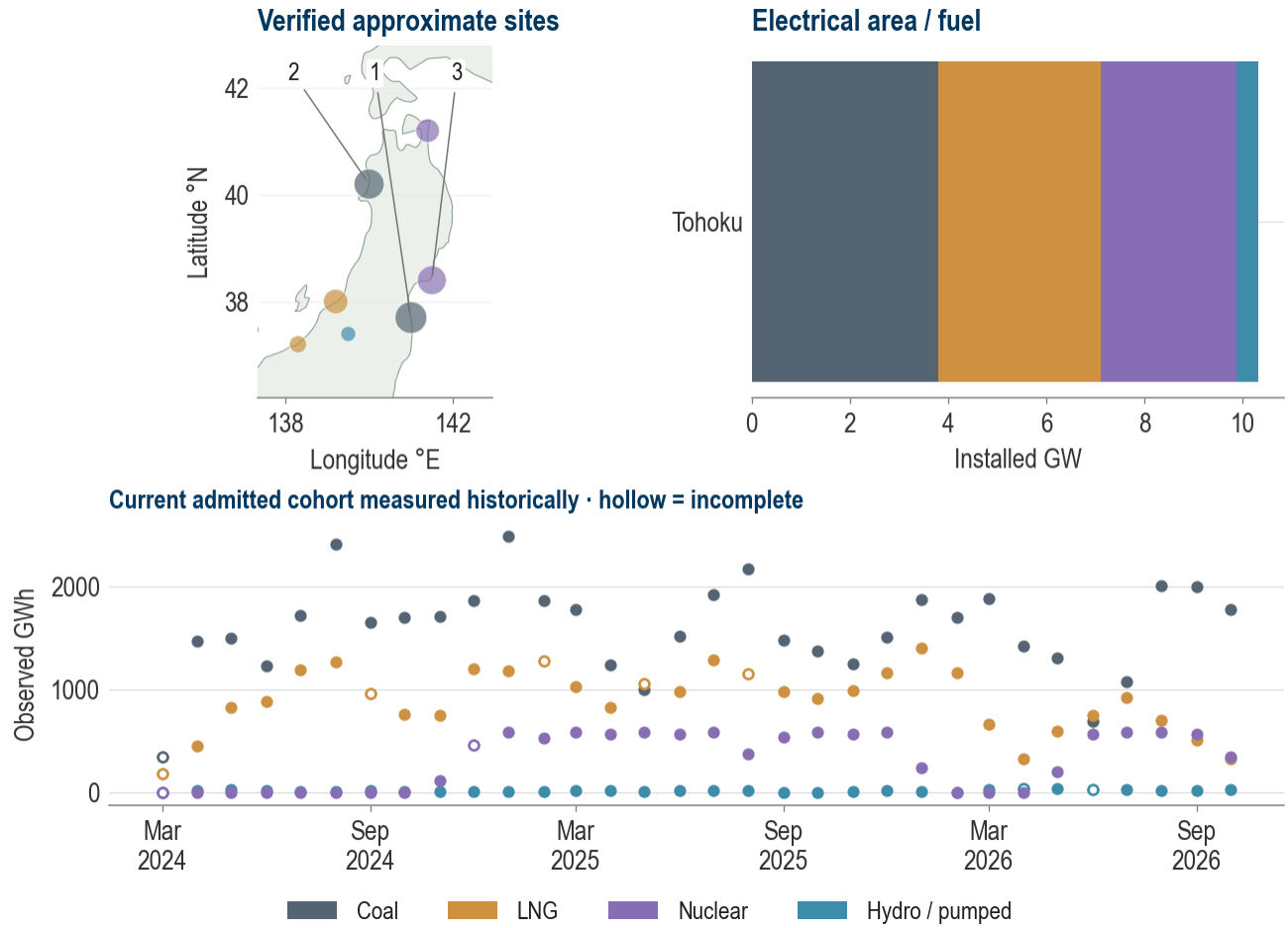
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Tohoku: two coasts and large thermal units

Pacific and Sea-of-Japan assets include Haramachi coal, LNG and Onagawa/Higashidori nuclear. Shin-Sendai and Sendai are distinct plants. Generation/sales equipment is separate from Network's island and network assets.

Admitted footprint: 10,323.0 MW in 16 records. Regional/fuel totals are shown below. Complete dated disclosure: 16,835.3 MW (generation/sales segment, not only direct units; 31 March 2025 unless stated), a different population. [347] [336]

Exhibit 76: Tohoku: sites, electrical footprint and measured operation



Note: 10,323.0 MW in the admitted footprint; 1,514.0 MW unmapped. Site locators: 1 Haramachi; 2 Noshiro; 3 Onagawa. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [347] · Chart data

Physical unit	MW	First commercial operation
Higashidori 1	1,100.0	2005-12-08
Haramachi 1	1,000.0	1997-07
Haramachi 2	1,000.0	1998-07
Onagawa 2	825.0	1995-07-28
Onagawa 3	825.0	2002-01-30

Haramachi 1/2 began July 1997/July 1998. The reactor units retain individual dates and operating states; suspended MW are not available output. Smaller facilities and unresolved duplicate identities are excluded from the admitted register. February 2026 overlapping support maxima are not additive delivered energy.

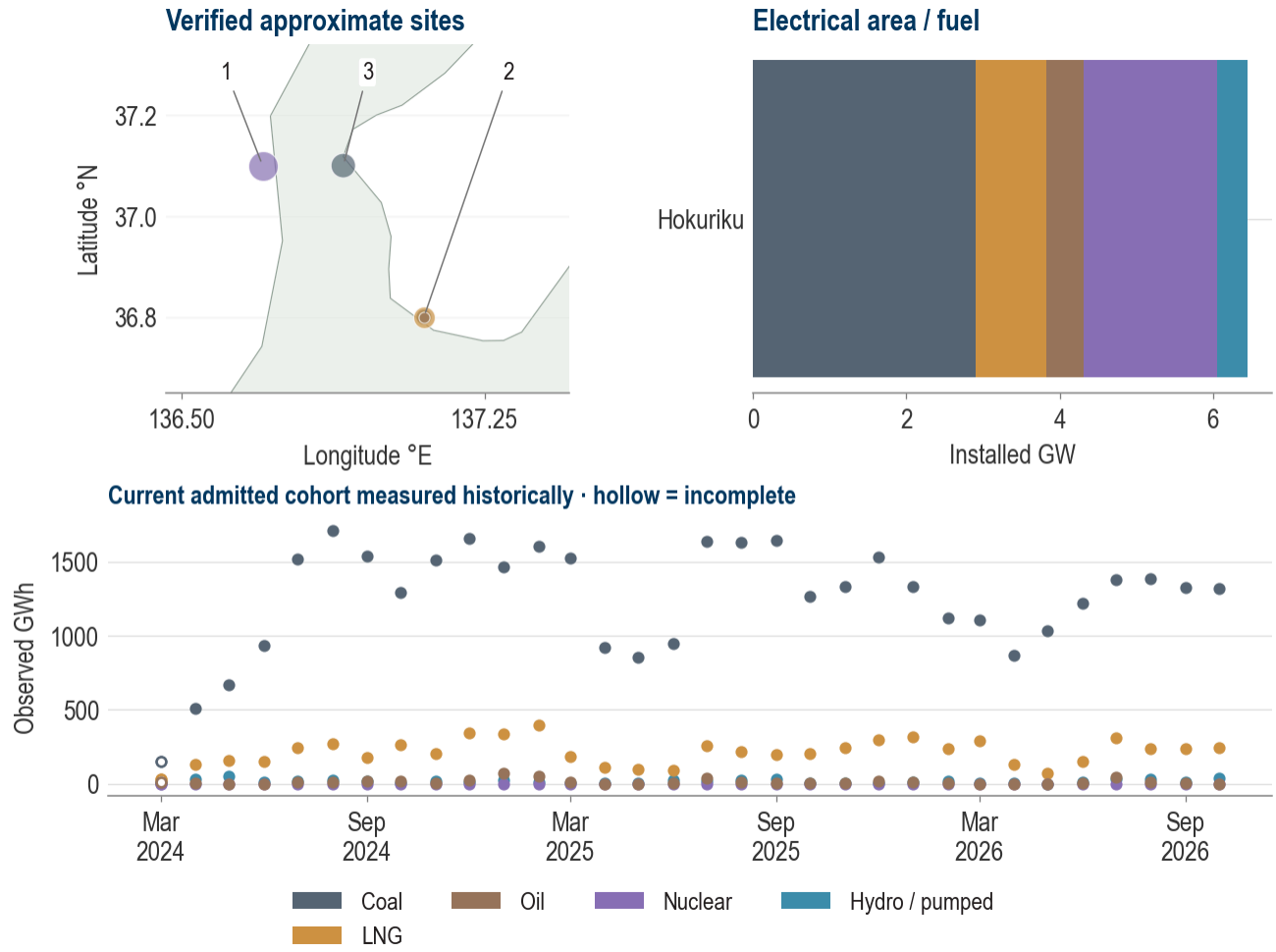
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Hokuriku: Shika, coal and hydro

Shika nuclear, Nanao-Ota/Tsuruga coal, Toyama-Shinko thermal and mountain hydro have different operating roles. Hokuriku's thermal Tsuruga is not JAPC's nuclear station.

Admitted footprint: 6,448.7 MW in 14 records. Regional/fuel totals are shown below. Complete dated disclosure: 8,281.6 MW (group hydro/steam/nuclear equipment; 31 March 2025 unless stated), a different population. [348] [336]

Exhibit 77: Hokuriku: sites, electrical footprint and measured operation



Note: 6,448.7 MW in the admitted footprint; 1,838.0 MW unmapped. Site locators: 1 Shika; 2 Toyama-Shinko; 3 Nanao-Ota. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [348] [349] · Chart data

Physical unit	MW	First commercial operation
Shika 2	1,206.0	2006-03-15
Nanao-Ota 2	700.0	1998-07
Tsuruga thermal (Hokuriku) 2	700.0	2000-09
Shika 1	540.0	1993-07-30
Nanao-Ota 1	500.0	1995-03

The March 2025 report lists 1,746 MW nuclear, 4,564.7 MW thermal and 1,970.93 MW hydro. Shika remains suspended. Nanao-Ota 1/2 began March 1995/July 1998; Tsuruga thermal 2 September 2000. Commercial-start vintage is not measured efficiency, remaining life or assured availability.

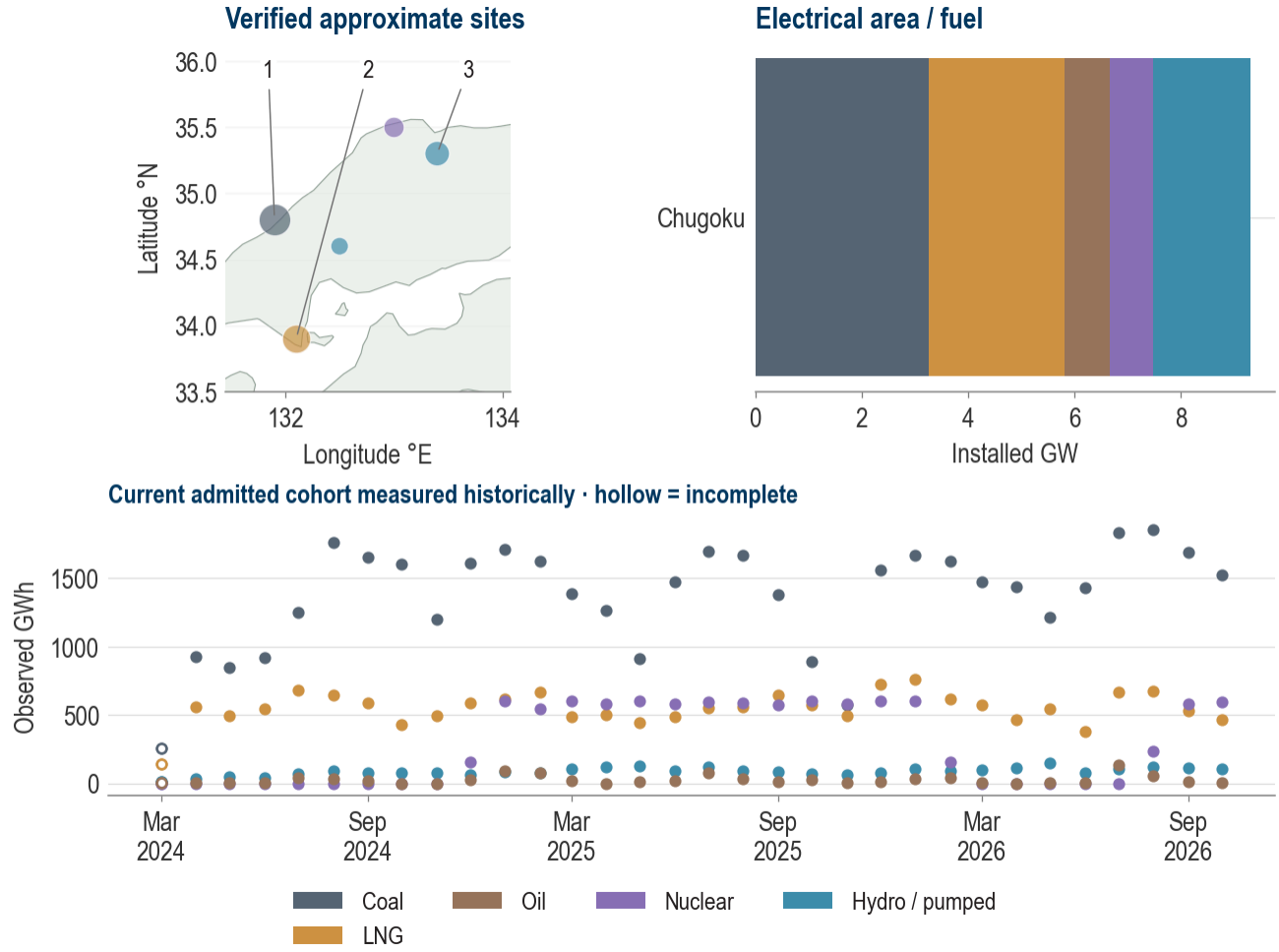
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Chugoku: Shimane and coastal thermal

Misumi 1/2 are separate 1,000 MW coal units. Own-equipment disclosure identifies Misumi, Shin-Onoda, Mizushima, Tamashima, Yanai and Shimane; adjacent regional systems remain important.

Admitted footprint: 9,302.0 MW in 27 records. Regional/fuel totals are shown below. Complete dated disclosure: 10,359.0 MW (own company; 31 March 2025 unless stated), a different population. [350] [336]

Exhibit 78: Chugoku: sites, electrical footprint and measured operation



Note: 9,302.0 MW in the admitted footprint; 3,084.0 MW unmapped. Site locators: 1 Misumi; 2 Yanai; 3 Matanogawa. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [350] · Chart data

Physical unit	MW	First commercial operation
Misumi 1	1,000.0	1998-06
Misumi 2	1,000.0	2022-11
Shimane 2	820.0	1989-02-10
Shin-Onoda 1	500.0	1986-04
Shin-Onoda 2	500.0	1987-01

Misumi 1/2 began June 1998/November 2022; Shin-Onoda 1/2 April 1986/January 1987. A later fuel conversion is not original commissioning. August 2026 changing nuclear output must be read alongside demand and solar, not as an isolated causal price experiment.

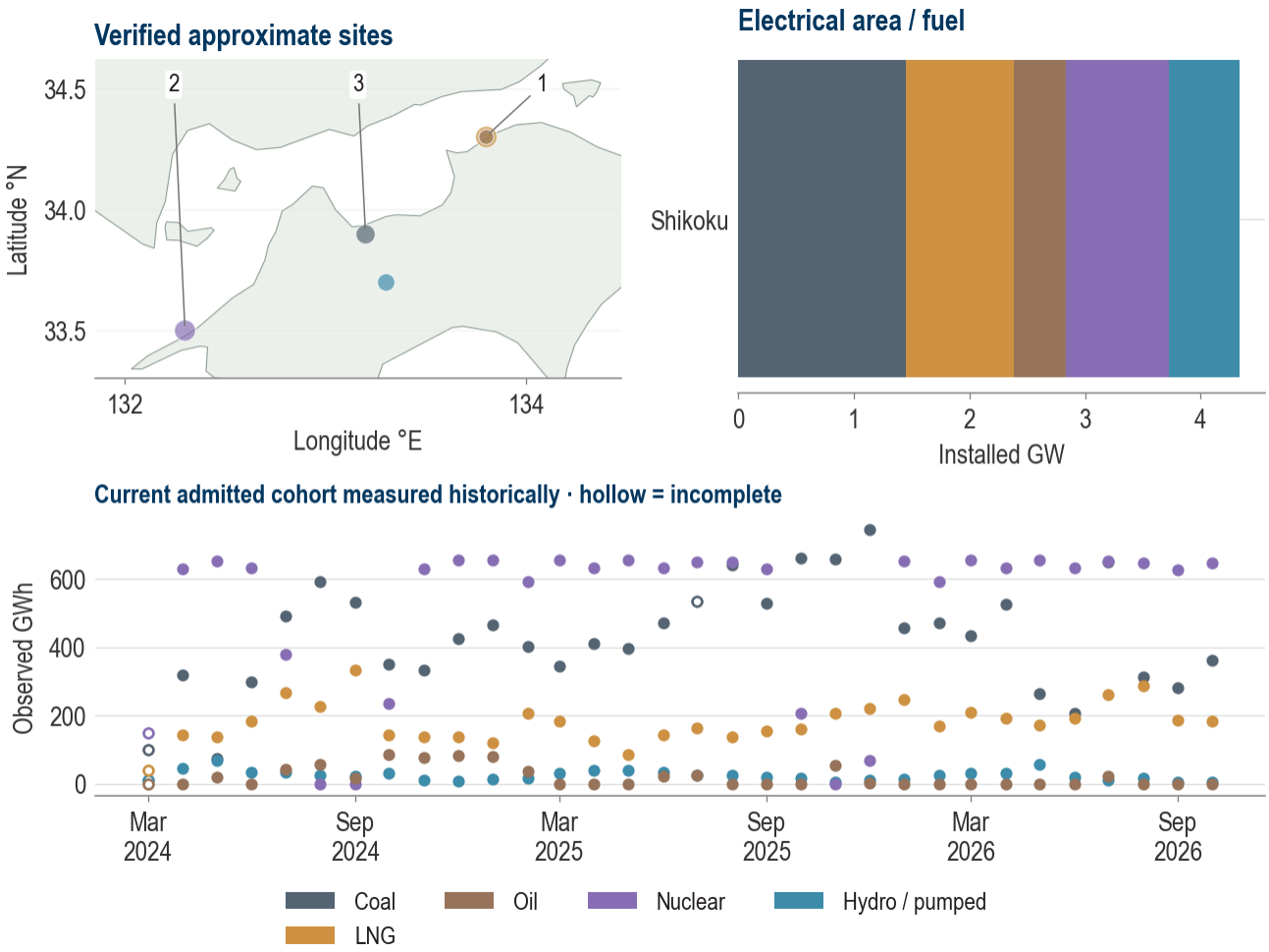
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Shikoku: Ikata and a compact thermal fleet

Ikata 3, Tachibanawan, Saijo, Sakaide and pumped storage define the footprint. Shikoku's 700 MW Tachibanawan unit is distinct from J-POWER's adjacent 1,050 MW units.

Admitted footprint: 4,340.0 MW in 10 records. Regional/fuel totals are shown below. Complete dated disclosure: 5,339.2 MW (own company; 31 March 2025 unless stated), a different population. [351] [336]

Exhibit 79: Shikoku: sites, electrical footprint and measured operation



Note: 4,340.0 MW in the admitted footprint; 700.0 MW unmapped. Site locators: 1 Sakaide; 2 Ikata; 3 Saijo. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [351–353] · Chart data

Physical unit	MW	First commercial operation
Ikata 3	890.0	1994-12-15
Tachibanawan (Shikoku)	700.0	2000-06
Saijo 1	500.0	2023-06-30
Sakaide 3	450.0	Unit date unresolved
Sakaide 4	350.0	Unit date unresolved

Own-equipment disclosure lists 890 MW nuclear, 3,285 MW thermal and 1,157.696 MW hydro plus small equipment. Saijo 1 is the replacement 500 MW unit commissioned June 30, 2023, not the predecessor with a changed rating. Sakaide's LNG/oil units require separate fuel attribution.

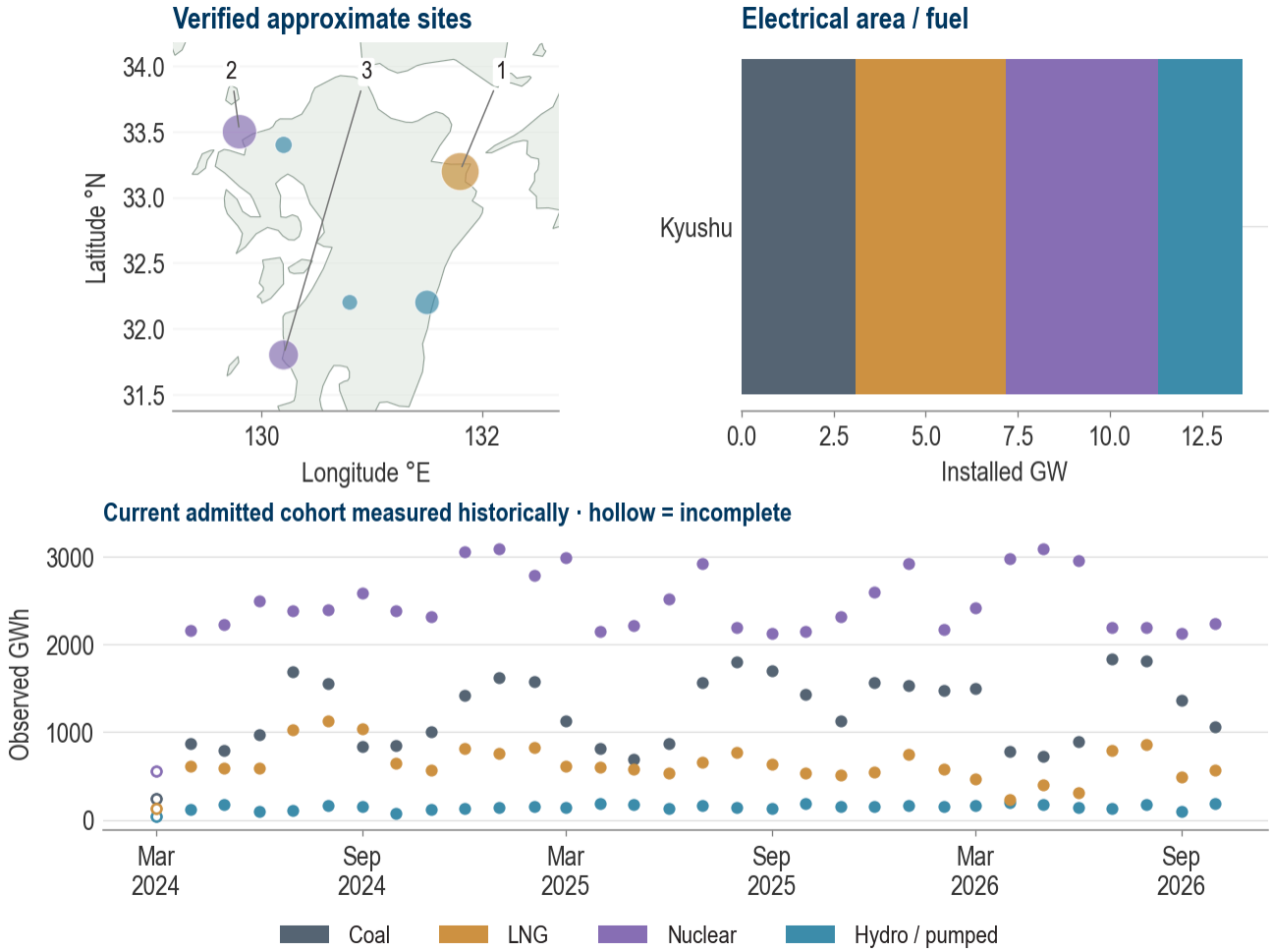
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Kyushu: nuclear depth and solar-shaped load

Genkai 3/4 and Sendai 1/2 total 4,140 MW, alongside coal, LNG and flexible hydro. Low midday residual demand and evening adequacy are different operating problems.

Admitted footprint: 13,615.0 MW in 32 records. Regional/fuel totals are shown below. Complete dated disclosure: 14,903.1 MW (own company hydro/steam/nuclear; 31 March 2025 unless stated), a different population. [354] [336]

Exhibit 80: Kyushu: sites, electrical footprint and measured operation



Note: 13,615.0 MW in the admitted footprint; 4,300.0 MW unmapped. Site locators: 1 Shin-Oita; 2 Genkai; 3 Sendai. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [341] [354] · Chart data

Physical unit	MW	First commercial operation
Genkai 3	1,180.0	1994-03-18
Genkai 4	1,180.0	1997-07-25
Matsuura 2	1,000.0	2019-12
Sendai 1	890.0	1984-07-04
Sendai 2	890.0	1985-11-28

Own-equipment disclosure distinguishes Kyushu Electric, Transmission and Distribution, and Kyuden Mirai Energy. Subsidiary renewable/island assets are not assigned to the parent. Matsuura 2 began December 2019. The May 2026 case separately measures curtailment, pumping and thermal response; regional PV is not parent-owned output.

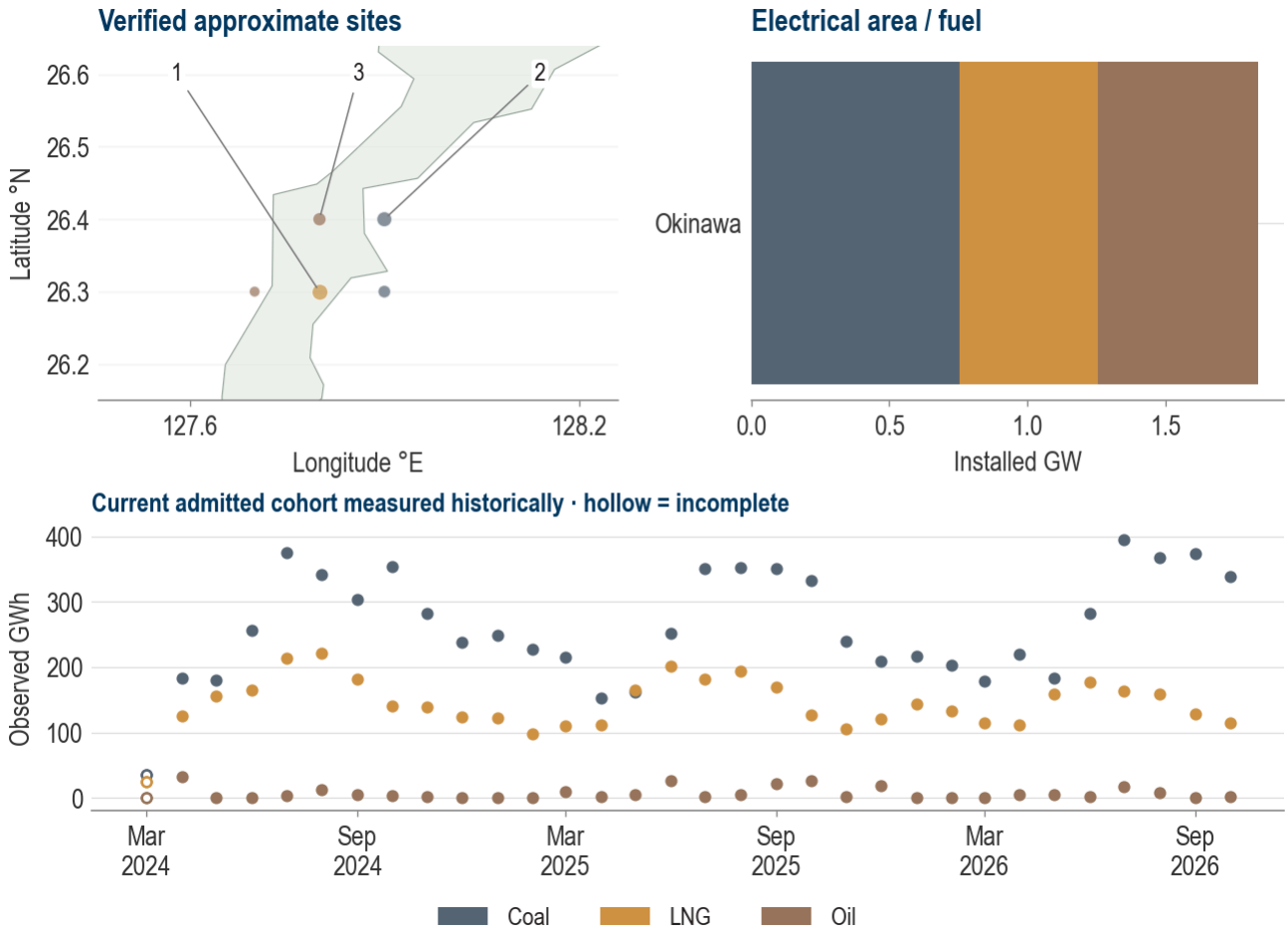
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Okinawa: an island fleet, not a JEPX area

Yoshinoura LNG, Kin/Gushikawa coal and Makiminato/Ishikawa thermal use corroborated station/municipality locations. J-POWER's Ishikawa remains separate. There is no tenth JEPX price or assumed mainland rescue.

Admitted footprint: 1,835.0 MW in 11 records. Regional/fuel totals are shown below. Complete dated disclosure: 2,209.9 MW (own company including island/small equipment; 31 March 2025 unless stated), a different population. [355] [336]

Exhibit 81: Okinawa: sites, electrical footprint and measured operation



Note: 1,835.0 MW in the admitted footprint; 0.0 MW unmapped. Site locators: 1 Yoshinoura; 2 Kin; 3 Ishikawa (Okinawa Electric). Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [335] [341] [355] · Chart data

Physical unit	MW	First commercial operation
Yoshinoura 1	251.0	2012-11
Yoshinoura 2	251.0	2013-05
Kin 1	220.0	2002-02
Kin 2	220.0	2003-05
Gushikawa 1	156.0	1994-03

The report includes 1,629 MW steam, 578.630 MW internal-combustion and 2.315 MW new energy; small/island/mobile equipment exceeds admitted coverage. Yoshinoura 1/2 began November 2012/May 2013; Kin 1/2 February 2002/May 2003; Gushikawa 1/2 March 1994/March 1995. Missing public output remains unknown.

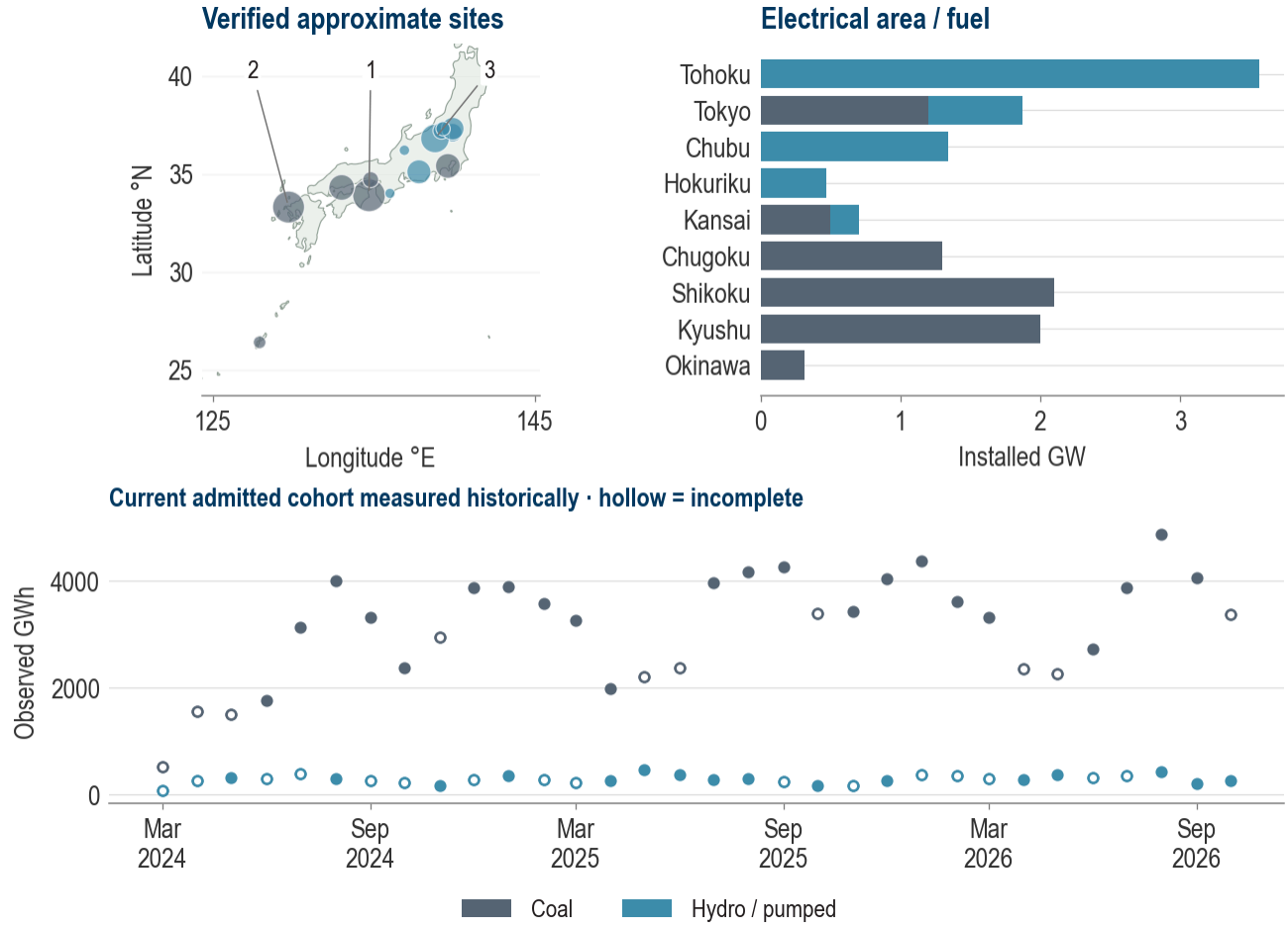
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

J-POWER: independent, cross-regional assets

J-POWER's domestic hydro/coal portfolio includes Okinawa. Its Tachibanawan and Matsuura are distinct from Shikoku/Kyushu utility stations. Transmission subsidiaries and overseas equity capacity are excluded.

Admitted footprint: 13,669.0 MW in 46 records. Regional/fuel totals are shown below. Complete dated disclosure: 17,009.3 MW (own domestic generation including stopped Matsushima; 31 March 2025 unless stated), a different population. [356] [336]

Exhibit 82: J-POWER: sites, electrical footprint and measured operation



Note: 13,669.0 MW in the admitted footprint; 470.0 MW unmapped. Site locators: 1 Tachibanawan (J-POWER); 2 Matsuura (J-POWER); 3 Okukiyotsu. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [341] [356] [357] · Chart data

Physical unit	MW	First commercial operation
Tachibanawan (J-POWER) 1	1,050.0	2000-07-27
Tachibanawan (J-POWER) 2	1,050.0	2000-12-15
Matsuura (J-POWER) 1	1,000.0	1990-06-29
Matsuura (J-POWER) 2	1,000.0	1997-07-04
Takehara 3	700.0	1983-03-18

The securities report identifies Electric Power Development as holder. Matsushima ceased operation at FY 2024-end: unit 1 retired, unit 2 suspended for GENESIS. Tachibanawan 1/2 began July/December 2000; Matsuura 1/2 June 1990/July 1997. Ownership is not offer-market share; storage output needs water/charging context.

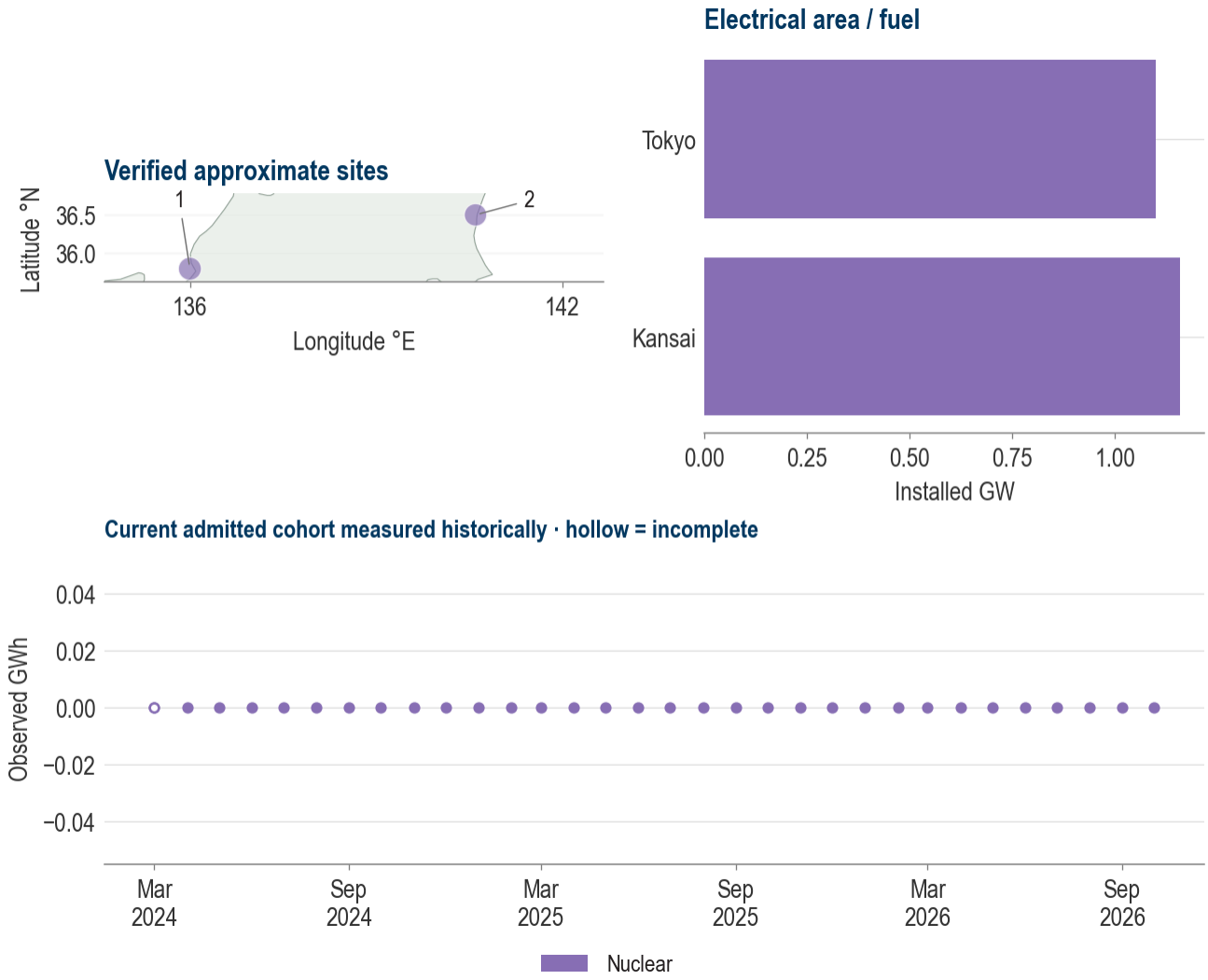
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

JAPC: two retained, suspended reactors

Only Tokai Daini 1,100 MW and Tsuruga 2 1,160 MW qualify: an honest ranking contains two, not five. Decommissioning Tokai/Tsuruga 1 and proposals are excluded.

Admitted footprint: 2,260.0 MW in 2 records. Regional/fuel totals are shown below. The March 2026 disclosure gives 2,260.0 MW for the same two retained reactors. [338] [336]

Exhibit 83: JAPC: sites, electrical footprint and measured operation



Note: 2,260.0 MW in the admitted footprint; 0.0 MW unmapped. Site locators: 1 Tsuruga; 2 Tokai Daini. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [332] [338] [341] · Chart data

Physical unit	MW	First commercial operation
Tsuruga 2	1,160.0	1987-02-17
Tokai Daini	1,100.0	1978-11-28

The FY 2025 overview lists own equipment and confirms the existing fleet stopped since May 2011. Genden Engineering's contracted O&M is not ownership; shareholders create no extra reactors. Commercial starts were November 28, 1978 and February 17, 1987. Restart requires separate licensing/works/notices, not an assumption from installed MW.

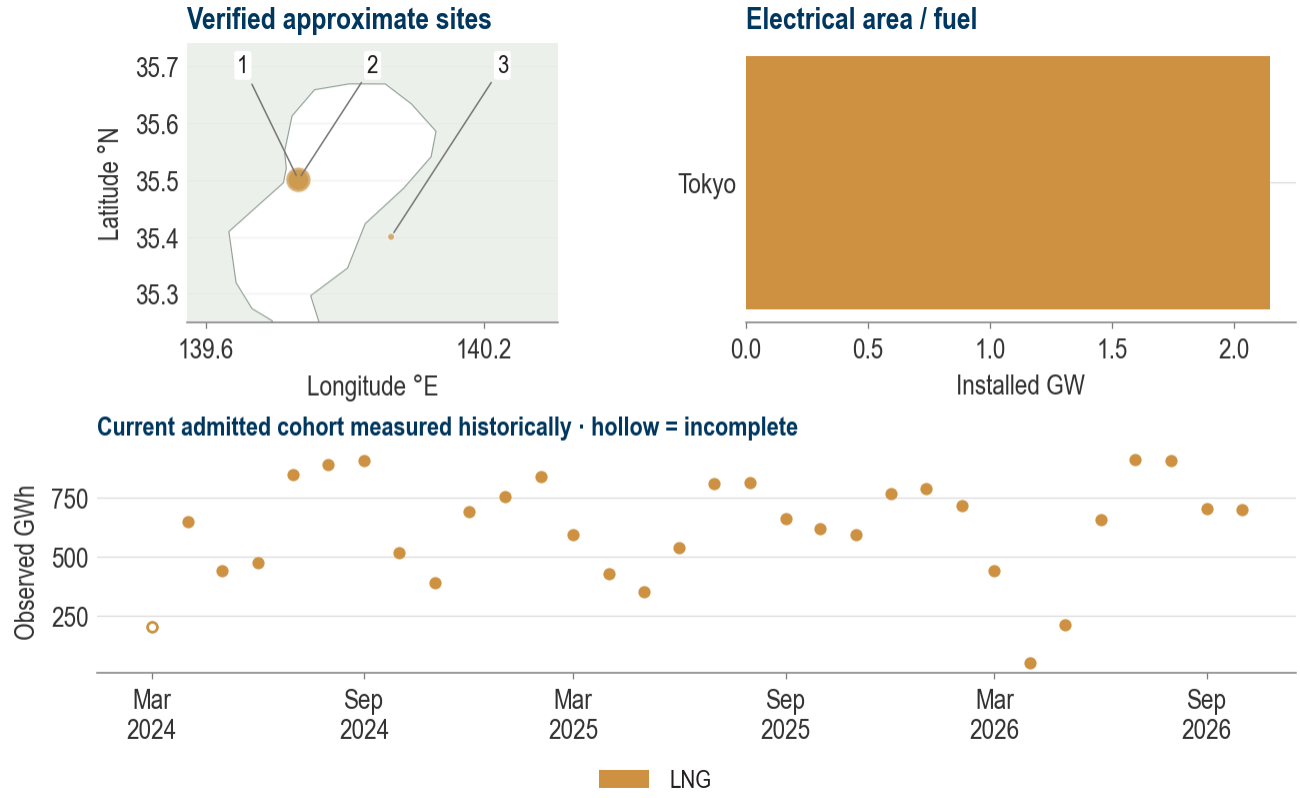
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [340]

Tokyo Gas: physical projects, not purchases

Ohgishima Power and Kawasaki Natural Gas are separate project companies, not JERA assets. Ten 9.78 MW Sodegaura engines total 97.8 MW, completed March 27, 2025; the later Colza succession agreement is not a verified completed transfer.

Admitted footprint: 2,145.1 MW in 15 records. Regional/fuel totals are shown below. [336]

Exhibit 84: Tokyo Gas: sites, electrical footprint and measured operation



Note: 2,145.1 MW in the admitted footprint; 0.0 MW unmapped. Site locators: 1 Ohgishima Power; 2 Kawasaki Natural Gas; 3 Sodegaura gas engine. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [341] [358] [359] [361] [362] · Chart data

Physical unit	MW	First commercial operation
Kawasaki Natural Gas 1	413.0	Unit date unresolved
Kawasaki Natural Gas 2	413.0	Unit date unresolved
Ohgishima Power 1	407.1	2010-03-31
Ohgishima Power 2	407.1	2010-07-12
Ohgishima Power 3	407.1	Unit date unresolved

The 2021 disclosure gives Kawasaki 840 MW and Ohgishima 1,221 MW, versus admitted 826 and 1,221.3 MW; rating vintage/scope remains unresolved. Its stakes were 49% TG/51% ENEOS and 75% TG/25% Idemitsu respectively, not asserted current. Gross project MW count once. Colza's 101 GWh FY2023 send-out lacks plant allocation; Mobara/Shiinomori capacities remain unresolved. Moka purchases and historical Baypower/Yokosuka are not current owned assets.

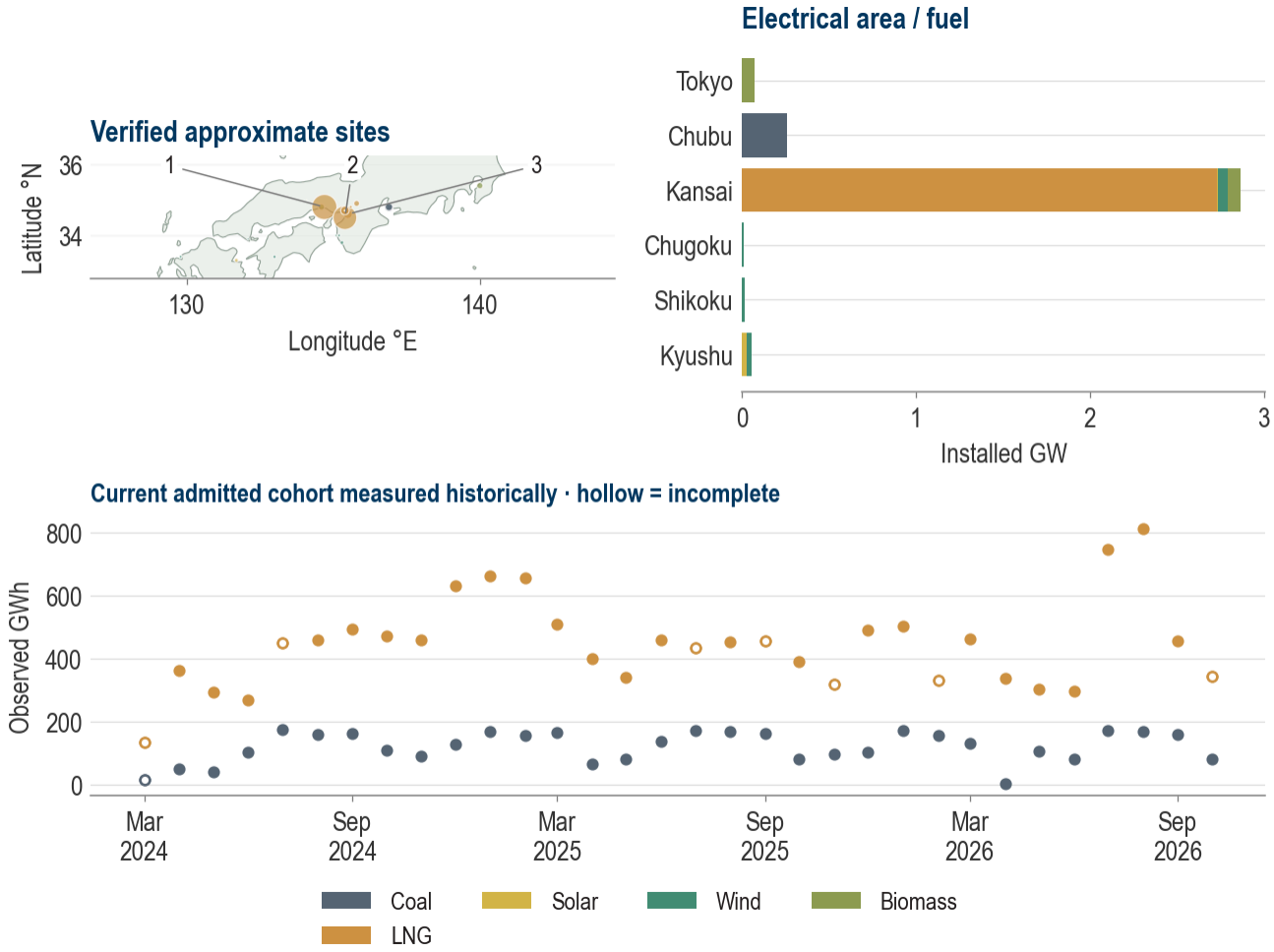
Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [358] [359] [360] [340]

Daigas: Himeji and a diversified portfolio

Himeji 1 actually began January 1, 2026; unit 2 began May 2026 (day unresolved). Their combined 1,245.2 MW is distinct from Kansai's Himeji; Senboku totals 1,109 MW. Torishima/Funamachi/Nagoya add thermal configurations.

Admitted footprint: 3,287.2 MW in 102 records. Regional/fuel totals are shown below. [336]

Exhibit 85: Daigas: sites, electrical footprint and measured operation



Note: 3,287.2 MW in the admitted footprint; 0.0 MW unmapped. Site locators: 1 Himeji Natural Gas; 2 Senboku Natural Gas; 3 Funamachi. Current cohort's observed history; hollow = incomplete. Map colour = fuel; circle area = MW; hydro includes pumped. Coast: Natural Earth.

Source: [6] [98] [331] [334] [341] [363–368] · Chart data

Physical unit	MW	First commercial operation
Himeji Natural Gas 1	622.6	2026-01-01
Himeji Natural Gas 2	622.6	2026-05
Senboku Natural Gas 3	277.5	Unit date unresolved
Senboku Natural Gas 4	277.5	Unit date unresolved
Senboku Natural Gas 1	277.0	Unit date unresolved

Himeji Natural Gas Power Generation is the project company, Osaka Gas its 100% parent and Daigas G&P Solution the O&M contractor. Other unresolved titles stay unknown. Documented wind/biomass/solar/cogeneration retain real scope; Shiribetsu is withheld because source totals conflict. Cogeneration nameplate is not assumed wholly exportable.

Output scope: current admitted cohort measured historically, not historical ownership or complete company output. Missing data stay missing; undated denominators yield no load factor. [331] [334] [363] [340]

Regional markets, prices and transmission

Regional markets and connected prices

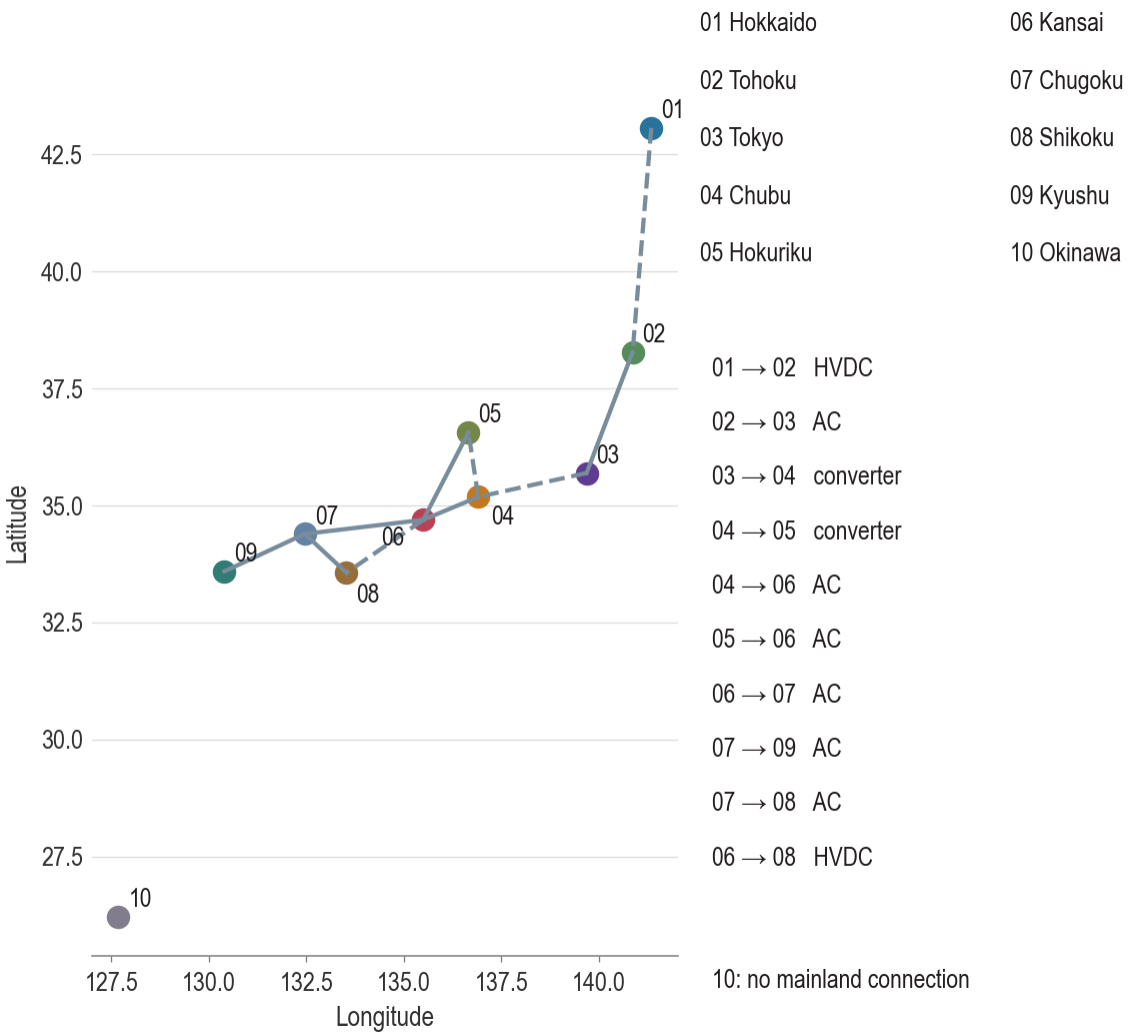
Tokyo averaged 20.38 JPY/kWh in summer 2026, against 19.48 in Chubu, 16.80 in Kansai and 12.83 in Shikoku. The resulting basis was not a uniform east–west premium: Hokkaido was cheaper than Tohoku in summer but dearer in winter 2026. Fleet availability, solar timing, heating and cooling exposure meet a network whose usable limits change with direction and operating configuration. [1]

- Tokyo: fleet and load geography
- Kansai: winter versus summer stress
- Reversals within the delivery day

Fleet and demand geography → price/physical histories → transmission and restrictions. Nine mainland price areas are joined by ten distinct routes; Okinawa has an isolated-system comparison, not an invented JEPX price.

Honshi is Chugoku–Shikoku AC; Anan–Kihoku is Kansai–Shikoku HVDC. The March 2025 asset inventory predates March 2026 central-loop/fence publication changes. A reporting fence is not another bilateral route. [278] [329] [279]

Exhibit 86: Nine coupled price areas; ten physical systems



Note: Schematic centres, not surveyed routes. March 2025 asset identities; dashed = DC/conversion. March 2026 fences are not extra routes. East is 50 Hz; Tokyo–Chubu converts frequency.
Source: [278] [279] [329] · Chart data

Tokyo: fleet and load geography

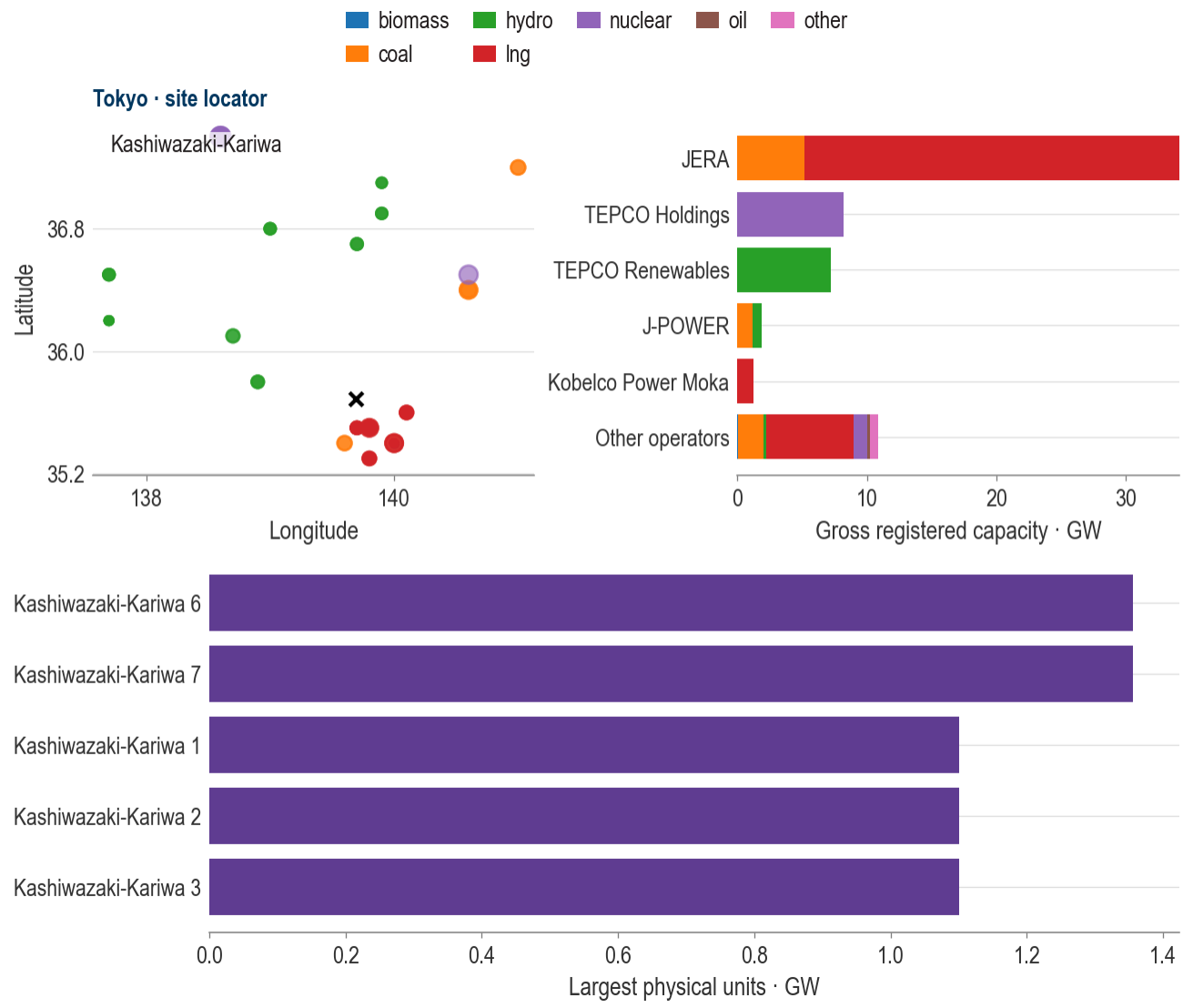
JERA's 34,139 MW is attributed separately from TEPCO Holdings and Renewable Power—not ownership or bidding by TEPCO Energy Partner. Tokyo Gas and independent/JV generators compete beyond the legacy utility. [331]

Tokyo–Yokohama/Kanto load meets Tokyo Bay LNG, northern Tohoku supply and western converters. Electrical destination, not prefecture, determines the balance. Mean demand was 34.90 GW in winter 2026, 32.16 in summer and 27.75 in spring; lighter spring demand still leaves solar-withdrawal exposure. [330] [2]

Registered operator	Gross MW
JERA	34,139
TEPCO Holdings	8,212
TEPCO Renewable Power	7,193
J-POWER	1,875
All other registered operators	12,090

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 87: Tokyo: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] [359] [364] · Chart data

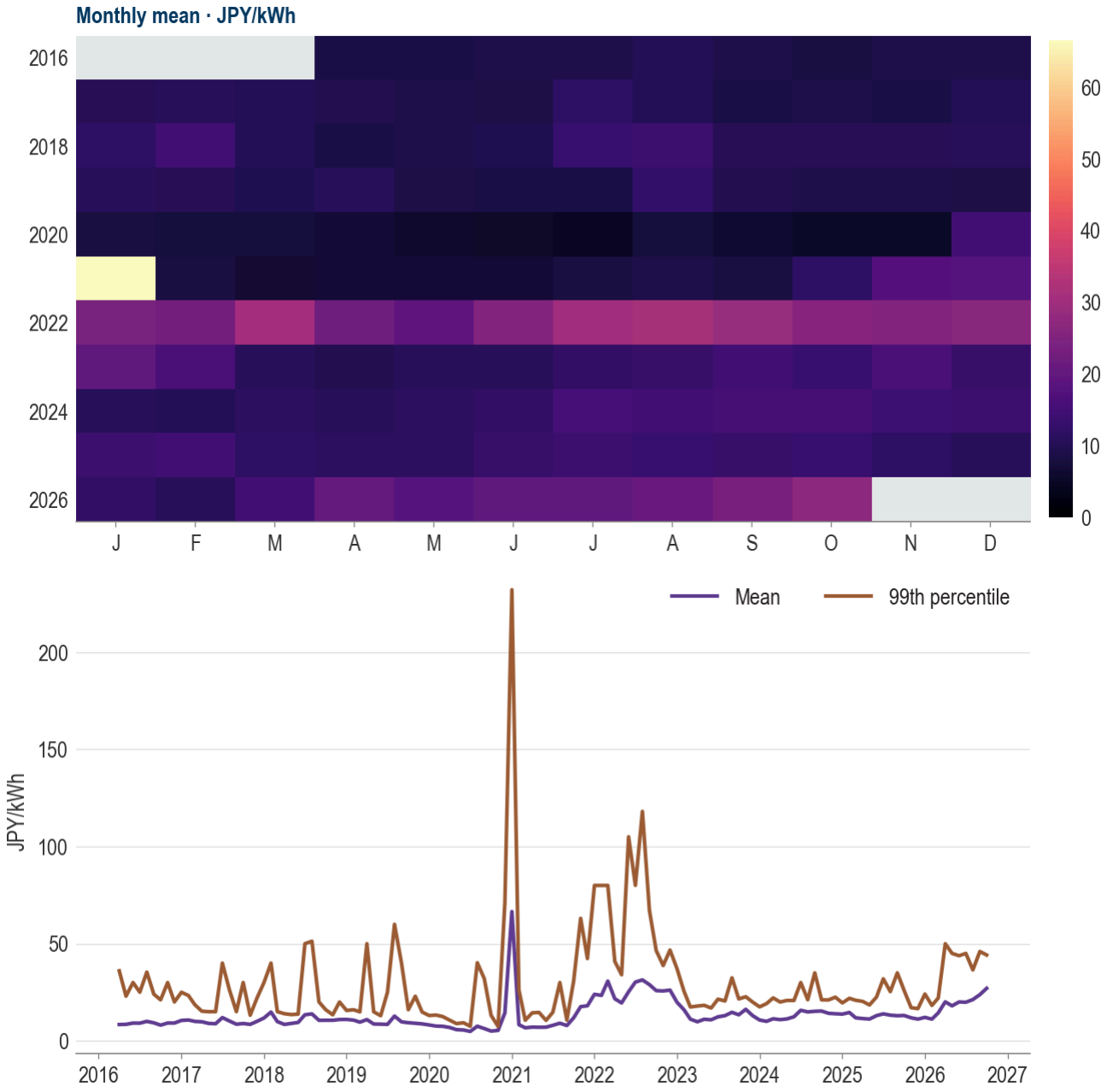
Tokyo: the long price history

January 2021 was the highest observed monthly mean in the April 2016–October 2026 history, at **66.53 JPY/kWh**. The uncapped common colour scale keeps that month visible beside subsequent years. True zero observations remain prices, while absent months are grey and partial-month means carry a marker. [1]

The eastern January 2021 shock combined prolonged fuel/energy stress with high winter consumption. A single monthly mean still understates the interval exposure: Tokyo reached 252.00 JPY/kWh on 15 January at 16:30 JST. The event was not merely one afternoon with insufficient turbine capacity; the case study follows fuel endurance and replenishment across the month. [1]

The monthly upper-tail trace distinguishes a costly month from scattered expensive intervals. Neither is a realised procurement cost without the buyer's actual timing and volume. The January 2021 case sets out the documented sequence and the competing physical explanations.

Exhibit 88: Tokyo: monthly price history and tails



Note: Common uncapped nine-area monthly colour scale; turquoise dots = partial months. Line panel retains monthly upper tails.

Source: [1] · Chart data

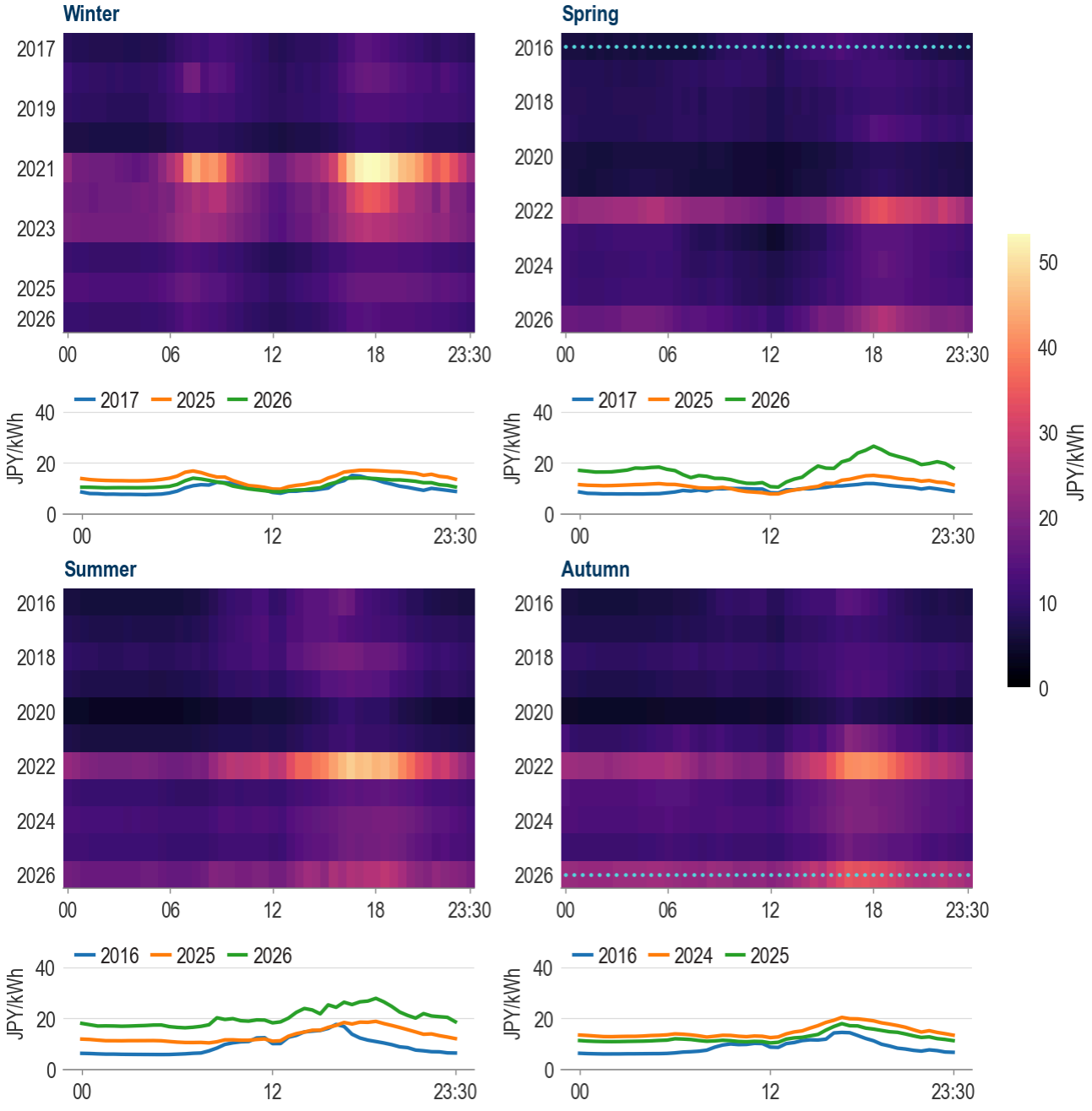
Tokyo: four-season price shape

Tokyo's evening–midday difference was 10.97 JPY/kWh in spring 2026 versus 4.84 in summer. Spring's solar-hour discount was larger; summer midday energy cost 20.39. A smaller daily swing did not mean cheaper energy. [1]

Season-year	Mean	Median	p90	Midday / evening
Winter 2026	11.48	10.90	15.08	9.47 / 13.74
Spring 2026	17.46	17.59	26.22	12.80 / 23.77
Summer 2026	20.38	19.18	29.78	20.39 / 25.23
Autumn 2025	12.61	11.51	16.47	11.54 / 15.65

JPY/kWh; midday 10:00–14:59, evening 17:00–20:59 JST starts; complete native price seasons. [1]

Exhibit 89: Tokyo: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

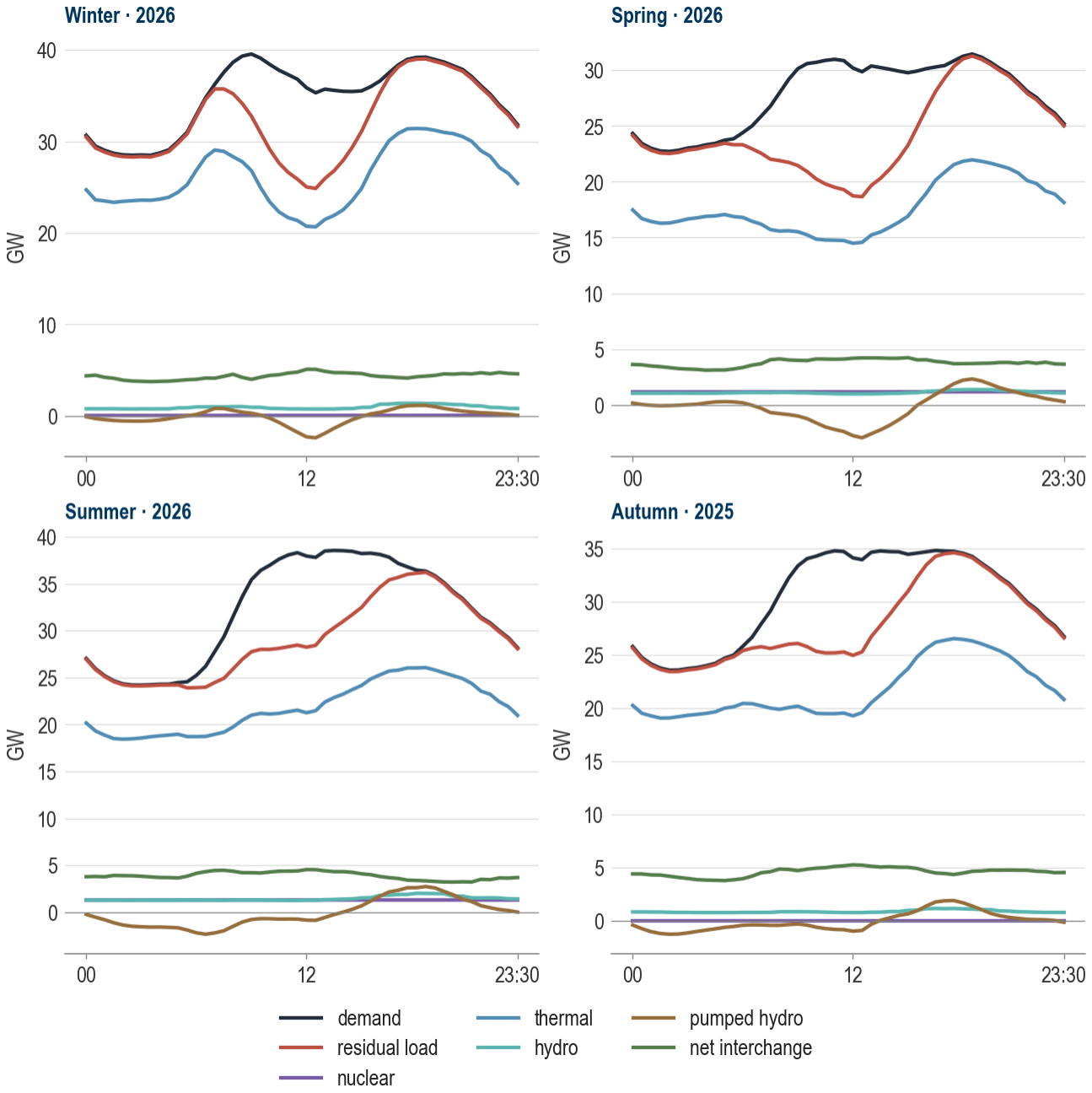
Tokyo: seasonal physical balance

Tokyo's residual demand averaged 32.09 GW in winter 2026 and 29.03 GW in summer, compared with 24.18 GW in spring. Thermal output moved from 26.10 GW in winter to 17.55 GW in spring and 21.79 GW in summer. Reported nuclear output rose from a 0.04 GW winter mean to 1.17 GW in spring and 1.30 GW in summer; these regional totals are not assigned mechanically to a named reactor. [2] [4]

Pumped-storage net output is signed; a small negative season mean can coexist with material daytime pumping and evening generation. The qualified interchange curve is positive for area net receipts, not a bilateral measurement. The demand chapter and supply chapter retain longer histories and separate thermal categories.

The clock profiles identify when energy must be replaced, but do not establish the marginal bidder. Weather, fuel opportunity cost, outages and adjoining areas remain competing contributors to prices.

Exhibit 90: Tokyo: seasonal physical balance



Note: Latest complete season-year per field, dated in extract; partial complete-day samples explicitly labelled. All admitted years remain in the extract. Interchange is qualified area net import, not bilateral flow.

Source: [2] [4] [10] · Chart data

Tokyo: neighbouring routes and basis

Tohoku→Tokyo carried a mean price difference of +4.05 JPY/kWh in summer 2026, with exact coupling in 53.9% of half-hours. Tokyo→Chubu was −0.90 and coupled in 57.9%. Tokyo's northern and western price exposures therefore cannot be compressed into a single import premium. In winter the corresponding mean differences were only +0.32 and −0.48. [1]

The March 2026 central-loop cutover ended updates to individual central-link operating-capacity publications and introduced separate Chubu/Kansai fence objects. Historical asset observations remain historical assets; the new fences are not relabelled line flows. [279]

Schedules retain the source clock and final-publication stage. Direction-specific annual restrictions and newly captured formulated day-ahead limits are separate evidence; an observed price split alone does not establish that any displayed route bound the auction. [280]

Exhibit 91: Tokyo: every connected neighbour

Note: Left: latest observed season-year price profile (partial seasons remain samples). Right: separate source-clock monthly schedules, never joined to prices. Both price orientations and all-year statistics in the boundary atlas.

Source: [1] [11] [279] · Chart data

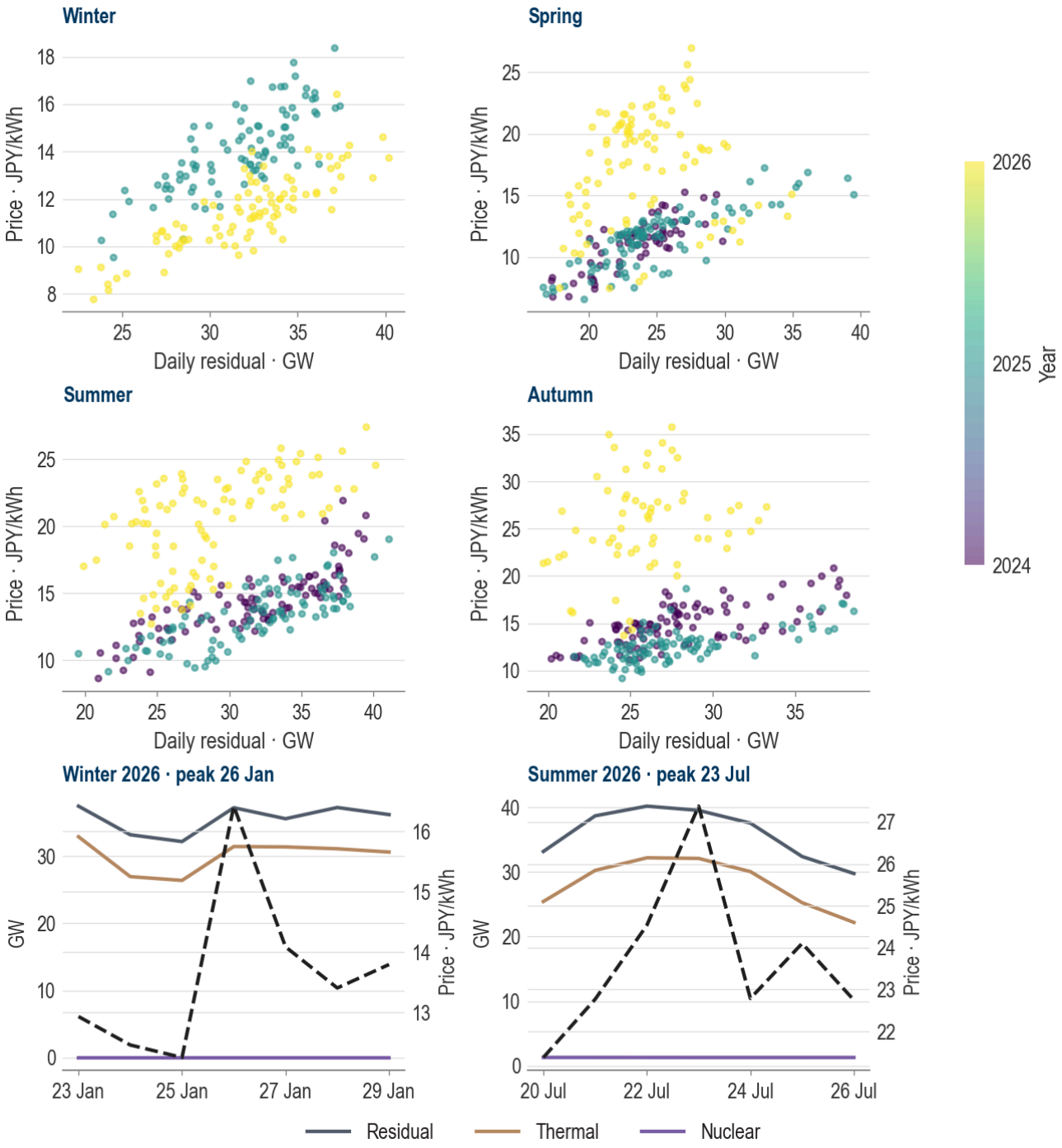
Tokyo: winter versus summer stress

Winter 2026 had higher mean residual demand but cheaper prices than summer: 11.48 versus 20.38 JPY/kWh. Fuel costs, supply availability, intraday scarcity and neighbour balances prevent a single-variable load explanation. [1] [4]

Winter's peak daily price, **26 January**, was 16.42 JPY/kWh at 37.24 GW residual demand, 31.45 GW thermal output and 3.9°C representative-area temperature. Summer's **23 July** peak reached 27.38 at 39.51 GW residual, 32.08 GW thermal and 32.2°C. Nuclear output rose from zero to 1.29 GW, yet summer was dearer: cooling, fuel and neighbouring balances remain competing contributors. [5] [1] [4]

Seven-day windows retain each peak's approach/retreat; restriction evidence does not establish auction binding.

Exhibit 92: Tokyo: seasonal populations and contrasting stress windows



Note: Complete days; common year colours. Bottom: seven days around peak daily price in the latest complete winter/summer. Dashed black = right-axis price. No causal or outage attribution.

Source: [1] [2] [4] [5] [10] · Chart data

Chubu: fleet and load geography

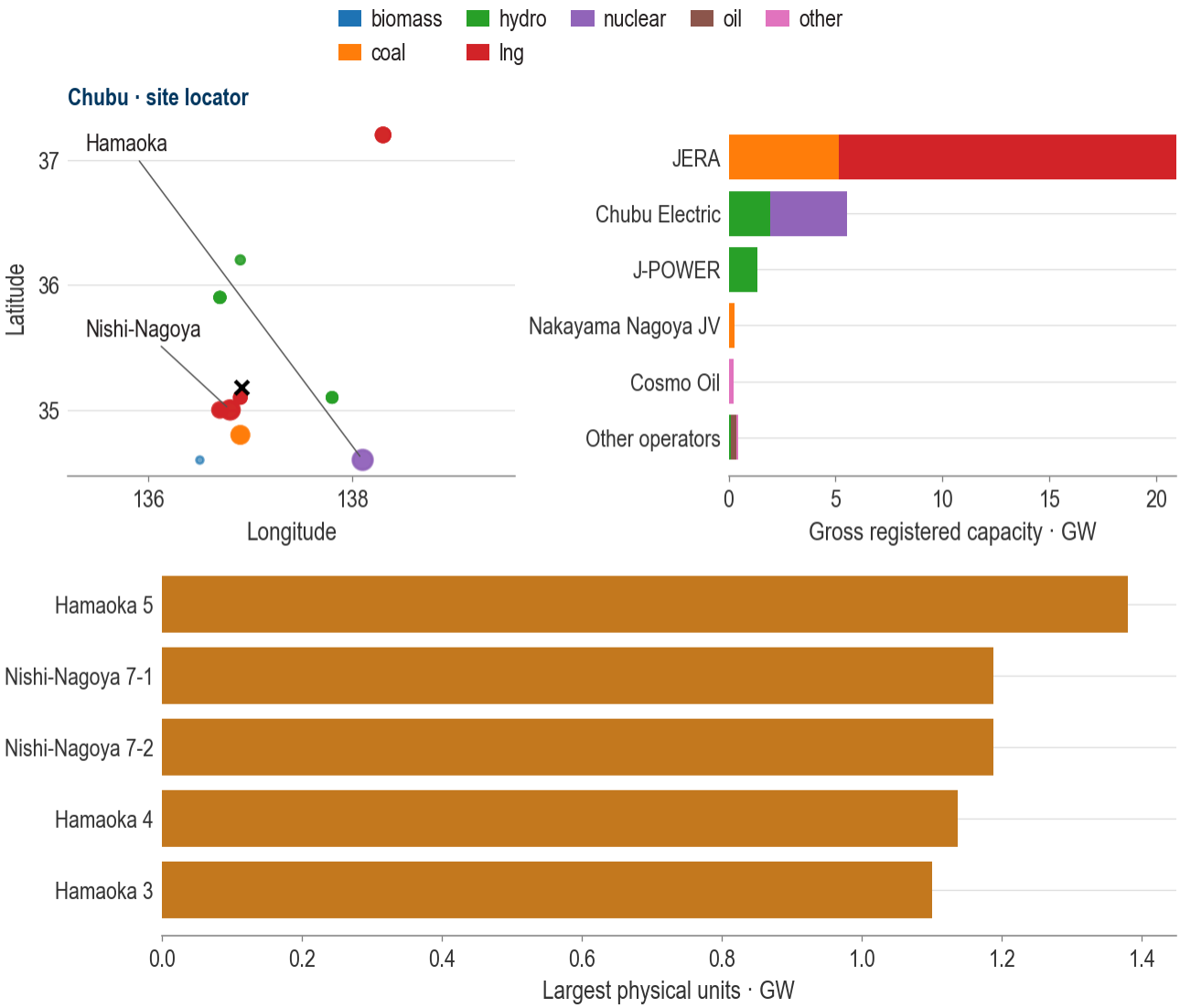
Nagoya–Ise Bay's industrial corridor meets JERA thermal generation, Chubu hydro/pumping and independents. JERA's 20,933 MW and Chubu's direct 5,544 MW are distinct; affiliation does not create another physical fleet. [331]

Holiday shutdowns alter daytime load. Mean demand was 15.79 GW in winter 2026, 13.20 in spring and 15.35 in summer. Nuclear output was zero in all four seasons: installed Hamaoka was not generating. Eastern converters and the western loop provide different exposures; geographic Chubu is not uniformly 60 Hz. [2] [4] [330]

Registered operator	Gross MW
JERA	20,933
Chubu Electric Power	5,544
J-POWER	1,340
Nakayama Nagoya Joint Power	259
All other registered operators	686

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 93: Chubu: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] [343] [364] · Chart data

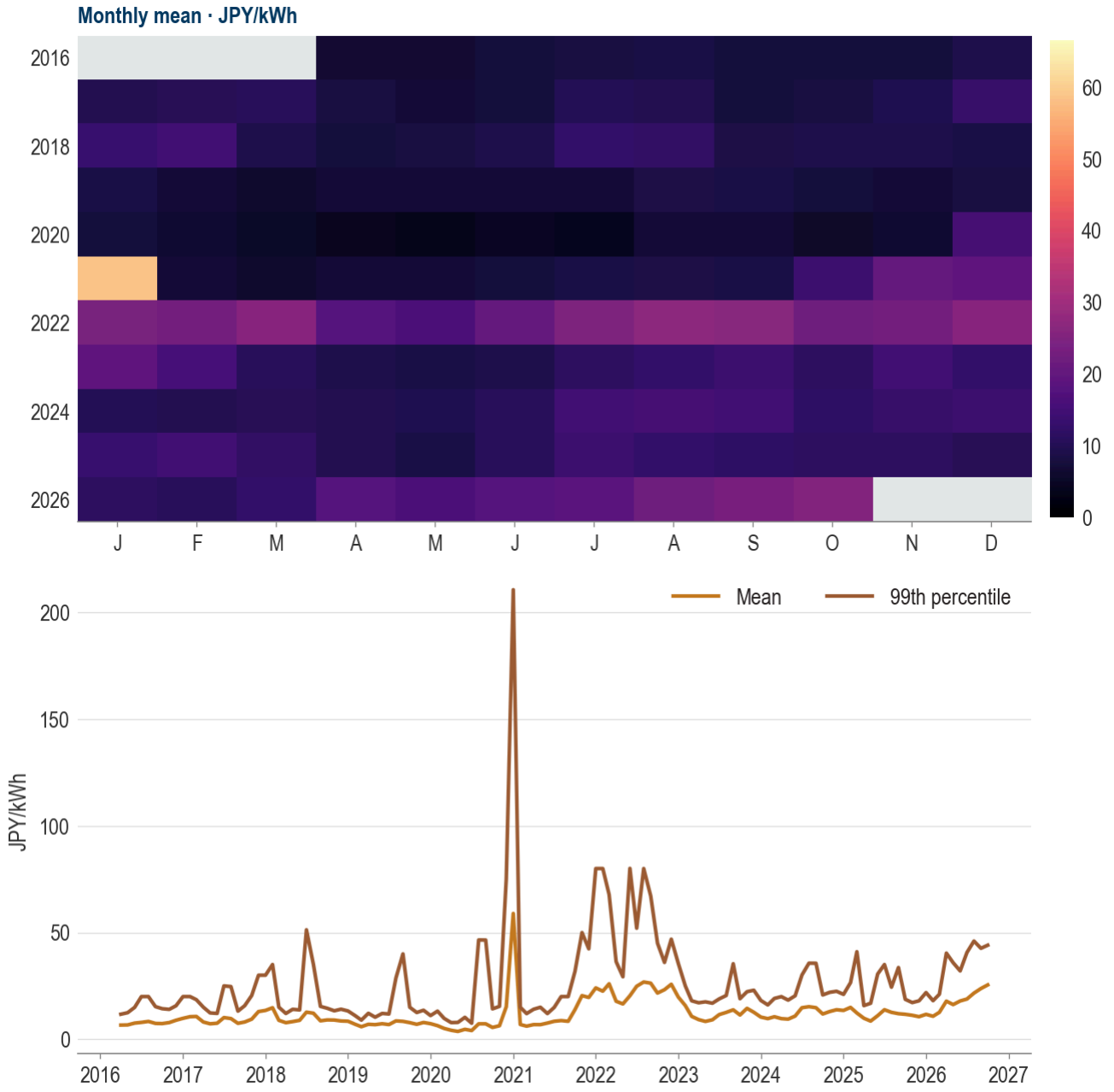
Chubu: the long price history

January 2021 was the highest observed monthly mean in the April 2016–October 2026 history, at **58.95 JPY/kWh**. The uncapped common colour scale keeps that month visible beside subsequent years. True zero observations remain prices, while absent months are grey and partial-month means carry a marker. [1]

Chubu was cheaper than Tokyo through much of the January 2021 event, but its own monthly mean still reached an exceptional level. Converter access did not transform the two areas into one unlimited pool. The all-year heatmap also separates the energy crisis from later fuel-cost and local-balance episodes, rather than treating every high-price year as the same mechanism. [1]

The monthly upper-tail trace distinguishes a costly month from scattered expensive intervals. Neither is a realised procurement cost without the buyer's actual timing and volume. The January 2021 case sets out the documented sequence and the competing physical explanations.

Exhibit 94: Chubu: monthly price history and tails



Note: Common uncapped nine-area monthly colour scale; turquoise dots = partial months. Line panel retains monthly upper tails.

Source: [1] · Chart data

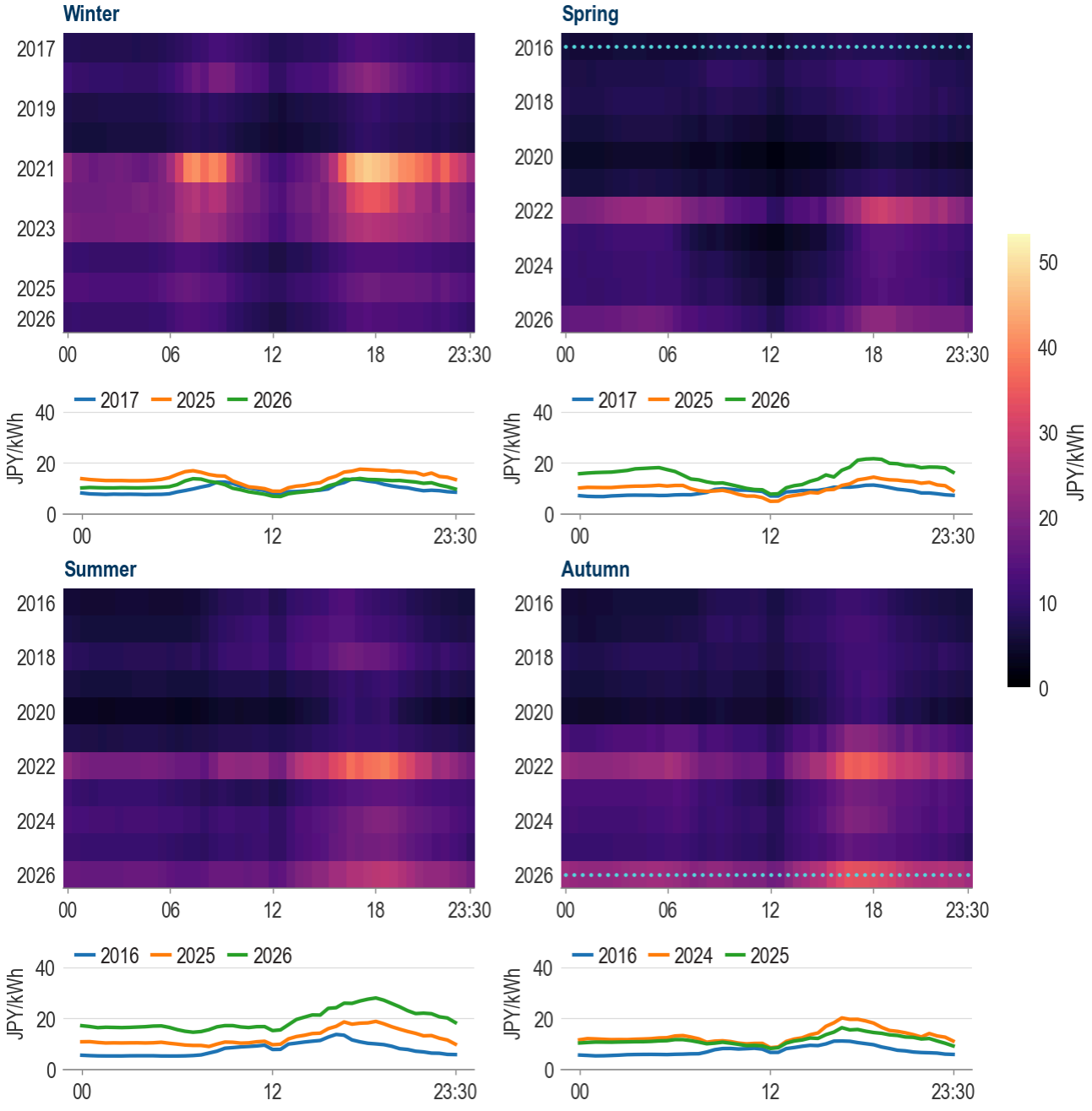
Chubu: four-season price shape

Chubu's summer 2026 evening mean, 26.15 JPY/kWh, exceeded Tokyo's 25.23 despite cheaper all-day prices: evening basis reversed the average exposure. Spring's evening–midday difference was 10.18, versus summer's 8.46. [1]

Season-year	Mean	Median	p90	Midday / evening
Winter 2026	11.00	10.70	15.11	8.25 / 13.39
Spring 2026	15.54	16.27	25.00	10.18 / 20.36
Summer 2026	19.48	18.30	30.00	17.69 / 26.15
Autumn 2025	11.62	11.16	15.68	10.05 / 14.36

JPY/kWh; midday 10:00–14:59, evening 17:00–20:59 JST starts; complete native price seasons. [1]

Exhibit 95: Chubu: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

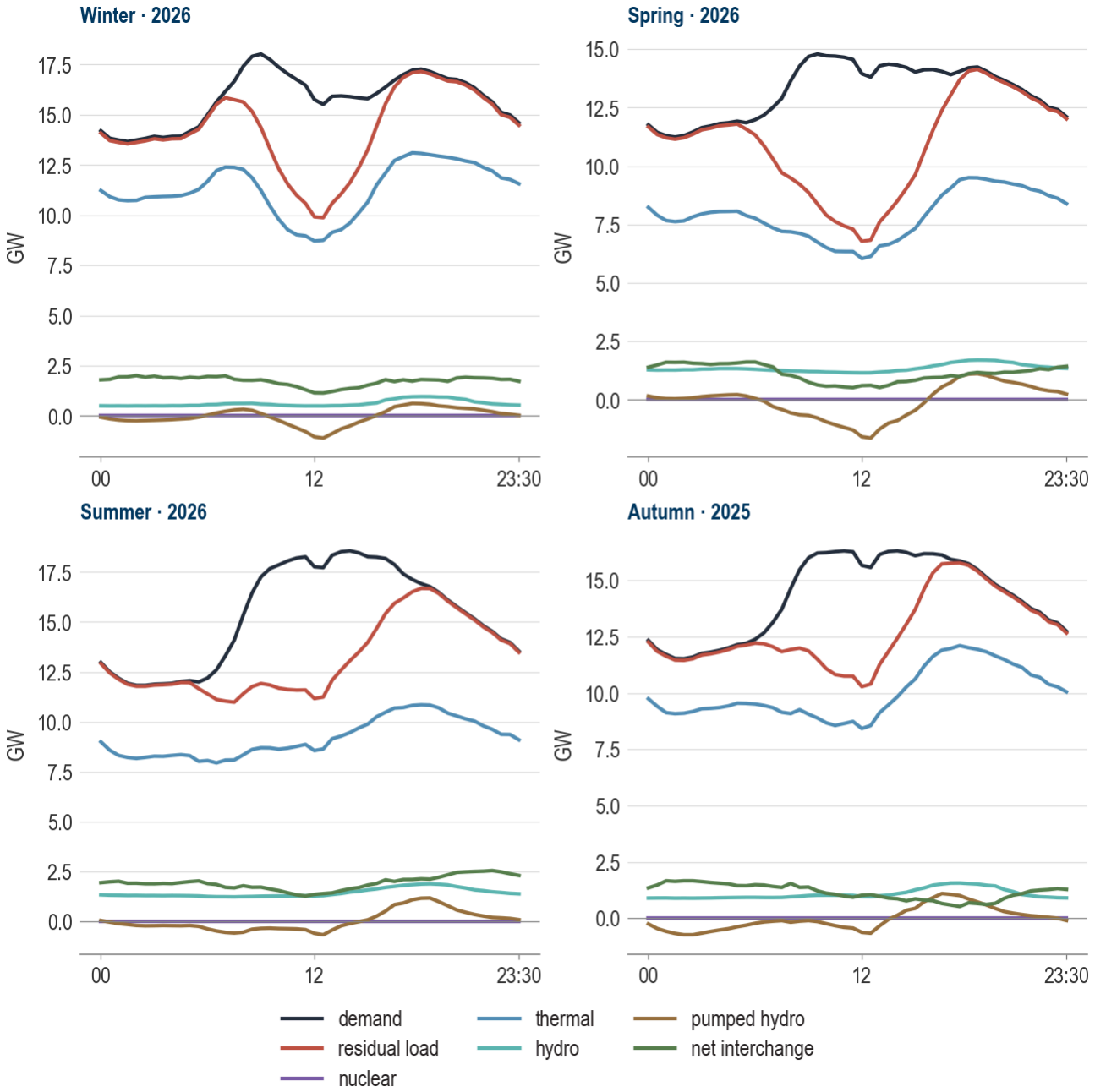
Chubu: seasonal physical balance

Chubu's accepted solar averaged 2.23 GW in spring 2026, reducing mean demand of 13.20 GW to residual load of 10.89 GW after wind. Summer demand rose to 15.35 GW and residual load to 13.15 GW, while solar averaged 2.15 GW. Thermal output was higher in winter (11.28 GW) than summer (9.16 GW), a useful contrast to summer's higher price level. [2] [4]

Pumped-storage net output is signed; a small negative season mean can coexist with material daytime pumping and evening generation. The qualified interchange curve is positive for area net receipts, not a bilateral measurement. The demand chapter and supply chapter retain longer histories and separate thermal categories.

The clock profiles identify when energy must be replaced, but do not establish the marginal bidder. Weather, fuel opportunity cost, outages and adjoining areas remain competing contributors to prices.

Exhibit 96: Chubu: seasonal physical balance



Note: Latest complete season-year per field, dated in extract; partial complete-day samples explicitly labelled. All admitted years remain in the extract. Interchange is qualified area net import, not bilateral flow.

Source: [2] [4] [10] · Chart data

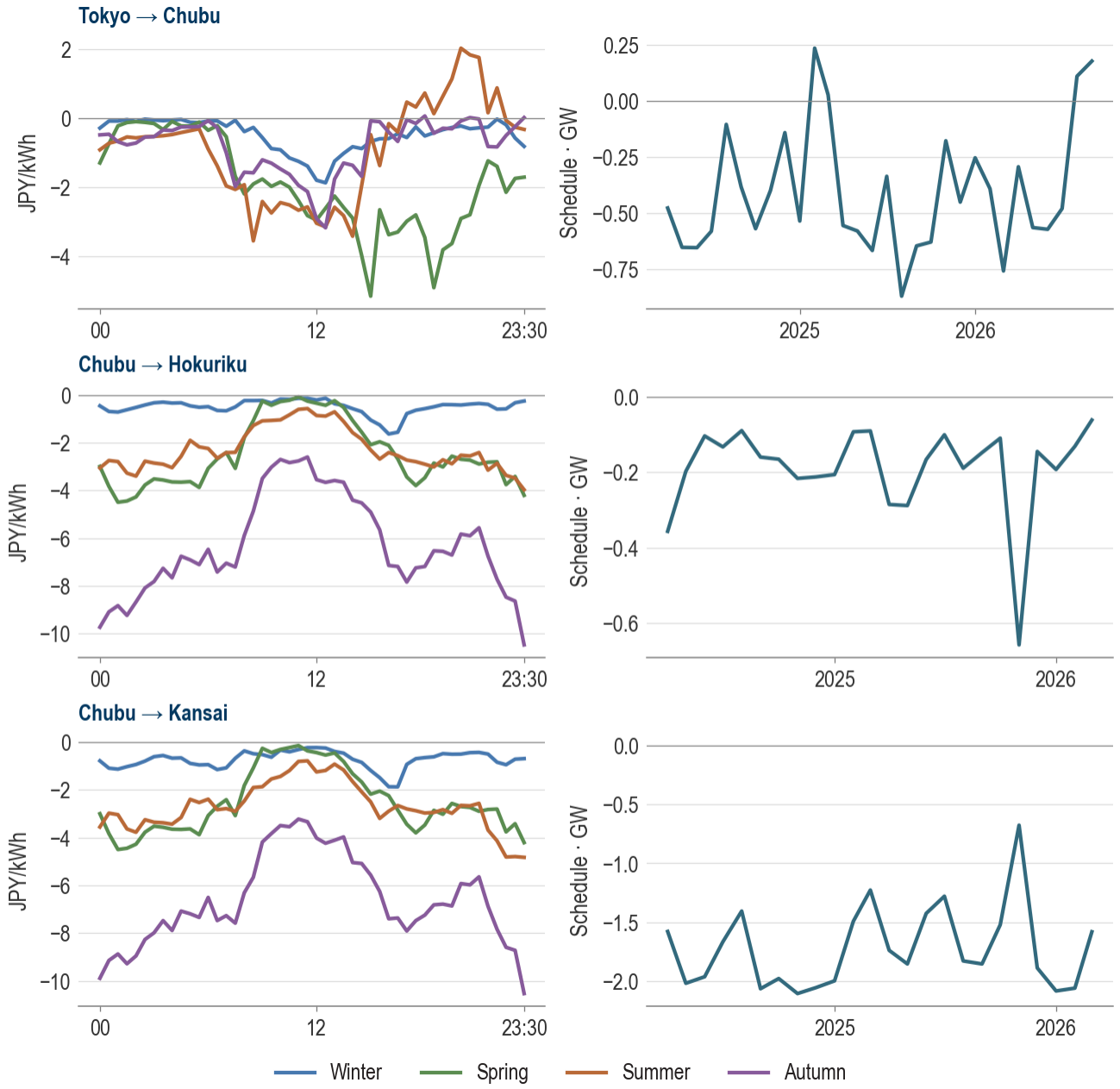
Chubu: neighbouring routes and basis

Chubu→Kansai averaged −2.67 JPY/kWh in summer 2026, with a −1.49 median and only 39.4% exact coupling. Chubu→Hokuriku averaged −2.29. Tokyo→Chubu averaged −0.90, so Chubu was cheaper than the east but dearer than both western neighbours on these seasonal averages. [1]

The March 2026 central-loop cutover ended updates to individual central-link operating-capacity publications and introduced separate Chubu/Kansai fence objects. Historical asset observations remain historical assets; the new fences are not relabelled line flows. [279]

Schedules retain the source clock and final-publication stage. Direction-specific annual restrictions and newly captured formulated day-ahead limits are separate evidence; an observed price split alone does not establish that any displayed route bound the auction. [280]

Exhibit 97: Chubu: every connected neighbour



Note: Left: latest observed season-year price profile (partial seasons remain samples). Right: separate source-clock monthly schedules, never joined to prices. Both price orientations and all-year statistics in the boundary atlas.

Source: [1] [11] [279] · Chart data

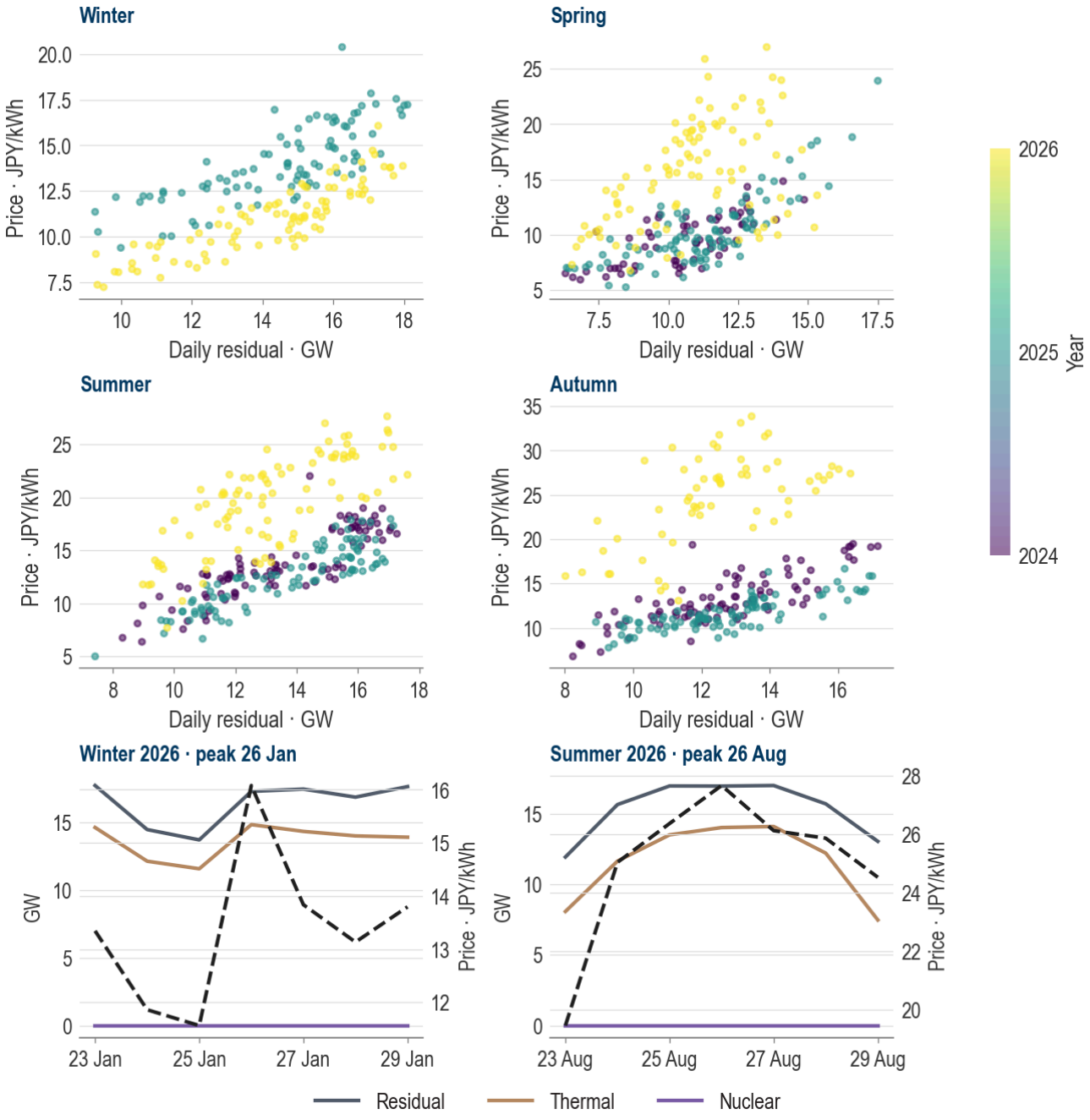
Chubu: winter versus summer stress

Summer 2026 was costlier despite lower mean demand and thermal output. Chubu's two-sided basis and industrial calendar matter; the 23 September 2024 holiday case retains both price legs without inventing a converter outage. [1] [4]

Winter's peak daily price, **16.08 JPY/kWh on 26 January**, accompanied 17.27 GW residual demand and 14.82 GW thermal output. Summer's **26 August** peak reached 27.66 despite lower residual/thermal output, 16.95/14.01 GW. Representative temperature rose from 1.6°C to 32.3°C; nuclear stayed zero. Fuel costs and interregional balance—not residual load alone—remain explanations. [5] [1] [4]

Seven-day contexts preserve the surrounding movement; restrictions and the holiday case supply separate network/calendar evidence.

Exhibit 98: Chubu: seasonal populations and contrasting stress windows



Note: Complete days; common year colours. Bottom: seven days around peak daily price in the latest complete winter/summer. Dashed black = right-axis price. No causal or outage attribution.

Source: [1] [2] [4] [5] [10] · Chart data

Kansai: fleet and load geography

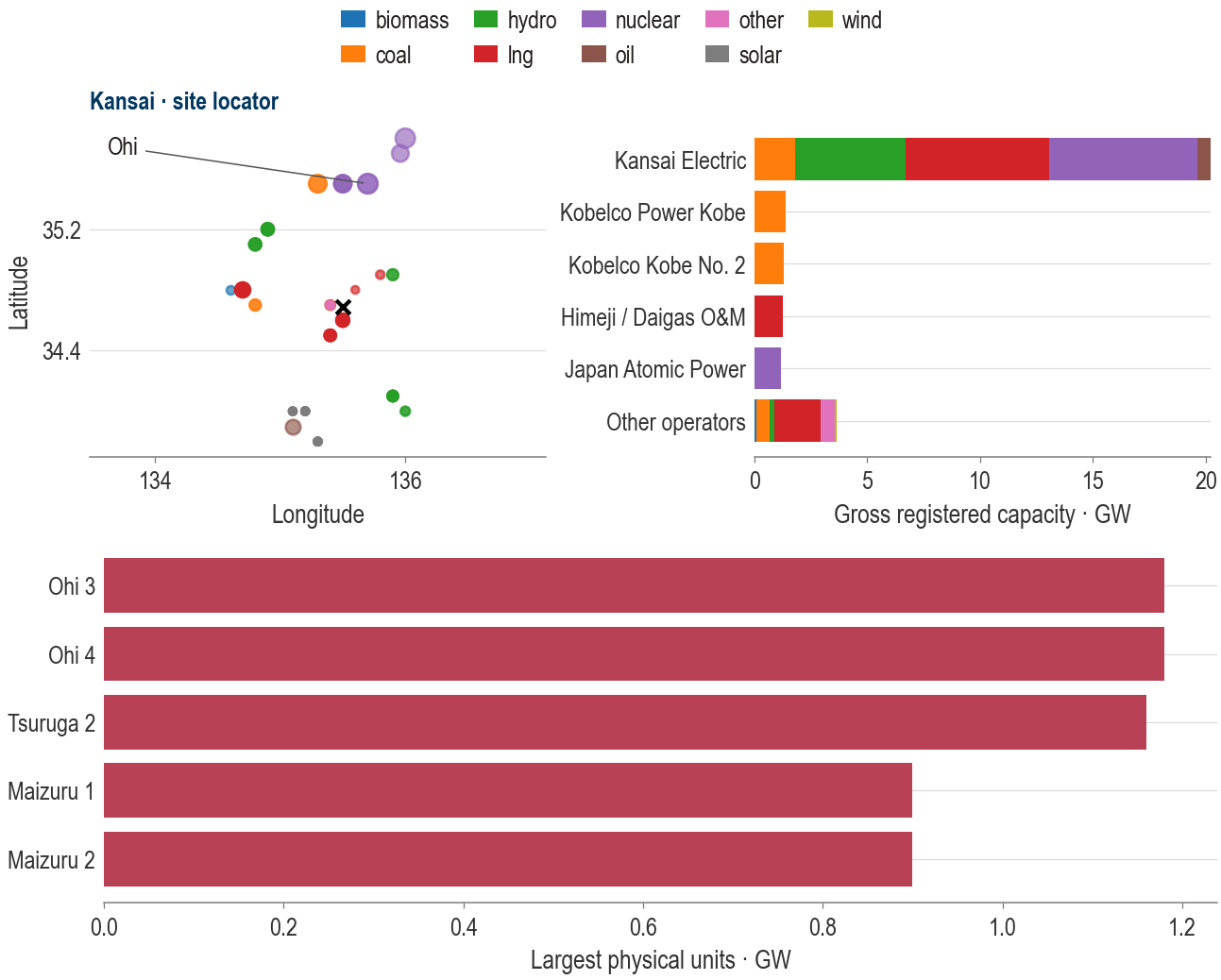
Osaka–Kobe–Kyoto load combines nuclear, coastal thermal and mountain hydro/pumping. Kansai Electric Power's regional 20,223 MW includes Mihama 3's 826 MW; its 20,605 MW company footprint also includes 116 MW in Chubu and 266 MW in Hokuriku. Kobe independents and Daigas-associated facilities are separate. [331]

Regional nuclear output averaged 6.24 GW in winter 2026, 3.93 in spring and 4.30 in summer—not reactor-level availability estimates. Replacement energy comes from thermal, pumping or neighbours: Chubu, Hokuriku, Chugoku and Shikoku. Shikoku's route is Anan–Kihoku, not Honshi; area net interchange cannot allocate individual-route support. [4] [329] [330]

Registered operator	Gross MW
Kansai Electric Power	20,223
Kobelco Power Kobe	1,400
Kobelco Power Kobe No. 2	1,300
Himeji Natural Gas Power; Daigas G&P Solution (O&M)	1,245
All other registered operators	4,791

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 99: Kansai: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [333] [334] [341] [363–368] · Chart data

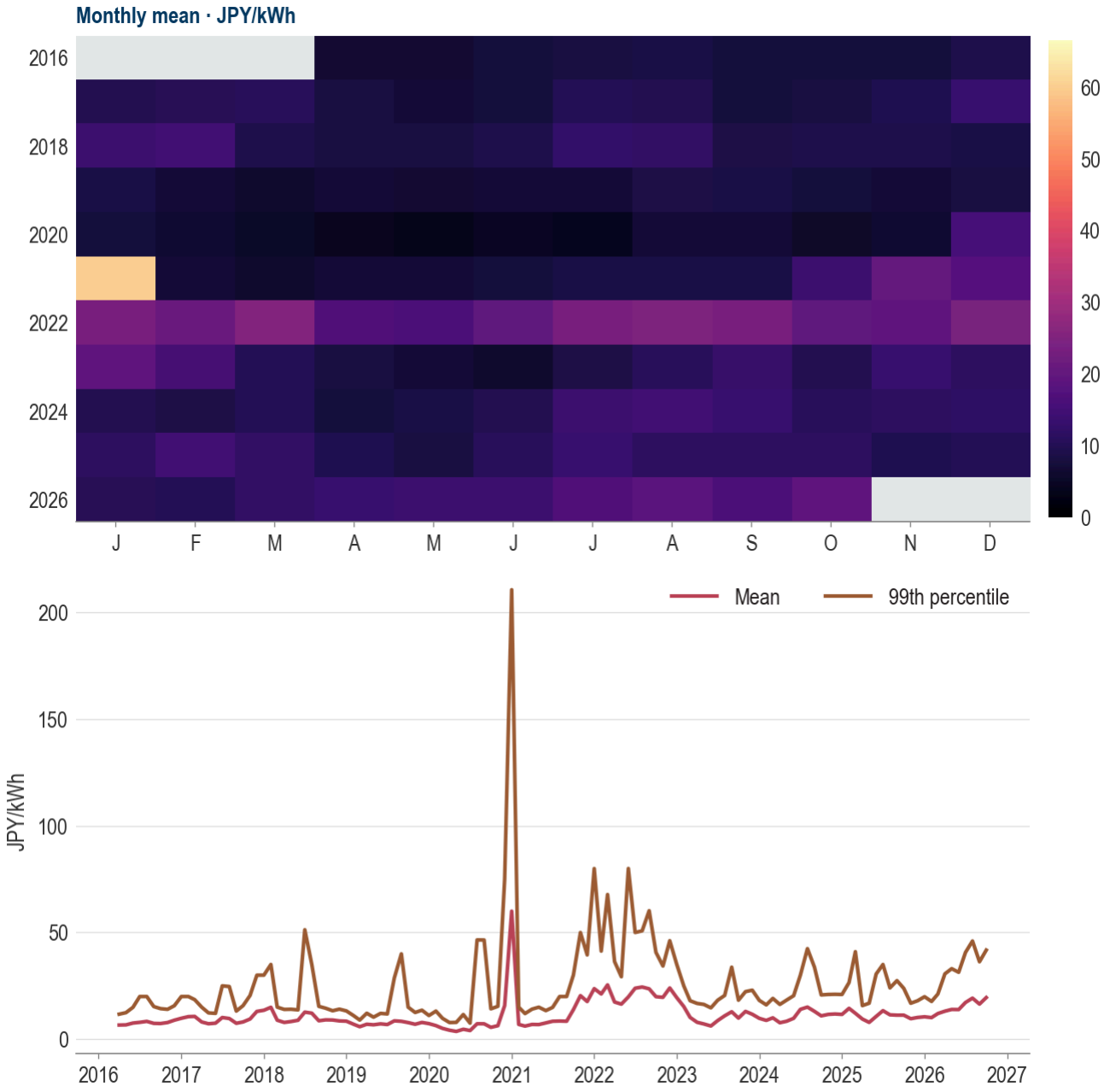
Kansai: the long price history

January 2021 was the highest observed monthly mean in the April 2016–October 2026 history, at **60.00 JPY/kWh**. The uncapped common colour scale keeps that month visible beside subsequent years. True zero observations remain prices, while absent months are grey and partial-month means carry a marker. [1]

Kansai shared the January 2021 western price shock despite its different fleet. Nuclear capability and pumping do not make a region immune to a sustained national energy shortage: inspection timing, stored energy, thermal replenishment and competing regional demand still matter. The interval maximum was 242.21 JPY/kWh on 15 January 2021 at 17:00 JST. [1]

The monthly upper-tail trace distinguishes a costly month from scattered expensive intervals. Neither is a realised procurement cost without the buyer's actual timing and volume. The January 2021 case sets out the documented sequence and the competing physical explanations.

Exhibit 100: Kansai: monthly price history and tails



Note: Common uncapped nine-area monthly colour scale; turquoise dots = partial months. Line panel retains monthly upper tails.

Source: [1] · Chart data

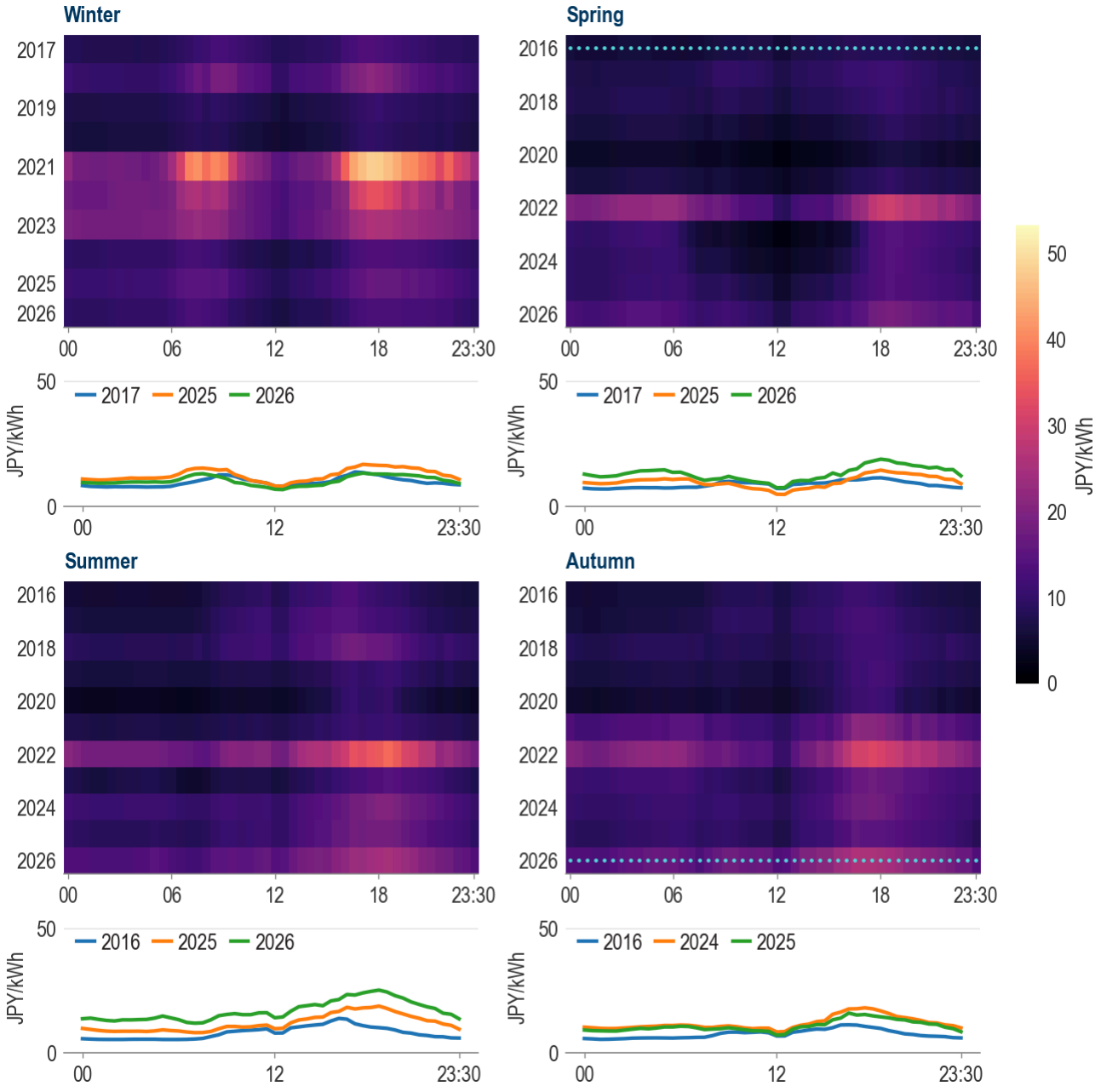
Kansai: four-season price shape

Kansai's spring 2026 midday/evening means were 9.55/17.29 JPY/kWh; summer raised them to 16.44/23.31. Nuclear substitutes for other supply, but does not remove cooling or solar-withdrawal exposure. Inspection cycles and neighbour balances vary too. [1]

Season-year	Mean	Median	p90	Midday / evening
Winter 2026	10.26	9.83	14.60	7.83 / 12.79
Spring 2026	13.02	13.65	19.77	9.55 / 17.29
Summer 2026	16.80	15.00	29.00	16.44 / 23.31
Autumn 2025	10.71	10.21	15.40	9.16 / 14.08

JPY/kWh; midday 10:00–14:59, evening 17:00–20:59 JST starts; complete native price seasons. [1]

Exhibit 101: Kansai: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

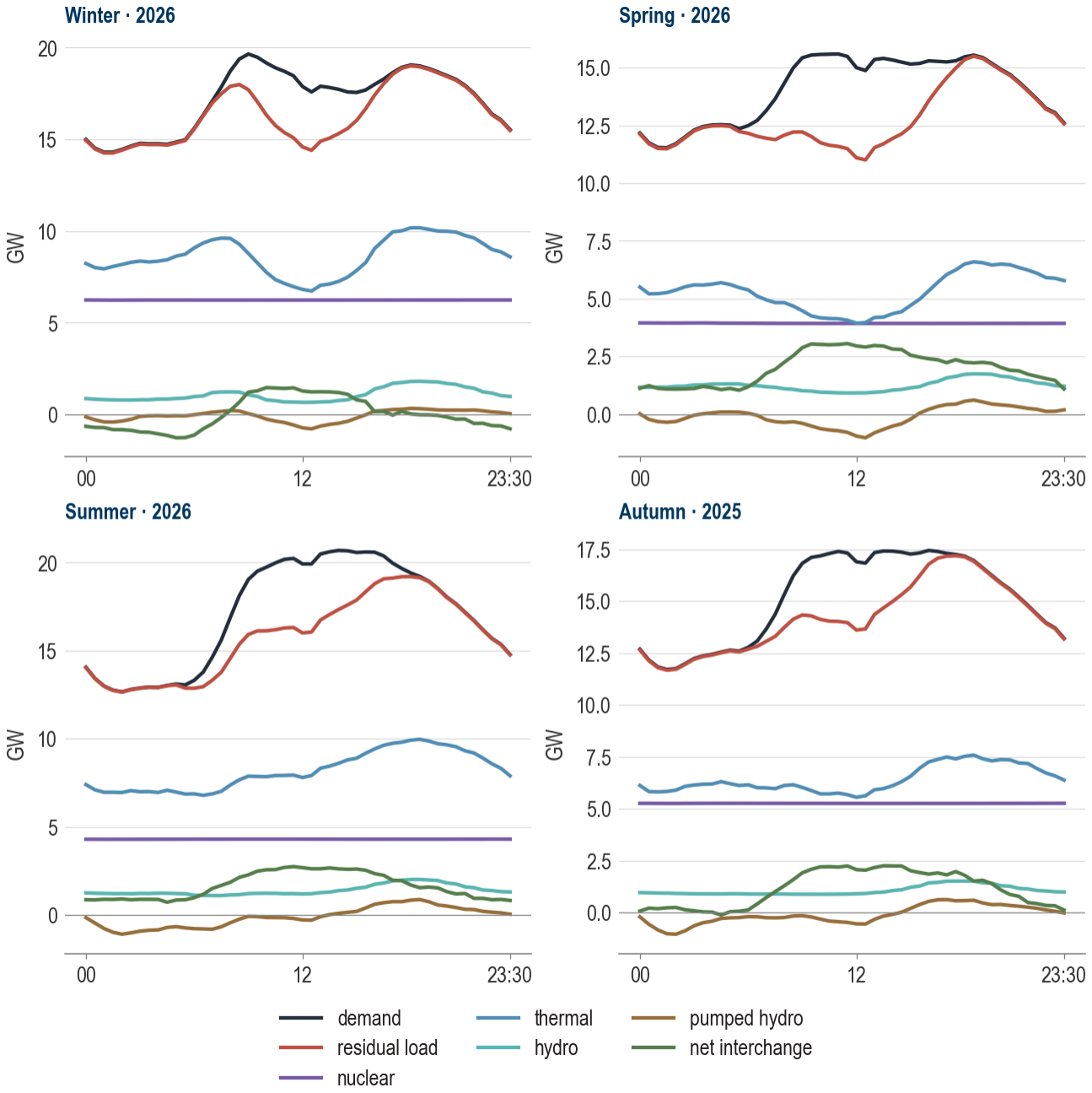
Kansai: seasonal physical balance

Kansai's winter and summer demand averages were almost identical: 17.16 and 17.06 GW. Nuclear output, however, differed by 1.94 GW (6.24 versus 4.30 GW), while accepted solar was higher in summer. Thus the similar headline loads conceal different thermal, storage and neighbour requirements. Spring demand was 14.06 GW and nuclear output 3.93 GW. [2] [4]

Pumped-storage net output is signed; a small negative season mean can coexist with material daytime pumping and evening generation. The qualified interchange curve is positive for area net receipts, not a bilateral measurement. The demand chapter and supply chapter retain longer histories and separate thermal categories.

The clock profiles identify when energy must be replaced, but do not establish the marginal bidder. Weather, fuel opportunity cost, outages and adjoining areas remain competing contributors to prices.

Exhibit 102: Kansai: seasonal physical balance



Note: Latest complete season-year per field, dated in extract; partial complete-day samples explicitly labelled. All admitted years remain in the extract. Interchange is qualified area net import, not bilateral flow.

Source: [2] [4] [10] · Chart data

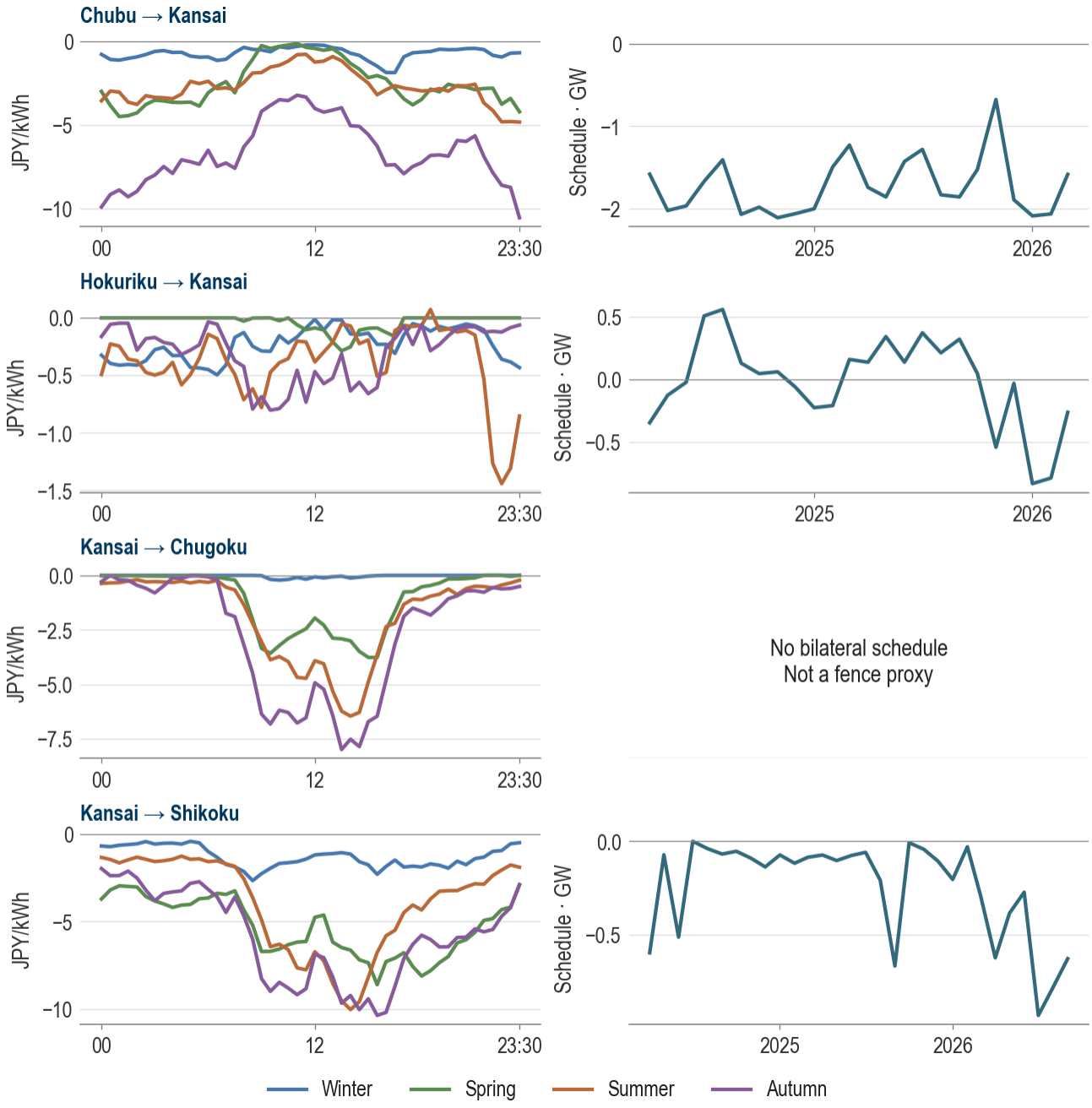
Kansai: neighbouring routes and basis

Hokuriku→Kansai was almost perfectly price-coupled in spring 2026 (99.3%), yet Kansai→Shikoku averaged −5.31 JPY/kWh with only 31.2% coupling. Physical proximity and a shared western frequency do not guarantee equivalent price exposure. Kansai→Chugoku was much closer, averaging −1.11 with 85.1% coupling. [1]

The March 2026 central-loop cutover ended updates to individual central-link operating-capacity publications and introduced separate Chubu/Kansai fence objects. Historical asset observations remain historical assets; the new fences are not relabelled line flows. [279]

Schedules retain the source clock and final-publication stage. Direction-specific annual restrictions and newly captured formulated day-ahead limits are separate evidence; an observed price split alone does not establish that any displayed route bound the auction. [280]

Exhibit 103: Kansai: every connected neighbour



Note: Left: latest observed season-year price profile (partial seasons remain samples). Right: separate source-clock monthly schedules, never joined to prices. Both price orientations and all-year statistics in the boundary atlas.

Source: [1] [11] [279] · Chart data

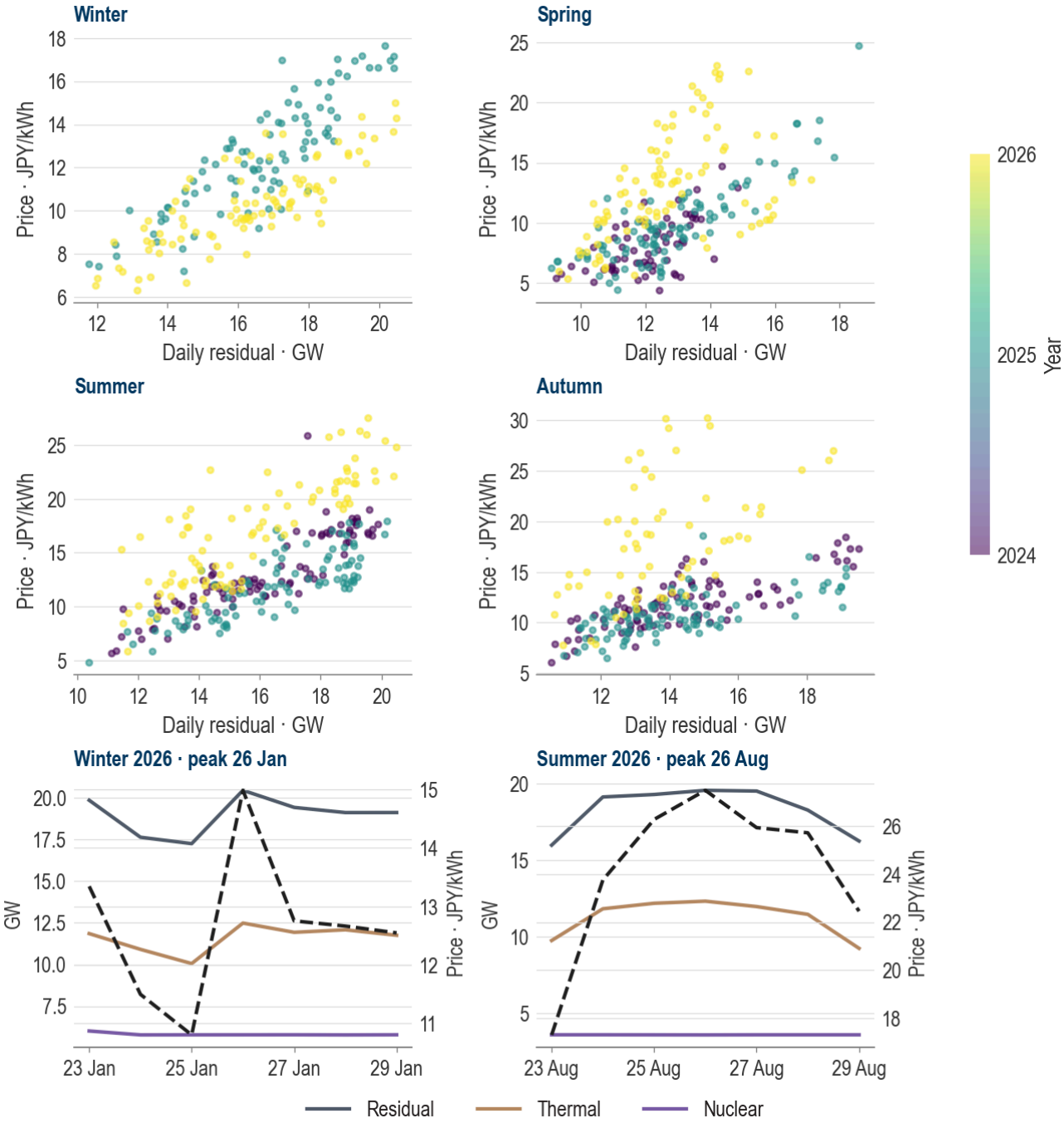
Kansai: winter versus summer stress

Winter 2026 combined more nuclear output with lower prices than summer. Fuel, cooling and neighbours prevent an isolated reactor-price attribution. The August heat and September Mihama cases retain actual operating chronologies, not nameplate steps. [1] [4]

Winter's **26 January** peak daily price was 14.98 JPY/kWh at 20.46 GW residual demand and 5.78 GW nuclear output. Summer's **26 August** peak reached 27.48 at lower residual demand, 19.57 GW, but only 3.60 GW nuclear. Thermal output was similar, 12.49/12.33 GW; representative temperatures were 3.0/31.5°C. Nuclear substitution matters, but fuel costs, heat and neighbours also changed. [5] [1] [4]

Physical-stage restrictions and the dated event cases qualify—not prove—the causal interpretation.

Exhibit 104: Kansai: seasonal populations and contrasting stress windows



Note: Complete days; common year colours. Bottom: seven days around peak daily price in the latest complete winter/summer. Dashed black = right-axis price. No causal or outage attribution.

Source: [1] [2] [4] [5] [10] · Chart data

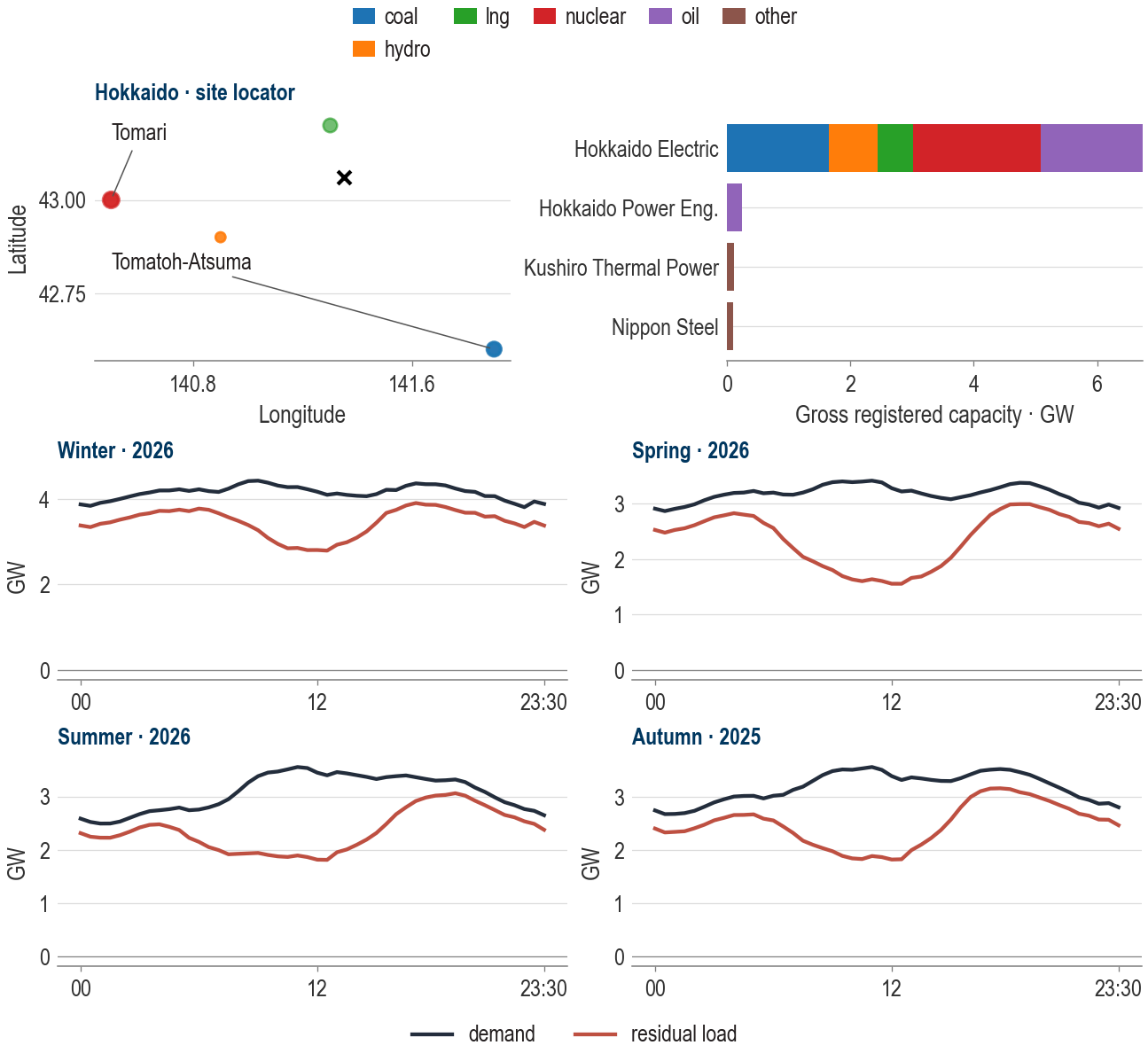
Hokkaido: winter and the mainland link

Hokkaido's winter 2026 demand averaged 4.15 GW against 3.07 GW in summer. Winter wind output averaged 0.47 GW, almost twice summer's 0.25 GW, while nuclear output remained reported zero. The region's cold-weather exposure therefore meets a thermal fleet and one aggregated HVDC mainland boundary, not the multiple import routes available to Kansai. [2] [4] [330]

Registered operator	Gross MW
Hokkaido Electric Power	6,739
Hokkaido Power Engineering	250
Kushiro Thermal Power	112
Nippon Steel	100
All other registered operators	0

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 105: Hokkaido: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] · Chart data

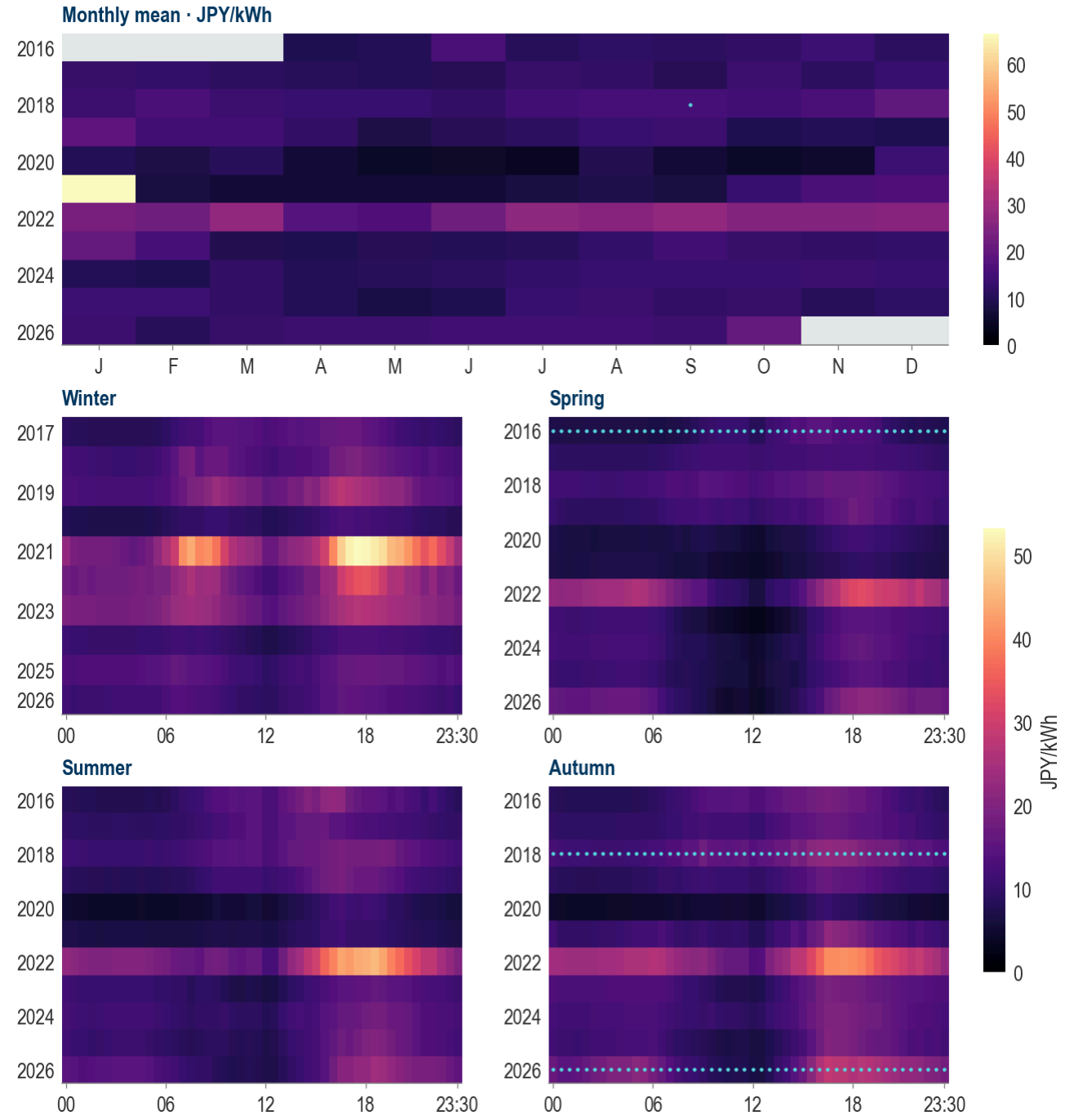
Hokkaido: price history and basis

Summer 2026 mean price was 14.38 JPY/kWh, above winter's 12.25 despite the much lighter load. Spring midday/evening prices were 5.77/20.24. Wind and solar timing, local thermal requirements and the northern gateway matter more than a blanket winter-premium assumption. [1]

Fixed origin → destination	Winter	Spring	Summer	Autumn
Hokkaido → Tohoku	-1.10	+0.15	+1.94	-0.67

Mean destination-minus-origin JPY/kWh; complete winter/spring/summer 2026 and autumn 2025. Both directions/distributions: boundary atlas. [1]

Exhibit 106: Hokkaido: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

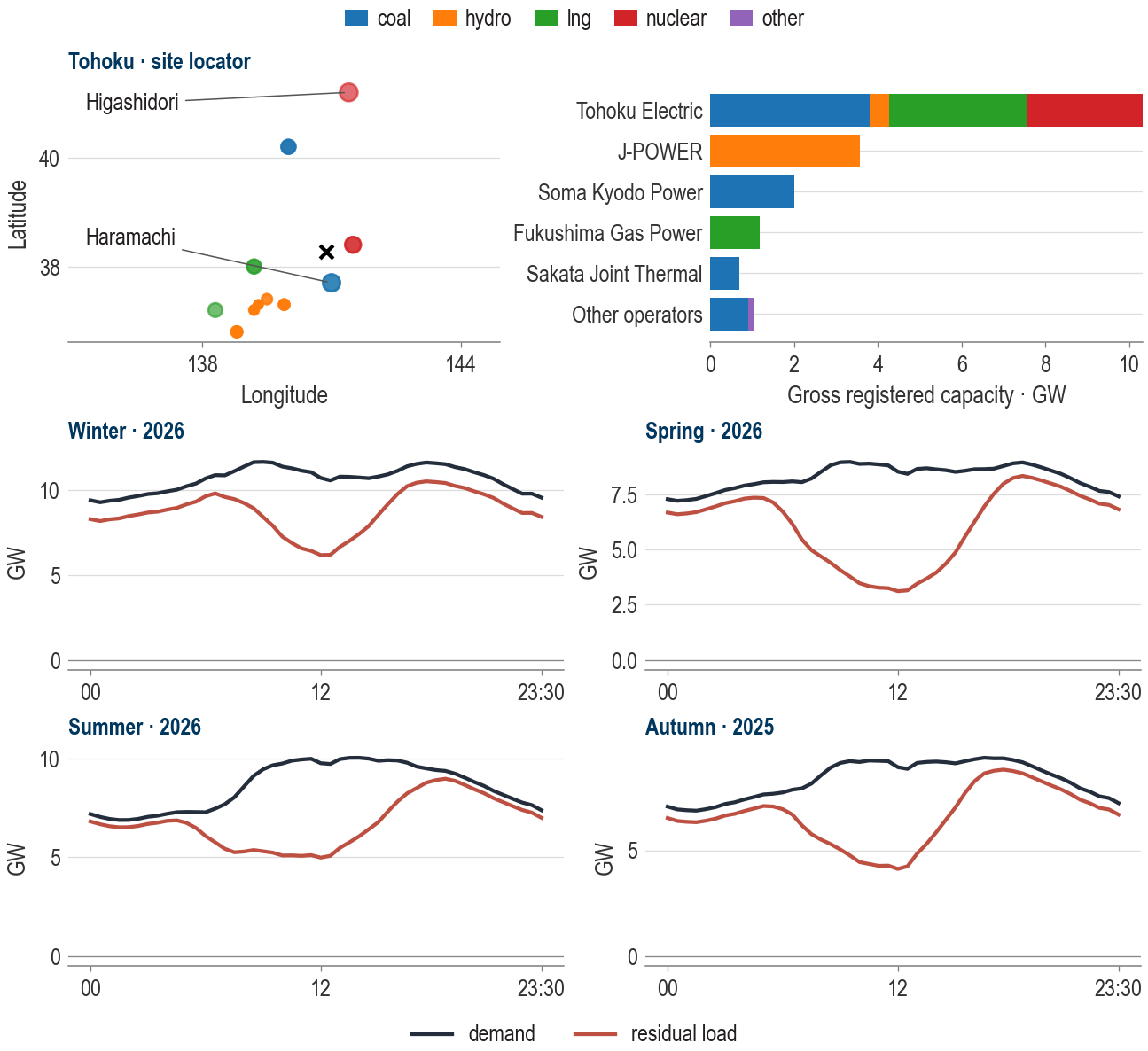
Tohoku: between two markets

Tohoku's winter 2026 mean thermal output was 10.04 GW against demand of 10.62 GW, alongside hydro, wind and other output. Spring thermal output fell to 6.56 GW; accepted solar rose from 0.84 GW in winter to 1.65 GW in spring. The area has northern HVDC connectivity and a southern AC interface to Tokyo, so whole-area interchange cannot identify which neighbour received the energy. [2] [4] [330]

Registered operator	Gross MW
Tohoku Electric Power	10,323
J-POWER	3,566
Soma Kyodo Power	2,000
Fukushima Gas Power	1,180
All other registered operators	1,720

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 107: Tohoku: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] · Chart data

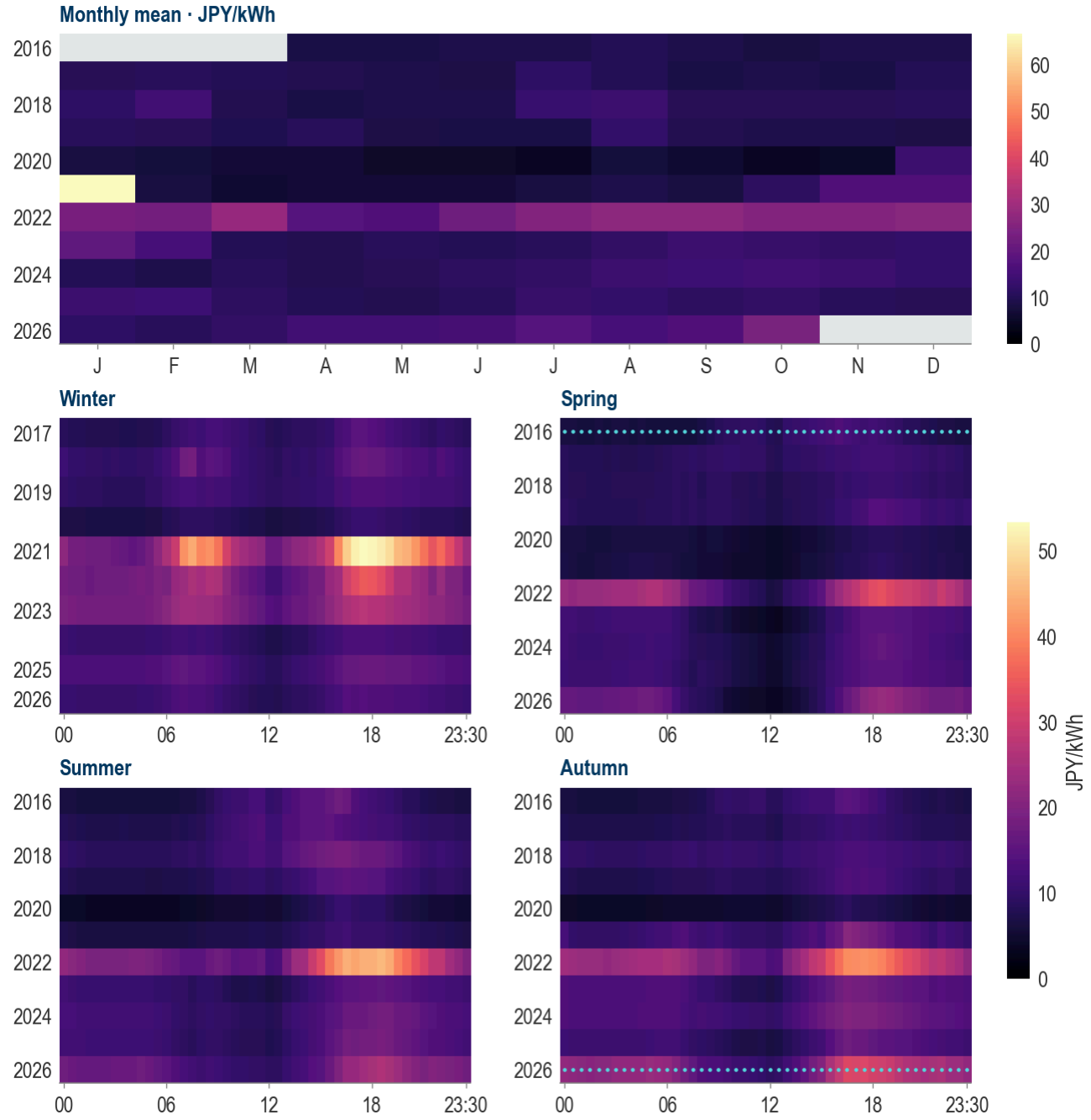
Tohoku: price history and basis

Tohoku's summer 2026 mean price was 16.33 JPY/kWh versus Tokyo's 20.38. Spring midday/evening means were 5.14/21.19: a deep solar-hour discount coexisted with expensive evenings. Nuclear output changed across the seasons, but regional aggregates alone do not identify a reactor-level price cause. [1]

Fixed origin → destination	Winter	Spring	Summer	Autumn
Hokkaido → Tohoku	-1.10	+0.15	+1.94	-0.67
Tohoku → Tokyo	+0.32	+3.81	+4.05	+1.20

Mean destination-minus-origin JPY/kWh; complete winter/spring/summer 2026 and autumn 2025. Both directions/distributions: boundary atlas. [1]

Exhibit 108: Tohoku: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

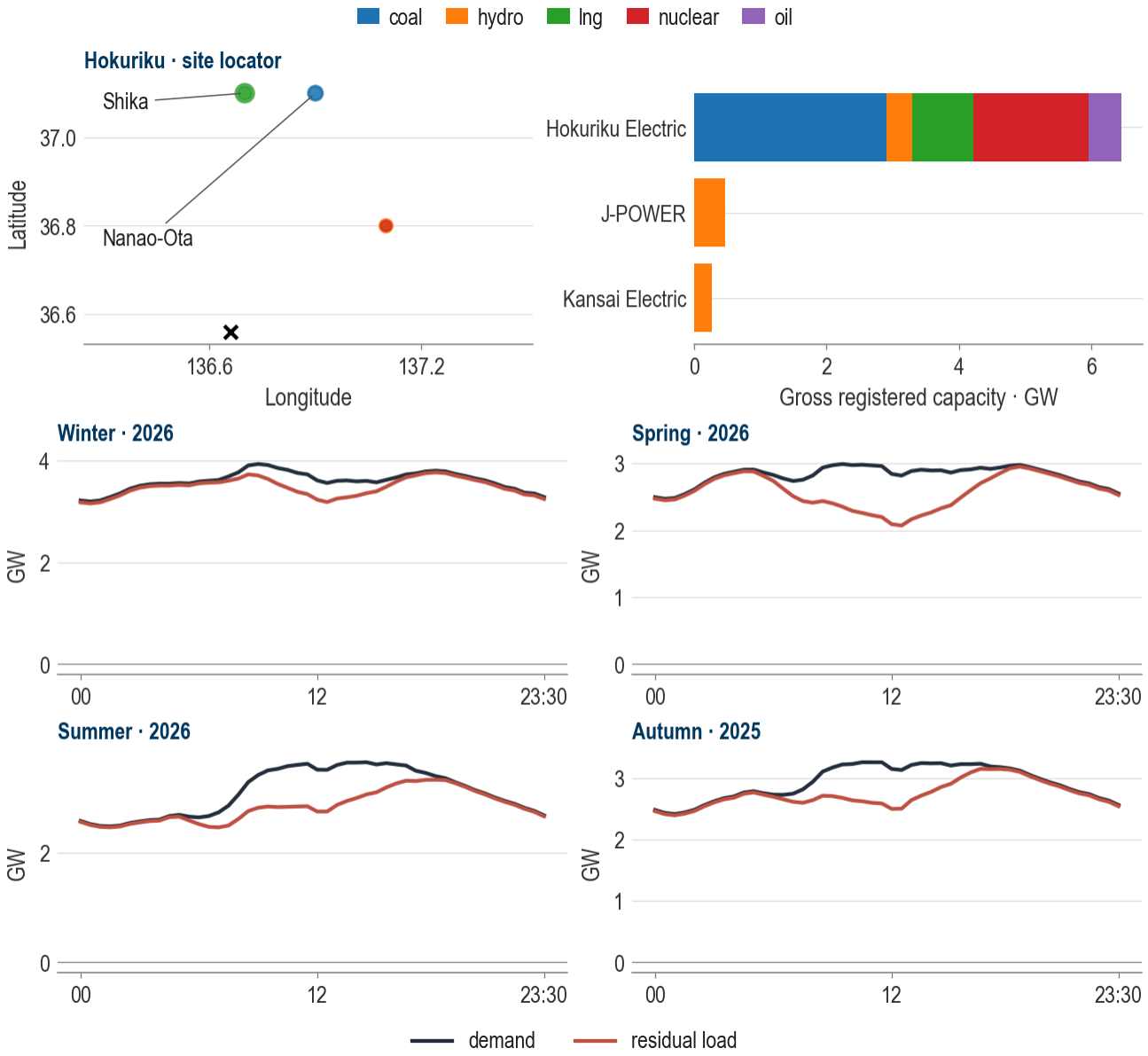
Hokuriku: hydro and the central loop

Hokuriku's winter 2026 mean demand was 3.60 GW and spring demand 2.82 GW. Accepted solar contributed only 0.08 GW in winter and 0.23 GW in spring; hydro conditions and thermal availability therefore remain material alongside renewable timing. Reported nuclear output was zero in the four comparison seasons. Its central neighbours are Chubu and Kansai, but fence management is not a new bilateral asset. [2] [4] [330]

Registered operator	Gross MW
Hokuriku Electric Power	6,449
J-POWER	470
Kansai Electric Power	266
All other registered operators	0

Covered installed-unit MW, rounded; not trading share. Chart names are abbreviated; source identities remain in the extract. [331]

Exhibit 109: Hokuriku: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] · Chart data

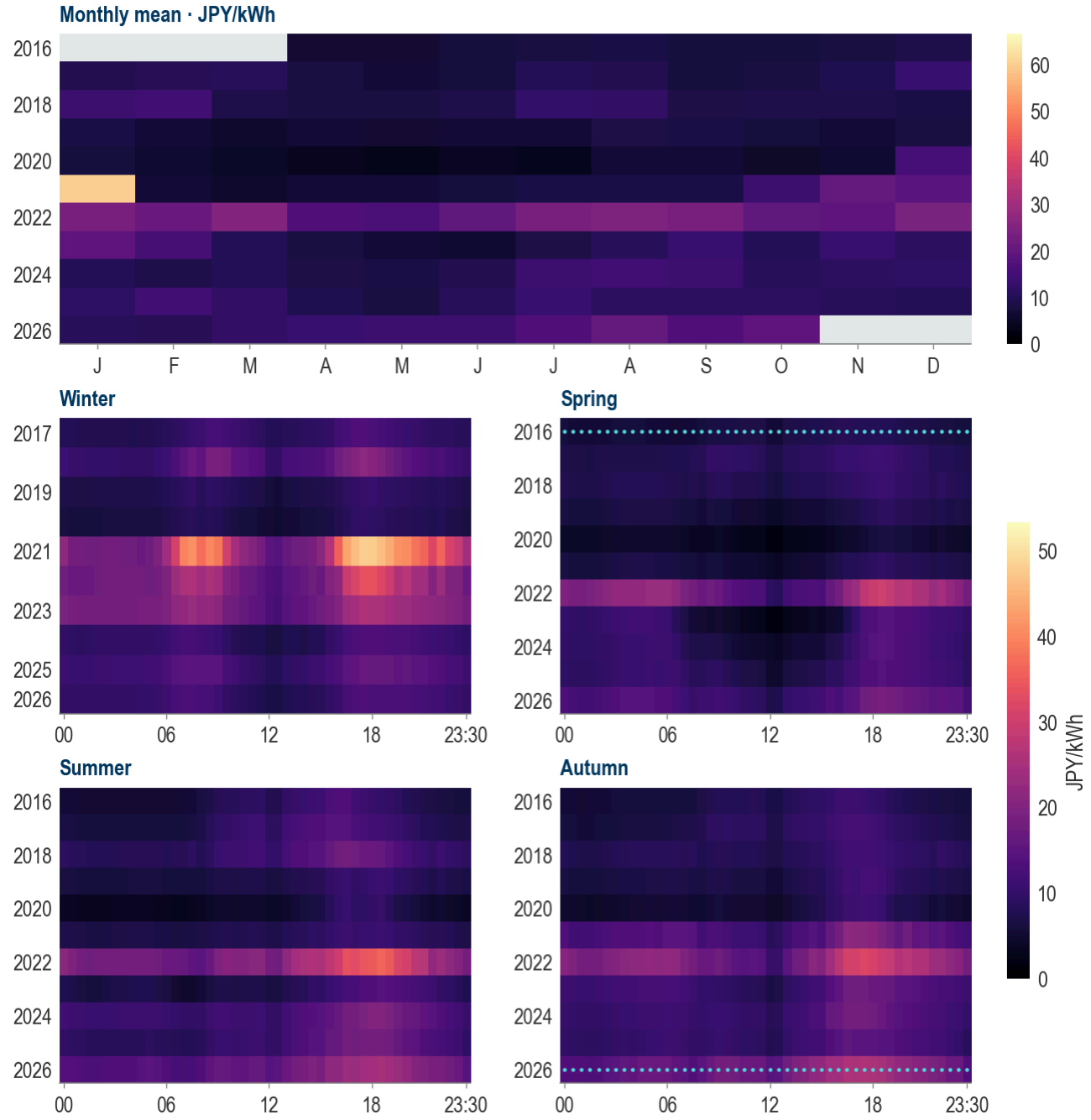
Hokuriku: price history and basis

Hokuriku and Kansai were exactly price-coupled in 99.3% of spring 2026 half-hours. Chubu was much less closely coupled. The summer Hokuriku mean was 17.19 JPY/kWh, compared with 13.06 in spring; summer evening prices averaged 23.38. Hydro or a shared frequency is not evidence of an invariant discount. [1]

Fixed origin → destination	Winter	Spring	Summer	Autumn
Chubu → Hokuriku	-0.51	-2.48	-2.29	-0.53
Hokuriku → Kansai	-0.23	-0.04	-0.38	-0.38

Mean destination-minus-origin JPY/kWh; complete winter/spring/summer 2026 and autumn 2025. Both directions/distributions: boundary atlas. [1]

Exhibit 110: Hokuriku: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

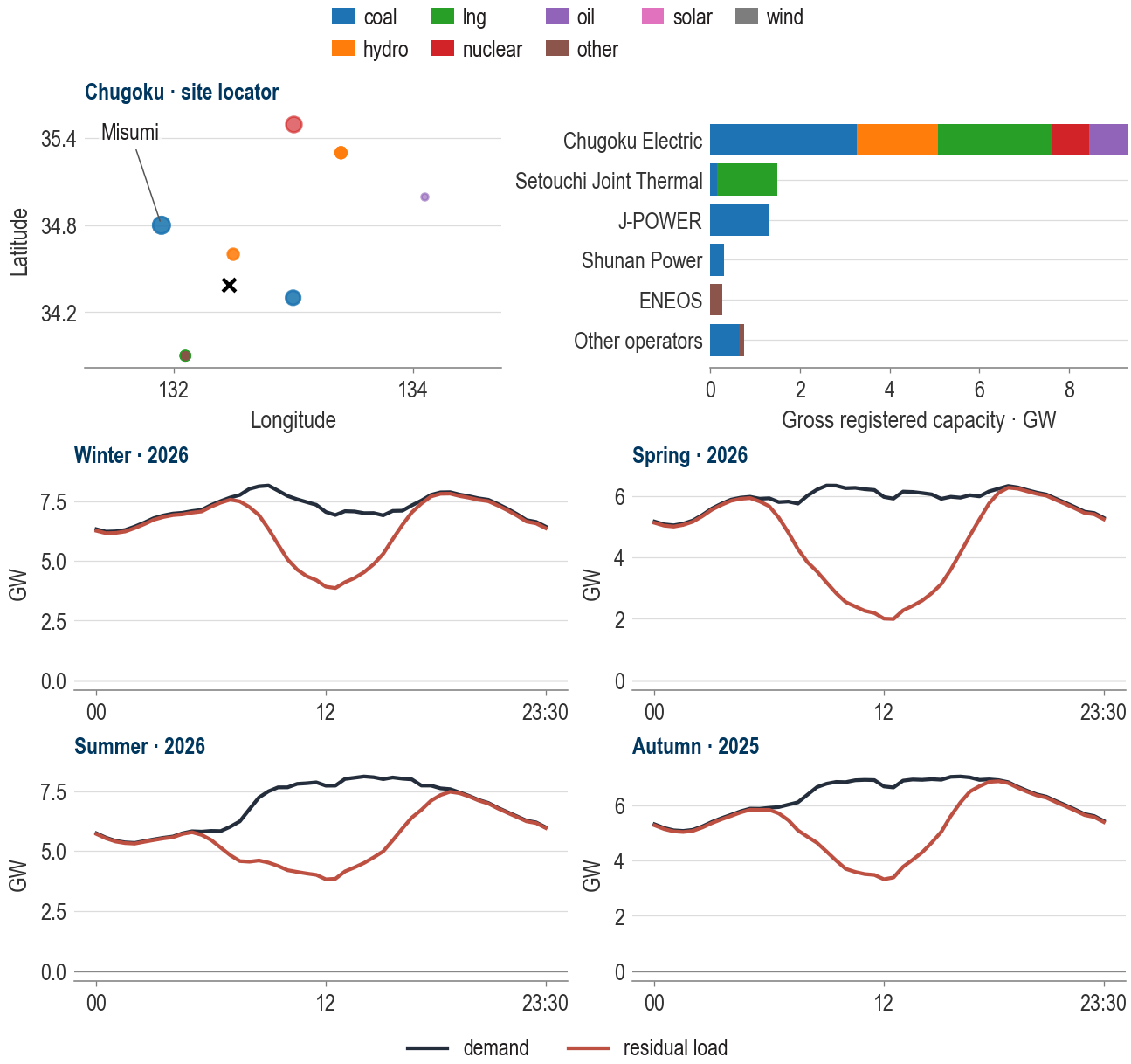
Chugoku: the western junction

Chugoku links Kansai, Kyushu and Shikoku. Coastal thermal/industrial exposure meets variable nuclear output: 0.63 GW in winter 2026, zero in spring, 0.11 in summer. Demand rose from 5.89 GW in spring to 6.85 in summer. Effects can propagate along three routes; area net interchange cannot identify each contribution. [2] [4] [330]

Registered operator	Gross MW
Chugoku Electric Power	9,302
Setouchi Joint Thermal Power	1,485
J-POWER	1,300
Shunan Power	300
All other registered operators	1,022

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 111: Chugoku: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] [366] [367] · Chart data

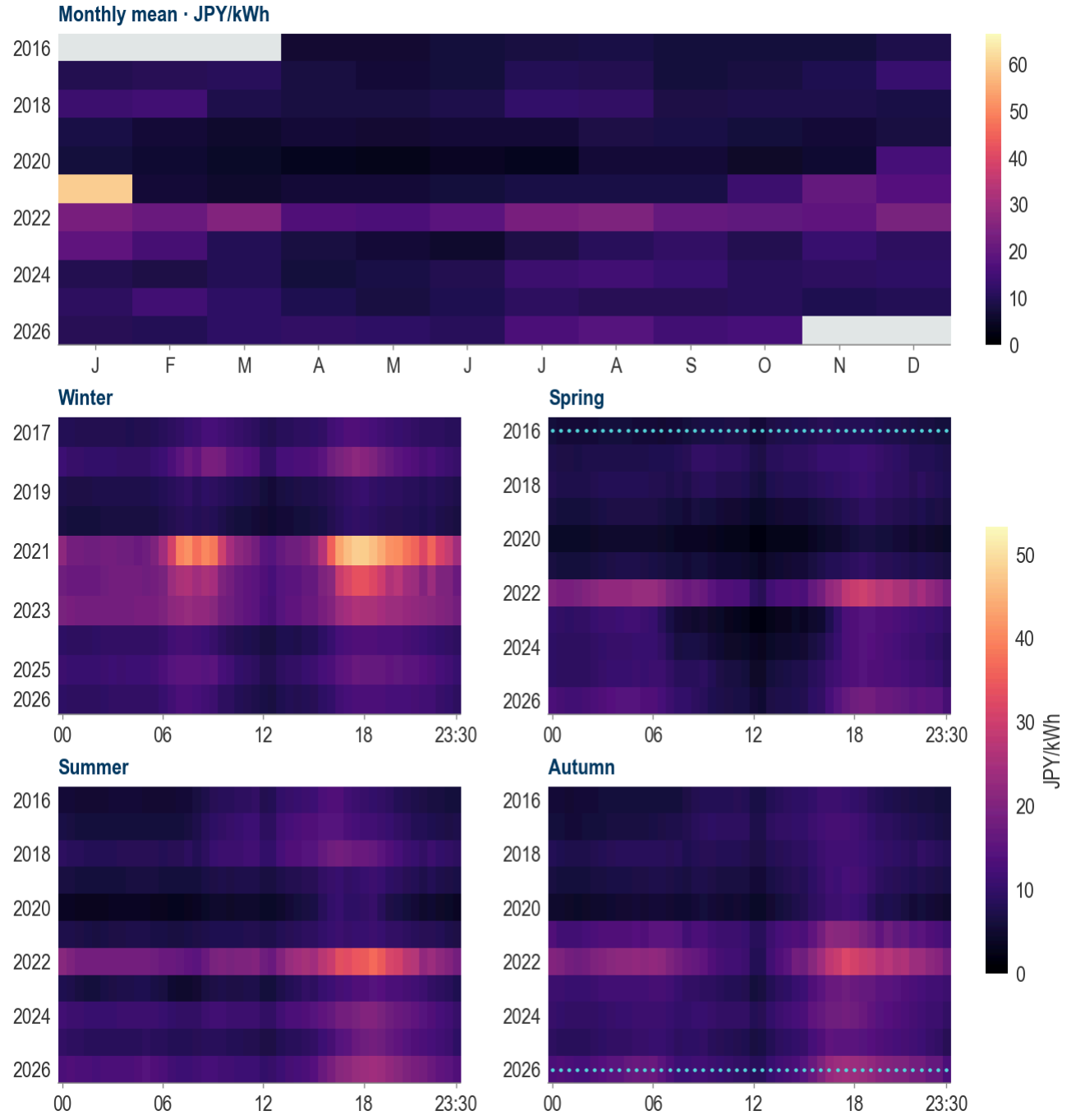
Chugoku: price history and basis

Exact Chugoku–Kansai coupling: 97.9% in winter 2026, 73.7% in summer. Chugoku's summer mean of 14.96 JPY/kWh masks 11.51 midday versus 22.38 evenings. Solar discounts did not remove evening replacement exposure. [1]

Fixed origin → destination	Winter	Spring	Summer	Autumn
Kansai → Chugoku	-0.03	-1.11	-1.84	-0.29
Chugoku → Kyushu	-0.25	-1.77	-0.86	-0.27
Chugoku → Shikoku	-1.27	-4.20	-2.14	-1.21

JPY/kWh, destination−origin; full winter–summer 2026 and autumn 2025. Atlas. [1]

Exhibit 112: Chugoku: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

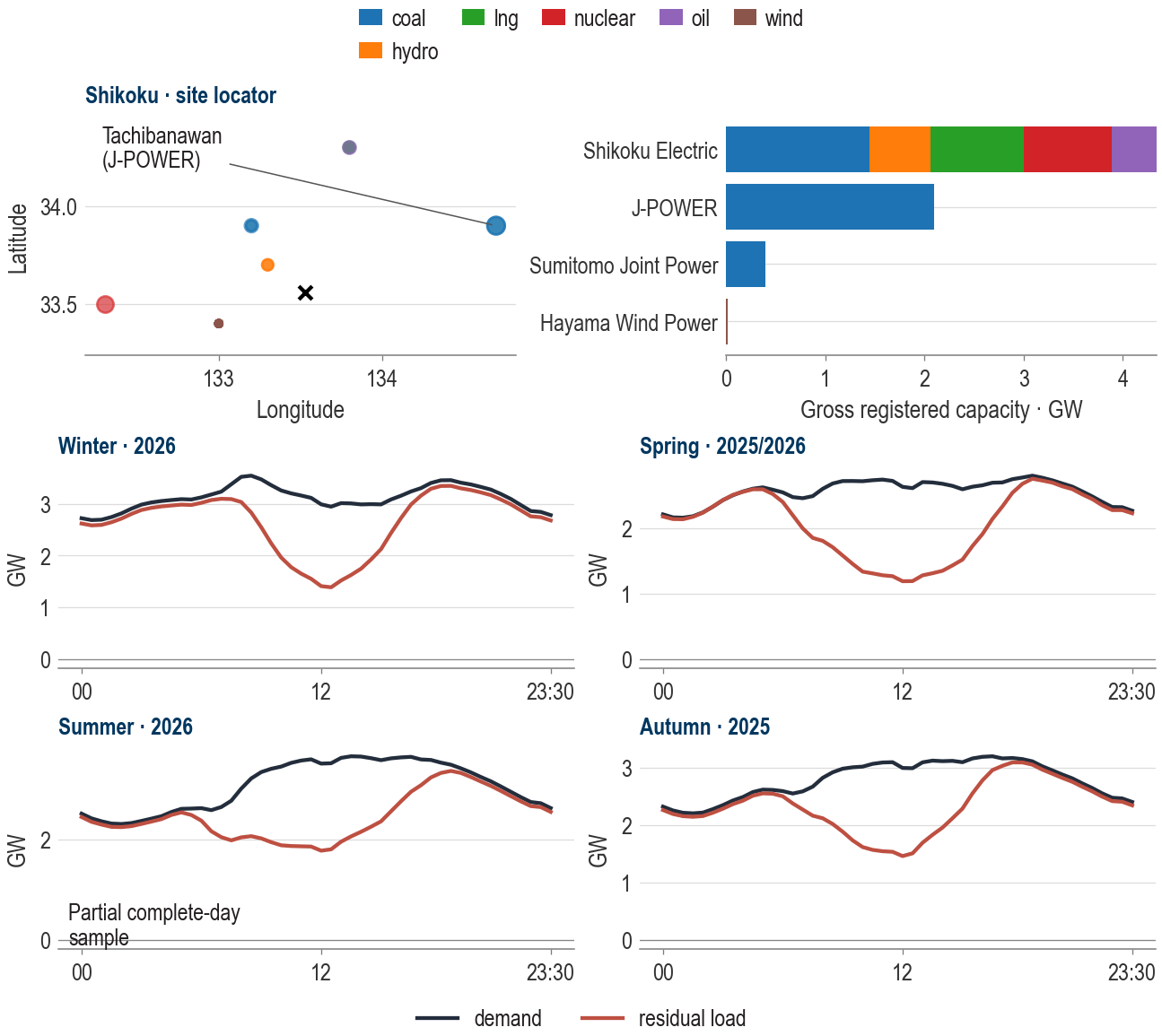
Shikoku: large units, small load

Shikoku's winter 2026 demand averaged 3.11 GW; spring observed demand averaged 2.56 GW. Regional nuclear output was 0.61 GW in winter and 0.88 GW in spring, substantial relative to the load. The full operator population includes J-POWER's thermal capacity rather than treating Shikoku Electric Power as the whole fleet. Honshi is the AC Chugoku route; Anan–Kihoku is the separate HVDC Kansai route. Spring and summer physical samples have missing half-hours. [2] [4] [330]

Registered operator	Gross MW
Shikoku Electric Power	4,340
J-POWER	2,100
Sumitomo Joint Electric Power	400
Hayama Wind Power Generation	20
All other registered operators	0

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 113: Shikoku: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] [352] [367] · Chart data

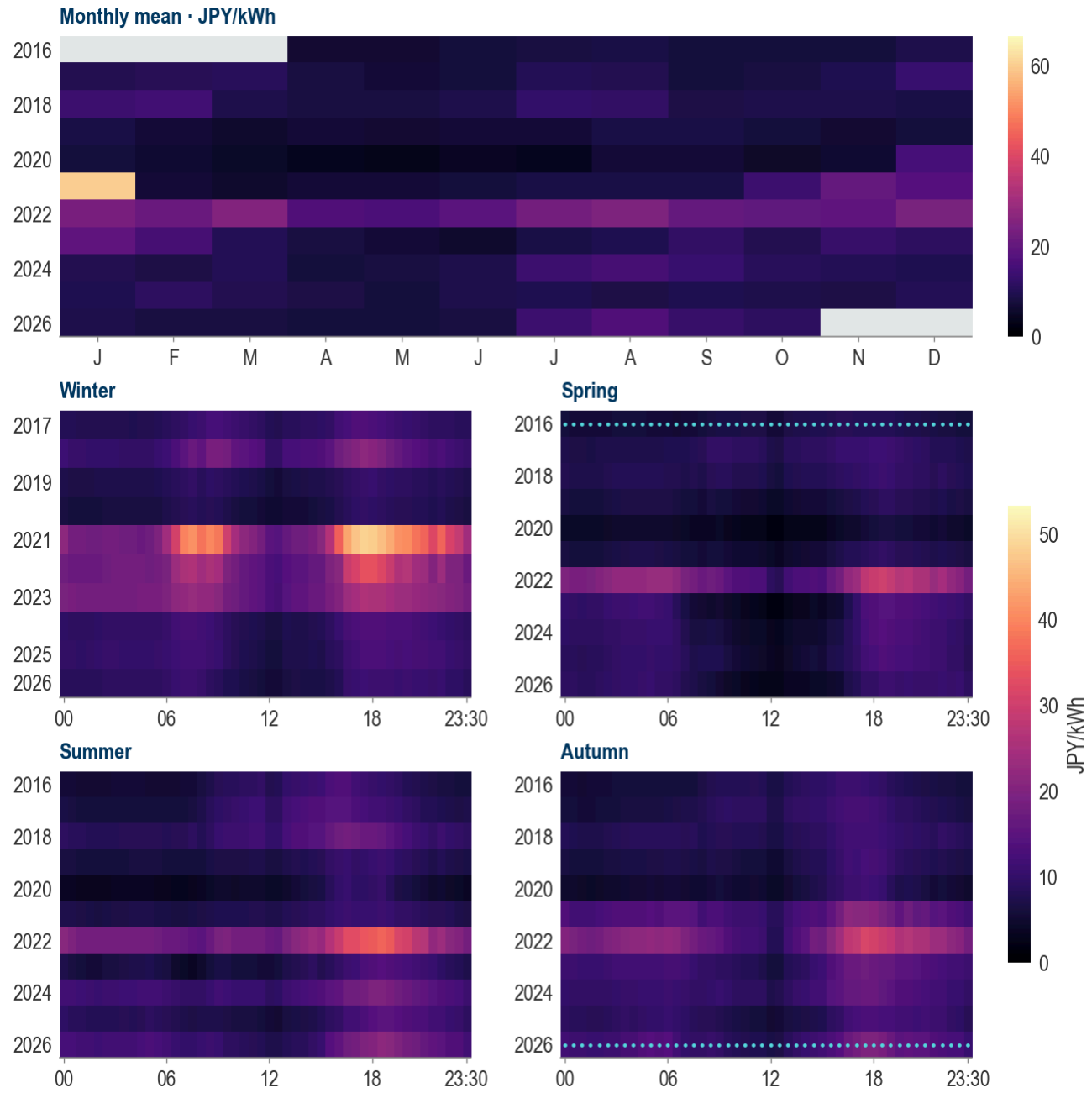
Shikoku: price history and basis

Shikoku's spring 2026 mean price was 7.71 JPY/kWh, below winter's 8.97 and summer's 12.83. The spring midday mean was 3.44, while the evening mean was 10.17. Cheap hours are consistent with a small residual market facing sizeable must-run/renewable output, but bilateral export constraints must be demonstrated from their own stage-specific records. [1]

Fixed origin → destination	Winter	Spring	Summer	Autumn
Chugoku → Shikoku	-1.27	-4.20	-2.14	-1.21
Kansai → Shikoku	-1.30	-5.31	-3.97	-1.50

Mean destination-minus-origin JPY/kWh; complete winter/spring/summer 2026 and autumn 2025. Both directions/distributions: boundary atlas. [1]

Exhibit 114: Shikoku: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

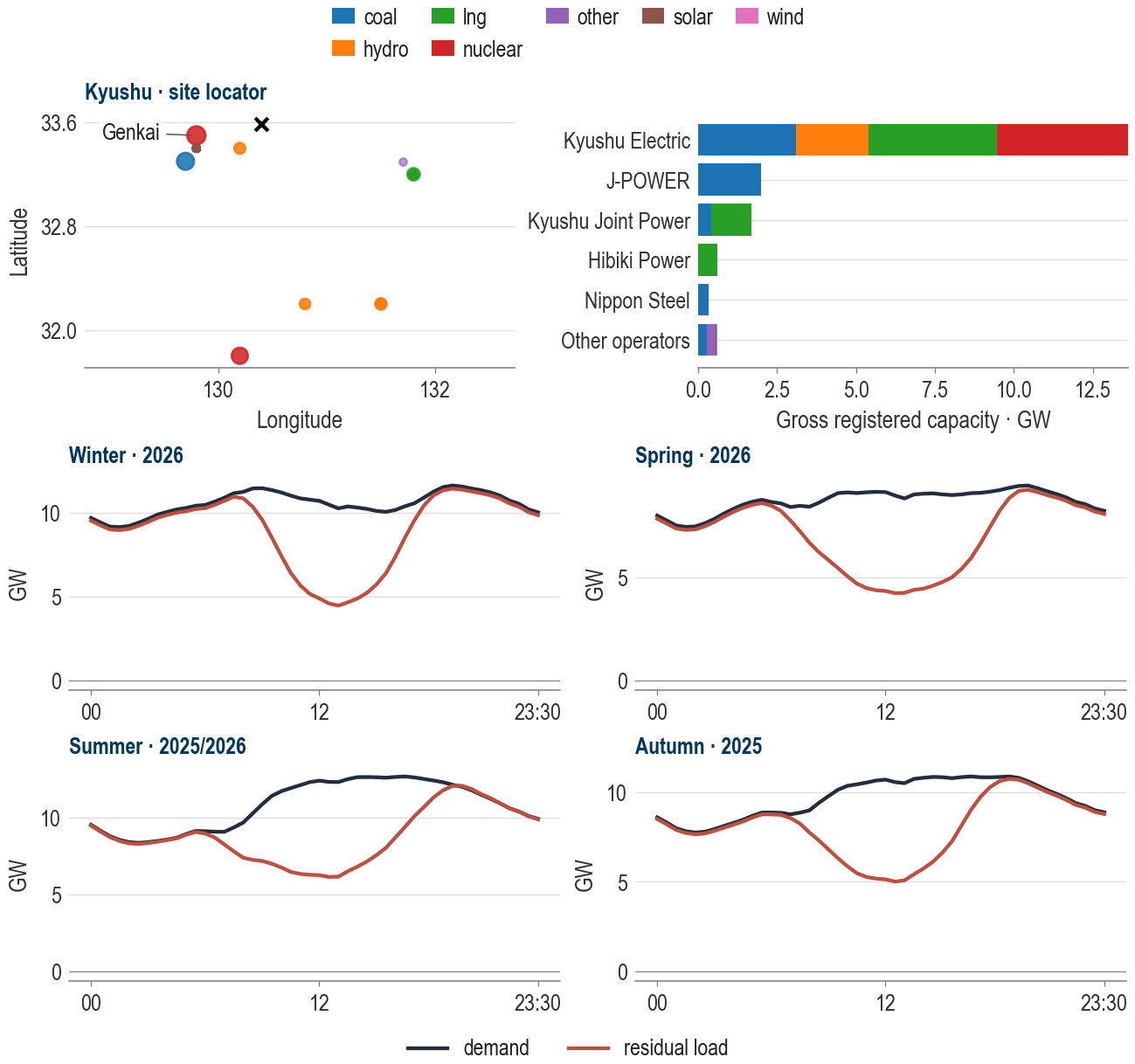
Kyushu: nuclear, solar and Kanmon

Winter-to-spring 2026 nuclear output rose from 3.56 to 3.84 GW; thermal fell from 5.43 to 3.41 GW as demand fell from 10.54 to 8.72 GW. Solar/control, pumping and exports adjust together. Kanmon connects to Chugoku, not directly to every western area. Summer retains two missing residual-load half-hours. [2] [4] [330]

Registered operator	Gross MW
Kyushu Electric Power	13,615
J-POWER	2,000
Kyushu Joint Power	1,697
Hibiki Power	623
All other registered operators	935

Rounded MW. Names are display translations/abbreviations, not verified legal names; originals in extract. [331]

Exhibit 115: Kyushu: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] [366] [367] · Chart data

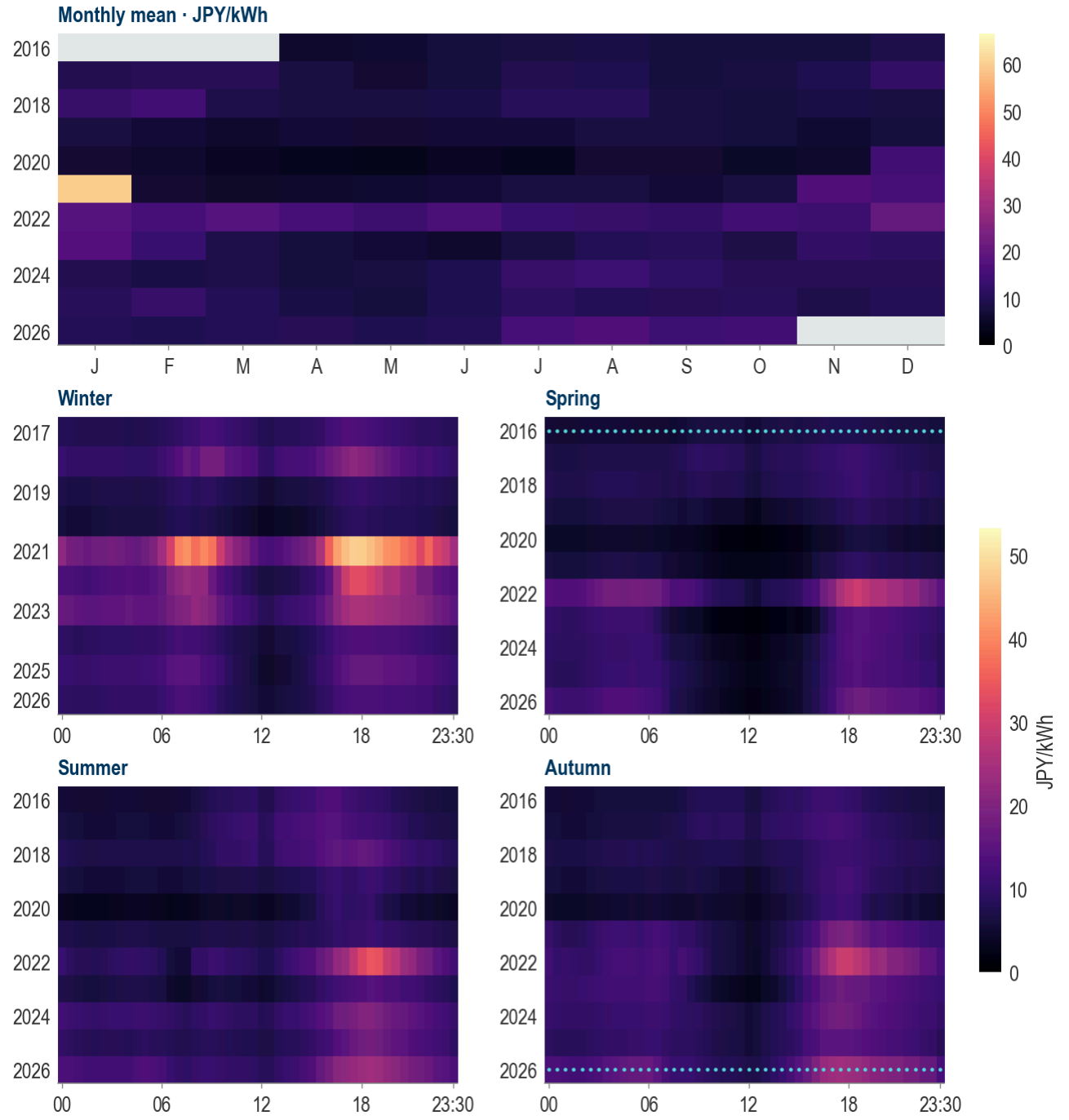
Kyushu: price history and basis

Kyushu's spring 2026 midday mean price was 3.36 JPY/kWh against 15.84 in the evening. Summer raised the corresponding means to 10.04 and 22.27. The all-day spring discount to Chugoku was 1.77 JPY/kWh, but median basis was zero; pronounced separated intervals coexist with frequent exact equality. [1]

Fixed origin → destination	Winter	Spring	Summer	Autumn
Chugoku → Kyushu	-0.25	-1.77	-0.86	-0.27

Mean destination-minus-origin JPY/kWh; complete winter/spring/summer 2026 and autumn 2025. Both directions/distributions: boundary atlas. [1]

Exhibit 116: Kyushu: four-season price atlas



Note: All season-years × 48 JST starts; common uncapped nine-area scale. Grey = absent; turquoise = partial. December joins the following winter.

Source: [1] · Chart data

Okinawa: isolated cooling demand

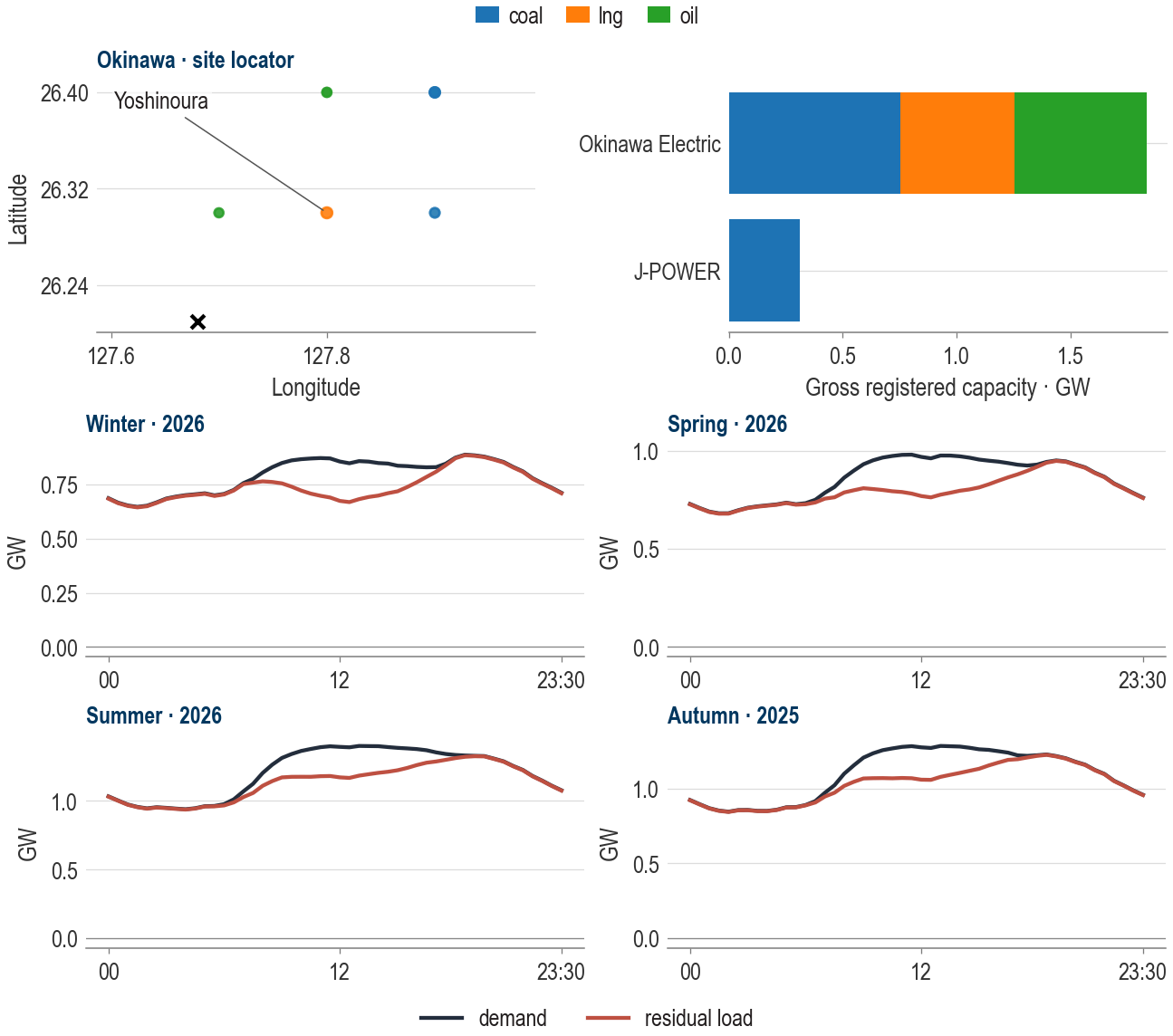
Okinawa's summer 2026 mean demand was 1.20 GW, compared with 0.79 GW in winter. Its residual-load means were 1.13 and 0.74 GW: accepted solar reduced the energy requirement, but did not make the hot-season load disappear. Thermal generation averaged 1.08 GW in summer and 0.69 GW in winter. These source-defined regional observations must not be assigned indiscriminately to every smaller island. [2] [4]

No mainland link or JEPX area price exists: local generation, fuel and reserves—not hypothetical Kyushu imports—support the island systems. The covered cohort includes Okinawa Electric Power and J-POWER. [331]

Registered operator	Gross MW
Okinawa Electric Power	1,835
J-POWER	312
All other registered operators	0

Covered installed-unit MW, rounded; not trading share. Chart names are abbreviated; source identities remain in the extract. [331]

Exhibit 117: Okinawa: operators, sites and physical character



Note: Covered physical-unit MW, not trading share; other operators stay in totals. Qualified site coordinates only; cross = demand centre.

Source: [2] [4] [10] [341] · Chart data

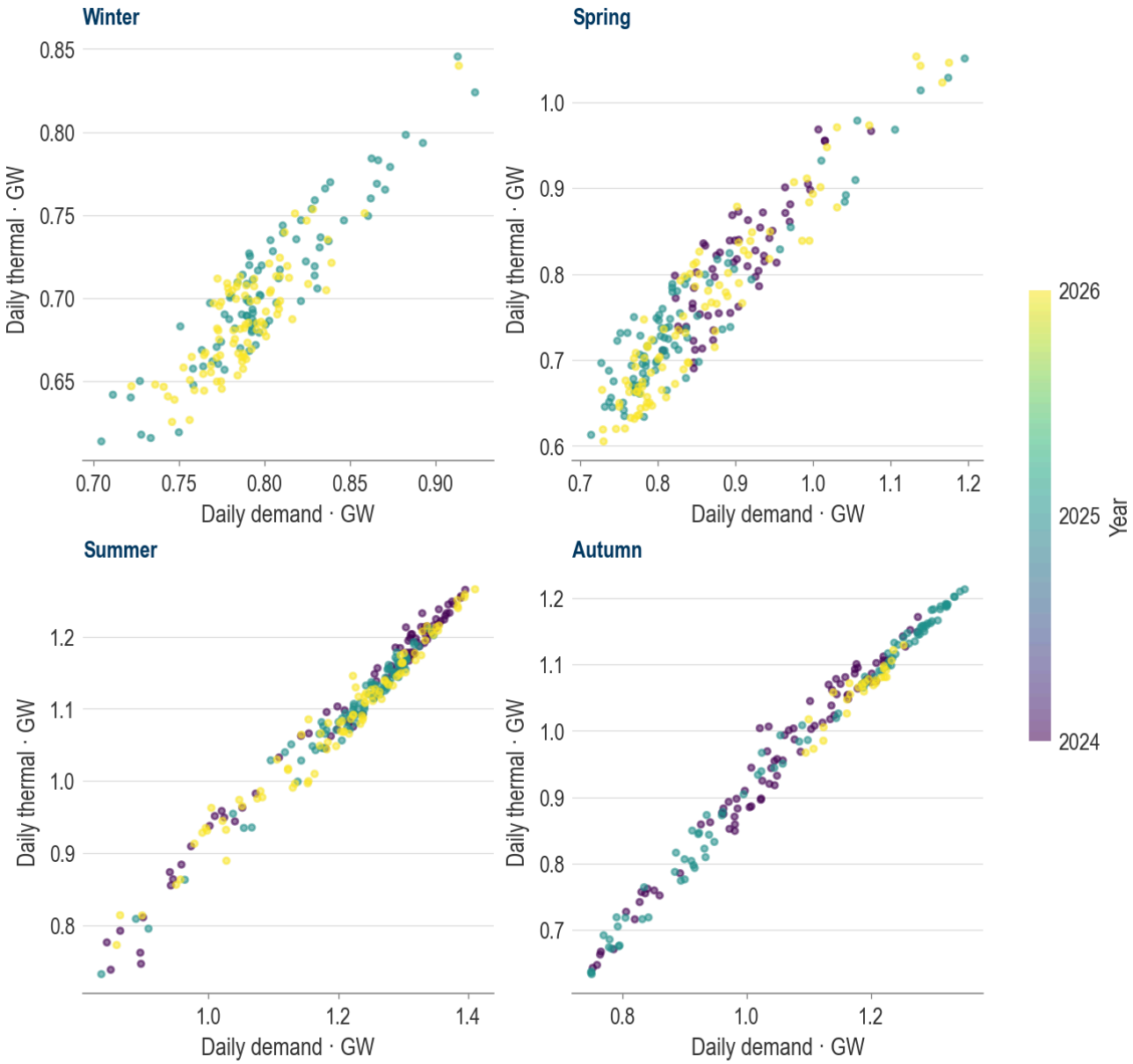
Okinawa: local fuel and operational stress

Okinawa's summer 2026 demand mean was roughly 52% above its winter mean, a sharper cooling-led contrast than the near-equal Kansai winter/summer means. The paired daily demand and thermal observations show the local energy response without inventing a price series. Wind is small in this source-defined balance; reported zero nuclear and pumping are observations, not missing imports filled with zero. [2] [4]

Fuel and storage exposures differ from those in the interconnected mainland. LNG and other thermal fuels must reach the relevant island supply chain; a national LNG stock observation is neither Okinawa's inventory nor evidence of cargo availability at a particular terminal. Terminal connections and native storage units are detailed in the fuel infrastructure chapter. [12]

The surrounding island systems have their own equipment and operating constraints. A regional aggregate cannot establish simultaneous reserves, unit-level available MW or the duration of fuel endurance. The absence of mainland support is a physical feature, not a predicted shortage. Local generation availability and cooling demand remain the commercially relevant stress dimensions.

Exhibit 118: Okinawa: seasonal populations and contrasting stress windows



Note: Complete days across all modern season-years; common year colours. No mainland JEPX price. Temperature and balance alternatives remain in the extract; no causal coefficient.

Source: [2] [4] [5] [10] · Chart data

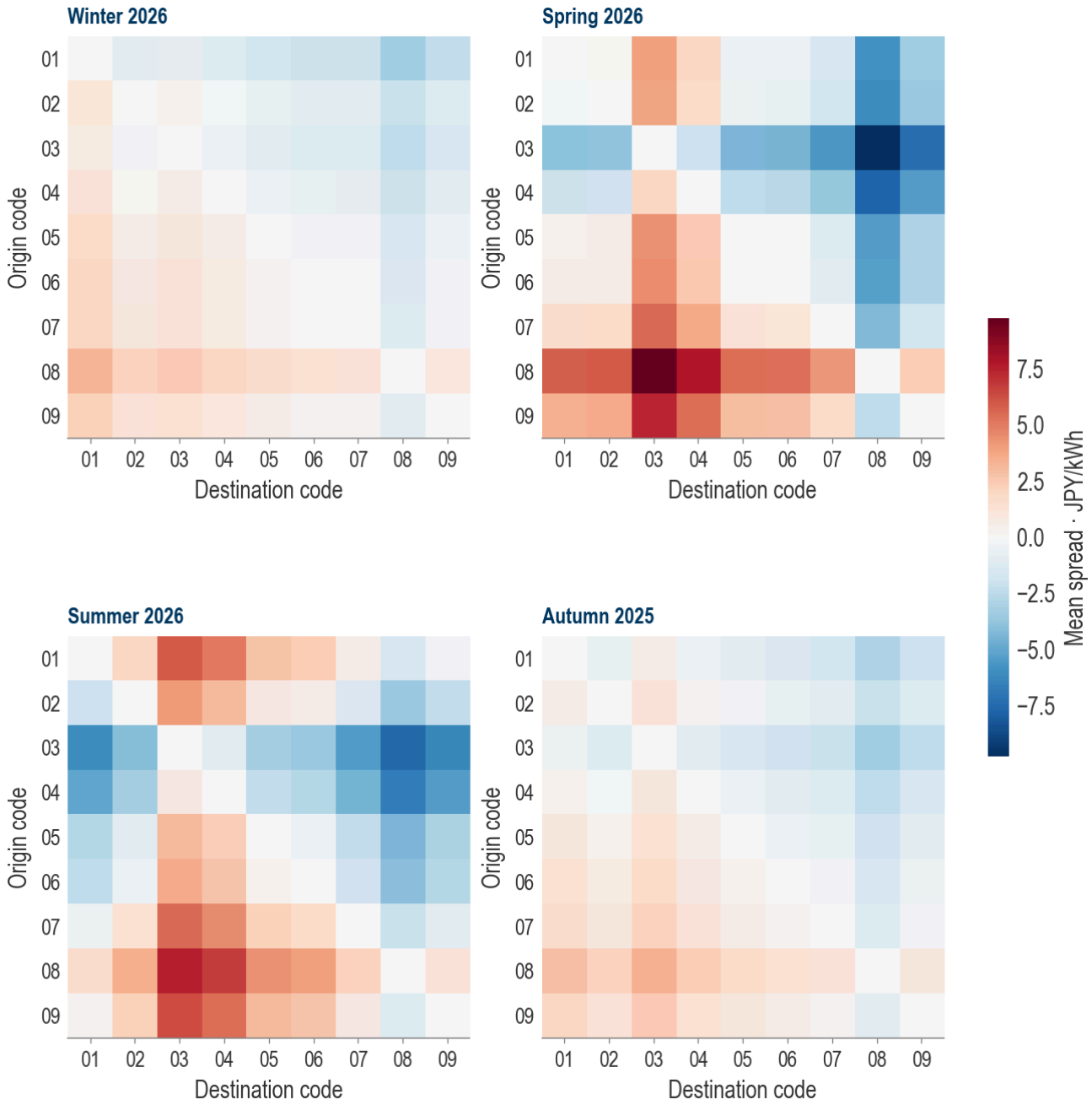
Seasonal basis: more than east–west

Summer 2026 Tokyo→Chubu averaged −0.90 JPY/kWh and Chubu→Kansai −2.67. On common timestamps they sum exactly to Tokyo→Kansai, −3.57: an economic comparison, not a direct line. Integer ticks preserve the identity. [1]

Winter 2026 was much closer: Tokyo→Chubu averaged −0.48 and Chubu→Kansai −0.74. Yet Hokkaido→Tohoku reversed its seasonal sign, from −1.10 in winter to +1.94 in summer. A single north–south or east–west premium would miss both differences.

Spring Kansai→Shikoku averaged −5.31, against just −0.04 for Hokuriku→Kansai. Shikoku's small residual market/large units, Kansai's nuclear cycle and renewable patterns offer competing mechanisms, not proof of line binding. These matrices share an all-nine population; pairwise extracts retain longer coverage.

Exhibit 119: Nine-area seasonal spread matrices



Note: Latest complete all-nine seasons; full history in extract. Rows = origin, columns = destination; codes 01–09 north to south. Tokyo–Kansai is not a physical line.

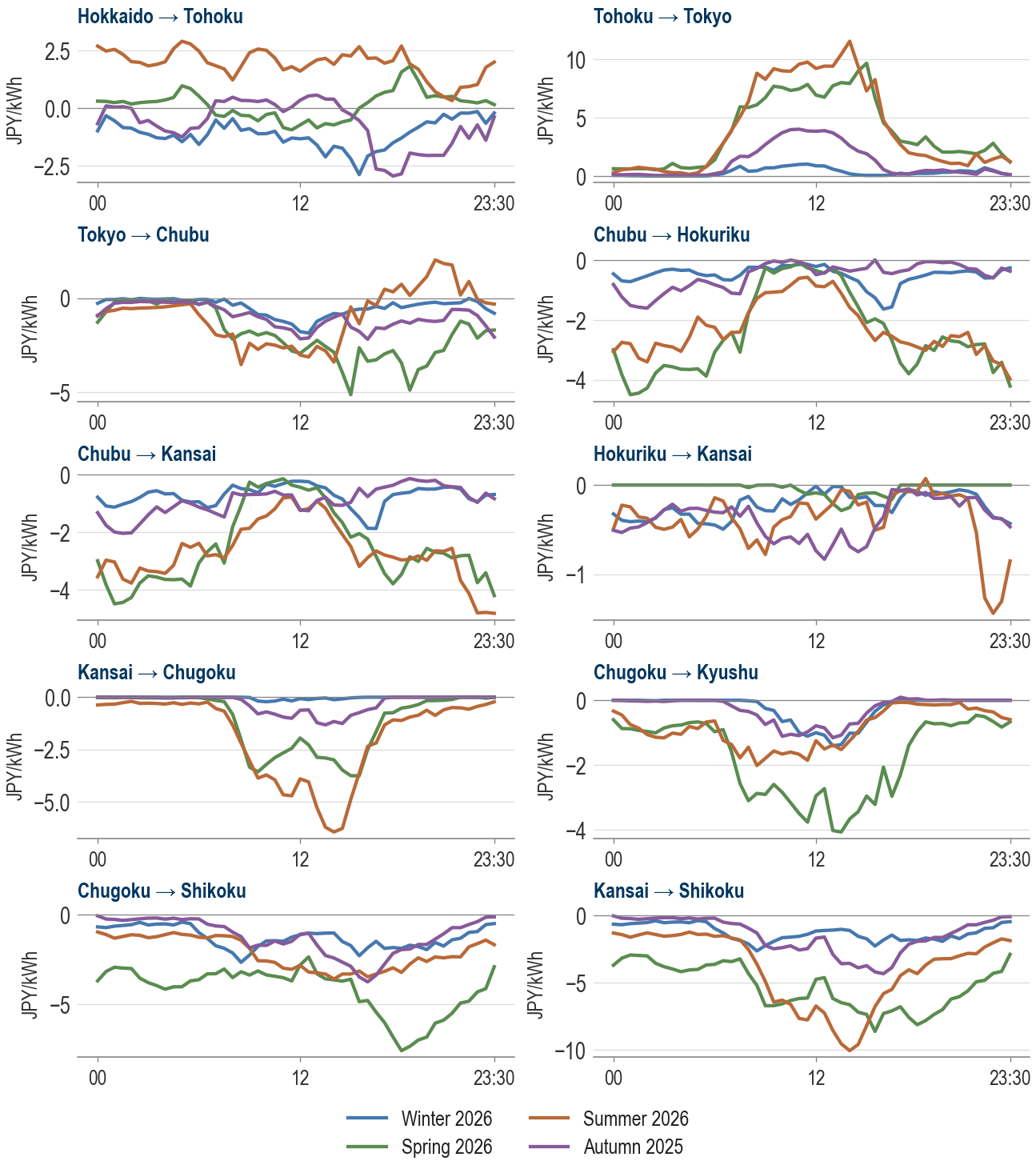
Source: [1] · Chart data

Ten boundaries, four seasonal profiles

Spring 2026 Hokuriku–Kansai prices were exactly equal in 99.3% of intervals, against 31.2% for Kansai–Shikoku. Identical quote precision applies throughout; exact equality is not the ±0.01 reversal deadband. [1]

Both directions and every season-year retain means, medians, p10/p90, absolute spread, coupling and direction frequencies by all-day/night/midday/evening band. Legend years identify the displayed seasons; clocks are JST. Honshi links Shikoku to **Chugoku** despite legacy ID `ks`; Anan–Kihoku links **Kansai**, ID `ks_anan_kihoku`. The shared island price does not make their flows interchangeable. [329] [330]

Exhibit 120: Every connected boundary: four-season diurnal spreads



Note: Latest complete seasons; all years/both orientations exported. Positive = destination dearer. Exact coupling uses identical ticks, not the ±0.01 reversal deadband.

Source: [1] · Chart data

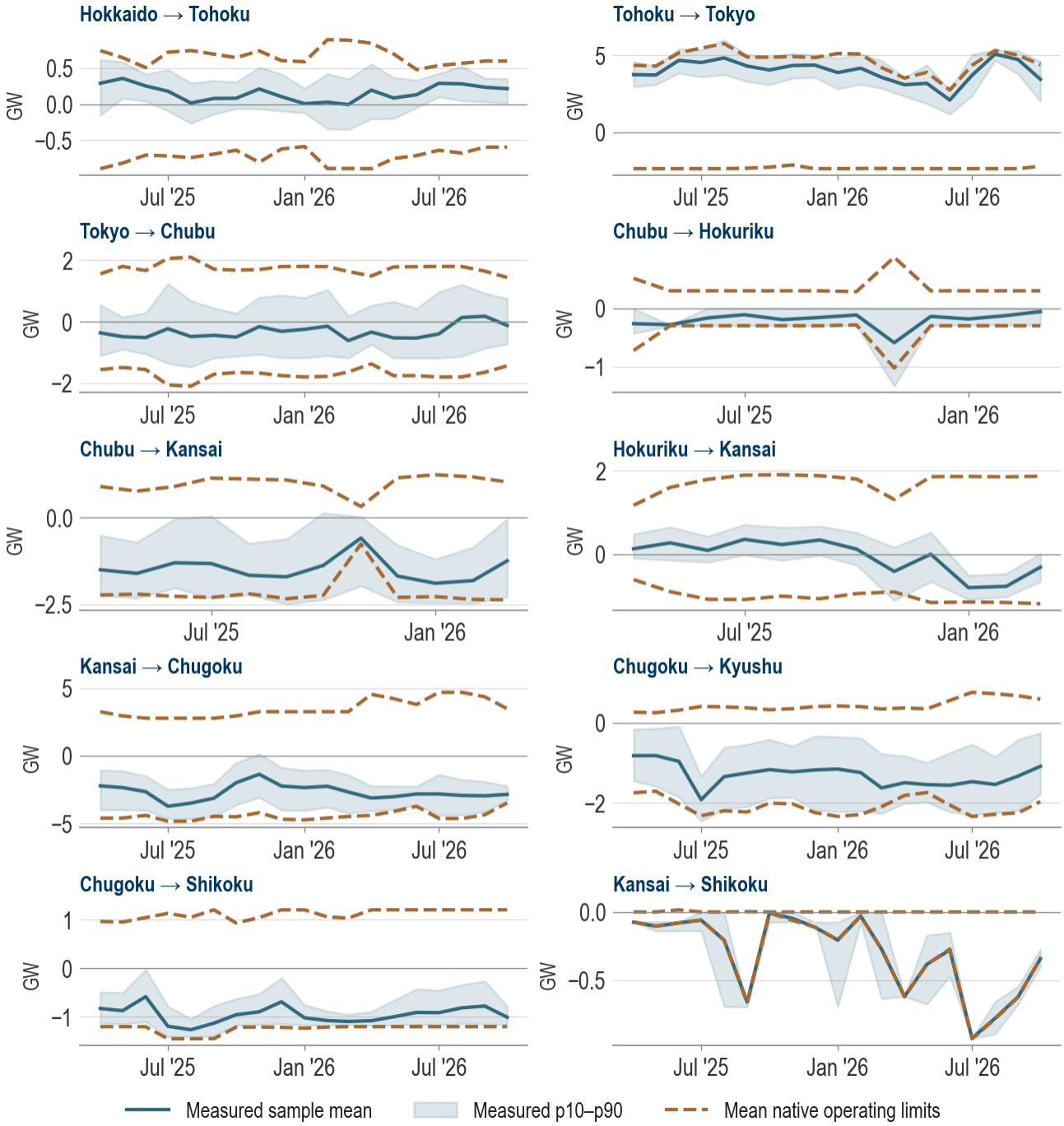
Transfers and operating limits

Tohoku→Tokyo sample flow averaged **4.135 GW in winter 2026**, against **3.642 GW in summer**; native forward capacity averaged 5.014/4.152 GW. Winter retains 4,317/4,320 complete six-sample half-hours; summer all 4,416. The corridor's usable capacity changed alongside the price incentive. [281]

Summer transfers averaged **1.525 GW Kyushu→Chugoku** and **0.261 GW Chubu→Tokyo**—reverse to the chart orientation, each with 4,415/4,416 observations. Northern, western and converter support were distinct exposures. [281]

March 2026 central fences are not spliced to individual-link histories. Formulated plans, final schedules and physical samples remain separate; later publications do not establish earlier auction constraints. Binding remains **not established**. [279] [282] [280]

Exhibit 121: Measured transfer samples and actual-stage directional limits



Note: Monthly six-sample mean/p10–p90; dashed = same-stage capacity, not nameplate. Expansion fields stay separate. Forward sign; central individual publication ends March 2026.

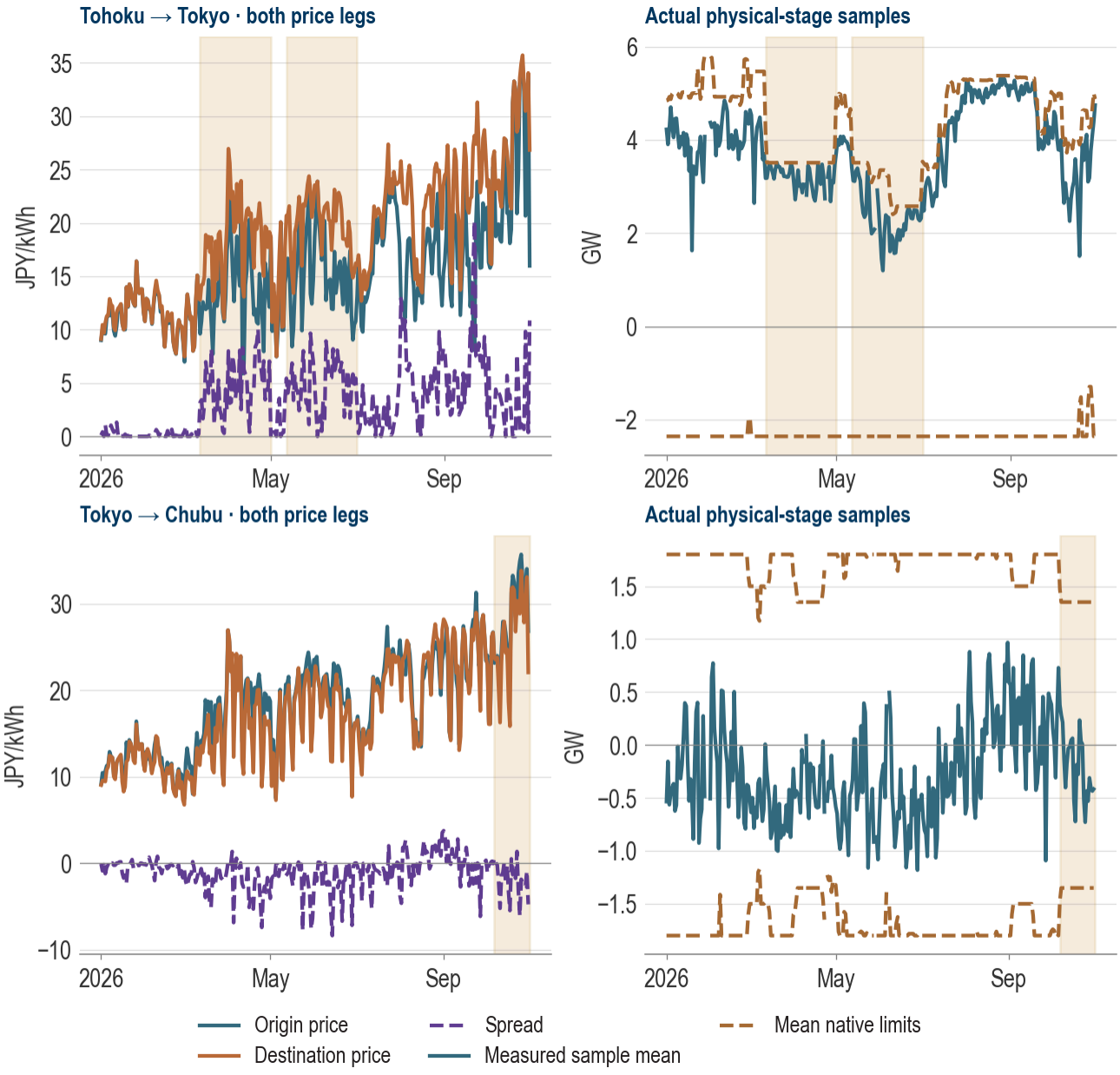
Source: [278–281] · Chart data

Restrictions: direction, equipment and publication vintage

For **12 May–30 June 2026**, the March plan placed Tohoku→Tokyo at 2.45–3.55 GW during transformer and line works. Actual-stage sample flow averaged **2.364 GW** and its published forward-capacity field **3.008 GW**, across 2,399/2,400 complete six-sample half-hours. Tokyo averaged **20.15 JPY/kWh**, Tohoku **15.31**: a **+4.84** basis. Their mean residual loads were 24.17 and 5.79 GW. Northern exports were important to Tokyo's larger residual requirement; that does not identify an auction-binding interval. [283] [281] [1] [4]

Converter work planned for **7 October–20 November** had different annual-plan bands: 0.90–1.35 GW Tokyo→Chubu and 0.84–1.31 GW reverse. In the observed **7–31 October** subset, native capacity averaged **1.355 GW each way**, while measured flow averaged **0.199 GW Chubu→Tokyo** over all 1,200 observations. Tokyo/Chubu prices averaged **28.20/26.30 JPY/kWh** and residual loads **24.09/11.53 GW**. The annual bands were not constant realised limits; the direction of physical exchange and the price premium are consistent, but causality remains unproved. [283] [281] [1] [4]

Exhibit 122: Restriction chronology: dated plans, prices and measured transfer context



Note: Shading is the 1 March 2026 annual plan. Daily panels retain both price legs, spread and six-sample mean physical transfers with same-source actual-stage operating capacities. Contemporaneous regional balance fields accompany the extract. No inference that each restriction occurred exactly as planned.

Source: [1] [2] [4] [281] [283] [284] · Chart data

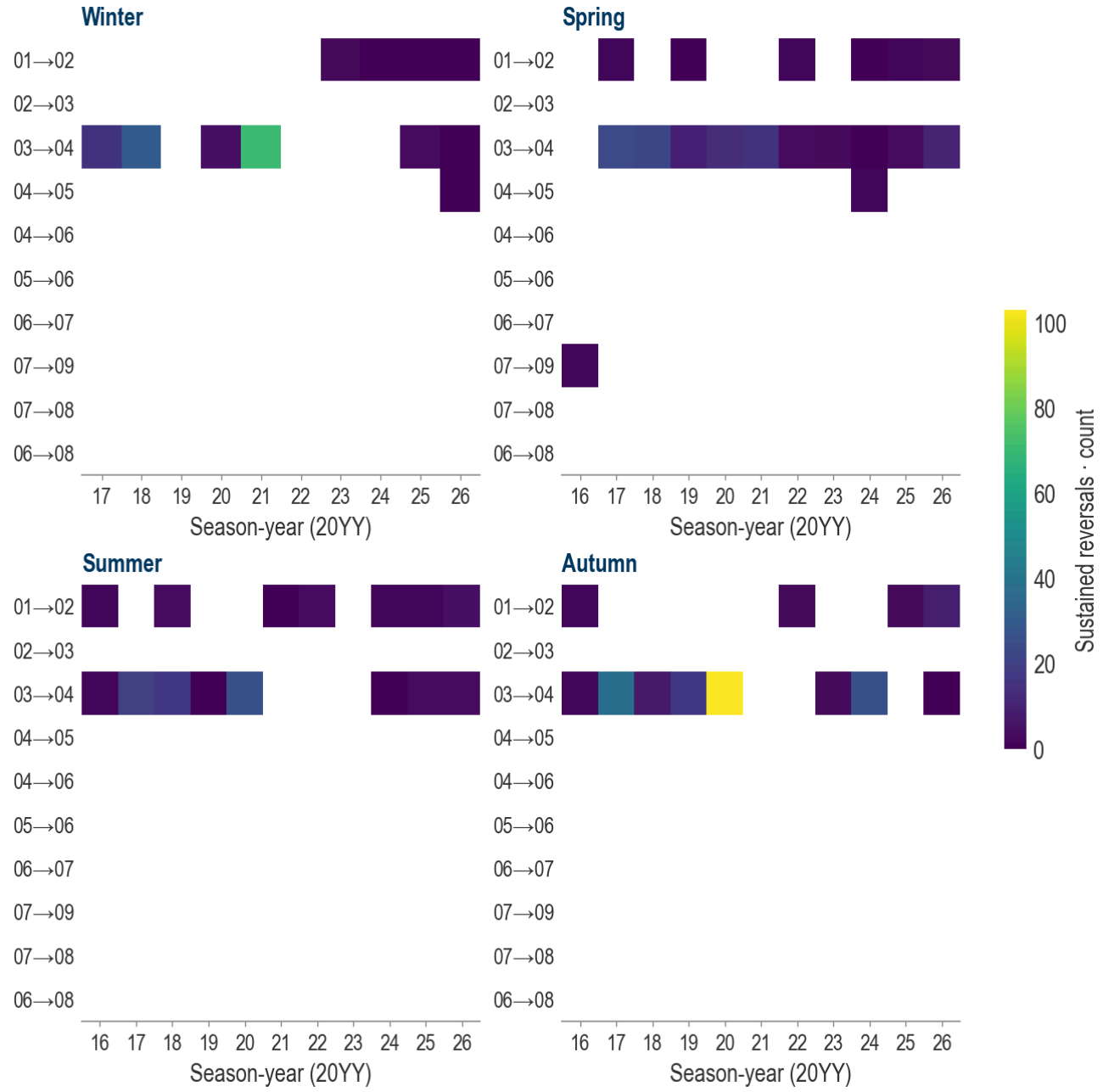
Reversals within the delivery day

Chubu's summer 2026 evening price exceeded Tokyo's even though the all-day seasonal average was lower. Kyushu's spring midday and evening prices differed by 12.48 JPY/kWh. These are concrete reasons why an all-day basis hedge and a daylight generation exposure can behave differently. Neither comparison is a realised trading profit. [1]

All ten boundaries, four seasons and observed years remain. Axis years are 20YY; grey cells have no counted event record, not a certified zero. Two consecutive half-hours establish each opposing state; missing prices break runs and short excursions remain separate. Neither coupling nor splitting establishes spare capacity or a binding line.

Nuclear cycles, solar withdrawal, calendar demand, wind and fuel availability can move the two legs differently. Restrictions need matched stage/clock/vintage before a binding claim. The next chapter connects these exposures to adjustment, balancing, capacity and contracts: transmission rights do not remove volume, shape or availability risk. [280]

Exhibit 123: Price-direction reversals across all connected boundaries



Note: Grey = no counted event record, not a certified zero. Sustained = two half-hours in each opposing state, at most two neutral intervals; no missing-price bridge. ±0.01 deadband is not coupling. Counts are not congestion frequencies.

Source: [1] · Chart data

Capacity, short-term products and commercial market structure

Capacity and market products: obligations survive the trade

Japan's commercial markets contract different services on different clocks. A capacity award is future expected capability under performance obligations; balancing procurement reserves qualified response; JEPX trades electricity; a certificate records environmental value. None removes the others' delivery and cash requirements. [217] [225] [219]

- Intraday adjustment is traded energy, not final imbalance
- Resource classes and performance
- Rights awards leave quantity and internal-network exposure

Market layer	Named actors	Commercial obligation that remains
Day-ahead / intraday	JEPX members, generators, retailers	Plans, network delivery, deviations and exchange settlement
Balancing capability / activation	EPRX, TSOs, qualified providers	Response speed, duration, monitoring and metered performance
Capacity main / additional	OCCTO, resource providers, retailers and networks	Expected-capacity qualification, availability and contribution payments
LTDA	OCCTO and awarded project entities	Investment and delivery conditions; other-market revenue return
Futures / cleared OTC	Exchanges, brokers, clearing members, CCPs	Area/shape/quantity basis and interim collateral
Indirect transmission rights	JEPX eligible participants	Specified directional financial settlement, not physical reservation

Chapter route

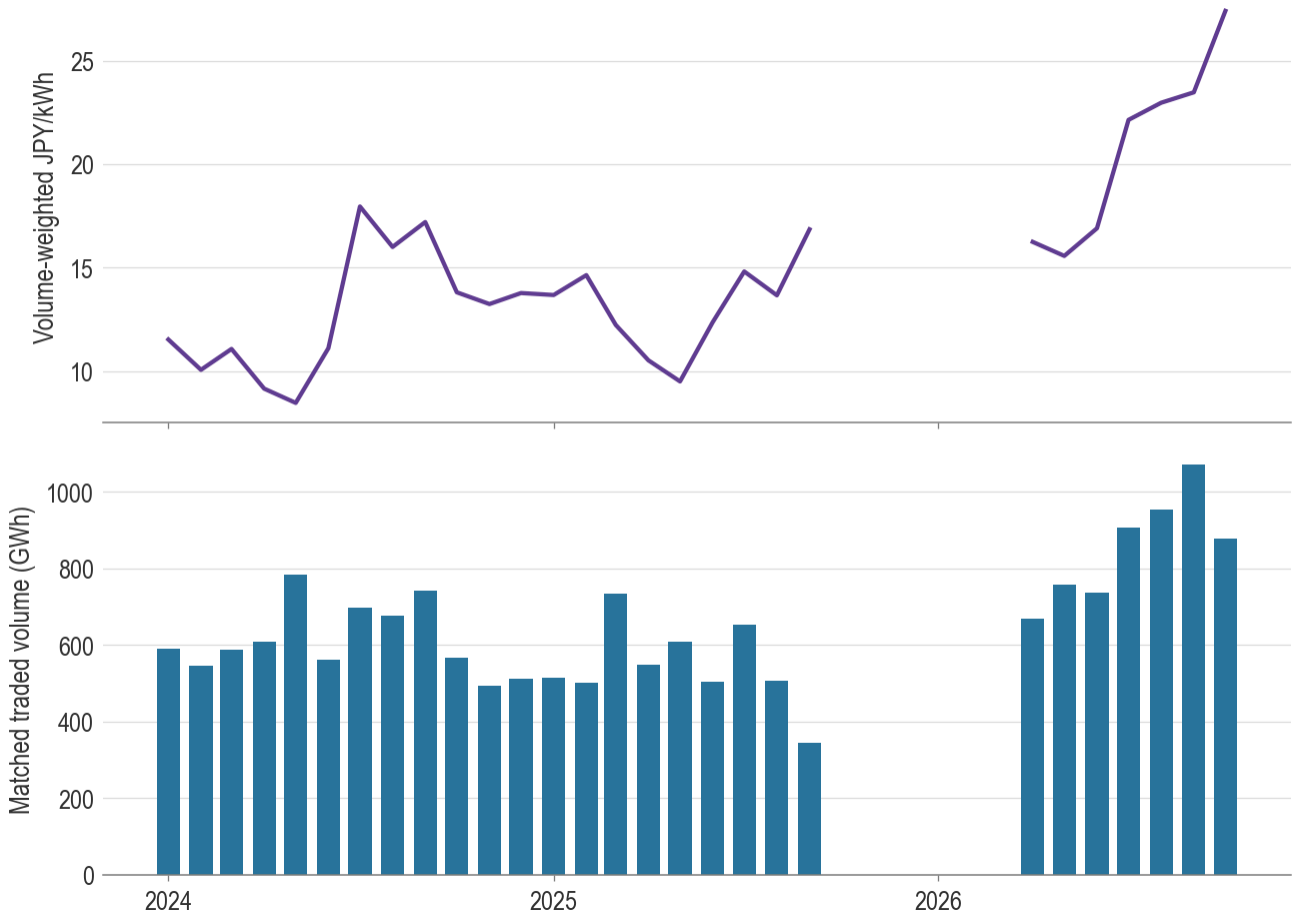
- Intraday adjustment, reserve procurement and imbalance settlement form the short-term operating interface.
- Capacity outcomes reveal regional deliverability and different resource obligations; collection follows its own calendar.
- Actual derivatives and rights activity reveal traded products, not confidential company hedge books.

Capacity price separation is especially important: the original FY2024 main auction cleared at JPY14,137/kW throughout the nine-area market, whereas the FY2029 result ranged from JPY12,388 to JPY15,112. This is not a set of ten independent regional auctions: OCCTO clears a national market and assesses regional reliability, with splitting and documented pricing exceptions. Okinawa is outside this ordinary capacity auction. [226] [227] [228]

Intraday adjustment is traded energy, not final imbalance

JEPX's continuous intraday market allows participants to adjust energy positions after the day-ahead auction. The relevant transaction is a delivered half-hour, not an annual capacity commitment; final deviation settlement remains separate. The captured trading rules govern matching, eligible products, closure and exchange payment, while a participant's latest plan does not guarantee its metered outcome. [219]

Exhibit 124: Observed intraday adjustment activity



Note: January 2024–October 2026: monthly volume and volume-weighted price use matched positive-volume intervals. Missing pairs are excluded; this system series is not a Tokyo adjustment premium.

Source: [1] · Chart data

The retained public history reports system-wide interval average prices and trading volume. Monthly prices weight only observed, matched, positive-volume intervals; monthly volume uses the same rows. A system-wide intraday average cannot establish Tokyo's adjustment premium over its area day-ahead price, and an absent or unmatched price is not a zero-price trade. [1]

Japan's short-term interface gives weather and outage updates commercial importance. A revised solar outlook changes the retailer's need to replace or sell energy; a generator outage changes the feasible delivery plan. Intraday trading can close part of that mismatch before gate closure, but EPRX capability procurement and TSO balancing activation address a different service: keeping the physical system balanced when actual conditions differ.

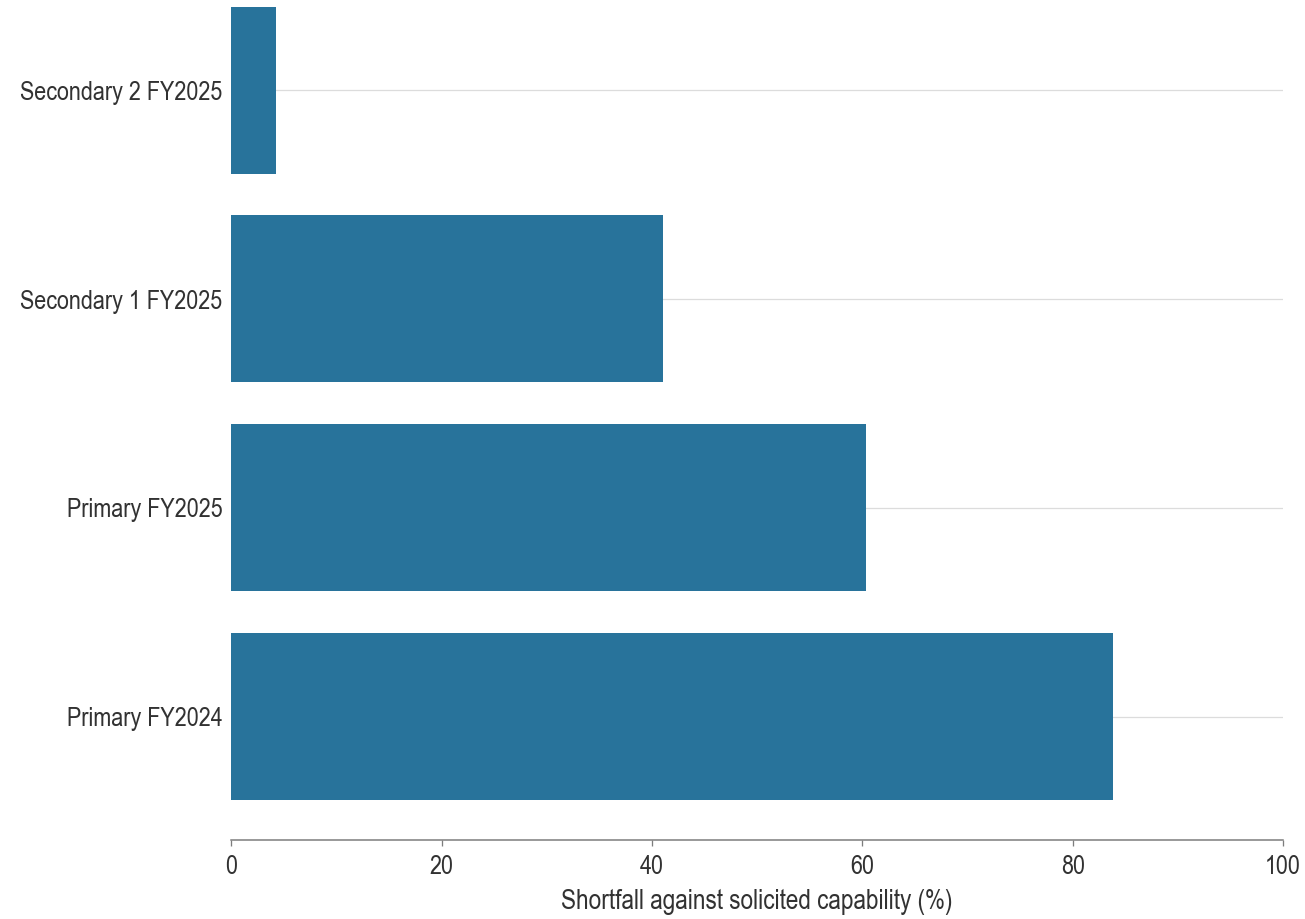
The four clocks remain distinct: day-ahead energy commitment → intraday adjustment → metered delivery → ex-post deviation settlement. An earlier plan guarantees neither plant availability nor the final rate.

Gross-bidding reform also affects exchange volume. Its April 2017 introduction, October 2023 suspension and June 2025 abolition change commercial routing; turnover should not be read as a stable fraction of Japanese consumption or independent-retailer dependence. [229] [230]

Balancing procurement: response capability can remain scarce

EPRX's FY2025 report records a primary-response procurement shortfall of **60.4%** of solicited capability, versus **83.8%** in FY2024. FY2025 secondary-1 was 41.1% and secondary-2 4.3%. These are shortfalls against market solicitation, excluding excess awards under the publisher's calculation, not percentages of consumers left without electricity. [231]

Exhibit 125: Balancing solicitation gaps are not unserved energy



Note: EPRX annual solicitation shortfall; excess awards excluded. FY2024 primary is the published comparator. Changed procurement arrangements prevent a reliability-rate reading.
Source: [231] · Chart data

The 13 March 2026 guide describes day-ahead procurement for all products, bids from 11:30 to 14:00 and clearing by 15:00. Historical observations must retain their earlier regime rather than inherit this timetable retrospectively. The guide does not establish an Okinawa balancing-market opening. [225]

Product	Response requirement	Sustained response
Primary / FCR	Within 10 seconds	At least 5 minutes
Secondary 1 / S-FRR	Within 5 minutes	30 minutes
Secondary 2 / FRR	Within 5 minutes	30 minutes
Tertiary 1 / RR	Within 15 minutes	30 minutes
Tertiary 2 / RR-FIT	Within 60 minutes	30 minutes

Qualification includes communications, monitoring and delivery, with the primary product's specified offline-monitoring exception. Batteries require duration and state of charge; thermal units require headroom, start capability and fuel. Awarded ΔkW is not observed activation MWh. Changed procurement rules also affect annual shortfalls; realized balancing profit requires matched instructions, metering and settlement.

Imbalance settlement: effective rules and publication vintages

Japan's April 2022 imbalance framework connects settlement to applicable marginal balancing-energy prices and a scarcity correction under specified conditions. Actual deviations are settled after physical balancing; they are not the same quantity as an awarded reserve or a JEPX area-price spread. Reserve-assessment blocks can differ from exchange price-coupling groups. [232] [21]

Dated evidence	Admitted mechanism	Boundary
April 2022 framework	Balancing-energy and scarcity components	Actual interval settlement needs its own record
METI 27 March 2026 response	Announced 1 October revision: C300, D50 JPY/kWh	Final operative notice not independently established here
Announced cumulative trigger	30 intervals at or above JPY200 in preceding seven days	Area spot prices; Okinawa uses imbalance prices
Announced temporary correction ceiling	JPY100/kWh from the following day	Not a cap on all wholesale prices
Announced release condition	Preceding seven-day count at or above JPY100 falls to zero	Separate from initial trigger

The official OCCTO committee material also describes C300/D50 as a planned change. It corroborates the policy direction, but it is not substituted for a final operator tariff or promulgated implementation notice. Direct official-notice searches and the later committee source did not establish that missing legal step; the October revision remains labelled announced in this edition. [233] [234] [235]

That qualification matters because the observation window reaches October 2026. Carrying old parameters forward as current law would be as misleading as treating an announcement as a verified settlement path. A price spike, support instruction and scarcity correction can share a stressed background without having the same geography or trigger.

For commercial exposure, preserve four times separately: the plan submitted, the information available before closure, the physical deviation and the rate ultimately published or revised. A historical final rate cannot be presented as a price a retailer could certainly transact at beforehand.

Regional capacity prices

Main-auction regional prices changed sharply across delivery years. FY2024's uniform JPY14,137/kW fell to JPY3,495 in the central areas for FY2025, while Hokkaido and Kyushu cleared at JPY5,242. By FY2029, Tohoku and Tokyo were JPY15,111, the central/western block JPY12,388, Hokkaido JPY14,972 and Kyushu JPY15,112. Auction year is four years before the labelled delivery year. [226] [236] [227]

Exhibit 126: Capacity prices separate by deliverable region

Note: Official main-auction area prices, delivery FY2024–2029; auction four years earlier. Okinawa excluded. Pricing exceptions and deductions mean individual payments can differ.
Source: [226] [227] [236–239] · Chart data

OCCTO first clears the national supply and demand curves, then tests regional supply reliability. It can add resources within a deficient block and remove resources from a surplus block where reliability permits. Thus the regional result depends on deliverability and the available resource set, not merely national aggregate nameplate. [228]

The pricing exception also matters. Where competition is limited after splitting, the applicable rules can set the area price at 1.5 times the adjacent-area price while paying higher-bid eligible resources their individual accepted prices. Transitional deductions further change contracted payment amounts. Multiplying every resource's MW by the displayed area price therefore need not reproduce the published total award yen.

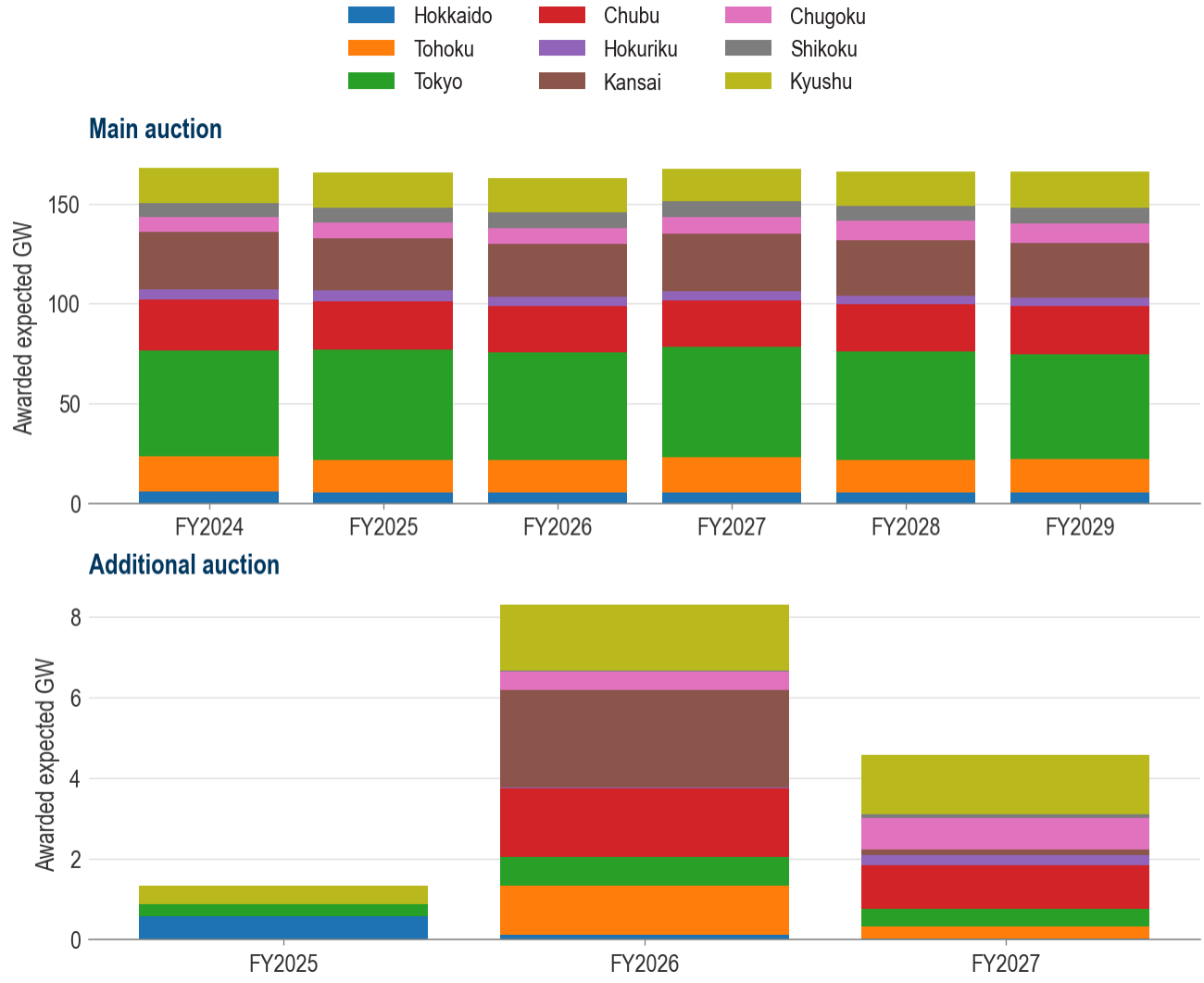
FY2026 illustrates why apparently small differences should not be rounded into one unconditional price: Tokyo was JPY5,834, Tohoku JPY5,833 and the central/western areas JPY5,832; Hokkaido and Kyushu were JPY8,749 and JPY8,748. These are actual published outcomes rather than a fitted regional fee. [237]

Capacity remuneration does not set a plant's half-hour spot offer or prove its fuel readiness. Regional fleet and outage patterns inform delivery risk, but a causal explanation for an auction-price change requires the auction's own supply, demand and qualification evidence.

Expected MW awards are not new plant

Main auctions awarded **167.692 GW** for FY2024 and **166.080 GW** for FY2029. Those are expected-capacity awards, not newly built plants, and the changing totals must not be appended to installed-capacity history. Additional auctions adjust the nearer delivery-year position: published awards were 1.333 GW for FY2025, 8.297 GW for FY2026 and 4.579 GW for FY2027. [226] [227] [240] [241] [242]

Exhibit 127: Main and additional awards have different clocks



Note: Expected GW, not operating nameplate. Main and additional panels use different scales. Original main awards cannot be added to later adjustments without exits and revisions.

Source: [226] [227] [236–242] · Chart data

The FY2027 additional auction awarded every submitted resource and reported JPY10,361/kW in all nine areas. Nevertheless, its clearing procedure distinguished Hokkaido from the other block. Equal displayed prices therefore do not prove that geography was irrelevant to the reliability assessment. The largest additional awards were Kyushu's 1,467.257 MW and Chubu's 1,094.291 MW; Hokkaido received only 28.792 MW. [242]

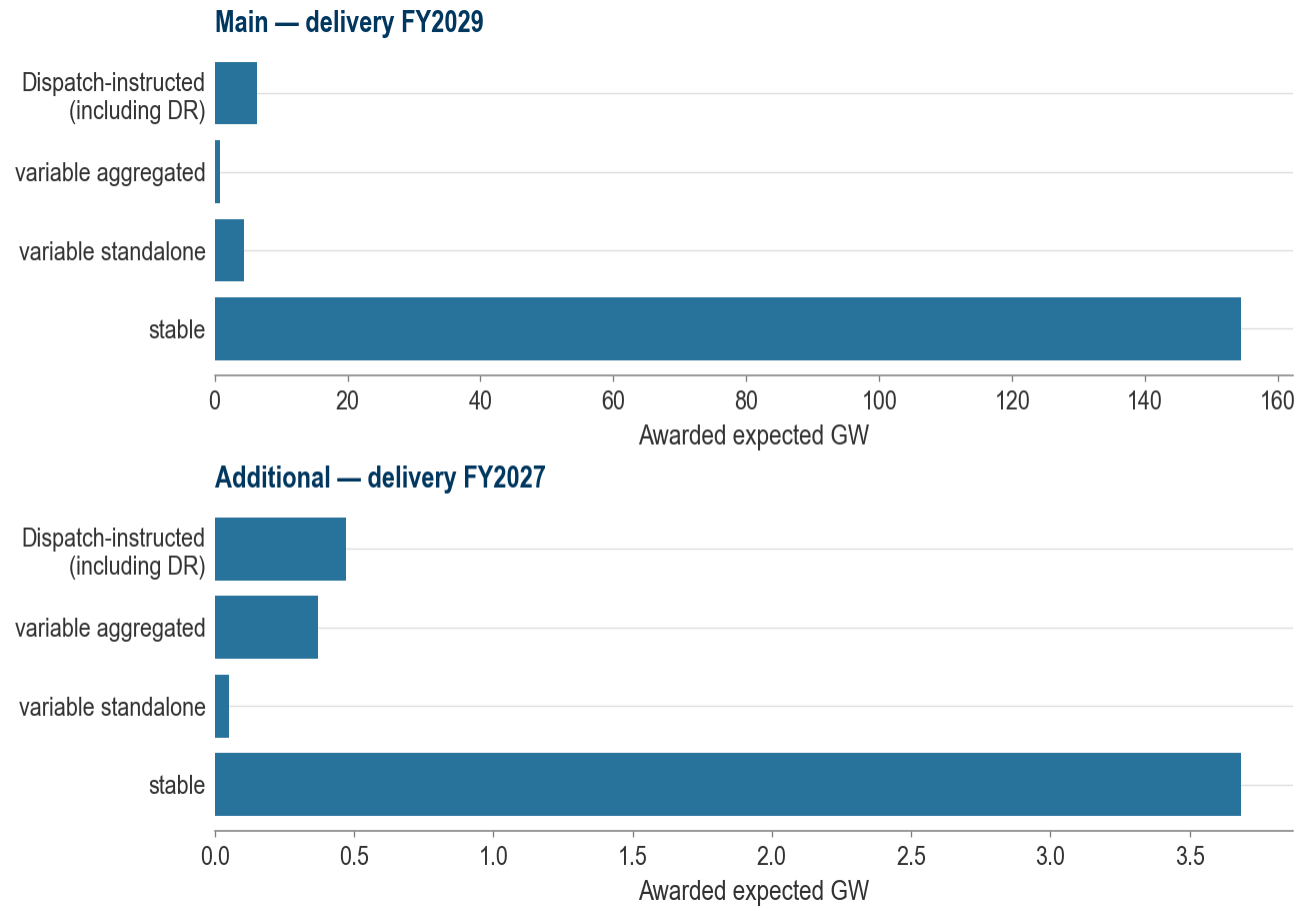
OCCTO's FY2027 report gives **167,402.399 MW** for combined main/additional contracted capacity using the main contract book at end-March 2026. This differs from simply adding the original main result to the new additional awards, because exits and contract changes intervene. FIT expected capacity and other outside-market supply are separately identified in the reliability assessment; they are not hidden additions to the auction quantity.

The four-year main commitment cannot establish future outages or fuel readiness. Additional procurement is conditional; dispatch-instructed resources require tests and accredited quantities, not theoretical demand reduction or generation capability.

Resource classes and performance

FY2029 main awards were **154.50 GW stable, 4.35 GW standalone variable, 0.84 GW aggregated variable and 6.39 GW dispatch-instructed**, at source precision. Acceptance was respectively 96.6%, 100%, 100% and 89.4%. The dispatch-instructed class includes demand response and qualifying other generation/aggregation, not DR alone. OCCTO suppresses regional class results revealing fewer than three participants. [243]

Exhibit 128: Accredited resource mix, not nameplate technology mix



Note: Published national resource classes at source rounding: FY2029 main / FY2027 additional. Stable includes qualifying storage; dispatch-instructed includes DR and other eligible resources. Confidential regional splits are not reconstructed.

Source: [242] [243] · Chart data

FY2027 additional-auction definitions place qualifying pumped storage and batteries with at least three hours' duration in the stable class, alongside qualifying generation. Expected capacity deducts station demand and output limits, with class-specific adjustment factors. Stable is not synonymous with fossil nameplate. [242]

Stable resources coordinate outages, maintain capability, offer unused capacity wholesale and respond to TSO instructions. Standalone variable resources retain maintenance obligations with separate natural-variability treatment; aggregated variable resources report their performance population. Dispatch-instructed resources face effectiveness tests and instructed delivery. [222]

The captured ordinary instruction framework specifies up to twelve annual activations, notice at least three hours ahead, weekdays 09:00–20:00 and a three-hour performance basis, with the applicable DR baseline. These are dated scheme requirements, not evidence that every awarded resource actually received twelve instructions. Shortfall, exit and failed tests can reduce payments or trigger penalties.

LTDA is separate: accepted investment bids receive conditional long-term support, in principle twenty years, under multi-price awards and other-market revenue-return requirements. An LTDA award is neither an ordinary annual capacity price nor a commissioned zero-emission asset. [244] [245]

Retailers fund capacity on peak allocation, not annual kWh

OCCTO allocates national capacity contributions across regions using H3 demand ratios. After applicable deductions and the network-business share, the regional retail burden is divided into monthly charges and assigned to retailers by the relevant peak-demand shares, with corrections for current-year changes. Annual customer kWh multiplied by a made-up national fee would not reproduce this allocation. [223]

Exhibit 129: Capacity delivery precedes collection and payment

-48 months	**Main auction**	OCCTO / qualified providers
-24 months	**Outage coordination / DR tests**	Providers / OCCTO / TSOs
-12 months	**Additional auction if required**	OCCTO / providers
+0 months	**Delivery year begins in April**	Providers deliver; TSOs assess
+3 months	**July: first April contribution invoice**	Retailers / networks pay OCCTO
+5 months	**September: first April capacity payment**	OCCTO pays providers

Note: OCCTO schedule captured 7 November 2026. April contributions first invoiced in July; April provider service first paid in September. Year offsets are indicative, not auction dates.
Source: [221–223] · Chart data

April–September retailer shares use the preceding summer peak evidence from July–September; October–March uses preceding winter December–February evidence. The current-year monthly contracted-demand comparison corrects for share changes. A retailer's expanding or shrinking customer book can therefore change allocation independently of the spot energy bought in a particular month.

The cash calendar introduces another separation. The first contribution invoice for April service is scheduled in July; the first provider payment for April service is scheduled in September, after actuals and collections. Monthly provider payment starts from one-twelfth of the applicable contract amount and deducts assessed penalties; sufficiently negative amounts can become a claim on the provider. These are OCCTO's captured ordinary schedules, not a promise that a given company's cash was received on a particular day. [222]

A bilateral power contract does not exempt the retailer from capacity contributions. OCCTO explicitly describes the need to review existing bilateral terms in light of separate generator capacity revenue and retailer contribution expense. Generator ownership and retail affiliation must therefore remain separate from the payment obligation.

The published FY2027 additional-result explanation states the relevant network burden basis using area main-auction price, H3 demand and an 8% share from FY2025 onward. Delivery-year manuals and invoice corrections govern the amount; the exhibit is not an estimate of a confidential retailer bill. [242]

Japanese wholesale access, bilateral supply and PPAs

An affiliated retailer, an independent supplier and a merchant generator do not obtain the same commercial exposure from identical physical output. Bilateral contracts, the JEPX baseload market, day-ahead/intraday purchases and PPAs allocate different delivery, replacement-energy and credit obligations. Public generation ownership does not disclose private procurement shares. [246]

Arrangement	Japan-specific commercial terms	Remaining obligation
Bilateral physical supply	Negotiated delivery, shape, revision and credit terms	Replacement supply, plans and deviation settlement
JEPX baseload market	Eligibility and one-/two-year delivery products; adjustment-coefficient variants	Not a financially settled baseload future
Physical PPA	Specified project/portfolio delivery and attribute provisions	Variable production, network service and balancing allocation
Financial PPA or swap	Index-linked cash settlement	Physical energy and scheduling still required
Retail supply	Customer contract, network use, capacity contribution and billing	Procurement cash can precede customer receipts

The dated baseload rules distinguish products with and without adjustment coefficients. Nondiscriminatory wholesale access commitments do not promise unlimited volume on identical duration, flexibility or credit terms. A route's existence is not evidence that every eligible retailer used it. [219]

Voluntary gross bidding began in April 2017, was suspended from 1 October 2023 and abolished in the June 2025 review. Internal commercial rerouting can therefore reduce exchange turnover without reducing end-use demand or generation. It cannot, on its own, prove either improving competition or deteriorating access. [229] [230]

Environmental rights require a separate contractual allocation. A PPA's renewable description does not establish certificate transfer, tracking or retirement; a financial PPA can settle cash without physically supplying the customer's region. [224] [220]

The fuel-adjustment lag on page 76, capacity collection on page 142 and exchange/clearing deadlines are separate funding clocks. No confidential procurement mix is imputed to close their cash-flow gap.

Listed contracts: Japan's area, profile and calendar basis

Japanese power futures settle financially against specified area-price averages. The venue, underlying area, eligible hours and clearing arrangement determine what remains unhedged; the contract does not reserve transmission or replace a physical procurement plan. [247] [248]

Dated family	Area / shape	Size and horizon	Clearing
TOCOM monthly, April 2026	Tokyo, Chubu, Kansai base/peak	100 kW × eligible hours; 24 months	JSCC
TOCOM weekly	Tokyo, Kansai base/peak	100 kW; five Saturday–Friday weeks	JSCC
TOCOM fiscal-year	Tokyo, Chubu, Kansai base/peak	100 kW; two fiscal years; cascades	JSCC
EEX futures, May 2026	Tokyo/Kansai short tenors; three areas monthly and longer	1 MW; day through fiscal-year set	ECC
EEX average-price options	Tokyo, Chubu, Kansai monthly base	1-MW equivalent; nine months including current	ECC
ICE JTB, captured November 2026	Tokyo all-hours base	1 MW × actual hours	ICE Clear Europe

EEX lists 34 days, five weekends, five weeks, ten months, seven quarters, eight seasons and six fiscal years, subject to area and contract settings; Chubu starts at monthly tenor. Its seasons are April–September and October–March, not the four meteorological seasons used for physical analysis. [249]

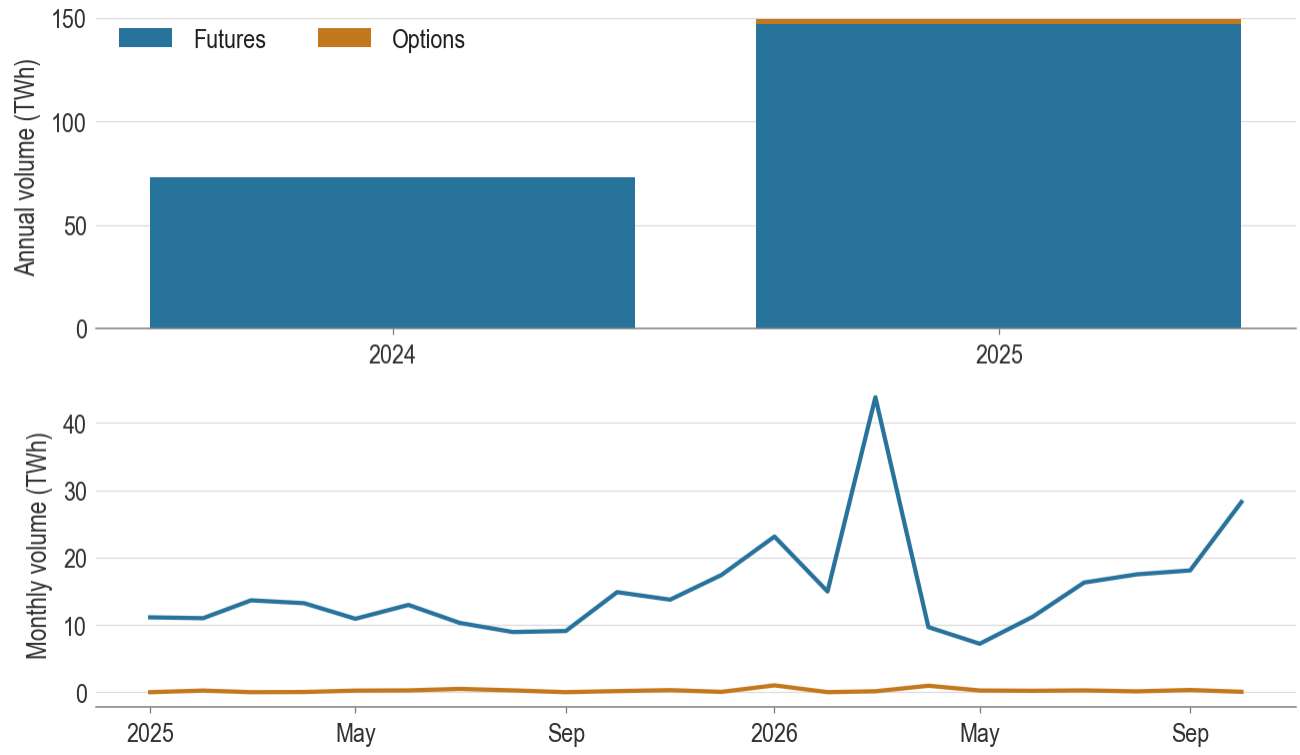
Standard peak uses 08:00–20:00 on eligible weekdays under the venue calendar; EEX day/weekend peak applies those hours every day. TOCOM's actual delivery-unit file gives 67,200 kWh per 100-kW February 2025 base contract and 74,400 kWh for March; their peak calendars contain eighteen and twenty eligible days. May has thirty-one days but eighteen eligible peak days. A fixed 720-hour conversion would be wrong. [250]

EEX options reference the monthly average rather than a selected extreme half-hour, and exercise creates the applicable underlying futures position. Listing terms establish permissible contracts, not evidence that every tenor actually traded. ICE's restricted historical activity is not replaced with invented volumes. [251] [252]

Exchange turnover is not consumption

EEX's annual tables distinguish **147.13 TWh of Japan power futures turnover and 2.02 TWh of options in 2025**. These are registered trading flows, not outstanding positions, plant generation or electricity finally delivered to consumers. Futures and options remain separate categories rather than being described as one physical energy market. [249]

Exhibit 130: EEX Japanese power: futures and options activity



Note: Official annual 2024–2025 and monthly January 2025–October 2026 registered turnover. Futures and options remain separate; turnover is neither open interest nor physical supply.
Source: [253–265] · Chart data

Monthly observations retain the official report vintages and later comparative columns used by the exchange. A growing trading series can reflect greater risk transfer, repeated trading of the same delivery exposure or changes in participation. It does not establish each retailer's dependence on exchange procurement or reveal a named generator's hedge book.

The May 2026 EEX overview records **119 participants at end-2025: 59 Japanese and 60 international**. This is a dated registered population, not 119 active daily market makers. Its statement that brokers represented over 90% of recent activity lacks a defined annual observation window and is therefore not treated as an annual market-share estimate. [249]

Execution and clearing are different events. An eligible bilateral derivative can be negotiated through a broker and then registered for exchange clearing; inclusion in registered turnover does not prove central order-book execution. Uncleared OTC deals are outside that observed exchange population, so no total-Japan OTC share is inferred by sub-

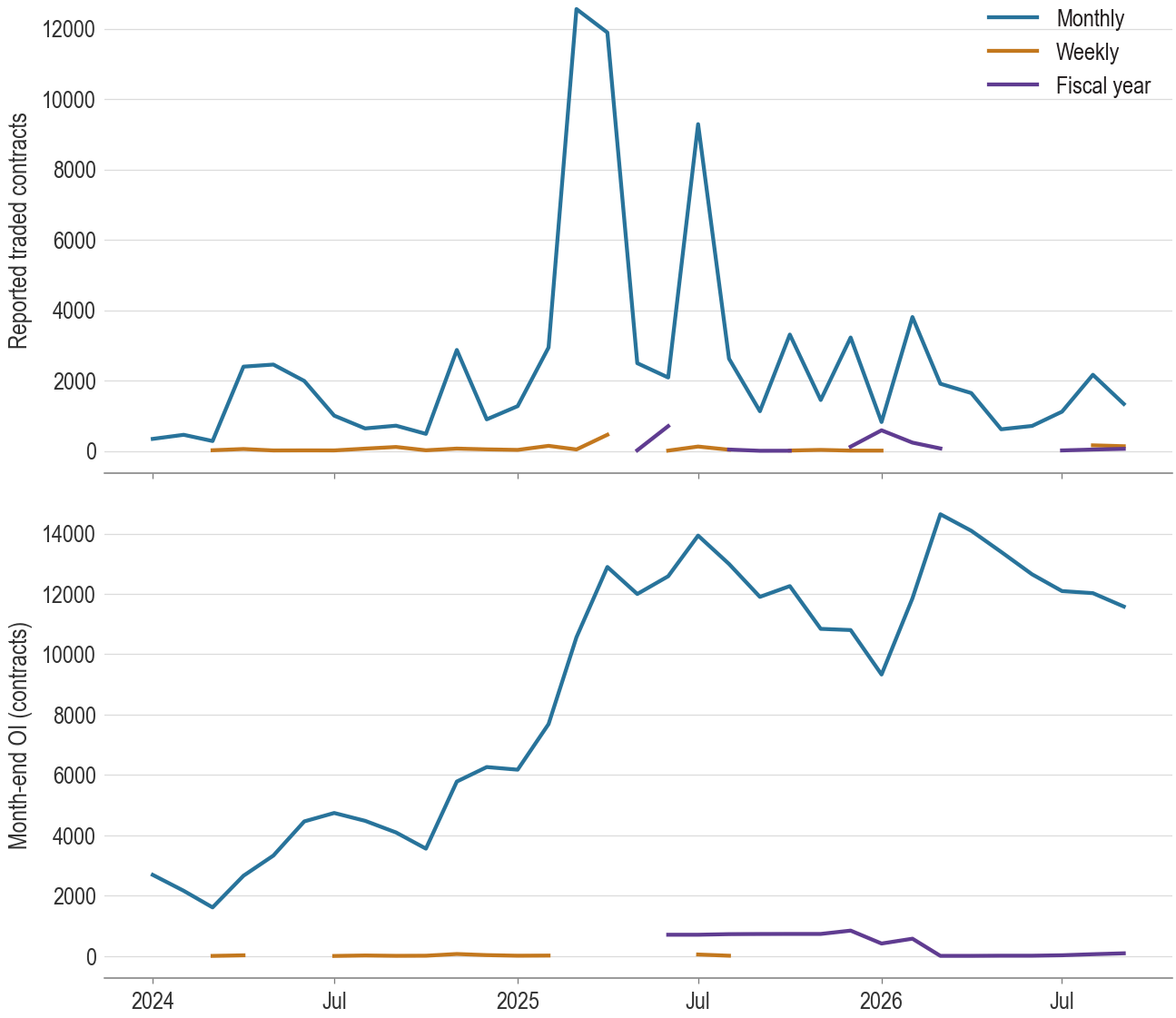
tracting one venue from another.

The regional contract matters as much as the venue total. A Tokyo monthly base position leaves differences in Chubu/Kansai price, customer shape, delivery volume and settlement timing unless those exposures are independently matched. Listing a Chubu contract does not show that every Chubu delivery month was liquid.

Open interest, clearing access and cash timing

JPX distinguishes monthly transactions from month-end open interest. Different maturities, base/peak calendars and cascading rules prevent a universal contracts-to-MWh conversion; blank source cells remain unquantified. [266] [250]

Exhibit 131: TOCOM turnover and month-end positions



Note: January 2024–September 2026: transactions during each month versus positions at month-end, by tenor. Source dashes remain unquantified; contracts are not a uniform MWh unit.
Source: [266] · Chart data

Brokers negotiate or register trades; clearing members provide CCP access and manage client collateral; CCPs supply a cleared counterparty framework. None establishes physical delivery, complete uncleared OTC notional or a universal client margin rate. Generator, retailer and trader participation does not reveal individual hedge books.

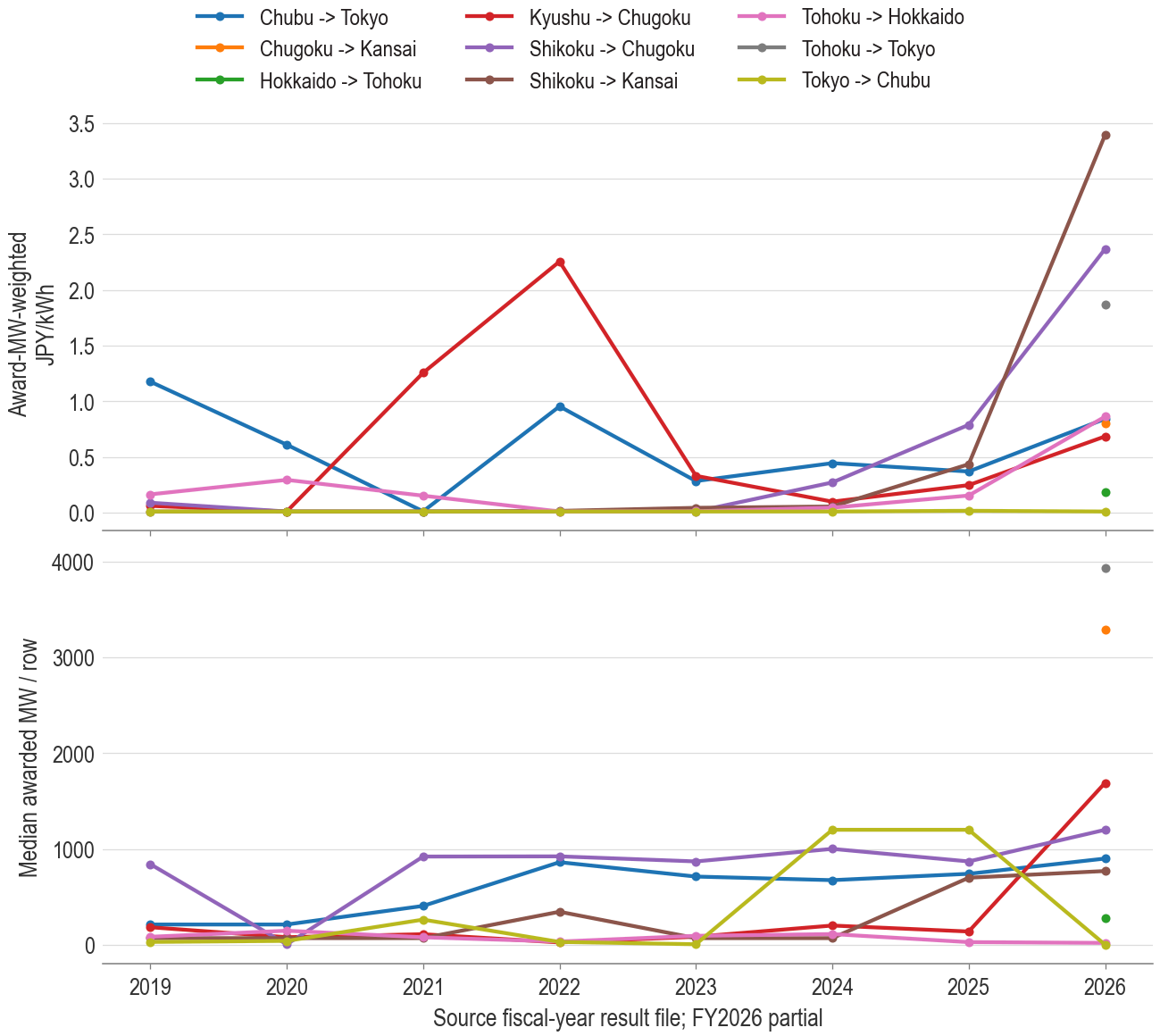
JPX's participant-volume publication excludes electricity and LNG futures, preventing named-firm electricity rankings. EEX registration totals likewise do not identify every end-client. Contract calendars on page 144 determine actual exposure. [267]

JSCC adopted historical-simulation VaR margin on **6 November 2023**. Initial margin is collateral, not automatically an expense; variation settlement and client terms create funding deadlines. JEPX ordinary spot settlement occurs two bank-business days after results. A hedge can protect eventual procurement value without supplying cash before the next invoice. No hypothetical calculation is presented as an observed Japanese margin call. [268] [219]

Directional transmission rights: actual auction history

JEPX indirect transmission rights settle specified directional area differences without reserving physical transfer. Result files retain trade dates, delivery windows, directions, offers, awards and auction prices—not measured interconnector flows. [269] [270]

Exhibit 132: Directional rights through changing auction years



Note: FY2019–partial FY2026, all reported directions. Award-MW-weighted price and median awarded MW per row; no invented energy denominator or physical-transfer entitlement.

Source: [269–276] · Chart data

The history preserves every reported direction, not a synthetic all-area matrix. A Tokyo–Kansai price spread implies neither a direct physical corridor nor a listed right; the FY2025 file has six directions and no Chubu–Kansai right. [271]

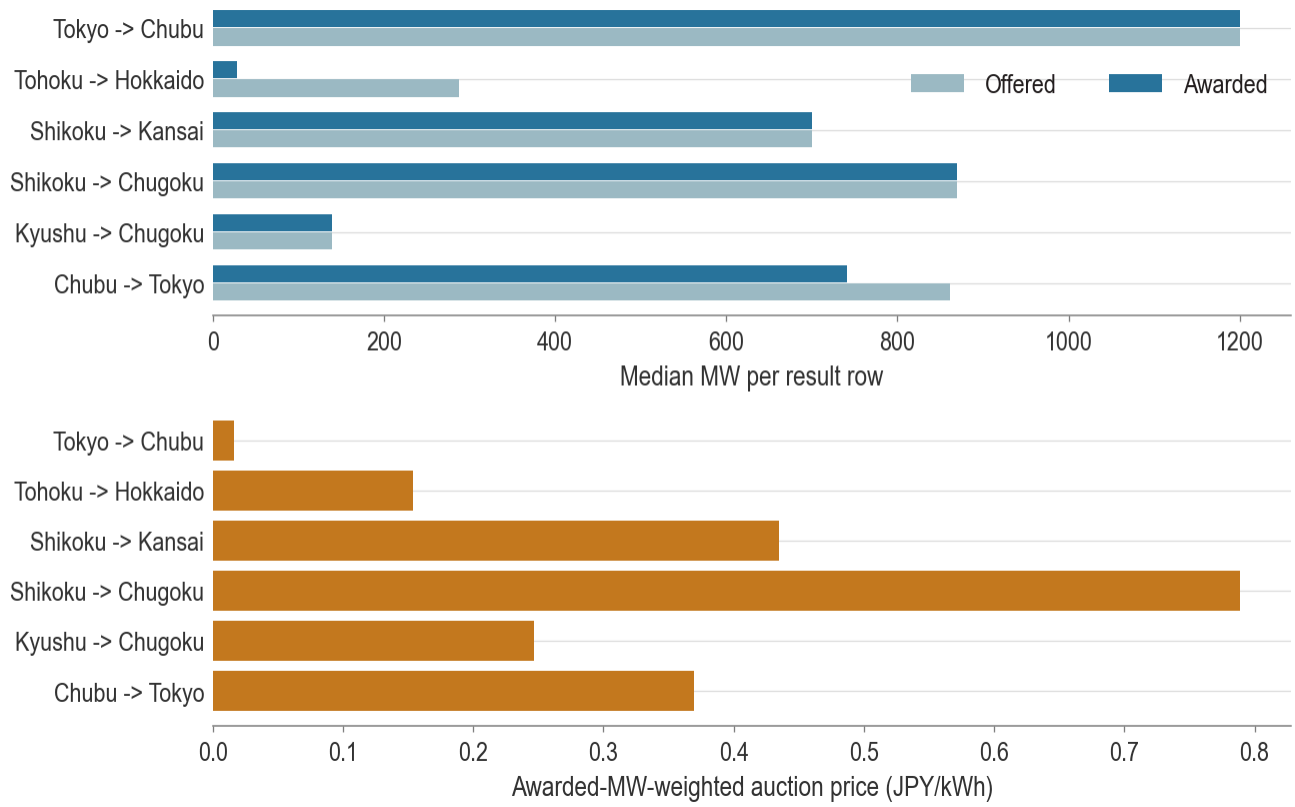
Offered MW is auction availability, not all-hour transfer capacity; **awarded MW** is accepted financial quantity, not matched customer load. **Auction JPY/kWh** is acquisition price, not the future realized area spread. Direction and delivery windows delimit eligible exposure.

Direction-year prices weight rows by awarded MW; quantities use median awards. Overlapping products cannot be summed into annual network capacity. Basis can reverse within a delivery window, while a retailer's shaped consumption need not match flat awards. Timing, quantity, fees and funding therefore separate observed spread from realized profit; historical prices do not prove an earlier executable hedge.

Rights awards leave quantity and internal-network exposure

FY2025 results compare all six published directions. Median offered/awarded MW describe auction rows, not annual capacity; the awarded-MW-weighted premium is not a realized transmission spread. [271]

Exhibit 133: Directional rights: actual auction outcomes



Note: FY2025, all six reported directions. Median MW per auction row; price weighted by awarded MW, not delivery energy. Trade dates and overlapping delivery windows remain separate in the extract.

Source: [271] · Chart data

A right leaves six exposures: **quantity** mismatches against actual net delivery; **hourly shape** outside eligible hours; **direction/region** mismatch; **internal-network constraints**; **plant availability**, including fuel loss; and **liquidity**, because premiums, settlement and collateral can precede customer receipts.

Financial area-difference settlement neither guarantees physical nominations nor settles final imbalance. JEPX prices are zonal: internal congestion does not create a plant-specific settlement point. [219] [277]

Connected-area spread evidence establishes sign and timing, not profit at an invented purchase price. Auction quantities remain separate from OCCTO transfer limits and TSO flows; matching their clocks and vintages is necessary before identifying a binding physical constraint.

Environmental procurement adds another requirement. A retailer may need electricity, risk-transfer contracts and eligible attributes, but purchasing one does not discharge the others. The next chapter separates national emissions duties, regional facility schemes and certificate ownership rather than treating every green-labelled purchase as compliance.

Environmental markets and obligations

Environmental obligations: national GX and electricity claims

Japan's statutory emissions-trading framework began with FY2026 under the GX Promotion Act as amended in June 2025. Eligibility is assessed at the level of the business entity using average direct CO_2 emissions over the preceding three fiscal years, and at least 100,000 tonnes brings the entity into scope; for the initial year, the assessment covers FY2023–FY2025. Emissions from purchased electricity do not count towards this direct-emissions test. [194] [195]

- Tokyo and Saitama use facility-level systems
- Allowances, credits and attributes serve different claims
- Compliance calendars and cost pass-through

FY2026 sequence	Obligation or legal treatment
From 1 April 2026	Calculate the relevant direct emissions
By 30 September 2026	Notify eligibility and submit the transition plan
By 30 September 2027	Initial-year allocation-related submission and emissions report under the special timetable
November 2027	Initial-year allocation and notification of the holding obligation under the published schedule
31 January 2028	Hold the required quantity for FY2026; the government extinguishes the obligation quantity

The initial-year timetable matters because the ordinary annual schedule cannot simply be shifted one year earlier. Allocation is initially free under prescribed rules, while the required holding quantity reflects verified emissions accounting and eligible adjustments. The official leaflet identifies J-Credit transfers and invalidations, and JCM invalidations, within that accounting, but it does not make every voluntary offset interchangeable with an allowance. [194] [196]

These are enacted and published implementation dates, and they carry no view on allowance prices. The captured price notice sets administrative price parameters and contains no transaction history, so the report does not manufacture a national allowance-price chart from J-Credit prices or from regulatory bounds. The compliance calendar on page 154 separates reporting, holding and retail pass-through clocks. [197]

A generation business can face direct-emissions accounting while its retail customers seek electricity attributes. Those obligations can coexist without being interchangeable. Nuclear generation can create non-fossil electricity value but not a renewable designation merely because it has low operational emissions. The same certificate cannot be reused to support incompatible claims by generator, retailer and consumer. [198]

Tokyo and Saitama use facility-level systems

The national direct-emissions threshold cannot be transferred to Tokyo's or Saitama's regional systems. Their large-facility rules use energy consumption in crude-oil-equivalent units and include electricity-related emissions within the applicable accounting boundary, so a facility can face regional reporting requirements even when its owner's national direct emissions fall below the GX threshold. [199] [200]

Feature	Tokyo	Saitama
Fourth period	FY2025–FY2029	FY2025–FY2029
Large-facility entry	1,500 kL/year crude-oil equivalent for three consecutive years for compliance designation	1,500 kL/year for three consecutive years for target-setting large facilities
Nature of requirement	Mandatory energy-related CO_2 reductions; reporting and verification	Target-setting reductions; mandatory plan/report requirements for specified businesses
Final timing	April 2030–September 2031 adjustment period	Target achievement by September 2031
Trading relationship	Eligible credits including qualifying Saitama credits	Eligible Tokyo-linked credits under stated restrictions

Tokyo distinguishes reporting designation, which follows the relevant prior-year consumption threshold, from compliance designation after three consecutive years. Saitama also has a broader business-level planning population that includes specified large retailers, and that population should not be confused with its large-facility trading population. [199] [200]

Both fourth-period documents describe expanded treatment of renewable electricity and certificates, yet eligibility, verification and transfer restrictions still apply, so buying a generic credit does not establish compliance. Tokyo's April 2025 leaflet explicitly places the fourth period through FY2029, with the adjustment period beginning in April 2030, which resolves an inconsistent date on the English landing page. These schedules define legal accounting periods and imply nothing about regional credit prices. [199]

For a Tokyo or Saitama commercial facility, purchased-electricity accounting therefore matters even where the national direct-emissions test does not. The contracting decision concerns eligible electricity and documentary evidence under the local scheme, not an inference that the physical electrons originate at a named power station. The regional rules' qualifying renewable-certificate treatment must be checked separately from a business's national GX holding obligation.

Allowances, credits and attributes serve different claims

A tonne-based instrument and an electricity attribute answer different questions. An allowance belongs to an emissions-accounting system, a project credit certifies an eligible reduction or removal against its methodology, and an electricity certificate records non-fossil attributes for a quantity of generation. Their units and cancellation rules determine which claim a buyer can make. [201] [198]

Instrument	Unit and basis	Use must be established under
National GX allowance	Scheme-defined tonnes of CO_2	Statutory holding and emissions accounting
GX League phase-one excess-reduction unit	Verified scheme-specific reduction	Voluntary phase-one rules and eligibility
J-Credit	Certified tCO_2 reduction or removal	Methodology, registry and receiving scheme
Tokyo/Saitama credit	Regional scheme-defined reduction	Regional eligibility and conversion rules
Non-fossil certificate	kWh of qualifying electricity attributes	Certificate category, tracking and retirement

GX League's voluntary phase-one framework preceded the statutory regime, so historical participation or a phase-one unit does not by itself establish a business's FY2026 legal obligation or the unrestricted usability of that unit in the new system. J-Credits also differ by project and category, and one quoted price cannot stand for all offset markets, still less for the statutory allowance price. [202] [194]

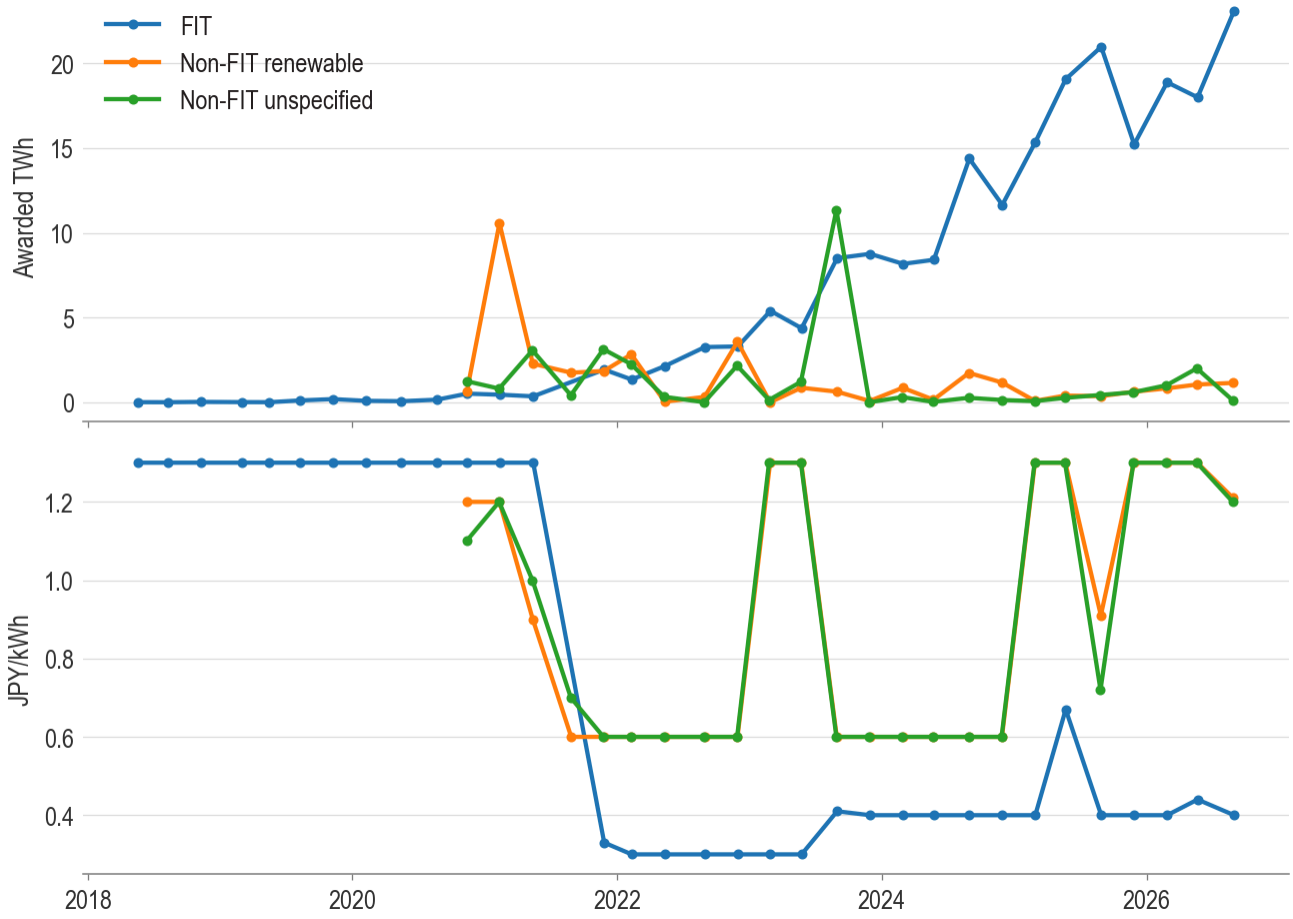
JPX's public daily carbon publication remains subject to its separate information-service and redistribution terms. Public visibility alone does not license external republication; no restricted transaction-price history or substitute J-Credit series is used as a national allowance price. The JEPX electricity-certificate auction evidence on the next page has its own source and retains its kWh unit. [203]

The transaction chain must end in the receiving scheme's recognized use: transfer to the proper registry account, verification of category and vintage, then holding, invalidation, retirement or allocation as required. An unretired certificate is not automatically a completed end-user claim. Contractual title, registry control and the right to make a public statement should agree; paying for energy alone does not establish any of them.

Non-fossil auctions reveal an attribute market

JEPX publishes separate auction results for FIT certificates and for non-FIT certificates with and without renewable designation, all of which trade electricity attributes rather than a carbon-price instrument. [198]

Exhibit 134: Non-fossil certificate auction observations



Note: The 81 dated auctions run from 18 May 2018 to 31 August 2026 across FIT and available non-FIT categories. The extract retains 21 undated placeholder rows separately; FY2026 remains partial. Prices concern electricity attributes, not emissions allowances.

Source: [204] [207–215] · Chart data

The FY2026 first-round results, dated 27, 28 and 31 August 2026, awarded about 0.115 TWh of non-FIT unspecified, 1.143 TWh of non-FIT renewable and 23.116 TWh of FIT certificates, at reported prices of JPY 1.20, 1.21 and 0.40/kWh respectively; FY2026 is incomplete and is not annualized. [204]

Because the FIT result also publishes maximum and minimum awards, its reported price should not automatically be read as one uniform price paid on every certificate. Changes in eligible buyers, categories and price rules complicate long comparisons, and the lines join discrete auctions rather than tracing a continuously traded market. [205] [206]

The price contrast across categories is an observed attribute-market outcome, not an emissions-abatement cost curve. FIT certificates, non-FIT renewable certificates and non-FIT unspecified certificates have different eligible populations and demand. An unspecified non-fossil certificate can include nuclear value, so a buyer seeking a renewable claim must select the appropriate category and evidence rather than simply the cheapest non-fossil label.

FIT, FIP and PPAs: who owns the claim?

Buying electricity and acquiring its environmental attributes are related but separate contractual acts. Under FIT, the support mechanism and certificate arrangements mean that a purchaser cannot assume the renewable claim travels with an anonymous exchange purchase. FIP and non-FIT contracting can leave more scope for the producer and buyer to agree how attributes are sold, but the agreement still has to identify the certificate, tracking and retirement pathway. [198] [224]

Because non-fossil electricity can include nuclear generation, the category alone does not establish a renewable claim; a buyer needs the appropriate renewable designation and supporting evidence as well as a contractual right to the attribute. [198]

Procurement route	Electricity or cash relationship	Attribute question
Anonymous spot purchase	Area electricity settlement	Which separately acquired certificate supports the claim?
FIT-supported generation	Purchase under support arrangements	Has the relevant FIT certificate been acquired and allocated?
FIP/non-FIT physical PPA	Contracted energy sale with specified terms	Are certificates retained, bundled or sold separately?
Financial PPA	Cash settlement against an agreed reference	Does a separate clause transfer valid attributes?
Retail renewable tariff	Retail supply contract	Which tracking, matching and cancellation evidence reaches the customer?

A financial hedge against power prices does not itself deliver renewable electricity attributes, and a certificate can support an eligible electricity claim without hedging the buyer's regional energy price. Even when sold together as one procurement package, the two contracts can therefore leave different basis, volume and accounting risks.

For compliance and disclosure, the attribute has to be followed from qualifying production through transfer to final retirement or allocation. The generation vintage and eligible consumption population must match under the receiving rules, and the same attribute must not support incompatible claims by several parties. Tokyo and Saitama's fourth-period documents permit specified certificate uses under their own accounting conditions, and those permissions should not be generalized into unrestricted substitution for national GX allowances. [199] [200]

Compliance calendars and cost pass-through

Electricity delivery, environmental reporting and legal discharge occur on different calendars. The responsible entity must identify the scheme, accounting boundary, eligible instrument and final-use step before assigning a cost to a trade or customer. An allowance's gross opportunity value, a cash acquisition requirement and a tariff adjustment are different measures. [194] [190]

Clock / responsible bodies	Operative milestone	Commercial consequence
National GX: METI and statutory scheme administrator	FY2026 emissions; eligibility/transition submission by 30 September 2026	Business-entity direct-emissions scope, not purchased-electricity entry threshold
National GX initial-year special schedule	Submission/report by 30 September 2027; allocation/holding notification November 2027; required holding 31 January 2028	Initial-year compliance cannot be moved forward using the ordinary annual template
Tokyo / Saitama governments	Fourth period FY2025–FY2029; final adjustment/achievement through September 2031 under each scheme	Facility obligations and qualifying linked-credit uses remain scheme-specific
JEPX non-fossil value market	Published auction date, generation vintage and applicable transfer/allocation rules	Electricity attributes do not settle energy delivery or national allowance obligations
FIT/FIP administration and PPA parties	Support eligibility, generation and attribute allocation under operative rules	Energy-price exposure and renewable claim can be owned by different parties
ANRE tariff / statutory fuel tax	Lagged regulated adjustment; fuel-specific statutory tax point	Customer pass-through need not coincide with spot or compliance cash cost

The GX timetable reflects the published initial-year implementation rather than a price forecast. Tokyo's fourth-period adjustment begins in April 2030; Saitama's target-setting scheme is not relabelled as Tokyo's mandatory reduction regime. [196] [199] [200]

The Electricity and Gas Market Surveillance Commission oversees the relevant electricity/gas trading conduct; JEPX administers its markets; METI/ANRE administer energy policy and the applicable national frameworks. NRA's nuclear-safety role is separate from emissions allowance allocation. No single trading venue discharges all these public obligations.

For generators, free allocation can reduce cash purchases without making emissions irrelevant at the margin. For retailers, certificate acquisition, capacity contributions and wholesale purchases can precede customer receipts. Contract terms determine whether and when those costs pass through. Statutory petroleum/coal tax already includes its climate component; adding it again would overstate costs. [191] [193]

The event studies now return to actual system conditions: fuel endurance, weather, fleet availability, renewable control and regional price separation. Legal and commercial mechanisms remain context, not invented causal explanations for each episode.

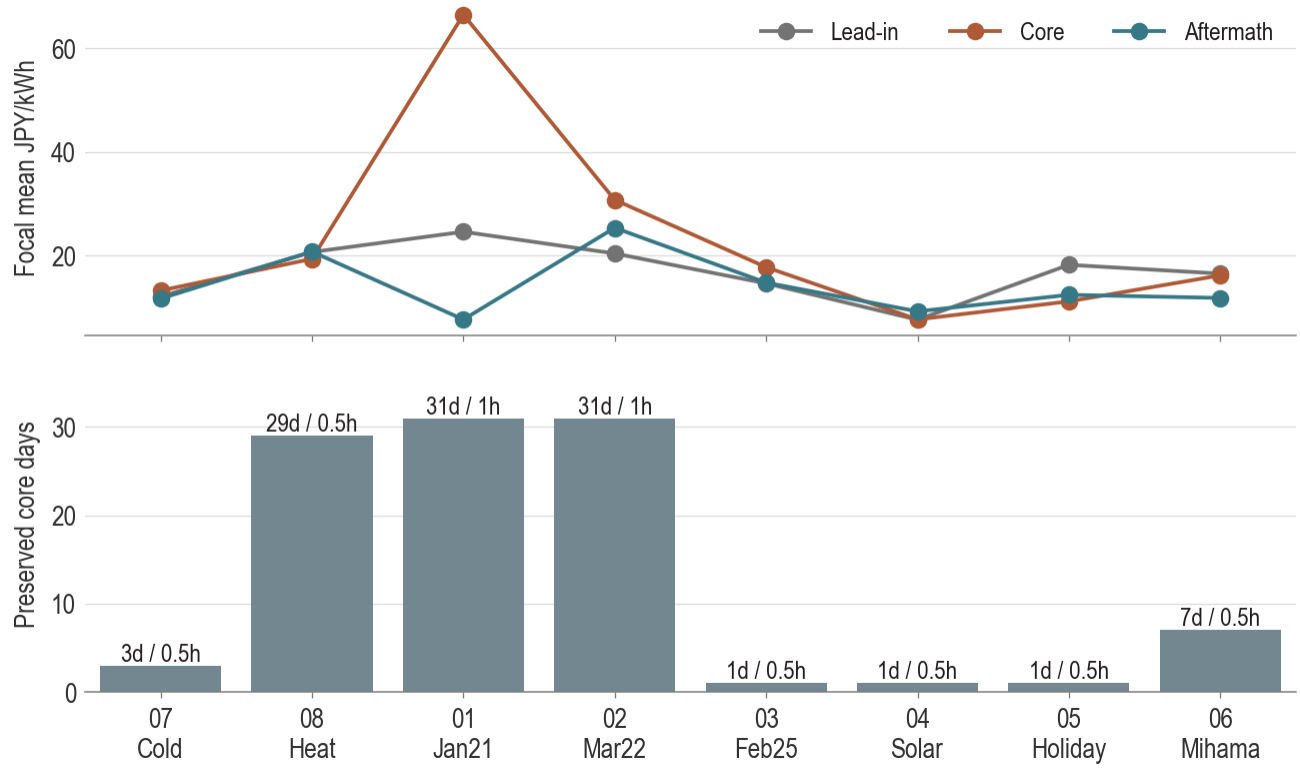
Historical event studies

Eight cases: evidence before explanation

Each preserved case has six parts: **A** preconditions, **B** chronology/actions, **C** physical response, **D** all-nine prices, **E** aftermath, **F** comparisons. Six distinct composites separate these questions; none substitutes an invented dispatch trace. [286]

- 07 · Feb 2026 cold · A. Preconditions
- 01 · Jan 2021 scarcity · A. Preconditions
- 06 · Sep 2026 Mihama · A. Preconditions

Exhibit 135: Eight preserved cases: price context and native clocks



Note: Upper: focal prices before/during/after. Lower: core duration / native physical interval; prices always half-hourly.

Source: [1] [286] · Chart data

Case / preserved core	Diagnostic	Case / preserved core	Diagnostic
07 · 7–9 Feb 2026	Northern cold/support	03 · 19 Feb 2025	Price-selected winter day
08 · 3–31 Aug 2026	Heat/Ohi 3 trip	04 · 10 May 2026	Solar curtailment/storage
01 · January 2021	Fuel scarcity	05 · 23 Sep 2024	Holiday spread reversal
02 · March 2022	Earthquake/cold alert	06 · 12–18 Sep 2026	Interrupted Mihama startup

Read in order 07,08,01–06. The atlas is not a severity ranking: focal areas and durations differ. Physical data are hourly for 2021/2022 and half-hourly for modern cases; prices are half-hourly, weather hourly. No finer history is synthesized. [285]

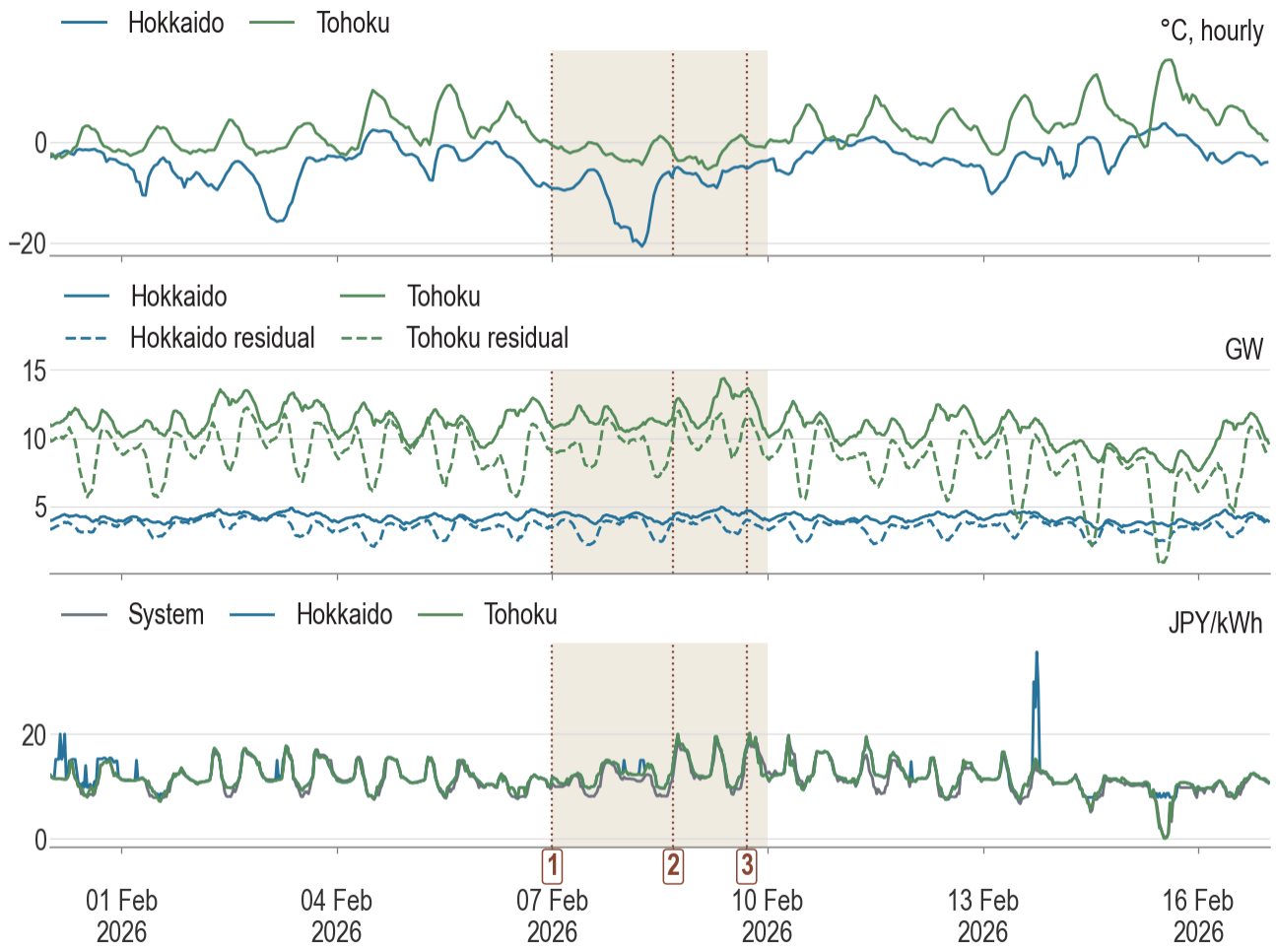
References preserve weekday/holiday class, 56-day pre-onset search, exclusions and complete native days. Alternative calendars are explicitly labelled. Instruction maxima are not delivered energy; observed associations do not identify causal fuel, unit or transmission effects.

Case 07 · February 2026 northern cold shock

07 · Feb 2026 cold · A. Preconditions

February 7–9 is a short northern cold incursion, not the whole month. JMA weather evidence and actual OCCTO support establish the selection independently of prices. The warm wider-month context does not erase three cold days. Hokkaido/Tohoku are focal; all nine prices remain in part D.

Exhibit 136: 7–9 February 2026: preconditions and onset



Note: 31 January–16 February 2026; shading marks 7–9 February. Hourly point weather sits above half-hour demand, dashed residual load and day-ahead prices. Numbered dates distinguish the cold onset and the two documented support dates.

Source: [1] [2] [4] [5] [287–290] · Chart data

Lead-in January 31–February 6 and aftermath February 10–16 are unchanged. The table quantifies colder weather and higher demand; it does not prove reserve exhaustion. Instructions, measured flows and prices retain separate evidence and clocks.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	12.084	4.236	-4.52
core	13.238	4.337	-8.98
aftermath	11.685	4.120	-2.36

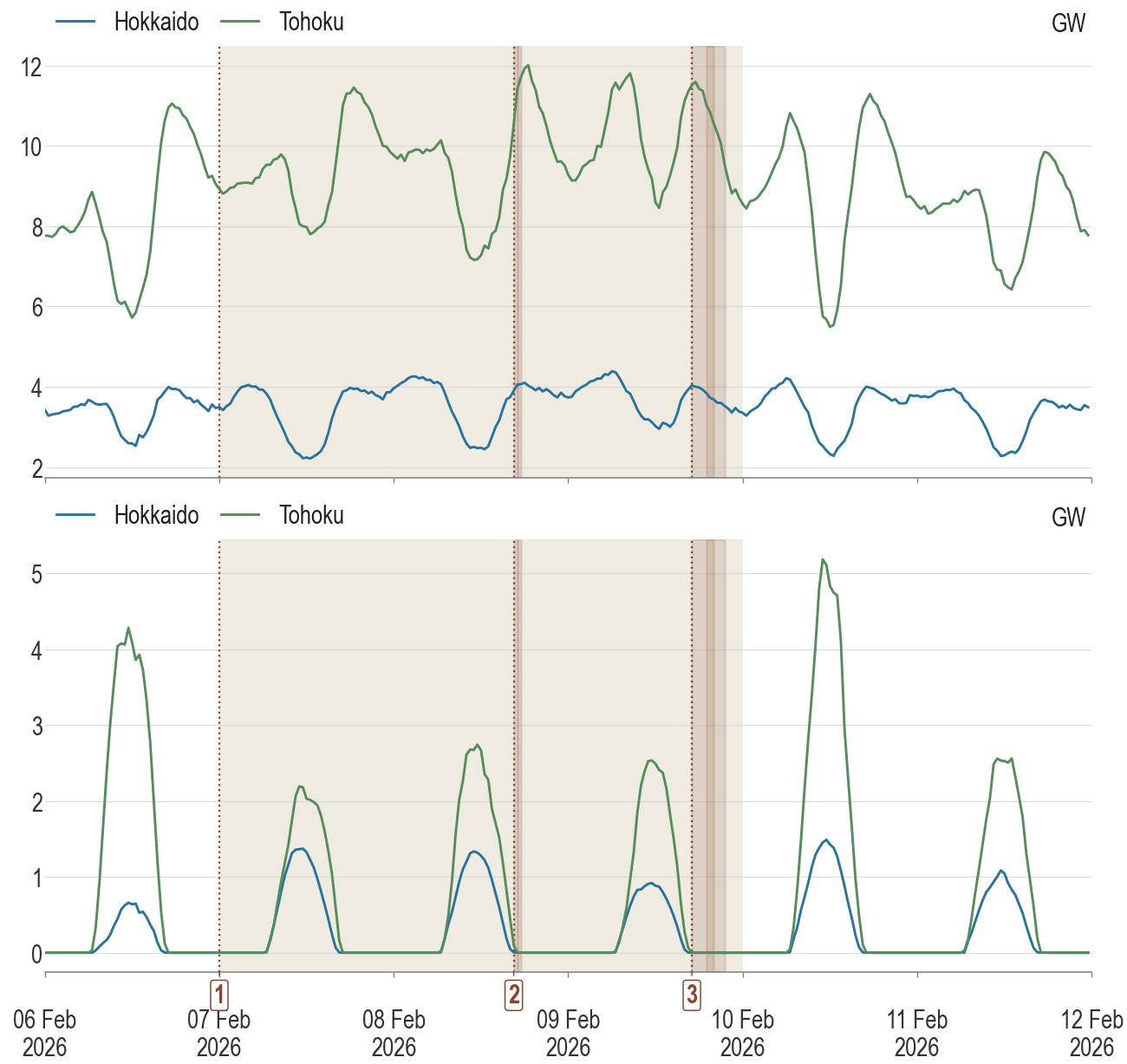
[287] [288] [289] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

07 · Feb 2026 cold · B. Chronology

Keep the four actual support spans separate. Hokkaido received instructions for 90 MW on February 8 at 16:30–17:00 and another 90 MW at 17:00–17:30. Tohoku's February 9 spans were 17:00–20:00 with a 463 MW maximum and 19:00–21:30 with a 406 MW maximum. The Tohoku spans overlap; their maxima must not be added as simultaneous import or converted to actual MWh.

Exhibit 137: 7–9 February 2026: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [4] [287–289] · Chart data

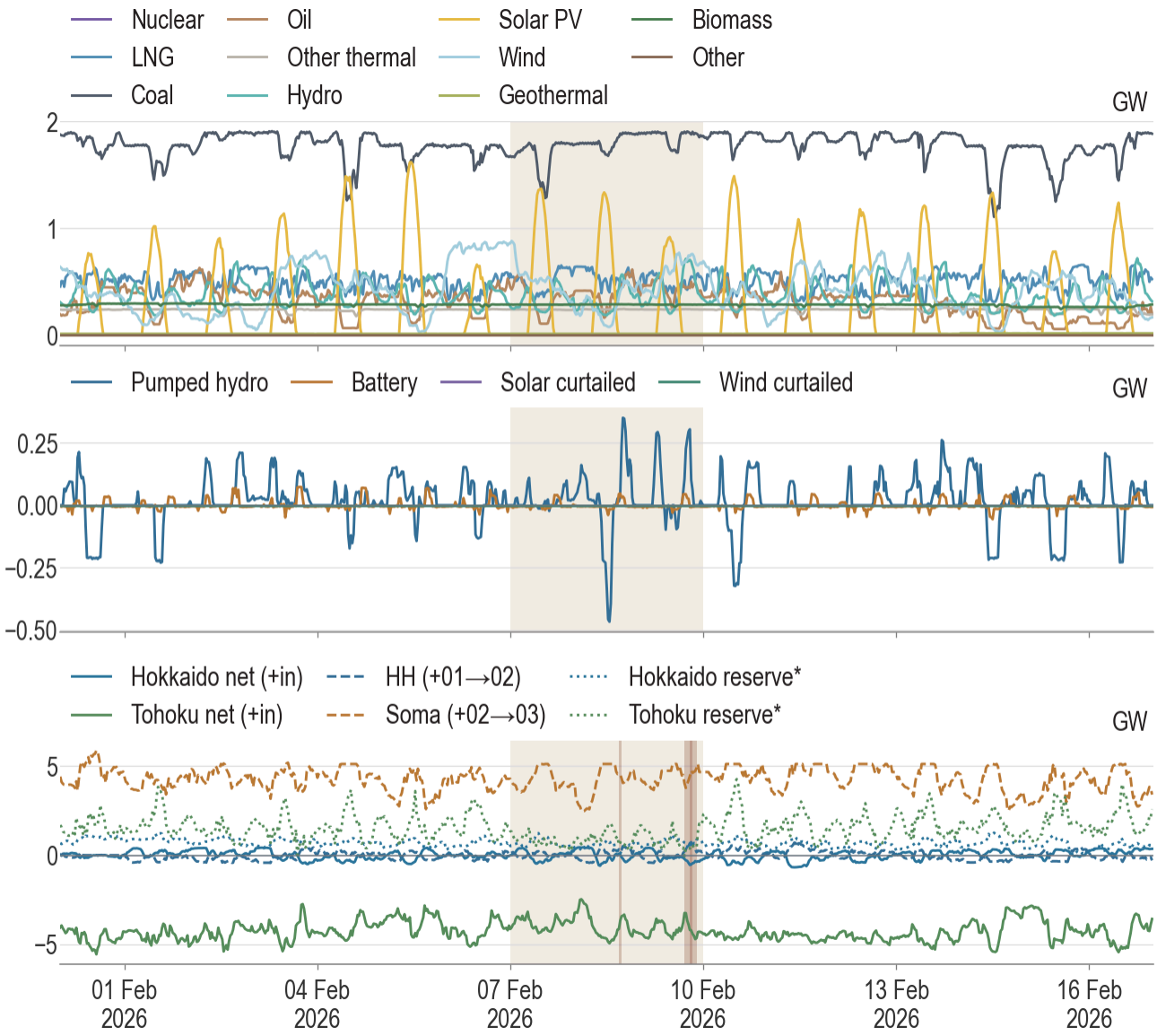
The chart aligns residual load and solar with dated actions, not with an invented instruction-MW trajectory. Native measured interconnector flow on the physical page is an independent observation. The preserved case selection rejects later February 15 support as a separate episode because the weather correspondence is weaker; this is not a claim that support never occurred outside the core. A trader should distinguish the instruction issue, the delivery interval, observed flow and the prices for those hours.

[287] [288] [289] [291] [285]

07 · Feb 2026 cold · C. Physical response

Hokkaido's mean thermal output rose from 2.852 GW in the lead-in to 2.935 GW in the core. Wind averaged 459 MW and solar 302 MW during the cold interval; both were higher than their lead-in means. That explains why a higher load did not translate one-for-one into higher residual load. Tohoku thermal output rose from 11.378 GW to 11.667 GW, alongside a stronger residual-load increase.

Exhibit 138: 7–9 February 2026: physical response



Note: Hokkaido generation and signed storage accompany regional interchange and final schedules. Shading marks the core; darker spans locate timed instructions. HH is Hokkaido→Tohoku, Soma is Tohoku→Tokyo. Schedules retain their source clock and are not measured deliveries. Dotted reserve* is the latest published area plan, not telemetry or an issue-time vintage.

Source: [4] [10] [11] [287–289] [293–295] · Chart data

Hokkaido pumped storage generated 3.4245 GWh and consumed 1.5345 GWh over the 72-hour core; batteries supplied 0.2855 GWh and charged 0.4350 GWh on the reported sign convention. These are observed interval-energy integrals, not attributable energy from OCCTO's support instructions. Gas, coal and other thermal contributions retain their native categories, and boundary sign must be read from its declared orientation rather than guessed from an area name.

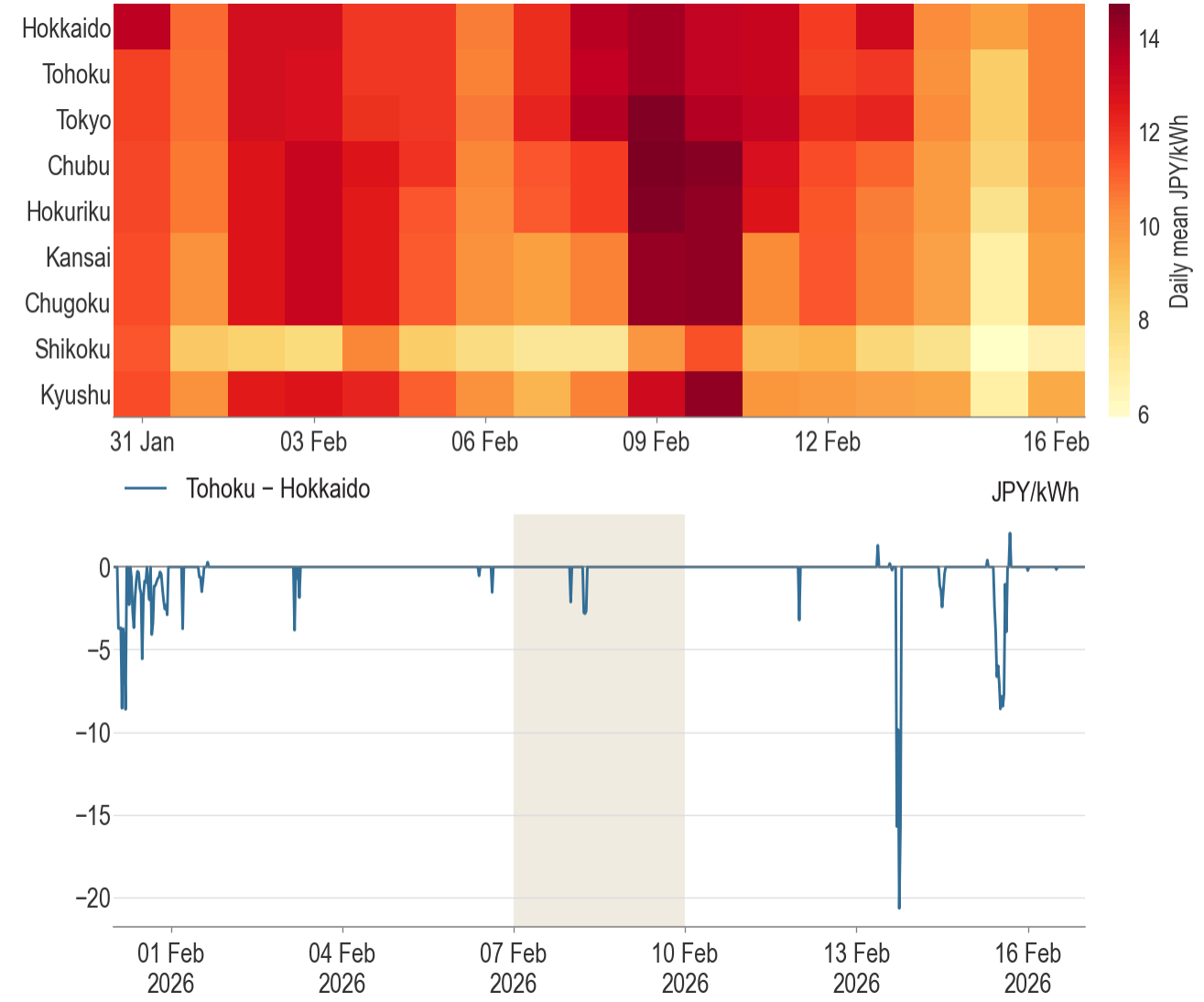
[287] [288] [289] [292] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

07 · Feb 2026 cold · D. Regional prices

The northern support event did not produce a uniform national scarcity price. Hokkaido and Tohoku core means were 13.238 and 13.166 JPY/kWh; Tokyo was 13.539, while Kansai and Chugoku were 11.512 and Shikoku 8.129. The full nine-area table prevents a focal-area narrative from becoming an unsupported national one.

Exhibit 139: 7–9 February 2026: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [287–289] · Chart data

Daily heat-map cells require all 48 valid half-hours. The lower panel retains native focal-area spreads, so an apparently modest three-day average cannot conceal short-lived separations. A spread alone does not identify which constraint, outage or offer set caused it. Hokkaido–Tohoku support is a documented system action, but the resulting prices also coexist with different renewable output, weather and demand across areas.

North/east	Central/west	West/south
Hokkaido **13.238**	Chubu **12.554**	Chugoku **11.512**
Tohoku **13.166**	Hokuriku **12.491**	Shikoku **8.129**
Tokyo **13.539**	Kansai **11.512**	Kyushu **10.833**

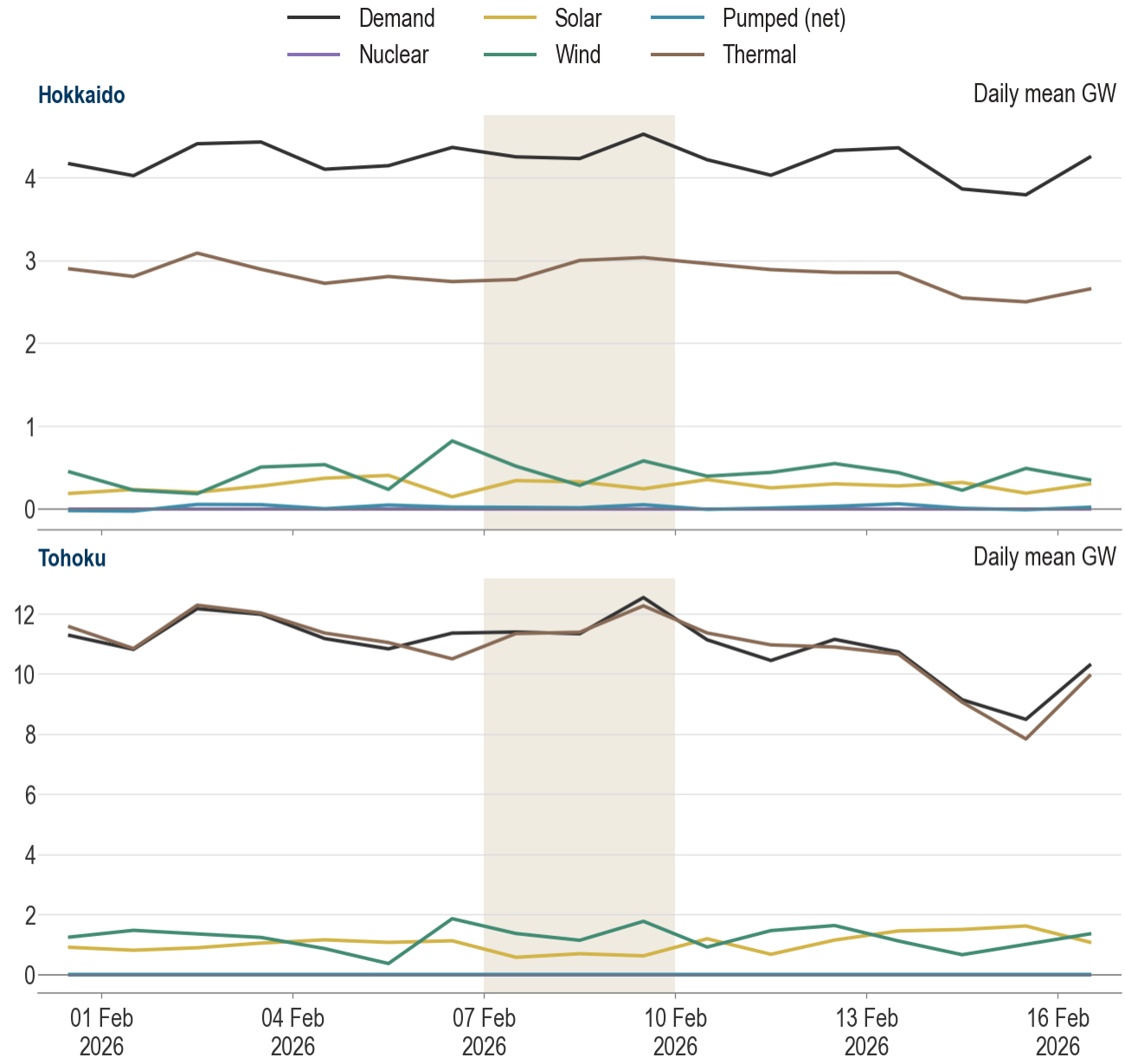
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[287] [288] [289] [296] [285]

07 · Feb 2026 cold · E. Aftermath

Recovery is an observed change in system conditions, not merely the last shaded day. Hokkaido aftermath temperature averaged−2.36°C, demand 4.120 GW and thermal output 2.752 GW. Tohoku demand fell to 10.199 GW and residual load to 7.804 GW, compared with 11.760 GW and 9.709 GW during the core. The seven-day aftermath therefore shows more than a price retracement.

Exhibit 140: 7–9 February 2026: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [4] [287–289] · Chart data

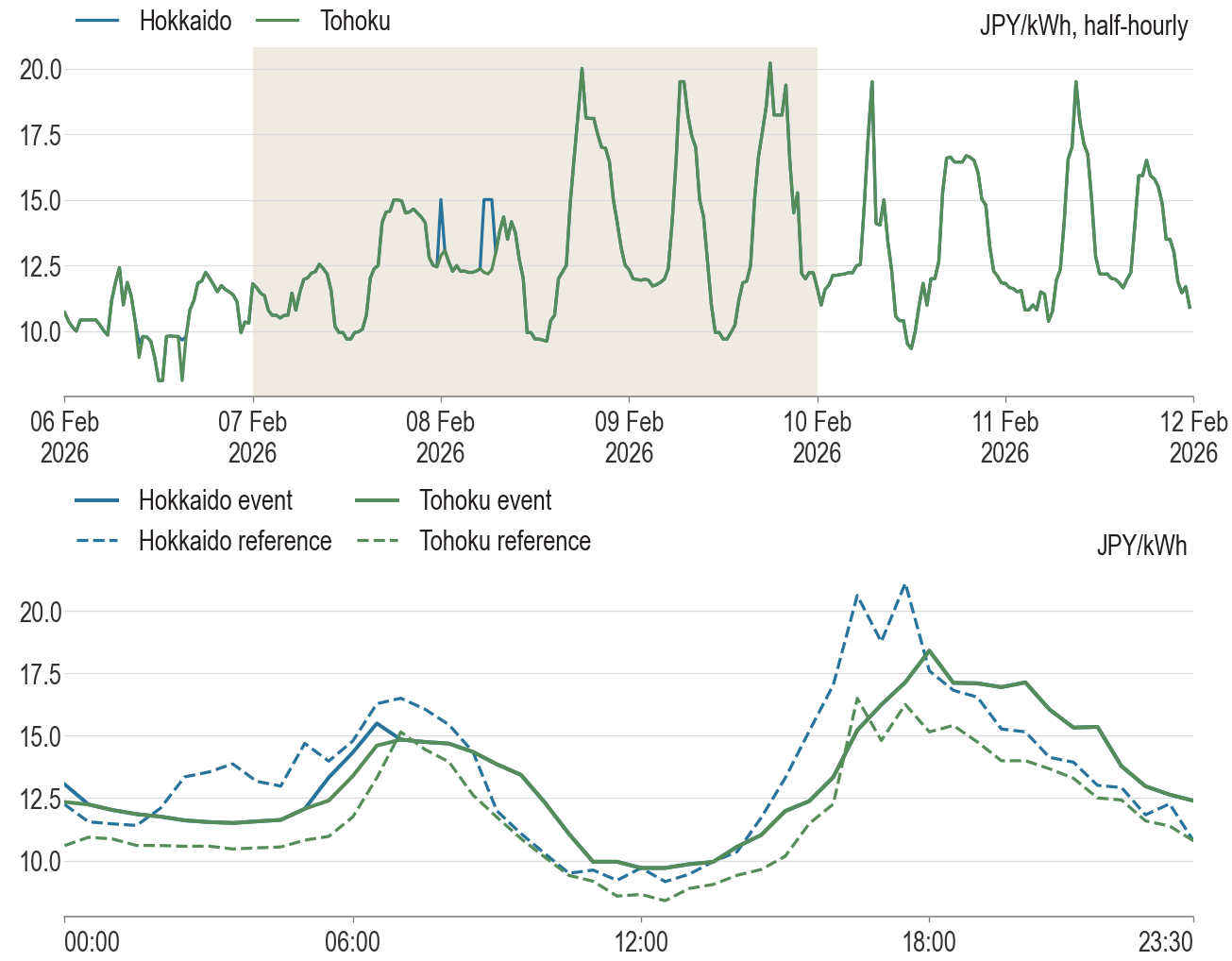
The physical panels retain daily demand, thermal, nuclear, solar, wind and the signed pumped-storage channel for both focal areas. Daily aggregation is used only on complete native observations. A support instruction ending does not certify every unit recovered, nor does weaker demand imply fuel inventories were restored. The evidence supports easing temperatures and load plus lower observed thermal production; it does not support an unobserved inventory or reserve-margin trajectory.

[287] [288] [289] [285]

07 · Feb 2026 cold · F. Comparisons

The original calendar comparison changes the interpretation. Hokkaido's core price was 13.238 JPY/kWh versus 13.458 on eligible same-weekday/holiday references, even though demand was 111 MW higher. Its residual load was 101 MW lower than that reference because wind and solar matter. Tohoku instead had a 1.425 JPY/kWh price premium and 754 MW more demand.

Exhibit 141: 7–9 February 2026: prices and matched references



Note: Half-hour prices around the cold spell sit above calendar-matched price shapes. Solid profiles are complete event days; dashed profiles average their eligible prior references. Support can be necessary without an unusually high daily price.

Source: [1] [2] [286–290] · Chart data

References are selected separately for each of the three event days from the nearest up tofour complete prior same-weekday, same-holiday-class days within 56 days of onset. Event days receive equal weight; a union of reference dates is not a second sample to average indiscriminately. The simple lead-in comparison is a separate diagnostic, not a substitute for this calendar rule. Different conclusions under these two comparisons demonstrate descriptive sensitivity, not proof that the instruction was ineffective.

Area	Event*	Reference*	Eligible core days
Hokkaido	13.238	13.458	3
Tohoku	13.166	11.741	3

*JPY/kWh; matched event population, not necessarily every day of the full core.

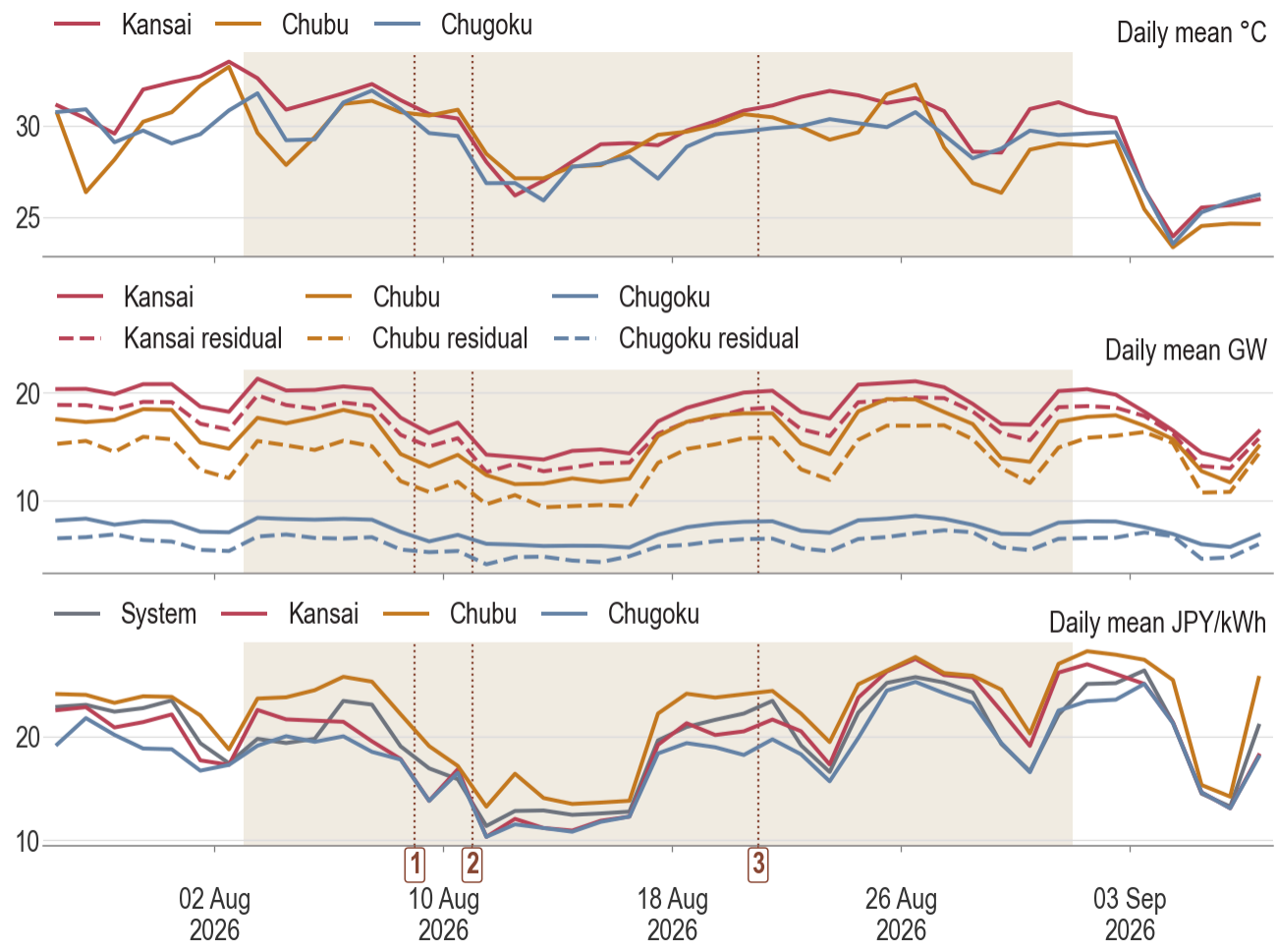
[287] [288] [289] [297]

Case 08 · August 2026 heat and support phases

08 · Aug 2026 heat · A. Preconditions

The August 3–31 core retains July 27–August 2 lead-in and September 1–7 aftermath. Kansai is focal, with Chubu/Chugoku neighbors. JMA heat, OCCTO support and the Ohi 3 operator notice are distinct evidence families.

Exhibit 142: 3–31 August 2026: preconditions and onset



Note: 27 July–7 September 2026, with the 3–31 August study shaded. Complete-day means preserve the cooler middle phase; demand is solid and residual load dashed. Markers locate Ohi 3's trip and the two calendar-phase boundaries, not support instructions.

Source: [1] [2] [4] [5] [290] [298–301] · Chart data

Mihama 3 was already stopped after the May 8 steam leak and June 16 inspection start; Ohi 3 tripped late August 8. Installed nuclear capacity therefore misstates availability. Part C separates early, middle and late phases hidden by the full-core means.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	20.679	19.884	31.69
core	19.355	18.200	30.28
aftermath	20.760	17.085	26.97

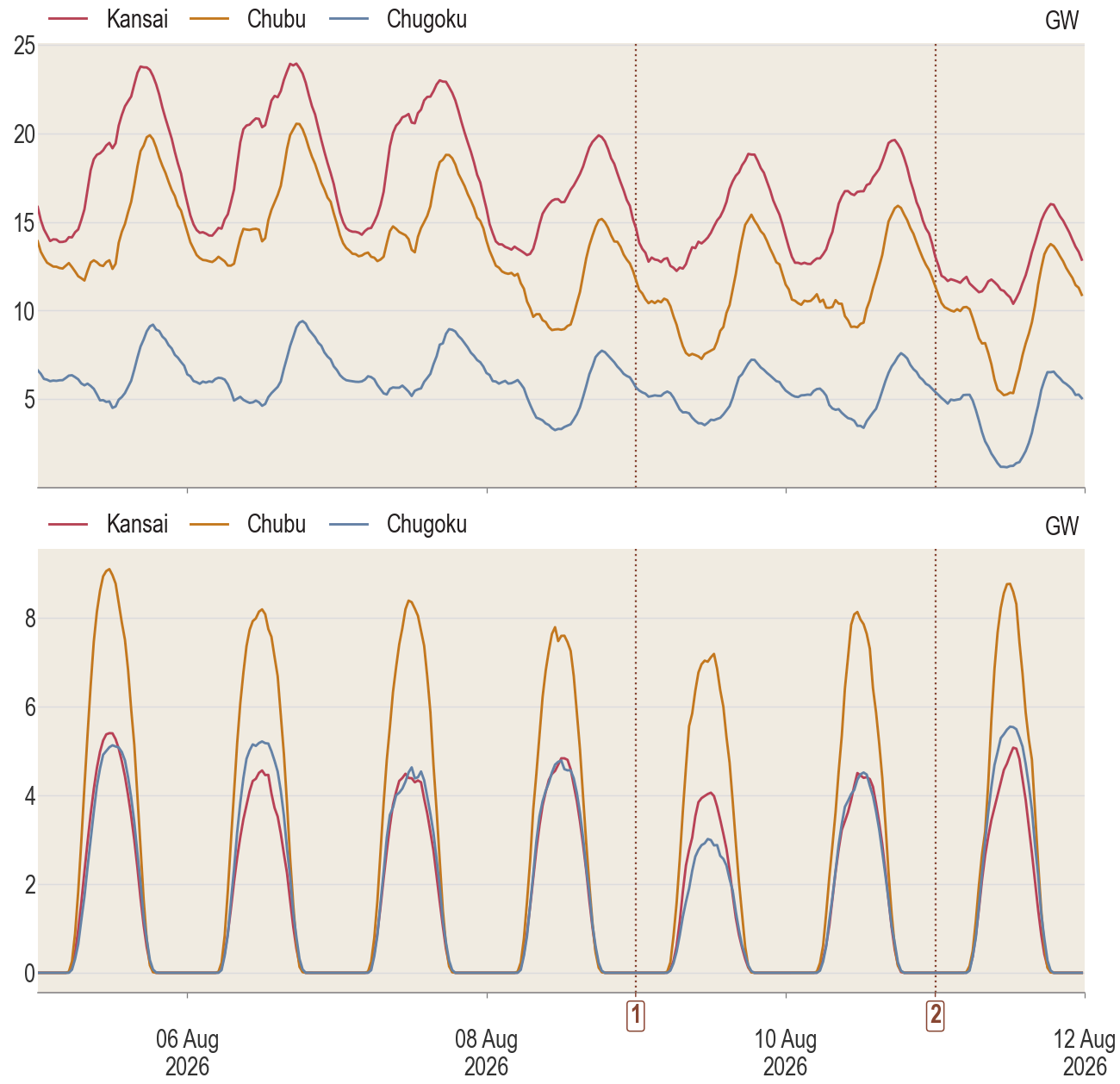
[298] [299] [300] [301] [285]

Native clocks: prices and physical series 30 min; weather hourly. Gaps are not zeros.

08 · Aug 2026 heat · B. Chronology

Ohi 3's documented trip occurred August 8 at 23:56. A daily mean for August 8 therefore contains almost a full day before the trip; it must not be interpreted as the post-trip level. Observed Kansai nuclear output averaged 4.726 GW on August 8 and 3.597 GW on August 9, but the same days also had different demand and weather. The timing is consistent with the unit event without constituting a controlled price experiment.

Exhibit 143: 3–31 August 2026: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [4] [298–301] · Chart data

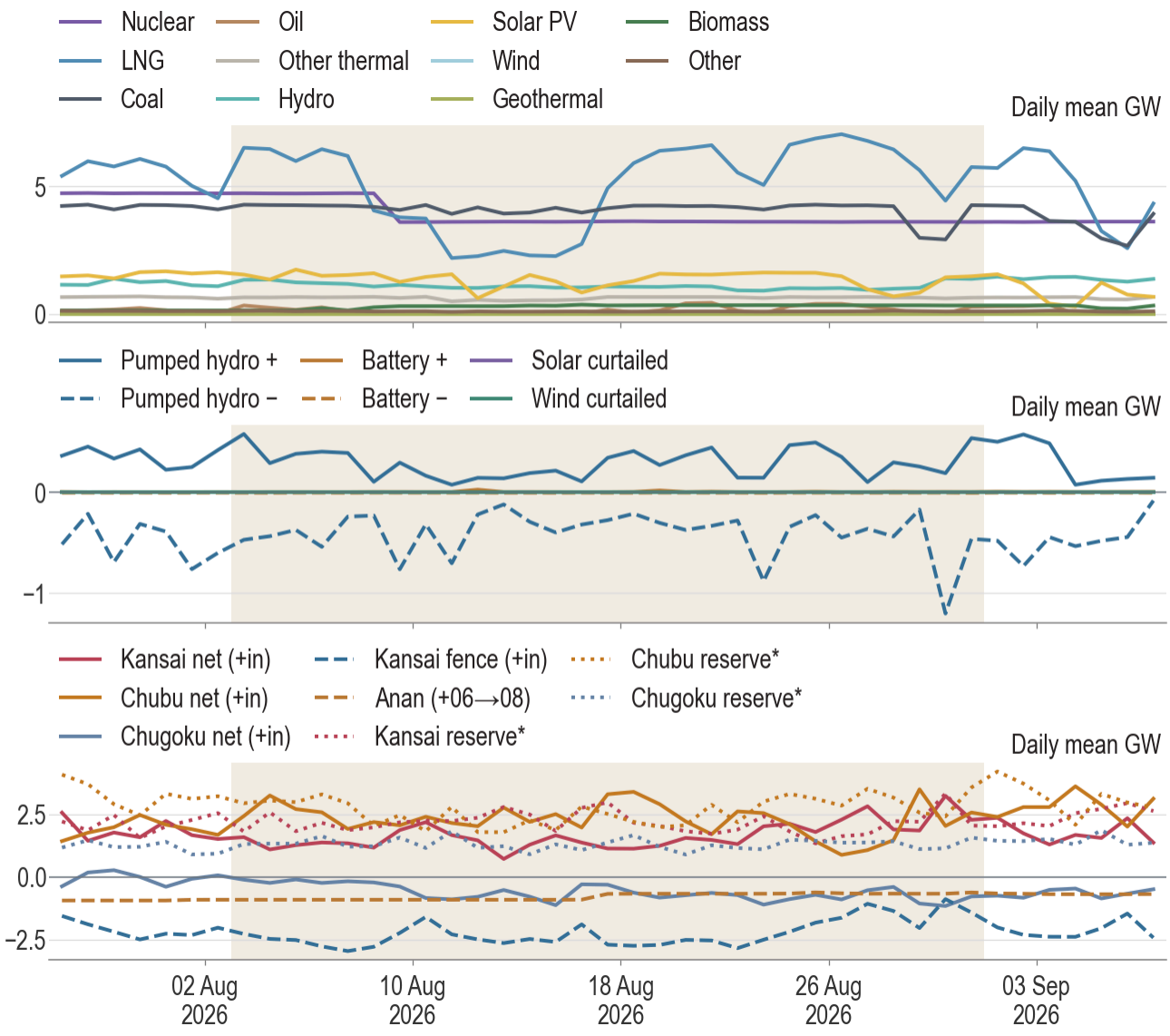
OCCTO reports ten Kansai support instructions during August 3–31, a maximum 1440 MW and longest duration 3 hours. The evidence does not provide a complete set of individual dispatch traces. Consequently the figure marks the documented episode and dated unit event but does not draw a synthetic 1440 MW rectangle or compute ten times a presumed energy volume. Separate unit loss, support action and observed interconnector behavior.

[298] [299] [300] [301] [291] [285]

08 · Aug 2026 heat · C. Physical response

Early(August 3–10), middle(August 11–20) and late(August 21–31) phases tell a stronger story than one average. Kansai demand was 19.247,16.121 and 19.328 GW; temperature 31.44,28.71 and 30.85°C. Nuclear output was 4.443,3.616 and 3.607 GW. Thermal output correspondingly averaged 10.407,8.542 and 10.920 GW: the cooler middle phase reduced load despite diminished nuclear supply.

Exhibit 144: 3–31 August 2026: physical response



Note: Kansai complete-day generation and separately averaged discharge/charge accompany qualified regional interchange and final schedules. The Kansai fence covers Chubu and Hokuriku boundaries; Anan is Kansai→Shikoku. No individual August instruction times or delivered support energy are inferred. Dotted reserve* is the latest published area plan, not telemetry or an issue-time vintage.

Source: [4] [10] [11] [294] [295] [298–303] · Chart data

Across the 29-day core, pumped storage generated 201.074 GWh while pumping consumed 280.8145 GWh on the regional sign convention. Battery discharge was 4.836 GWh and charging 0.9725 GWh. These reported aggregates are not a closed storage-loss calculation unless energy boundaries and opening/closing state of charge match. The actual physical levers are thermal, storage, renewable output and inter-area movement, not an invented instruction-MW path.

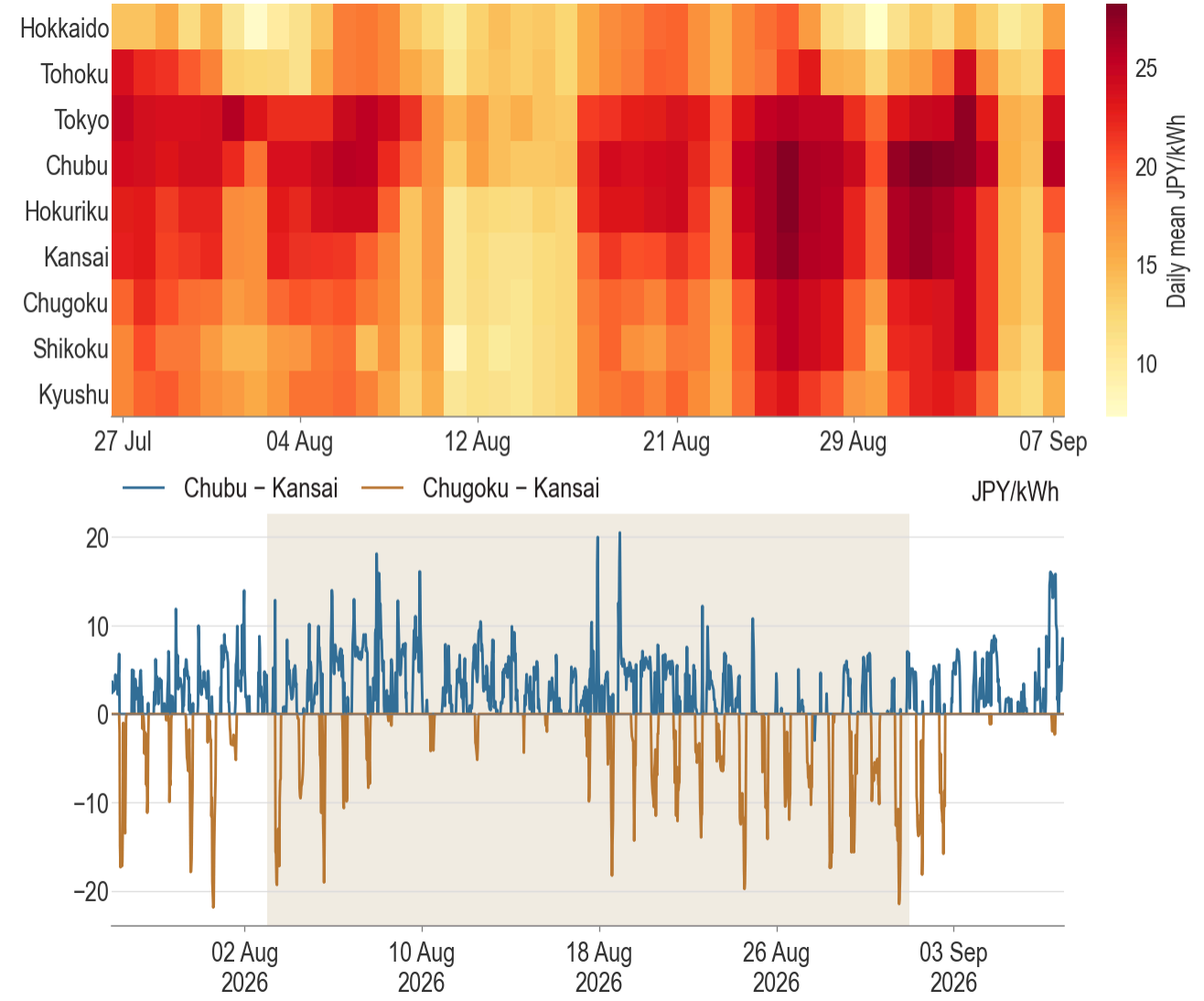
[298] [299] [300] [301] [292] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

08 · Aug 2026 heat · D. Regional prices

The all-area comparison separates Kansai's issue from national conditions. Chubu's core mean 21.697 JPY/kWh exceeded Kansai 19.355; Tokyo was 21.008 and Chugoku 17.835. Hokkaido averaged 14.702. Thus neither the operator trip nor western support makes Kansai the universally highest-priced area throughout the month.

Exhibit 145: 3–31 August 2026: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [298–301] · Chart data

Within Kansai the early/middle/late mean prices were 19.399, 14.979 and 23.302 JPY/kWh. Late-period prices exceeded early prices even though nuclear output was already depressed during the cooler middle period. That is evidence of interacting load and supply conditions, not a numerical estimate of the price effect of Ohi 3. The native spreads in the lower panel preserve evening and intraday differences that a daily heat map smooths.

North/east	Central/west	West/south
Hokkaido **14.702**	Chubu **21.697**	Chugoku **17.835**
Tohoku **15.887**	Hokuriku **20.448**	Shikoku **16.953**
Tokyo **21.008**	Kansai **19.355**	Kyushu **16.819**

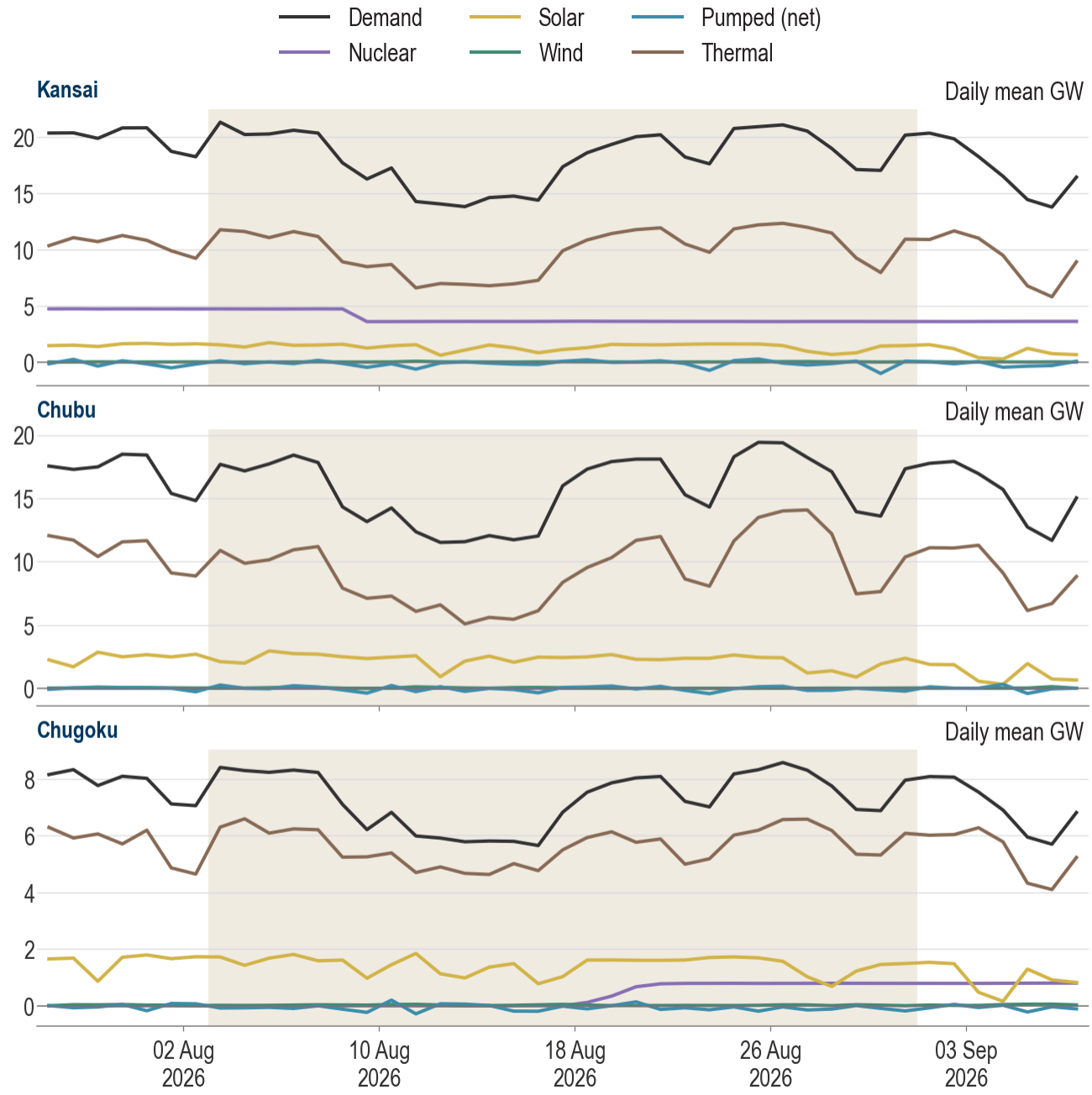
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[298] [299] [300] [301] [296] [285]

08 · Aug 2026 heat · E. Aftermath

Kansai's seven-day aftermath was cooler at 26.97°C and lower demand at 17.085 GW, but its mean price 20.760 JPY/kWh was above the full-core 19.355. Nuclear output remained 3.610 GW and solar fell to 864 MW mean, compared with 1,346 MW in the core. The last event date therefore does not imply complete physical or price normalization.

Exhibit 146: 3–31 August 2026: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [4] [298–301] · Chart data

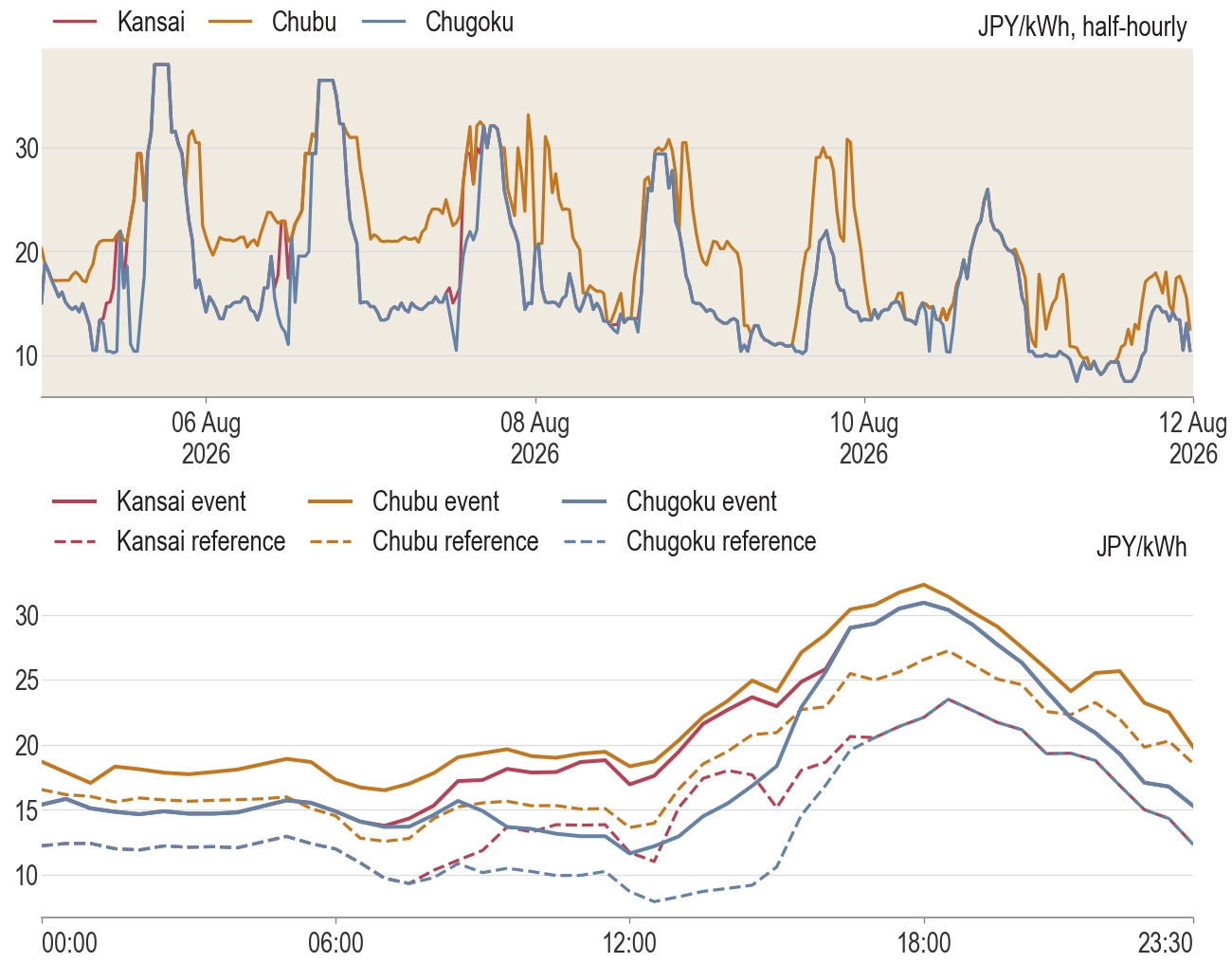
Chubu aftermath demand was 15.401 GW and price 23.452; Chugoku demand 7.008 GW and price 19.865. Chugoku nuclear output averaged 795 MW in the aftermath after rising during August. The neighboring physical panels expose these different recovery paths. Lower demand alone cannot establish easing residual supply or a completed unit return, and no unobserved fuel-stock recovery is claimed.

[298] [299] [300] [301] [285]

08 · Aug 2026 heat · F. Comparisons

The retained reference rule admits 28 of 29 core days. August 11 is a national holiday without an eligible prior matching holiday within the selected window; it is omitted from the matched statistic, not quietly replaced with an ordinary Tuesday. This explains why matched-event price 19.678 JPY/kWh differs from the all 29-day 19.355 mean.

Exhibit 147: 3–31 August 2026: prices and matched references



Note: The native-price window isolates early August; the lower profiles cover all complete core days with references. Solid and dashed lines compare event and prior-calendar prices, not realised and hypothetical prices without heat or support.

Source: [1] [2] [286] [290] [298–301] · Chart data

On the 28-day matched population Kansai's reference price is 15.085 JPY/kWh, demand 17.465 GW and residual load 16.052 GW, against event 18.340 GW and 16.983 GW. The price premium is descriptive and combines many changing conditions. The separately displayed phase comparison is essential: early and late have similar demand but different nuclear output; the middle phase is cooler. None of these comparisons removes all weather, fuel, outages and transmission confounders.

Area	Event*	Reference*	Eligible core days
Kansai	19.678	15.085	28
Chubu	21.999	18.512	28
Chugoku	18.103	13.624	28

*JPY/kWh; matched event population, not necessarily every day of the full core.

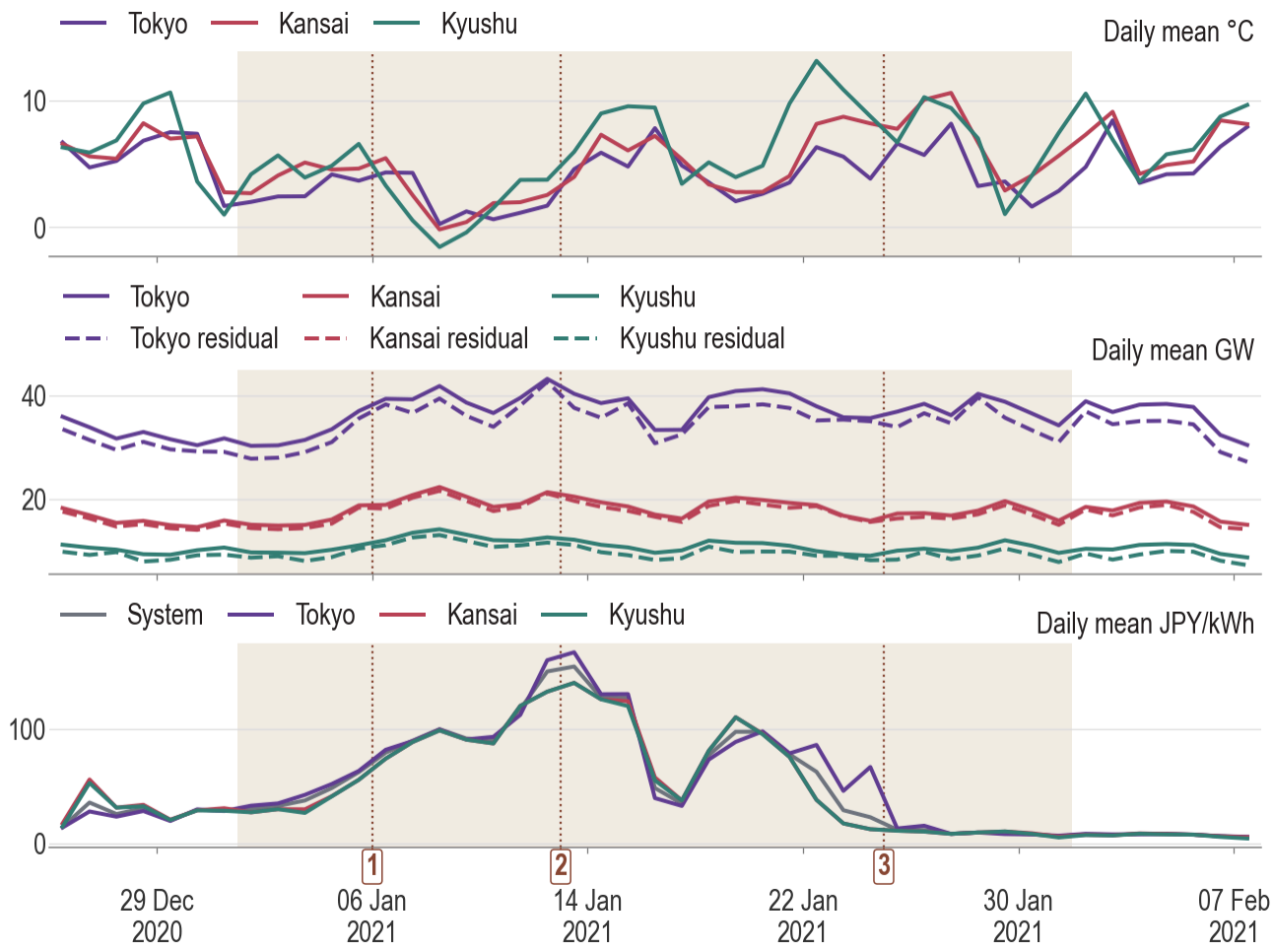
[298] [299] [300] [301] [297]

Case 01 · January 2021 energy and fuel stress

01 · Jan 2021 scarcity · A. Preconditions

January 2021 is sustained scarcity, not a single demand peak. The full January core retains December 25–31 lead-in and February 1–7 aftermath. Official reviews describe winter weather, constrained fuel availability and implemented responses.

Exhibit 148: January 2021: preconditions and onset



Note: 25 December 2020–7 February 2021; January is shaded. Complete-day means preserve the month-long sequence without compressing hourly observations. Demand is solid, residual load dashed; the numbered chronology is documentary, published after the crisis.

Source: [1–3] [5] [290] [304] [305] · Chart data

Physical response uses the native hourly historical dataset, not modern unit data. Tokyo's higher residual demand co-existed with zero regional nuclear output; Kansai and Kyushu differed. Installed MW alone cannot explain the price tail or establish usable fuel and supply.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	24.594	32.549	5.72
core	66.533	37.328	3.73
aftermath	7.645	36.078	5.63

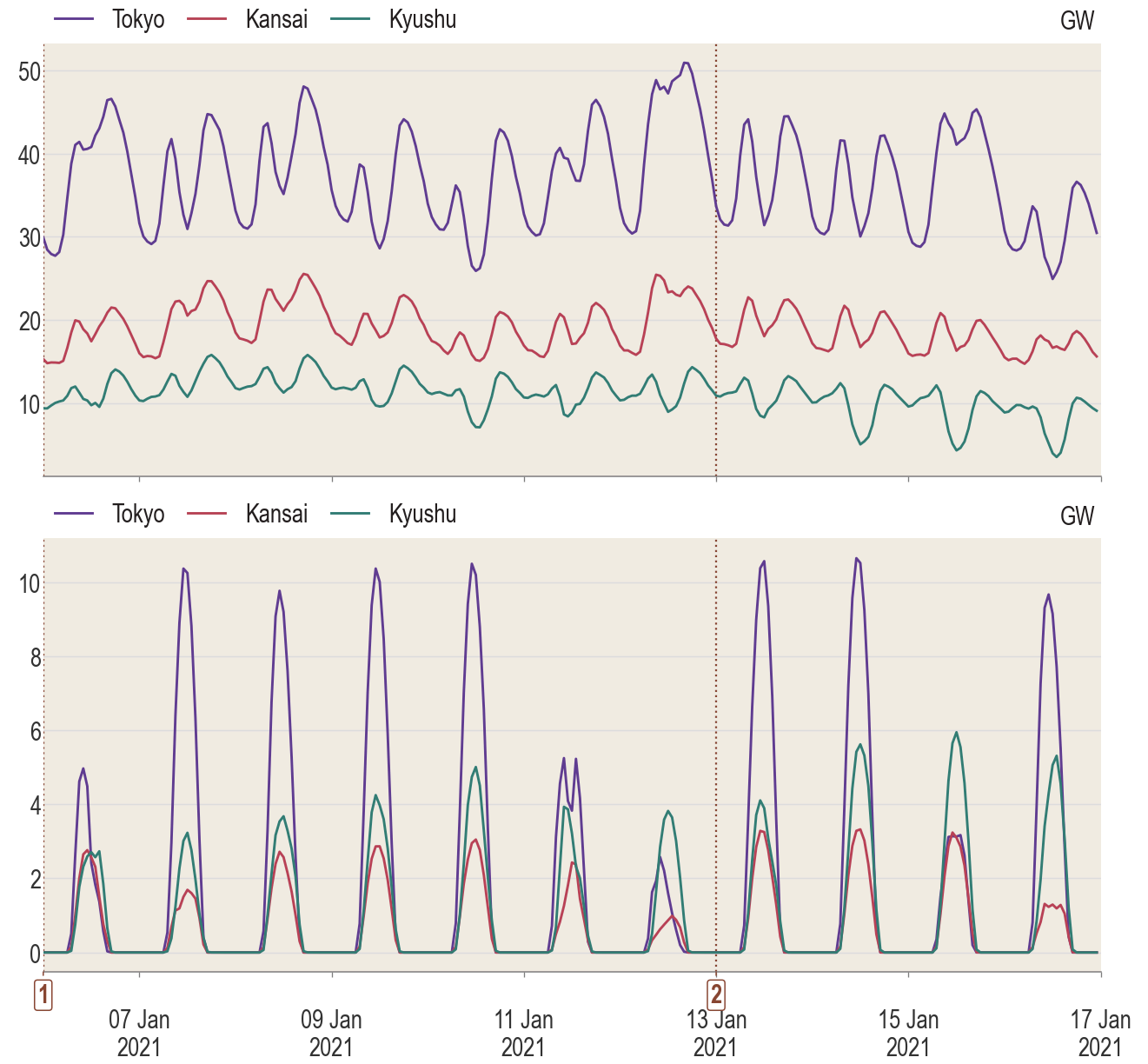
[304] [305] [285]

Native clocks: prices 30 min; physical series and weather hourly. Gaps are not zeros.

01 · Jan 2021 scarcity · B. Chronology

The public review places support instructions from December 15, before the plotted seven-day lead-in. Generation instructions ran January 6–26 and interconnector-capacity extensions began January 8. These are documentary dates, not a reconstructed continuous instruction volume. The chronology panel keeps them separate from observed demand, solar and residual load.

Exhibit 149: January 2021: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [3] [304] [305] · Chart data

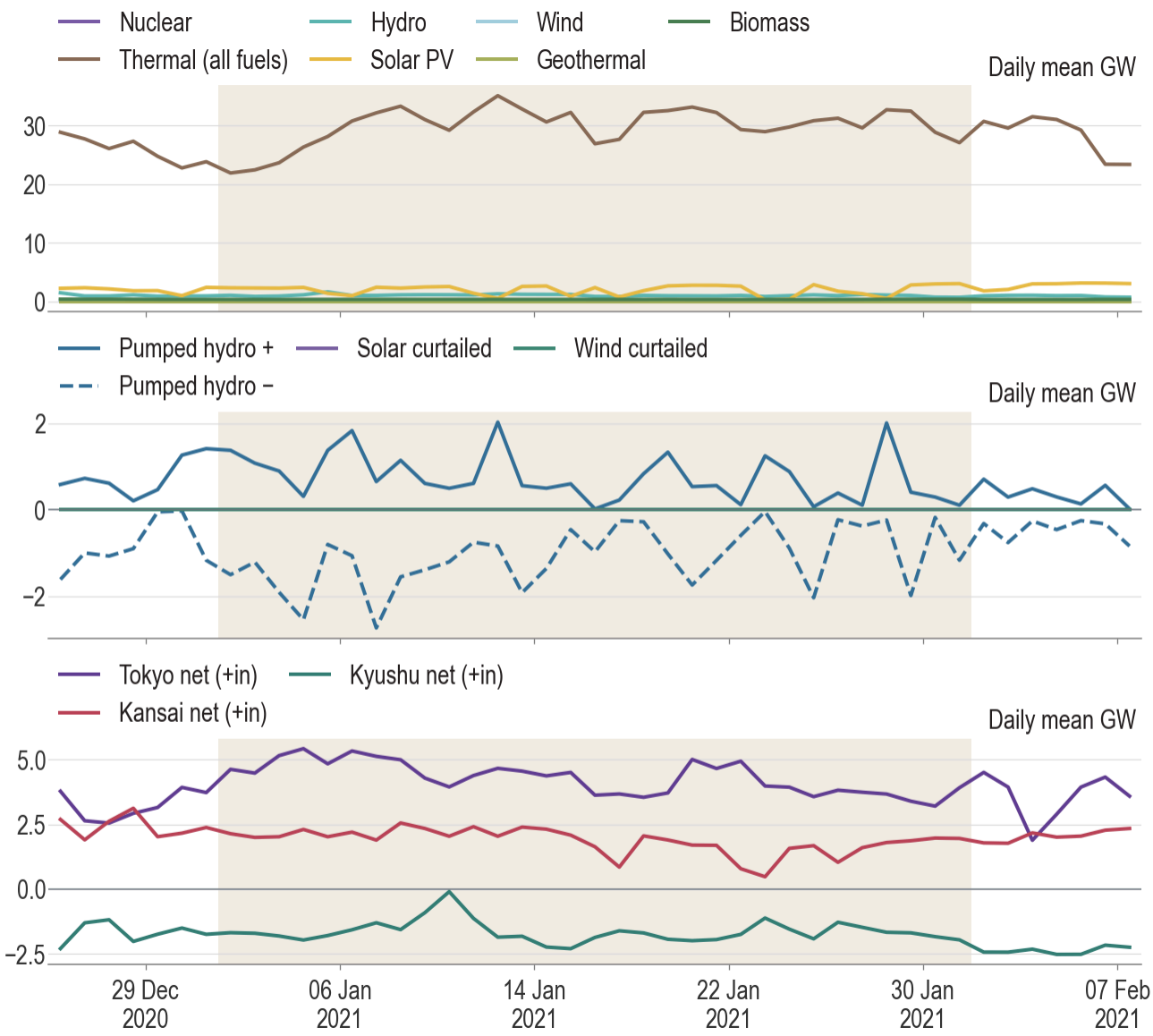
Tokyo's January 8 daily mean price was 99.598 JPY/kWh with 41.748 GW mean demand and 0.25°C temperature. January 15 averaged 130.541 JPY/kWh and reached 252.00 in one half-hour, despite lower mean demand 39.388 GW. That difference rules out a simple one-variable narrative in which the coldest day must be the most expensive. Fuel inventories, unit constraints, trading conditions and forecasts require their own evidence; the figure does not assign causal shares.

[304] [305] [291] [285]

01 · Jan 2021 scarcity · C. Physical response

Observed Tokyo thermal production averaged 29.920 GW in January versus 25.904 GW in the lead-in. The increase demonstrates actual response, not unlimited spare thermal capacity: a unit can be installed while fuel or operational constraints restrict its usable output. The aggregate historical dataset identifies thermal as a broad category rather than manufacturing a modern LNG/coal split.

Exhibit 150: January 2021: physical response



Note: Tokyo generation retains the legacy hourly thermal aggregate; complete-day means do not retrospectively split it into fuels. Pumped-storage charge/discharge are averaged separately. Regional interchange is qualified independently; no historical bilateral schedule or fleet battery series is invented.

Source: [3] [294] [295] [304–307] · Chart data

Tokyo pumped storage generated 555.880 GWh and consumed 825.470 GWh across the 744 hourly core observations; solar generation was 1,460.180 GWh. Hydro timing and imports could relieve particular hours without eliminating sustained fuel scarcity. Kansai's mean nuclear contribution was 489 MW during January and Kyushu's 2.992 GW, reinforcing the need for regional physical comparisons. Actual support-energy volumes cannot be inferred from documentary instruction counts or maxima.

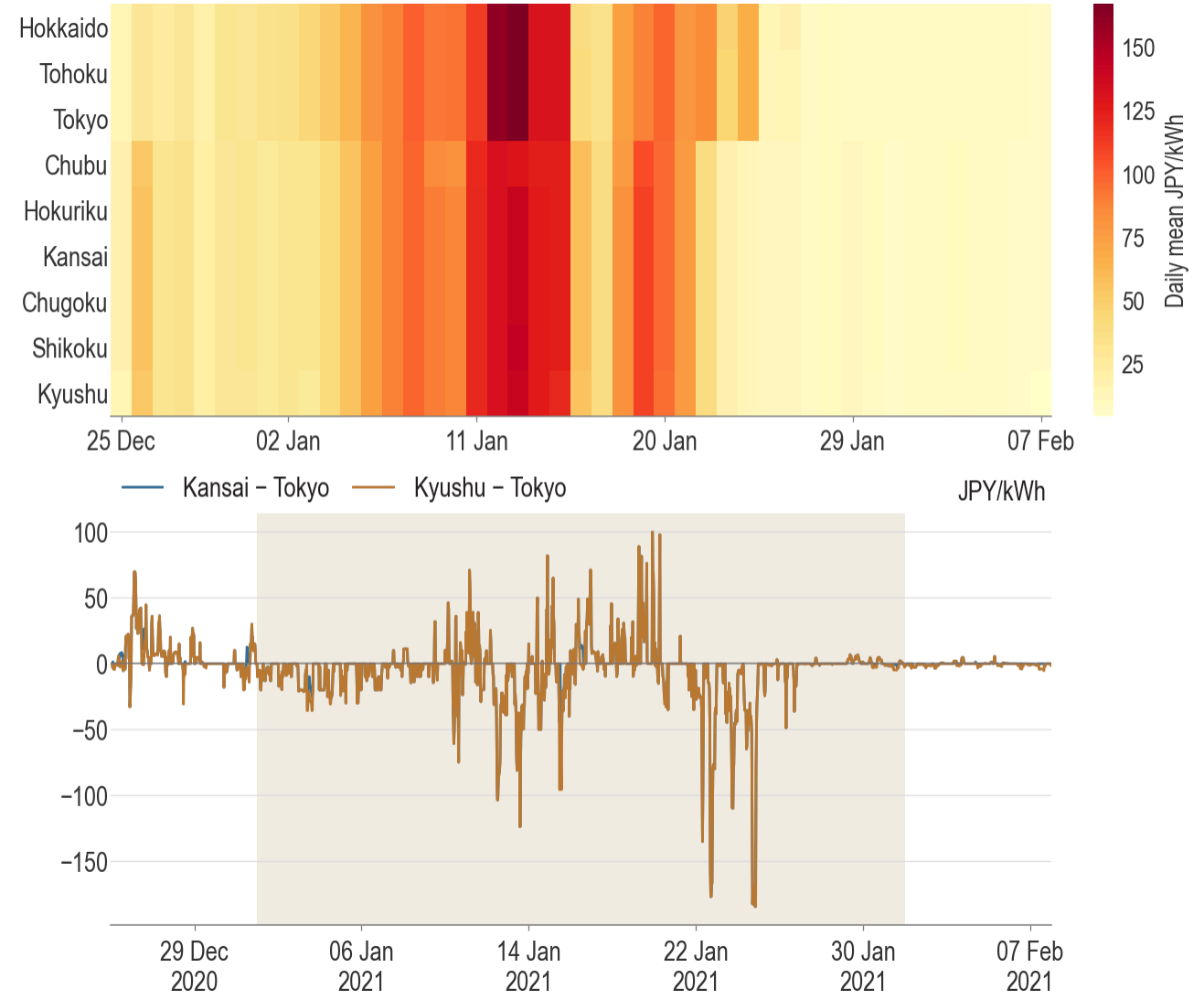
[304] [305] [292] [285]

Observation clock: prices 30 minutes; physical series 60 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

01 · Jan 2021 scarcity · D. Regional prices

The price elevation was broad but not uniform. Tokyo, Tohoku and Hokkaido core means were 66.533,66.513 and 66.595 JPY/kWh. Chubu averaged 58.950, Kansai 60.004 and Kyushu 59.618. These are all half-hour observations within the same month, not a mix of daily peaks and averages.

Exhibit 151: January 2021: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [304] [305] · Chart data

Daily heat-map shading exposes the changing spatial pattern while native focal spreads retain shorter separations. Similar monthly means do not prove that areas cleared together every hour, and a shared high-price episode does not identify one common physical constraint. The observed price tail and the unusually high number of expensive intervals must be read alongside the actual physical response, not labelled purely as transmission congestion.

North/east	Central/west	West/south
Hokkaido **66.595**	Chubu **58.950**	Chugoku **60.004**
Tohoku **66.513**	Hokuriku **60.004**	Shikoku **60.077**
Tokyo **66.533**	Kansai **60.004**	Kyushu **59.618**

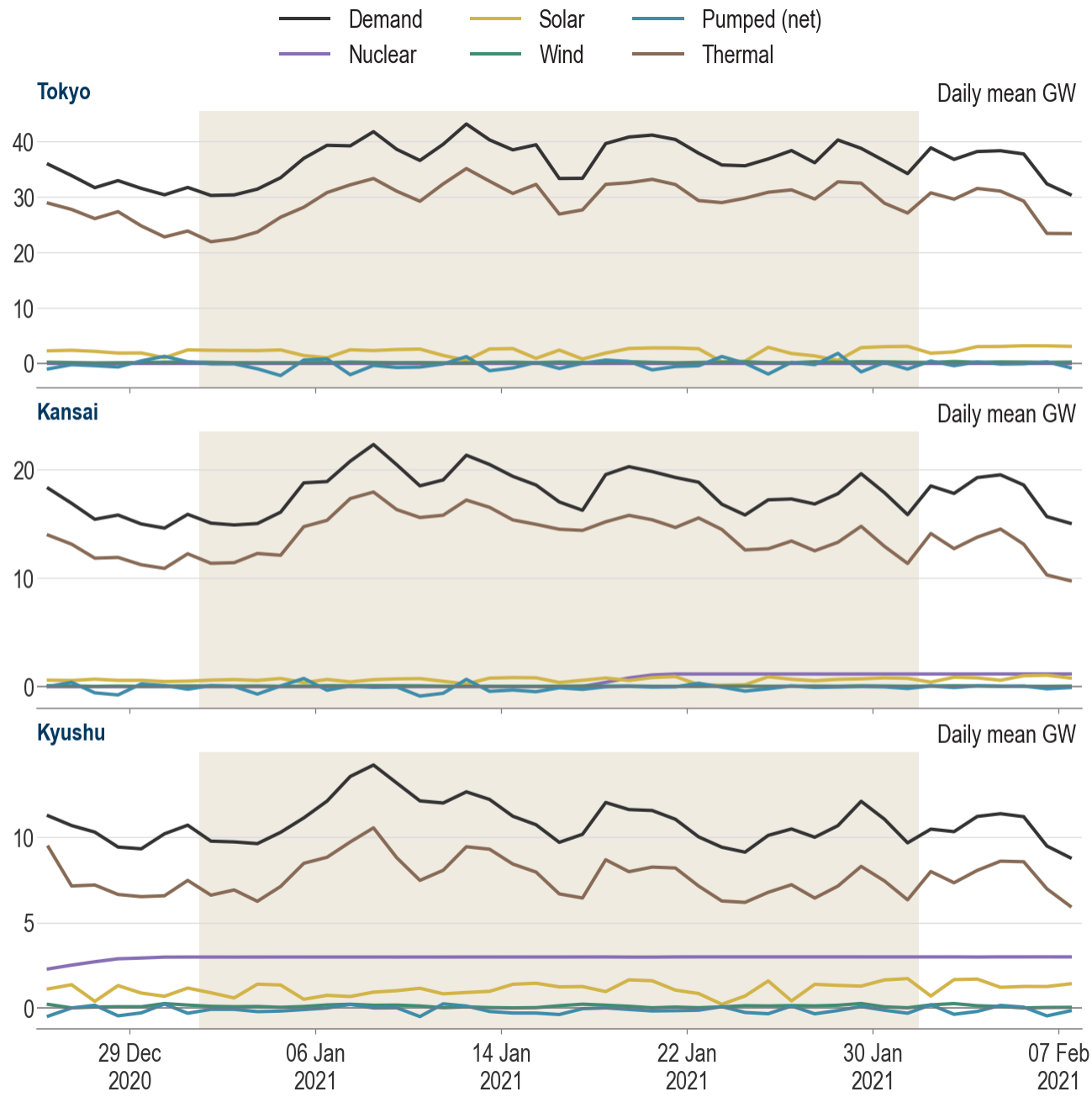
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[304] [305] [296] [285]

01 · Jan 2021 scarcity · E. Aftermath

The aftermath is a sharp price retracement with incomplete physical normalization. Tokyo's February 1–7 mean price fell to 7.645 JPY/kWh, but demand remained 36.078 GW, above the 32.549 GW lead-in. Thermal production was 28.404 GW, also above lead-in. A price collapse therefore did not require demand returning to its starting level.

Exhibit 152: January 2021: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [3] [304] [305] · Chart data

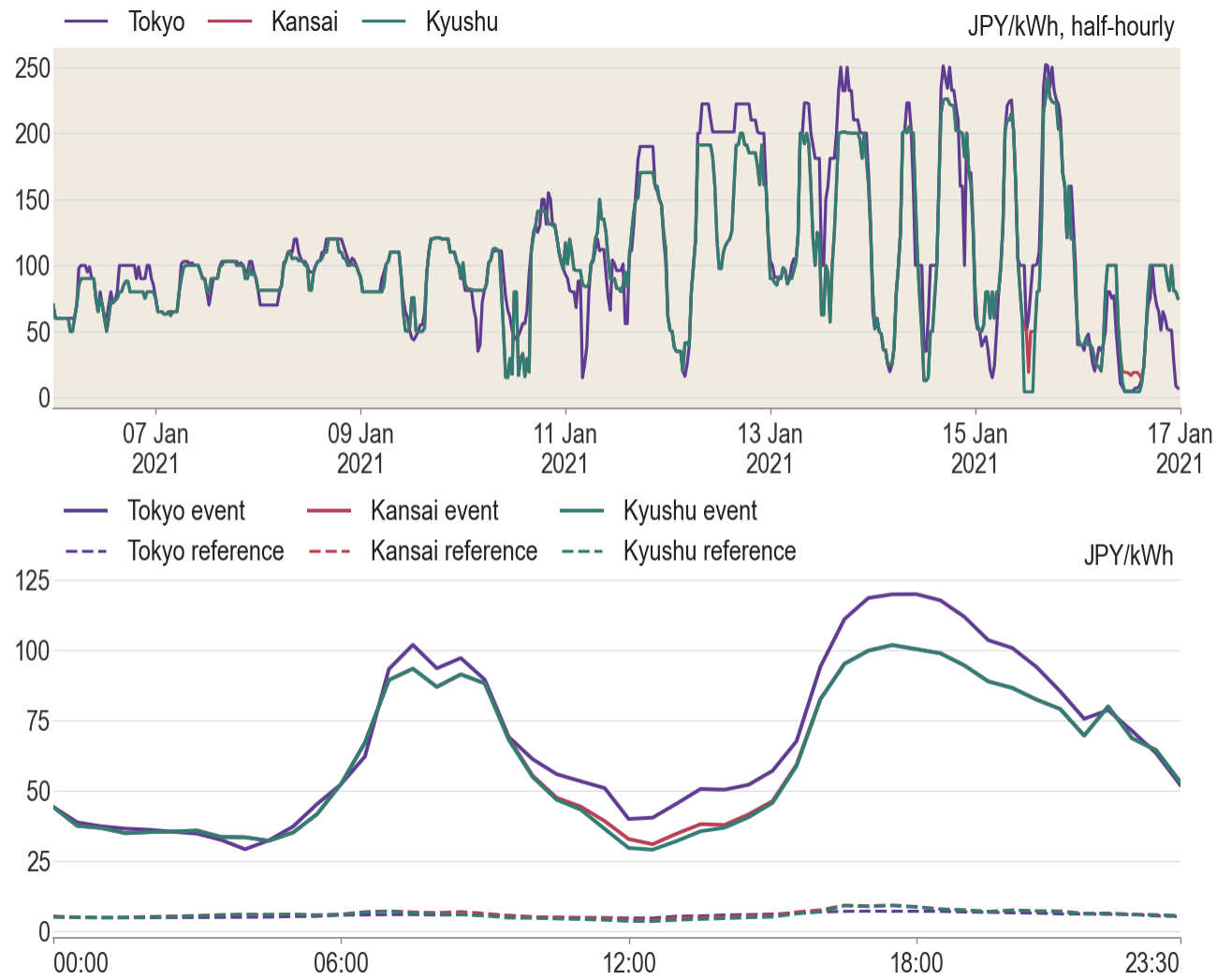
Kansai aftermath nuclear output averaged 1.173 GW versus 489 MW during January; Kyushu remained near 2.996 GW. Solar rose in all three focal physical traces. These observations are consistent with changing supply/demand conditions, but they do not measure fuel-stock replenishment or isolate which intervention lowered prices. The documentary reviews support the action chronology; the daily panels support only the measured production and demand trajectory.

[304] [305] [285]

01 · Jan 2021 scarcity · F. Comparisons

The preserved same-calendar comparison admits 30 of 31 January days: January 1 lacks an eligible prior holiday-Friday reference in the 56-day rule. Tokyo's matched-event mean is 67.648 JPY/kWh versus 5.777 on eligible references. This is not the same population as the complete-month 66.533 mean, and the difference must remain explicit.

Exhibit 153: January 2021: prices and matched references



Note: Half-hour prices show the January stress at native frequency; lower profiles use 30 complete calendar-matched days. January 1 remains in the observed chronology but has no eligible holiday-class reference and is excluded from both profile means.

Source: [1] [3] [286] [290] [304] [305] · Chart data

Matched Tokyo demand is 37.564 GW versus 31.808 GW; residual load 35.474 GW versus 30.088 GW. The reference dates end before the documented mid December stress exclusion. Consequently these are cleaner pre-stress calendar comparators but are also earlier-season weather and fuel conditions. The original comparison is retained, not redefined to make the result smaller; neither it nor the simple lead-in identifies a causal price effect of fuel alone.

Area	Event*	Reference*	Eligible core days
Tokyo	67.648	5.777	30
Kansai	61.094	6.296	30
Kyushu	60.694	5.939	30

*JPY/kWh; matched event population, not necessarily every day of the full core.

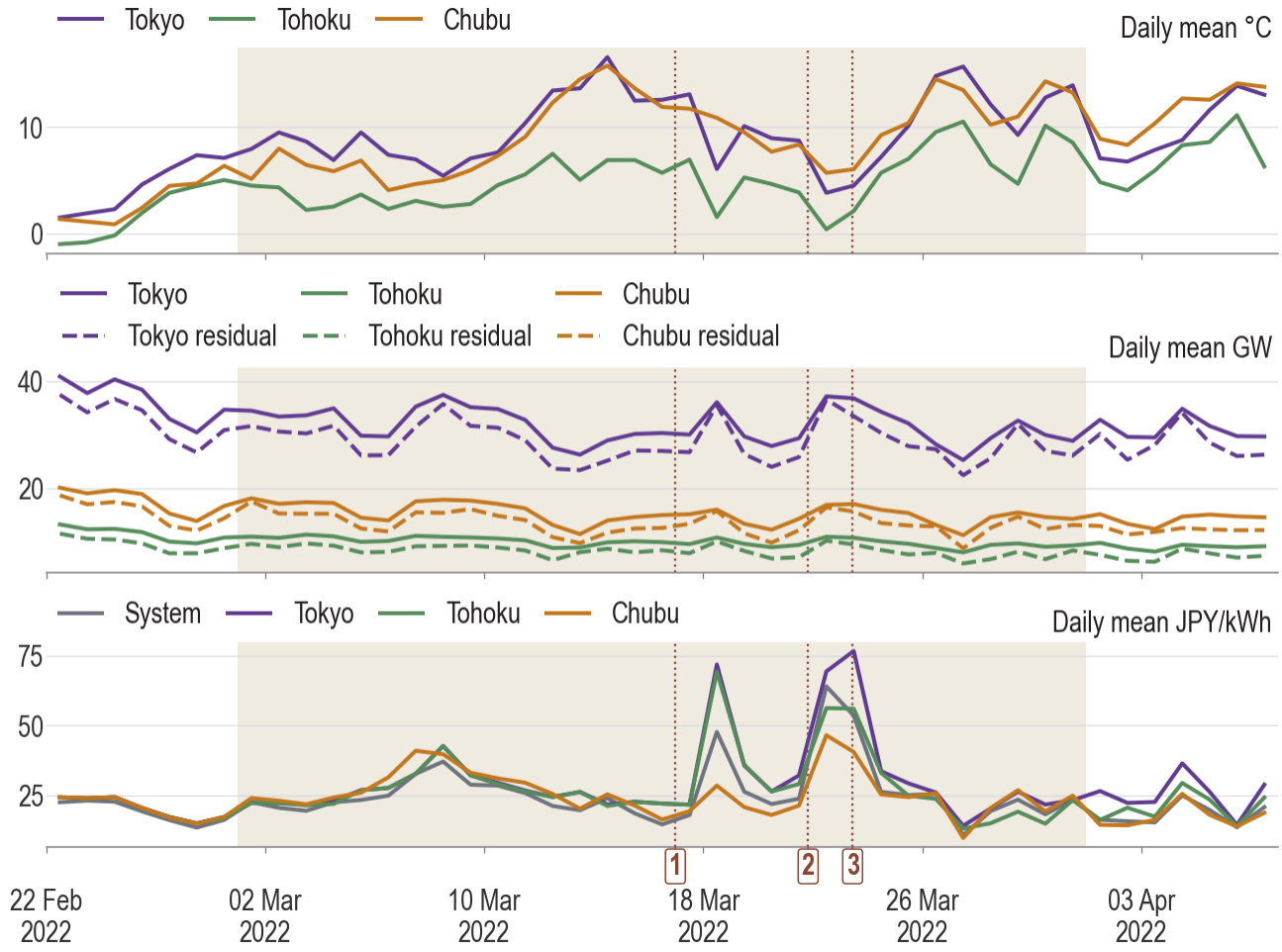
[304] [305] [297]

Case 02 · March 2022 compound eastern shock

02 · Mar 2022 alert · A. Preconditions

The core remains all March 2022, with February 22–28 lead-in and April 1–7 aftermath. The operational alert concentrated around March 22, after the March 16 earthquake. Tokyo/Tohoku/Chubu are focal; all nine prices remain visible.

Exhibit 154: March 2022: preconditions and onset



Note: 22 February–7 April 2022; March is shaded. Daily means distinguish the selected high-price month from the shorter earthquake/cold emergency. Solid demand and dashed residual load retain complete hourly populations; prices require all 48 half-hours.

Source: [1–3] [5] [290] [308–310] · Chart data

Tokyo's monthly demand fell while price rose, so a monthly average cannot diagnose the short alert. Generation losses preceded cold and weak solar. Parts B/C retain that event order and focal physical window without silently replacing the accepted monthly comparison. Whole-month pricing and the shorter operational crisis answer different questions.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	20.402	36.474	4.42
core	30.760	31.682	9.91
aftermath	25.357	31.111	9.86

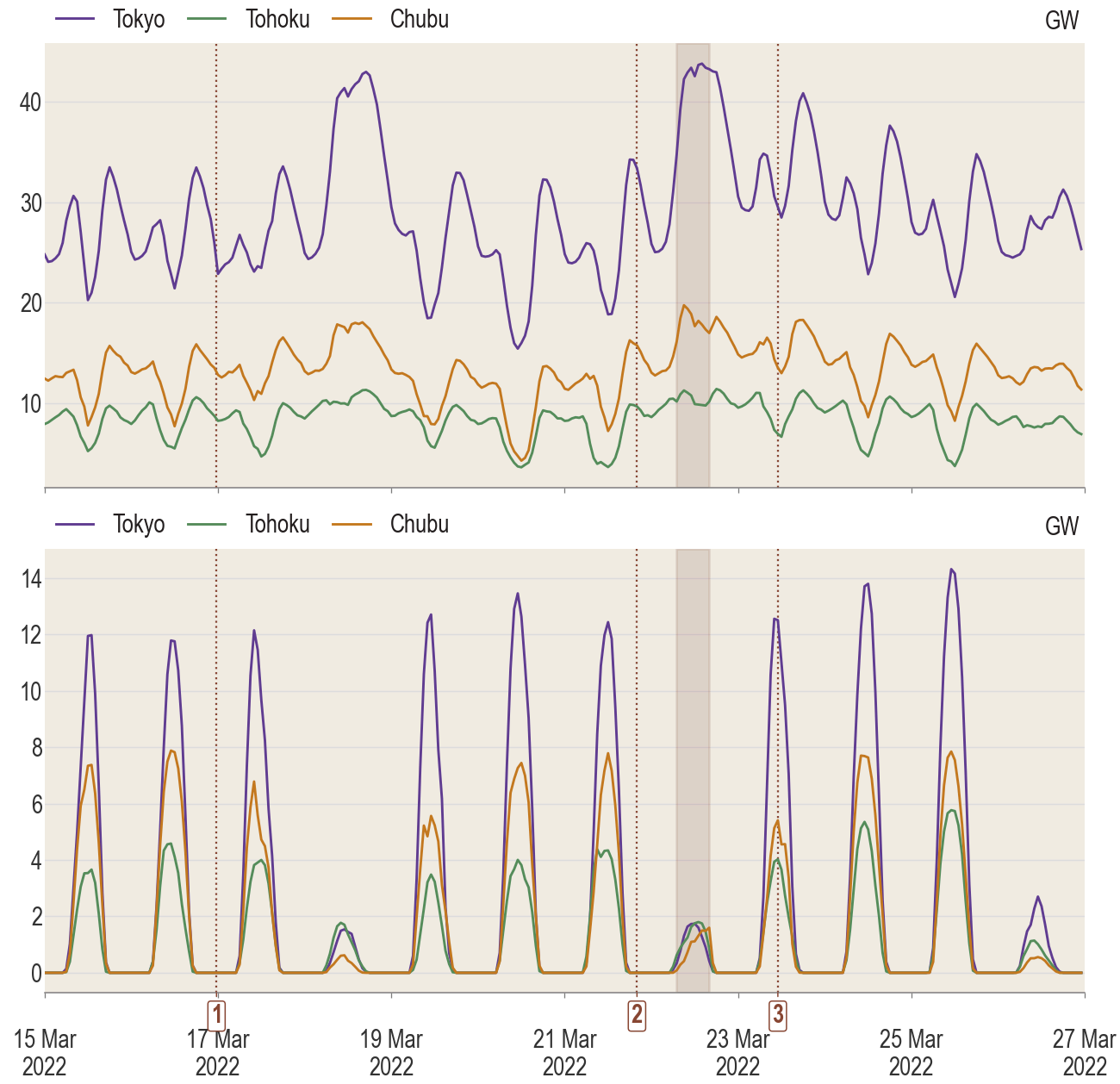
[308] [309] [310] [285]

Observation clock: prices 30 minutes; physical series 60 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

02 · Mar 2022 alert · B. Chronology

The earthquake occurred March 16 at 23:36. The government tightness alert began March 21 at 20:00 and ended March 23 at 11:00. A Tokyo support instruction was issued March 22 at 05:59 for 07:00–16:00 with a maximum 1417.8 MW. These times have different meanings: shock, public alert, instruction issue and delivery span. The instruction maximum is not a measured nine-hour energy rectangle.

Exhibit 155: March 2022: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [3] [308–310] · Chart data

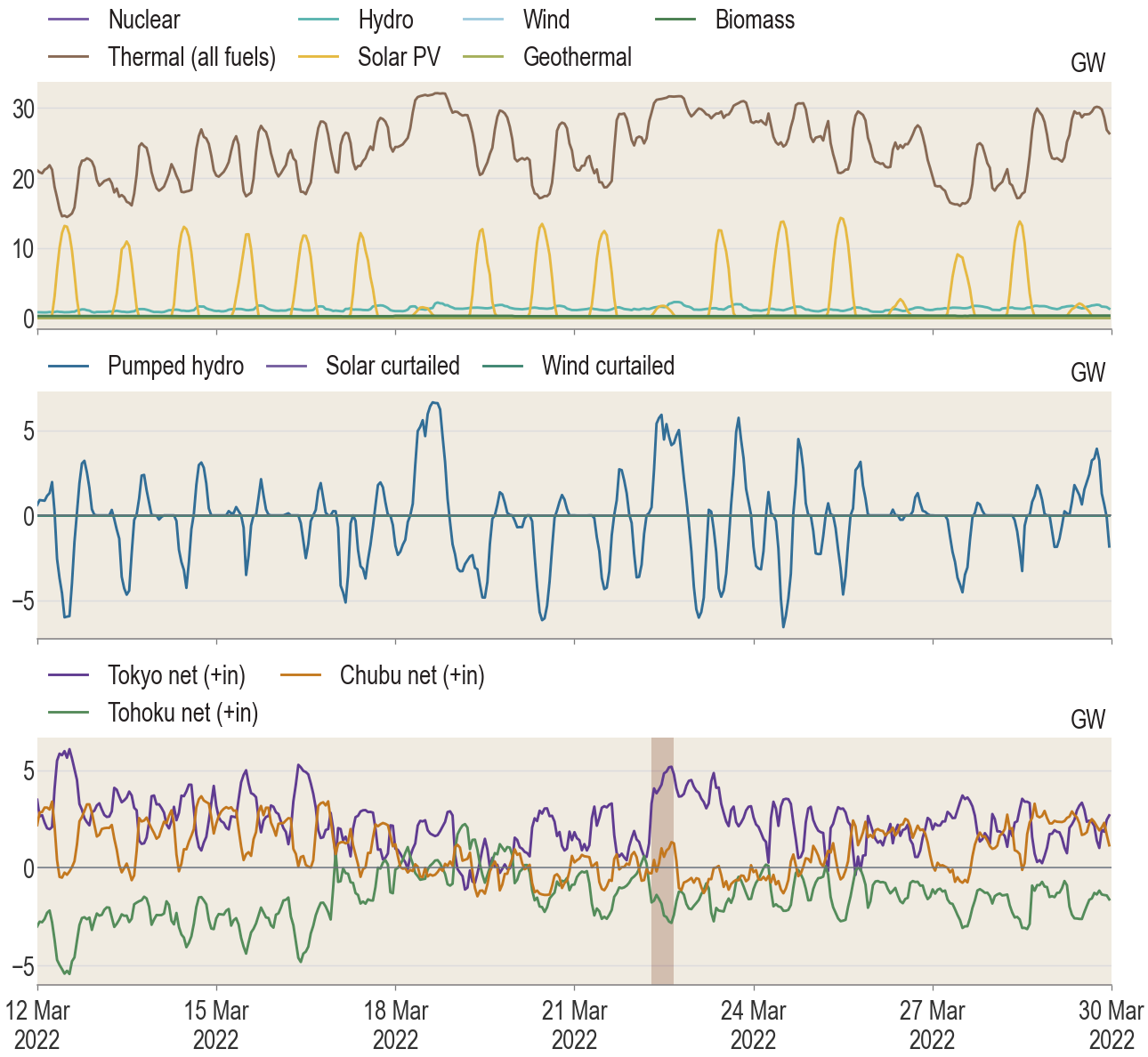
On March 22 Tokyo temperature averaged 3.84°C, demand 37.144 GW and solar 495 MW, compared with March 16's 12.56°C, 30.318 GW and 3.227 GW. The earthquake-day daily mean mostly precedes the late-night shock. Interpreting it as the post-earthquake supply state would invert the chronology. Neighboring production, imports and pumping therefore remain visible at native historical resolution.

[308] [309] [310] [291] [285]

02 · Mar 2022 alert · C. Physical response

The physical problem cannot be summarized as "demand was higher in March". Whole-month demand was lower than the winter lead-in, yet March 22 had a cold demand surge and very weak solar. Tokyo's peak historical hourly demand that day was 45.340 GW; maximum solar was only 1.740 GW. The alert followed earlier generation losses rather than a contemporaneous increase in installed capacity.

Exhibit 156: March 2022: physical response



Note: 12–29 March 2022 at native hourly resolution for Tokyo supply, pumped storage and qualified regional interchange. Thermal remains an aggregate. The darker span marks the 22 March instruction window, not metered support or an observed reservoir trajectory.

Source: [3] [294] [295] [308–312] · Chart data

Across March, Tokyo pumped storage generated 446.600 GWh and consumed 606.880 GWh. Those monthly totals are not the energy remaining in reservoirs at the alert peak. Storage state and actual support dispatch are not reconstructed from instructions. The physical figures show observed regional demand, aggregate thermal, renewable output and correctly oriented boundary flow; no modern half-hour fuel disaggregation is manufactured for 2022.

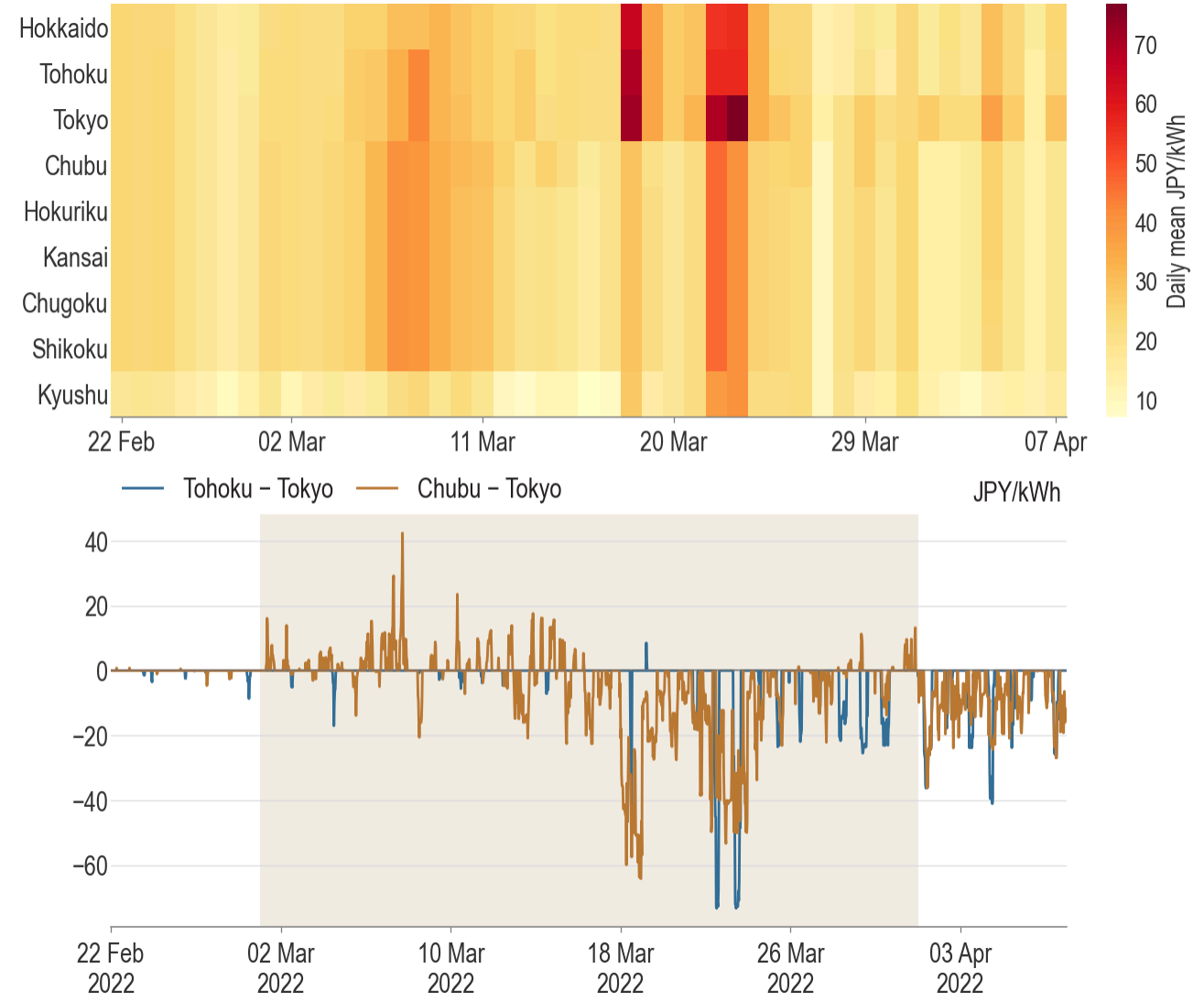
[308] [309] [310] [292] [285]

Observation clock: prices 30 minutes; physical series 60 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

02 · Mar 2022 alert · D. Regional prices

Tokyo's March mean 30.760 JPY/kWh exceeded Tohoku 28.513 and Chubu 25.988; Kyushu was 18.074. The all-nine view therefore shows a stronger eastern price exposure, but not a direct measurement of a specific bottleneck's marginal cost. Native paired spreads are retained rather than comparing independently sampled daily averages.

Exhibit 157: March 2022: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [308–310] · Chart data

Tokyo's March 22 daily mean was 69.424 JPY/kWh and March 23's 76.726, even though the public alert ended at 11:00 on March 23. An alert is an operational/public communication, not a promise that prices instantly revert at its end. Day-ahead prices and contemporaneous operating actions have different clocks; a high settlement after an alert boundary is not by itself contradictory.

North/east	Central/west	West/south
Hokkaido **27.552**	Chubu **25.988**	Chugoku **25.433**
Tohoku **28.513**	Hokuriku **25.441**	Shikoku **25.427**
Tokyo **30.760**	Kansai **25.441**	Kyushu **18.074**

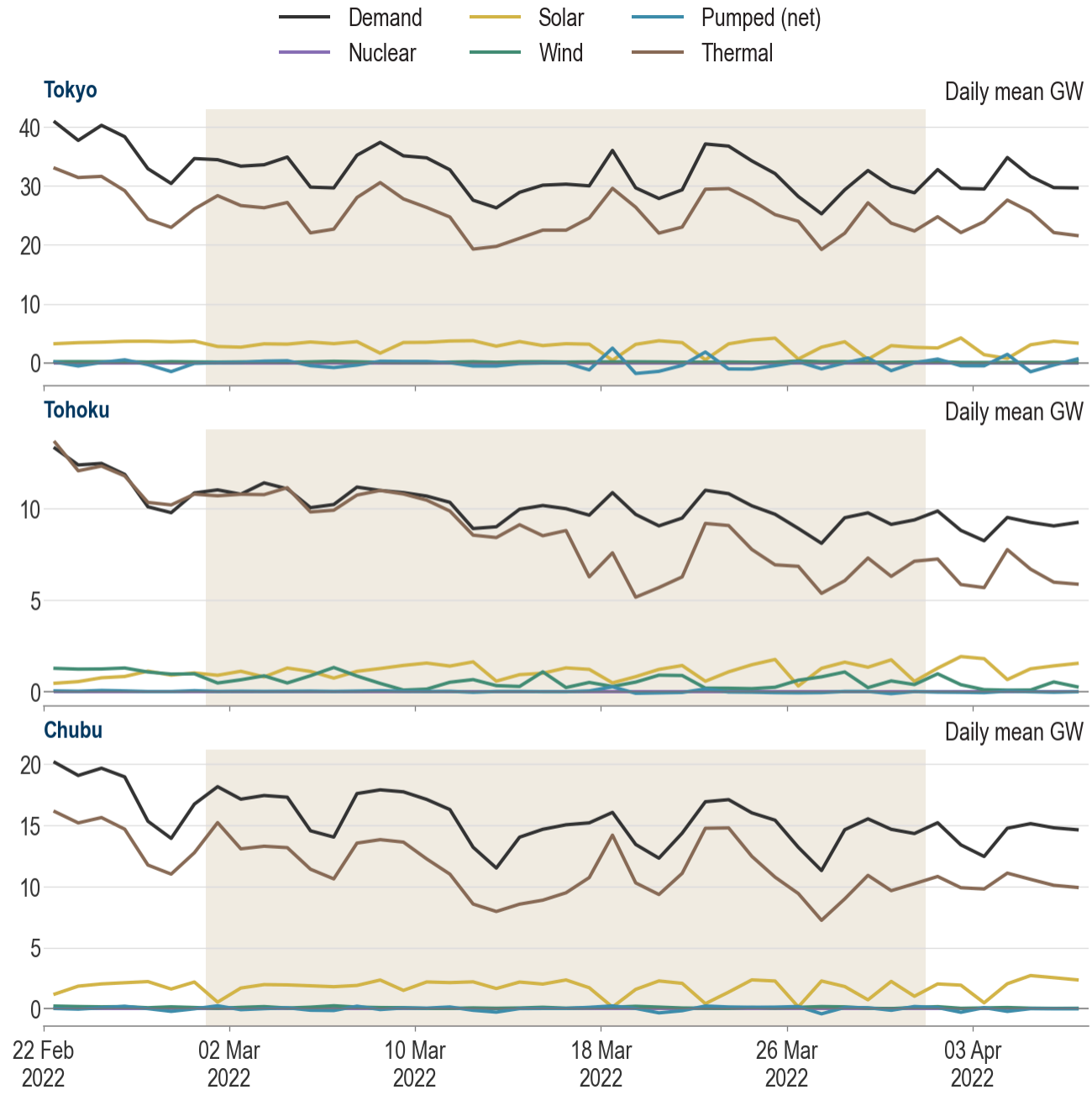
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[308] [309] [310] [296] [285]

02 · Mar 2022 alert · E. Aftermath

Tokyo aftermath price averaged 25.357 JPY/kWh, below March but above the 20.402 lead-in. Demand was 31.111 GW and thermal output 23.963 GW. Tohoku's aftermath demand 9.145 GW and thermal 6.441 GW were also lower than their March means. These describe observed easing, not a certified restoration date for every earthquake-affected unit.

Exhibit 158: March 2022: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [3] [308–310] · Chart data

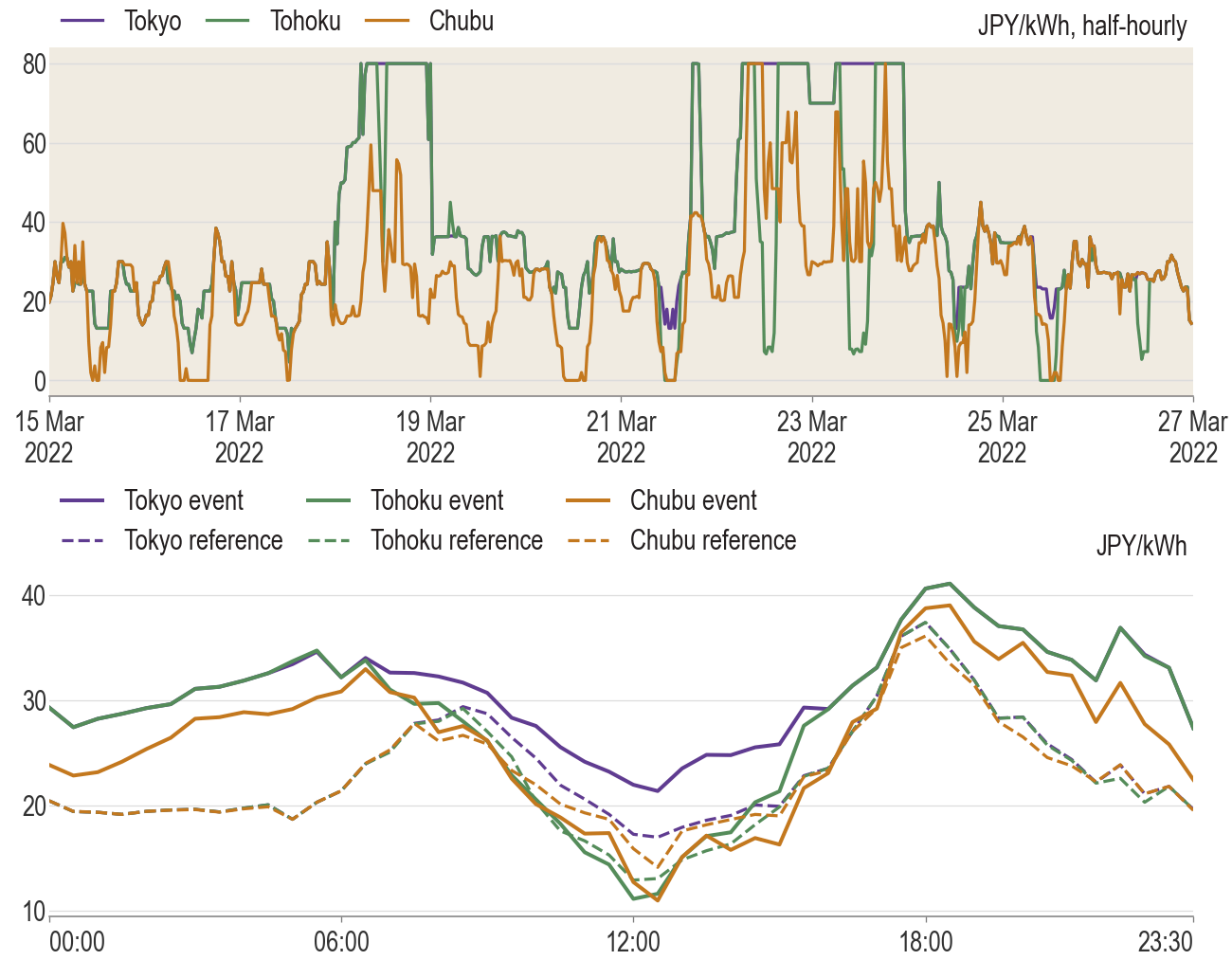
The recovery panels retain the full neighboring-system context rather than presenting only Tokyo's price. Weather, solar availability, unit return and system scheduling can change at different speeds. The March 23 alert withdrawal is a dated public milestone; the April 1–7 aftermath is a separate fixed comparison window. Neither licenses an invented reserve-margin series or an assumption that all affected plants were restored by midnight March 23.

[308] [309] [310] [285]

02 · Mar 2022 alert · F. Comparisons

All 31 March days qualify under the original calendar rule. Tokyo's matched price is 30.760 JPY/kWh versus 23.370 on prior eligible same-weekday/holiday-class days. Yet matched demand is 31.682 GW versus 38.647 GW and residual load 28.686 GW versus 35.952 GW. The positive price premium alongside lower average load is a descriptive warning against using demand alone to explain the month.

Exhibit 159: March 2022: prices and matched references



Note: Half-hour prices focus on the earthquake and subsequent cold; lower profiles average the full March case against eligible prior-calendar references. A monthly profile can dilute a brief operational emergency, so the two panels answer different questions.

Source: [1] [3] [286] [290] [308–310] · Chart data

The baseline is seasonal winter data within 56 days before March onset; it is not an earthquake counterfactual with identical temperatures, fuel costs and generating units. The separate March 16/March 22/March 23 observations isolate the chronology but remain observational. Keep both scales: the accepted monthly comparison for reproducibility, and the focal physical window for understanding why an alert occurred despite a lower monthly mean.

Area	Event*	Reference*	Eligible core days
Tokyo	30.760	23.370	31
Tohoku	28.513	22.502	31
Chubu	25.988	22.660	31

*JPY/kWh; matched event population, not necessarily every day of the full core.

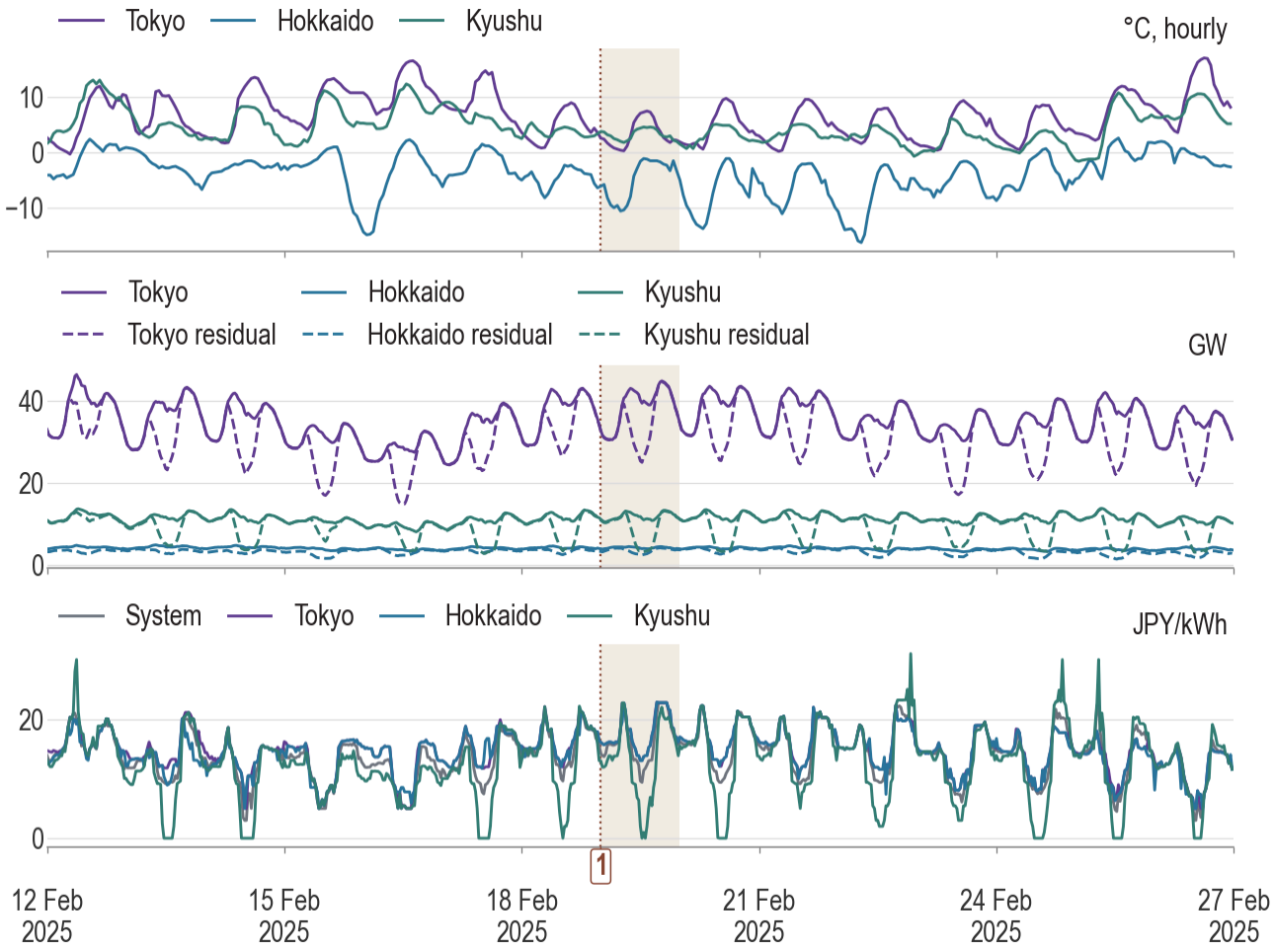
[308] [309] [310] [297]

Case 03 · February 2025 winter price episode

03 · Feb 2025 winter day · A. Preconditions

February 19, 2025 is a reproducibly price-selected day, not an evidenced emergency declaration. Lead-in February 12–18 and aftermath February 20–26 remain fixed. Tokyo is focal; Hokkaido/Kyushu provide contrasting northern/western systems.

Exhibit 160: 19 February 2025: preconditions and onset



Note: 12–26 February 2025; the selected 19 February day is shaded. Hourly point temperatures, solid demand, dashed residual load and half-hour prices show three regions. A price-selected day does not by itself demonstrate a physical shortage.

Source: [1] [2] [4] [5] [290] [313] · Chart data

Colder weather and higher Tokyo demand/residual load establish the physical setting. They do not establish reserve exhaustion or justify invented support instructions. The day's solar, wind, nuclear and neighboring conditions distinguish candidate explanations; the calendar references in part F are not weather-matched counterfactuals.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	14.617	34.584	7.76
core	17.751	38.806	3.38
aftermath	14.769	35.821	5.80

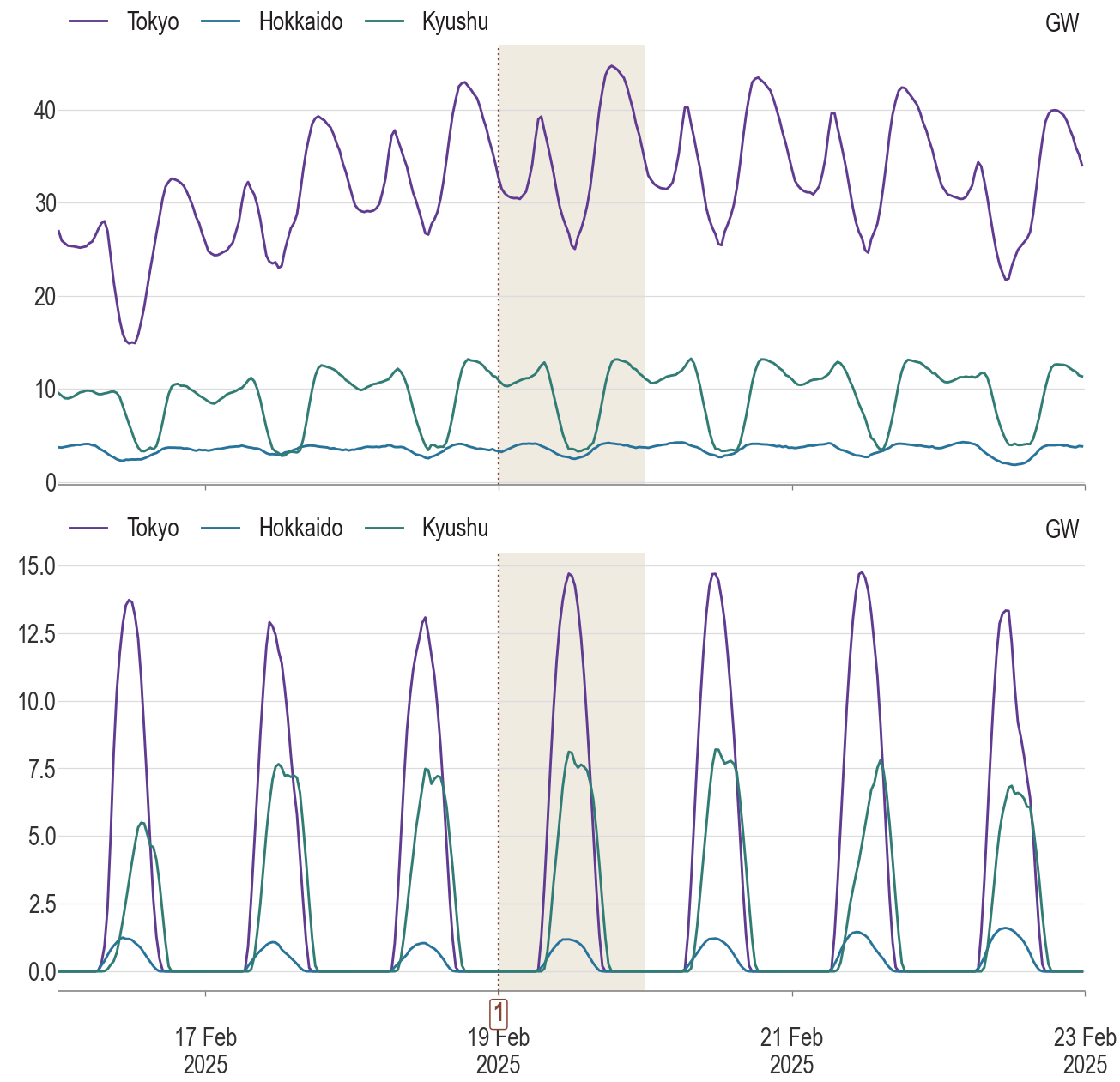
[313] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

03 · Feb 2025 winter day · B. Chronology

This case has a clock of observed changes, not a dated support-order chronology. The day is shown at native half-hour physical and price resolution, with hourly weather left on its own clock. Solar's daytime shape and the residual-load ramp are aligned without upsampling historical observations or pretending a daily weather average identifies the tightest half-hour.

Exhibit 161: 19 February 2025: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [4] [313] · Chart data

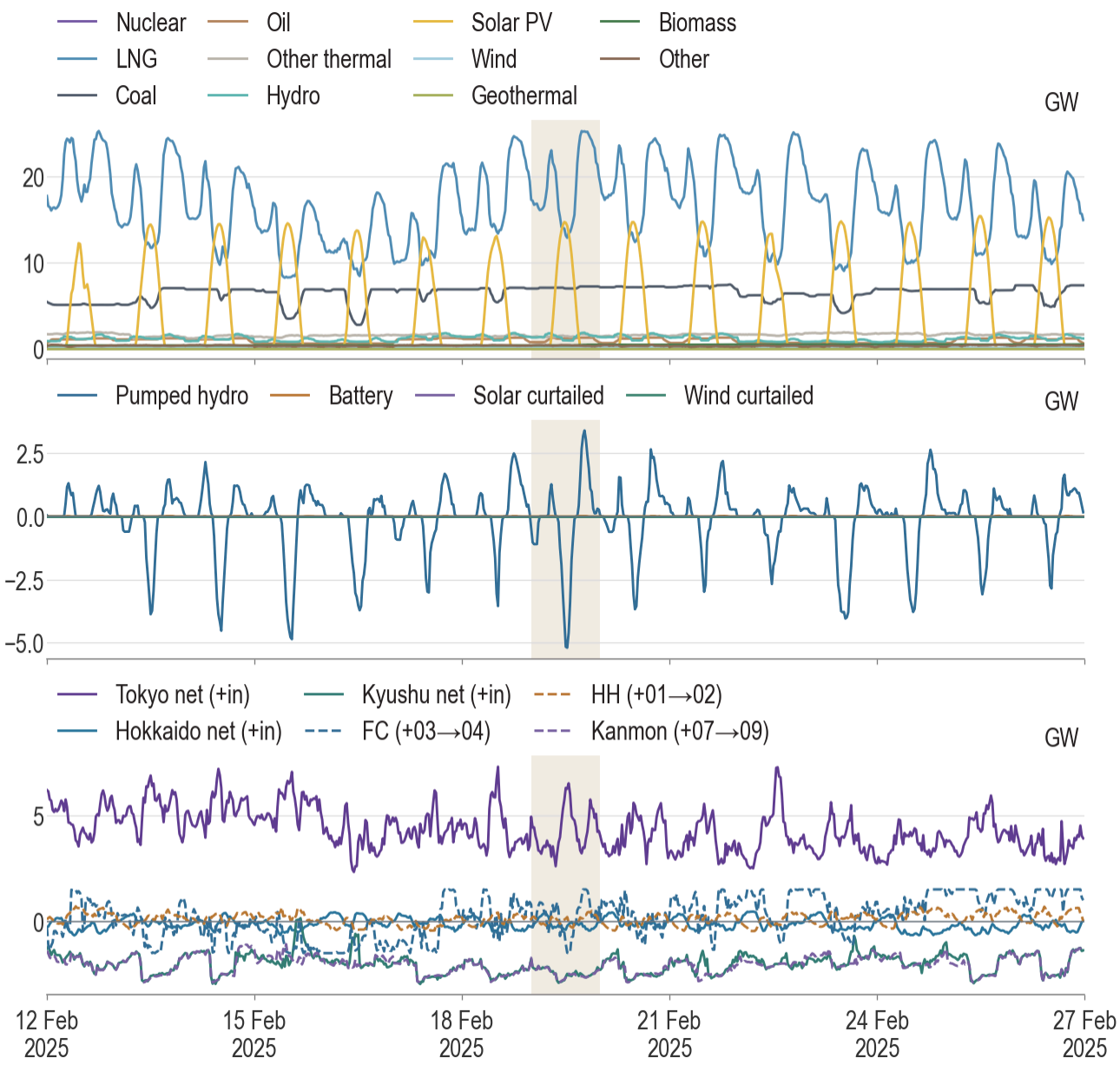
The weather narrative is supported by JMA's February report, while the exact price selection is a local reproducible observation. Hokkaido demand was 4.219 GW and Kyushu 11.862 GW on the day, but their renewable and nuclear settings differed markedly from Tokyo. The absence of a verified action notice is an evidence limit, not proof that no operator took any action; it simply prohibits inventing one for the page.

[313] [291] [285]

03 · Feb 2025 winter day · C. Physical response

Tokyo's mean thermal output rose to 28.682 GW from 24.568 GW in the lead-in. Solar averaged 3.875 GW and wind 154 MW. Thus the price increase coexisted with more accepted solar than the lead-in; "low solar caused the day" is not supported by the whole-day comparison. Timing and residual load matter more than a single daily renewable total.

Exhibit 162: 19 February 2025: physical response



Note: Tokyo's reported generation and storage are shown with qualified area interchange and separate final schedules. HH is Hokkaido→Tohoku, FC Tokyo→Chubu and Kanmon Chugoku→Kyushu. Neither a schedule level nor price separation certifies an auction constraint.

Source: [4] [10] [11] [294] [295] [313] [314] · Chart data

Tokyo pumped storage supplied 12.542 GWh and consumed 18.3415 GWh, with 65 MWh of reported battery discharge. Kyushu, by contrast, averaged 4.156 GW nuclear, 2.314 GW solar and 200 MW wind, leaving 9.348 GW residual demand despite higher total load than its lead-in. Hokkaido wind averaged 357 MW. These physical contrasts are the useful explanation candidates; the charts do not allocate a causal price contribution to each.

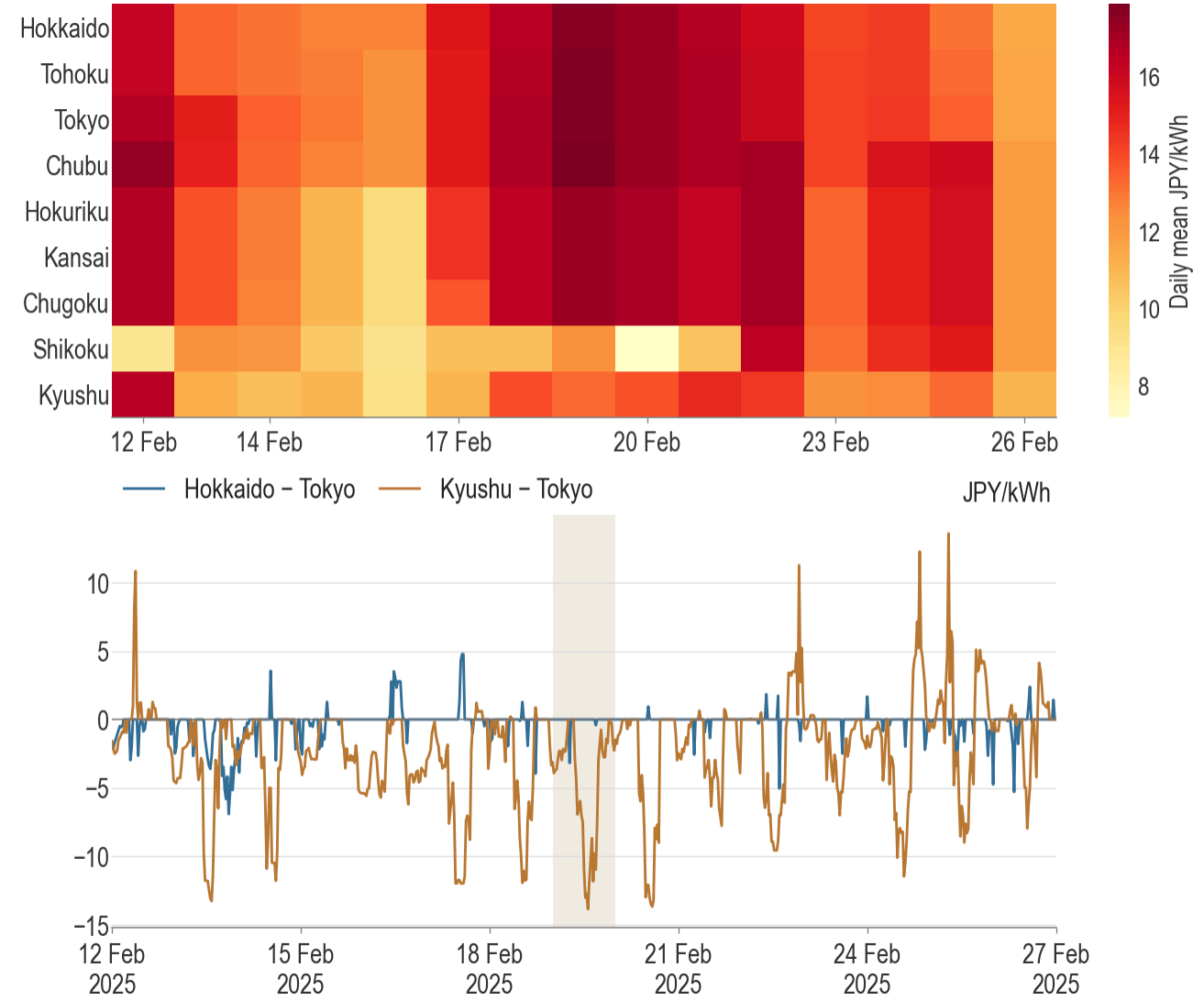
[313] [292] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

03 · Feb 2025 winter day · D. Regional prices

Tokyo and Tohoku both averaged 17.751 JPY/kWh; Hokkaido 17.676 and Chubu 17.859. Kyushu was 13.243 and Shikoku 12.249. The case is therefore an eastern/high-relative-price configuration, not an equally priced national event. All nine areas are shown so that the reader can see the western divergence.

Exhibit 163: 19 February 2025: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [313] · Chart data

The native spread panel prevents an equality of daily means from being mistaken for equality in every half-hour. A one-day local-price selection is not evidence of an annual record or a reliability failure. Compare price levels, tails and directional spreads on the same observed population, then check physical supply and weather rather than importing an unsupported emergency narrative.

North/east	Central/west	West/south
Hokkaido **17.676**	Chubu **17.859**	Chugoku **17.158**
Tohoku **17.751**	Hokuriku **17.158**	Shikoku **12.249**
Tokyo **17.751**	Kansai **17.158**	Kyushu **13.243**

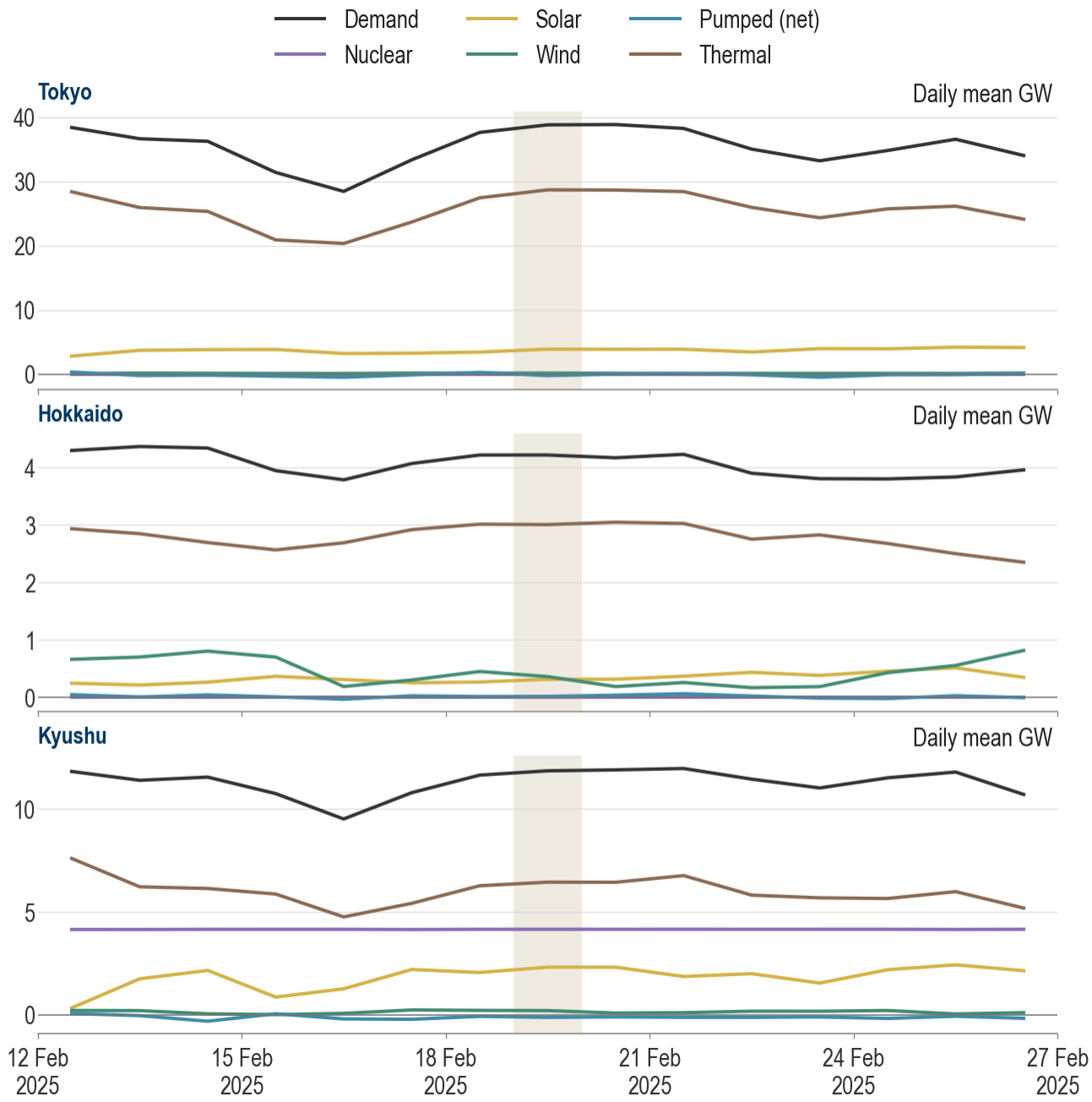
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[313] [296] [285]

03 · Feb 2025 winter day · E. Aftermath

In the following week Tokyo demand averaged 35.821 GW, thermal 26.191 GW and price 14.769 JPY/kWh. Those values eased from the event but remained above some lead-in levels; "recovery" here means the measured aftermath comparison, not withdrawal of an alert that was never evidenced.

Exhibit 164: 19 February 2025: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [4] [313] · Chart data

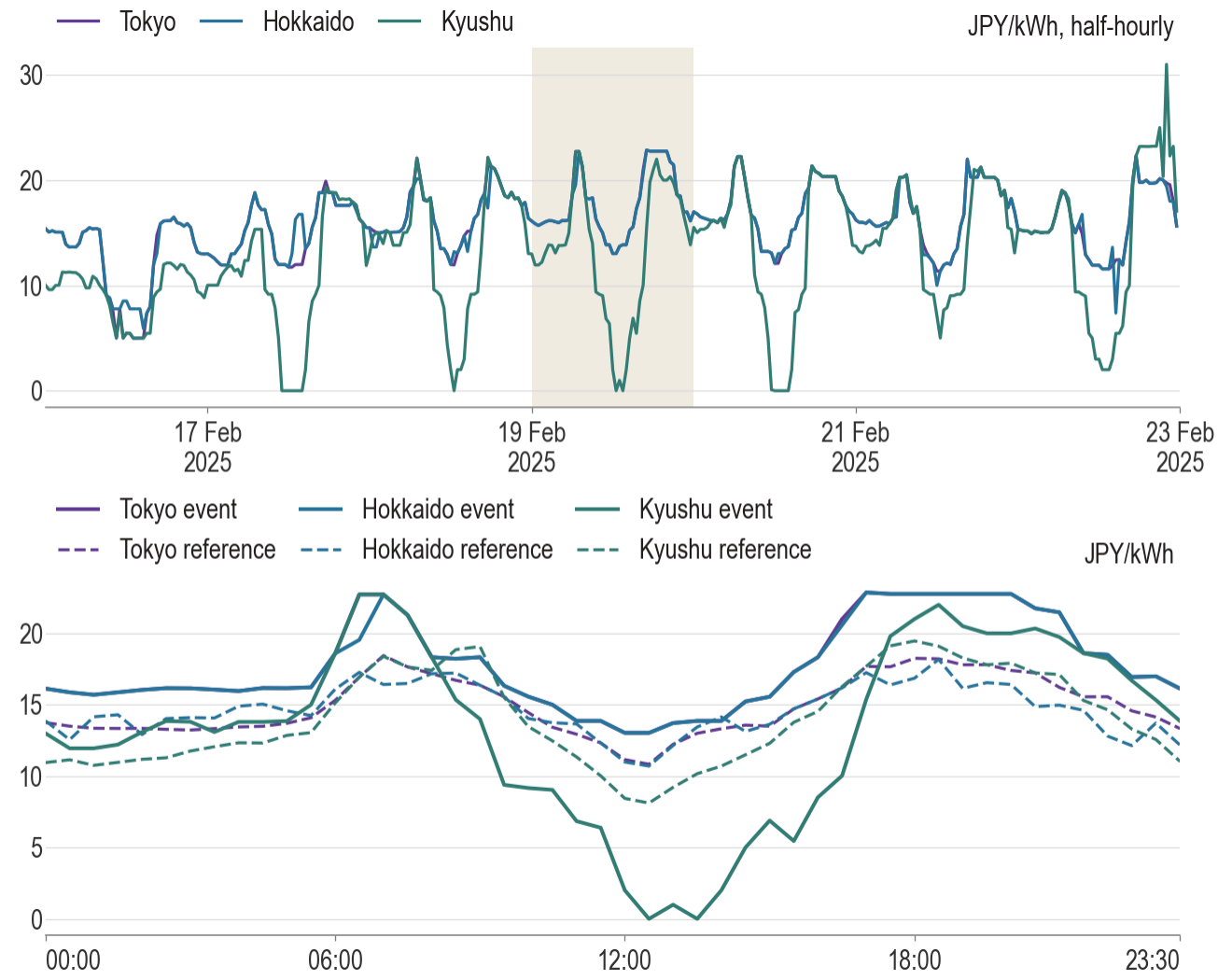
Hokkaido aftermath demand averaged 3.958 GW and wind 368 MW, while Kyushu nuclear remained near 4.154 GW and solar averaged 2.066 GW. Different physical paths can coexist with a broad price retracement. The complete-day generation panels show actual changes across those areas; missing data would remain gaps, not zero output. A trader can use this case to test whether the local morning/evening premium persists after load normalizes without asserting that one technology alone drove it.

[313] [285]

03 · Feb 2025 winter day · F. Comparisons

The preserved reference days are January 22,January 29,February 5 and February 12: four prior non-holiday Wednesdays. Tokyo's event price 17.751 JPY/kWh compares with 14.885; demand 38.806 GW with 37.422 GW; residual 34.777 GW with 34.198 GW. The matched load difference is much smaller than the simple lead-in comparison.

Exhibit 165: 19 February 2025: prices and matched references



Note: Native prices around 19 February and the four-reference intraday comparison retain regional differences. The reference Wednesdays match calendar class, not weather, fuel procurement, outages or network availability.

Source: [1] [2] [286] [290] [313] · Chart data

Kyushu's event price 13.243 JPY/kWh was below its 14.011 reference, while residual demand 9.348 GW was below 10.964 GW. Hokkaido's price rose from 14.639 reference to 17.676 despite residual load being slightly lower. These opposite regional outcomes are the main lesson. Calendar matching improves comparability but leaves weather, renewables, outages and offers uncontrolled; it supports a measured contrast rather than a causal conclusion.

Area	Event*	Reference*	Eligible core days
Tokyo	17.751	14.885	1
Hokkaido	17.676	14.639	1
Kyushu	13.243	14.011	1

*JPY/kWh; matched event population, not necessarily every day of the full core.

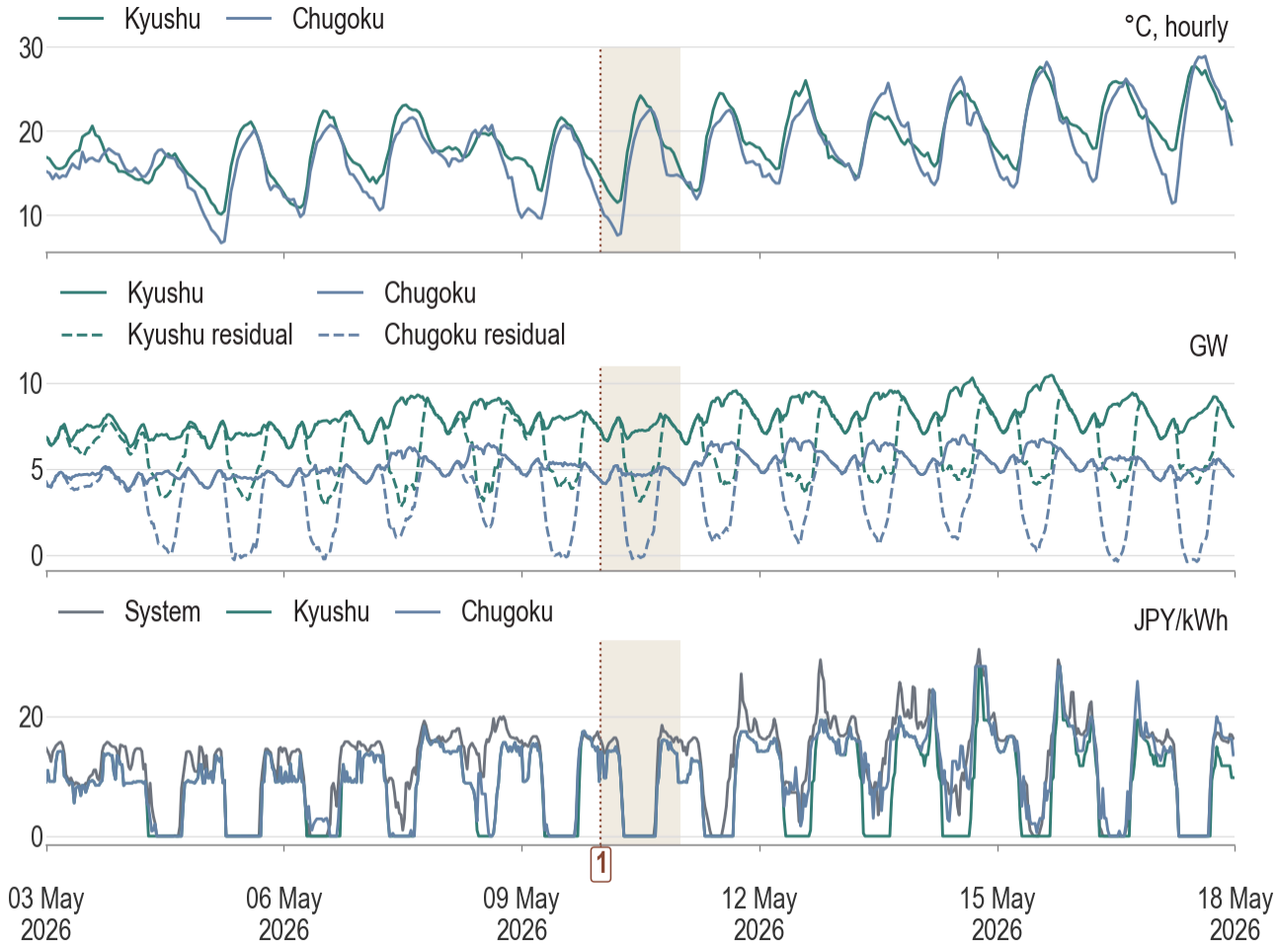
[313] [297]

Case 04 · May 2026 Kyushu solar control

04 · May 2026 curtailment · A. Preconditions

May 10, 2026 is a Sunday solar-curtailment case, not a claim that every sunny day trades at the floor. May 3–9/May 11–17 remain lead-in/aftermath. Kyushu is focal; Chugoku supplies neighboring physical context.

Exhibit 166: 10 May 2026: preconditions and onset



Note: 3–17 May 2026; 10 May is shaded. Demand and residual load use native half-hours, weather hourly representative points. The charts distinguish low residual demand from the amount of renewable generation actually accepted.

Source: [1] [2] [4] [5] [290] [315–317] · Chart data

Accepted solar is output after curtailment, not available sunlight-driven production. The separate curtailment field, utility action record and weather evidence are therefore essential. Nuclear, thermal, pumping and exports coexist with surplus pressure; the daily price can conceal both low solar-window prices and expensive non-curtailment hours.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	7.596	7.610	16.98
core	7.657	7.381	17.85
aftermath	9.211	8.470	20.67

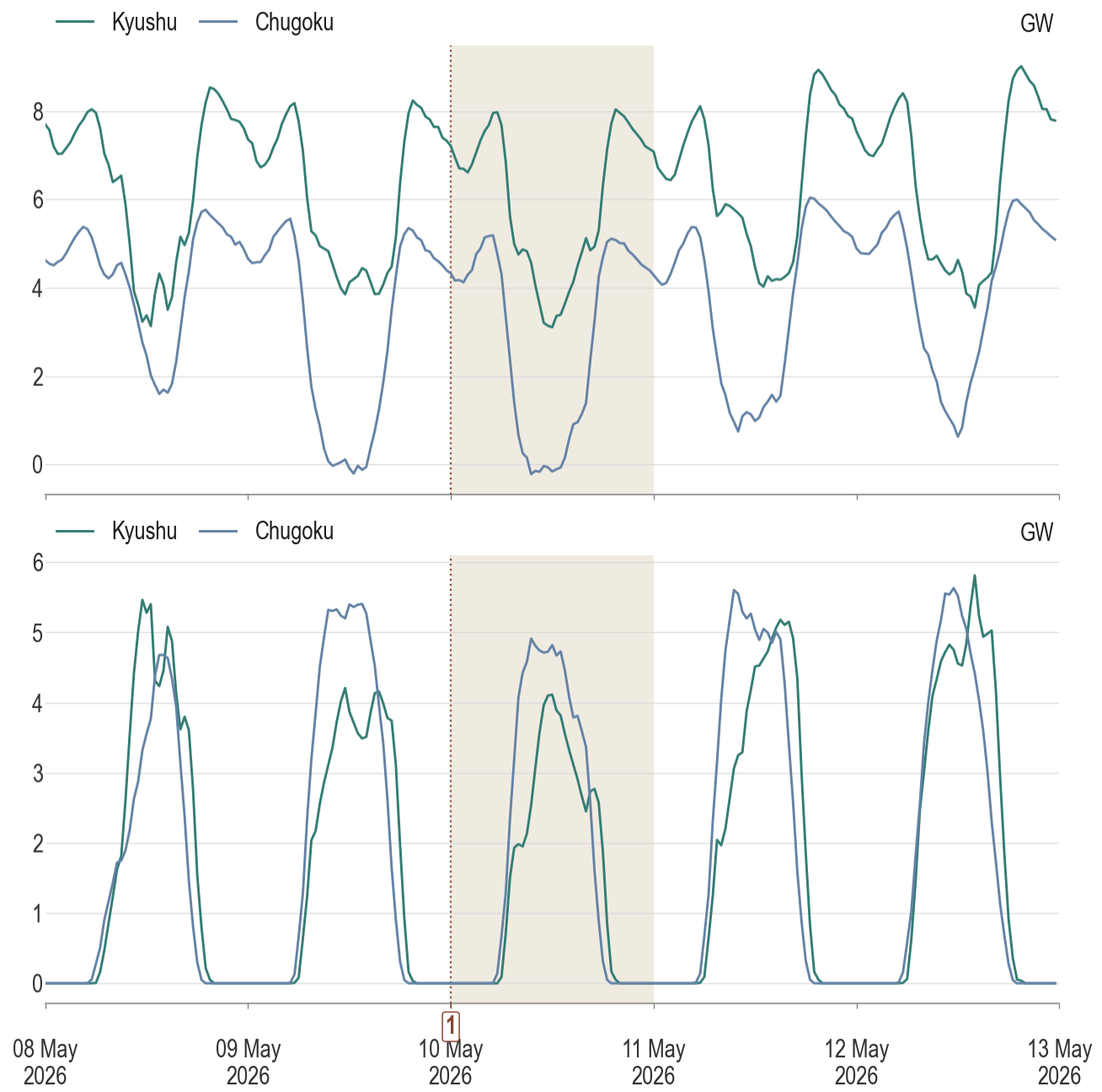
[315] [316] [317] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

04 · May 2026 curtailment · B. Chronology

Positive reported solar curtailment spans 20 native half-hours, starting 07:30 and ending with the 17:00 interval. These observed intervals are not inferred from a generic noon profile. The utility's dated curtailment evidence is a separate documentary record; a reported field and an instruction can differ in scope and timing.

Exhibit 167: 10 May 2026: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [4] [315–317] · Chart data

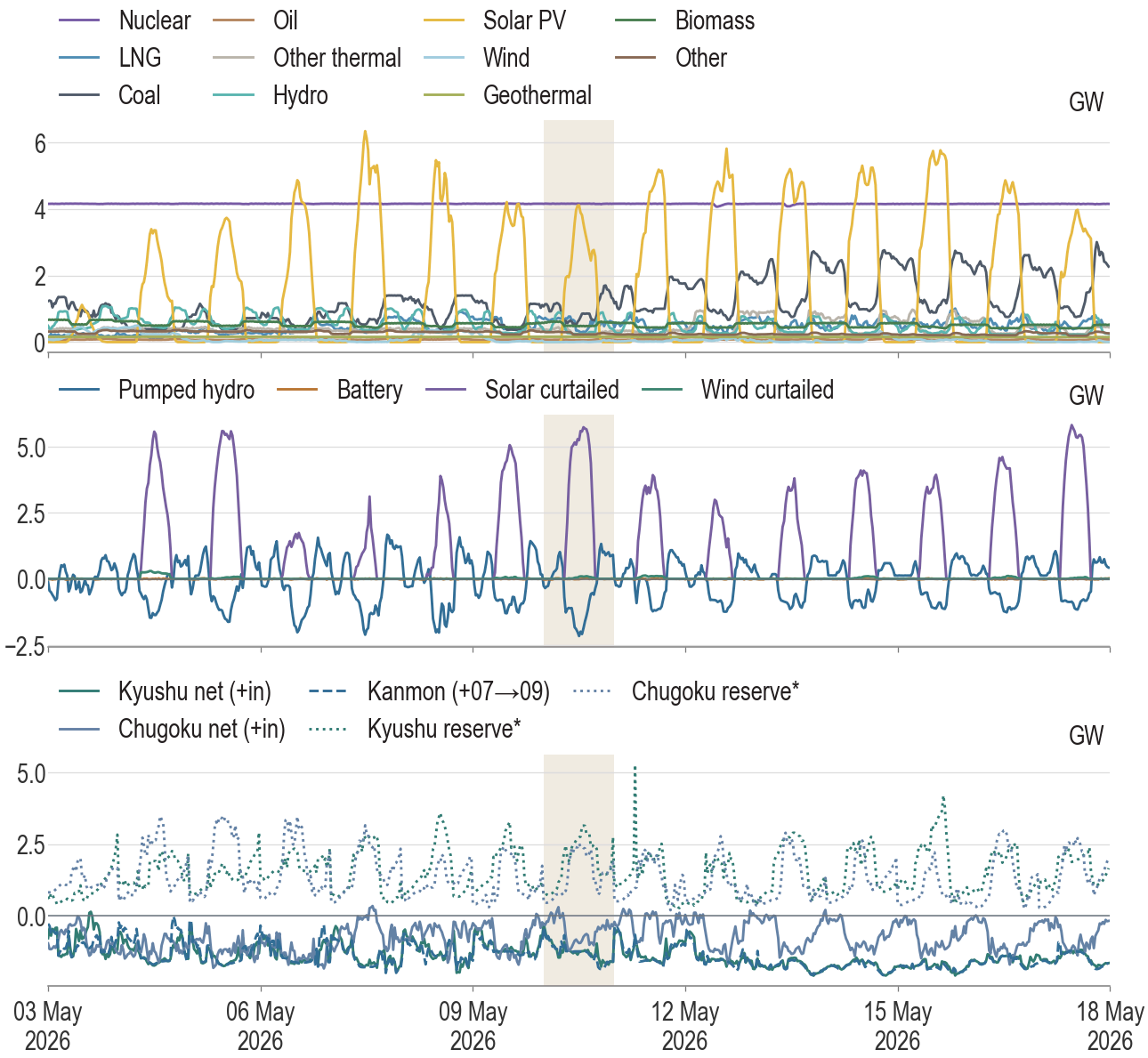
During those positive-curtailment intervals Kyushu mean price was 0.4215 JPY/kWh; outside them it was 12.8254. The full-day mean 7.6571 therefore conceals both near-floor daytime pricing and substantially higher non-curtailment hours. This calculation is an association on paired native observations, not proof that curtailment mechanically sets price. Demand, baseload supply, storage and export capability coexist in the balance.

[315] [316] [317] [291] [285] [318]

04 · May 2026 curtailment · C. Physical response

Integrating the 48 half-hours gives 44.9745 GWh curtailed solar and 34.1915 GWh accepted solar. Reconstructed available solar is 79.166 GWh; curtailed divided by accepted-plus-curtailed is 56.81%. That denominator is explicit and differs from curtailment divided by accepted output. It is not an independent irradiance-model estimate.

Exhibit 168: 10 May 2026: physical response



Note: Kyushu generation, signed storage and reported curtailment share the same native clock. Regional interchange is positive for imports after source-sign qualification; Kanmon's final schedule is positive Chugoku→Kyushu and is not a measured flow or a capacity limit. Dotted reserve* is the latest published area plan, not telemetry or an issue-time vintage.

Source: [4] [10] [11] [294] [295] [315–317] [319] · Chart data

Pumped storage consumed 15.025 GWh and generated 7.298 GWh; batteries charged 0.150 GWh and discharged 0.122 GWh. Thermal still averaged 1.943 GW and nuclear 4.155 GW. Storage and inter-area movement can absorb surplus at some hours without eliminating all curtailment. Do not compare a storage-power maximum with curtailed daily energy, or call the net pumping difference a round-trip loss without matching state of charge and measurement boundaries.

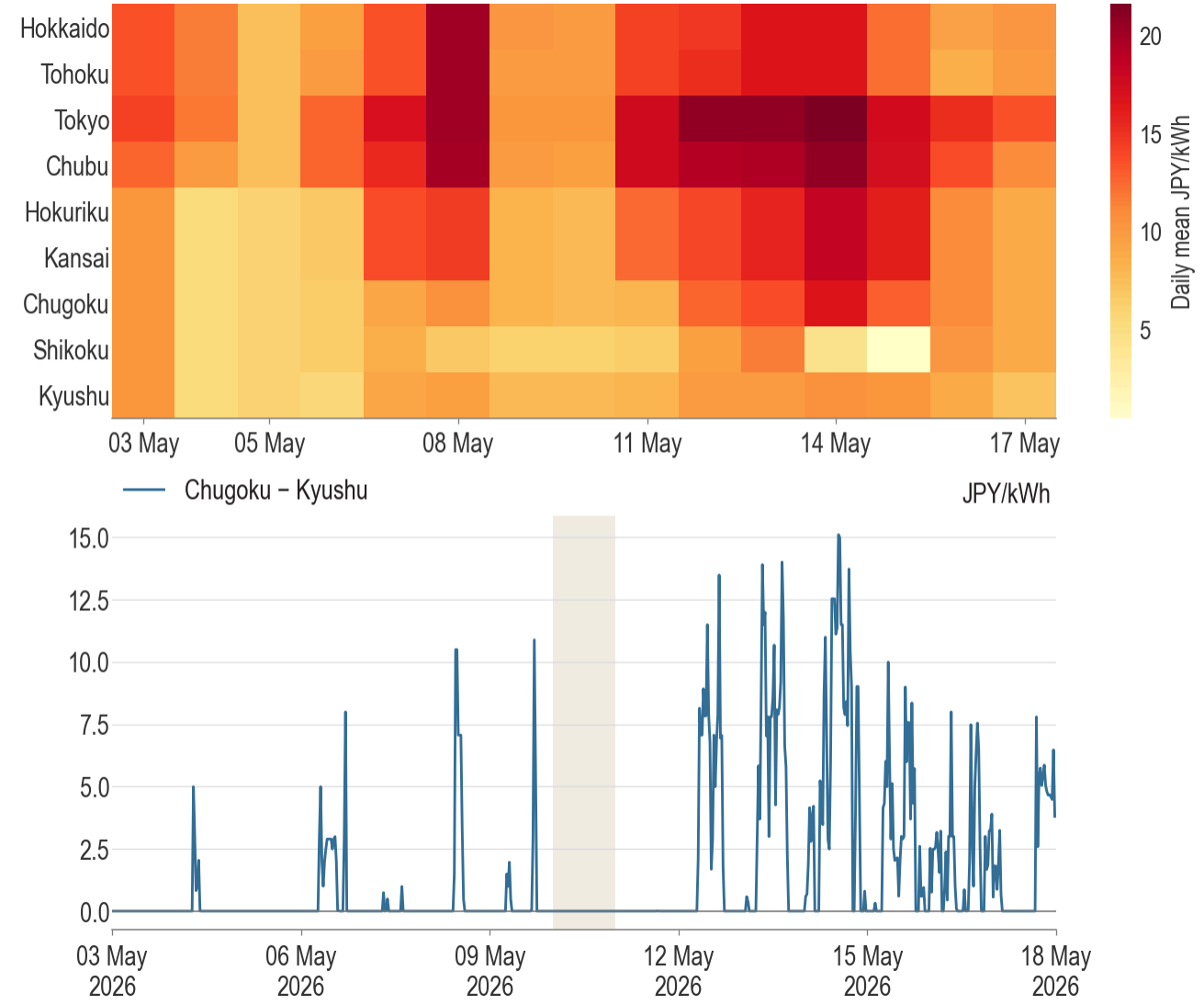
[315] [316] [317] [292] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

04 · May 2026 curtailment · D. Regional prices

Kyushu's daily mean 7.657 JPY/kWh equalled Kansai, Hokuriku and Chugoku in the rounded table, while Shikoku was 6.188 and Tokyo 10.329. Equality of daily averages is not proof of identical half-hour prices. The native spread panel preserves any intraday separation; all nine areas establish the broader spatial pattern.

Exhibit 169: 10 May 2026: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [315–317] · Chart data

A curtailed day can have a positive full-day price and can even average above a matched baseline. Here the full-day mean is slightly above the reference despite very low prices during curtailment. That is precisely why a trading explanation must distinguish the solar window from evening and overnight hours, rather than using the existence of curtailment as a complete forecast for every product on that delivery date.

North/east	Central/west	West/south
Hokkaido **9.940**	Chubu **9.597**	Chugoku **7.657**
Tohoku **9.965**	Hokuriku **7.657**	Shikoku **6.188**
Tokyo **10.329**	Kansai **7.657**	Kyushu **7.657**

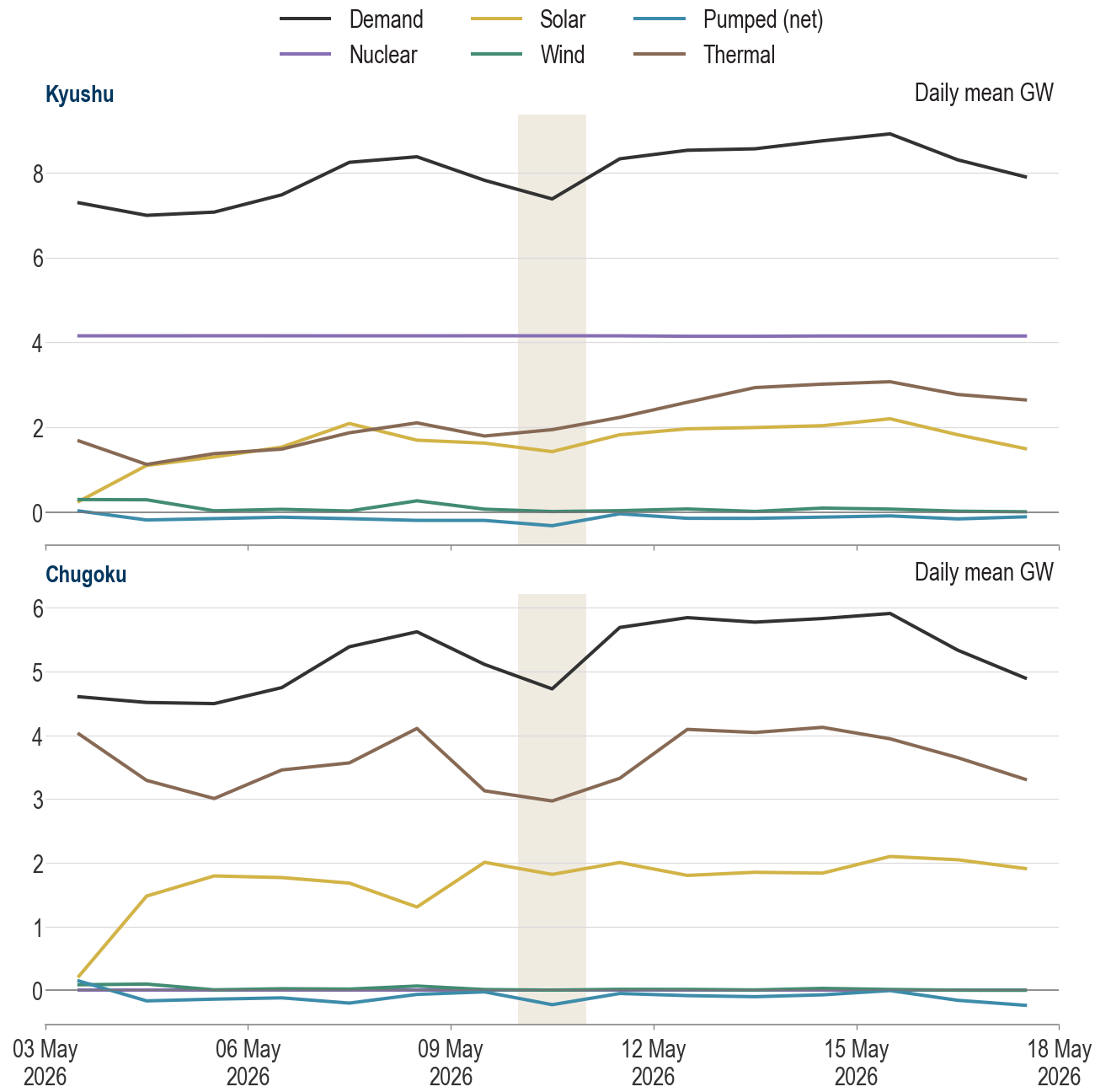
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[315] [316] [317] [296] [285]

04 · May 2026 curtailment · E. Aftermath

The aftermath has higher demand and solar output: Kyushu mean load 8.470 GW, accepted PV 1.904 GW and thermal 2.751 GW, compared with 7.381 GW, 1.425 GW and 1.943 GW on May 10. Nuclear remains near 4.149 GW. Mean price rises to 9.211 JPY/kWh. A higher post-event price is not evidence that the curtailment instruction was wrongly timed.

Exhibit 170: 10 May 2026: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [4] [315–317] · Chart data

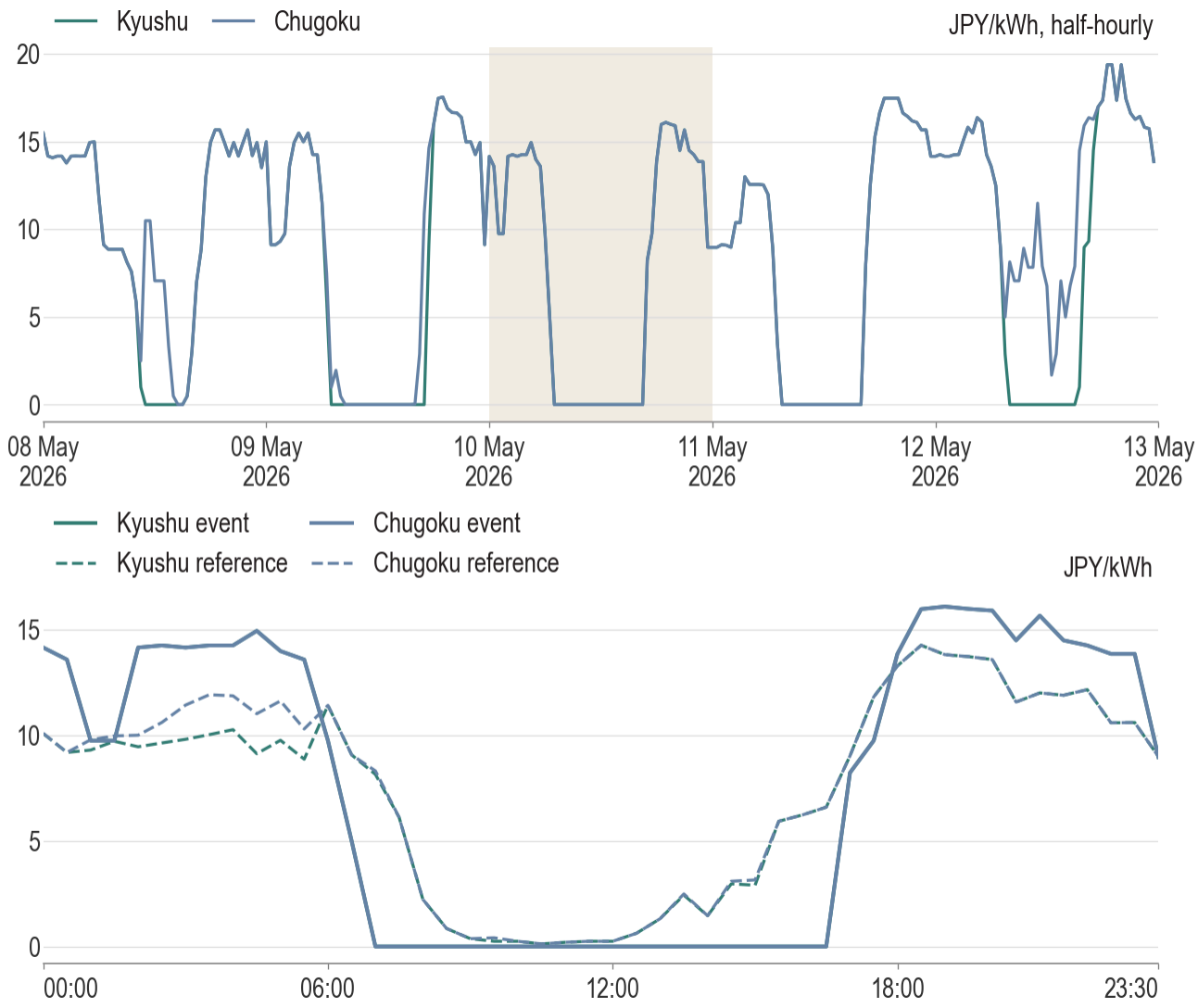
Chugoku's aftermath demand 5.614 GW and thermal 3.786 GW differ from its low-load core 4.730 GW and 2.971 GW. Recovery here is a move away from the specific surplus-day configuration, not a repaired failed asset. The daily neighbor panels preserve accepted renewables and pumped-storage sign; they do not silently replace missing curtailment data with zeros or claim every subsequent day was uncurtailed.

[315] [316] [317] [285]

04 · May 2026 curtailment · F. Comparisons

The original references are April 5,12,19 and 26, four eligible non-holiday Sundays. May 3 is excluded by holiday class. Kyushu event price 7.657 JPY/kWh is above the 7.358 reference by 0.299, while demand is 51 MW lower and residual load 381 MW lower. The result is retained even though it is less tidy than "curtailment means a cheaper day".

Exhibit 171: 10 May 2026: prices and matched references



Note: Half-hour area prices show the daylight/evening pattern, while the profile compares 10 May with four eligible April Sundays. Their calendar similarity does not control sunlight or commitment, and curtailed energy does not mechanically determine the daily price.

Source: [1] [2] [286] [290] [315–317] · Chart data

Chugoku's price is 7.657 versus 7.636 reference, but residual demand 2.905 GW versus 3.861 GW. The day-average comparison and the within-day positive-curtailment comparison answer different questions. The former preserves calendar reproducibility; the latter explains product-hour exposure. Neither by itself determines an interconnector's shadow price or a counterfactual with unlimited storage.

Area	Event*	Reference*	Eligible core days
Kyushu	7.657	7.358	1
Chugoku	7.657	7.636	1

*JPY/kWh; matched event population, not necessarily every day of the full core.

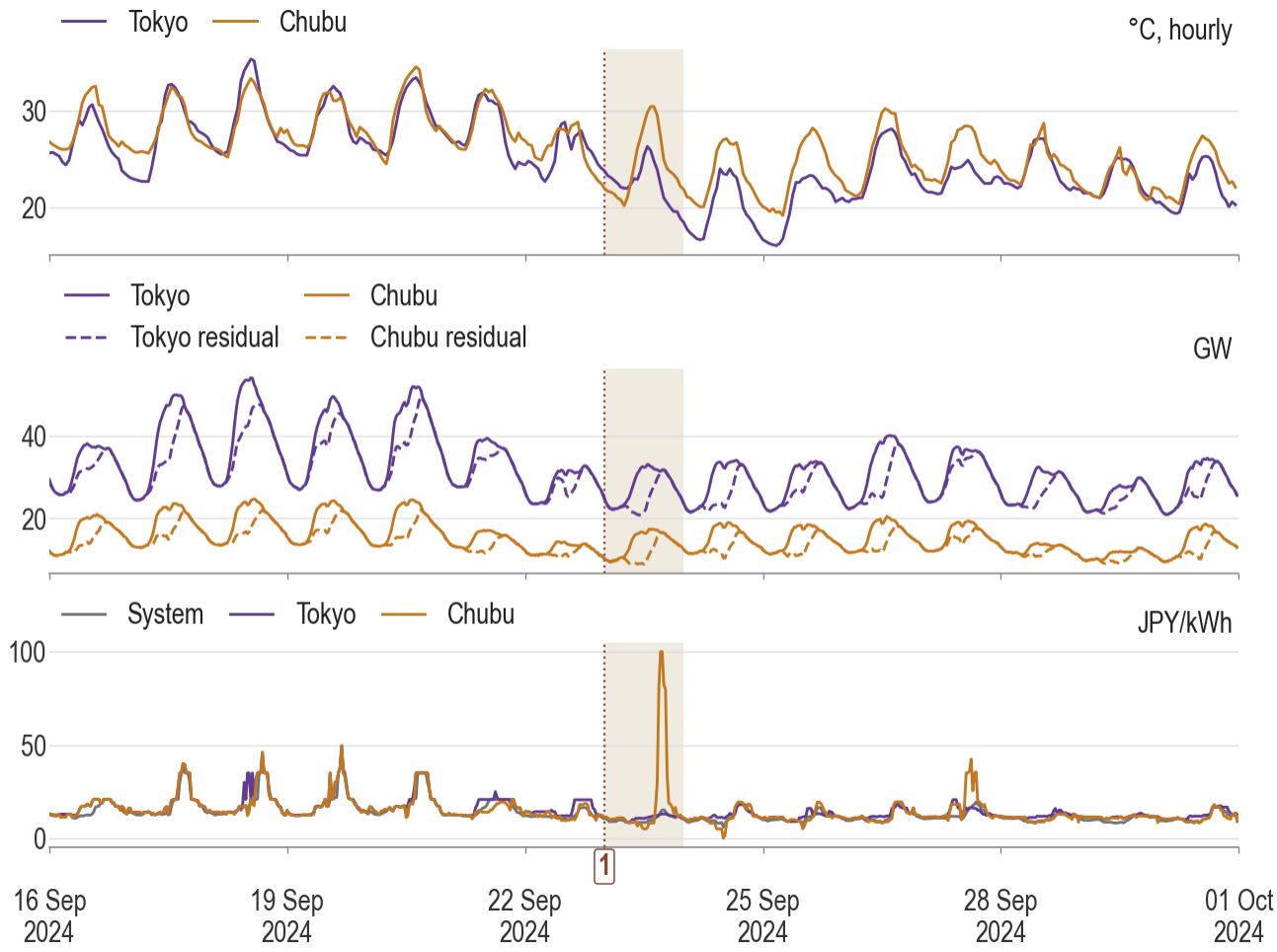
[315] [316] [317] [297]

Case 05 · September 2024 holiday basis

05 · Sep 2024 holiday · A. Preconditions

September 23, 2024 is a national-holiday Monday. The one-day core retains September 16–22 lead-in and September 24–30 aftermath. Tokyo/Chubu straddle the 50/60 Hz interface; this is a divergent-price case, not an evidenced converter outage.

Exhibit 172: 23 September 2024: preconditions and onset



Note: 16–30 September 2024; the substitute-holiday Monday is shaded. Tokyo and Chubu demand, dashed residual load and prices use matching native clocks. The preserved selection is not reranked over the longer edition.

Source: [1] [2] [4] [5] [290] [320] [321] · Chart data

Both areas' demand fell, but prices moved oppositely. A national holiday/weather label cannot determine a directional spread. Frequency conversion is a plausible system channel, not a proved cause without dated operational evidence. Part F preserves the sole eligible holiday reference and separately labels ordinary-Monday sensitivity.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	18.210	35.823	27.75
core	11.144	27.856	22.58
aftermath	12.414	28.905	22.19

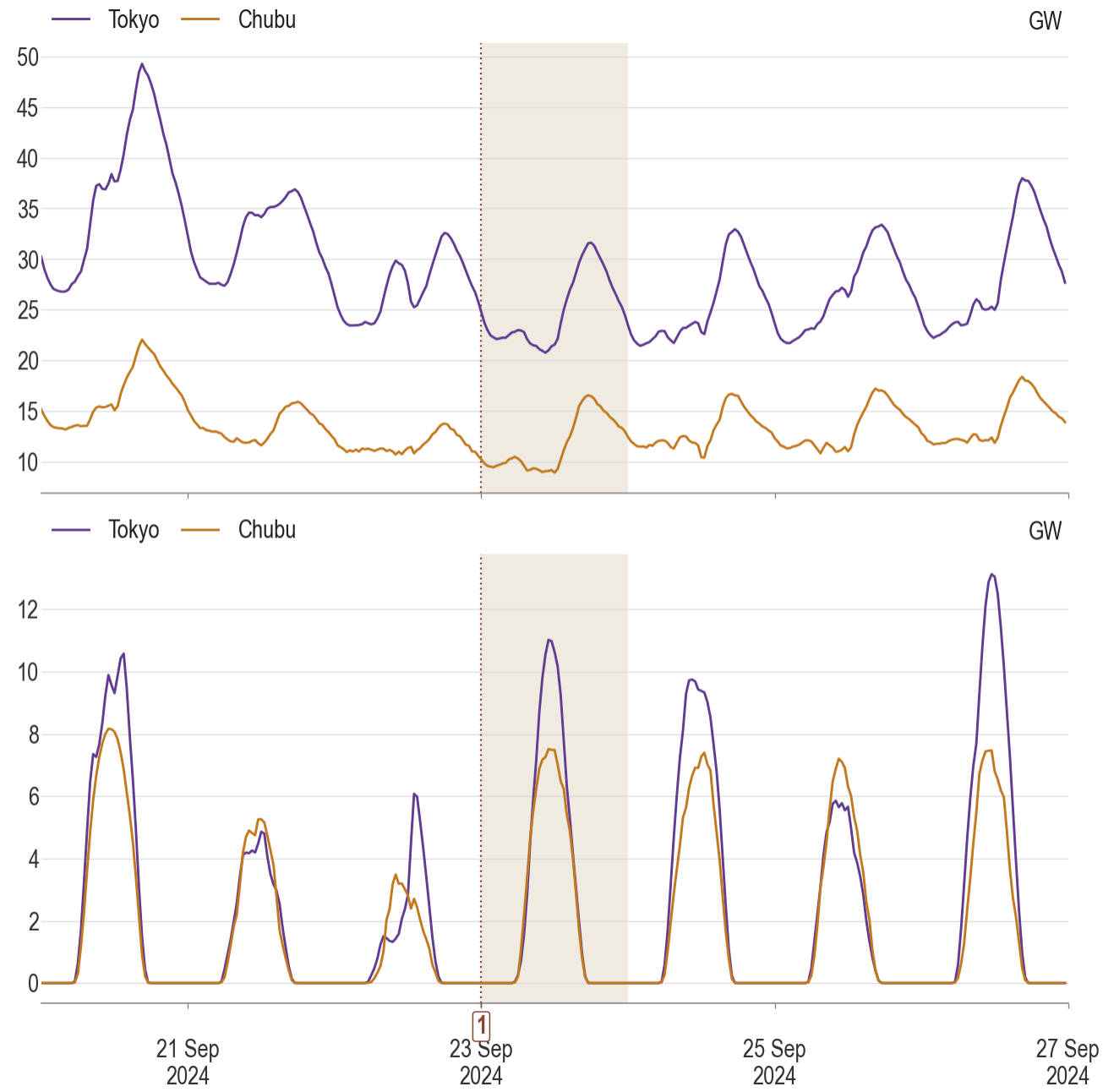
[320] [321] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

05 · Sep 2024 holiday · B. Chronology

The primary chronology is intraday. Chubu-minus-Tokyo price was negative in part of the day and switched directly from negative to positive at 15:30. Its minimum was−5.88 JPY/kWh and maximum 87.04. A single signed daily average therefore hides both direction changes and a pronounced positive tail.

Exhibit 173: 23 September 2024: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [4] [320] [321] · Chart data

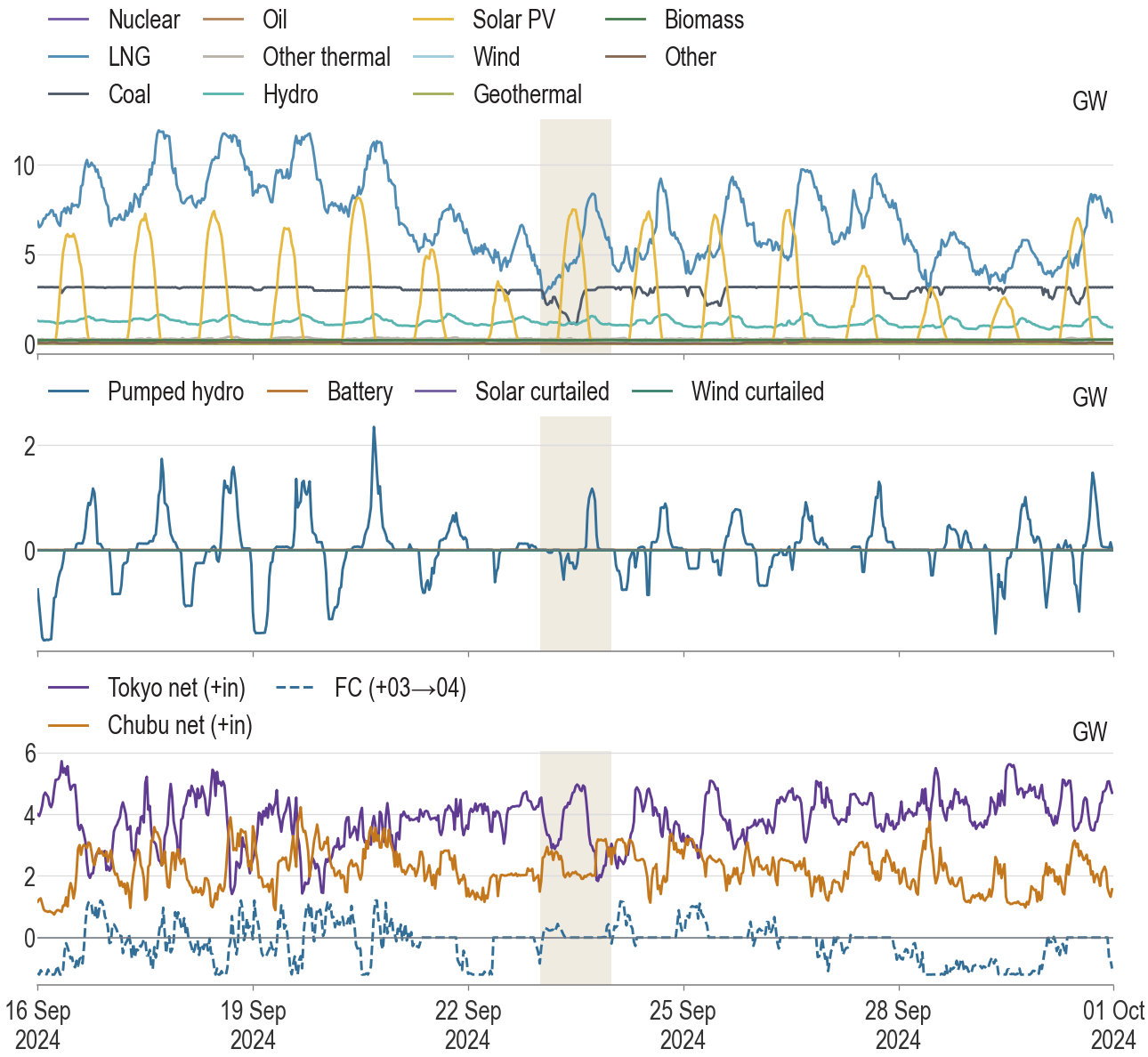
No dated converter-outage notice is established in the accepted evidence. The chart accordingly aligns residual load and solar with actual prices and measured boundary flow rather than drawing an invented outage band. Holiday classification is factual; an assumed mechanism based only on the spread is not. Demand, solar timing, available offers and the boundary's actual operating conditions remain separate explanatory candidates.

[320] [321] [291] [285]

05 · Sep 2024 holiday · C. Physical response

Tokyo's mean thermal output was 19.201 GW versus 27.494 GW lead-in, while accepted solar averaged 2.785 GW and wind 130 MW. Chubu thermal was 7.853 GW versus 11.504 GW lead-in, with solar 2.221 GW and wind 87 MW. Both areas used less thermal on average, so "more thermal generation caused Chubu's higher price" is not supported by these day totals.

Exhibit 174: 23 September 2024: physical response



Note: Chubu generation and storage accompany both areas' qualified net interchange and the FC final schedule. FC's positive direction is Tokyo→Chubu. Separate source-clock schedules cannot identify converter utilisation, a binding auction limit or physical reversal.

Source: [4] [10] [11] [294] [295] [320–322] · Chart data

Tokyo pumped storage generated 3.463 GWh and consumed 7.1555 GWh; the core accepted solar energy was 66.832 GWh. The native physical panels retain the paired areas and signed converter flow. A flow close to an observed limit may be relevant, but neither price separation nor a line on this chart identifies a particular outage, offer constraint or direction of causality without corroboration.

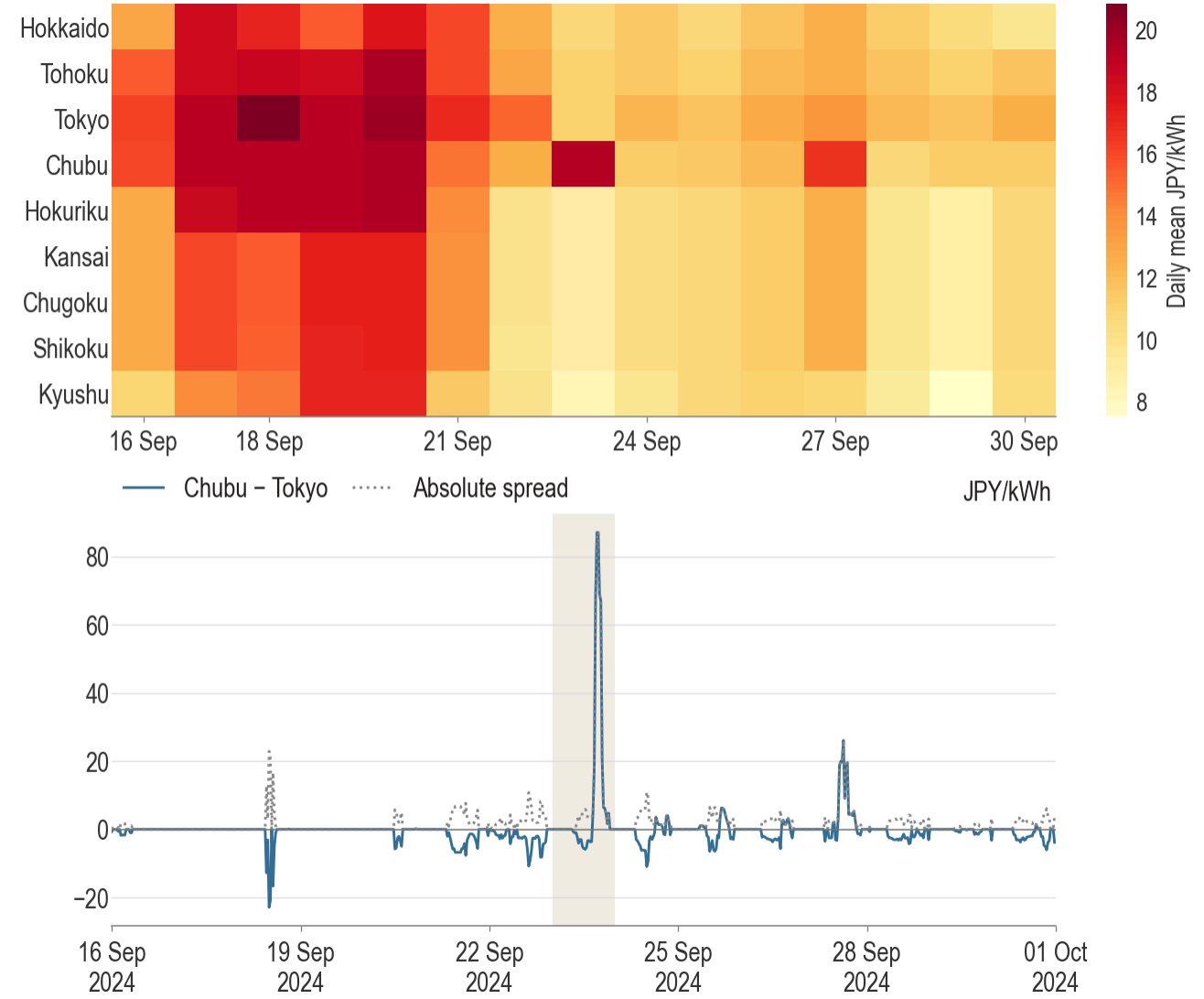
[320] [321] [292] [285]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

05 · Sep 2024 holiday · D. Regional prices

Tokyo averaged 11.144 JPY/kWh while Chubu averaged 19.369; the signed Chubu-minus-Tokyo spread was 8.2248. Mean absolute spread was 10.3410, larger because opposite-signed intervals partially cancel in the signed measure. The positive tail reached 87.04, so the mean alone understates intraday risk.

Exhibit 175: 23 September 2024: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [320] [321] · Chart data

All nine prices show that Chubu was the clear daily outlier: Kansai/Hokuriku/Chugoku were 8.957, Shikoku 8.956 and Kyushu 8.372. That does not prove unlimited western energy could cross the interface or identify a broken converter. It establishes a spatial price configuration requiring separate physical investigation. The chart preserves both the whole-day map and native paired differences rather than mixing independent sample means.

North/east	Central/west	West/south
Hokkaido **10.677**	Chubu **19.369**	Chugoku **8.957**
Tohoku **11.144**	Hokuriku **8.957**	Shikoku **8.956**
Tokyo **11.144**	Kansai **8.957**	Kyushu **8.372**

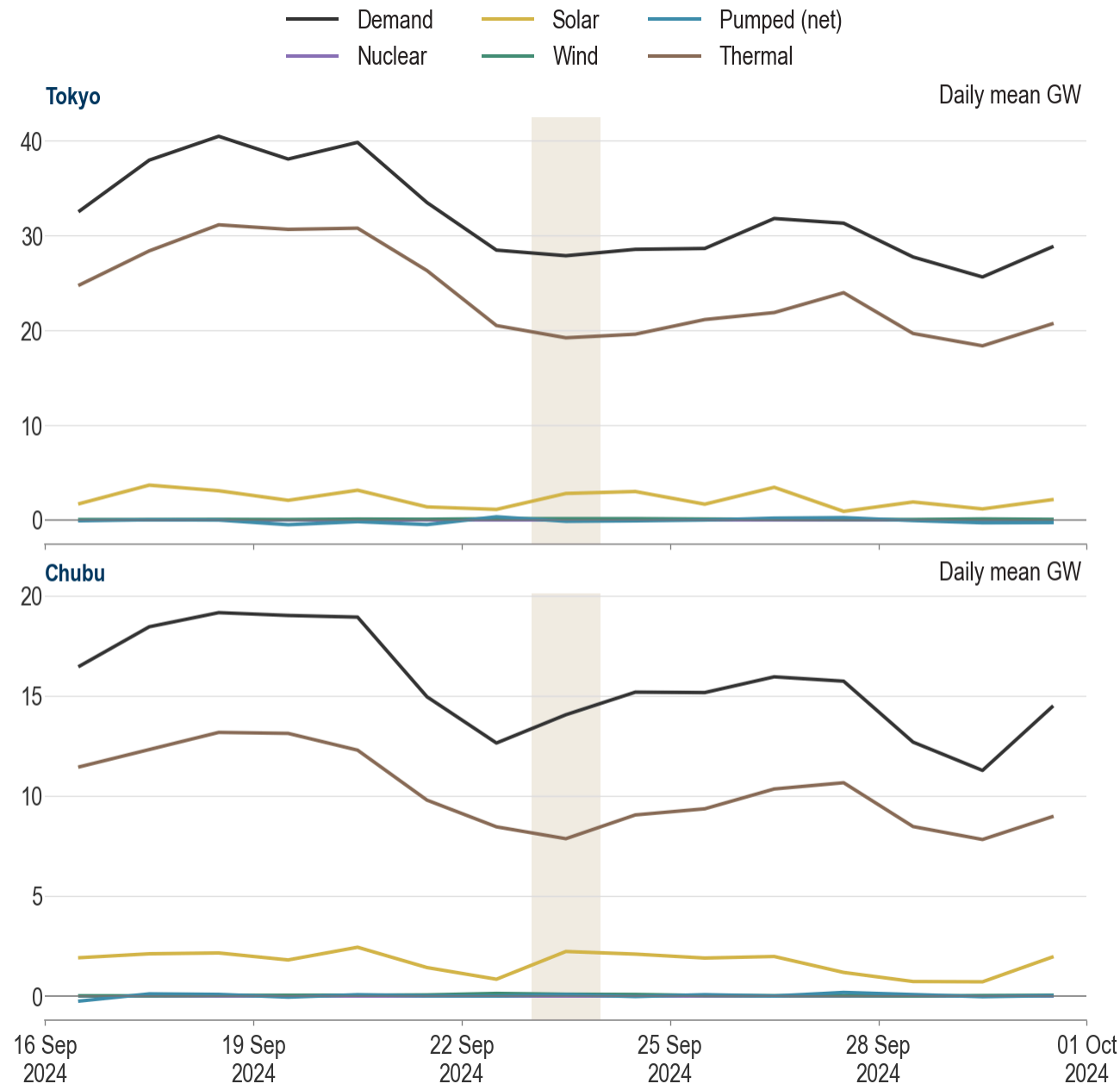
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[320] [321] [296] [285]

05 · Sep 2024 holiday · E. Aftermath

The following week's Tokyo price averaged 12.414 JPY/kWh and Chubu 12.077. Tokyo demand 28.905 GW and Chubu 14.336 GW were only moderately above the holiday, while thermal output rose to 20.743 GW and 9.228 GW respectively. The extraordinary directional spread did not persist in the simple aftermath mean.

Exhibit 176: 23 September 2024: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [4] [320] [321] · Chart data

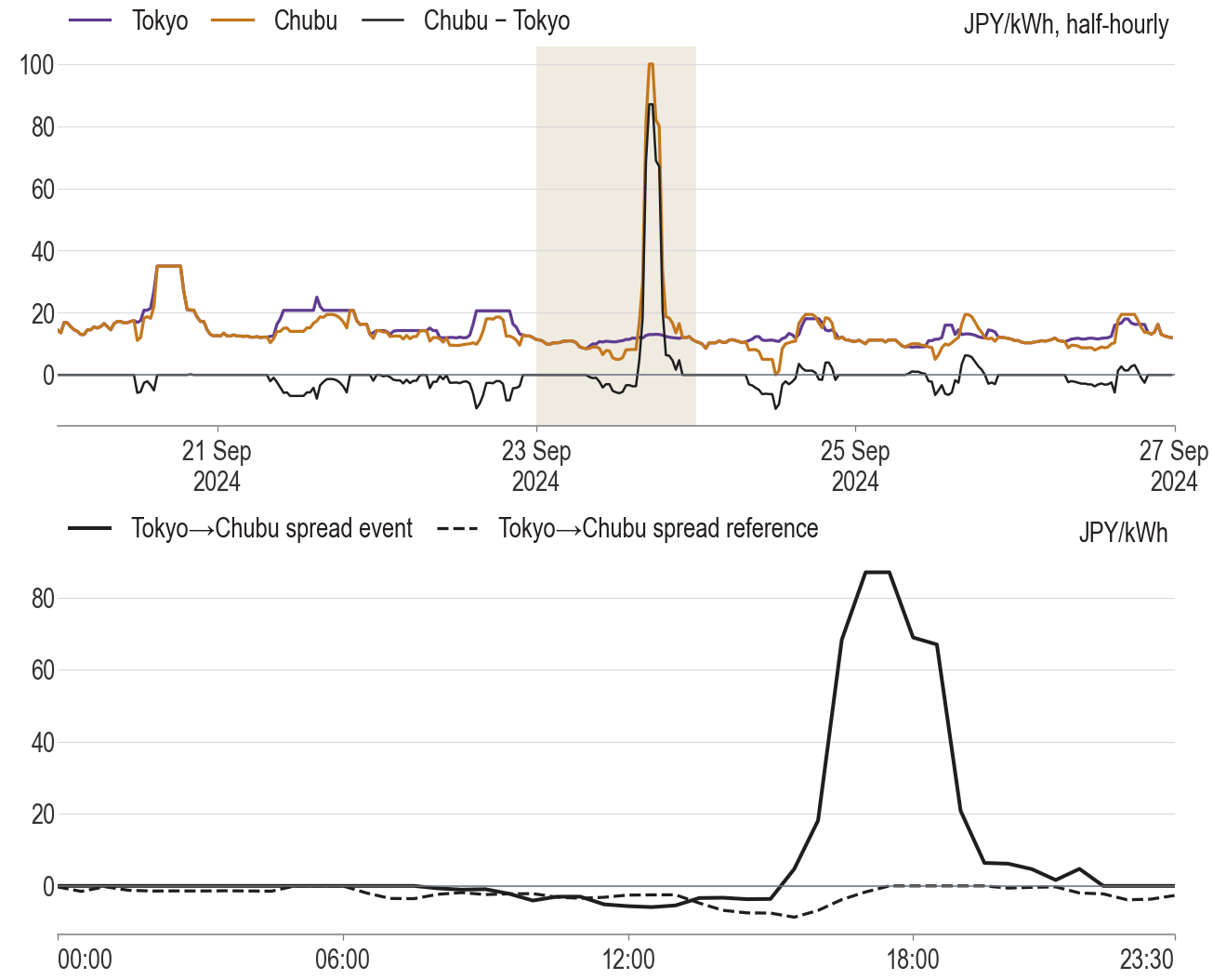
The recovery chart compares both systems' demand and generation, not a supposed repair date. There is no evidenced converter outage to declare fixed. Lower spreads after the event are consistent with a changed hourly balance but do not prove that transmission capacity increased. The maintained comparison window provides a reproducible endpoint; a trader should still inspect individual hours rather than treating a seven-day mean as a hedge for a specific delivery block.

[320] [321] [285]

05 · Sep 2024 holiday · F. Comparisons

The only eligible prior holiday Monday under the preserved rule is August 12. Tokyo event price 11.144 JPY/kWh compares with 14.967; Chubu 19.369 with 12.636. The matched directional spread therefore moves from−2.331 to+8.225. It would be misleading to silently replace that one holiday reference with ordinary Mondays and still call the result calendar matched.

Exhibit 177: 23 September 2024: prices and matched references



Note: Black is Chubu minus Tokyo; lower profiles compare 23 September with its sole eligible holiday reference, 12 August. Signed and absolute spreads differ; a price reversal is not a measured flow reversal.

Source: [1] [2] [286] [290] [320] [321] · Chart data

An explicitly separate sensitivity uses August 19,August 26,September 2 and September 9. Their mean signed spread is+0.557 JPY/kWh and mean absolute spread 1.969, versus event+8.225 and 10.341. This corroborates an unusual separation under another comparison but has a different calendar population. The holiday reference remains the primary comparison; neither baseline isolates a converter outage or controls for weather and unit conditions.

Area	Event*	Reference*	Eligible core days
Tokyo	11.144	14.967	1
Chubu	19.369	12.636	1

*JPY/kWh; matched event population, not necessarily every day of the full core.

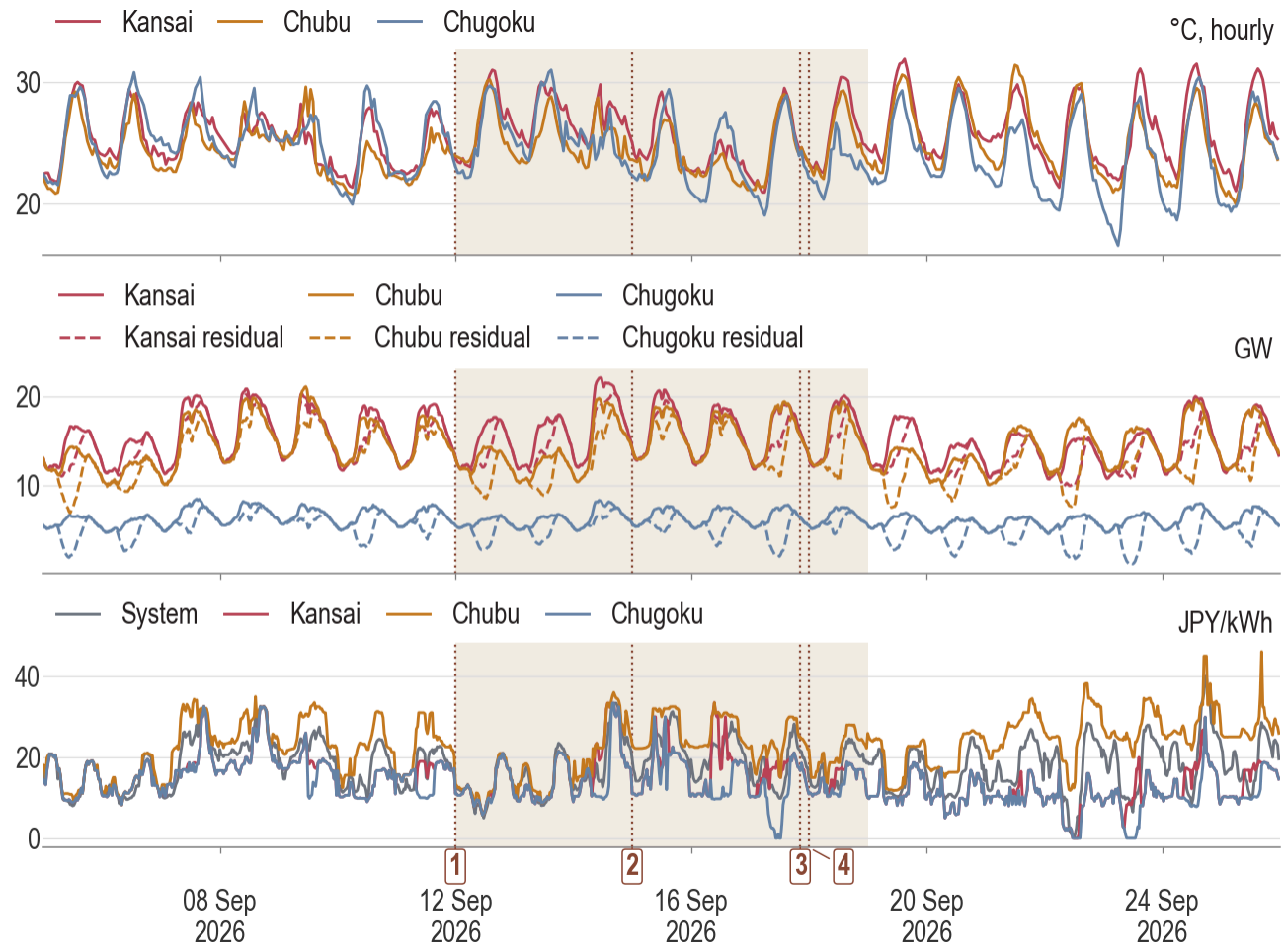
[320] [321] [297] [323] [290]

Case 06 · September 2026 Mihama return and interruption

06 · Sep 2026 Mihama · A. Preconditions

September 12–18 is Mihama 3's interrupted startup, not completed commercial return. September 5–11/September 19–25 remain lead-in/aftermath, with Kansai focal and Chubu/Chugoku neighbors.

Exhibit 178: 12–18 September 2026: preconditions and onset



Note: 5–25 September 2026; 12–18 September is shaded. Markers locate reactor startup, first generation, load reduction and stop. Regional demand and prices are contemporaneous observations; the confirming operator notices were published later.

Source: [1] [2] [4] [5] [290] [301] [324–326] · Chart data

The 826 MW reactor's May 8 steam-leak stop and June 16 inspection preceded the core. Kansai nuclear averaged 3.620 GW before and 3.701 GW during—81 MW higher over seven days, consistent with brief adjustment rather than a flat 826 MW injection. October 10 startup and November 4 commercial return update the documentary endpoint, not the original window.

Focal period	Price JPY/kWh	Demand GW	Temperature °C
lead-in	16.512	15.644	25.17
core	16.169	16.096	25.75
aftermath	11.808	14.432	26.05

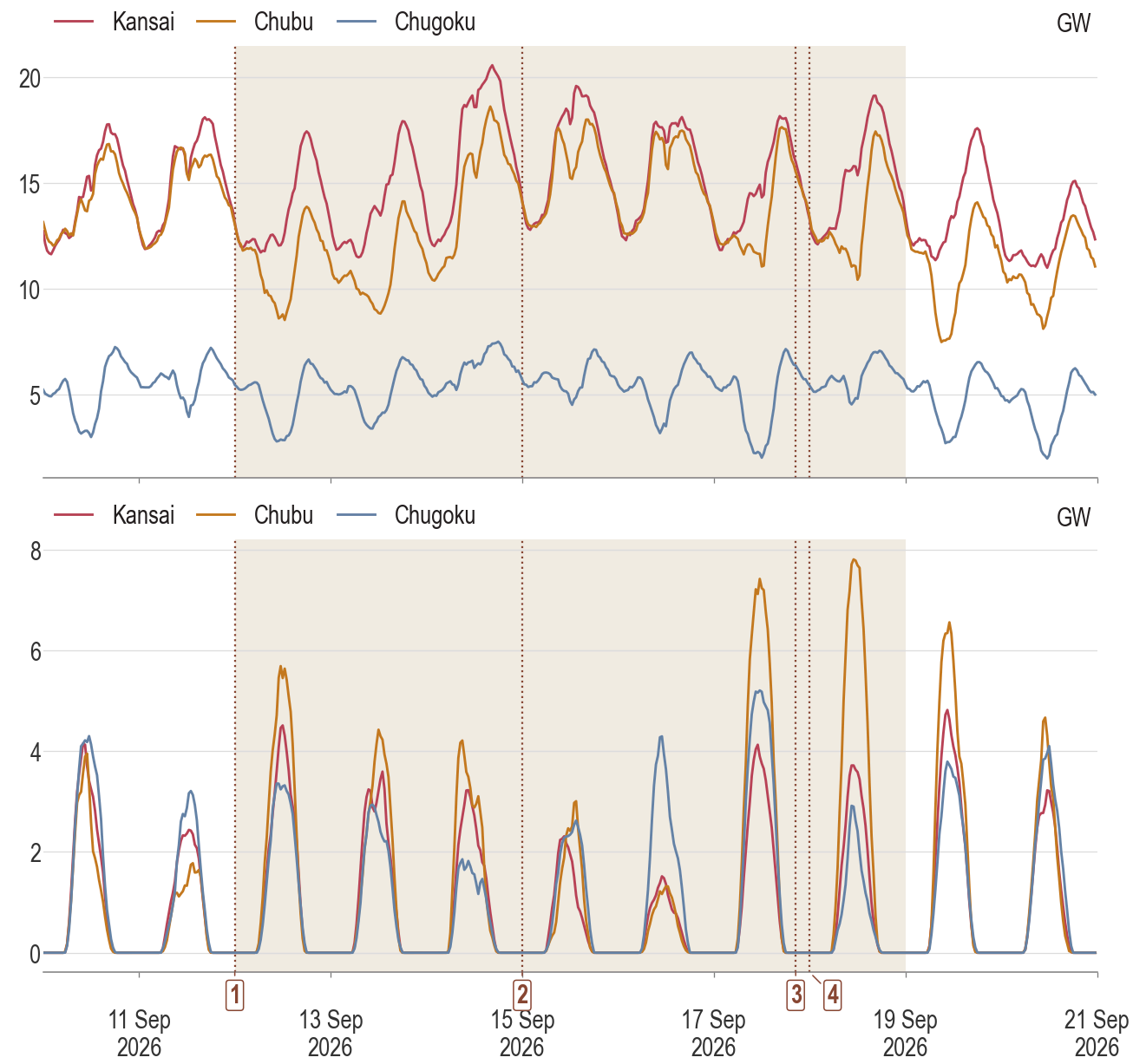
[324] [325] [301] [326] [285] [333]

Native clocks: prices and physical series 30 min; weather hourly. Gaps are not zeros.

06 · Sep 2026 Mihama · B. Chronology

The operator chronology is precise: reactor startup and criticality September 12; adjustment operation/first generation September 15; water dripping identified September 17 at 15:45, load reduction from 20:30; reactor manually stopped September 18. Discovery, load reduction and reactor stop are three different milestones. Day-ahead price and actual operating events also have different information clocks.

Exhibit 179: 12–18 September 2026: native chronology



Note: Native replay retains neighbouring systems. Numbered milestones correspond to the dated chronology; shading is not an inferred response.

Source: [2] [4] [301] [324–326] · Chart data

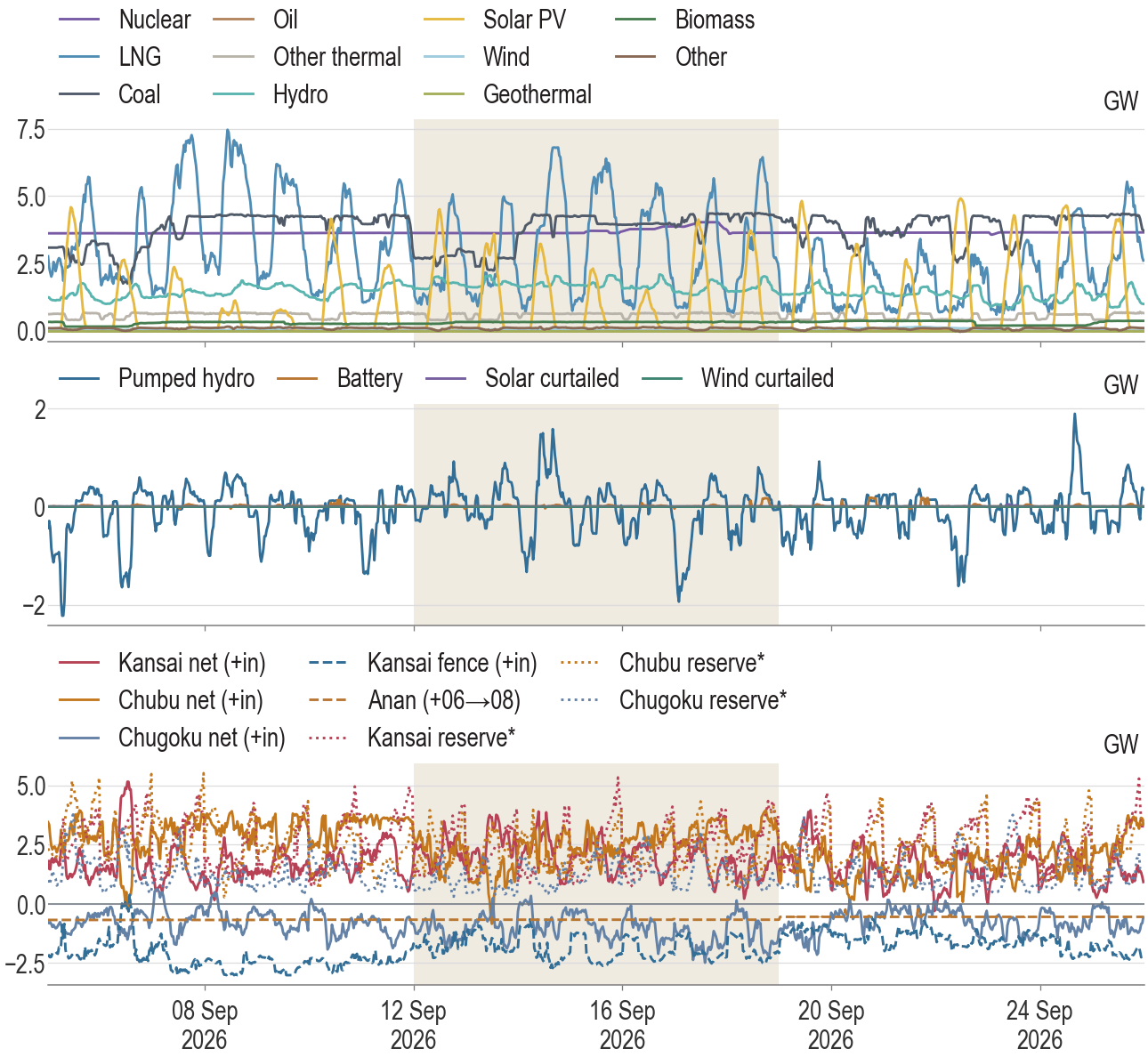
Later evidence updates the documentary endpoint without moving the original core: startup, criticality and adjustment operation resumed October 10, and full commercial operation resumed November 4 at 15:00 after the integrated load-performance inspection. These are actual milestones in the November 4 retrospective, not September forecasts treated as outcomes. The chronology plot marks dated events; it does not fabricate a flat 826 MW trace between them.

[324] [325] [301] [326] [291] [285] [333]

06 · Sep 2026 Mihama · C. Physical response

Kansai regional nuclear output averaged 3.662 GW on September 15, 3.946 GW on September 17 and 3.635 GW on September 18; the core maximum was 4.027 GW. Regional output includes other reactors, so changes are not automatically a unit-level Mihama series. JAIF's September operating record reports 12,996 MWh and 61 operating hours for Mihama 3, supporting brief operation rather than a month of full output.

Exhibit 180: 12–18 September 2026: physical response



Note: Kansai nuclear output is an area total, alongside reported thermal categories, renewables, storage and qualified interchange. The Kansai fence includes Chubu and Hokuriku boundaries; Anan is Kansai→Shikoku. Neither series is Mihama 3 telemetry. Dotted reserve* is the latest published area plan, not telemetry or an issue-time vintage.

Source: [4] [10] [11] [294] [295] [301] [324–327] · Chart data

The seven-day core also had 16.096 GW demand, 7.350 GW thermal and 894 MW solar on average. Pumped storage generated 34.439 GWh and consumed 37.055 GWh. These observed levers coexist with startup changes and warmer/different daily load. A causal claim that every thermal reduction was displaced by Mihama would require matched unit, demand and import counterfactuals that the evidence does not supply.

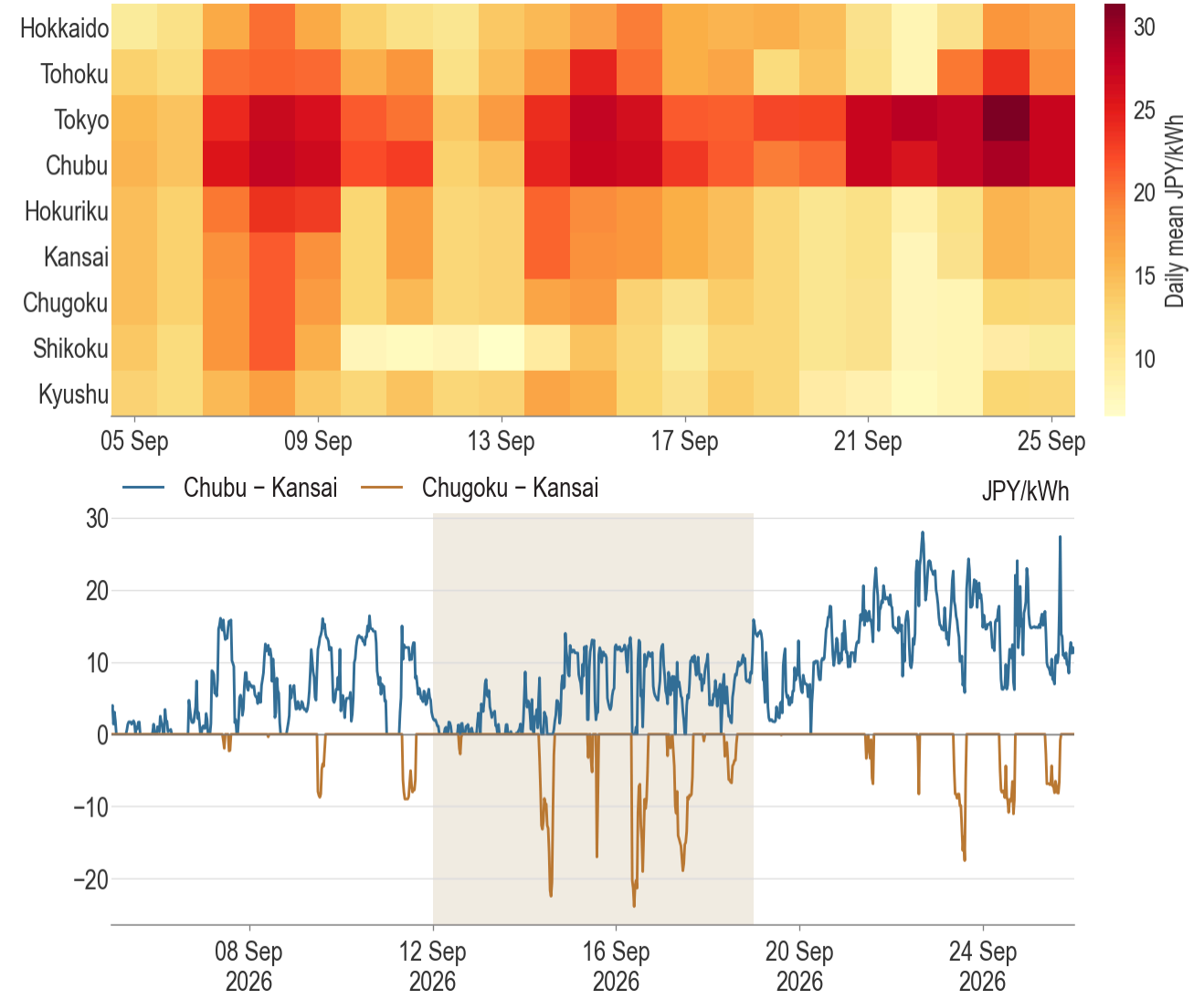
[324] [325] [301] [326] [292] [285] [333]

Observation clock: prices 30 minutes; physical series 30 minutes; weather 60 minutes. Incomplete observations remain gaps, not zero supply.

06 · Sep 2026 Mihama · D. Regional prices

Kansai's core mean 16.169 JPY/kWh was below Chubu 21.463 and Tokyo 21.570, but above Chugoku 13.821 and Shikoku 10.385. The all-nine map prevents a single-reactor narrative from standing in for the national market. Chubu-minus-Kansai and Chugoku-minus-Kansai spreads retain native half-hour signs and tails.

Exhibit 181: 12–18 September 2026: all-nine prices and focal basis



Note: All nine JEPX areas retain the full context; the colour scale includes the actual extreme. Signed spreads use destination minus origin.

Source: [1] [301] [324–326] · Chart data

Kansai daily price on September 15 was 18.296, September 17 16.029 and September 18 14.670 JPY/kWh. The price did not simply jump upward on the stop day. Weather, demand, renewable output, adjacent systems and the day-ahead information set also changed. This observation is a warning against equating a reactor milestone with an identified price effect, not proof that the unit had no commercial value.

North/east	Central/west	West/south
Hokkaido **15.316**	Chubu **21.463**	Chugoku **13.821**
Tohoku **17.367**	Hokuriku **16.257**	Shikoku **10.385**
Tokyo **21.570**	Kansai **16.169**	Kyushu **13.566**

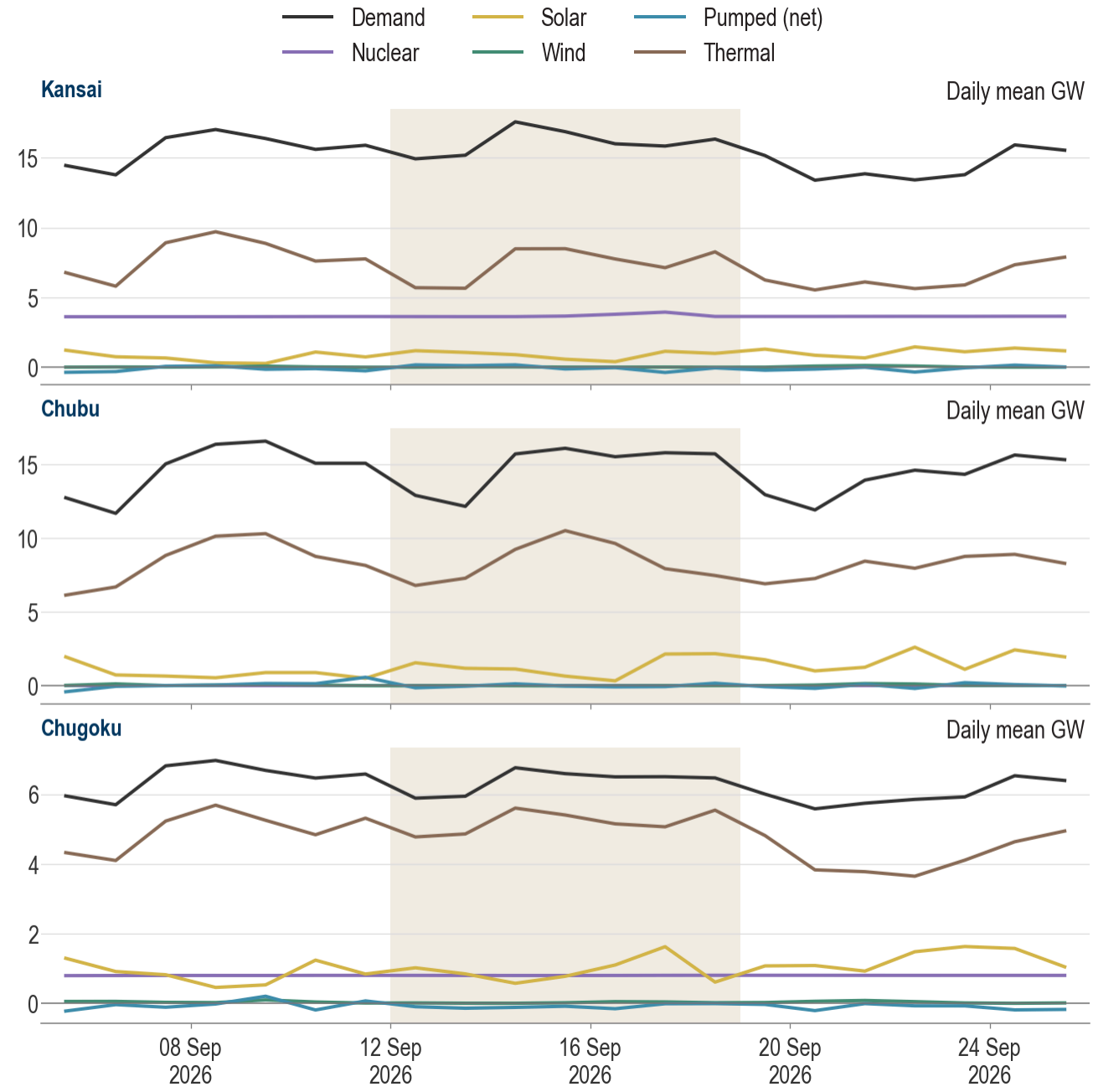
Core means in JPY/kWh; all nine use the same event dates. Daily heat-map cells require 48 valid prices.

[324] [325] [301] [326] [296] [285] [333]

06 · Sep 2026 Mihama · E. Aftermath

September 19–25 is the fixed aftermath, not the later full-commercial return. Kansai nuclear output averaged 3.641 GW, demand 14.432 GW and thermal 6.373 GW. Price fell to 11.808 JPY/kWh even though the interrupted reactor had not completed its later commercial return. Lower demand and stronger solar are visible alongside the lower thermal total.

Exhibit 182: 12–18 September 2026: neighbouring systems and recovery



Note: Focal and neighbouring demand and supply remain visible through the aftermath; lower demand, different weather or other outages can contradict a single-asset price story.

Source: [2] [4] [301] [324–326] · Chart data

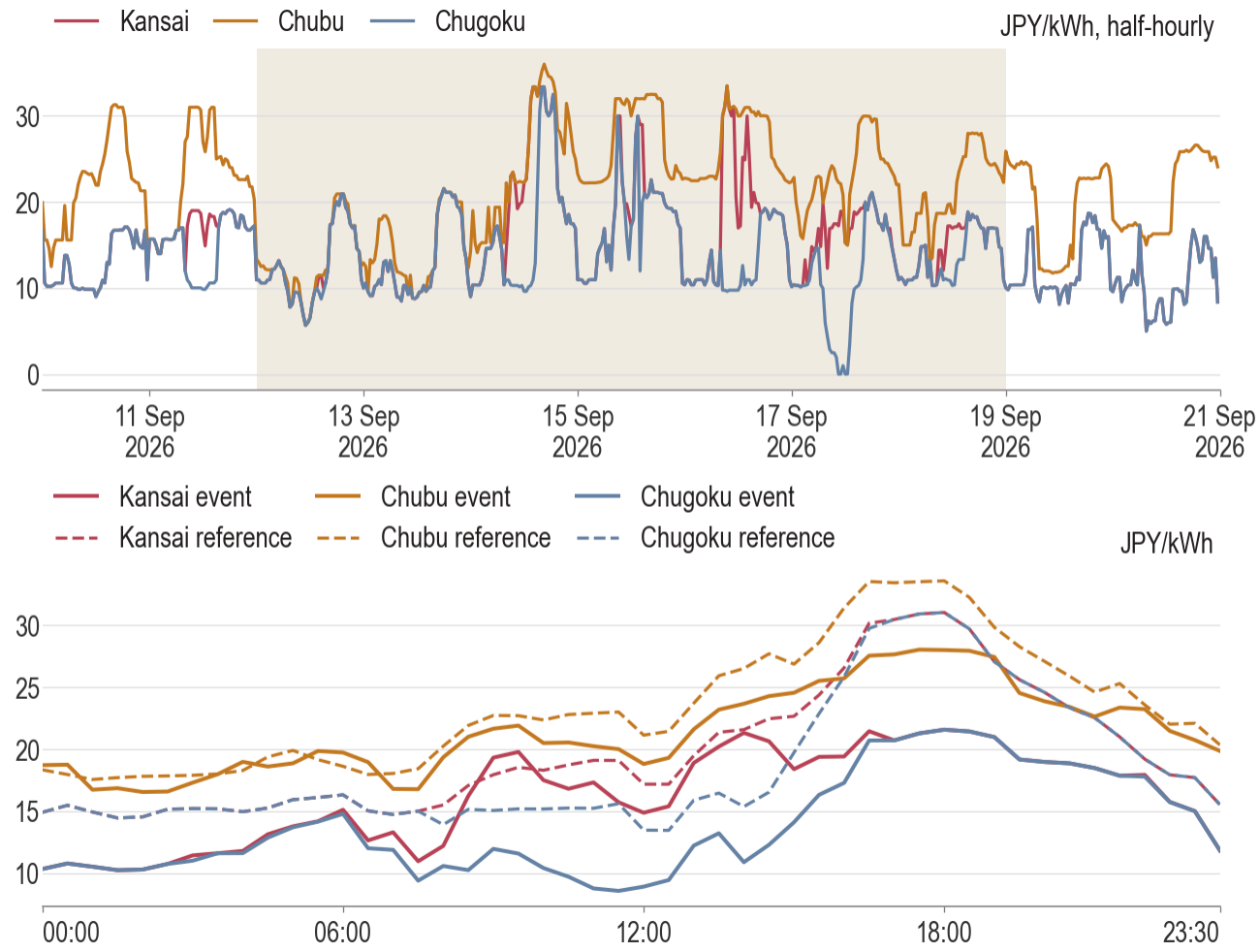
Chubu aftermath price rose to 25.217 JPY/kWh while Chugoku fell to 10.614. Thus regional recovery was not uniform. The separate October 10 and November 4 milestones are documentary updates beyond the observed core, not data spliced into its seven-day comparison. This preserves the historical experiment's boundaries and avoids claiming that an originally forecast return had already happened in September.

[324] [325] [301] [326] [285] [333]

06 · Sep 2026 Mihama · F. Comparisons

All seven core days have eligible original-rule references, although reference counts vary by day and exclude other documented stress periods. Kansai matched price 16.169 JPY/kWh compares with 19.744; demand 16.096 GW with 17.717 GW; residual 15.189 GW with 16.600 GW. A lower price alongside lower load cannot be attributed solely to a brief reactor startup.

Exhibit 183: 12–18 September 2026: prices and matched references



Note: Native-price context surrounds the interrupted return; calendar-matched profiles cover all seven core days, with fewer than four eligible references where stress exclusions require it. Later price declines cannot be assigned to a reactor that had already stopped.

Source: [1] [2] [286] [290] [301] [324–326] · Chart data

The matched reference, simple lead-in and documented unit chronology are complementary, not interchangeable. The lead-in nuclear mean 3.620 GW rises only modestly to the core 3.701 GW; the aftermath falls to 3.641 GW. The current conclusion is therefore an interrupted adjustment-operation episode with later actual resumption, not a completed September 826 MW shock. Keeping native observations, dated notices and explicit comparison populations separate is the transferable trading discipline.

Area	Event*	Reference*	Eligible core days
Kansai	16.169	19.744	7
Chubu	21.463	23.350	7
Chugoku	13.821	18.549	7

*JPY/kWh; matched event population, not necessarily every day of the full core.

[324] [325] [301] [326] [297] [333]

Cross-case observations: stress does not have one price signature

The eight cases show different kinds of exposure: sustained fuel/energy scarcity, a compound generation-and-weather shock, regional winter demand, solar absorption, holiday basis and interrupted nuclear recovery. Their daily-average prices do not order those mechanisms on one reliability scale. [304] [308] [289] [299]

Case / focal area	Lead-in	Core	Aftermath
7–9 February 2026 · Hokkaido	12.08	13.24	11.69
3–31 August 2026 · Kansai	20.68	19.36	20.76
January 2021 · Tokyo	24.59	66.53	7.65
March 2022 · Tokyo	20.40	30.76	25.36
19 February 2025 · Tokyo	14.62	17.75	14.77
10 May 2026 · Kyushu	7.60	7.66	9.21
23 September 2024 · Tokyo	18.21	11.14	12.41
12–18 September 2026 · Kansai	16.51	16.17	11.81

Mean prices are JPY/kWh. Lead-in/aftermath are the stated seven-day contexts; core durations range from one day to a month, so these are within-case comparisons, not a severity ranking. The full windows and matched references remain with each case. [285] [328] [1]

January 2021 stands apart in both level and persistence. Tokyo's core-month mean of **66.53** followed an already expensive **24.59** lead-in, then fell to **7.65** in the following week. That pattern fits the importance of fuel endurance and replenishment, not merely the height of one demand peak. March 2022, by contrast, contains a shorter earthquake/cold/low-solar emergency within a wider expensive month. [304] [308]

The August 2026 case reverses a simple "cooler means cheaper" reading: Kansai's aftermath price exceeded its core-period average even as mean demand fell. The changing nuclear/thermal and neighbouring balances therefore remain part of the explanation. Support actions demonstrate an operational response, but do not imply a uniform full-day price premium. [299] [300]

Cross-case observations: the adjustment lever determines the exposure

Duration separates fuel stress from a short ramp

January 2021's energy endurance problem gave replenishment and sustained generation a central role. March 2022 brought damaged generation, cold and weak solar together: storage could shift energy into a critical window, but turbine MW alone did not establish how long that response could last. The recurring exposure is the duration of the deficit and the route by which support can reach it. [304] [308]

Mechanism across cases	Market implication
Cold or heat raises the regional requirement	The neighbouring fleet and feasible support routes determine whether stress spreads or remains local.
Strong solar meets low daytime demand	Curtailment, pumping and export opportunities shape midday capture; the evening can remain a different market.
Holiday load changes both price legs	Basis can widen or reverse without a verified converter outage.
Nuclear return is interrupted or coincides with lower demand	Regional prices cannot be attributed to the headline restart date alone.

Solar and holidays change shape as well as level

Kyushu's 10 May 2026 daily average was not materially below its lead-in average despite solar control. Its 20 positive-curtailment half-hours averaged **JPY 0.42/kWh**, versus **12.83** in the other half-hours: low daylight prices and an expensive remainder can coexist. The September 2024 holiday likewise requires both Tokyo and Chubu prices, not one

price leg or a generic "low demand" label. [318] [290]

Recovery is usually a combination of changes

The Mihama sequence separates startup, generation, reduced load, interruption and full commercial operation on **4 November 2026**. Kansai's lower September aftermath price occurred alongside lower demand and changes elsewhere in the fleet; it is not an isolated estimate of one reactor's price effect. The February 2025 price episode likewise does not become a reliability emergency simply because it ranks highly in a selected price comparison. Across the cases, the strongest explanations join a dated physical mechanism to the regional price pattern while retaining the contemporaneous changes that could also explain it. [325] [333] [326]

Specialist glossary: networks and resource status

These terms describe Japanese arrangements or distinctions that materially affect the analysis. Standard power-market vocabulary is not repeated. [23] [16]

Term	Meaning in this publication
Electrical area	The system/market geography receiving an asset's output; it is not inferred solely from prefecture or parent-company name.
Frequency-converter boundary	The controlled exchange between the 50 Hz and 60 Hz systems, distinct from an ordinary AC interconnector.
Operating fence	A restriction applying to a combination of paths; not another physical cable or additive capacity.
Non-firm connection	Conditional network access under which output may be controlled for congestion.
Prescribed-order redispatch	Japan's congestion-management arrangement applying the specified sequence to affected generation; broader than control of only newly connected plants.
N-1 generation control	An arrangement using rapid designated generation reduction after a contingency to permit greater normal network use.
Online proxy control	Online facilities reduce output on behalf of other facilities; deemed-generation settlement reallocates the economic consequences.
Wide-area reserve block	The source-defined group used in a reserve assessment; it is not necessarily a JEPX equal-price group.
FIT/FIP purchase-start capacity	Capacity for which purchases under the relevant scheme have begun; not all certified projects or all installed renewable capacity.
Connected / commissioned battery	Connected MW and actual commercial operation are dated states; applications and auction awards are not operating assets.
Reactor restart milestone	Approval, startup, first generation and return to full operation are separate dated events.
Reported outage exposure	Notice-derived unavailable or reduced capability over a stated period; not proof of all remaining plant being available.

Regional net interchange, final schedules and sampled bilateral flows retain their distinct boundaries. A quoted limit must also identify its direction, delivery period and operating or auction stage before it can support a binding claim. [330] [369]

Specialist glossary: Japanese market arrangements

Term	Meaning and commercial boundary
System price	JEPX's unsplit auction calculation, not an average of area prices.
Area price	The settlement price for the delivery area after constrained clearing.
Block order	Linked acceptance across consecutive half-hour products; a profitable-looking block may still be rejected under the rules.
Planning code	The registered code connecting an exchange order to the participant's applicable delivery plans.
Baseload market	The specified Japanese procurement channel for baseload access; distinct from an arbitrary flat bilateral contract.
Balancing-capability procurement	Purchase of qualified response capability through the applicable EPRX products; not the same quantity as activated balancing energy.
Imbalance settlement	The applicable settlement of deviations from plan; an announced formula is not treated as effective before implementation.
Indirect transmission right	Conditional financial settlement of an eligible directional area-price difference, not a privately reserved path.
Capacity main / additional auction	Distinct procurement stages for a named delivery year; auction year is not delivery year.
Expected capacity	The qualifying contribution admitted under capacity-market rules, not automatically the plant's nameplate MW.
Long-term decarbonisation auction	A separate investment-support mechanism with its own term and revenue conditions; awarded projects are not commissioned capacity.
Non-fossil certificate	An environmental attribute with category-specific permitted use and retirement rules; not a substitute for energy.
FIT / FIP	Fixed-price purchase support / market-linked premium support, with different market exposure and attribute arrangements.
Gross operated / equity capacity	Physical operating MW / ownership-weighted MW with verified dated shares; neither measure is automatically traded market share.

The official rules govern the products. Futures settlement indices, physical procurement, capacity contributions and environmental claims remain separate exposures even when their costs ultimately enter the same retail bill. [16] [217] [225] [220] [224]

Methods: clocks, seasons and complete comparisons

The standard modern observation window is 1 April 2024–31 October 2026. The long historical window begins in April 2016 where a qualified series exists. Neither date implies that every region, fuel or weather series has complete observations throughout; exhibit dates and the coverage register identify the actual populations. [2] [3]

Convention	Application
Clock	JST, interval start, only where the publisher's convention is established.
Native resolution	Half-hour modern power observations; hourly legacy history.
Hourly modern comparison	Both valid half-hours are required; an old hourly observation is not duplicated into two half-hours.
Spring / summer / autumn	MAM / JJA / SON.
Winter year Y	December Y−1 through February Y, including leap day when applicable.
Complete season-year	The expected native interval population is present for the metric's required fields.
Partial sample	Kept visibly separate from complete-season rankings; not extrapolated into a seasonal total.

Observed energy is interval-average power integrated over its actual interval duration:

$$E = \sum_t P_t \Delta t.$$

With power in MW and duration in hours, energy is MWh. The energy in a complete month is not estimated from a few observed days. Where only a partial month survives, the chart uses observed mean power or explicitly observed-period energy. Missing values remain missing; reported and validated zeros remain zero. National or multi-region totals require simultaneous common observations, not the sum of separately occurring regional peaks.

Price means are unweighted means of valid half-hour prices unless a different weighting is explicitly stated. JPY/kWh is multiplied by 1,000 before combining with MWh in monetary calculations. [29]

Methods: demand shapes, weather and holidays

Daily-normalized demand divides every interval by that day's own mean before seasonal averaging. This prevents large-demand days from receiving an unintended extra weight in a comparison of shape:

$$\widetilde{D}_{d,h} = D_{d,h} / \overline{D}_d.$$

Absolute GW remains beside the normalized profile. The maximum of the mean daily profile is not the same statistic as the mean of daily peaks. Empirical p10/p50/p90 distributions describe observed daily variation, not confidence intervals. [29]

Weather scatters use one observation per eligible ordinary weekday: mean demand and mean representative-point temperature over 13:00–16:59 JST. Region and season are separate panels, and year is retained. Two-degree temperature-bin medians and p10/p90 bands require at least ten distinct days. The priority-region night comparison uses 01:00–04:59 on matching day/season-year populations. A reanalysis point is not a measured area-average temperature, and the association is not a fitted causal load coefficient. [5]

Holiday classification uses the retained official calendar, with statutory holidays before weekend classification and separately identified customary closures. Golden Week combines its actual statutory and surrounding dates; Obon is not declared a universal statutory shutdown. Daily MWh and same-month ordinary-weekday indices show the regional outcomes, with shutdown/reopening profiles retaining their actual years. A once-a-year event is not rejected merely because it cannot supply ten repetitions in that year. [97]

The selected-year curves use complete comparable seasons and keep the historical reporting break explicit. Gaps do not become flat segments, and no thermal-fuel split is backcast into an aggregate legacy series.

Methods: residual load, ramps, storage and capture

Residual load is demand minus accepted solar and wind on identical observations:

$$RL_t = D_t - S_t - W_t, \qquad \Delta RL = \Delta D - \Delta S - \Delta W.$$

The evening decomposition uses the stated 14:00–20:00 JST window. Adjacent changes do not bridge missing observations. Modern half-hour changes are reported in GW/30 min; historical hourly changes remain GW/hour. Daily residual range, the 10:00–14:59 trough and peak timing are distinct statistics. Tied peaks use the earliest interval; clock times are not averaged as ordinary linear numbers. [29]

Demand load factor is the daily mean divided by the daily maximum. A residual-load factor is not calculated where its maximum is nonpositive; residual range remains interpretable in that case. A combined regional curve uses simultaneous observations rather than separately observed peaks.

Battery positive and negative contributions share the same eligible interval denominator. An inactive contribution is zero only when the underlying observation exists. If the source is regional net output, these are net-positive and net-negative profiles—not reconstructed simultaneous gross charging and discharging. Conditional active-interval means are different statistics. Pumped storage remains separate. [4]

Generation-weighted capture price is:

$$P_g^{cap} = \frac{\sum_t P_t G_{g,t} \Delta t}{\sum_t G_{g,t} \Delta t}.$$

Price and generation must refer to the same supported intervals and area. Reported curtailment fractions require matched technology and population in both curtailed and accepted output. Observed charging and discharging price context is not realised profit, efficiency or state of charge; the regional net series cannot reveal a private battery's operating strategy. [29]

Methods: spreads, reversals and transfer constraints

Directional price spreads consistently use destination minus origin:

$$\Delta P_t^{A\to B} = P_{B,t} - P_{A,t}.$$

Calculations use native JPY 0.01/kWh integer ticks. On common timestamps, Tokyo→Kansai equals Tokyo→Chubu plus Chubu→Kansai; the first is an economic comparison, not a direct physical line. Pairwise summaries otherwise use the pair's common observations, while explicitly all-area exhibits use the common nine-area population. Exact equal-price coupling is separate from an analytical reversal deadband. [1]

Typical-spread summaries retain signed mean, median, p10/p90, mean absolute spread, exact coupling and direction frequencies by season/year and clock band. Missing observations interrupt reversal runs. The existing sustained-reversal convention requires two consecutive valid intervals in each state, at most two neutral intervals between them, and no gap bridging; sensitivity thresholds remain labelled rather than silently changing the event definition. [29]

Comparison	Required agreement
Schedule against limit	Boundary, direction, delivery clock, publication vintage and market stage.
Measured flow against limit	Physical boundary, sign, interval basis and applicable operating limit.
Binding indicator	Official indicator, or a match at the published precision explicitly called "at the reported limit".
Fence history	Retains its grouped constraint identity; never spliced into a discontinued individual link.

A static nameplate rating or ATC median is not a substitute for an applicable historical limit. Complete six-sample half-hour means remain sample means unless their energy-metering basis is established. Unsupported binding statistics are not produced from regional net interchange or an arbitrary 95% threshold. [330] [369]

Methods: fleet identity, capacity and outage exposure

A physical unit is counted once. Operating entity, legal owner, parent and joint-venture stake are separate attributes. JERA's generation is not added again in full under both TEPCO and Chubu. Equity-attributed MW requires a sourced share and effective date; otherwise ownership remains unknown rather than being copied from the operator field. Commissioning, retirement and rating changes are applied at their actual dates. [29]

Capacity history comes from dated operating assets or qualified official statistics, not today's fleet projected backwards. Installed, suspended, retired, contracted and actually generated quantities remain separate. Age uses commercial-start dates; an uprate does not reset a unit's age and a replacement is not its retired predecessor. Regional assignment follows the verified electrical destination, not a geographical guess.

Outage exposure first resolves notices and revisions to physical units or blocks. Overlapping component and whole-block notices cannot create more lost capability than the time-valid block rating. A derating uses the reported reduction and scope; a full stop uses qualified affected MW. Planned/unplanned source categories remain separate from long-term suspension flags. [8]

A target restoration date is not a confirmed return. Day-only targets generate start/end-of-day scenarios; their total physical exposure is bounded, but overlap reassignment can lower an individual category's allocation in the later-end scenario. Unresolved ends are censored at the cutoff. Capture/revision time is not original publication time. Unit-year rates require a known observed population and time-valid denominator.

Finally, installed MW minus reported outage MW is not guaranteed available supply. Unit production and official operating evidence are used separately, and short measured samples are not annualised into claims about typical company performance. Unknown fuel, ownership and age stay explicit in the factual registers.

Methods: fuel economics, cash flows and event comparisons

Fuel-cost comparisons retain the actual fuel reference, currency, observation month and thermal/electrical basis. A generic cost identity is:

$$MC_f = (p_f + \tau_f + ce_f)/\eta_f + v_f.$$

Here fuel price and tax are JPY/MWh-thermal, carbon cost is JPY/tCO_2, emissions are tCO_2/MWh-thermal, efficiency is dimensionless and variable operating cost is JPY/MWh-electric. Illustrative efficiencies are not observed bids. Climate tax must not be added twice. Import averages, disclosed spot surveys and JKM are not interchangeable; missing fuel prices are not carried forward as observations. [29]

An illustrative average-price hedge leaves both shape and area exposure unless quantities and settlement match. Transmission-right proceeds use eligible settlement MW, applicable reductions and directional prices; acquisition charges apply to held rights, and negative spreads can create liabilities. These identities do not establish a participant's private contract, hedge book or realised profit. [24] [219]

Historical event references retain up to four prior same-weekday/holiday-class dates within 56 days, the specified stress exclusions and complete metric-specific days. Event-day weights are equal. Alternative calendar or weather comparisons are shown separately, not substituted silently or labelled causal counterfactuals. January 2021's preceding week is already stressed, not an unaffected normal period. [29]

Event time, announcement, original publication, first capture and revision are distinct. Final reserve plans remain plans; aggregate instruction counts and maximum MW do not become delivered energy. Revised observations can explain a historical episode but do not reconstruct the information available to a trader before its auction. Documentary chronology and measured physical response are therefore kept separate when assessing the levers used to resolve stress.

Reactor register: Hokkaido through Hokuriku

These non-retired reactors retain their individual ratings and first-operation dates. Status is the NRA snapshot dated **29 October 2026**, not a promise of continuous output or a diagnosis of why a unit is stopped. Retired/decommissioning units remain in the downloadable register but are excluded from this equipment cohort. [38] [39]

Reactor	Electrical area	MW	First operation	NRA status
Tomari 1	Hokkaido	579	1989-06-22	Suspended
Tomari 2	Hokkaido	579	1991-04-12	Suspended
Tomari 3	Hokkaido	912	2009-12-22	Suspended
Higashidori 1	Tohoku	1,100	2005-12-08	Suspended
Onagawa 2	Tohoku	825	1995-07-28	Operation
Onagawa 3	Tohoku	825	2002-01-30	Suspended
Kashiwazaki-Kariwa 1	Tokyo	1,100	1985-09-18	Suspended
Kashiwazaki-Kariwa 2	Tokyo	1,100	1990-09-28	Suspended
Kashiwazaki-Kariwa 3	Tokyo	1,100	1993-08-11	Suspended
Kashiwazaki-Kariwa 4	Tokyo	1,100	1994-08-11	Suspended
Kashiwazaki-Kariwa 5	Tokyo	1,100	1990-04-10	Suspended
Kashiwazaki-Kariwa 6	Tokyo	1,356	1996-11-07	Operation
Kashiwazaki-Kariwa 7	Tokyo	1,356	1997-07-02	Suspended
Tokai Daini	Tokyo	1,100	1978-11-28	Suspended
Hamaoka 3	Chubu	1,100	1987-08-28	Suspended
Hamaoka 4	Chubu	1,137	1993-09-03	Suspended
Hamaoka 5	Chubu	1,380	2005-01-18	Suspended
Shika 1	Hokuriku	540	1993-07-30	Suspended
Shika 2	Hokuriku	1,206	2006-03-15	Suspended

"First operation" follows the NRA specification's date basis, not the most recent restart. Electrical-area attribution follows the qualified connection, not the operator's headquarters or the plant's prefecture. Unit identity, operator, explicit ownership gaps, coordinates and later milestones remain in the full reactor register. [39]

Reactor register: Kansai through Kyushu

These non-retired reactors retain their individual ratings and first-operation dates. Status is the NRA snapshot dated **29 October 2026**, not a promise of continuous output or a diagnosis of why a unit is stopped. Retired/decommissioning units remain in the downloadable register but are excluded from this equipment cohort. [38] [39]

Reactor	Electrical area	MW	First operation	NRA status
Tsuruga 2	Kansai	1,160	1987-02-17	Suspended
Mihama 3	Kansai	826	1976-12-01	Operation
Ohi 3	Kansai	1,180	1991-12-18	Suspended
Ohi 4	Kansai	1,180	1993-02-02	Operation
Takahama 1	Kansai	826	1974-11-14	Operation
Takahama 2	Kansai	826	1975-11-14	Operation
Takahama 3	Kansai	870	1985-01-17	Suspended
Takahama 4	Kansai	870	1985-06-05	Operation
Shimane 2	Chugoku	820	1989-02-10	Operation
Ikata 3	Shikoku	890	1994-12-15	Operation
Genkai 3	Kyushu	1,180	1994-03-18	Operation
Genkai 4	Kyushu	1,180	1997-07-25	Operation
Sendai 1	Kyushu	890	1984-07-04	Operation
Sendai 2	Kyushu	890	1985-11-28	Operation

"First operation" follows the NRA specification's date basis, not the most recent restart. Electrical-area attribution follows the qualified connection, not the operator's headquarters or the plant's prefecture. Unit identity, operator, explicit ownership gaps, coordinates and later milestones remain in the full reactor register. [39]

Mihama 3 reached full commercial operation on **4 November 2026 at 15:00** after its interrupted September startup and October resumption. That later event does not alter the date of this NRA snapshot. [333]

Reactor milestones: physical dates and reporting dates are not interchangeable

Recent changes affect different electrical areas and different phases of operation. The table records documented milestones, not forecasts of another restart or estimates of a reactor's isolated price effect. Times shown are JST; a date without a clock retains day precision. [44] [41] [40]

Physical milestone	Reactor / rating	What changed
26 Dec 2024, 16:00	Onagawa 2 · 825 MW	Commercial operation resumed in Tohoku.
10 Jan 2025, 15:10	Shimane 2 · 820 MW	Commercial operation resumed in Chugoku.
16 Apr 2026, 16:00	Kashiwazaki-Kariwa 6 · 1,356 MW	Commercial return serving Tokyo.
8 Aug 2026, 23:56	Ohi 3 · 1,180 MW	Automatic trip during the prolonged heat episode.
12 Sep 2026	Mihama 3 · 826 MW	Reactor startup and criticality.
15 Sep 2026	Mihama 3	Adjustment operation / first generation.
17 Sep 2026, 20:30	Mihama 3	Load reduction after water-drop discovery at 15:45.
18 Sep 2026	Mihama 3	Reactor stopped; this is not the 17 September load-reduction time.
10 Oct 2026	Mihama 3	Startup, criticality and adjustment operation resumed.
4 Nov 2026, 15:00	Mihama 3	Full commercial operation after the integrated-load inspection.

The first three operator releases were published on their stated commercial-return dates. The **4 November Mihama retrospective** establishes the completed sequence; it must not be treated as information already known at each September auction. The contemporary interruption notice and the later completion release serve different evidential purposes. [300] [325] [333]

Kansai's observed September nuclear output is not an 826 MW flat trace. Unit and regional records have different coverage; regional totals also change with other reactors. The interrupted-startup case keeps demand, thermal output, storage, neighbouring prices and those alternative changes visible.

Company register: gross assets, date evidence and mapped coverage

The admitted register contains **596 additive asset records and 195,690.674 MW**. This is a source-qualified cohort, not the complete Japanese generating fleet. Gross equipment is attributed once to its mapped group; joint-venture assets are not multiplied by an unsupported equity share or counted again under a shareholder. [331]

Profile group	Records	Gross MW	Start known, MW %	Mapped, MW %
Hokkaido Electric	18	6,739.4	63.6	69.6
Tohoku Electric	16	10,323.0	46.0	85.3
TEPCO group	32	15,405.0	53.3	100.0
Chubu Electric	12	5,544.0	65.2	92.3
Hokuriku Electric	14	6,448.7	64.3	71.5
Kansai Electric	44	20,605.0	40.7	91.1
Chugoku Electric	27	9,302.0	41.1	66.8
Shikoku Electric	10	4,340.0	36.6	83.9
Kyushu Electric	32	13,615.0	37.8	68.4
Okinawa Electric	11	1,835.0	94.4	100.0
JERA	119	55,072.1	36.5	82.5
J-POWER	46	13,669.0	48.3	96.6
JAPC	2	2,260.0	100.0	100.0
Tokyo Gas	15	2,145.1	38.0	100.0
Osaka Gas / Daigas	102	3,287.2	53.9	100.0

The fifteen groups total **170,590.514 MW**; the other **96 records / 25,100.160 MW** remain in regional competitor denominators. Displayed one-decimal MW may not sum to the full-precision total. [336]

"Start known" requires a sourced commissioning date at its retained day/month/year precision. It does not make the first day of a month an observed date. "Mapped" requires coordinates with a corroborating source; unmapped MW stays in the capacity total. Dated legal equipment holders remain separate from operators, group attribution and present-day title.

The full asset register retains those fields. The top-five physical-unit register excludes unresolved station/series aggregates from individual-unit rankings rather than presenting a multi-unit plant as one machine. [340]

Capacity register: official equipment and admitted units are different populations

OCCTO's FY2025 supply-plan compilation reports **324,160 MW** for FY2024 end. These are equipment quantities, including idle capacity; they are not a measured available-supply stack. The compilation includes non-utility/FIT resources under its stated reporting and estimation rules. Future-year columns in the original are plans, not additions already commissioned. [27]

Non-overlapping category	FY2024-end MW
Coal	52,460
LNG	79,980
Oil and other thermal	15,520
Nuclear	33,080
Conventional hydro	22,030
Pumped hydro	27,340
Wind	5,940
Solar	76,700
Geothermal	500
Biomass	6,550
Waste	1,340
Batteries	360
Hydrogen	0
Ammonia	0
Other	2,360

Values convert the publisher's units of 10,000 kW to MW. Component rounding can differ from the separately published total; thermal aggregate is not added again to coal, LNG and oil. The nuclear population includes **33 non-retired reactors**, including reactors without a restart date. [27]

The accompanying ten-area composition file preserves the source's one-decimal shares without renormalising them. It does **not** infer regional MW from national totals. The company-unit register has a narrower, identity-qualified population: its sum is therefore neither this national denominator nor a substitute for unavailable historical area inventories. [28]

Download the dated regional composition. Capacity, age, observed output and outage exposure retain their own populations and dates in the supply and company chapters.

LNG register: eastern and central receiving systems

Storage ratings retain the publisher's **mass or volume unit, scope and date**. They are not stocks currently in tanks. JERA's rows use its 6 December 2021 disclosure; joint-site quantities must not be added to Tokyo Gas figures as if they were separate physical terminals. [99]

Receiving system / disclosed scope	Electrical destination	Storage rating / date
JERA Futtsu	Tokyo	612,000 tonnes · 2021
JERA Higashi-Ohgishima	Tokyo	243,000 tonnes · 2021
JERA Sodegaura	Tokyo	536,000 tonnes · 2021
JERA Negishi	Tokyo	61,000 tonnes · 2021
JERA Chita	Chubu	702,000 tonnes · 2021
JERA Kawagoe	Chubu	378,000 tonnes · 2021
JERA Yokkaichi	Chubu	144,000 tonnes · 2021
JERA Joetsu	Chubu	243,000 tonnes · 2021
Tokyo Gas Hitachi	Tokyo network	460,000 kL · Sep 2026
Tokyo Gas Negishi / Sodegaura	Tokyo network	Current totals not established
Tokyo Gas Ohgishima · tank 4 only	Tokyo network	250,000 m³ · Apr 2026 source
Ishikari · Hokkaido Gas tanks 1–2	Hokkaido	380,000 kL · Jul 2018 scope
Nihonkai LNG Niigata	Tohoku	720,000 kL · Dec 2024
Shin-Sendai	Tohoku	320,000 kL · Dec 2024
INPEX Naoetsu	No named power edge established	360,000 kL · captured Nov 2026

Tokyo Gas's network membership does not establish an exclusive terminal-to-plant flow. Ohgishima's 2026 tank recognition is **not** a commissioning date. Ishikari's figure excludes separate utility tanks. [112] [109] [105] [111]

Joetsu's physical location in Niigata does not make its electrical destination Tohoku. INPEX Naoetsu's domestic pipeline system is not automatically the fuel supply of the neighbouring JERA station. Its disclosed maximum gas production of **8.25 million m³/day**, revised April 2024, is not a nominated power-plant flow. [100] [101]

LNG register: western systems and dated gas-company equipment

These operator disclosures establish different equipment scopes. Missing quantities remain unknown; a named supply connection is not evidence of inventory ownership, current stock or unrestricted fuel transfer between companies. [107] [108] [102]

System / scope	Region	Storage rating / source date
Toyama Shinko · one LNG tank	Hokuriku	180,000 m³ · captured Nov 2026
KEPCO Sakai · four tanks	Kansai	560,000 kL · captured Nov 2026
KEPCO Himeji · seven tanks	Kansai	520,000 kL · May 2026
Daigas Senboku / Himeji	Kansai	Not quantified in admitted plant sources
Yanai · six tanks	Chugoku	480,000 kL · Jul 2018
Sakaide	Shikoku	Current total not established
Hibiki · two operating tanks	Kyushu	360,000 kL · captured Nov 2026
Yoshinoura · two tanks	Okinawa	280,000 kL · May 2020

Hibiki's planned 230,000-kL tank is excluded. KEPCO Himeji's disclosed **19,000 m³/h** is spare gas-production capability in May 2026, not total send-out. Yanai's **275 t/h** is a dated power-generation vaporizer specification. [104] [106] [113] [110] [103] [114]

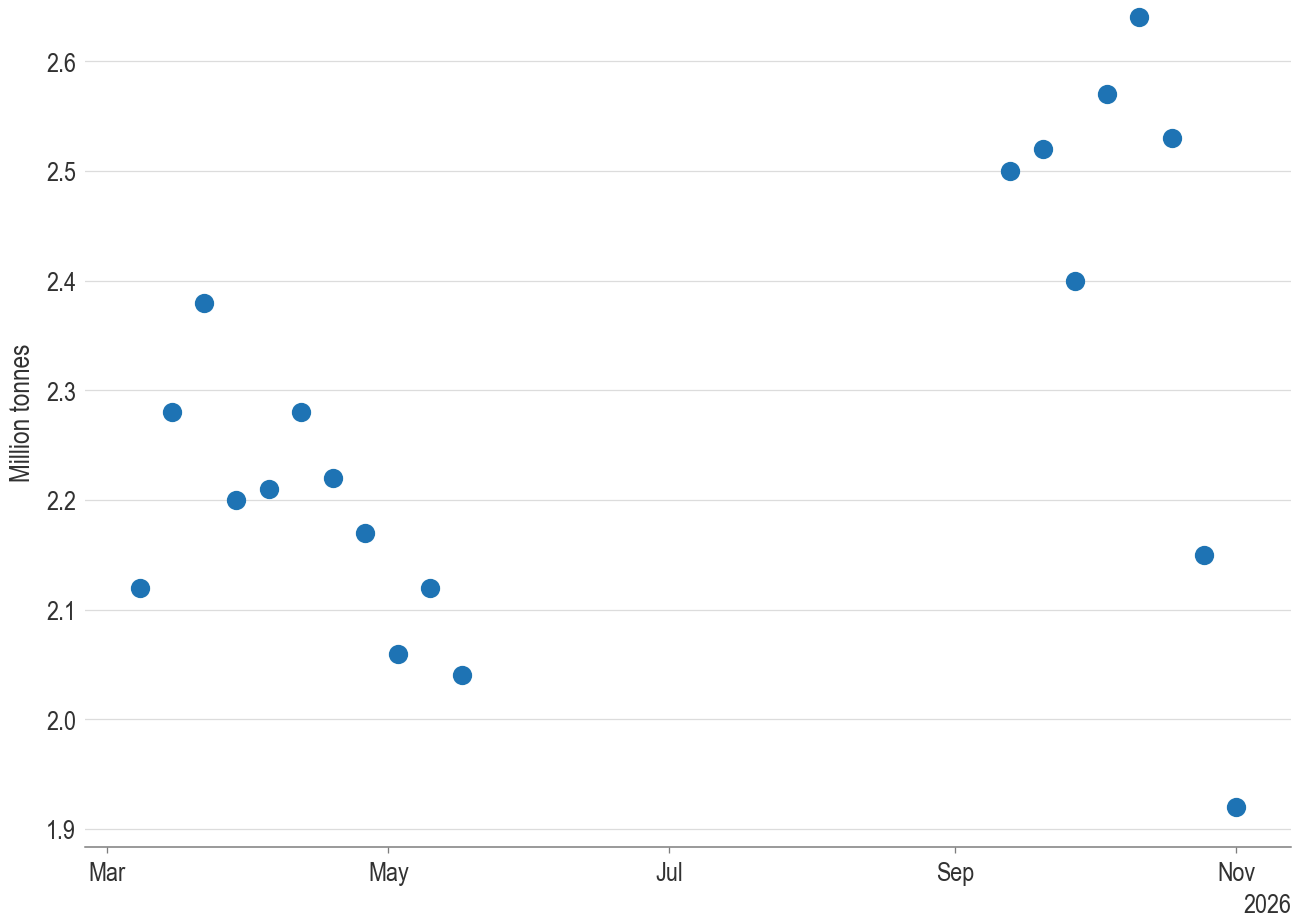
Historical Tokyo Gas equipment—not current capacity

Terminal	Storage, kL	Vaporization, t/h
Negishi	1,040,000	835
Ohgishima	850,000	1,434
Sodegaura	1,350,000	1,445
Hitachi	230,000	750

These are the **2020** Tokyo Gas equipment disclosures, not 2026 inventory. Hitachi's later published 460,000-kL scope provides a separate dated observation; the difference does not identify a commissioning day. Joint-site boundaries still prevent summing these figures with JERA's mass-based ratings. [216] [112]

LNG stocks and infrastructure reconciliation

Exhibit 184: Sparse observations of utility LNG stocks



Note: 19 observed national utility-stock dates span 2026-03-08–2026-11-01. The 4 November release revises 13 September from 2.49 to 2.50 million tonnes, and the extract retains both values. Points stay unjoined because no stock path between dates is observed.

Source: [13] [118] · Chart data

The stock observations describe the **reported national aggregate of major power utilities**, with the publisher's treatment of dead stock. They are neither an inventory for every receiving terminal nor the gas-company working stock available to every generator. Sparse observations stay as points: interpolation would invent an unobserved replenishment path. [118]

A storage rating measures equipment; a stock observation measures a dated quantity. Neither alone produces reliable days of cover. That calculation would also need a matching accessible-inventory population, contractual availability and a defensible consumption denominator. LNG receipts, vaporization and generator burn are different flows.

The infrastructure register retains **25 terminal/operator scopes and 35 relationship records**. Repeated terminal ratings describe several disclosed supply edges; they are not extra tanks. The historical equipment table keeps the separately dated gas-company capacities.

Geographic generation-site locators in the fuel chapter use corroborated plant locations. Schematic terminal/network connections do not assert surveyed pipeline routes or precise terminal coordinates. A source supporting a network relationship is not evidence that a specific cargo reached a particular unit. [99] [111] [112]

Boundary register: ten connections, with direction and aggregation retained

The book's economic spread is **destination price minus origin price**. The orientation below fixes that convention; it does not overwrite the publisher's native forward/reverse definitions. Each route is examined in both directions. Okinawa has no mainland boundary. [278]

Book orientation	Boundary identifier	Physical / publication identity
Hokkaido → Tohoku	`hh`	Hokkaido–Honshu DC connection group
Tohoku → Tokyo	`tt_soma`	Soma–Futaba / Tohoku–Tokyo corridor
Tokyo → Chubu	`fc`	50/60 Hz frequency-converter group
Chubu → Hokuriku	`ch_minami_fukumitsu`	Minami-Fukumitsu individual-link series
Chubu → Kansai	`ck_mie_higashiomi`	Mie–Higashiomi individual-link series
Hokuriku → Kansai	`hk_echizen_reinan`	Echizen–Reinan individual-link series
Kansai → Chugoku	`kc`	Western AC boundary; retain operating-fence scope
Chugoku → Kyushu	`km`	Kanmon connection
Chugoku → Shikoku	`honshi`	Honshi AC route; raw schedule key `ks`
Kansai → Shikoku	`anan_kihoku`	Anan–Kihoku HVDC; schedule key `ks_anan_kihoku`

The central publication break is not a new bilateral cable

The March 2026 central-network change introduces operating-fence publications including **`ck_chubu_fence`**, **`ck_kansai_fence`** and **`hk_fence`**. These aggregates must remain separate from discontinued individual-link series. Appending a fence to a former bilateral trace would change the measured object while hiding the break. [279]

Honshi and Anan–Kihoku are different connections to Shikoku. Tokyo–Kansai is an economic spread comparison, not a direct physical line. Regional net imports combine several routes and cannot replace any one bilateral observation.

The regional and boundary atlas contains seasonal prices, transfer histories and source-qualified restrictions. The restriction register retains direction, date band, market stage, publication vintage and the distinction between an asset and a boundary aggregate.

Limit register: an annual restriction is not an auction-binding observation

A useful limit record identifies the **object, direction, delivery interval, market stage, publication vintage and native clock**. Equipment ratings, operating limits, available capacity, planned transfers and measured transfers are different quantities. The publisher's source fields remain separate. [280]

Dated annual-plan example	Direction	Published constrained range
10 May–6 Jun 2025 · Sakuma / Shin-Shinano works	Chubu → Tokyo	1,200 MW
22 Jul–22 Aug 2025 · Anan–Kihoku bipolar outage	Shikoku → Kansai	0 MW on the stated asset
Same period · associated Honshi constraint	Chugoku → Shikoku	180–1,200 MW
12 Mar–30 Apr 2026 · Nishi-Sendai / Soma–Futaba works	Tohoku → Tokyo	3,450–3,550 MW
12 May–30 Jun 2026 · overlapping eastern works	Tohoku → Tokyo	2,450–3,550 MW
7 Oct–20 Nov 2026 · converter and related works	Tokyo → Chubu	900–1,350 MW
Same period	Chubu → Tokyo	840–1,310 MW

The first three rows are the **1 March 2025** plan; the remaining rows are the **1 March 2026** plan. Displayed dates are inclusive; the CSV's stop is exclusive. These are planned restriction bands, not claims that every outage occurred exactly as planned or that the range applied uniformly to every half-hour. [284] [283]

Formulated day-ahead export records retain their native clock and latest-update timestamp. An update after the 10:00 JEPX deadline is not information known before that auction. Zero available capacity also need not mean scheduled transfer equals physical rating: margins and other allocations remain relevant.

Native five-minute measurements can form a half-hour mean only with six distinct valid samples. Same-row physical-stage limits support a physical comparison, not an earlier auction-vintage claim. **Auction-binding frequency remains unestablished where those alignments are not certified**; the full seasonal price-spread analysis remains valid on its own qualified population. [281] [280]

Coverage register: complete demand and residual seasons are counted separately

Each cell counts **complete demand / complete residual-load season-years**, from April 2016 through October 2026. Complete means every expected interval on one native basis; December belongs to the following winter year. These counts do not turn partially observed seasons into full-year comparisons. [2] [4] [3] [32]

Physical area	Winter	Spring	Summer	Autumn
Hokkaido	10 / 10	9 / 9	10 / 10	9 / 9
Tohoku	9 / 9	9 / 9	11 / 11	10 / 10
Tokyo	9 / 9	9 / 9	11 / 11	10 / 10
Chubu	10 / 10	9 / 9	11 / 11	10 / 10
Hokuriku	10 / 10	9 / 9	11 / 11	10 / 10
Kansai	10 / 10	9 / 9	11 / 11	10 / 10
Chugoku	10 / 10	9 / 9	11 / 11	10 / 10
Shikoku	10 / 10	9 / 8	11 / 8	10 / 10
Kyushu	10 / 7	8 / 6	11 / 7	10 / 7
Okinawa	1 / 1	1 / 1	1 / 1	1 / 1

The native hourly/half-hour boundary makes spring 2024 a mixed-basis season, excluded from these complete-season counts. An hourly shape produced by explicitly aggregating modern observations is a separately declared analysis basis, not a silent replacement for native-interval ramp evidence.

Okinawa has one complete admitted modern season-year per season: autumn 2025 and winter, spring and summer 2026. Its earlier complete-day samples are dated as samples. Chugoku's official FY2016–2023 hourly files repair the earlier historical gap. Shikoku's negative reported wind observations prevent some complete residual seasons; Kyushu also has fewer complete residual than demand populations. Missing or rejected renewable output is not zero generation. [32]

The analysis coverage ledger records each region, season, year, basis, expected population and exclusion reason. Representative-point weather remains reanalysis, not an area-average measurement. Price-area coverage excludes Okinawa by market design, not by dropping its physical-system analysis. [5]

Evidence register: claims not established

Every region, company profile and event remains in the book. The limits below concern specific claims, not permission to replace missing observations with zeros or to describe a narrower cohort as a complete system. The coverage ledger retains the corresponding populations, dates and reasons.

Claim not established	Qualified evidence retained
Complete historical electrical-area installed MW	Official FY2024-end regional composition, national equipment totals and dated admitted units. The current fleet is not projected backward.
Historical outage reliability rates by fuel or company	Revision-qualified expected notice exposure. First capture begins in September 2026; historical unit-year denominators and confirmed restoration times remain unavailable.
Two complete recent seasons for every Okinawa comparison	One complete modern year for each season, plus explicitly dated complete-day samples. Earlier incomplete seasons are not ranked as complete.
Earlier auction-binding frequency from later transfer records	Seasonal price spreads, separate native-clock plans, measured-transfer samples and same-row physical-stage limits. Stage and publication vintage are not interchangeable.
Complete terminal stocks, surveyed pipelines or every legal owner	Dated equipment ratings, source-certified supply edges and the utility national stock aggregate. Unknown quantities and title remain unknown.
Origin composition at every Japanese LNG port	National import composition and the admitted Kawasaki port evidence; neither is silently assigned to another terminal.
Regional resource-class capacity auction awards	Published regional prices/awarded MW and national resource classes. Suppressed regional classes are not reconstructed from totals.
Final effective implementation of the October 2026 imbalance change	The verified announced treatment is labelled announced, not an implemented operating rule.
Complete national or regional battery MWh inventory	Dated operating-project MW/MWh where disclosed and separately scoped connected-MW statistics. No standard duration converts missing energy ratings.
Continuous ownership history and every commissioning date	Dated legal equipment holders, source-qualified start precision and explicit unknown/ambiguous age buckets. Gross capacity is not an unsupported equity allocation.

The electricity-survey workbook route was checked through the publisher and browser, then against official alternatives. e-Stat's accessible older database does not provide the required modern area history; OCCTO's report supports the separately labelled composition snapshot. An inaccessible original is not treated as numerical evidence. [27] [28]

Original-source rights remain with their publishers. Captures, calculations and citations establish provenance; they do not assert permission for unrestricted public republication of third-party images.

Sources and reading guide

Sources are grouped into a reading route through the evidence. Publication, effective and retrieval dates are listed separately because a later lookup cannot establish what participants knew during an event. Full records, including local snapshot paths, hashes and usage restrictions, are in sources.csv.

Observed system, price and transfer data

The primer's numerical exhibits come from a frozen local snapshot of the ten transmission utilities' area supply–demand files, JEPX spot and intraday results, OCCTO interconnection publications and a representative-point weather reanalysis. Each record gives the snapshot file and capture date; the figure ledger and extracts shipped with the report give the exact population behind every chart. Readers who want the originals should start from OCCTO's links to the ten utilities' publications and JEPX's market-data pages, noting that revisions, reporting boundaries and time conventions differ between publishers.

1. Local collection (data/database), primer snapshot of prices.parquet. JEPX spot and intraday market results, parsed
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/prices.parquet
2. Local collection (data/database), primer snapshot of demand.parquet. TSO area supply-demand files (ten general transmission and distribution utilities), parsed area demand
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/demand.parquet
3. Local collection (data/database), primer snapshot of generation_history.parquet. TSO legacy hourly supply-demand files (FY2016–FY2023), parsed
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/generation_history.parquet
4. Local collection (data/database), primer snapshot of generation.parquet. TSO area supply-demand files, parsed generation categories, storage and curtailment
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/generation.parquet
5. Local collection (data/database), primer snapshot of weather.parquet. Open-Meteo reanalysis at one representative point per area
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/weather.parquet
6. Local collection of official OCCTO/HJKS originals. OCCTO measured unit generation with reported and capacity-qualified values
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/generator_units/unit_generation.parquet
7. Local collection of official OCCTO/HJKS originals. HJKS notice capture/revision history, raw dates and withdrawal status
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/generator_units/unit_outage_versions.parquet
8. Local collection (data/database), primer snapshot of generator_units/unit_outages.parquet. HJKS (power-generation information disclosure) unit outage notices, parsed
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/generator_units/unit_outages.parquet
9. Local collection (data/database), primer snapshot of fundamental/fuel_prices_monthly.parquet. World Bank Pink Sheet monthly commodity prices and ECB EXR USD/JPY, compiled
   Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/fundamental/fuel_prices_monthly.parquet
10. Local collection (data/database), primer snapshot of imports.parquet. TSO area supply-demand files, parsed area interconnection/net-import column
    Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/imports.parquet
11. Local collection (data/database), primer snapshot of interconnector_actual_flows.parquet. OCCTO interconnection actual-snapshot publication, parsed
    Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/interconnector_actual_flows.parquet
12. Local collection (data/database), primer snapshot of lng_terminals.csv. LNG terminal list compiled from public sources
    Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/lng_terminals.csv
13. Local collection (data/database), primer snapshot of lng_stock_national.parquet. METI/ANRE LNG stock of major power utilities, discrete observations
    Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/local/lng_stock_national.parquet

Institutions, networks and reliability

Institutional and network sources explain who balances the system, how areas connect and how reserves and curtailment are organised. Maps and planning documents are dated: a master-plan scenario describes a proposal at its publication date rather than commissioned capacity, and operating limits must be read from the operator's current publications.

14. FEPC. Interconnected transmission network and frequency geography
    Read: 2026-11-07. Captured publisher page; dates and rule scope retained
15. Okinawa Electric Power. 会社説明会 — 2026年3月26日
    Published: 2026-03-26; Effective: 2026-03-26; Read: 2026-11-06. PDF/printed pp.6–7: island geography and small independent systems, no mainland interconnection
16. JEPX. 一般社団法人日本卸電力取引所 取引規程 — captured original
    Published: 2026-08-31; Read: 2026-11-06. Arts3,8–11,12–28,78–89; pp.1–8,19–22; pp.28–29 revision list
17. Japan Electric Power Information Center (JEPIC). Electric Power Industry in Japan 2025 — network foundation
    Published: 2025-07; Effective: 2025-07; Read: 2026-11-06. Printed/PDF pp.34–36: Bulk Transmission System; Reinforcement Plan; Structure of the Power System
18. OCCTO. Role of OCCTO in the Electricity System Reform
    Read: 2026-11-07. Captured publisher page; dates and rule scope retained
19. EPRX. About the Electric Power Reserve Exchange
    Read: 2026-11-07. Captured publisher page; dates and rule scope retained
20. Japan Electric Power Information Center (JEPIC). Electric Power Industry in Japan 2025
    Published: 2025-07; Read: 2026-11-06. Printed/PDF pp. 1–24; p. 5 energy self-sufficiency, pp. 7–14 policy, p. 16 history, pp. 18–22 institutions/markets, p. 24 demand
21. OCCTO. 地内系統混雑を踏まえた当面の需給運用 — 第109回調整力及び需給バランス評価等に関する委員会 資料2

Published: 2025-06-24; Read: 2026-11-06. Pp2,4,11–14,16,19–27: area/wide-area reserves, notional equalisation vs KJC, internal limitations

22. JEPX. Trading Rules — official original verified 7 November 2026
Read: 2026-11-07. Articles 11–28; byte-identical to the previously retained October capture
23. OCCTO. Grid connection and use rules: non-firm access and redispatch
Read: 2026-11-07. Captured publisher page; dates and rule scope retained
24. OCCTO. 間接オークションの概要
Read: 2026-11-06. 間接オークションとは; rule-change overview
25. METI Agency for Natural Resources and Energy. なるほど！グリッド — 出力制御について
Effective: Various explicitly dated historical implementations; Read: 2026-11-06. Sections1–3: energy versus network curtailment, priority dispatch, online proxy control and compensation, Connect&Manage, nonfirm access, N-1, redispatch
26. Federation of Electric Power Companies of Japan. 全国を連携する送電線
Read: 2026-11-06. お客さまへ電気をお届けするまで; 電力の安定供給に向けた日常の取り組み
27. OCCTO. FY2025 Annual Report — dated generating-capacity composition
Published: 2025-12; Read: 2026-11-07. PDF pages 138–140 / printed IV-18–20; Table 3-1 and Figure 3-2
28. OCCTO; primer transcription. Ten-area FY2024-end capacity composition transcribed from Figure 3-2
Read: 2026-11-07. 100 region/fuel records; source PDF physical page 140, printed IV-20
29. Japan Power Handbook. Edition methods and explicitly hypothetical teaching examples
Published: 2026-11-06; Effective: Edition cutoff 2026-11-05; common descriptive window 2024-04-01–2026-09-27; Read: 2026-11-06. PLAN.md, Chapters05/33–36, snapshot/figure/evidence ledgers

Demand, weather, generation and storage

These sources support the demand, renewable, nuclear and storage chapters. Capacity statistics follow the publisher's own population: FIT/FIP 'introduced' capacity means purchases under the scheme have started, and grid-connected battery figures are dated power ratings distinct from applications, awards or energy capacity.

30. Agency for Natural Resources and Energy (ANRE), METI. 燃料費調整制度について
Published: 2023-12-22; Effective: Mechanism explanation updated 2023-12-22; monthly mechanism introduced FY2009; Read: 2026-11-06. 燃料費調整制度とは; 燃料費調整単価の算定方法; 制度見直しの概要（平成21年度）
31. Japan Meteorological Agency. Overview of Japan's climate
Effective: Climatological explanation; not an event chronology; Read: 2026-11-06. Winter, Spring, Summer and Autumn subsections
32. Japan power primer; official-source transcription. Chugoku hourly historical repair from FY2016–2023 official CSVs
Read: 2026-11-07. hourly_history_repair.csv
33. Okinawa Electric Power. Management reference materials, May 2025
Read: 2026-11-07. Whole original; publisher date/units preserved
34. Cabinet Office, Government of Japan. 国民の祝日・休日月日 (national holidays and substitute/citizens' holidays), CSV
Read: 2026-11-07. Whole file; columns 国民の祝日・休日月日 (date) and 国民の祝日・休日名称 (name); Shift_JIS encoded; HTTP Last-Modified Mon, 02 Feb 2026 00:30:17 GMT
35. Shikoku Electric Power Transmission & Distribution. Supply-demand publication handling and disclaimer
Read: 2026-11-07. Whole original; publisher date/units preserved
36. Shikoku Electric Power Transmission & Distribution. Native supply-demand actuals 202406
Read: 2026-11-07. Whole original; publisher date/units preserved
37. Shikoku Electric Power Transmission & Distribution. Native supply-demand actuals 202506
Read: 2026-11-07. Whole original; publisher date/units preserved
38. NRA. Present states of operation
Published: 2026-10-29; Read: 2026-11-07. Saved original; curated records retain field scope
39. NRA. Draft tenth national report under Convention on Nuclear Safety; reactor appendix pp127-129
Published: 2025-09; Read: 2026-11-07. Saved original; curated records retain field scope
40. TEPCO. Kashiwazaki Kariwa 6 commercial operation
Published: 2026-04-16; Read: 2026-11-07. Saved original; curated records retain field scope
41. Chugoku Electric. Shimane 2 commercial operation resumes
Published: 2025-01-10; Read: 2026-11-07. Saved original; curated records retain field scope
42. OCCTO. Annual supply demand report and electrical territories
Published: 2026-02-06; Read: 2026-11-07. Saved original; curated records retain field scope
43. Primer factual transcription from named official/operator originals. Dated commercial reactor status specifications and confirmed milestones
Read: 2026-11-07. reactor_history.csv
44. Tohoku Electric. Onagawa 2 commercial operation resumes
Published: 2024-12-26; Read: 2026-11-07. Saved original; curated records retain field scope
45. Primer factual transcription from named official/operator originals. Qualified regional FIT/FIP purchase-start trajectories
Read: 2026-11-07. regional_vre_capacity.csv
46. Agency for Natural Resources and Energy / METI. FIT/FIP introduced-capacity publication and download index
Read: 2026-11-07. Publication notices; table definition and quarterly A-workbook links
47. Agency for Natural Resources and Energy / METI. Quarterly prefectural introduced renewable capacity: 36 releases, 2017-06 to 2026-03 [47–82]
Each release is cited separately; its URL, hash and observation date are listed in sources.csv. Read: 2026-11-07. Tables A②-1 and A②-2; prefecture rows and total row; solar <10 / ≥10 kW, wind <20 / ≥20 kW
83. Kyushu Electric Power Transmission and Distribution. Renewable output control under priority dispatch

Read: 2026-11-07. Saved original; curated records retain field scope

84. Kyushu Electric Power Transmission and Distribution. Kyushu mainland renewable control times FY2026
Published: 2026-03-24; Read: 2026-11-07. Saved original; curated records retain field scope
85. ENEOS Power / EGC. ENEOS Power battery business
Published: 2024-06-11; Read: 2026-11-07. Saved original; curated records retain field scope
86. ORIX. Kinokawa battery begins operation
Published: 2024-11-29; Read: 2026-11-07. Saved original; curated records retain field scope
87. Kansai Electric. Kansai integrated report 2025; commissioned Kinokawa
Published: 2025; Read: 2026-11-07. Saved original; curated records retain field scope
88. Hexa Energy Services. Tagawa BESS operational completion
Published: 2026-03-24; Read: 2026-11-07. Saved original; curated records retain field scope
89. JAPEX. Mihama battery commercial operation
Published: 2025-08-01; Read: 2026-11-07. Saved original; curated records retain field scope
90. Tohoku Electric. Nishi Sendai battery commercial operation
Published: 2015-02-20; Read: 2026-11-07. Saved original; curated records retain field scope
91. Kyushu Electric. Buzen storage operation begins
Published: 2016-03-03; Read: 2026-11-07. Saved original; curated records retain field scope
92. TEPCO / NTT Anode Energy. Tsumagoi battery commercial operation
Published: 2025-05-15; Read: 2026-11-07. Saved original; curated records retain field scope
93. erex. Kushima battery commercial operation
Published: 2026-04-07; Read: 2026-11-07. Saved original; curated records retain field scope
94. Japan Electric Power Exchange. HJKS operation manual: output reductions and date precision
Published: 2020-09-23; Effective: 2020-10-01; Read: 2026-11-07. manual pp.1,3,12; current index footer link
95. Japan Electric Power Exchange. HJKS current public entry and currently linked operating manual
Read: 2026-11-07. current manual link and publication policy
96. Primer factual transcription from named official/operator originals. Supply requested-analysis evidence qualifications
Read: 2026-11-07. analysis_coverage.csv
97. Cabinet Office, Government of Japan. 国民の祝日について
Effective: 2026 annual calendar and Holiday Act Articles 2–3; Read: 2026-11-06. 令和8年（2026年）の国民の祝日・休日; 国民の祝日に関する法律 Articles 2–3; 2020/2021 special moves

Fuels, legislation and environmental markets

Fuel references, statutes and environmental-market documents are listed with publication, effective and retrieval dates. Statutory obligations, implementing notices, voluntary schemes and policy proposals are different kinds of evidence; where a publisher restricts redistribution of prices, the primer relies on permitted aggregates or mechanism explanations.

98. Natural Earth. Natural Earth 1:50m countries: Japan geographic context
Read: 2026-11-08. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/ne_50m_admin_0_countries.geojson
99. JERA. LNG terminal storage scope, connected plants and pipeline groups, 6 December 2021
Published: 2021-12-06; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-jera-terminal.html
100. Chubu Electric. Joetsu line commissioned to Shin-Hokushin substation, 8 June 2011
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-joetsu-grid.html
101. INPEX. Naoetsu LNG terminal
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-inpex-naoetsu.html
102. Daigas Gas and Power Solution. Daigas operated thermal power stations
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-daigas-plants.html
103. Hibiki LNG. Hibiki LNG terminal specifications
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-hibiki.html
104. Hokuriku Electric. Toyama Shinko LNG facilities
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-hokuriku.html
105. Hokkaido Gas. Ishikari LNG terminal equipment and shared site
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-ishikari.html
106. Kansai Electric. Himeji LNG terminal supplies Himeji No.1 and No.2 power stations (reader text)
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-kepco-himeji.reader.txt
107. Kansai Electric. Sakai LNG terminal and connected station
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-kepco-sakai.html
108. Kansai Electric. Published LNG storage and vaporizer capacities (reader proxy text)
Published: 2026-05; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-kepco-storage.reader.txt
109. Tokyo Gas. Ohgishima tank 4: 250000 cubic metres; recognition date not commissioning
Published: 2026-04-20; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-ohgishima-tank4.html
110. Sakaide LNG. Sakaide LNG terminal specifications
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-sakaide.html
111. Tohoku Electric. LNG operating lower limits and connected generation
Published: 2024-12-17; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-tohoku.pdf

112. Tokyo Gas. Four-terminal Tokyo Gas pipeline system
Published: 2026-09-15; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-tokyogas-network.html

113. Chugoku Electric. Yanai LNG access and facility specifications July 2018
Published: 2018-07; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-yanai.pdf

114. Okinawa Electric. May 2020 management reference: Yoshinoura LNG terminal
Published: 2020-05; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-yoshinoura.pdf

115. METI. Calendar 2024 LNG imports by country and month (reader proxy text)
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-import-origin.reader.txt

116. Yokohama Customs. Kawasaki port calendar 2024 trade results
Published: 2025-03-05; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-customs-kawasaki.pdf

117. JOGMEC. Spot LNG price survey 202609
Read: 2026-11-07. Monthly contract-based and arrival-based survey table

118. Agency for Natural Resources and Energy (ANRE), Electricity Infrastructure Division. Power-generation LNG inventory status, 4 November 2026
Published: 2026-11-04; Read: 2026-11-07. p1 actual week-end stock rows 13September–1November; later blank dates excluded

119. JOGMEC. Spot LNG price survey: 64 releases, 2021-05 to 2026-08 [119–182]
Each release is cited separately; its URL, hash and observation date are listed in sources.csv. Read: 2026-11-07. Monthly contract-based and arrival-based survey table

183. Agency for Natural Resources and Energy (ANRE), METI. LNG market developments and Strategic Buffer LNG, 1 December 2023
Published: 2023-12-01; Read: 2026-11-07. Document pp5–8: SBL arrangements; third-party outlook pages not used

184. Japan Organization for Metals and Energy Security (JOGMEC). Additional JOGMEC operations following legal amendment: emergency fuel procurement and risk finance
Read: 2026-11-07. News release: emergency fuel procurement under government request

185. Government of Japan; e-Gov Laws published by the Digital Agency. ガス事業法 (昭和二十九年法律第五十一号)
Published: 2024-06-26; Effective: 2025-12-25; Read: 2026-11-07. e-Gov consolidated law text, revision 329AC0000000051_20251225_506AC0000000067; operative provisions and supplementary commencement clauses

186. Government of Japan; e-Gov Laws published by the Digital Agency. 電気事業法 (昭和三十九年法律第百七十号)
Published: 2026-07-24; Effective: 2026-08-03; Read: 2026-11-07. e-Gov consolidated law text, revision 339AC0000000170_20260803_508AC0000000068; operative provisions and supplementary commencement clauses

187. Government of Japan; e-Gov Laws published by the Digital Agency. 経済施策を一体的に講ずることによる安全保障の確保の推進に関する法律 (令和四年法律第四十三号)
Published: 2026-06-17; Effective: 2026-09-18; Read: 2026-11-07. e-Gov consolidated law text, revision 504AC0000000043_20260918_508AC0000000038; operative provisions and supplementary commencement clauses

188. Government of Japan; e-Gov Laws published by the Digital Agency. 独立行政法人エネルギー・金属鉱物資源機構法 (平成十四年法律第九十四号)
Published: 2024-05-24; Effective: 2026-05-22; Read: 2026-11-07. e-Gov consolidated law text, revision 414AC0000000094_20260522_506AC0000000038; operative provisions and supplementary commencement clauses

189. Government of Japan; e-Gov Laws published by the Digital Agency. 石油の備蓄の確保等に関する法律 (昭和五十年法律第九十六号)
Published: 2022-06-17; Effective: 2025-06-01; Read: 2026-11-07. e-Gov consolidated law text, revision 350AC0000000096_20250601_504AC0000000068; operative provisions and supplementary commencement clauses

190. Agency for Natural Resources and Energy (ANRE), METI. Fuel-cost adjustment system for electricity tariffs
Read: 2026-11-07. Regulated-tariff calculation: base/actual fuel cost, conversion coefficients, three-month window and billing lag

191. Government of Japan; e-Gov Laws published by the Digital Agency. 石油石炭税法 (昭和五十三年法律第二十五号)
Published: 2024-06-14; Effective: 2026-05-25; Read: 2026-11-07. e-Gov consolidated law text, revision 353AC0000000025_20260525_506AC0000000052; operative provisions and supplementary commencement clauses

192. Government of Japan; e-Gov Laws published by the Digital Agency. 租税特別措置法 (昭和三十二年法律第二十六号)
Published: 2025-03-31; Effective: 2026-11-01; Read: 2026-11-07. e-Gov consolidated law text, revision 332AC0000000026_20261101_507AC0000000013; operative provisions and supplementary commencement clauses

193. Ministry of the Environment, Japan. Climate tax: fuel-specific rates and phased introduction
Effective: 2016-04-01; Read: 2026-11-07. Rate graphic: base rates plus climate additions for oil, gas and coal

194. METI. GX Promotion Act emissions trading: FY2026 introductory leaflet
Published: 2026; Effective: 2026-04-01; Read: 2026-11-07. pp1–4: 100,000-tonne threshold, first-year special timetable, verified emissions and holdings

195. Government of Japan; e-Gov Laws published by the Digital Agency. 脱炭素成長型経済構造への円滑な移行の推進に関する法律 (令和五年法律第三十二号)
Published: 2025-06-04; Effective: 2026-04-01; Read: 2026-11-07. e-Gov consolidated law text, revision 505AC0000000032_20260401_507AC0000000052; operative provisions and supplementary commencement clauses

196. METI. METI Notice No.29: implementation guidelines for emissions trading
Read: 2026-11-07. Notice and attached implementation guidelines under GX Act Article32(1)

197. METI. METI Notice No.27: reference upper and adjustment reference trading prices
Published: 2026-03-30; Effective: 2026-04-01; Read: 2026-11-07. Articles1–2 and supplementary provisions; 30 March2026 notice

198. Japan Electric Power Exchange (JEPX). Outline of non-fossil certificate trading
Read: 2026-11-07. Trading products, participants and auction arrangements

199. Tokyo Metropolitan Government, Bureau of Environment. Tokyo Cap-and-Trade Program for Large Facilities: fourth compliance period
Published: 2025-04; Effective: 2025-04-01; Read: 2026-11-07. April2025 leaflet pp1–4: coverage, reporting/compliance designation, periods and eligible instruments

200. Saitama Prefecture, Global Warming Measures Division. Climate Change Planning and Target-Setting Emissions Trading: fourth-period leaflet

Published: 2025-03; Effective: 2025-04-01; Read: 2026-11-07. March2025 leaflet: planning population, three-year 1,500kL facility threshold, periods and eligible credits

201. J-Credit Scheme Secretariat (METI, Ministry of the Environment and MAFF). About the J-Credit Scheme
Read: 2026-11-07. Scheme purpose, certification of reductions/removals and uses

202. GX League Secretariat. Rules for the first phase of GX-ETS
Published: 2023-02; Read: 2026-11-07. February2023 rules: targets, reporting and excess-reduction framework

203. JPX. Carbon-credit external price redistribution restriction
Published: 2026-11-06; Read: 2026-11-07. Four-Price Information service paragraph; alternate https://www.jpx.co.jp/markets/paid-info-equities/reference/10.html

204. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2026
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

205. Japan Electric Power Exchange (JEPX). Non-fossil Value Trading Regulations
Read: 2026-11-07. Trading products, auctions, settlement and Article11 price restrictions

206. Japan Electric Power Exchange (JEPX). Non-fossil trading price limits, 10 July 2023
Read: 2026-11-07. Product minimum/maximum table under trading rule Article11

207. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2017
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

208. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2018
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

209. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2019
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

210. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2020
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

211. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2021
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

212. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2022
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

213. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2023
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

214. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2024
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

215. Japan Electric Power Exchange (JEPX). Non-fossil certificate auction results FY2025
Read: 2026-11-07. Original CSV: fiscal year, round, product, auction date, awarded kWh, reported price and award extrema; hyphens are missing

216. Tokyo Gas. Tokyo Gas Investors Guide 2020 terminal capacity scope
Published: 2020; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-fuels-tokyogas-storage2020.pdf

Market rules, products and exchanges

Rulebooks, product specifications and exchange reports define how prices settle and what a contract covers. Product terms change: check the effective date of each specification, and keep exchange volume, open interest and bilateral activity separate when comparing venues.

217. OCCTO. 容量市場メインオークションとは
Read: 2026-11-06. Mechanism, participants, four-year main and one-year additional auctions

218. jepx_stack research. Clearing-engine replication results
Published: 2026-09-29; Read: 2026-11-06. Sections1–2 and5; lines48,63,126,136

219. JEPX. 取引規程 — frozen official original
Published: 2026-08-31; Read: 2026-11-06. Articles3,5,11,14–18,21–22,27–28,46–57,62–68,78–89

220. JEPX. 非化石価値取引市場データ
Read: 2026-11-06. FIT/non-FIT renewable designation categories; displayed units and weighted-average price label

221. OCCTO. Capacity market actors and main-auction calendar
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-about.html

222. OCCTO. Capacity provider delivery-year obligations
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-after.html

223. OCCTO. Retail contribution collection calendar
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-contributions.html

224. METI / ANRE. 再エネを日本の主力エネルギーに！FIP制度が2022年4月スタート
Published: 2021-08-03; Effective: 2022-04 start; Read: 2026-11-06. FIT/FIP comparison, reference price, monthly premium, aggregation

225. Electric Power Reserve Exchange. 商品要件一覧第6版
Published: 2026-03-13; Read: 2026-11-06. Pages2–4: product requirements, timing, revision history

226. OCCTO. Capacity main auction results delivery FY2024
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2024.zip

227. OCCTO. Capacity main auction results delivery FY2029
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2029.zip

228. OCCTO. Regional clearing and market splitting
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-split.html

229. Electricity and Gas Market Surveillance Commission. グロス・ビディングに関する検討
Published: 2025-06-27; Read: 2026-11-06. Printed pp2–4,9

230. Electricity and Gas Market Surveillance Commission. 2025年4月～6月期電力取引報告の結果
Published: 2025-09-24; Read: 2026-11-06. Printed pp5–7 quarterly table and gross-bidding abolition footnote

231. Electric Power Reserve Exchange. FY2025 balancing-market trading results
Published: 2026-06-18; Read: 2026-11-07. Pages5,8,16,24: members and product procurement-shortfall rates

232. METI / ANRE. インバランス料金制度の見直しに関する意見募集に対する回答
Published: 2026-03-27; Effective: 2026-10-01 planned; Read: 2026-11-06. Pages2–7,11–12: response, current framework, announced1October2026 revision

233. METI / ANRE. Reader-proxy text of METI 005_09_00.pdf; direct GET blocked by AWS WAF human verification
Published: 2026-03-27; Read: 2026-11-06. Document dated 27 March 2026, pages 2–7 and 11–12

234. OCCTO. Balancing committee 110: imbalance-price regime review
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-markets-imbalance-review.pdf

235. TEPCO Power Grid / Okinawa Electric Power. Imbalance notices: attempted implementation verification
Read: 2026-11-06. Notice index; TEPCO counterpart https://www.tepco.co.jp/pg/consignment/workshop/procedure/fees-terms/imbalance/index-j.html

236. OCCTO. Capacity main auction results delivery FY2025
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2025.zip

237. OCCTO. Capacity main auction results delivery FY2026
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2026.zip

238. OCCTO. Capacity main auction results delivery FY2027
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2027.zip

239. OCCTO. Capacity main auction results delivery FY2028
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2028.zip

240. OCCTO. Capacity additional auction results delivery FY2025
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-additional-2025.zip

241. OCCTO. Capacity additional auction results delivery FY2026
Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-additional-2026.zip

242. OCCTO. Additional capacity auction delivery FY2027 results 6 August 2026
Published: 2026-08-06; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-additional-2027.pdf

243. OCCTO. Main capacity delivery FY2029 result report corrected 23 January 2026
Published: 2026-01-23; Read: 2026-11-07. reports/japan_power_market_primer/data/snapshot/online/trader_fuels_markets/trader-capacity-main-2029-report.pdf

244. OCCTO. 長期脱炭素電源オークションとは
Effective: FY2023 start; Read: 2026-11-06. Introduction and in-principle20-year period

245. OCCTO. 長期脱炭素電源オークションの仕組み
Read: 2026-11-06. Multi-price auction, JPY/kW/year, other-market revenue return

246. METI / ANRE. 令和6年度エネルギーに関する年次報告: 第2部第6章第1節
Published: 2025; Read: 2026-11-06. Sections2(3)–(6),3(2)–(3): capacity awards, non-fossil market, wholesale access, futures, balancing

247. JPX / TOCOM. Electricity futures: monthly contract specifications
Published: 2026-04-13; Read: 2026-11-06. Monthly base/peak Tokyo,Kansai,Chubu; size, tick,24months, trading hours, settlement,JSCC

248. European Energy Exchange. Reference Japanese translation; German governing text prevails. Pages 7–16 and 25 distinguish monthly average-price options and futures-style premium.
Published: 2026-05-29; Read: 2026-11-07. PDF member pages 5–16,25; contract-details spreadsheet

249. European Energy Exchange. Japan Power Overview, May 2026
Published: 2026-05-29; Read: 2026-11-06. Pages 2–8

250. JPX / TOCOM. Electricity futures official contract units and delivery calendars
Read: 2026-11-07. Named base/peak contract-unit and eligible-day columns

251. ICE Futures Europe. Japanese Tokyo Area Power Financial Base Futures
Read: 2026-11-06. JTB: contract size, quote, cash settlement, underlying index, expiry, trading hours,ICEU

252. ICE Data Services. Report Center access and redistribution conditions
Read: 2026-11-07. Report304 Japanese power route; access notice and Report Center disclaimer

253. European Energy Exchange. EEX Group monthly volume table for 2025-11
Published: 2025-12-02; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

254. European Energy Exchange. EEX Group annual volume table 2025 vs 2024
Published: 2026-01-13; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

255. European Energy Exchange. EEX Group monthly volume table for 2025-12
Published: 2026-01-13; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

256. European Energy Exchange. EEX Group monthly volume table for 2026-01
Published: 2026-02-04; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

257. European Energy Exchange. EEX Group monthly volume table for 2026-02
Published: 2026-03-06; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

258. European Energy Exchange. EEX Group monthly volume table for 2026-03
Published: 2026-04-08; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

259. European Energy Exchange. EEX Group monthly volume table for 2026-04

Published: 2026-05-08; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

260. European Energy Exchange. EEX Group monthly volume table for 2026-05
Published: 2026-06-05; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

261. European Energy Exchange. EEX Group monthly volume table for 2026-06
Published: 2026-07-09; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

262. European Energy Exchange. EEX Group monthly volume table for 2026-07
Published: 2026-08-06; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

263. European Energy Exchange. EEX Group monthly volume table for 2026-08
Published: 2026-09-04; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

264. European Energy Exchange. EEX Group monthly volume table for 2026-09
Published: 2026-10-07; Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

265. European Energy Exchange. EEX Group monthly volume table for 2026-10
Read: 2026-11-06. Page 1, Japan Power Futures / Japan Power Options

266. JPX / TOCOM. Summary of Statistics — month
Read: 2026-11-07. BO_DM0040/41/42: product/year/month, total volume, J-NET and end-period open interest columns

267. JPX / TOCOM. JPX trading-participant volume: electricity/LNG exclusion
Read: 2026-11-07. HTML notes: excluded products

268. Japan Securities Clearing Corporation. VaR margin methodology
Effective: 2023-11-06; Read: 2026-11-06. Historical-simulation electricity margin;6November2023 transition

269. JEPX. JEPX indirect transmission rights auction results FY2019 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

270. JEPX. JEPX indirect transmission rights auction results FY2026 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

271. JEPX. JEPX indirect transmission rights auction results FY2025 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

272. JEPX. JEPX indirect transmission rights auction results FY2020 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

273. JEPX. JEPX indirect transmission rights auction results FY2021 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

274. JEPX. JEPX indirect transmission rights auction results FY2022 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

275. JEPX. JEPX indirect transmission rights auction results FY2023 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

276. JEPX. JEPX indirect transmission rights auction results FY2024 (CSV served by the market-data page via csv_read.php; plain GET with Referer header)
Read: 2026-11-06. All CSV rows; trade date, direction, delivery start/end retained

277. JEPX. 取引ガイド第2.00版
Published: 2019; Read: 2026-11-06. Spot bidding and market explanation

Regions, interchange and transfers

Regional interchange definitions and OCCTO transfer publications determine what a signed quantity measures. Final schedules, measured samples, operating limits and auction-stage inputs are distinct quantities, and their publication formats changed during the edition window.

278. OCCTO. FY2024 electricity supply/demand and system report: interconnector directions
Published: 2025-09-10; Effective: 2025-03-31; Read: 2026-11-07. Printed page23, Table2-1: all ten forward/reverse regional pairs and named equipment

279. OCCTO. Central-region AC-loop operating-capacity publication changes (February2026 notice)
Published: 2026-02-13; Read: 2026-11-07. PDF pages2–4: display switch March11; release/new-fence values early March13; individual series remain historical

280. OCCTO. System Information Service operating manual (2026-03-13 revision)
Published: 2026-03-13; Read: 2026-11-07. Section2.4 formulated versus updated and day-ahead CSV; Section6.3 Table6-2 measured-flow MW fields; Section24 information download

281. OCCTO; primer source curation. Native measured-flow and physical-capacity history
Read: 2026-11-07. Every row preserves original file/source ID, native timestamp, raw clock, physical boundary/fence and publication eligibility

282. OCCTO; primer source curation. Native formulated day-ahead directional limit and planned-transfer history
Read: 2026-11-07. Every row preserves original file/source ID, native timestamp, raw clock, physical boundary/fence and publication eligibility

283. OCCTO. Interconnector capacity impacts of scheduled work, FY2026–2027 (March2026 vintage)
Published: 2026-03-01; Read: 2026-11-07. PDF pages5–6; Tohoku–Tokyo and converter work windows

284. OCCTO. Interconnector capacity impacts of scheduled work, FY2025–2026 (March2025 vintage)
Published: 2025-03-01; Read: 2026-11-07. PDF pages9,10,12,14; exact row locators retained in boundary_limits.csv

Event investigations and chronologies

Official weather summaries, supply–demand instructions, outage notices and investigation reports document what happened during each case. Publication dates matter: information released after an event can explain it but was not available to participants at the time.

285. Primer research computation. Eight case lead/core/aftermath observations
Read: 2026-11-07. event_phase_findings.csv

286. Local frozen JEPX/TSO source collection. Frozen event selection populations

Read: 2026-11-06. prices.parquet productday_ahead/half_hourly_spot;generation.parquet Kyushu solar_curtailment_mw

287. JMA. 日々の天気図 2026年2月 No289
Read: 2026-11-06. PDFp1 header and7–9,15February entries

288. JMA. 2026年2月の天候
Published: 2026-03-02; Read: 2026-11-06. Overview and旬別 upper/middle thirds

289. OCCTO. 2025年度冬季における広域機関の需給対応
Published: 2026-03-19; Read: 2026-11-06. Section3.1 Hokkaido;3.2 Tohoku

290. Cabinet Office. 国民の祝日・休日 CSV
Read: 2026-11-06. Date/name columns; publication-side subset2020–2026

291. Primer research computation. Dated daily observations within original case windows
Read: 2026-11-07. event_daily_findings.csv

292. Primer research computation. Physical case energy integrals
Read: 2026-11-07. event_energy_findings.csv

293. OCCTO. Published reserve-block archive: case-07 2026-01-31–2026-02-16
Read: 2026-11-07. CSV first-line latest update; target date/time/area and area reserve MW columns

294. OCCTO. Reserve clock qualification browser evidence
Read: 2026-11-07. Browser-rendered text and acquisition decision

295. OCCTO. 広域予備率Web公表システム 操作マニュアル 一般利用者用
Published: 2026-03-13; Read: 2026-11-07. pp12,15,46–48; date retention, latest update field, CSV columns

296. Primer research computation. All-nine area case price observations
Read: 2026-11-07. event_nine_area_findings.csv

297. Primer research computation. Eight case preserved same-calendar references
Read: 2026-11-07. event_reference_findings.csv

298. JMA. 2026年8月の天候
Published: 2026-09-01; Read: 2026-11-06. 旬別 upper/middle/lower thirds; western lateAugust+2.5°C

299. OCCTO. 2026年度夏季における広域機関の需給対応
Published: 2026-10-28; Read: 2026-11-06. Section3.5 Kansai; section3 later candidates

300. Kansai Electric Power. 大飯発電所 3 号機の原子炉自動停止
Published: 2026-08-09; Effective: 2026-08-08; Read: 2026-11-06. p1 actual generator/reactor stop8August23:56 following loss-of-field alarm

301. Fukui Prefecture Nuclear Safety Measures Division. 美浜発電所 3 号機の第29回定期検査開始について
Published: 2026-06-15; Effective: 2026-06-16; Read: 2026-11-06. pp1–3 May8steamleak stop,June16inspection; September/October milestones are plans

302. OCCTO. Published reserve-block archive: case-08 2026-07-27–2026-08-26
Read: 2026-11-07. CSV first-line latest update; target date/time/area and area reserve MW columns

303. OCCTO. Published reserve-block archive: case-08 2026-08-27–2026-09-07
Read: 2026-11-07. CSV first-line latest update; target date/time/area and area reserve MW columns

304. ANRE. 2021年初頭、電力供給が大ピンチに。どうやって乗り切った？（前編）
Published: 2021-11-15; Read: 2026-11-06. Sections December2020–January2021 chronology; 全国の電気事業者の協力; recovery

305. OCCTO. 今冬の需給ひっ迫への対応の概要について
Published: 2021-02-04; Read: 2026-11-07. pp3–10; response chronology, support and pumped-storage mechanisms

306. OCCTO. Reserve archive retention rejection: case-01 2020-12-25–2021-01-24
Read: 2026-11-07. HTML validation message: 期間は前年度4月以降を指定してください。

307. OCCTO. Reserve archive retention rejection: case-01 2021-01-25–2021-02-07
Read: 2026-11-07. HTML validation message: 期間は前年度4月以降を指定してください。

308. OCCTO. 需給状況改善のための指示の実施について 東京 3月22日5時59分
Published: 2022-03-22; Read: 2026-11-06. Sections2–3: instruction maxima/windows and reasons

309. ANRE / Cabinet Office. 2022年3月の東日本における電力需給ひっ迫について
Published: 2022-04-25; Read: 2026-11-07. pp1–6; chronology, affected facilities and one-day generation totals

310. METI / ANRE. 2022年3月の東日本における電力需給ひっ迫に係る検証 取りまとめ（案）
Published: 2022-05-27; Read: 2026-11-07. May 27 committee draft; printed pp3–7,27–33

311. OCCTO. Reserve archive retention rejection: case-02 2022-02-22–2022-03-24
Read: 2026-11-07. HTML validation message: 期間は前年度4月以降を指定してください。

312. OCCTO. Reserve archive retention rejection: case-02 2022-03-25–2022-04-07
Read: 2026-11-07. HTML validation message: 期間は前年度4月以降を指定してください。

313. Japan Meteorological Agency. 2025年2月の天候
Read: 2026-11-07. Overview and early/middle/late-month discussion

314. OCCTO. Reserve archive retention rejection: case-03 2025-02-12–2025-02-26
Read: 2026-11-07. HTML validation message: 期間は前年度4月以降を指定してください。

315. ANRE. 出力制御について
Read: 2026-11-06. Sections1–3; priority dispatch and proxy control

316. Kyushu Electric Power Transmission and Distribution. 九州本土における再エネ出力制御時間について
Published: 2026-03-24; Read: 2026-11-07. Main text and change table

317. Japan Meteorological Agency. 2026年5月の天候
Read: 2026-11-07. Overview and early/middle/late-month discussion

318. Primer research computation. Kyushu2026-05-10 paired native solar curtailment and prices

Read: 2026-11-07. event_curtailment_intervals.csv

319. OCCTO. Published reserve-block archive: case-04 2026-05-03–2026-05-17
Read: 2026-11-07. CSV first-line latest update; target date/time/area and area reserve MW columns

320. OCCTO. 2024年度夏季における広域機関の需給対応
Published: 2024-10-17; Read: 2026-11-06. Sections4–6 support, cooperation and maintenance adjustment

321. Japan Meteorological Agency. 2024年9月の天候
Read: 2026-11-07. Overview and early/middle/late-month discussion

322. OCCTO. Reserve archive retention rejection: case-05 2024-09-16–2024-09-30
Read: 2026-11-07. HTML validation message: 期間は前年度4月以降を指定してください。

323. Primer research computation. Explicit ordinary-Monday sensitivity beside preserved holiday reference
Read: 2026-11-07. event_holiday_sensitivity.csv

324. Kansai Electric Power. 美浜発電所 3 号機の本格運転再開
Published: 2026-11-04; Effective: 2026-11-04; Read: 2026-11-06. p1 actual-results chronology: September12 startup,15 generation,18 stop; October10 restart; November4 15:00 full operation

325. Kansai Electric Power. 美浜発電所 3 号機 空気抜き弁周辺からの水の滴下に関する調査状況
Published: 2026-09-25; Effective: 2026-09-17; Read: 2026-11-06. pp1–2 discovery17September15:45, load reduction20:30; piping investigation andSeptember12 startup mention

326. Japan Atomic Industrial Forum. 日本の原子力発電所の運転実績 2026年9月
Published: 2026-10-09; Effective: 2026-09-30; Read: 2026-11-06. p1 allreactor table; Mihama3 12996MWh,61h,September15parallel/18stop; allotherplants720or0h

327. OCCTO. Published reserve-block archive: case-06 2026-09-05–2026-09-25
Read: 2026-11-07. CSV first-line latest update; target date/time/area and area reserve MW columns

328. Handbook frozen-data derivation. Observed event pre/core/post periods and instruction-window extracts
Read: 2026-11-06. observed_period_summaries.csv; restart_period_summaries.csv; cold_instruction_observations.csv; derived from data/regional_observations.parquet and weather_hourly.parquet

Companies, assets and fleets

Company and asset sources identify operators, parents and joint ventures. Fleet totals in the primer count eligible physical units once and are not equity-attributed capacity or available supply.

329. J-POWER Transmission Network. Transmission business overview
Effective: retrieved2026-11-06; structural facts and stated plan; Read: 2026-11-06. Sections Sakuma frequency converter and Honshi line:300000kW to600000kW plan; Honshi500kV Okayama/Kagawa

330. OCCTO charts / dated shadow direction ledger. Dated source direction and boundary identity
Effective: 2026-09-29/2026-10-25; Read: 2026-11-06. 2026-10-25.json orientation rows01–15; direct visual inspection2026-10-24_09.png Honshi positiveShikoku/negativeChugoku

331. Primer research computation. Admitted physical fleet register, dated owners, exact units and map coordinates
Read: 2026-11-07. fleet_units.csv

332. FEPC. Electricity Review Japan 2025: major plant map
Published: 2025-09-29; Read: 2026-11-07. operator asset tables / dated release

333. Kansai Electric Power. Mihama 3 actual operating milestones confirmed 4 November 2026
Published: 2026-11-04; Read: 2026-11-07. operator asset tables / dated release

334. Daigas G&P Solution. Operating thermal power plants
Read: 2026-11-07. operator asset tables / dated release

335. Okinawa Prefecture. Yoshinoura environmental assessment location: Nakagusuku Tomari
Published: 2026-07-07; Read: 2026-11-07. operator asset tables / dated release

336. Primer research computation. Admitted physical fleet regional fuel totals
Read: 2026-11-07. company_region_fuel.csv

337. JERA / Singapore Exchange. JERA 2025 offering circular: owned domestic thermal facilities
Read: 2026-11-07. operator asset tables / dated release

338. Japan Atomic Power Company. JAPC FY2025 company overview: own reactors and annual generation
Published: 2026-03-31; Read: 2026-11-07. operator asset tables / dated release

339. tepco / METI gBiz. tepco FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates

340. Primer research computation. Top-five admitted physical units by company
Read: 2026-11-07. company_top_physical_units.csv

341. JEPX HJKS / local collected register. Frozen unit master and explicit publication eligibility
Effective: 2026-11-05; Read: 2026-11-06. unit_id/source_unit_key/status/capacity_mw/operator; publication asset_enrichment.csv and company_mapping.csv

342. chubu / METI gBiz. chubu FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates

343. Chubu Electric Power / JERA. Joetsu Unit1-1 commercial operation announcement and station directory
Published: 2012-06-29; Effective: dated structural connection; station directory retrieved2026-11-06; Read: 2026-11-06. 2012-06-29 press release main text and attached station overview: four595000kW blocks, Nagano supply; corroborating JERA station page maximum2380MW

344. kansai / METI gBiz. kansai FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates

345. hokkaido / METI gBiz. hokkaido FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
346. Hokkaido Electric. Ishikariwan Shinko LNG plant outline and location
Read: 2026-11-07. operator asset tables / dated release
347. tohoku / METI gBiz. tohoku FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
348. hokuriku / METI gBiz. hokuriku FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
349. Hokuriku Electric. Nanao-Ota thermal station location and unit specifications
Read: 2026-11-07. operator asset tables / dated release
350. chugoku / METI gBiz. chugoku FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
351. shikoku / METI gBiz. shikoku FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
352. Shikoku Electric Power. Commercial operation of replacement Saijo Unit1
Published: 2023-06-30; Effective: 2023-06-30; Read: 2026-11-06. page1 new/oldUnit1 comparison and operating milestones
353. Shikoku Electric. Shikoku owned thermal facilities, unit outputs and locations
Read: 2026-11-07. operator asset tables / dated release
354. Kyushu Electric / METI gBiz. Kyushu FY2024 securities report: equipment holders and annual generation
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
355. Okinawa Electric / METI gBiz. Okinawa securities report: owned generation equipment and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
356. jpower / METI gBiz. jpower FY2024 securities report: legal asset holders, own-generation and history
Effective: 2025-03-31; Read: 2026-11-07. Major equipment/PPE and named station tables;history section for dates
357. J-POWER. J-POWER equipment and domestic site map (April 2023 scope)
Published: 2023-04-30; Read: 2026-11-07. operator asset tables / dated release
358. Tokyo Gas. Investor guide 2021: named generating units and ownership
Published: 2021-08; Read: 2026-11-07. operator asset tables / dated release
359. Tokyo Gas. Sodegaura gas-engine plant completion, ten 9.78 MW engines
Published: 2025-03-27; Read: 2026-11-07. operator asset tables / dated release
360. Tokyo Gas / Nagoya Stock Exchange. Company division disclosure including Colza FY2023 annual operating report
Published: 2025-05-27; Read: 2026-11-07. operator asset tables / dated release
361. Tokyo Gas. Ohgishima unit1 and2 commercial starts and planned3 (not actual3)
Read: 2026-11-07. operator asset tables / dated release
362. Tokyo Gas. Tokyo Gas current group-operated plant list and FY2024 retail disclosure
Read: 2026-11-07. operator asset tables / dated release
363. Osaka Gas. Himeji 1 commercial start 1 January and wholly owned project company
Published: 2026-01-05; Read: 2026-11-07. operator asset tables / dated release
364. Daigas G&P Solution. Daigas biomass operating facilities and project operators
Read: 2026-11-07. operator asset tables / dated release
365. Daigas G&P Solution. Daigas cogene operating facilities and project operators
Read: 2026-11-07. operator asset tables / dated release
366. Daigas G&P Solution. Daigas solar operating facilities and project operators
Read: 2026-11-07. operator asset tables / dated release
367. Daigas G&P Solution. Daigas wind operating facilities and project operators
Read: 2026-11-07. operator asset tables / dated release
368. Osaka Gas. Himeji 2 scheduled commercial start 1 May; corroborated by later operator directory
Published: 2026-04-30; Read: 2026-11-07. operator asset tables / dated release
369. OCCTO / bounded shadow capture. Native five-minute samples and complete six-sample means
Effective: 2026-09-29/2026-10-25; Read: 2026-11-06. flowobs_261029/native_samples.parquet and sampled_half_hours.parquet; native_count/expected_native_count/orientation_evidence